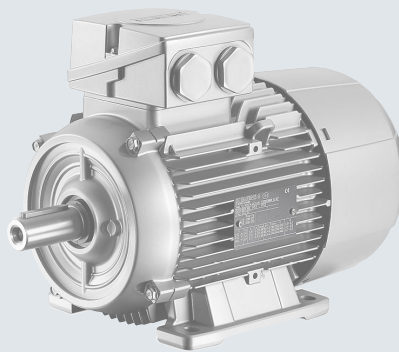
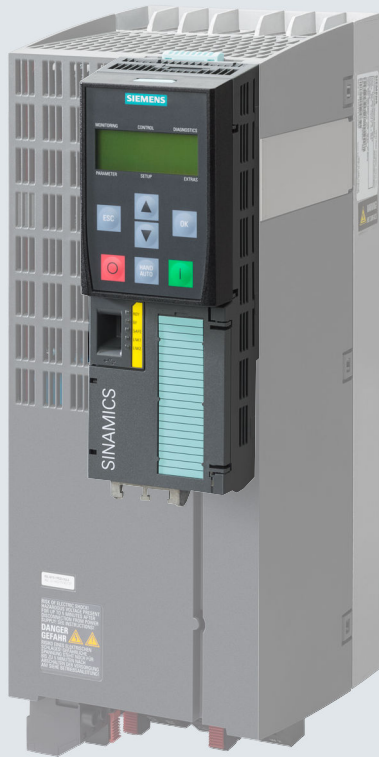


SIEMENS



Operating instructions

SINAMICS

SINAMICS G120

Low voltage converters
Chassis devices with CU240B-2 and CU240E-2
Control Units

Edition

01/2017

www.siemens.com/drives

SINAMICS

SINAMICS G120 Converter with the CU240B-2 and CU240E-2 Control Units

Operating Instructions

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Edition 01/2017, Firmware V4.7 SP6

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING

Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

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Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

Changes in the current manual

Notable changes since the 01/2016 edition of the manual

New hardware

- New PM240P-2 Power Modules, FSD ... FSF
 Power Module (Page 33)
- New version of the PM230 Power Modules, FSA ... FSC, with article numbers 6SL32 ... - ... - ... G .
 Detailed technical data, PM230 (Page 461)

Hardware that is no longer described in the manual

The following Power Modules are no longer described in the current edition of the manual:

- PM240 Power Module
- PM340 Power Module
- PM260 Power Module

Additional information is provided on the Internet:



PM240 Power Modules mounting instructions (<https://support.industry.siemens.com/cs/ww/en/view/109738501>)

Installation Guide for the PM260 Power Module (<https://support.industry.siemens.com/cs/ww/en/view/79109730>)

Corrections

- The 1PH8 induction motor may no longer be operated with the PM230 Power Module. The encoderless permanently-excited SIMOTICS S 1FK7 synchronous motor and the SIMOTICS 1FG1 may be operated on the inverter only on request.
 Motors and multi-motor drives that can be operated (Page 49)
- We recommend that you also activate the friction characteristic when you use the "moment of inertia estimator" function. An appropriate note has been added to the description of the "moment of inertia estimator" function.
 Moment of inertia estimator (Page 293)
- After selecting the "Standard Drive Control" application class, the inverter activates the slip compensation. An appropriate note has been added to the description of the "U/f control" function.
 V/f control (Page 276)

- After selecting the "Standard Drive Control" application class, only the settings p1900 = 0 or 2 are possible for the motor data identification. The settings p1900 = 1, 3, 11 and 12 have been removed.
 -  Standard Drive Control (Page 158)
 -  Standard Drive Control (Page 138)
- For an inverter replacement, the permissible minimum ratio of the motor/inverter rated power is changed from 1/8 to 1/4.
 -  Replacing inverter components (Page 396)

Further revised chapters

- Summary of the directives and standards relevant for the inverter
 -  Directives and standards (Page 29)
- Description of the differences for star and delta connection of an induction motor augmented
 -  Connecting the motor to the inverter in a star or delta connection (Page 84)
- New designations -X... of the terminals on the Control Unit augmented
 -  Overview of the interfaces (Page 91)
- Sorting of the inverter functions changed, new "Energy saving" and "Availability" categories
 -  Overview of the inverter functions (Page 173)
- Graphic augmented
 -  Motor and inverter protection by limiting the voltage (Page 322)
- LED symbols augmented, tables revised
 -  Operating states indicated on LEDs (Page 374)
- Descriptions shortened and graphics simplified
 -  Alarms, alarm buffer, and alarm history (Page 379)
 -  Faults, alarm buffer and alarm history (Page 382)
- Fault and alarm lists combined to form a common list
 -  List of alarms and faults (Page 386)
- The description for the firmware upgrade or downgrade extended, graphics revised
 -  Firmware upgrade and downgrade (Page 409)
- The description for the setting of the resistance braking revised
 -  Dynamic braking (Page 309)
- The description for the "Calculating the energy saving for fluid flow machines" function revised
 -  Calculating the energy saving for fluid flow machines (Page 343)
- The description for the know-how protection revised
 -  Know-how protection (Page 366)
- The special requirements placed on the EMC-conform installation for use of fail-safe digital inputs are augmented.
 -  Connecting a fail-safe digital input (Page 496)

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Fundamental safety instructions

1.1 General safety instructions



DANGER

Danger to life due to live parts and other energy sources

Death or serious injury can result when live parts are touched.

- Only work on electrical devices when you are qualified for this job.
- Always observe the country-specific safety rules.

Generally, six steps apply when establishing safety:

1. Prepare for shutdown and notify all those who will be affected by the procedure.
2. Disconnect the machine from the supply.
 - Switch off the machine.
 - Wait until the discharge time specified on the warning labels has elapsed.
 - Check that it really is in a no-voltage condition, from phase conductor to phase conductor and phase conductor to protective conductor.
 - Check whether the existing auxiliary supply circuits are de-energized.
 - Ensure that the motors cannot move.
3. Identify all other dangerous energy sources, e.g. compressed air, hydraulic systems, or water.
4. Isolate or neutralize all hazardous energy sources by closing switches, grounding or short-circuiting or closing valves, for example.
5. Secure the energy sources against switching on again.
6. Ensure that the correct machine is completely interlocked.

After you have completed the work, restore the operational readiness in the inverse sequence.



WARNING

Danger to life through a hazardous voltage when connecting an unsuitable power supply

Touching live components can result in death or severe injury.

- Only use power supplies that provide SELV (Safety Extra Low Voltage) or PELV- (Protective Extra Low Voltage) output voltages for all connections and terminals of the electronics modules.



! WARNING

Danger to life when live parts are touched on damaged devices

Improper handling of devices can cause damage.

For damaged devices, hazardous voltages can be present at the enclosure or at exposed components; if touched, this can result in death or severe injury.

- Ensure compliance with the limit values specified in the technical data during transport, storage and operation.
- Do not use any damaged devices.



! WARNING

Danger to life through electric shock due to unconnected cable shields

Hazardous touch voltages can occur through capacitive cross-coupling due to unconnected cable shields.

- As a minimum, connect cable shields and the conductors of power cables that are not used (e.g. brake cores) at one end at the grounded housing potential.



! WARNING

Danger to life due to electric shock when not grounded

For missing or incorrectly implemented protective conductor connection for devices with protection class I, high voltages can be present at open, exposed parts, which when touched, can result in death or severe injury.

- Ground the device in compliance with the applicable regulations.



! WARNING

Danger to life due to electric shock when opening plug connections in operation

When opening plug connections in operation, arcs can result in severe injury or death.

- Only open plug connections when the equipment is in a no-voltage state, unless it has been explicitly stated that they can be opened in operation.



! WARNING

Danger to life through electric shock due to the residual charge of the power component capacitors

Because of the capacitors, a hazardous voltage is present for up to 5 minutes after the power supply has been switched off. Contact with live parts can result in death or serious injury.

- Wait for 5 minutes before you check that the unit really is in a no-voltage condition and start work.

NOTICE**Material damage due to loose power connections**

Insufficient tightening torques or vibrations can result in loose electrical connections. This can result in damage due to fire, device defects or malfunctions.

- Tighten all power connections with the specified tightening torques, e.g. line supply connection, motor connection, DC link connections.
- Check all power connections at regular intervals. This applies in particular after transport.

**WARNING****Danger to life due to fire spreading if housing is inadequate**

Fire and smoke development can cause severe personal injury or material damage.

- Install devices without a protective housing in a metal control cabinet (or protect the device by another equivalent measure) in such a way that contact with fire is prevented.
- Ensure that smoke can only escape via controlled and monitored paths.

**WARNING****Danger to life from electromagnetic fields**

Electromagnetic fields (EMF) are generated by the operation of electrical power equipment, such as transformers, converters, or motors.

People with pacemakers or implants are at particular risk in the immediate vicinity of this equipment.

- If you have a heart pacemaker or implant, maintain a minimum distance of 2 m from electrical power equipment.

**WARNING****Danger to life through unexpected movement of machines when using mobile wireless devices or mobile phones**

Using mobile wireless devices or mobile phones with a transmit power > 1 W closer than approx. 2 m to the components may cause the devices to malfunction, influence the functional safety of machines therefore putting people at risk or causing material damage.

- Switch the wireless devices or mobile phones off in the immediate vicinity of the components.



WARNING

Danger to life due to the motor catching fire in the event of insulation overload

There is higher stress on the motor insulation through a ground fault in an IT system. If the insulation fails, it is possible that death or severe injury can occur as a result of smoke and fire.

- Use a monitoring device that signals an insulation fault.
- Correct the fault as quickly as possible so the motor insulation is not overloaded.



WARNING

Danger to life due to fire if overheating occurs because of insufficient ventilation clearances

Inadequate ventilation clearances can cause overheating of components with subsequent fire and smoke. This can cause severe injury or even death. This can also result in increased downtime and reduced service lives for devices/systems.

- Ensure compliance with the specified minimum clearance as ventilation clearance for the respective component.



WARNING

Danger of an accident occurring due to missing or illegible warning labels

Missing or illegible warning labels can result in accidents involving death or serious injury.

- Check that the warning labels are complete based on the documentation.
- Attach any missing warning labels to the components, in the national language if necessary.
- Replace illegible warning labels.

NOTICE

Device damage caused by incorrect voltage/insulation tests

Incorrect voltage/insulation tests can damage the device.

- Before carrying out a voltage/insulation check of the system/machine, disconnect the devices as all converters and motors have been subject to a high voltage test by the manufacturer, and therefore it is not necessary to perform an additional test within the system/machine.

**WARNING****Danger to life when safety functions are inactive**

Safety functions that are inactive or that have not been adjusted accordingly can cause operational faults on machines that could lead to serious injury or death.

- Observe the information in the appropriate product documentation before commissioning.
- Carry out a safety inspection for functions relevant to safety on the entire system, including all safety-related components.
- Ensure that the safety functions used in your drives and automation tasks are adjusted and activated through appropriate parameterizing.
- Perform a function test.
- Only put your plant into live operation once you have guaranteed that the functions relevant to safety are running correctly.

Note**Important safety notices for Safety Integrated functions**

If you want to use Safety Integrated functions, you must observe the safety notices in the Safety Integrated manuals.

**WARNING****Danger to life or malfunctions of the machine as a result of incorrect or changed parameterization**

As a result of incorrect or changed parameterization, machines can malfunction, which in turn can lead to injuries or death.

- Protect the parameterization (parameter assignments) against unauthorized access.
- Respond to possible malfunctions by applying suitable measures (e.g. EMERGENCY STOP or EMERGENCY OFF).

1.2 Handling electrostatic sensitive devices (ESD)

Electrostatic sensitive devices (ESD) are individual components, integrated circuits, modules or devices that may be damaged by either electric fields or electrostatic discharge.



NOTICE

Damage through electric fields or electrostatic discharge

Electric fields or electrostatic discharge can cause malfunctions through damaged individual components, integrated circuits, modules or devices.

- Only pack, store, transport and send electronic components, modules or devices in their original packaging or in other suitable materials, e.g. conductive foam rubber or aluminum foil.
- Only touch components, modules and devices when you are grounded by one of the following methods:
 - Wearing an ESD wrist strap
 - Wearing ESD shoes or ESD grounding straps in ESD areas with conductive flooring
- Only place electronic components, modules or devices on conductive surfaces (table with ESD surface, conductive ESD foam, ESD packaging, ESD transport container).

1.3 Industrial security

Note

Industrial security

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens products and solutions only represent one component of such a concept.

The customer is responsible for preventing unauthorized access to its plants, systems, machines and networks. Systems, machines and components should only be connected to the enterprise network or the internet if and to the extent necessary and with appropriate security measures (e.g. use of firewalls and network segmentation) in place.

Additionally, Siemens' guidance on appropriate security measures should be taken into account. For more information about industrial security, please visit:

Industrial security (<http://www.siemens.com/industrialsecurity>).

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends to apply product updates as soon as available and to always use the latest product versions. Use of product versions that are no longer supported, and failure to apply latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed at:

Industrial security (<http://www.siemens.com/industrialsecurity>).



WARNING

Danger to life as a result of unsafe operating states resulting from software manipulation

Software manipulations (e.g. viruses, trojans, malware or worms) can cause unsafe operating states in your system that may lead to death, serious injury, and property damage.

- Keep the software up to date.
- Incorporate the automation and drive components into a holistic, state-of-the-art industrial security concept for the installation or machine.
- Make sure that you include all installed products into the holistic industrial security concept.
- Protect files stored on exchangeable storage media from malicious software by with suitable protection measures, e.g. virus scanners.

1.4 Residual risks of power drive systems

When assessing the machine- or system-related risk in accordance with the respective local regulations (e.g., EC Machinery Directive), the machine manufacturer or system installer must take into account the following residual risks emanating from the control and drive components of a drive system:

1. Unintentional movements of driven machine or system components during commissioning, operation, maintenance, and repairs caused by, for example,
 - Hardware and/or software errors in the sensors, control system, actuators, and cables and connections
 - Response times of the control system and of the drive
 - Operation and/or environmental conditions outside the specification
 - Condensation/conductive contamination
 - Parameterization, programming, cabling, and installation errors
 - Use of wireless devices/mobile phones in the immediate vicinity of electronic components
 - External influences/damage
 - X-ray, ionizing radiation and cosmic radiation
2. Unusually high temperatures, including open flames, as well as emissions of light, noise, particles, gases, etc., can occur inside and outside the components under fault conditions caused by, for example:
 - Component failure
 - Software errors
 - Operation and/or environmental conditions outside the specification
 - External influences/damage
3. Hazardous shock voltages caused by, for example:
 - Component failure
 - Influence during electrostatic charging
 - Induction of voltages in moving motors
 - Operation and/or environmental conditions outside the specification
 - Condensation/conductive contamination
 - External influences/damage
4. Electrical, magnetic and electromagnetic fields generated in operation that can pose a risk to people with a pacemaker, implants or metal replacement joints, etc., if they are too close
5. Release of environmental pollutants or emissions as a result of improper operation of the system and/or failure to dispose of components safely and correctly
6. Influence of network-connected communication systems, e.g. ripple-control transmitters or data communication via the network

For more information about the residual risks of the drive system components, see the relevant sections in the technical user documentation.

Introduction

2.1 About the Manual

Who requires the operating instructions and what for?

These operating instructions primarily address fitters, commissioning engineers and machine operators. The operating instructions describe the devices and device components and enable the target groups being addressed to install, connect-up, set, and commission the converters safely and in the correct manner.

What is described in the operating instructions?

These operating instructions provide a summary of all of the information required to operate the converter under normal, safe conditions.

The information provided in the operating instructions has been compiled in such a way that it is sufficient for all standard applications and enables drives to be commissioned as efficiently as possible. Where it appears useful, additional information for entry level personnel has been added.

The operating instructions also contain information about special applications. Since it is assumed that readers already have a sound technical knowledge of how to configure and parameterize these applications, the relevant information is summarized accordingly. This relates, e.g. to operation with fieldbus systems and safety-related applications.

What is the meaning of the symbols in the manual?

 Reference to further information in the manual

 1. An operating instruction starts here.
2.

 This concludes the operating instruction.

 Download from the Internet

 DVD that can be ordered

 Examples of inverter function symbols



2.2 Guide through the manual

Section	In this section you will find answers to the following questions:
 Description (Page 27)	• How is the inverter marked? • Which components make up the inverter? • Which optional components are available for the inverter? • What is the purpose of the optional components? • Which motors can be fed from the inverter? • Which commissioning tools are there?
 Installing (Page 51)	• Which sequence is recommended when installing the inverter? • What does EMC-compliant installation actually mean? • Which options are available to install optional components below the inverter? • What are the inverter dimensions? • Which mounting and installation materials are required when installing the inverter? • To which line supplies can the inverter be connected? • How is the inverter connected to the line supply? • How is the braking resistor connected to the inverter? • Which terminals and fieldbus interfaces does the inverter have? • What are the interface functions?
 Commissioning (Page 129)	• Which motor data is required for commissioning • How is the inverter set in the factory? • What is the commissioning procedure? • How do you restore the inverter factory settings?
 Advanced commissioning (Page 173)	• Which functions are included in the inverter firmware? • How do the functions interoperate with one another? • How are the functions set?
 Saving the settings and series commissioning (Page 347)	• Why is it necessary to back up the inverter settings? • Which options are available to back up the settings? • How does the data backup function? • How do you prevent the inverter settings from being changed? • How do you prevent the inverter settings from being read out?
 Corrective maintenance (Page 395)	• How are inverter components replaced? • How do you change the firmware version of the inverter?
 Alarms, faults and system messages (Page 373)	• What is the meaning of the LEDs provided on the inverter? • How does the system runtime respond? • How does the inverter save alarms and faults? • What do the inverter alarms and faults mean? • How are inverter faults resolved? • Which I&M data is saved in the inverter?

Section	In this section you will find answers to the following questions:
 Technical data (Page 419)	• What is the inverter technical data? • What do "High Overload" and "Low Overload" mean?
 Appendix (Page 477)	• What are the new functions of the current firmware? • What are the most important inverter parameters? • How is the inverter operated using the BOP-2 Operator Panel? • How does the device trace function in STARTER? • How can signal interconnections be changed in the inverter firmware? • What does "BiCo technology" mean? • Where can you find additional manuals and information about the inverter?

Description

Use for the intended purpose

The inverter described in this manual is a device to control a three-phase motor. The inverter is designed for installation in electrical installations or machines.

It has been approved for industrial and commercial use on industrial networks. Additional measures have to be taken when connected to public grids.

The technical specifications and information about connection conditions are indicated on the rating plate and in the operating instructions.

Use of third-party products

This document contains recommendations relating to third-party products. Siemens accepts the fundamental suitability of these third-party products.

You can use equivalent products from other manufacturers.

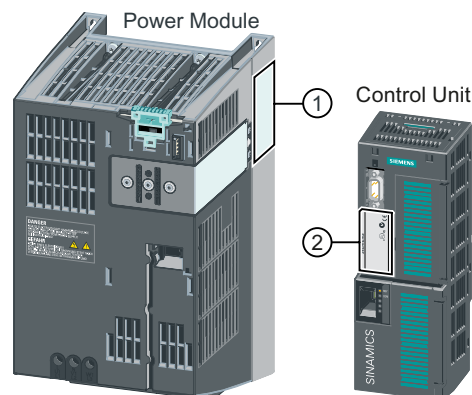
Siemens does not accept any warranty for the properties of third-party products.

3.1 Identifying the converter

Main components of the inverter

Each SINAMICS G120 inverter comprises a Control Unit and a Power Module.

- The Control Unit controls and monitors the connected motor.
- The Power Module provides the connections for line supply and motor.



The following data is provided on the Power Module type plate (①):

- Designation, e.g. PM240-2 Power Module
- Technical specifications: voltage and current
- Article number, e.g. 6SL3210-1PE21-1UL0
- Version, e.g. A02

The following data can be found on the Control Unit type plate (②):

- Designation, e.g. Control Unit CU240E-2 DP-F
- Article number, e.g. 6SL3244-0BB13-1PA0
- Version, e.g. A02 (hardware), 4.7 (firmware)

Further inverter components

The following components are available so that you can adapt the inverter to different applications and ambient conditions:

-  Line filter (Page 39)
-  Line reactor (Page 40)
-  Output reactor (Page 42)
-  Sine-wave filter (Page 45)
-  Braking resistor (Page 46)
-  Brake relay to control a motor holding brake (Page 48)

3.2 Directives and standards

Relevant directives and standards

The following directives and standards are relevant for the inverters:



European Low Voltage Directive

The inverters fulfil the requirements stipulated in the Low-Voltage Directive 2014/35/EU, if they are covered by the application area of this directive.

European Machinery Directive

The inverters fulfil the requirements stipulated in the Machinery Directive 2006/42/EU, if they are covered by the application area of this directive.

However, the use of the inverters in a typical machine application has been fully assessed for compliance with the main regulations in this directive concerning health and safety.

European EMC Directive

By completely complying with IEC/EN 61800-3, it has been proven that the inverter is in compliance with directive 2004/108/EC or 2014/30/EU.



Underwriters Laboratories (North American market)

Inverters provided with one of the test symbols displayed fulfil the requirements stipulated for the North American market as a component of drive applications, and are appropriately listed.



EMC requirements for South Korea

The inverters with the KC marking on the rating plate satisfy the EMC requirements for South Korea.



Eurasian conformity

The inverters comply with the requirements of the Russia/Belarus/Kazakhstan customs union (EAC).



Australia and New Zealand (RCM formerly C-Tick)

The inverters showing the test symbols fulfil the EMC requirements for Australia and New Zealand.

Specification for semiconductor process equipment voltage drop immunity

The inverters comply with the requirements of standard SEMI F47-0706.

Quality systems

Siemens AG employs a quality management system that meets the requirements of ISO 9001 and ISO 14001.

Certificates for download



- EC Declaration of Conformity: (<https://support.industry.siemens.com/cs/ww/de/view/58275445>)
- Certificates for the relevant directives, prototype test certificates, manufacturers declarations and test certificates for functions relating to functional safety ("Safety Integrated"): (<http://support.automation.siemens.com/WW/view/en/22339653/134200>)
- Certificates of products that were certified by UL: (<http://database.ul.com/cgi-bin/XYV/template/LISEXT/1FRAME/index.html>)
- Certificates of products that were certified by TÜV SÜD: (https://www.tuev-sued.de/industrie_konsumprodukte/zertifikatsdatenbank)

Standards that are not relevant



China Compulsory Certification

The inverters do not fall in the area of validity of the China Compulsory Certification (CCC).

3.3 Overview of Control Units

Table 3-1 Control Units CU240B-2 ...

	The CU240B-2 Control Units differ with regard to the type of fieldbus.		
	Designation	CU240B-2	CU240B-2 DP
	Article number	6SL3244-0BB00-1BA1	6SL3244-0BB00-1PA1
	Fieldbus	USS, Modbus RTU	PROFIBUS DP

Table 3-2 Control Units CU240E-2 ...

	The CU240E-2 Control Units have, in comparison to the CU240B-2, an extended terminal strip with integrated safety functions. The CU240E-2 Control Units differ with regard to the type of fieldbus and the scope of the integrated safety functions.		
	Designation	CU240E-2	CU240E-2 DP
	Article number	6SL3244-0BB12-1BA1	6SL3244-0BB12-1PA1
	Fieldbus	USS, Modbus RTU	PROFIBUS DP
Safety Integrated	Basic functions		
	Designation	CU240E-2 F	CU240E-2 DP-F
	Article number	6SL3244-0BB13-1BA1	6SL3244-0BB13-1PA1
	Fieldbus	USS, Modbus RTU	PROFIBUS DP
Safety Integrated	Extended functions		
	Designation	CU240E-2 PN	CU240E-2 PN-F
	Article number	6SL3244-0BB12-1FA0	6SL3244-0BB12-1FA0
	Fieldbus	PROFINET IO, EtherNet/IP	PROFINET IO, EtherNet/IP

Shield connection kit for the Control Unit

The shield connection kit is an optional component. The shield connection kit comprises the following components:

- Shield plate
- Elements for optimum shield support and strain relief of the signal and communication cables

Table 3-3 Article Nos.

Shield connection kit 2 for CU240B-2 and CU240E-2 Control Units with all fieldbus interfaces with the exception of PROFINET.	6SL3264-1EA00-0HA0
Shield connection kit 3 for the CU230P-2 and CU240E-2 Control Units with PROFINET interface.	6SL3264-1EA00-0HB0

Adapter for operating a PM230 IP55 Power Module



For operating the Control Unit with a PM230 IP55 Power Module, FSA ... FSC, an adapter is required between the Control Unit and operator panel (BOP-2 or IOP).

The adapter, which is included in the scope of delivery of the Power Module, is too short for the CU240E-2 Control Unit. The matching adapter for the CU240E-2 Control Unit can be ordered through the KnorrTec company.

Article number 10055500

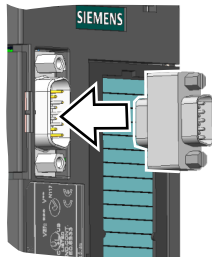


Figure 3-1 Insert the KnorrTec adapter on the interface for the operator panel

3.4 Power Module

Important data on the Power Modules is provided in this section. Further information is contained in the Hardware Installation Manual of the Power Module.

 Overview of the manuals (Page 504)

All power data refers to rated values or to power for operation with low overload (LO).

Which Power Module can I use with the Control Unit?

Table 3-4 Permitted combinations of Control Unit and Power Module

Control Unit	Power Module				
	PM230		PM240-2	PM240P-2	PM250
	IP20 and Push-Through technology	IP55			
CU240B-2...	✓	---	✓	✓	✓
CU240E-2...	✓	✓	✓	✓	✓

3.4.1 Power Module with IP20 degree of protection



Figure 3-2 Examples of Power Modules with IP20 degree of protection

PM240-2 for standard applications

The PM240-2 Power Module is available without a filter or with an integrated class A line filter. The PM240-2 permits dynamic braking via an external braking resistor.

Table 3-5 1-phase/3-phase 200 VAC ... 240 VAC, article number 6SL3210-1PB... and 6SL3210-1PC...

Frame size	FSA	FSB	FSC	FSD	FSE	FSF
Power (kW)	0.55 ... 0.75	1.1 ... 2.2	3.0 ... 4.0	11 ... 18.5	22 ... 30	37 ... 55

Table 3-6 3-phase 380 VAC ... 480 VAC, article number 6SL3210-1PE...

Frame size	FSA	FSB	FSC	FSD	FSE	FSF
Power (kW)	0.55 ... 3.0	4.0 ... 7.5	11 ... 15	18.5 ... 37	45 ... 55	75 ... 132

Table 3-7 3-phase 500 VAC ... 690 VAC, article number 6SL3210-1PH...

Frame size	FSD	FSE	FSE
Power (kW)	11 ... 37	45 ... 55	75 ... 132

PM240P-2 for standard applications

The PM240P-2 Power Module is available without a filter or with an integrated class A line filter.

Table 3-8 3-phase 380 VAC ... 480 VAC, article number 6SL3210-1RE...

Frame size	FSD	FSE	FSF
Power (kW)	22 ... 37	45 ... 55	75 ... 132

Table 3-9 3-phase 500 VAC ... 690 VAC, article number 6SL3210-1RH...

Frame size	FSD	FSE	FSE
Power (kW)	11 ... 37	45 ... 55	75 ... 132

PM230, 3-phase 400 VAC in IP20 degree of protection for pump and fan applications

The PM230 Power Module in IP20 degree of protection is available without a filter or with an integrated class A line filter.

Table 3-10 3-phase 380 VAC ... 480 VAC, article numbers: 6SL3210-1NE...

Frame size	FSA	FSB	FSC	FSD	FSE	FSF
Power (kW)	0.37 ... 3	4 ... 7.5	11 ... 18.5	22 ... 37	45 ... 55	75 ... 90

PM250 for standard applications with energy recovery

The PM250 Power Module is available without a filter or with integrated class A line filter. The PM250 permits dynamic braking with energy recovery into the line supply.

Table 3-11 3-phase 380 VAC ... 480 VAC, article number 6SL3225-0BE...

Frame size	FSC	FSD	FSE	FSF
Power (kW)	7.5 ... 15	18.5 ... 30	37 ... 45	55 ... 90

3.4.2 Power Module with Push-Through technology



Figure 3-3 Examples of Power Modules with Push-Through technology FSA ... FSC

PM240-2 with Push-Through technology for standard applications

The PM240-2 Power Module is available with Push-Through technology without a filter or with an integrated class A line filter. The PM240-2 allows dynamic braking using an external braking resistor.

Table 3-12 1-phase/3-phase 200 VAC ... 240 VAC, article number 6SL3211-1PB...

Frame size	FSA	FSB	FSC
Power (kW)	0.75	2.2	4.0

Table 3-13 3-phase 380 VAC ... 480 VAC, article number 6SL3211-1PE...

Frame size	FSA	FSB	FSC
Power (kW)	3.0	7.5	15

PM230 in Push-Through technology for pump and fan applications

The PM230 Power Module is available without a filter or with integrated class A line filter.

Table 3-14 3-phase 380 VAC ... 480 VAC, article number 6SL3211-1NE...

Frame size	FSA	FSB	FSC
Power (kW)	3	7.5	18.5

3.4.3 Power Module in IP55 degree of protection / UL Type 12



Figure 3-4 PM230, 3-phase 400 VAC, degree of protection IP55 / UL Type 12

PM230 for pumps and fan applications

The PM230 Power Module is suitable for cabinet-free installation.

Table 3-15 3-phase 380 VAC ... 480 VAC, article number 6SL3223-0DE...

Frame size		FSA	FSB	FSC	FSD	FSE	FSF
Power (kW)	Filter Class A	0.37 ... 3	4 ... 7.5	11 ... 18.5	22 ... 30	37 ... 45	55 ... 90
	Filter Class B	0.37 ... 3	4 ... 7.5	11 ... 15	18.5 ... 30	37 ... 45	55 ... 90

3.5 Components for the Power Modules

3.5.1 Accessories for shielding

Shield connection kit

Establish the shield and strain relief for the power connections using the shield connection kit.

The shield connection kit comprises a shield plate and serrated strips with screws.

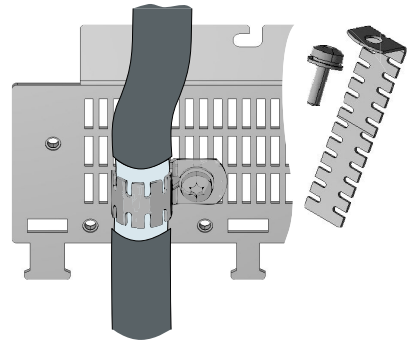


Table 3-16 Article numbers for the shield connection kit

Frame size	PM250 Power Module	PM230, PM240-2	PM240P-2
FSA	6SL3262-1AA00-0BA0	The shield connection kit is included in the scope of delivery	---
FSB	6SL3262-1AB00-0DA0		
FSC	6SL3262-1AC00-0DA0		
FSD	6SL3262-1AD00-0DA0		
FSE	6SL3262-1AD00-0DA0		The lower shield connection kit is included in the scope of delivery
FSF	6SL3262-1AF00-0DA0		

3.5.2 Line filter

With a line filter, the inverter can achieve a higher radio interference class. An external filter is not required for inverters with integrated line filter.

NOTICE

The line filter is damaged when operated on inadmissible line supplies

The line filter is only suitable for operation on TN or TT line supplies with a grounded neutral point. The line filter is damaged if operated on all other line supplies.

- For inverters equipped with line filter, only connect to TN or TT line supplies with a grounded neutral point.

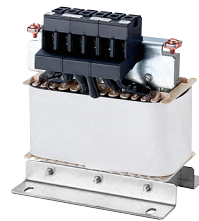
External line filters for PM250

Power Module		Power	Class B line filter according to EN55011: 2009
FSC	6SL3225-0BE25-5AA0, 6SL3225-0BE27-5AA0, 6SL3225-0BE31-1AA0	7.5 kW ... 15.0 kW	6SL3203-0BD23-8SA0

3.5.3 Line reactor

The line reactor supports the overvoltage protection, smoothes the harmonics in the line supply and bridges commutation dips. For the Power Modules subsequently listed, a line reactor is suitable in order to dampen the specified effects.

The figure on the right-hand side shows as example the line reactors for the PM240-2 Power Modules, FSB.



The line requirements for deployment of a line reactor depends on the Power Module:



AUTOHOTSPOT

See also

Technical data (Page 419)

NOTICE

Damage to inverter as a result of a missing line reactor

Depending on the Power Module and line supply, if a line reactor is not used, this can result in damage to the inverter and other components in the electrical plant or system.

- Install a line reactor if the relative short-circuit voltage of the line supply is below 1%

Line reactors for PM240-2, 380 V ... 480 V

Power Module		Power	Line reactor
FSA	6SL3210-1PE11-8 . L1, 6SL3210-1PE12-3 . L1, 6SL3210-1PE13-2 . L1	0.55 kW ... 1.1 kW	6SL3203-0CE13-2AA0
FSB	6SL3210-1PE14-3 . L1, 6SL321 . -1PE16-1 . L1, 6SL321 . -1PE18-0 . L1	1.5 kW ... 3 kW	6SL3203-0CE21-0AA0
FSC	6SL3210-1PE21-1 . L0, 6SL3210-1PE21-4 . L0, 6SL321 . -1PE21-8 . L0	4 kW ... 7.5 kW	6SL3203-0CE21-8AA0
	6SL3210-1PE22-7 . L0, 6SL321 . -1PE23-3 . L0	11 kW ... 15 kW	6SL3203-0CE23-8AA0

FSD ... FSF: A line reactor is not required.

Line reactors for PM240-2 Power Module, 500 V ... 690 V

A line reactor is not required.

Line reactors for PM240-2, 200 V ... 240 V

Power Module		Power	Line reactor
FSA	6SL3210-1PB13-0 . L0, 6SL3210-1PB13-8 . L0	0.55 kW ... 0.75 kW	6SL3203-0CE13-2AA0
FSB	6SL3210-1PB15-5 . L0, 6SL3210-1PB17-4 . L0, 6SL321 . -1PB21-0 . L0	1.1 kW ... 2.2 kW	6SL3203-0CE21-0AA0
FSC	6SL3210-1PB21-4 . L0, 6SL321 . -1PB21-8 . L0	3 kW ... 4 kW	6SL3203-0CE21-8AA0
	6SL321 . -1PC22-2 . L0, 6SL3210-1PC22-8 . L0	5.5 kW ... 7.5 kW	6SL3203-0CE23-8AA0

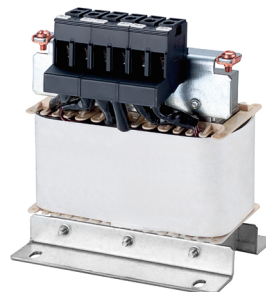
FSD ... FSF: A line reactor is not required.

Line reactors for PM240-2 Power Modules

A line reactor is not required.

3.5.4 Output reactor

Output reactors reduce the voltage stress on the motor windings and the load placed on the inverter as a result of capacitive recharging currents in the cables. An output reactor is required for shielded motor cables longer than 50 m or unshielded motor cables longer than 100 m. The figure on the right-hand side shows as example the output reactors for PM240-2 Power Modules, FSB and FSC.



NOTICE

The output reactor is damaged if the inverter pulse frequency is too high

The output reactors are designed for pulse frequencies of 4 kHz. The output reactor can overheat if the inverter is operated with pulse frequencies > 4 KHz. Excessively high temperatures damage the output reactor.

- Operate the inverter with an output reactor with a maximum pulse frequency of 4 kHz.

Output reactors for PM240-2 Power Modules, 380 V ... 480 V

Power Module		Power	Output reactor
FSA	6SL3210-1PE11-8 . L1, 6SL3210-1PE12-3 . L1, 6SL3210-1PE13-2 . L1, 6SL3210-1PE14-3 . L1, 6SL3210-1PE16-1 . L1	0.55 kW ... 2.2 kW	6SL3202-0AE16-1CA0
	6SL321 . -1PE18-0UL1	3 kW	6SL3202-0AE18-8CA0
FSB	6SL3210-1PE21-1 . L0, 6SL3210-1PE21-4 . L0, 6SL321 . -1PE21-8 . L0	4 kW ... 7.5 kW	6SL3202-0AE21-8CA0
FSC	6SL3210-1PE22-7 . L0, 6SL321 . -1PE23-3 . L0	11 kW ... 15 kW	6SL3202-0AE23-8CA0

FSD ... FSF: An output reactor is not required.

Output reactors for PM240-2 Power Modules, 500 V ... 690 V

An output reactor is not required.

Output reactors for PM240-2 Power Modules, 200 V ... 240 V

Power Module		Power	Output reactor
FSA	6SL3210-1PB13-0 . L0, 6SL321 . -1PB13-8 . L0	0.55 kW ... 0.75 kW	6SL3202-0AE16-1CA0
FSB	6SL3210-1PB15-5 . L0	1.1 kW	
	6SL3210-1PB17-4 . L0	1.5 kW	6SL3202-0AE18-8CA0
FSB	6SL321 . -1PB21-0 . L0	2.2 kW	6SL3202-0AE21-8CA0
FSC	6SL3210-1PB21-4 . L0, 6SL321 . -1PB21-8 . L0	3 kW ... 4 kW	
	6SL321 . -1PC22-2 . L0, 6SL3210-1PC22-8 . L0	5.5 kw ... 7.5 kW	6SL3202-0AE23-8CA0

FSD ... FSF: An output reactor is not required.

Output reactors for PM240P-2 Power Modules, 380 V ... 480 V

Power Module		Power	Output reactor
FSD	6SL3210-1RE24-5 . L0	22 kW	6SE6400-3TC03-8DD0
	6SL3210-1RE26-0 . L0	30 kW	6SE6400-3TC07-5ED0
	6SL3210-1RE27-5 . L0	37 kW	
FSE	6SL3210-1RE28-8 . L0	45 kW	6SE6400-3TC14-5FD0
	6SL3210-1RE31-1 . L0	55 kW	
FSF	6SL3210-1RE31-5 . L0	75 kW	
	6SL3210-1RE31-8 . L0	90 kW	6SL3000-2BE32-1AA0
	6SL3210-1RE32-1 . L0	110 kW	
	6SL3210-1RE32-5 . L0	132 kW	

Output reactors for PM250 Power Module

Power Module		Power	Output reactor
FSC	6SL3225-0BE25-5 . A0, 6SL3225-0BE27-5 . A0, 6SL3225-0BE31-1 . A0	7.5 kW ... 15.0 kW	6SL3202-0AJ23-2CA0
FSD	6SL3225-0BE31-5 . A0	18.5 kW	6SE6400-3TC05-4DD0
	6SL3225-0BE31-8 . A0	22 kW	6SE6400-3TC03-8DD0
	6SL3225-0BE32-2 . A0	30 kW	6SE6400-3TC05-4DD0
FSE	6SL3225-0BE33-0 . A0	37 kW	6SE6400-3TC08-0ED0
	6SL3225-0BE33-7 . A0	45 kW	6SE6400-3TC07-5ED0
FSF	6SL3225-0BE34-5 . A0	55 kW	6SE6400-3TC14-5FD0
	6SL3225-0BE35-5 . A0	75 kW	6SE6400-3TC15-4FD0
	6SL3225-0BE37-5 . A0	90 kW	6SE6400-3TC14-5FD0

Output reactors for PM230 Power Modules (IP20)

Power Module		Power	Output reactor
FSA	6SL3210-1NE11-3 . G1 6SL3210-1NE11-7 . G1 6SL3210-1NE12-2 . G1 6SL3210-1NE13-1 . G1 6SL3210-1NE14-1 . G1 6SL3210-1NE15-8 . G1	0.37 kW ... 2.2 kW	6SL3202-0AE16-1CA0
	6SL3210-1NE17-7 . G1	3.0 kW	6SL3202-0AE18-8CA0
FSB	6SL3210-1NE21-0 . G1 6SL3210-1NE21-3 . G1 6SL3210-1NE21-8 . G1	4.0 kW ... 7.5 kW	6SL3202-0AE21-8CA0
	6SL3210-1NE22-6 . G1 6SL3210-1NE23-2 . G1 6SL3210-1NE23-8 . G1	11.0 kW ... 18.5 kW	6SL3202-0AE23-8CA0
FSD	6SL3210-1NE24-5 . L0	22 kW	6SE6400-3TC03-8DD0
	6SL3210-1NE26-0 . L0	30 kW	6SE6400-3TC05-4DD0
FSE	6SL3210-1NE27-5 . L0	37 kW	6SE6400-3TC08-0ED0
	6SL3210-1NE28-8 . L0	45 kW	6SE6400-3TC07-5ED0
FSF	6SL3210-1NE31-1 . L0	55 kW	6SE6400-3TC14-5FD0
	6SL3210-1NE31-5 . L0	75 kW	6SE6400-3TC15-4FD0

Output reactors for PM230 push-through Power Modules

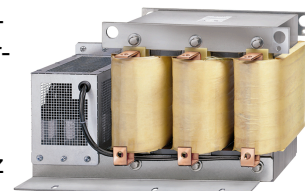
Power Module		Power	Output reactor
FSA	6SL3211-1NE17-7 . G1	3.0 kW	6SL3202-0AE18-8CA0
FSB	6SL3211-1NE21-8 . G1	7.5 kW	6SL3202-0AE21-8CA0
FSC	6SL3211-1NE23-8 . G1	18.5 kW	6SL3202-0AE23-8CA0

3.5.5 Sine-wave filter

The sine-wave filter at the inverter output limits the voltage rate-of-rise and the peak voltages at the motor winding. The maximum permissible length of motor feeder cables is increased to 300 m.

The following applies when using a sine-wave filter:

- Operation is only permissible with pulse frequencies from 4 kHz to 8 kHz.
From 110 kW power rating of the Power Modules (according to the type plate) only 4 kHz is permissible.
- The inverter power is reduced by 5%.
- The maximum output frequency of the inverter is 150 Hz at 380 V to 480 V.
- Operation and commissioning may only be performed with the motor connected, as the sine-wave filter is not no-load proof.
- An output reactor is superfluous.



Sine-wave filter for PM250 Power Module

Power Module		Power	Sine-wave filter
FSC	6SL3225-0BE25-5 . A0	7.5 kW	6SL3202-0AE22-0SA0
	6SL3225-0BE27-5 . A0, 6SL3225-0BE31-1 . A0	11.0 kW ... 15.0 kW	6SL3202-0AE23-3SA0
FSD	6SL3225-0BE31-5 . A0, 6SL3225-0BE31-8 . A0	18.5 kW ... 22 kW	6SL3202-0AE24-6SA0
	6SL3225-0BE32-2 . A0	30 kW	6SL3202-0AE26-2SA0
FSE	6SL3225-0BE33-0 . A0, 6SL3225-0BE33-7 . A0	37 kW ... 45 kW	6SL3202-0AE28-8SA0
FSF	6SL3225-0BE34-5 . A0, 6SL3225-0BE35-5 . A0	55 kW ... 75 kW	6SL3202-0AE31-5SA0
	6SL3225-0BE37-5 . A0	90 kW	6SL3202-0AE31-8SA0

Sine-wave filter for PM240P-2 Power Module, 380 V ... 480 V

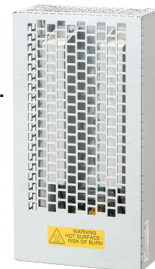
Power Module		Power	Sine-wave filter
FSD	6SL3210-1RE24-5 . L0	22 kW	6SL3202-0AE26-2SA0
	6SL3210-1RE26-0 . L0	30 kW	6SL3202-0AE28-8SA0
	6SL3210-1RE27-5 . L0	37 kW	
FSE	6SL3210-1RE28-8 . L0	45 kW	6SL3202-0AE31-5SA0
	6SL3210-1RE31-1 . L0	55 kW	
FSF	6SL3210-1RE31-5 . L0	75 kW	6SL3202-0AE31-8SA0
	6SL3210-1RE31-8 . L0	90 kW	6SL3000-2CE32-3AA0
	6SL3210-1RE32-1 . L0	110 kW	
	6SL3210-1RE32-5 . L0	132 kW	6SL3000-2CE32-8AA0

3.5.6 Braking resistor

The braking resistor allows loads with a high moment of inertia to be quickly braked.

The Power Module controls the braking resistor via its integrated braking module.

The figure shown on the right-hand side shows as example the braking resistor for a PM240-2 Power Module, FSB.



Braking resistors for PM240-2, 380 V ... 480 V

Power Module		Power	Braking resistor
FSA	6SL3210-1PE11-8 . L1, 6SL3210-1PE12-3 . L1, 6SL3210-1PE13-2 . L1, 6SL3210-1PE14-3 . L1	0.55 kW ... 1.5 kW	6SL3201-0BE14-3AA0
	6SL321 . -1PE16-1 . L1, 6SL321 . -1PE18-0 . L1	2.2 kW ... 3.0 kW	6SL3201-0BE21-0AA0
FSB	6SL3210-1PE21-1 . L0, 6SL3210-1PE21-4 . L0, 6SL321 . -1PE21-8 . L0	4 kW ... 7.5 kW	6SL3201-0BE21-8AA0
FSC	6SL3210-1PE22-7 . L0, 6SL321 . -1PE23-3 . L0	11 kW ... 15 kW	6SL3201-0BE23-8AA0
FSD	6SL3210-1PE23-8 . L0, 6SL3210-1PE24-5 . L0	18.5 kW ... 22 kW	JJY:023422620001
	6SL3210-1PE26-0 . L0, 6SL3210-1PE27-5 . L0	30 kW ... 37 kW	JJY:023424020001
FSE	6SL3210-1PE28-8 . L0, 6SL3210-1PE31-1 . L0	45 kW ... 55 kW	JJY:023434020001
FSF	6SL3210-1PE31-5 . L0, 6SL3210-1PE31-8 . L0,	75 kW ... 90 kW	JJY:023454020001
	6SL3210-1PE32-1 . L0, 6SL3210-1PE32-5 . L0	90 kW ... 132 kW	JJY:023464020001

Braking resistors for PM240-2, 500 V ... 690 V

Power Module		Power	Braking resistor
FSD	6SL3210-1PH21-4 . L0, 6SL3210-1PH22-0 . L0, 6SL3210-1PH22-3 . L0, 6SL3210-1PH22-7 . L0, 6SL3210-1PH23-5 . L0, 6SL3210-1PH24-2 . L0	11 kW ... 37 kW	JJY:023424020002
FSE	6SL3210-1PH25-2 . L0, 6SL3210-1PH26-2 . L0	45 kW ... 55 kW	JJY:023434020002

Power Module		Power	Braking resistor
FSF	6SL3210-1PH28-0 . L0, 6SL3210-1PH31-0 . L0,	75 kW ... 90 kW	JJY:023464020002
	6SL3210-1PH31-2 . L0, 6SL3210-1PH31-4 . L0	110 kW ... 132 kW	JJY:023464020002

Braking resistors for PM240-2, 200 V ... 240 V

Power Module		Power	Braking resistor
FSA	6SL3210-1PB13-0 . L0, 6SL321 . -1PB13-8 . L0	0.55 kW ... 0.75 kW	JJY:023146720008
FSB	6SL3210-1PB15-5 . L0, 6SL3210-1PB17-4 . L0, 6SL321 . -1PB21-0 . L0	1.1 kW ... 2.2 kW	JJY:023151720007
FSC	6SL3210-1PB21-4 . L0, 6SL321 . -1PB21-8 . L0	3 kW ... 4 kW	JJY:02 3163720018
	6SL3210-1PC22-2 . L0, 6SL3210-1PC22-8 . L0	5.5 kW ... 7.5 kW	JJY:023433720001
FSD	6SL3210-1PC24-2UL0, 6SL3210-1PC25-4UL0, 6SL3210-1PC26-8UL0	11 kW ... 18.5 kW	JJY:023422620002
FSE	6SL3210-1PC28-0UL0, 6SL3210-1PC31-1UL0	22 kW ... 30 kW	JJY:023423320001
FSF	6SL3210-1PC31-3UL0, 6SL3210-1PC31-6UL0, 6SL3210-1PC31-8UL0	37 kW ... 55 kW	JJY:023434020003

3.5.7 Brake Relay



The Brake Relay has a switch contact (NO contact) for controlling a motor holding brake.

Article number: 6SL3252-0BB00-0AA0

The following Power Modules have a connection possibility for the Brake Relay:

- PM240-2
- PM240P-2
- PM250

3.5.8 Safe Brake Relay



The Safe Brake Relay controls a 24 V motor brake and monitors the brake control for a short-circuit or interrupted cable.

Article number: 6SL3252-0BB01-0AA0

The following Power Modules have a connection possibility for the Brake Relay:

- PM240-2
- PM240P-2
- PM250

3.6 Motors and multi-motor drives that can be operated

Siemens motors that can be operated

You can connect standard induction motors to the inverter.



You can find information on further motors on the Internet:

Motors that can be operated (<https://support.industry.siemens.com/cs/ww/en/view/100426622>)

Third-party motors that can be operated

You can operate standard asynchronous motors from other manufacturers with the inverter:

NOTICE

Motor damage due to the use of an unsuitable third-party motor

A higher load occurs on the motor insulation in inverter mode than with mains operation. Damage to the motor winding may occur as a result.

- Please observed the notes in the System Manual "Requirements for third-party motors"



Additional information is provided on the Internet:

Requirements for third-party motors (<https://support.industry.siemens.com/cs/ww/en/view/79690594>)

Multi-motor drive

A multi-motor drive involves simultaneously operating several motors from one inverter. For standard induction motors, multi-motor drives are permissible in installations that are compliant with IEC.



Further information is provided in the Internet:

Multi-motor drive (<http://support.automation.siemens.com/WW/view/en/84049346>)

Multi-motor drives are not permissible for installations that are compliant with UL.

Description

3.6 Motors and multi-motor drives that can be operated

4.1 EMC-compliant setup of the machine or plant

The inverter is designed for operation in industrial environments where strong electromagnetic fields are to be expected.

Reliable and disturbance-free operation is only guaranteed for EMC-compliant installation.

To achieve this, subdivide the control cabinet and the machine or system into EMC zones:

EMC zones

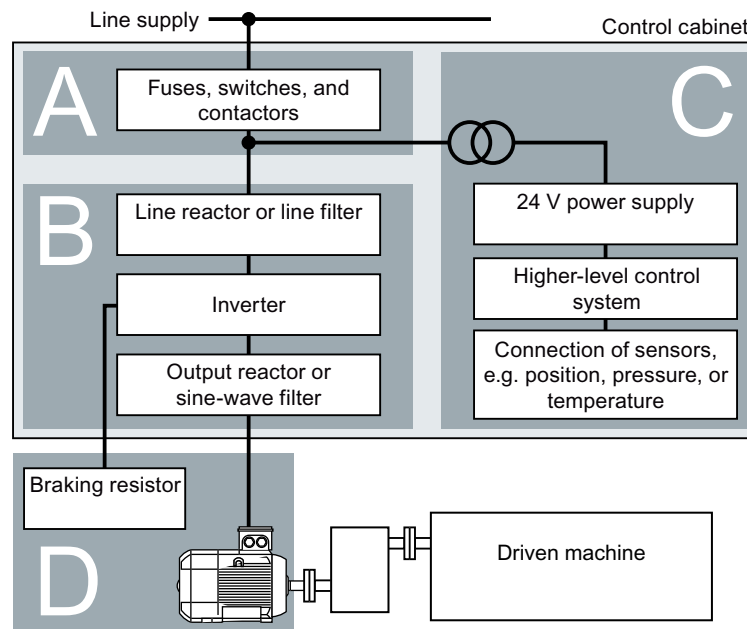


Figure 4-1 Example of the EMC zones of a plant or machine

Inside the control cabinet

- Zone A: Line supply connection
- Zone B: Power electronics
Devices in Zone B generate energy-rich electromagnetic fields.
- Zone C: Control and sensors
Devices in Zone C do not generate any energy-rich electromagnetic fields themselves, but their functions can be impaired by electromagnetic fields.

Outside the control cabinet

- Zone D: Motors, braking resistors
Devices in Zone D generate electromagnetic fields with a significant amount of energy

4.1.1 Control cabinet

- Assign the various devices to zones in the control cabinet.
- Electromagnetically uncouple the zones from each other by means of one of the following actions:
 - Side clearance ≥ 25 cm
 - Separate metal enclosure
 - Large-area partition plates
- Route cables of various zones in separate cable harnesses or cable ducts.
- Install filters or isolation amplifiers at the interfaces of the zones.

Control cabinet assembly

- Connect the door, side panels, top and base plate of the control cabinet with the control cabinet frame using one of the following methods:
 - Electrical contact surface of several cm^2 for each contact location
 - Several screw connections
 - Short, finely stranded, braided copper wires with cross-sections $\geq 95 \text{ mm}^2 / 000 (3/0) (-2) \text{ AWG}$
- Install a shield support for shielded cables that are routed out of the control cabinet.
- Connect the PE bar and the shield support to the control cabinet frame through a large surface area to establish a good electrical connection.
- Mount the control cabinet components on a bare metal mounting plate.
- Connect the mounting plate to the control cabinet frame and PE bar and shield support through a large surface area to establish a good electrical connection.
- For screw connections onto painted or anodized surfaces, establish a good conductive contact using one of the following methods:
 - Use special (serrated) contact washers that cut through the painted or anodized surface.
 - Remove the insulating coating at the contact locations.

Measures required for several control cabinets

- Install equipotential bonding for all control cabinets.
- Screw the frames of the control cabinets together at several locations through a large surface area using serrated washers to establish a good electrical connection.
- In plants and systems where the control cabinets are lined up next to one another, and which are installed in two groups back to back, connect the PE bars of the two cabinet groups at as many locations as possible.

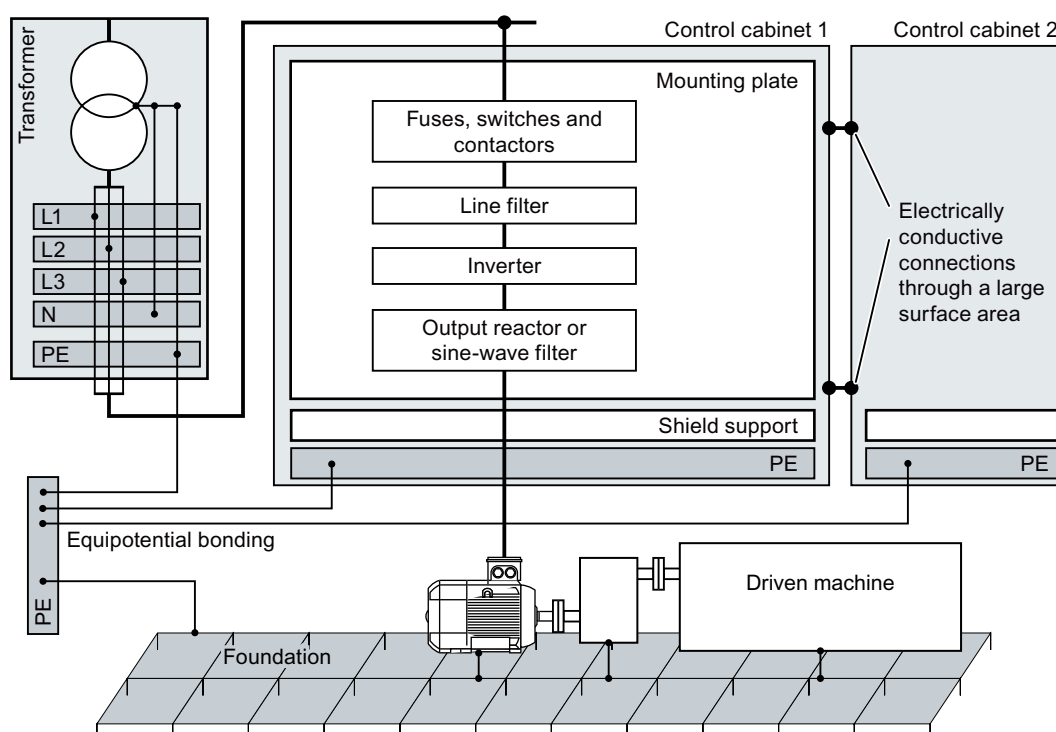


Figure 4-2 Grounding and high-frequency equipotential bonding measures in the control cabinet and in the plant/system

Further information



Additional information about EMC-compliant installation is available in the Internet:

EMC installation guideline (<http://support.automation.siemens.com/WW/view/en/60612658>)

4.1.2 Cables

Cables with a high level of interference and cables with a low level of interference are connected to the inverter:

- Cables with a high level of interference:
 - Cable between the line filter and inverter
 - Motor cable
 - Cable at the inverter DC link connection
 - Cable between the inverter and braking resistor
- Cables with a low level of interference:
 - Cable between the line and line filter
 - Signal and data cables

Cable routing inside the cabinet

- Route the power cables with a high level of interference so that there is a minimum clearance of 25 cm to cables with a low level of interference.
If the minimum clearance of 25 cm is not possible, insert separating metal sheets between the cables with a high level of interference and cables with a low level of interference. Connect these separating metal sheets to the mounting plate to establish a good electrical connection.
- Cables with a high level of interference and cables with a low level of interference may only cross over at right angles:
- Keep all of the cables as short as possible.
- Route all of the cables close to the mounting plates or cabinet frames.
- Route signal and data cables - as well as the associated equipotential bonding cables - parallel and close to one another.
- Twist incoming and outgoing unshielded individual conductors.
Alternatively, you can route incoming and outgoing conductors in parallel, but close to one another.
- Ground any unused conductors of signal and data cables at both ends.
- Signal and data cables must only enter the cabinet from one side, e.g. from below.
- Using shielded cables for the following connections:
 - Cable between the inverter and line filter
 - Cable between the inverter and output reactor or sine-wave filter

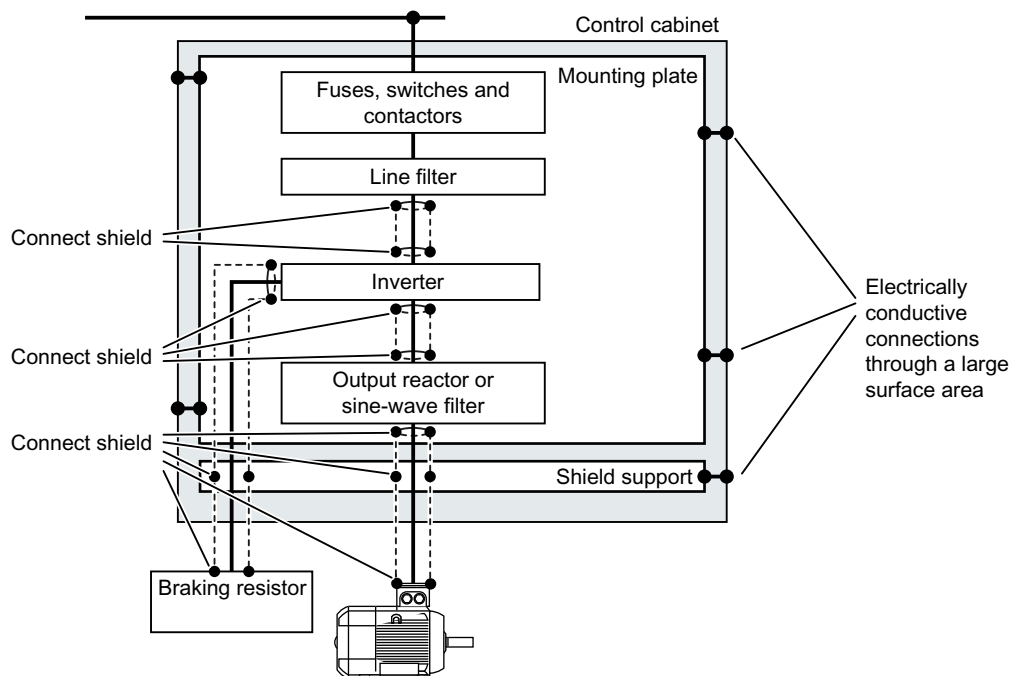


Figure 4-3 Routing inverter cables inside and outside a control cabinet

Routing cables outside the control cabinet

- Maintain a minimum clearance of 25 cm between cables with a high level of interference and cables with a low level of interference.
- Using shielded cables for the following connections:
 - Inverter motor cable
 - Cable between the inverter and braking resistor
 - Signal and data cables
- Connect the motor cable shield to the motor enclosure using a PG gland that establishes a good electrical connection.

Requirements relating to shielded cables

- Use cables with finely-stranded, braided shields.
- Connect the shield to at least one end of the cable.

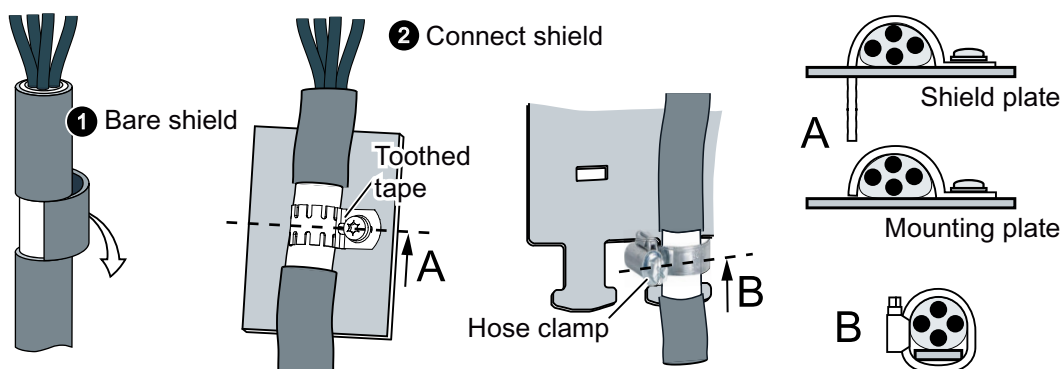


Figure 4-4 Examples for EMC-compliant shield support

- Attach the shield to the shield support directly after the cable enters the cabinet.
- Do not interrupt the shield.
- Only use metallic or metallized plug connectors for shielded data cables.

4.1.3 Electromechanical components

Radio interference suppression

- Connect interference suppression elements to the following components:
 - Coils of contactors
 - Relays
 - Solenoid valves
 - Motor holding brakes
- Connect the interference suppression element directly at the coil.
- Use RC elements or varistors for AC-operated coils and freewheeling diodes or varistors for DC-operated coils.

4.2 Installing reactors, filters and braking resistors

Installing reactors, filters and braking resistors

The following supplementary components may be required depending on the Power Modules and the particular application:

- Line reactors
- Filter
- Braking resistors
- Brake Relay

Installing these components is described in the documentation provided.

4.3 Installing Power Modules

4.3.1 Basic installation rules

Protection against the spread of fire

The built-in units may be operated only in closed housings or in higher-level control cabinets with closed protective covers, and when all of the protective devices are used. The installation of the built-in units in a metal control cabinet or protection with another equivalent measure must prevent the spread of fire and emissions outside the control cabinet.

Wall mounting units may be operated outside the control cabinet.

Protection against condensation or electrically conductive contamination

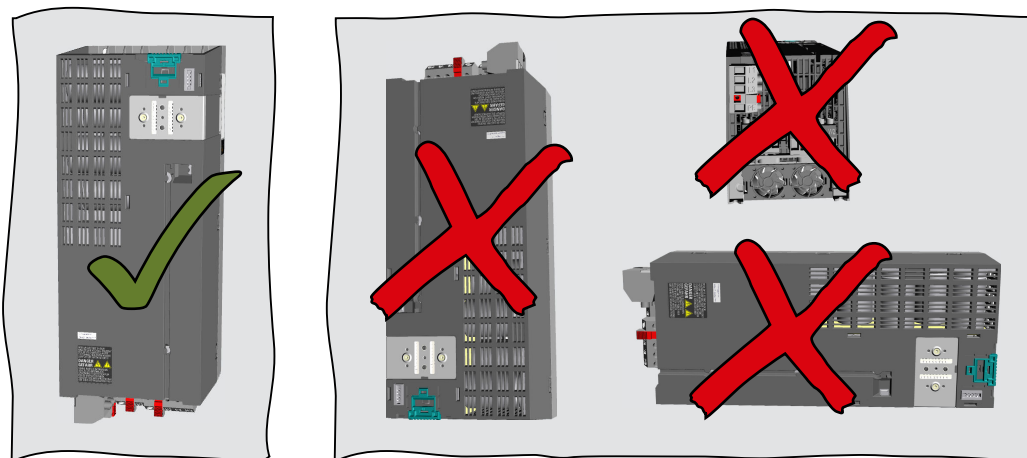
Protect the inverter, e.g. by installing it in a control cabinet with degree of protection IP54 according to IEC 60529 or NEMA 12. Further measures may be necessary for particularly critical operating conditions.

If condensation or conductive pollution can be excluded at the installation site, a lower degree of control cabinet protection may be permitted.

Installing Power Modules

The following is required to correctly install a Power Module:

- Install the Power Module vertically with the motor connections facing downwards.



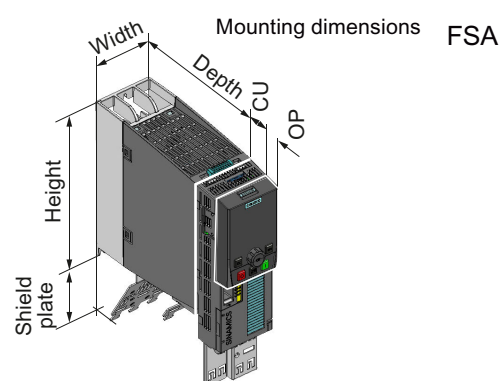
- Comply with the installation regulations specified in the following sections:
 - Minimum clearances to other components
 - Fixing elements
 - Tightening torques for fixing elements

4.3.2 Dimensioned drawings, drilling dimensions for the PM240-2 Power Module, IP20

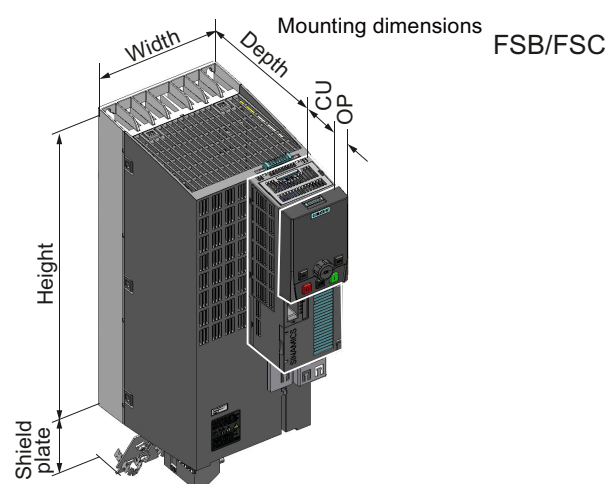
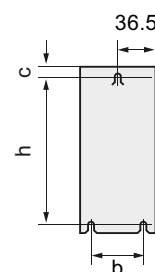
Table 4-1 Mounting dimensions without Control Unit [mm]

Frame size	Width	Height		Depth
		Power Module	Shield plate	
FSA	73	196	80	165
FSB	100	292	78	165
FSC	140	355	77	165
FSD	200	472	152	237
FSE	275	551	177	237
FSF	305	708	257	357

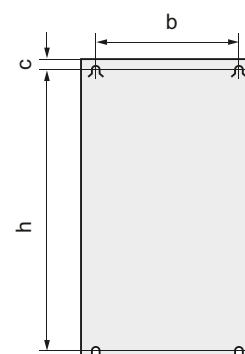
The following dimension drawings and drilling patterns are not to scale.



Drilling dimensions



Drilling dimensions



4.3 Installing Power Modules

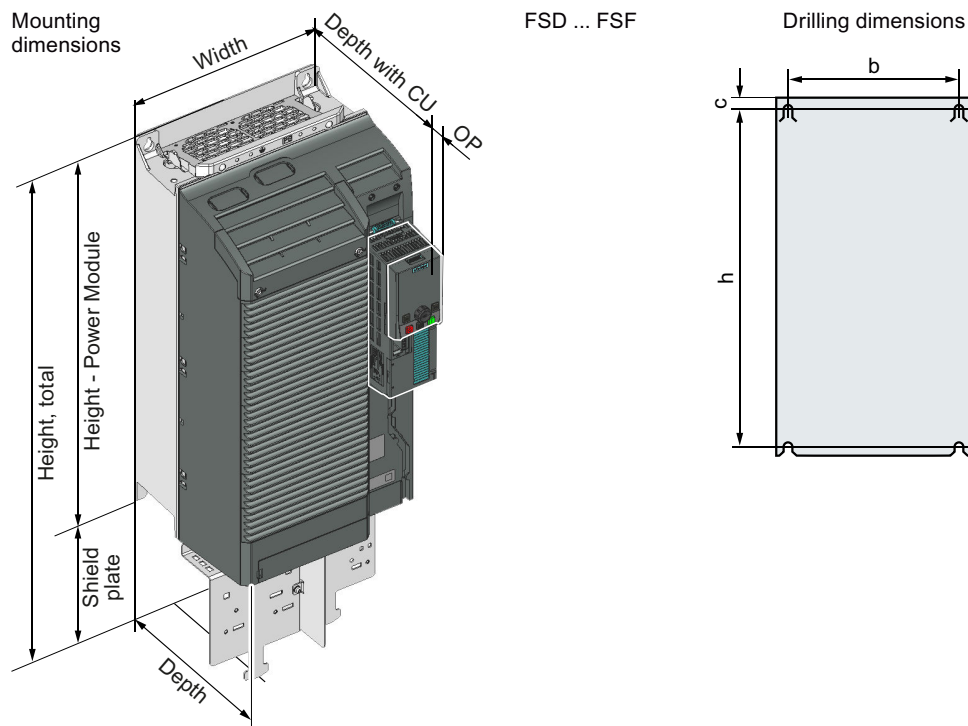


Table 4-2 Additional depth with Control Unit (CU) and BOP-2 or IOP Operator Panel [mm]

Frame size	with CU	with CU and blanking cover or BOP-2	with CU and IOP
FSA ... FSC	+ 41	+ 52	+ 63
FSD ... FSF	+ 0	+ 10	+ 21

Table 4-3 Drilling dimensions, cooling air clearances [mm] and fixing [Nm]

Frame size	Drilling dimensions			Cooling air clearances ¹⁾			Fixing/torque
	h	b	c	Top	Bottom	Front	
FSA	186	62.3	6	80	100	100	3 x M4 / 2.5
FSB	281	80	6	80	100	100	4 x M4 / 2.5
FSC	343	120	6	80	100	100	4 x M5 / 3.5
FSD	430	170	7	300	350	100	4 x M5 / 6.0
FSE	509	230	8.5	300	350	100	4 x M6 / 10
FSF	680	270	13	300	350	100	4 x M8 / 25

¹⁾ The Power Modules are designed for mounting without any lateral cooling air clearance. For tolerance reasons, we recommend a lateral clearance of approx. 1 mm.

4.3.3 Dimension drawings, drilling dimensions for PM240P-2, IP20

Table 4-4 Mounting dimensions

Frame size	Width ¹⁾ [mm]	Height [mm]		Depth [mm]
		Power Module	Shield plate at the bottom	
FSD	200	472	152	237
FSE	275	551	177	237
FSF	305	708	257	357

¹⁾ The Power Modules can be mounted and operated side-by-side. For tolerance reasons, we recommend a lateral clearance of approx. 1 mm.

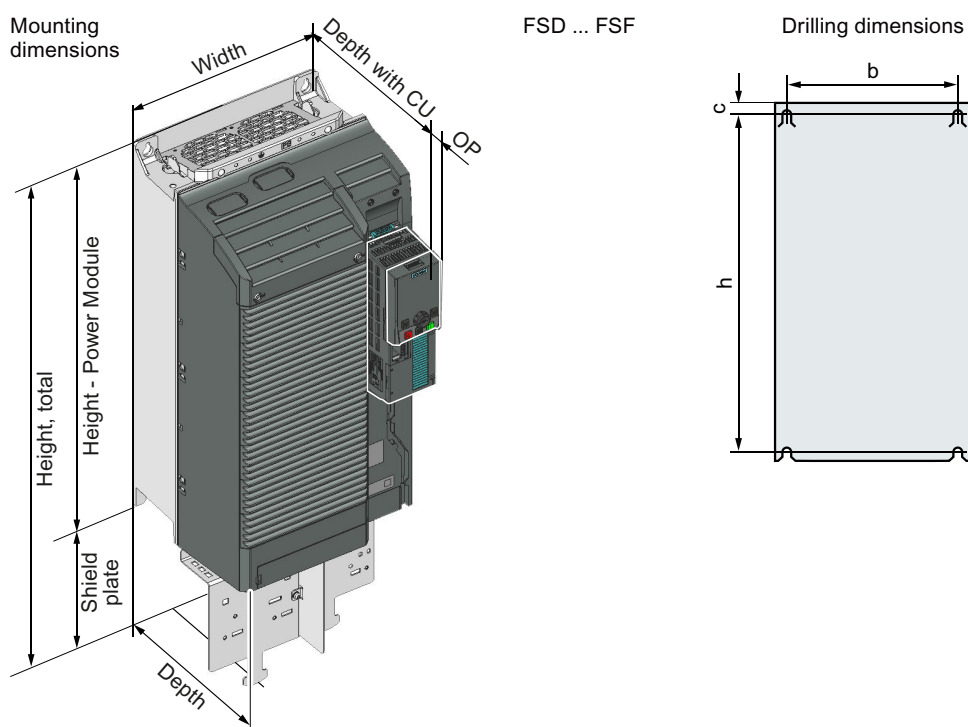


Table 4-5 Additional depth with Control Unit (CU) and BOP-2 or IOP Operator Panel [mm]

with CU	with CU and blanking cover or BOP-2	with CU and IOP
+ 0	+ 10	+ 21

4.3 Installing Power Modules

Table 4-6 Drilling dimensions, cooling air clearances [mm] and fixing [Nm]

Frame size	Drilling dimensions			Cooling air clearances ¹⁾			Mounting
	h	b	c	Top	Bottom	Front	Screws/torque
FSD	430	170	15	300	350	100	4 x M5 / 6
FSE	509	230	11	300	350	100	4 x M6 / 10
FSF	680	270	13	300	350	100	4 x M8 / 25

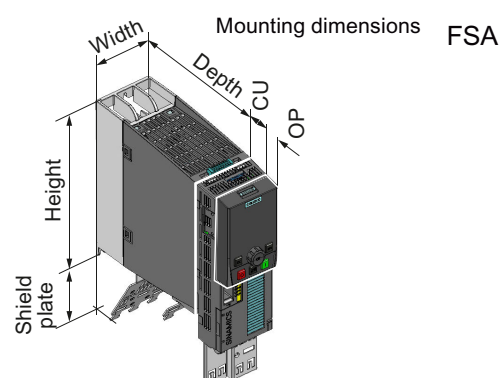
¹⁾ You can mount the Power Modules without any lateral cooling air clearance. For tolerance reasons, we recommend a lateral clearance of approx. 1 mm.

4.3.4 Dimension drawings, drilling dimensions for the PM230 Power Module, IP20

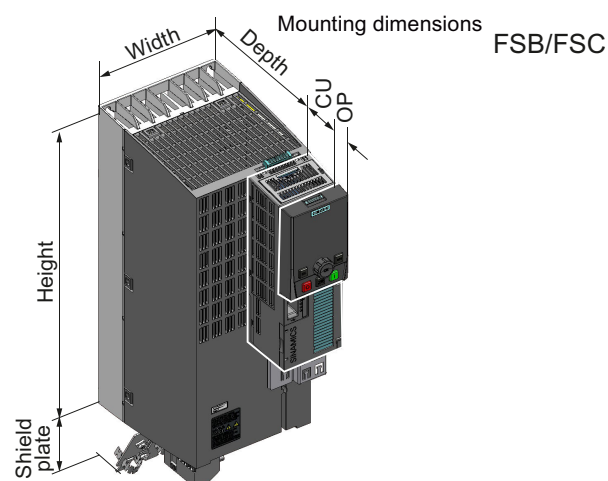
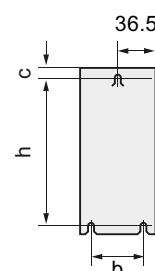
Table 4-7 Mounting dimensions without Control Unit [mm]

Frame size	Width	Height		Depth without Control Unit (CU)
		without shield plate	with shield plate	
FSA	73	196	276	165
FSB	100	292	370	165
FSC	140	355	432	165
FSD without filter	275	419	542	204
FSD with filter	275	512	635	204
FSE without filter	275	499	622	204
FSE with filter	275	635	758	204
FSF without filter	350	634	792	316
FSF with filter	350	934	1092	316

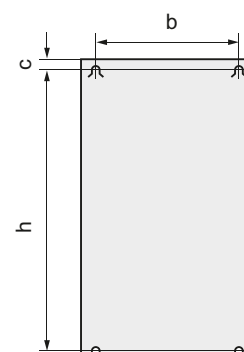
The following dimension drawings and drilling patterns are not to scale.



Drilling dimensions



Drilling dimensions



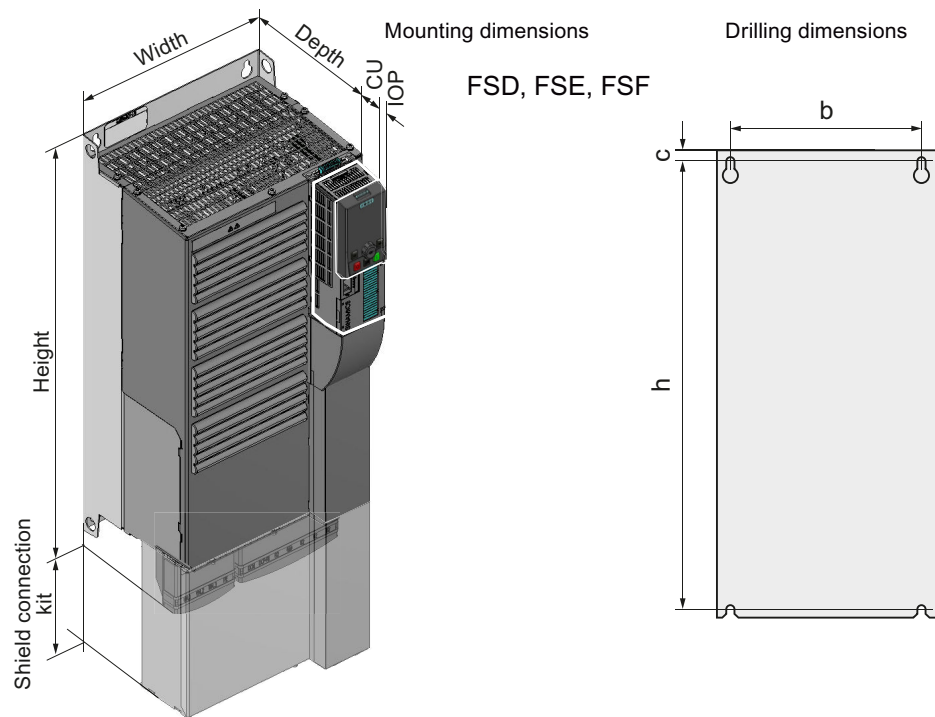


Table 4-8 Additional depth with Control Unit (CU) and BOP-2 or IOP Operator Panel [mm]

with CU	with CU and blanking cover or BOP-2	with CU and IOP
+ 41	+ 52	+ 63

Table 4-9 Drilling dimensions, cooling air clearances [mm] and fixing [Nm]

Frame size	Drilling dimensions			Cooling air clearances ¹⁾			Mounting
	b	h	c	Top	Bottom	Front	Screws/torque
FSA	62.3	186	6	80	100	---	2 x M4 / 2.5
FSB	80	281	6	80	100	---	4 x M4 / 2.5
FSC	120	343	6	80	100	---	4 x M5 / 3.0
FSD without filter	235	325	11	300	300	100	4 x M6 / 6.0
FSD with filter	235	419	11	300	300	100	4 x M6 / 6.0
FSE without filter	235	405	11	300	300	100	4 x M6 / 10
FSE with filter	235	451	11	300	300	100	4 x M6 / 10
FSF without filter	300	598	11	350	350	100	4 x M8 / 13
FSF with filter	300	899	11	350	350	100	4 x M8 / 13

¹⁾ You can mount the Power Modules without any lateral cooling air clearance. For tolerance reasons, we recommend a lateral clearance of approx. 1 mm.

4.3.5 Dimension drawings, drilling dimensions for PM230 and PM240-2 Power Modules utilizing push-through technology

Table 4-10 Mounting dimensions without Control Unit (CU) [mm]

Frame size	Width	Height		Depth ¹⁾		
		without shield plate	with shield plate	T1 +T2	T1	T2
FSA	126	238	322	171	118	53
FSB	154	345	430	171	118	53
FSC	200	411	500	171	118	53

¹⁾ Panel thickness of the control cabinet ≤ 3.5 mm

The following dimension drawings and drilling patterns are not to scale.

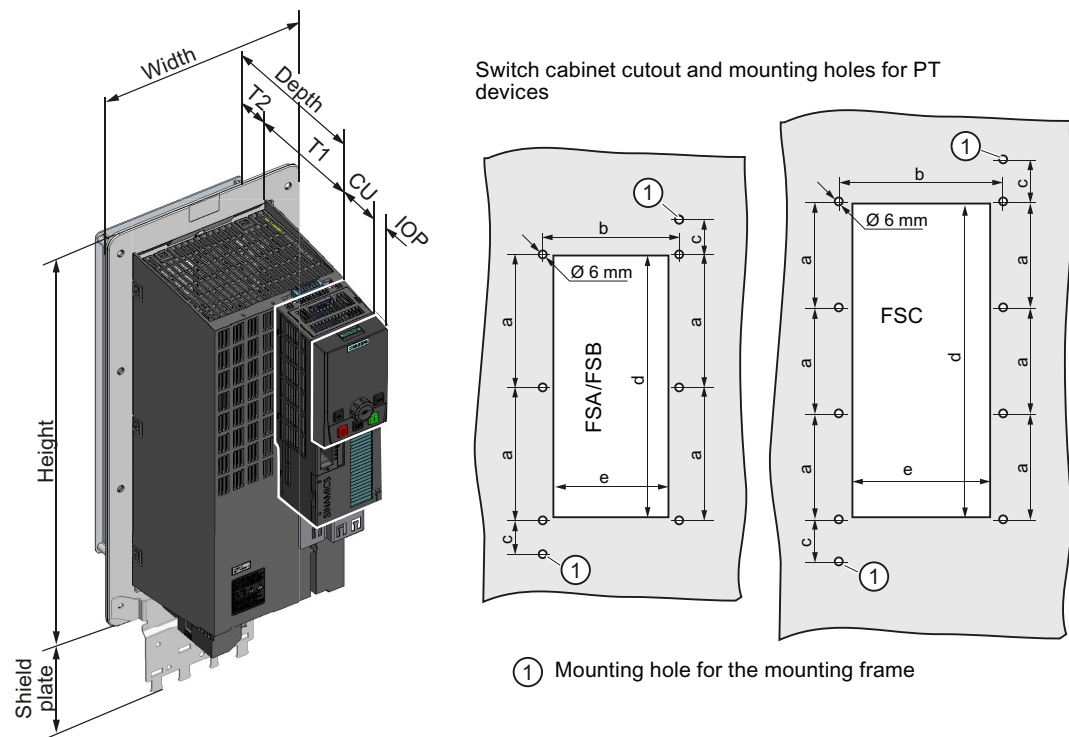


Table 4-11 Additional depth with Control Unit (CU) and BOP-2 or IOP Operator Panel [mm]

with CU	with CU and blanking cover or BOP-2	with CU and IOP
+ 41	+ 52	+ 63

Table 4-12 Drilling dimensions, cooling clearances and fixing

Frame size	Control cabinet cutout [mm]					Cooling air clearances [mm] ¹⁾			Fixing/torque
	a	b	c	d	e	Top	Bottom	Front	
FSA	103	106	27	198	88	80	100	100	8 x M5 / 3.5 Nm
FSB	148	134	34.5	304	116	80	100	100	8 x M5 / 3.5 Nm
FSC	123	174	30.5	365	156	80	100	100	10 x M5 / 3.5 Nm

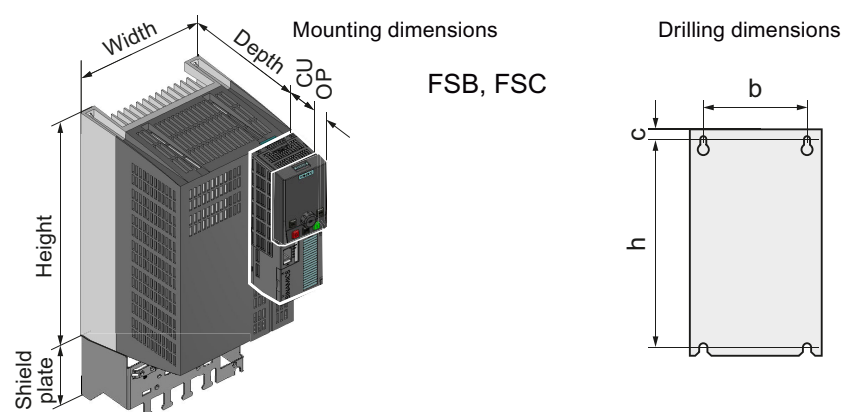
¹⁾ The Power Modules are designed for mounting without any lateral cooling air clearance. For tolerance reasons, we recommend a lateral clearance of 1 mm.

4.3.6 Dimensioned drawings, drilling dimensions for the PM250 Power Module

Table 4-13 Mounting dimensions without Control Unit (CU) [mm]

Frame size	Width	Height		Depth
		without shield plate	with shield plate	
FSC	189	334	432	185
FSD without filter	275	419	542	204
FSD with filter	275	512	635	204
FSE without filter	275	499	622	204
FSE with filter	275	635	758	204
FSF without filter	350	634	792	316
FSF with filter	350	934	1092	316

The following dimension drawings and drilling patterns are not to scale.



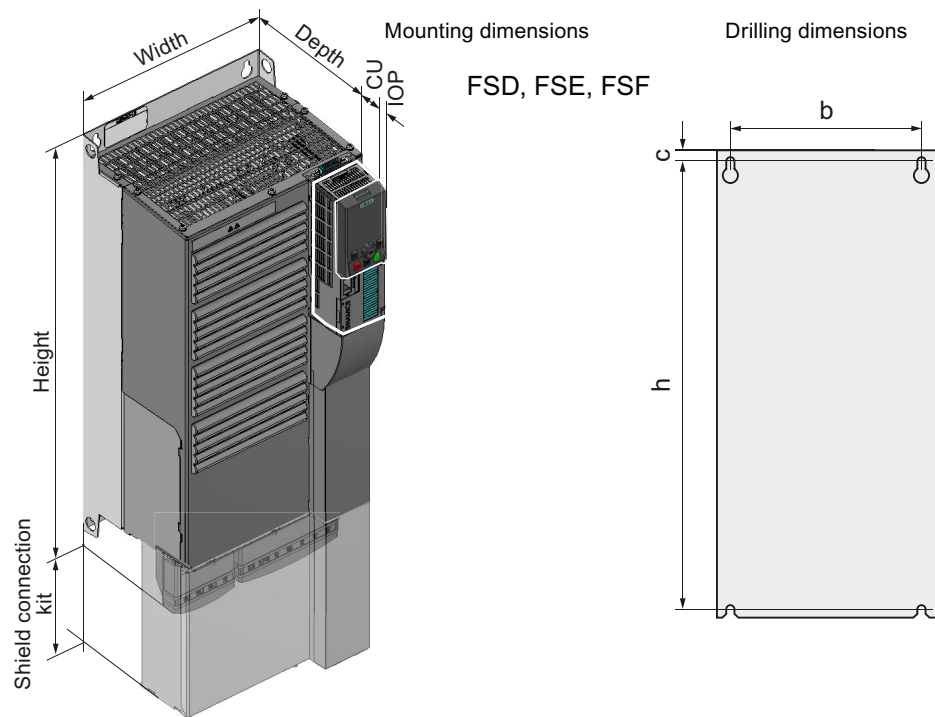


Table 4-14 Additional depth with Control Unit (CU) and BOP-2 or IOP Operator Panel [mm]

with CU	with CU and blanking cover or BOP-2	with CU and IOP
+ 41	+ 52	+ 63

Table 4-15 Drilling dimensions, cooling air clearances [mm] and fixing [Nm]

Frame size	Drilling dimensions			Cooling air clearances ¹⁾			Mounting
	b	h	c	Top	Bottom	Front	Screws/torque
FSC	167	323	6	125	125	65	4 x M5 / 3
FSD without filter	235	325	11	300	300	65	4 x M6 / 6
FSD with filter	235	419	11	300	300	65	4 x M6 / 6
FSE without filter	235	405	11	300	300	65	4 x M6 / 6
FSE with filter	235	541	11	300	300	65	4 x M6 / 6
FSF without filter	300	598	11	350	350	65	4 x M8 / 13
FSF with filter	300	898	11	350	350	65	4 x M8 / 13

¹⁾ You can mount the Power Modules without any lateral cooling air clearance. For tolerance reasons, we recommend a lateral clearance of approx. 1 mm.

4.4 Connecting the line supply, motor, and inverter components

WARNING

Danger to life through electric shock as well as fire hazard due to protective devices that either do not trip or trip too late

Overcurrent protective equipment that trips too late or not at all can cause electric shock or fire.

- In the case of a conductor-conductor or conductor-ground short-circuit, ensure that the short-circuit current at the point where the inverter is connected to the line supply corresponds as a minimum to the requirements of the protective equipment used.
- You must additionally use a residual-current protective device (RCD) if, for a conductor-ground short circuit, the required short-circuit current is not reached. Especially for TT line systems, the required short-circuit can be too low.
- It is not permissible that the short-circuit current exceeds the SCCR or the I_{CC} of the inverter and the disconnecting capacity of the protective equipment.



DANGER

Danger to life as a result of a hazardous voltage at the motor connections

As soon as the inverter is connected to the line supply, the motor connections of the inverter may carry dangerous voltages. When the motor is connected to the inverter, there is danger to life through contact with the motor terminals if the terminal box is open.

- Close the terminal box of the motor before connecting the inverter to the line supply.

Restrictions for installation altitudes above 2000 m

Above an installation altitude of 2000 m, the permissible line supplies are restricted.



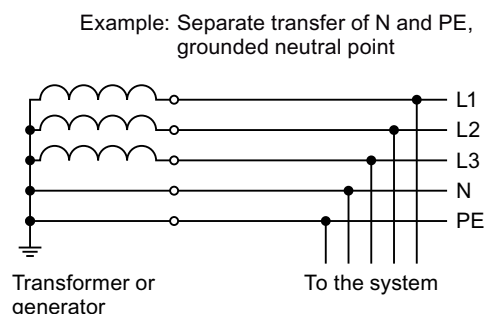
Restrictions for special ambient conditions (Page 474)

4.4.1 TN line system

A TN line system transfers the PE protective conductor to the installed plant or system using a cable.

Generally, in a TN line system the neutral point is grounded. There are versions of a TN system with a grounded line conductor, e.g. with grounded L1.

A TN line system can transfer the neutral conductor N and the PE protective conductor either separately or combined.



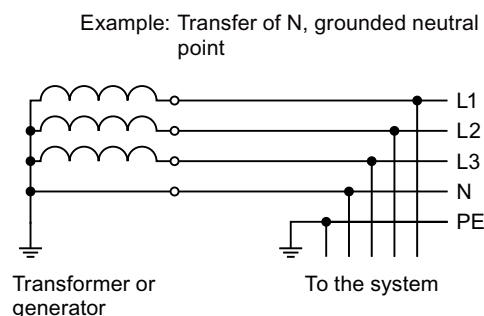
Inverter operated on a TN line system

- Inverter with integrated or external line filter:
 - Operation on TN line systems with grounded neutral point permissible.
 - Operation on TN line systems with grounded line conductor not permissible.
- Inverter without line filter:
 - Operation on all TN line systems ≤ 600 V permissible
 - Operation on TN line systems > 600 V and grounded neutral point permissible.
 - Operation on TN line systems > 600 V and grounded line conductor not permissible.

4.4.2 TT line system

In a TT line system, the transformer grounding and the installation grounding are independent of one another.

There are TT line supplies where the neutral conductor N is either transferred – or not.



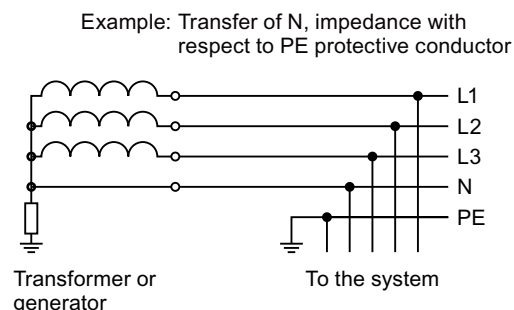
Inverter operated on a TT line system

- Inverter with integrated or external line filter:
 - Operation on TT line systems with grounded neutral point permissible.
 - Operation on TT line systems without grounded neutral point not permissible.
- Inverter without line filter:
 - Operation on all TT line systems permissible.
- For installations in compliance with IEC, operation on a TT line system is permissible. For installations in compliance with UL, operation on a TT line system is not permissible.

4.4.3 IT system

In an IT line system, all of the conductors are insulated with respect to the PE protective conductor – or connected to the PE protective conductor through an impedance.

There are IT systems with and without transfer of the neutral conductor N.



Inverter operated on an IT line system

- Inverters with integrated line filter:
 - Operation on IT line systems not permissible.
- Inverter without line filter:
 - Operation on all IT line systems permissible.

Behavior of the inverter when a ground fault occurs

In some instances, even for a ground fault, the inverter should still remain functional. In cases such as these, you must install an output reactor. This prevents an overcurrent trip or damage to the drive.

4.4.4 Protective conductor



WARNING

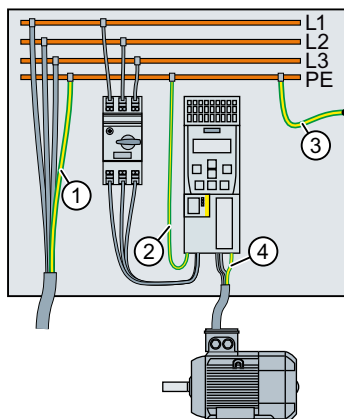
Danger to life caused by high leakage currents for an interrupted protective conductor

The drive components conduct a high leakage current via the protective conductor. Touching conductive parts when the protective conductor is interrupted can result in death or serious injury.

- Dimension the protective conductor as stipulated in the appropriate regulations.

Dimensioning the protective conductor

Observe the local regulations for protective conductors subject to an increased leakage current at the site of operation.



- ① Protective conductor for line feeder cables
- ② Protective conductor for inverter line feeder cables
- ③ Protective conductor between PE and the electrical cabinet
- ④ Protective conductor for motor feeder cables

The minimum cross-section of the protective conductor ① ... ④ depends on the cross-section of the line or motor feeder cable:

- Line or motor feeder cable $\leq 16 \text{ mm}^2$
 $\Rightarrow$ Minimum cross-section of the protective conductor = cross-section of the line or motor feeder cable
- $16 \text{ mm}^2 < \text{line or motor feeder cable} \leq 35 \text{ mm}^2$
 $\Rightarrow$ Minimum cross-section of the protective conductor = 16 mm^2
- Line or motor feeder cable $> 35 \text{ mm}^2$
 $\Rightarrow$ Minimum cross-section of the protective conductor = $\frac{1}{2}$ cross-section of the line or motor feeder cable

Additional requirements placed on the protective conductor ①:

- For permanent connection, the protective conductor must fulfill at least one of the following conditions:
 - The protective conductor is routed so that it is protected against damage along its complete length.
 Cables routed inside electrical cabinets or enclosed machine housings are considered to be adequately protected against mechanical damage.
 - As a conductor of a multi-conductor cable, the protective conductor has a cross-section $\geq 2.5 \text{ mm}^2 \text{ Cu}$.
 - For an individual conductor, the protective conductor has a cross-section $\geq 10 \text{ mm}^2 \text{ Cu}$.
 - The protective conductor consists of two conductors with the same cross-section.
- When connecting a multi-core cable using an industrial plug connector according to EN 60309, the protective conductor must have a cross-section of $\geq 2.5 \text{ mm}^2 \text{ Cu}$.

4.4.5 Connecting an inverter with the PM240-2 Power Module

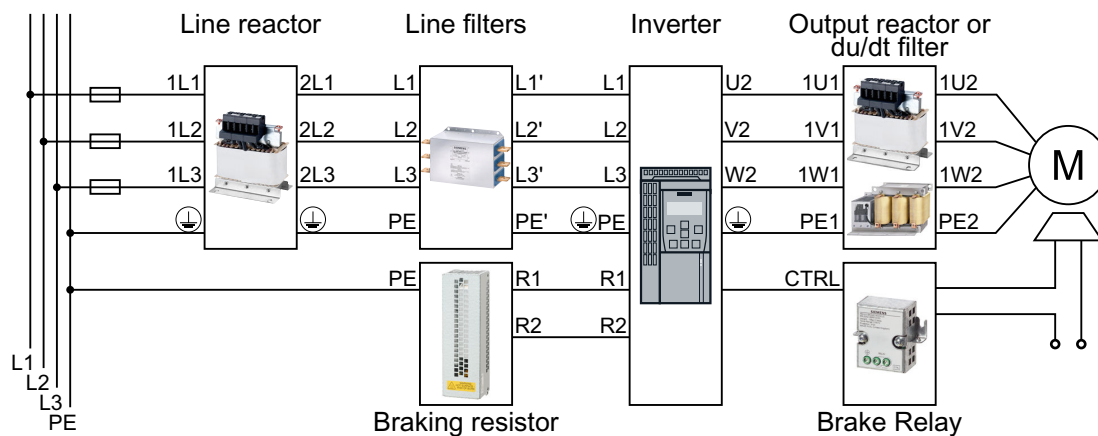


Figure 4-5 Connection of the PM240-2 Power Module, 3-phase AC

A line reactor is not required for the FSD ... FSF Power Modules.

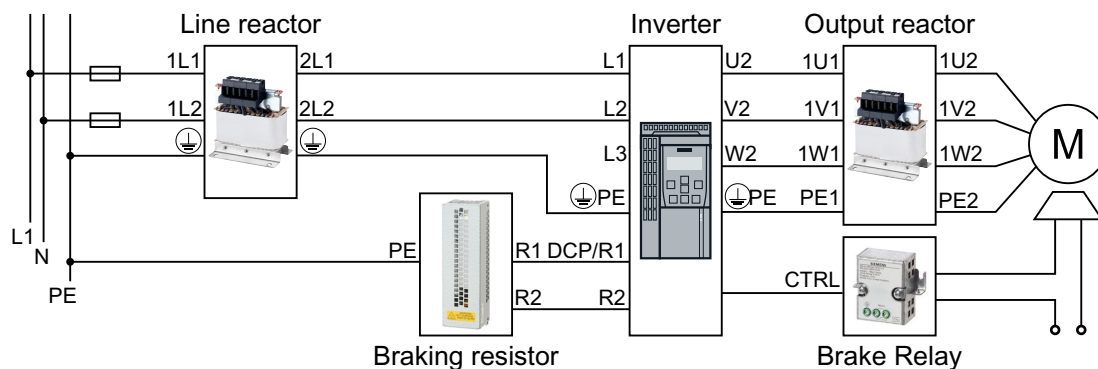


Figure 4-6 Connection of the PM240-2 Power Module, 1-phase 200 VAC

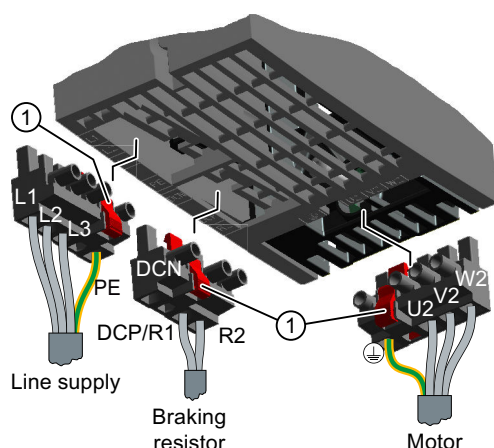
Table 4-16 Connection, cross-section and tightening torque for PM240-2 Power Modules

Inverters	Connection		Cross-section, tightening torque		Stripped insulation length
			Metric	Imperial	
FSA	Line supply, motor, DC link and braking resistor		1.5 ... 2.5 mm ² , 0.5 Nm	16 ... 14 AWG, 4.5 lbf in	8 mm
FSB			1.5 ... 6 mm ² , 0.6 Nm	16 ... 10 AWG, 5.5 lbf in	8 mm
FSC			6 ... 16 mm ² , 1.3 Nm	10 ... 6 AWG, 12 lbf in	10 mm
FSD	Line supply, motor and DC link	Screw-type terminal	10 ... 35 mm ² , 2.5 ... 4.5 Nm	20 ... 10 AWG, 22 lbf in 8 ... 2 AWG, 40 lbf in	18 mm
	Braking resistor		2.5 ... 16 mm ² , 1.2 ... 1.5 Nm	20 ... 6 AWG, 15 lbf in	10 mm

4.4 Connecting the line supply, motor, and inverter components

Inverters	Connection		Cross-section, tightening torque		Stripped insulation length
			Metric	Imperial	
FSE	Line supply, motor and DC link	Screw-type terminal	25 ... 70 mm ² , 8 ... 10 Nm	6 ... 3/0 AWG, 88.5 lbf in	25 mm
	Braking resistor		10 ... 35 mm ² , 2.5 ... 4.5 Nm	20 ... 10 AWG, 22 lbf in 8 ... 2 AWG, 40 lbf in	18 mm
FSF	Line supply, motor and DC link	 Cable lug according to SN71322	35 ... 2 × 120 mm ² , 22 ... 25 Nm	1 ... 2 × 4/0 AWG, 210 lbf.in	--
	Braking resistor	Screw-type terminal	25 ... 70 mm ² , 8 ... 10 Nm	6 ... 3/0 AWG, 88.5 lbf in	25 mm

Connections for frame sizes FSA ... FSC



The Power Modules are equipped with withdrawable plug connectors that cannot be inadvertently interchanged.

To remove a plug connector, you must release it by pressing on the red lever.

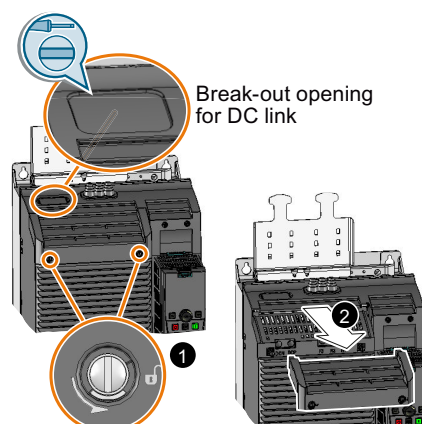
① Release lever

Connections for frame sizes FSD ... FSF

You must remove the covers from the connections in order to connect the line supply, braking resistor and motor to the inverter.

In addition, for frame sizes FSD and FSE, release the two terminal screws on the connections for the motor and braking resistor, and remove the dummy plug.

For frame size FSF you must breakout the openings from the connection cover for the power connections. Use side cutters or a fine saw blade.



FSD ... FSF: remove the top covers

4.4 Connecting the line supply, motor, and inverter components

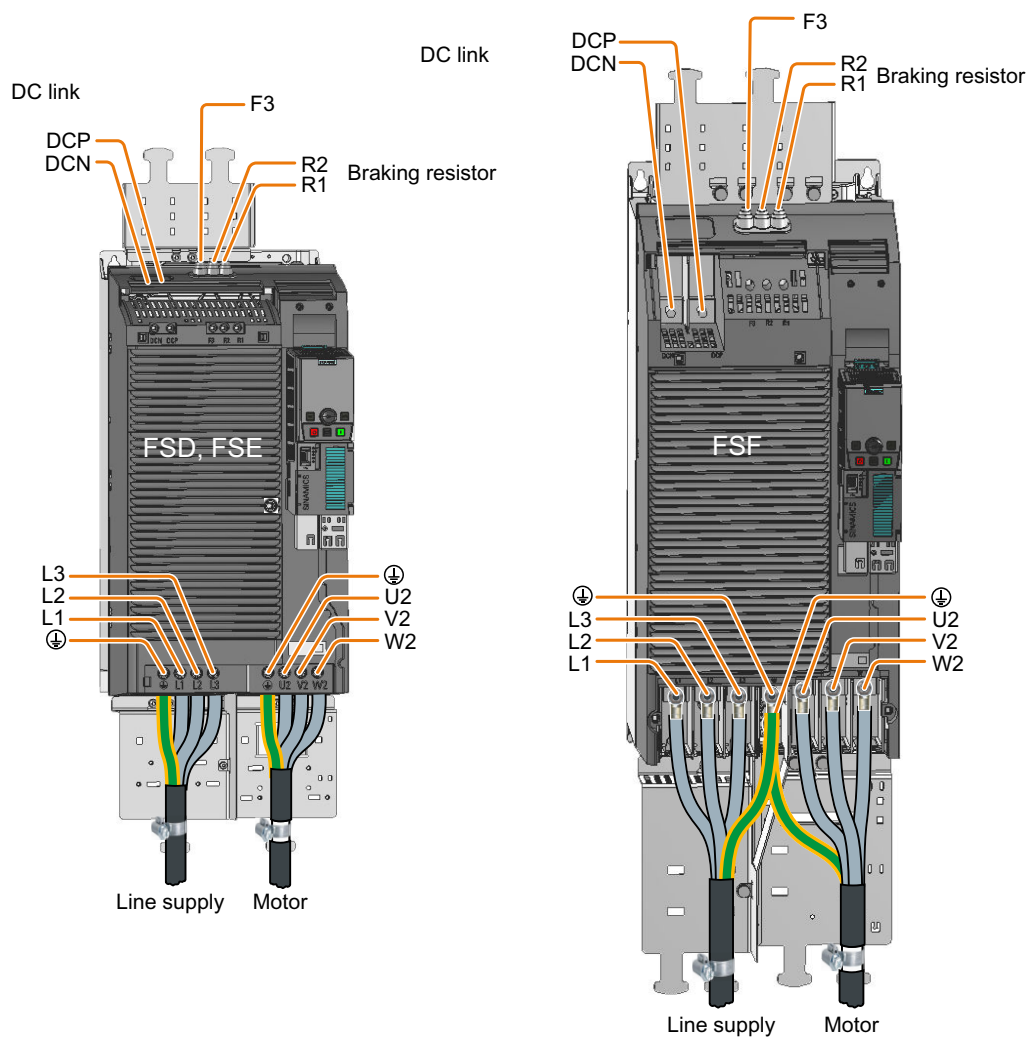
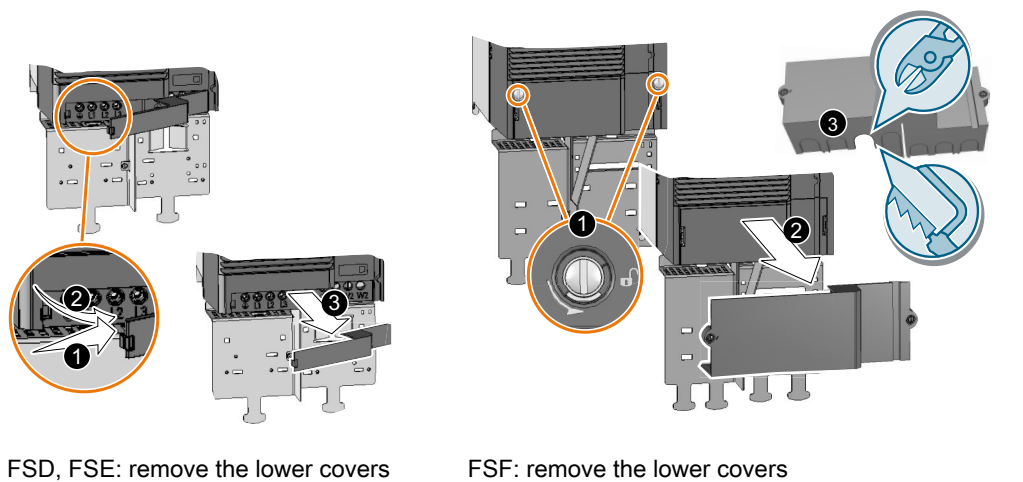


Figure 4-7 Connections for the line supply, motor and braking resistor

4.4 Connecting the line supply, motor, and inverter components

You must re-attach the connection covers in order to re-establish the touch protection of the inverter after it has been connected up.

4.4.6 Connecting the inverter with the PM230 Power Module

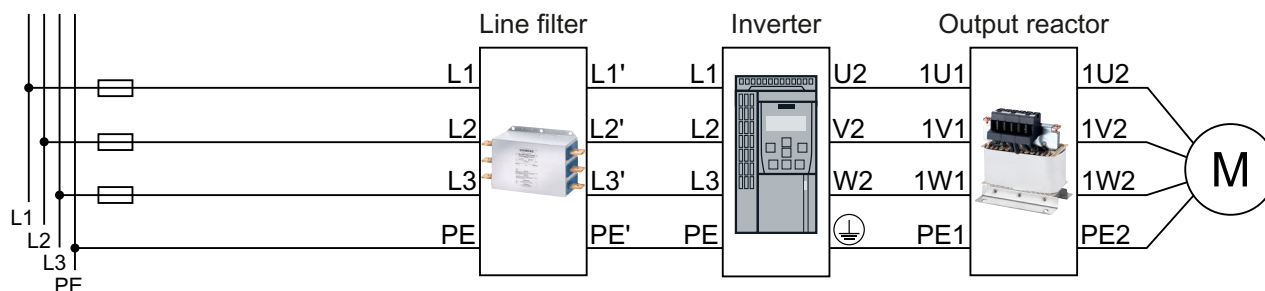
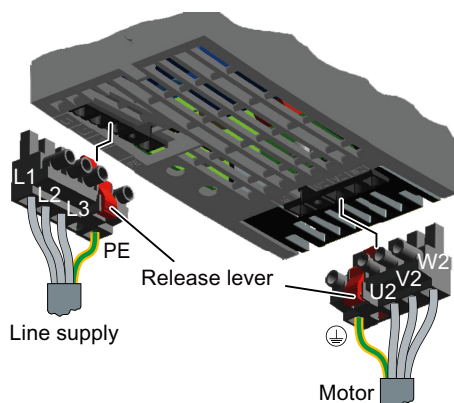


Figure 4-8 PM230 Power Module connection overview

Table 4-17 Connection, cross-section and tightening torque for PM230 Power Modules

Inverters	Connection		Cross-section, tightening torque		Stripped insulation length
			Metric	Imperial	
FSA	Line supply, motor		1 ... 2.5 mm ² , 0.5 Nm	16 ... 14 AWG, 4.5 lbf in	8 mm
FSB			1.5 ... 6 mm ² , 0.6 Nm	16 ... 10 AWG, 5.5 lbf in	8 mm
FSC			6 ... 16 mm ² , 1.3 Nm	10 ... 6 AWG, 12 lbf in	10 mm
FSD	Line supply, motor		10 ... 35 mm ² , 6 Nm	7 ... 2 AWG, 53 lbf in	--
FSE			25 ... 50 mm ² , 6 Nm	3 ... 1 AWG, 53 lbf in	--
FSF			35 ... 120 mm ² , 13 Nm	2 ... 4/0 AWG, 115 lbf in	--

Connections for frame sizes FSA ... FSC

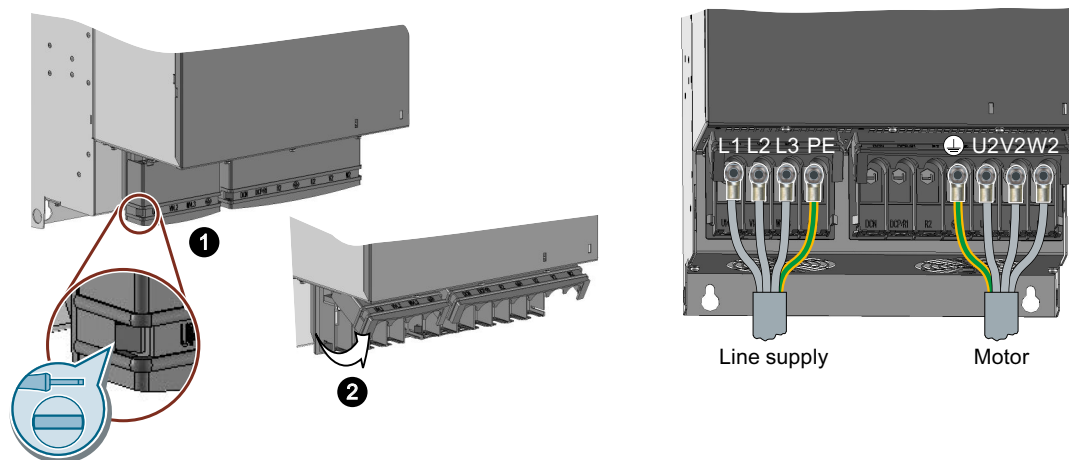


The Power Modules are equipped with withdrawable plug connectors that cannot be inadvertently interchanged.

To remove a plug connector, you must release it by pressing on the red lever.

Connections for frame sizes FSD ... FSF

The line and motor connections have covers to prevent them from being touched.



You must open the cover to connect the line and motor:

1. Release the catches on both sides of the covers using a screwdriver.
2. Swivel the covers upwards.

Close the covers once you have connected the line and motor.

4.4.7 Connecting the inverter with the PM240P-2 Power Module

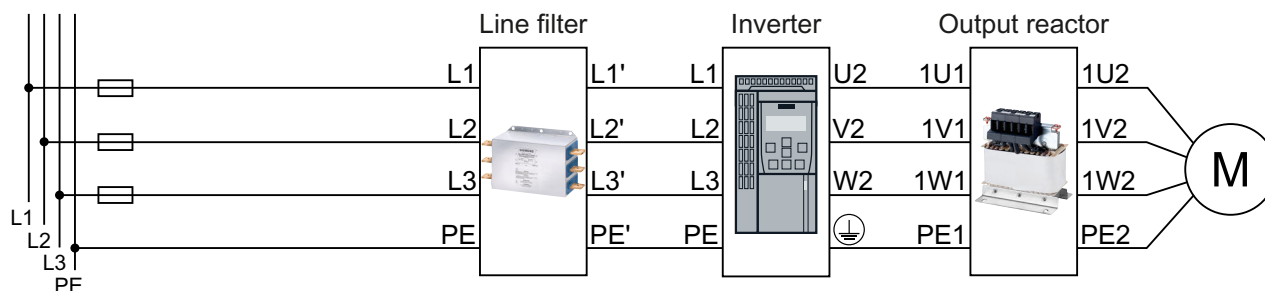


Figure 4-9 PM240P-2 Power Module connection overview

Table 4-18 Connection, cross-section and tightening torque for PM240P-2 Power Modules

Inverters	Connection		Cross-section, tightening torque		Stripped insulation length
			Metric	Imperial	
FSD	Line supply, motor	Screw-type terminal	10 ... 35 mm ² , 2.5 ... 4.5 Nm	20 ... 10 AWG, 22 lbf in 8 ... 2 AWG, 40 lbf in	18 mm
FSE			25 ... 70 mm ² , 8 ... 10 Nm	6 ... 3/0 AWG, 88.5 lbf in	25 mm
FSF	Line supply, motor	Cable lug according to SN71322	35 ... 2 × 120 mm ² , 22 ... 25 Nm	1 ... 2 × 4/0 AWG, 210 lbf in	--

Connections, frame sizes FSD ... FSF

You must remove the covers from the connections in order to connect the line supply and motor cables to the inverter.

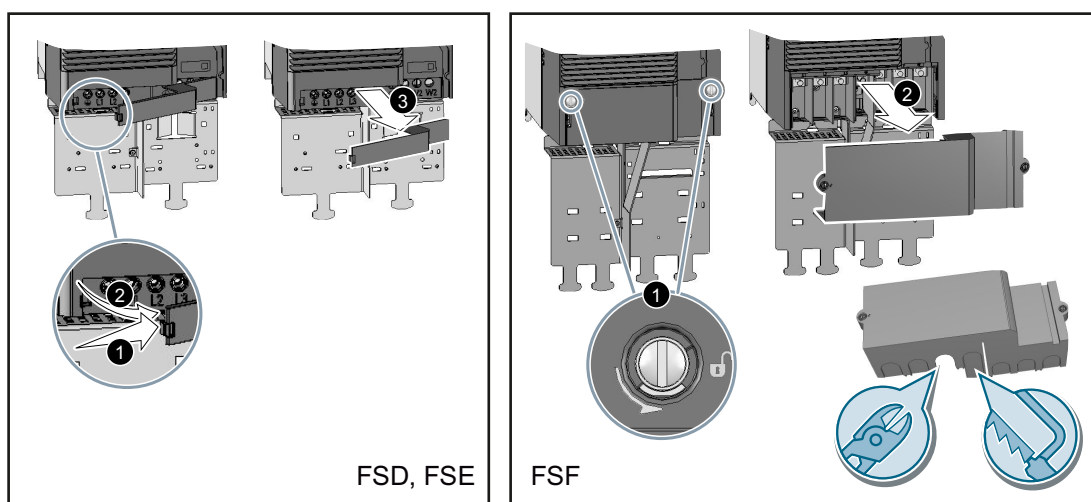


Figure 4-10 Remove the connection covers

4.4 Connecting the line supply, motor, and inverter components

In addition, for frame sizes FSD and FSE, release the two terminal screws on the connections for the motor and remove the dummy plug.

For frame size FSF you must breakout the openings from the connection cover for the power connections. Use side cutters or a fine saw blade.

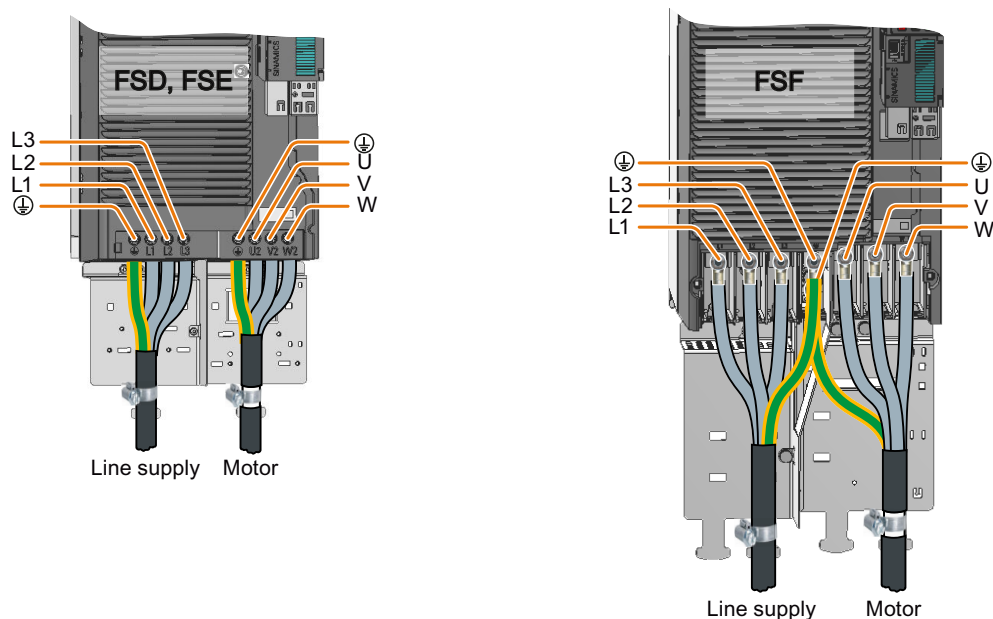


Figure 4-11 Line and motor connections

You must re-attach the connection covers in order to re-establish the touch protection of the inverter after it has been connected up.

4.4.8 Connecting the inverter with the PM250 Power Module

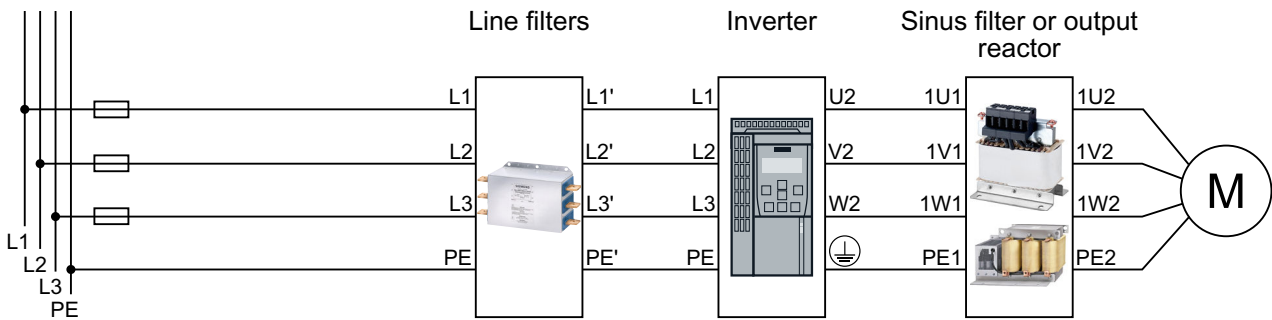
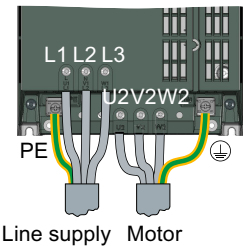


Figure 4-12 Connecting the PM250 Power Module

Table 4-19 Connection, cross-section and tightening torque for PM250 Power Modules

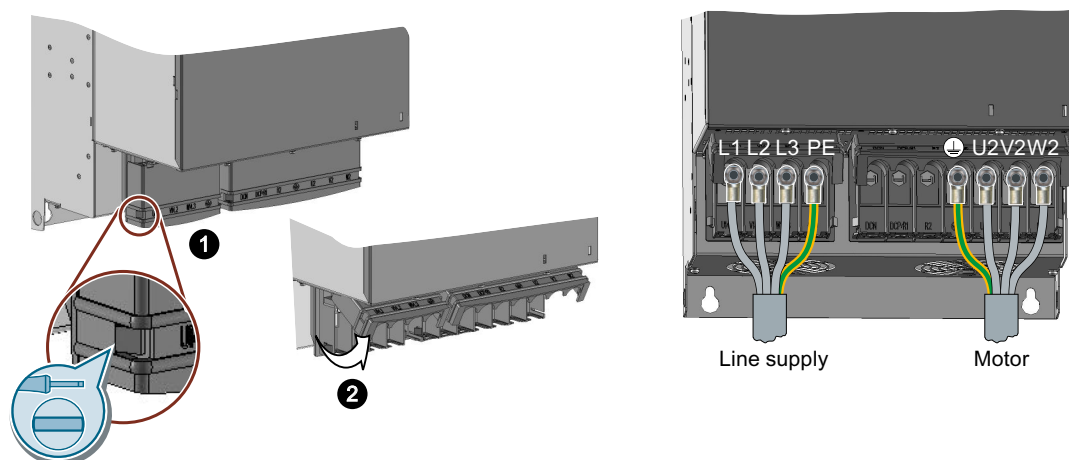
Inverters	Line supply and motor connection	Cross-section and tightening torque		Stripped insulation length
		Metric	Imperial	
FSC	Screw-type terminal	4 ... 10 mm ² , 2.3 Nm	12 ... 8 AWG, 20 lbf in	10 mm
FSD	 Cable lug	10 ... 35 mm ² , 6 Nm	7 ... 2 AWG, 53 lbf in	--
FSE		25 ... 50 mm ² , 6 Nm	3 ... 1 AWG, 53 lbf in	--
FSF		35 ... 120 mm ² , 13 Nm	2 ... 4/0 AWG, 115 lbf in	--

Connections for frame size FSC



Connections for frame sizes FSD ... FSF

The line and motor connections have covers to prevent them from being touched.



You must open the cover to connect the line and motor:

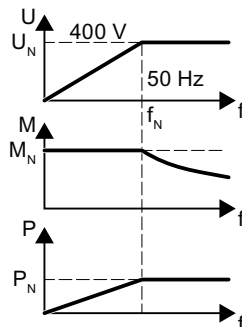
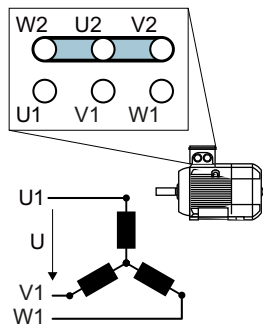
1. Release the catches on both sides of the covers using a screwdriver.
2. Swivel the covers upwards.

Close the covers once you have connected the line and motor.

4.4.9 Connecting the motor to the inverter in a star or delta connection

Standard induction motors with a rated power of approximately ≤ 3 kW are normally connected in a star/delta connection (Y/ Δ) at 400 V/230 V. For a 400-V line supply, you can connect the motor to the inverter either in a star or in a delta connection.

Operating the motor in a star connection

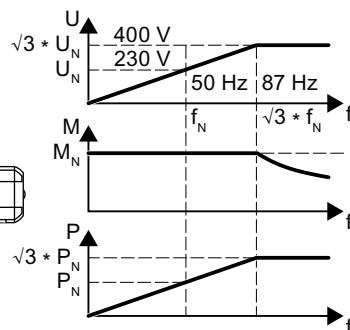
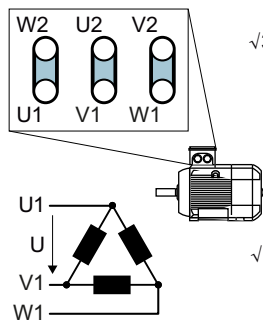


In a star connection, the motor can provide its rated torque M_N in the range 0 ... rated frequency f_N .

Rated voltage $U_N = 400$ V is available at a rated frequency $f_N = 50$ Hz.

The motor goes into field weakening above the rated frequency. In field weakening, the available motor torque decreases linearly with $1/f$. In field weakening, the available power remains constant.

Operating the motor in a delta connection with 87 Hz characteristic



In a delta connection, the motor is operated with a voltage and frequency above its rated values. As a consequence, the motor power is increased by a factor $\sqrt{3} \approx 1.73$.

In the range $f = 0 \dots 87$ Hz, the motor can output its rated torque M_N .

The maximum voltage $U = 400$ V is available at a frequency of $f = \sqrt{3} \times 50$ Hz ≈ 87 Hz.

The motor only goes into field weakening above 87 Hz.

The higher motor power when operated with an 87 Hz characteristic has the following disadvantages:

- The inverter must supply approximately 1.73x current. Select an inverter based on its rated current - and not its rated power.
- The motor temperature increases more significantly than when operated with $f \leq 50$ Hz.
- The motor must have windings that are approved for a voltage $>$ rated voltage U_N .
- As the fan impeller rotates faster, the motor has a higher noise level than operation with $f \leq 50$ Hz.

4.4.10 Connecting a motor holding brake

The inverter uses the Brake Relay to control the motor holding brake. Two types of Brake Relay exist:

- The Brake Relay controls the motor holding brake
- The Safe Brake Relay controls a 24 V motor holding brake and monitors the brake control for short-circuit or cable breakage.

Note

Brake Relay and Safe Brake Relay

The Brake Relay and the Safe Brake Relay do not differ from each other in the installation and the connection to the inverter.

Connection to the inverter

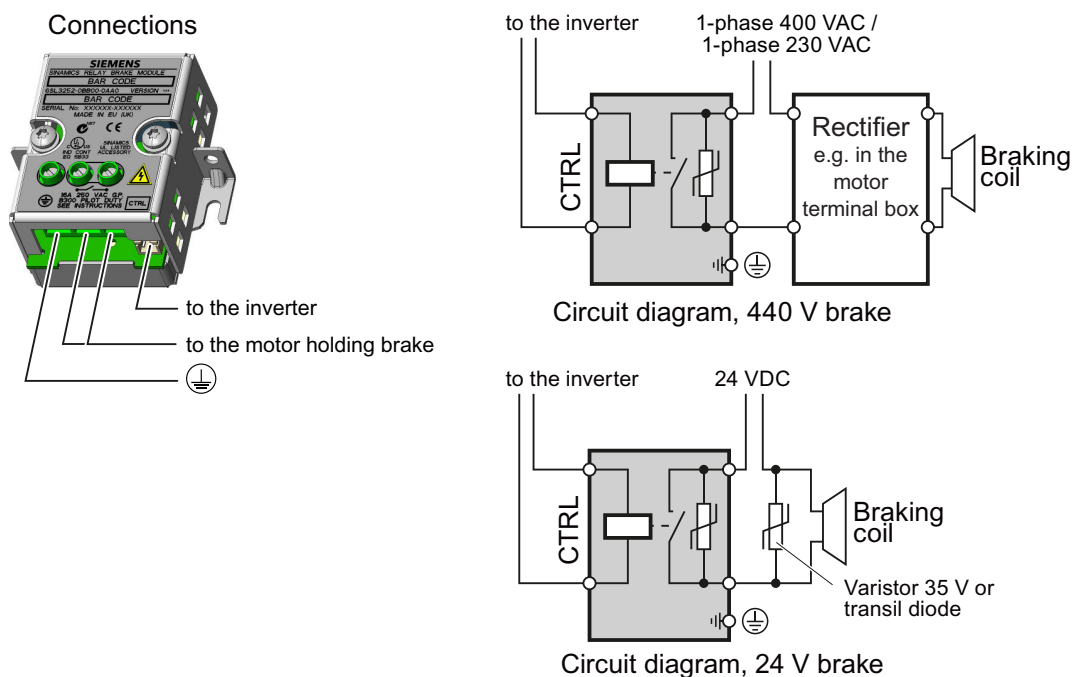
To ensure that you have the correct cable for connecting the Brake Relay irrespective of the inverter size, you are supplied with two preassembled cables with different lengths. Connect the appropriate cable to the Brake Module and to the inverter as shown below.

If you are using your own cable, ensure that the cable is insulated and rated for 600 V.

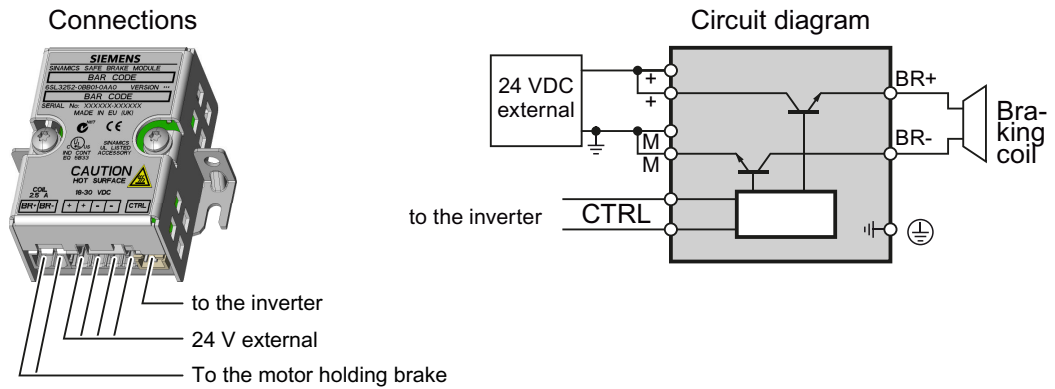
Connecting the motor holding brake via a PELV circuit

The Brake Relay must be connected to the protective conductor if the motor brake is supplied from a PELV circuit.

Brake Relay



Safe Brake Relay



4.4.10.1 Installing a Brake Relay - PM250 Power Module

Installing the Brake Relay

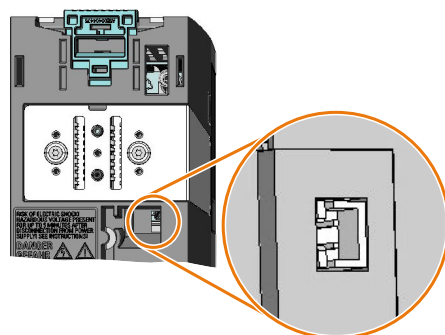
If you use the optional shield plate, install the Brake Relay on the shield plate of the Power Module.

If you do not use the shield plate, install the Brake Relay as close as possible to the Power Module.

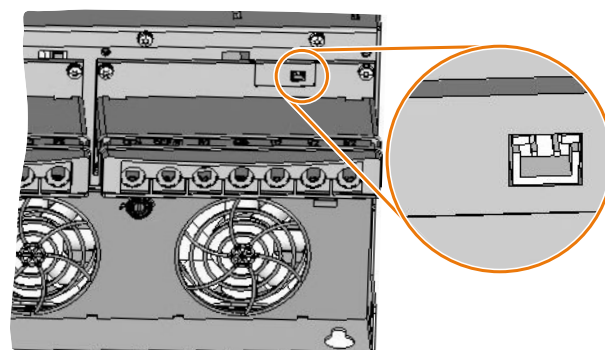
Connecting the Brake Relay to the inverter

The connector for the Brake relay is located at the front of the Power Module for the FSA ... FSC frame sizes. These Power Modules have a cable entry for the connecting cable to the Brake Relay.

The connector for the Brake Relay is located at the bottom of the Power Module for the FSD ... FSF frame sizes.



Brake relay connector for FSA ... FSC
Power Modules



Brake relay connector for FSD ... FSF
Power Modules

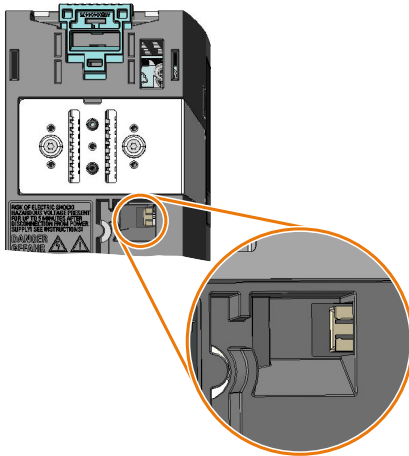
4.4.10.2 Mounting and connecting the brake relay

Installing the Brake Relay

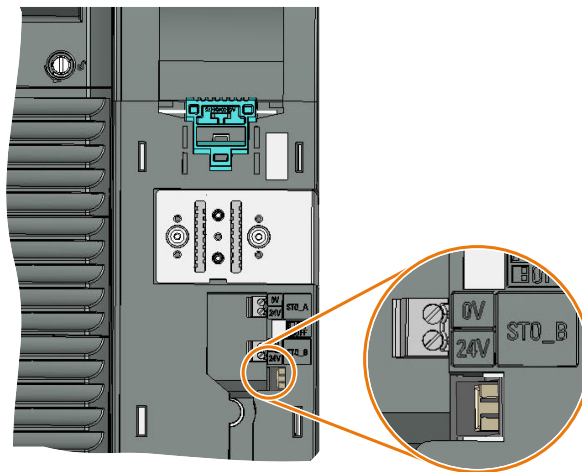
- **FSA ... FSC:** Install the Brake Relay next to the Power Module.
- **FSD ... FSF:** Install the Brake Relay at the rear of the lower shield plate. Attach the Brake Relay before you install the shield plate.

Connecting the Brake Relay to the inverter

The connector for the Brake Relay is located at the front of the Power Module. Lay the cable harness for the Brake Relay in the cable routing.



Brake relay connector for FSA ... FSC
Power Modules without STO terminals



Brake relay connector for FSD ... FSF
Power Modules with STO terminals

4.5 Installing Control Unit

Installing the Control Unit - General

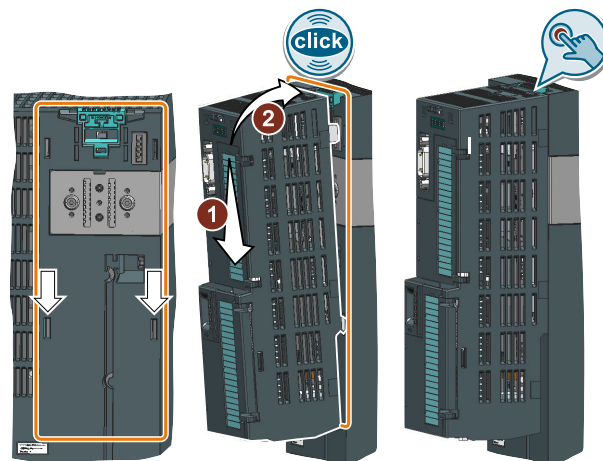
Each Power Module has an appropriate holder for the Control Unit and a release mechanism.

Inserting the Control Unit



1. Proceed as follows to plug the Control Unit onto a Power Module:
- 2.

1. Place the two catches of the Control Unit in the matching grooves of the Power Module.
2. Press the Control Unit onto the Power Module until you hear that it latches.



You have now plugged the Control Unit onto the Power Module.

Removing the Control Unit

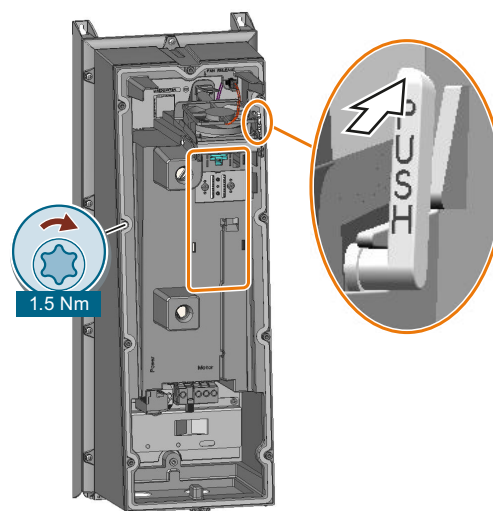
Remove the Control Unit from the Power Module by pressing the release mechanism.

Special features for the PM230 Power Module IP55, FSA ... FSC

To insert or detach the Control Unit, you must release eight or ten fixing screws of the cover and then remove the cover.

The Power Module release mechanism is shown in the diagram.

Attach the cover again before you commission the inverter. Do not damage the seal of the cover when attaching it.



Adapter for operating a PM230 IP55 Power Module



For operating the Control Unit with a PM230 IP55 Power Module, FSA ... FSC, an adapter is required between the Control Unit and operator panel (BOP-2 or IOP).

The adapter, which is included in the scope of delivery of the Power Module, is too short for the CU240E-2 Control Unit. The matching adapter for the CU240E-2 Control Unit can be ordered through the KnorrTec company.

Article number 10055500

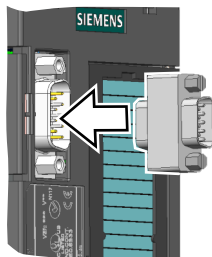
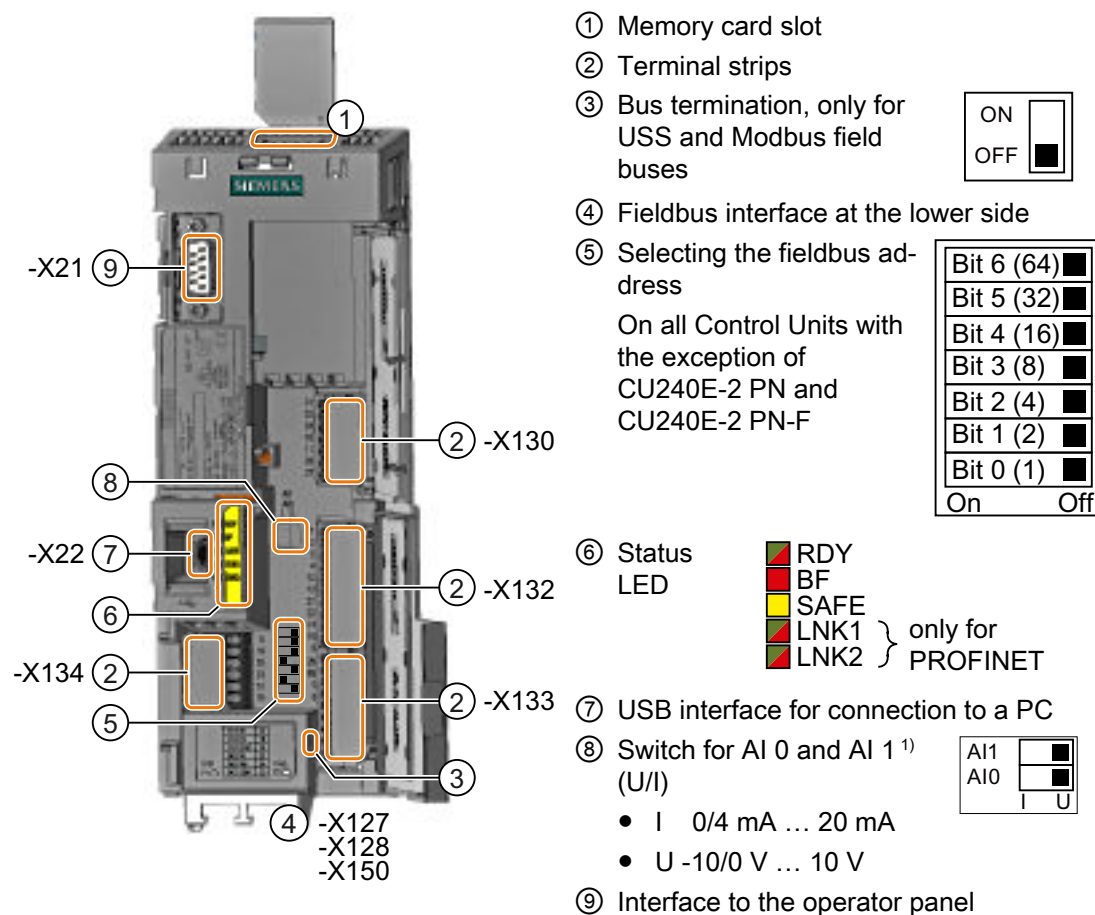


Figure 4-13 Insert the KnorrTec adapter on the interface for the operator panel

4.5.1 Overview of the interfaces

Interfaces at the front of the Control Unit

To access the interfaces at the front of the Control Unit, you must lift the Operator Panel (if one is being used) and open the front doors.



¹⁾ AI 1 is not available on the CU240B-2 Control Unit

Table 4-20 Number of inputs and outputs

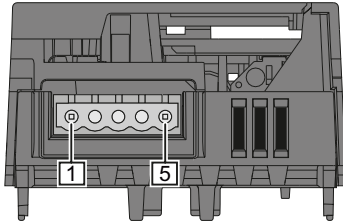
	Digital inputs DI	Digital outputs DO	Analog inputs AI	Analog outputs AO	Fail-safe digital inputs F-DI ¹⁾
CU240B-2, CU240B-2 DP	4	1	1	1	0
CU240E-2, CU240E-2 DP, CU240E-2 PN	6	3	2	2	1
CU240E-2 F, CU240E-2 DP-F, CU240E-2 PN-F	6	3	2	2	3

¹⁾ Each fail-safe digital input F-DI used takes up two digital inputs DI

4.5.2 Fieldbus interface allocation

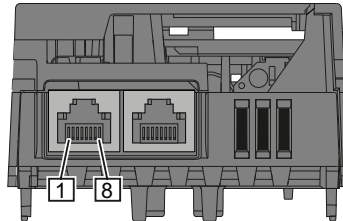
Interfaces at the lower side of the CU240B-2 and CU240E-2 Control Units

RS485 plug for
USS and Modbus RTU (X128)



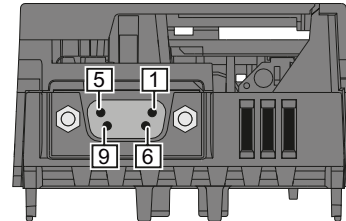
- Pin
- 1 0 V, reference potential
 - 2 RS485P, receive and transmit (+)
 - 3 RS485N, receive and transmit (-)
 - 4 Cable shield
 - 5 Not connected

RJ45 connector socket for PROFINET
IO (X150 P1, X150 P2)



- Pin
- 1 RX+, receive data +
 - 2 RX-, receive data -
 - 3 TX+, Transmit data +
 - 4 Not assigned
 - 5 Not assigned
 - 6 TX-, transmit data -
 - 7 Not assigned
 - 8 Not assigned

SUB-D socket for PROFIBUS DP
(X126)



- Pin
- 1 Shield, grounding connection
 - 2 Not assigned
 - 3 RxD/TxD-P, receive and transmit (B/B')
 - 4 CNTR-P, control signal
 - 5 DGND, reference potential for data (C/C')
 - 6 VP, supply voltage
 - 7 Not assigned
 - 8 RxD/TxD-N, receive and transmit (A/A')
 - 9 Not assigned

4.5.3 Terminal strips on CU240B-2 Control Units

Terminal strips with wiring example

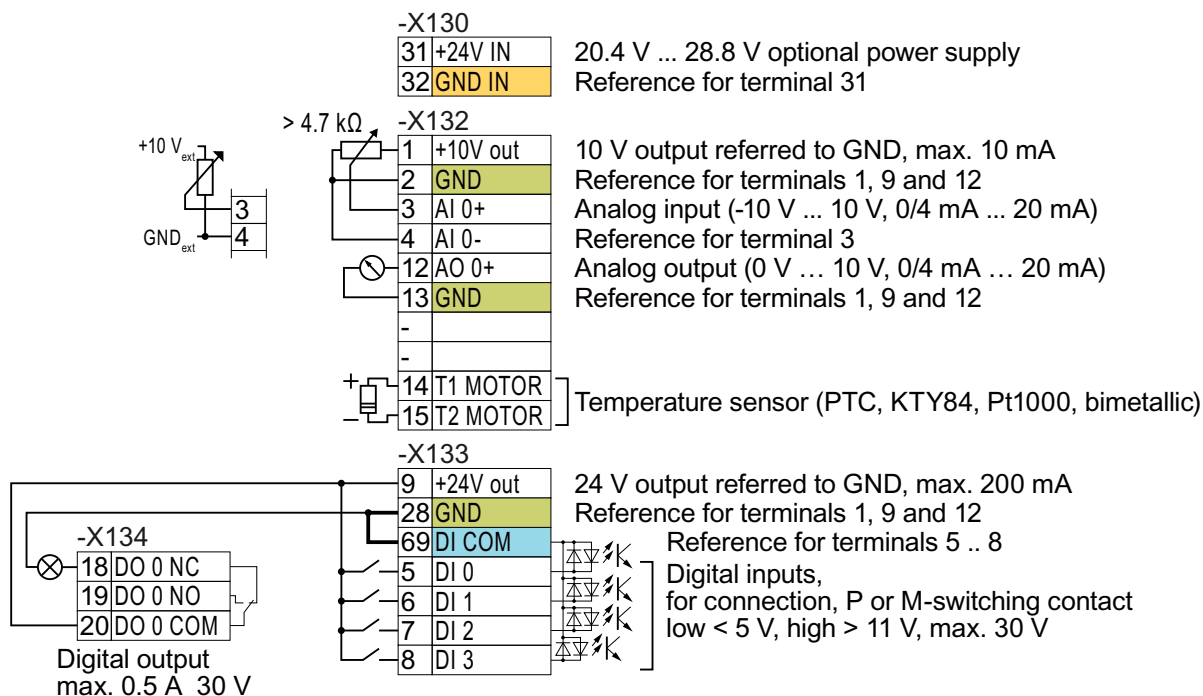


Figure 4-14 Wiring example of the digital inputs with the internal inverter 24 V power supply

GND All terminals labelled with reference potential "GND" are connected internally in the inverter.

DI COM Reference potential "DI COM" is electrically isolated from "GND".

→ if, as described above, you use the 24-V power supply from terminal 9 to supply the digital inputs, then you must connect "GND" with "DI COM" at the terminals.

31 +24 V IN
32 GND IN When an optional 24-V power supply is connected to terminals 31, 32, the Control Unit remains in operation even after the Power Module has been disconnected from the line supply. The Control Unit thus maintains the fieldbus communication, for example.

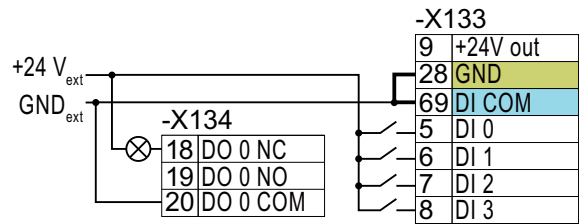
→ At terminals 31, 32, only connect a power supply that is in accordance with SELV (Safety Extra Low Voltage) or PELV (Protective Extra Low Voltage).

→ if you also wish to use the power supply at terminals 31, 32 for the digital inputs, then you must connect "DI COM" and "GND IN" with one another at the terminals.

3 AI 0+
4 AI 0- You may use the internal 10V power supply or an external power supply for the analog input.

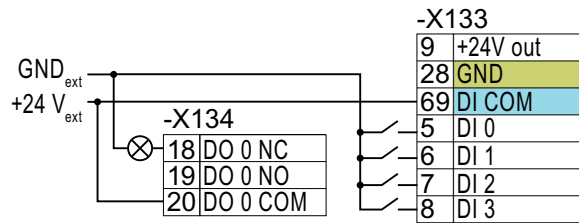
→ If you use the internal 10 V power supply, you must connect AI 0- to GND.

Additional options for wiring the digital inputs



You must remove the jumper between terminals 28 and 69 if it is necessary to have electrical isolation between the external power supply and the internal inverter power supply.

Connecting P-switching contacts with an external power supply



It is not permissible that terminals 28 and 69 are connected with one another.

Connecting M-switching contacts with an external power supply

4.5.4 Factory setting of the CU240B-2 interfaces

The factory setting of the terminals depends on which fieldbus the Control Unit supports.

Control Units with PROFIBUS interface

The function of the fieldbus interface and digital inputs DI 0, DI 1 depends on DI 3.

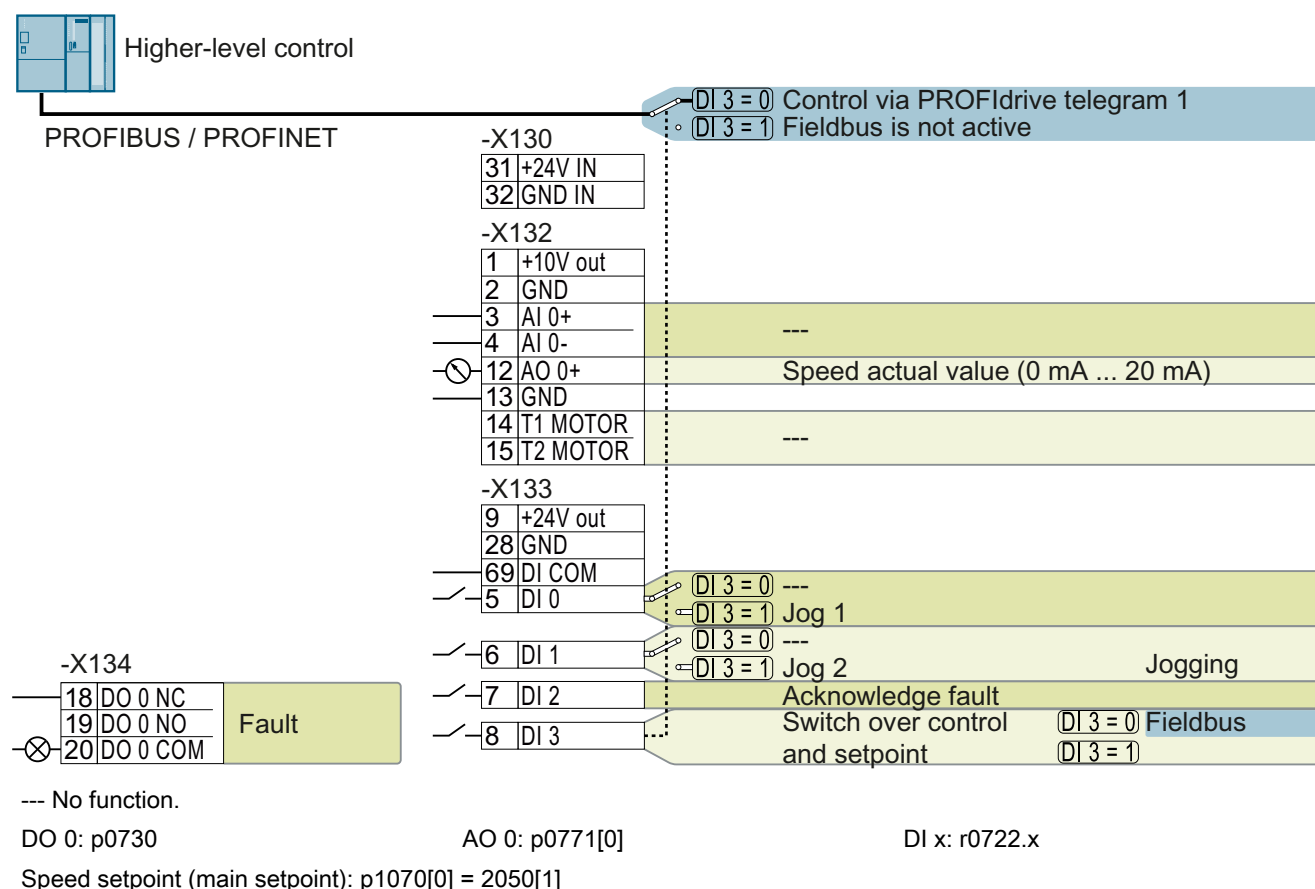
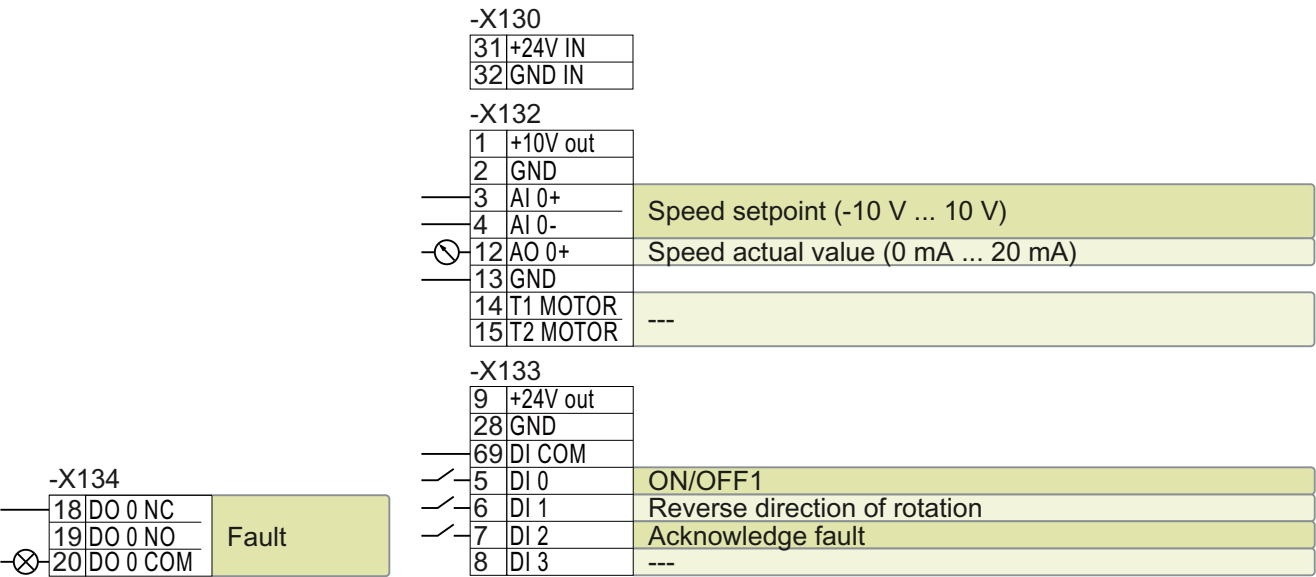


Figure 4-15 Factory setting of the CU240B-2 DP Control Unit

Control Units with USS interface

The fieldbus interface is not active.



--- No function.

DO 0: p0730

AO 0: p0771[0]

DI x: r0722.x

AI 0: r0755[0]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Figure 4-16 Factory setting of the CU240B-2 Control Unit

4.5.5 Default settings of the CU240B-2 interfaces

The function of the terminals and fieldbus interface can be set.

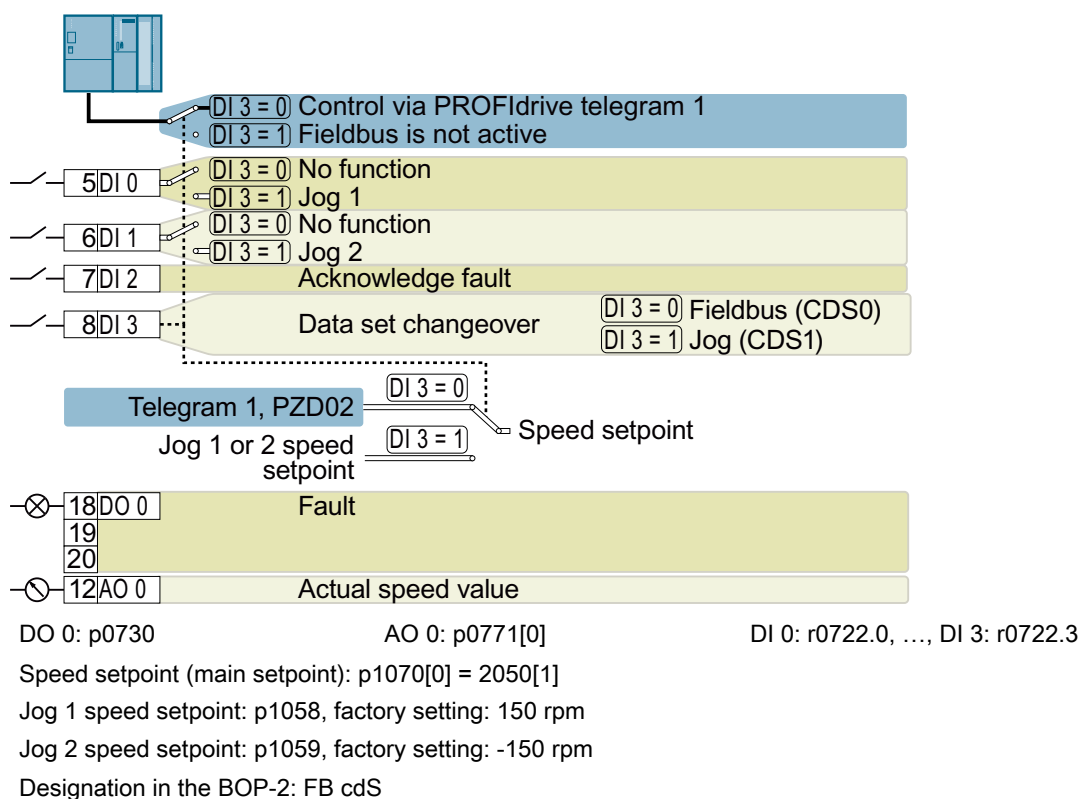
In order that you do not have to successively change terminal for terminal, several terminals can be jointly set using default settings ("p0015 Macro drive unit").

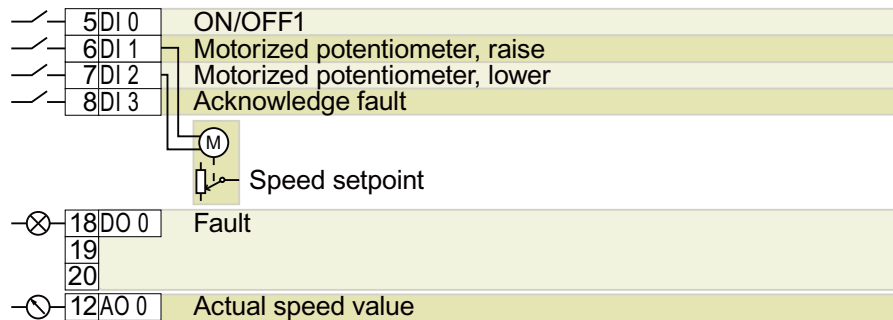
The terminal settings made in the factory described above correspond to the following default settings:

- Default setting 12 (p0015 = 12): "Standard I/O with analog setpoint"
- Default setting 7 (p0015 = 7): "Fieldbus with data set switchover"

Default setting 7: "Fieldbus with data set switchover"

Factory setting for inverters with PROFIBUS interface



Default setting 9: "Standard I/O with MOP"

DO 0: p0730

AO 0: p0771[0]

DI 0: r0722.0, ..., DI 3: r0722.3

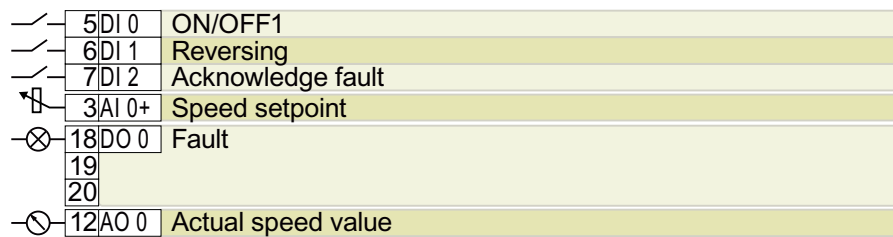
Motorized potentiometer setpoint after ramp-function generator: r1050

Speed setpoint (main setpoint): p1070[0] = 1050

Designation in the BOP-2: Std MoP

Default setting 12: "Standard I/O with analog setpoint"

Factory setting for inverters with USS interface



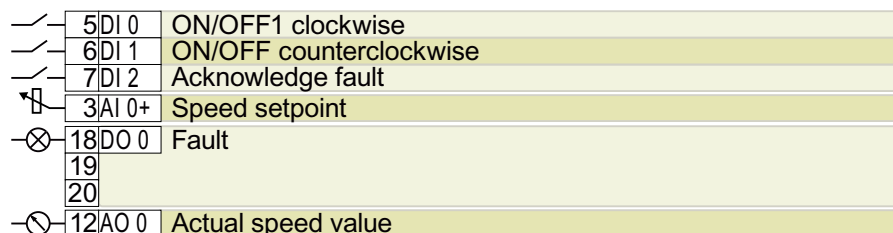
DO 0: p0730

AO 0: p0771[0]

DI 0: r0722.0, ..., DI 2: r0722.2
AI 0: r0755[0]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: Std ASP

Default setting 17: "2-wire (forw/backw1)"

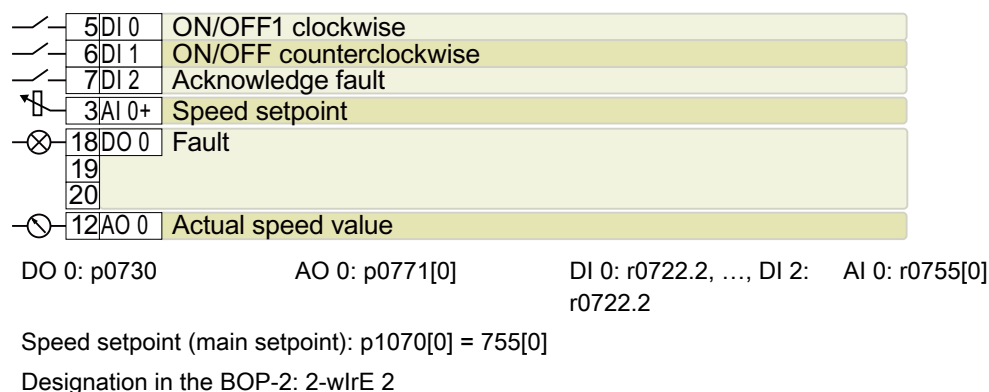
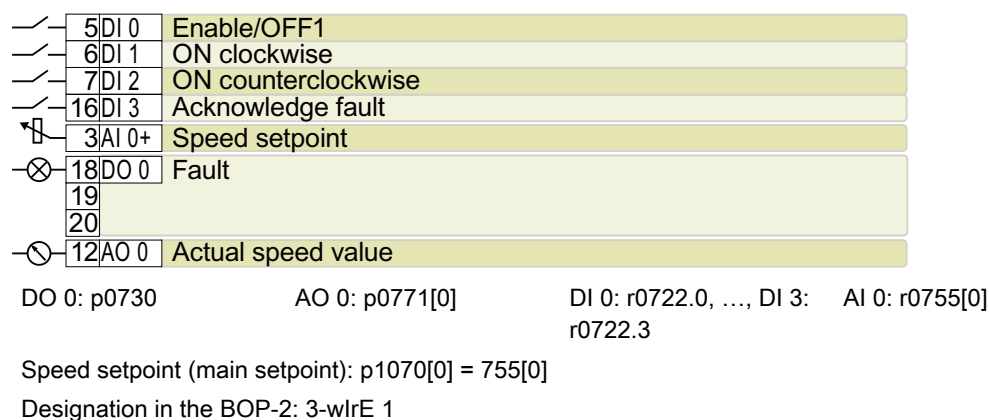
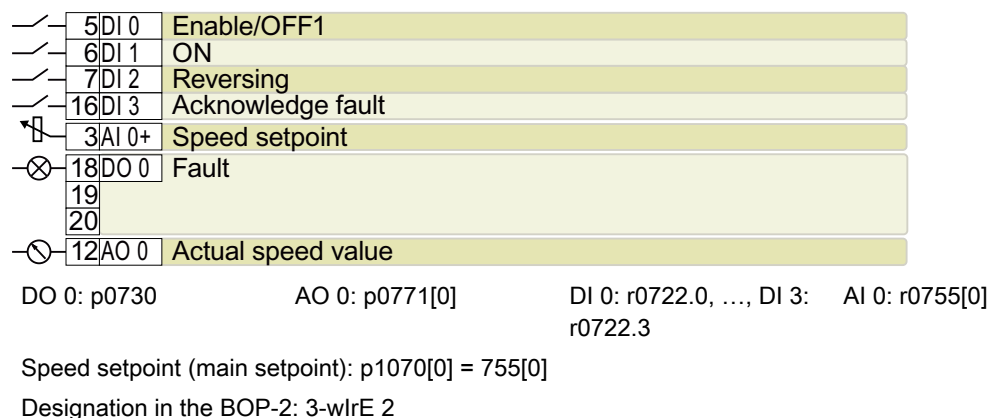
DO 0: p0730

AO 0: p0771[0]

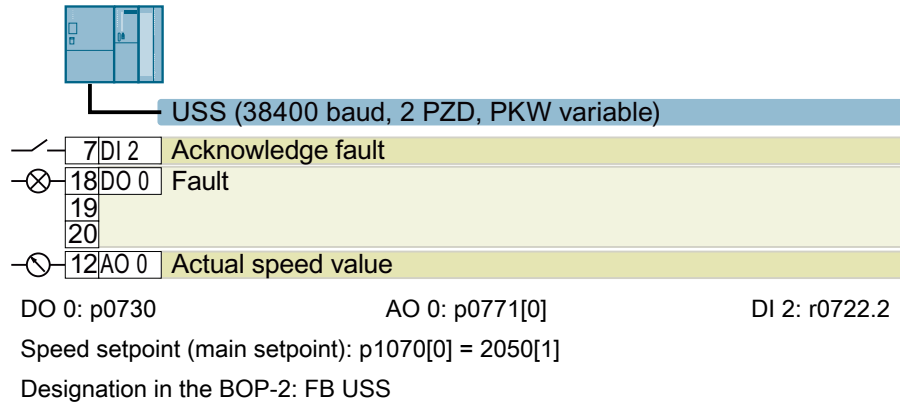
DI 0: r0722.2, ..., DI 2: r0722.2
AI 0: r0755[0]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: 2-wlrE 1

Default setting 18: "2-wire (forw/backw2)"**Default setting 19: "3-wire (enable/forw/backw)"****Default setting 20: "3-wire (enable/on/reverse)"**

Default setting 21: "USS fieldbus"



4.5.6 Terminal strips on CU240E-2 Control Units

Terminal strips with wiring example

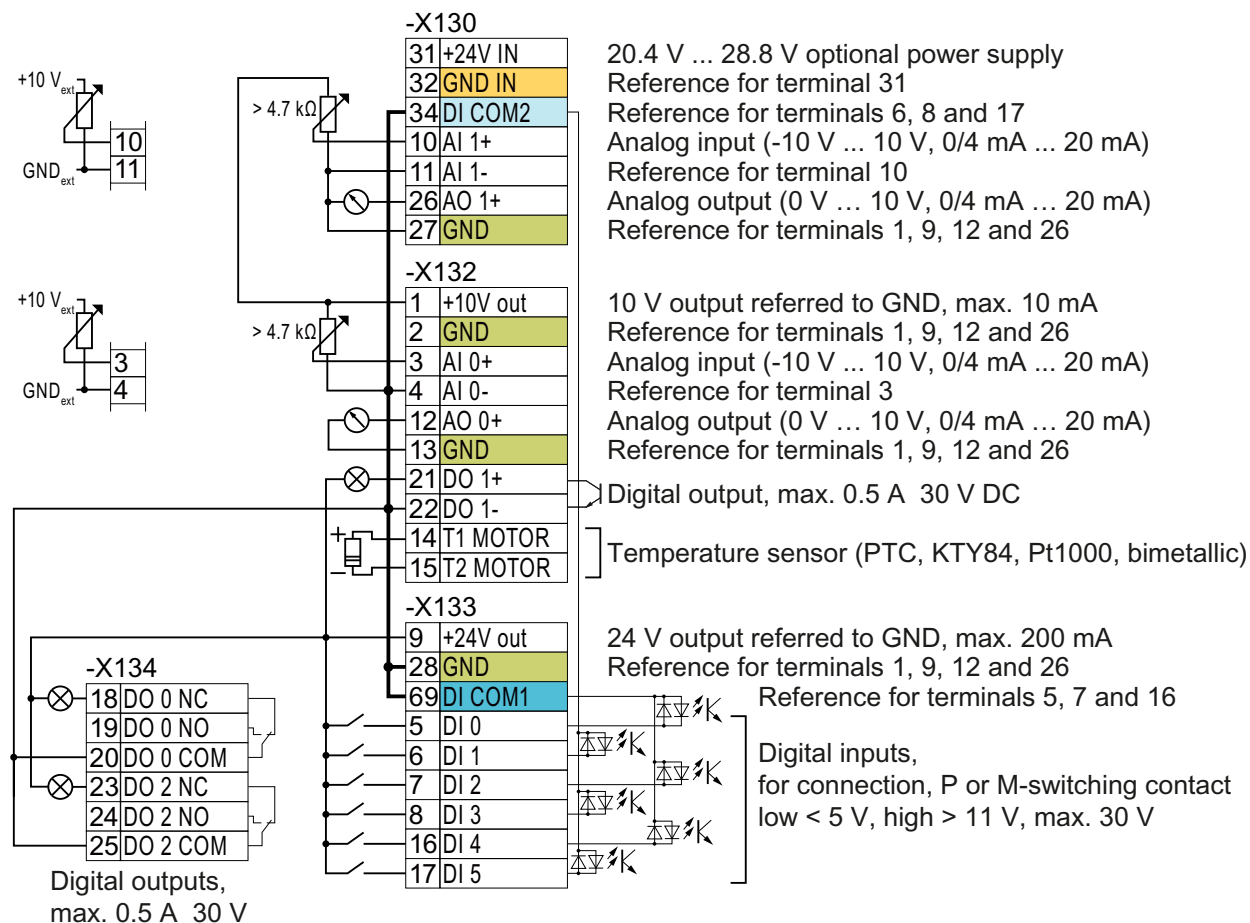


Figure 4-17 Wiring example of the digital inputs with the internal inverter 24 V power supply

- GND** All terminals labelled with reference potential "GND" are connected internally in the inverter.
- DI COM1** Reference potentials "DI COM1" and "DI COM2" are electrically isolated from "GND".
- DI COM2** → if, as described above, you use the 24-V power supply from terminal 9 to supply the digital inputs, then you must connect "GND" with "DI COM1" and "DI COM2" at the terminals.

31 +24 V IN
32 GND IN When an optional 24-V power supply is connected to terminals 31, 32, the Control Unit remains in operation even after the Power Module has been disconnected from the line supply. The Control Unit thus maintains the fieldbus communication, for example.

→ At terminals 31, 32, only connect a power supply that is in accordance with SELV (Safety Extra Low Voltage) or PELV (Protective Extra Low Voltage).

→ if you also wish to use the power supply at terminals 31, 32 for the digital inputs, then you must connect "DI COM1/2" and "GND IN" with one another at the terminals.

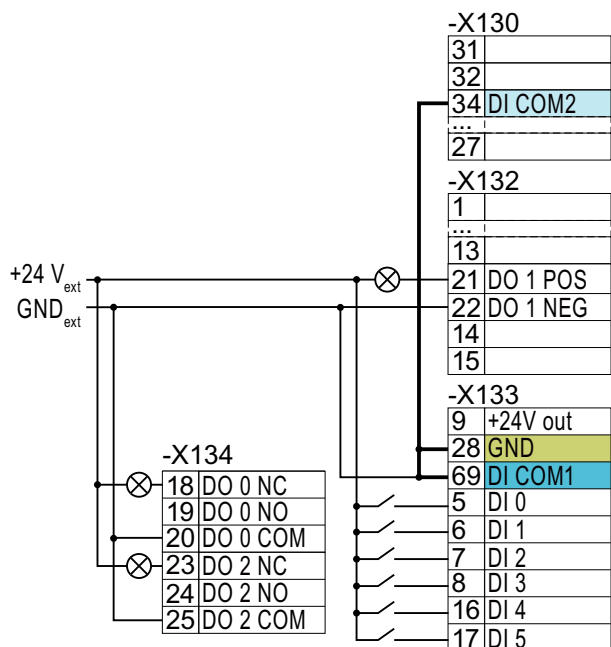
4.5 Installing Control Unit

10	AI 1+
11	AI 1-
3	AI 0+
4	AI 0-

You may use the internal 10V power supply or an external power supply for the analog inputs.

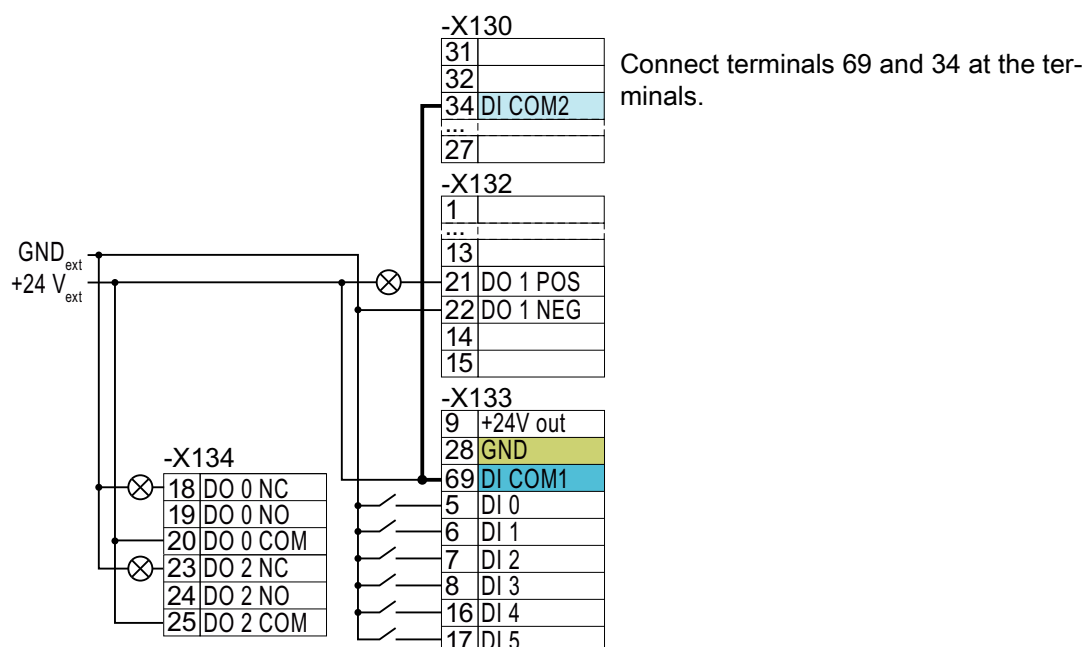
→ If you use the internal 10 V power supply, you must connect AI 0- or AI 1- to GND.

Additional options for wiring the digital inputs



If you wish to connect the external and the internal inverter power supply voltages with one another, then you must connect "GND" with terminals 34 and 69 at the terminals.

Connecting P-switching contacts with an external power supply



Connecting M-switching contacts with an external power supply

NOTICE

Damage to the CU240E-2 PN and CU240E-2 PN-F Control Units in the event of a short-circuit of the 24 V output

It is possible that the Control Units are defective if the following conditions occur simultaneously:

1. A short-circuit at the 24 V output occurs at terminal 9 when the inverter is operational.
2. The ambient temperature is at the upper permitted limit.
3. You have connected an external 24 V supply to terminals 31 and 32, and the voltage at terminal 31 is at the upper permitted limit.

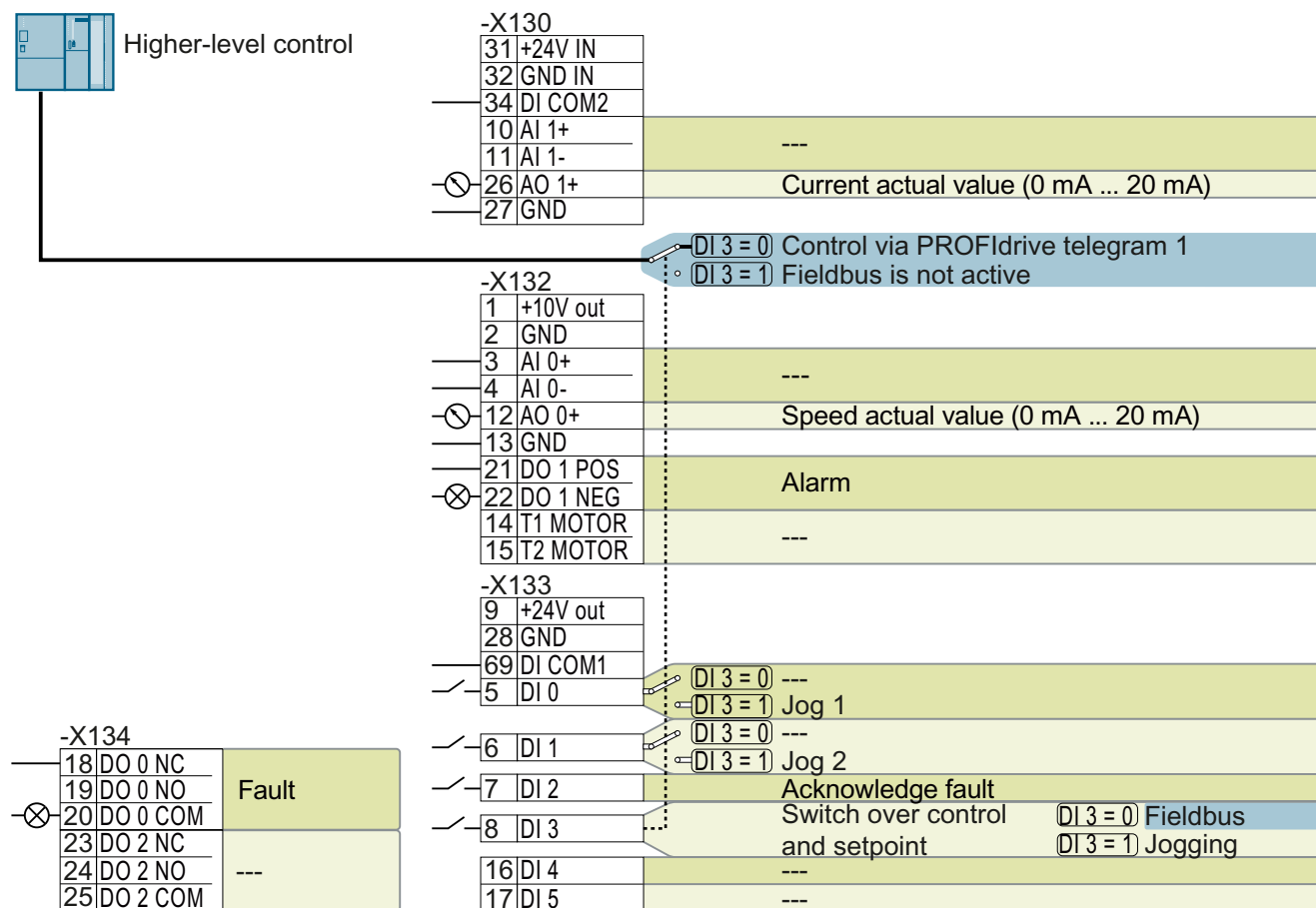
In order to rule out damage to the Control Units, you have to prevent all three conditions occurring simultaneously.

4.5.7 Factory setting of the CU240E-2 interfaces

The factory setting of the terminal strip depends on the Control Unit.

Control Units with PROFIBUS or PROFINET interface

The function of the fieldbus interface and digital inputs DI 0, DI 1 depends on DI 3.



DO x: p073x

AO 0: p0771[0]

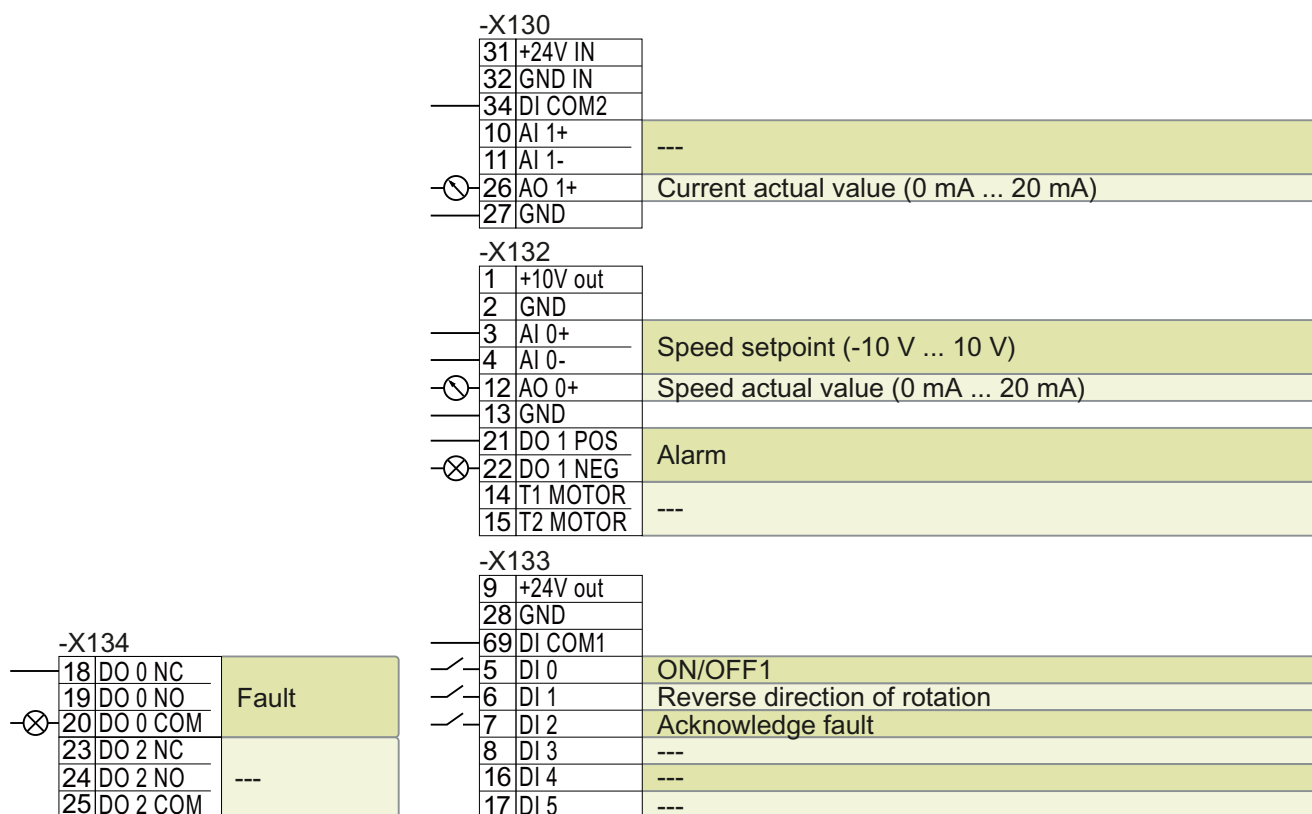
DI x: r0722.x

Speed setpoint (main setpoint): p1070[0] = 2050[1]

Figure 4-18 Factory setting of CU240E-2 DP(-F) and CU240E-2 PN(-F) Control Units

Control Units with USS interface

The fieldbus interface is not active.



--- No function.

DO x: p073x

AO 0: p0771[0]

DI x: r0722.x

AI 0: r0755[0]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Figure 4-19 Factory setting of the CU240E-2 and CU240E-2 F Control Units

4.5.8 Default settings of the CU240E-2 interfaces

The function of the terminals and fieldbus interface can be set.

In order that you do not have to successively change terminal for terminal, several terminals can be jointly set using default settings ("p0015 Macro drive unit").

The terminal settings made in the factory described above correspond to the following default settings:

- Default setting 12 (p0015 = 12): "Standard I/O with analog setpoint"
- Default setting 7 (p0015 = 7): "Fieldbus with data set switchover"

Default setting 1: "Conveyor technology with 2 fixed frequencies"

5	DI 0	ON/OFF1 clockwise
6	DI 1	ON/OFF1 counterclockwise
7	DI 2	Acknowledge fault
16	DI 4	Fixed speed setpoint 3:
17	DI 5	Fixed speed setpoint 4
18	DO 0	Fault
19		
20		
21	DO 1	Warning
22		
12	AO 0	Actual speed value
26	AO 1	Actual current value

DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 0: r0722.0, ..., DI 5: r0722.5

Fixed speed setpoint 3: p1003, fixed speed setpoint 4: p1004, fixed speed setpoint active: r1024

Speed setpoint (main setpoint): p1070[0] = 1024

DI 4 and DI 5 = high: The inverter adds both fixed speed setpoints

Designation in the BOP-2: coN 2 SP

Default setting 2: "Conveyor systems with Basic Safety"

5	DI 0	ON/OFF1 with fixed speed setpoint 1
6	DI 1	Fixed speed setpoint 2:
7	DI 2	Acknowledge fault
16	DI 4	} Reserved für a safety function
17	DI 5	
18	DO 0	Fault
19		
20		
21	DO 1	Warning
22		
12	AO 0	Actual speed value
26	AO 1	Actual current value

DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 0: r0722.0, ..., DI 5: r0722.5

Fixed speed setpoint 1: p1001, fixed speed setpoint 2: p1002, fixed speed setpoint active: r1024

Speed setpoint (main setpoint): p1070[0] = 1024

DI 0 and DI 1 = high: The inverter adds both fixed speed setpoints.

Designation in the BOP-2: coN SAFE

Default setting 3: "Conveyor technology with 4 fixed frequencies"

—	5	DI 0	ON/OFF1 with fixed speed setpoint 1
—	6	DI 1	Fixed speed setpoint 2
—	7	DI 2	Acknowledge fault
—	16	DI 4	Fixed speed setpoint 3
—	17	DI 5	Fixed speed setpoint 4
⊗	18	DO 0	Fault
	19		
	20		
⊗	21	DO 1	Warning
	22		
⊗	12	AO 0	Actual speed value
⊗	26	AO 1	Actual current value

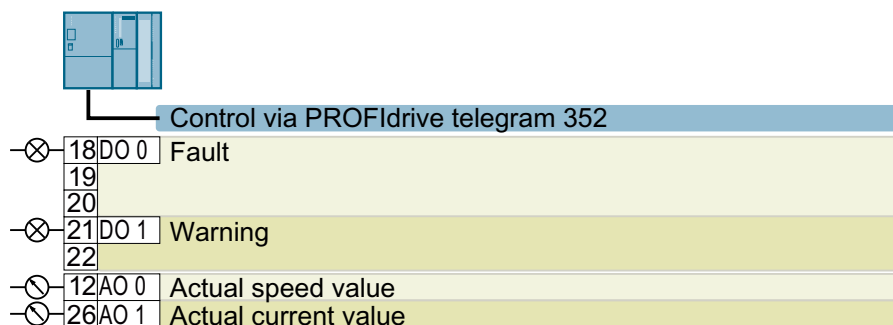
DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 0: r0722.0, ..., DI 5: r0722.5

Fixed speed setpoint 1: p1001, ... fixed speed setpoint 4: p1004, fixed speed setpoint active: r1024

Speed setpoint (main setpoint): p1070[0] = 1024

Several DI 0, DI 1, DI 4 and DI 5 = high: The inverter adds the corresponding fixed speed setpoints.

Designation in the BOP-2: coN 4 SP

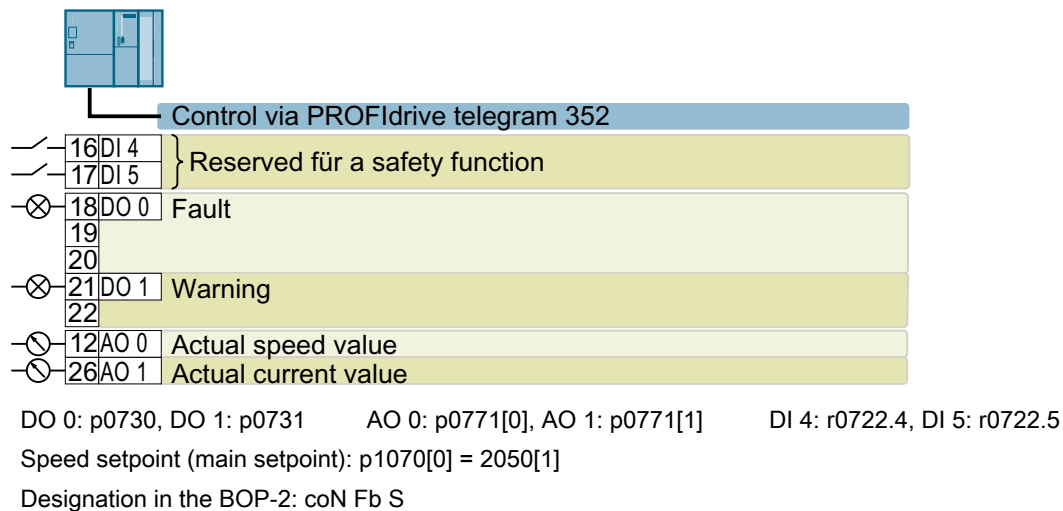
Default setting 4: "Conveyor systems with fieldbus"

DO 0: p0730, DO 1: p0731

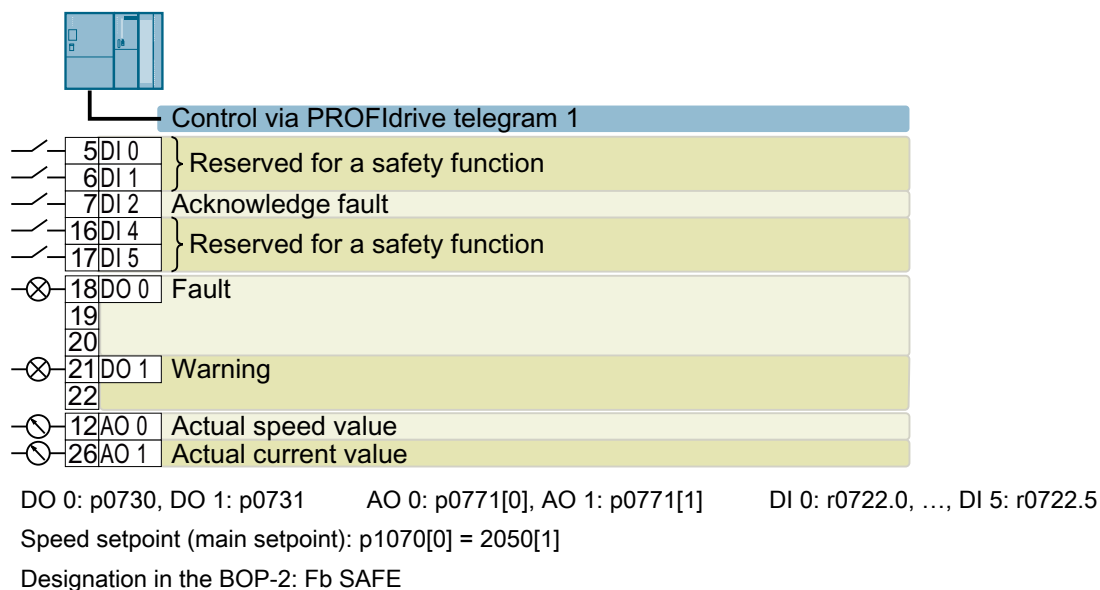
AO 0: p0771[0], AO 1: p0771[1]

Speed setpoint (main setpoint): p1070[0] = 2050[1]

Designation in the BOP-2: coN Fb

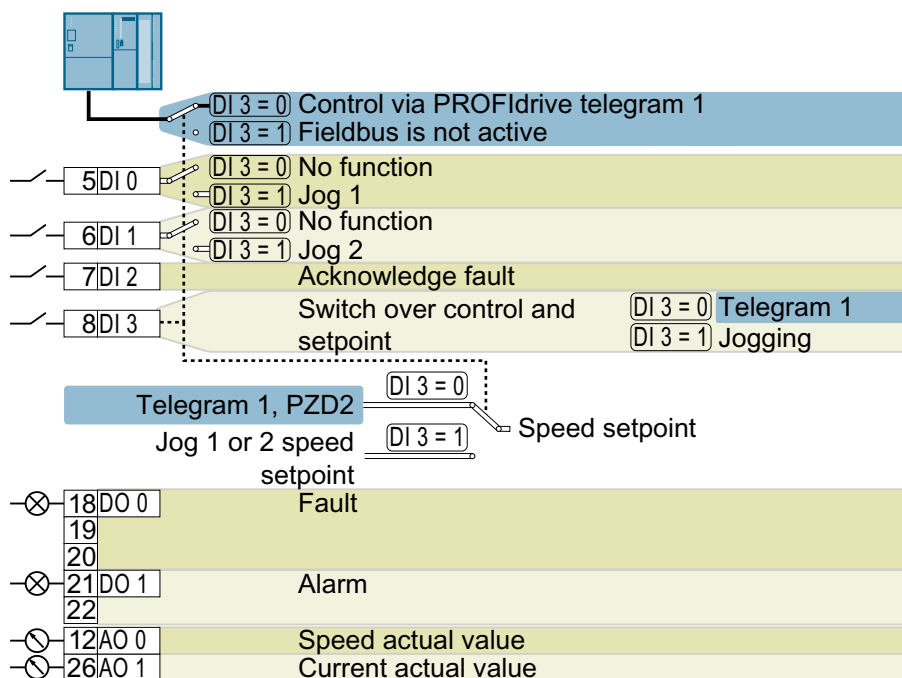
Default setting 5: "Conveyor systems with fieldbus and Basic Safety"**Default setting 6: "Fieldbus with Extended Safety"**

Only with Control Units CU240E-2 F, CU240E-2 DP-F, and CU240E-2 PN-F.



Default setting 7: "Fieldbus with data set switchover"

Factory setting for inverters with PROFIBUS or PROFINET interface



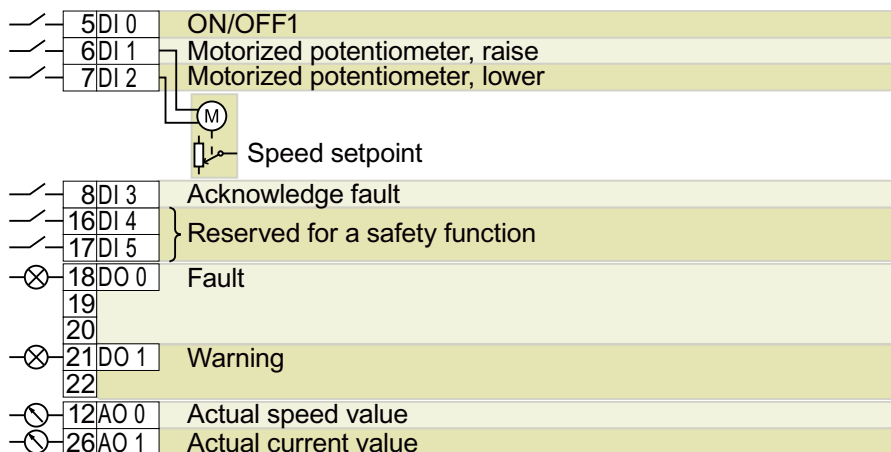
DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 0: r0722.0, ..., DI 3: r0722.3

Speed setpoint (main setpoint): p1070[0] = 2050[1]

Jog 1 speed setpoint: p1058, factory setting: 150 rpm

Jog 2 speed setpoint: p1059, factory setting: -150 rpm

Designation in the BOP-2: FB cdS

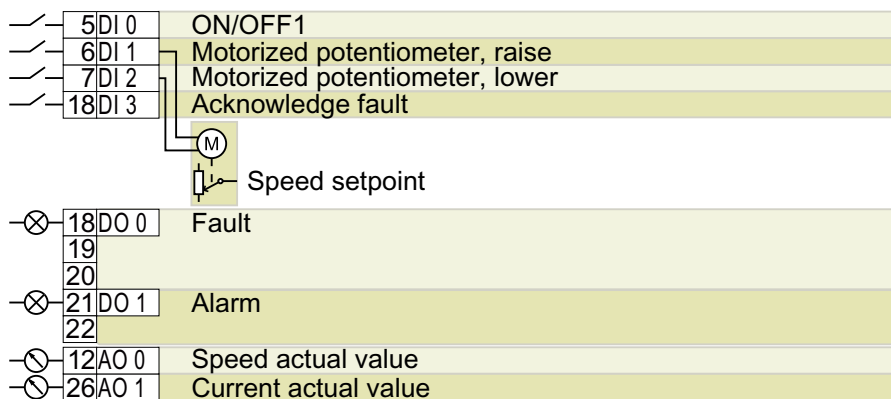
Default setting 8: "MOP with Basic Safety"

DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 0: r0722.0, ..., DI 5: r0722.5

Motorized potentiometer setpoint after ramp-function generator: r1050

Speed setpoint (main setpoint): p1070[0] = 1050

Designation in the BOP-2: MoP SAFE

Default setting 9: "Standard I/O with MOP"

DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 0: r0722.0, ..., DI 3: r0722.3

Motorized potentiometer setpoint after ramp-function generator: r1050

Speed setpoint (main setpoint): p1070[0] = 1050

Designation in the BOP-2: Std MoP

Default setting 12: "Standard I/O with analog setpoint"

5	DI 0	ON/OFF1
6	DI 1	Reversing
7	DI 2	Acknowledge fault
3	AI 0+	Speed setpoint
18	DO 0	Fault
19		
20		
21	DO 1	Alarm
22		
12	AO 0	Speed actual value
26	AO 1	Current actual value

DO 0: p0730, DO 1: AO 0: p0771[0], DI 0: r0722.0, ..., DI 2: r0722.2 AI 0: r0755[0]
 p0731 AO 1: p0771[1]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: Std ASP

Default setting 13: "Standard I/O with analog setpoint and safety"

5	DI 0	ON/OFF1
6	DI 1	Reversing
7	DI 2	Acknowledge fault
16	DI 4	} Reserved für a safety function
17	DI 5	
3	AI 0+	Speed setpoint
18	DO 0	Fault
19		
20		
21	DO 1	Warning
22		
12	AO 0	Actual speed value
26	AO 1	Actual current value

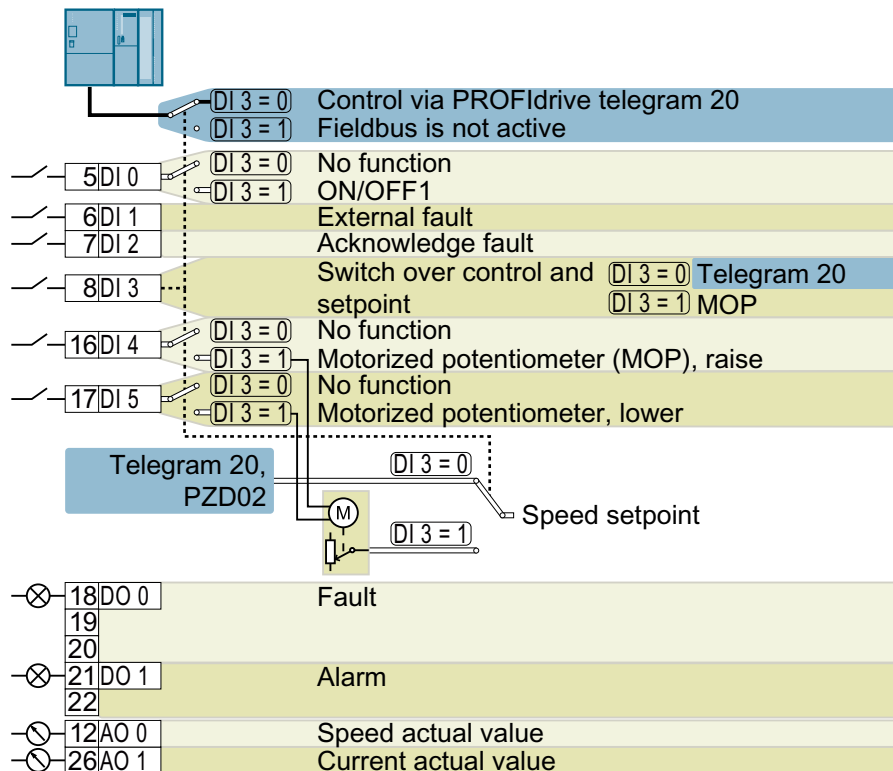
DO 0: p0730, DO 1: AO 0: p0771[0], DI 0: r0722.0, ..., DI 5: r0722.5 AI 0: r0755[0]
 p0731 AO 1: p0771[1]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: ASPS

Default setting 14: "Process industry with fieldbus"

PROFIdrive telegram 20



MOP = motorized potentiometer

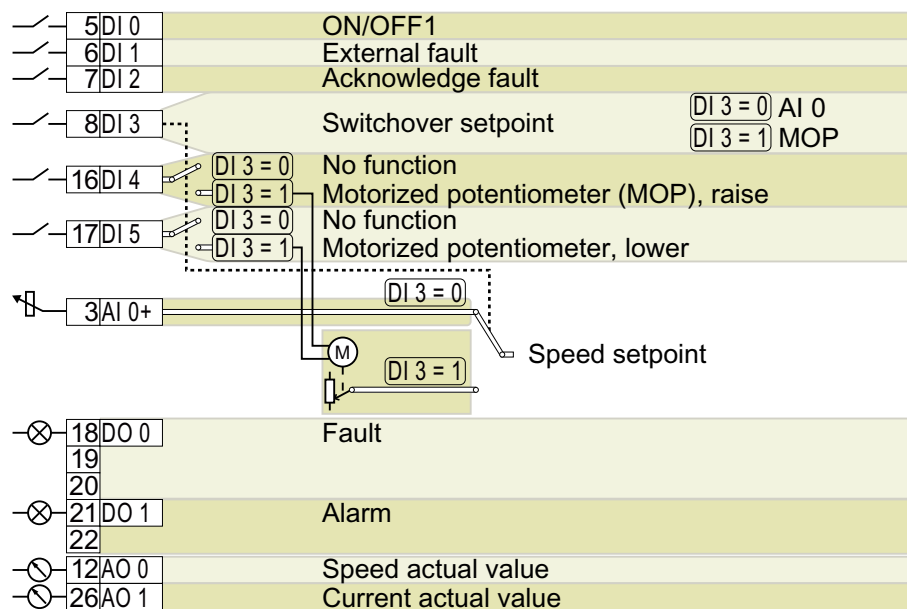
DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 0: r0722.0, ..., DI 5: r0722.5

Motorized potentiometer setpoint after ramp-function generator: r1050

Speed setpoint (main setpoint): p1070[0] = 2050[1], p1070[1] = 1050

Designation in the BOP-2: Proc Fb

Default setting 15: "Process industry"



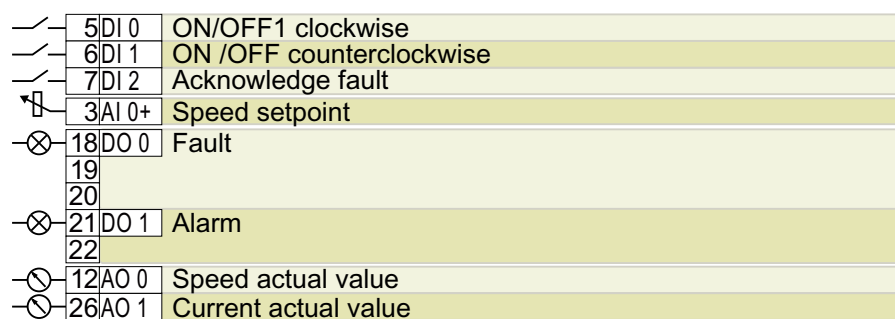
DO 0: p0730, DO 1: AO 0: p0771[0], DI 0: r0722.5, ..., DI 4: r0722.5 AI 0: r0755[0]
 p0731 AO 1: p0771[1]

Motorized potentiometer setpoint after ramp-function generator: r1050

Speed setpoint (main setpoint): p1070[0] = 755[0], p1070[1] = 1050

Designation in the BOP-2: Proc

Default setting 17: "2-wire (forw/backw1)"



DO 0: p0730, DO 1: AO 0: p0771[0], DI 0: r0722.2, ..., DI 2: r0722.2 AI 0: r0755[0]
 p0731 AO 1: p0771[1]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: 2-wlrE 1

Default setting 18: "2-wire (forw/backw2)"

5	DI 0	ON/OFF1 clockwise
6	DI 1	ON /OFF counterclockwise
7	DI 2	Acknowledge fault
3	AI 0+	Speed setpoint
18	DO 0	Fault
19		
20		
21	DO 1	Alarm
22		
12	AO 0	Speed actual value
26	AO 1	Current actual value

DO 0: p0730, DO 1: AO 0: p0771[0], DI 0: r0722.0, ..., DI 2: r0722.2 AI 0: r0755[0]
 p0731 AO 1: p0771[1]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: 2-wlrE 2

Default setting 19: "3-wire (enable/forw/backw)"

5	DI 0	Enable / OFF1
6	DI 1	ON clockwise
7	DI 2	ON counterclockwise
16	DI 4	Acknowledge fault
3	AI 0+	Speed setpoint
18	DO 0	Fault
19		
20		
21	DO 1	Alarm
22		
12	AO 0	Speed actual value
26	AO 1	Current actual value

DO 0: p0730, DO 1: AO 0: p0771[0], DI 0: r0722.0, ..., DI 4: r0722.4 AI 0: r0755[0]
 p0731 AO 1: p0771[1]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: 3-wlrE 1

Default setting 20: "3-wire (enable/on/reverse)"

5	DI 0	Enable / OFF1
6	DI 1	ON
7	DI 2	Reversing
16	DI 4	Acknowledge fault
3	AI 0+	Speed setpoint
18	DO 0	Fault
19		
20		
21	DO 1	Alarm
22		
12	AO 0	Speed actual value
26	AO 1	Current actual value

DO 0: p0730, DO 1: p0731 AO 0: p0771[0], DI 0: r0722.0, ..., DI 4: r0722.4 AI 0: r0755[0]
 p0731 AO 1: p0771[1]

Speed setpoint (main setpoint): p1070[0] = 755[0]

Designation in the BOP-2: 3-wlrE 2

Default setting 21: "USS fieldbus"

		USS (38400 baud, 2 PZD, PKW variable)
7	DI 2	Acknowledge fault
18	DO 0	Fault
19		
20		
21	DO 1	Alarm
22		
12	AO 0	Speed actual value
26	AO 1	Current actual value

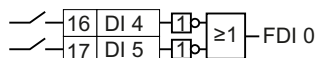
DO 0: p0730, DO 1: p0731 AO 0: p0771[0], AO 1: p0771[1] DI 2: r0722.2

Speed setpoint (main setpoint): p1070[0] = 2050[1]

Designation in the BOP-2: FB USS

4.5.9 Safety input of the CU240E-2

To enable a safety function via the terminal strip of the inverter, you need a fail-safe digital input.



For specific default settings of the terminal strip, e.g. default setting 2, the inverter combines two digital inputs to form one fail-safe digital input FDI 0.

Which devices are you allowed to connect?

The fail-safe digital input is designed for the following devices:

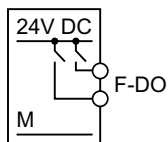
- Connection of safety sensors, e.g. emergency stop command devices or light curtains.
- Connection of pre-processing devices, e.g. fail-safe control systems and safety relays.

Signal state

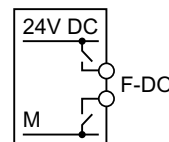
The inverter expects signals with the same state at its fail-safe digital input:

- High signal: The safety function is deselected.
- Low signal: The safety function is selected.

Connecting P/P and P/M-switching fail-safe digital outputs



PP-switching digital output



PM-switching digital output

It is permissible to connect PP and PM-switching safe outputs to a fail-safe digital input.



Connecting a fail-safe digital input (Page 496)

Fault detection

The inverter compares the two signals of the fail-safe digital input. The inverter thus detects, for example the following faults:

- Cable break
- Defective sensor

The inverter cannot detect the following faults:

- Cross-circuit of the two cables
- Short-circuit between signal cable and 24 V power supply

Special measures to prevent cross-circuits and short-circuits

The routing of cables over longer distances, e.g. between remote control cabinets, increases the risk of damaging cables. Damaged cables raise the risk of an undetected cross-circuit with power-conducting cables laid in parallel. A cross-circuit can cause interruption to the transfer of safety-related signals.

To reduce the risk of cable damage, you need to lay signal lines in steel pipes.

Special requirements placed on EMC-compliant installation

Use shielded signal cables. Connect the shield at both conductor ends.

In order to connect two or more inverter terminals, use the shortest possible jumpers directly at the terminals themselves.

Bright and dark test

The inverter filters signal changes using bright and dark tests at the fail-safe digital input using an adjustable software filter.

4.5.10 Wiring the terminal strip

**! WARNING**

Danger to life as a result of hazardous voltages when connecting an unsuitable power supply

Death or serious injury can result when live parts are touched in the event of a fault.

- For all connections and terminals of the electronic boards, only use power supplies that provide PELV (Protective Extra Low Voltage) or SELV (Safety Extra Low Voltage) output voltages.

**! WARNING**

Danger to life due to electric shock caused by sparkovers to the motor temperature sensor cable

Voltage flashovers to the electronics of the inverter can occur in motors without safe electrical separation of the temperature sensors in accordance with IEC 61800-5-1 when the motor develops a fault.

- Install a temperature monitoring relay 3RS1... or 3RS2...
- Evaluate the temperature monitoring relay output using a digital input of the inverter, e.g. using the "External fault" function.



You can find additional information about the temperature monitoring relay on the Internet:

Manual 3RS1 / 3RS2 temperature monitoring relays (<https://support.industry.siemens.com/cs/ww/en/view/54999309>)

Note

Malfunction caused by incorrect switching states as the result of diagnostic flows in the off state (logical state "0")

In contrast to mechanical switching contacts, e.g. emergency stop switches, diagnostic flows can also flow with semiconductor switches in the off state. If interconnection with digital inputs is faulty, the diagnostic flows can lead to incorrect switching states and thus to a malfunction of the drive.

- Observe the conditions for digital inputs and digital outputs specified in the relevant manufacturers documentation.
- Check the conditions of the digital inputs and digital outputs in regard to the flows in off state. If applicable, connect the digital inputs with suitably dimensioned, external resistors to protect against the reference potential of the digital inputs.

Note

If your application requires UL certification, please note that the power supply of the digital output must comply with specific specifications.



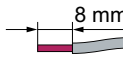
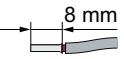
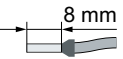
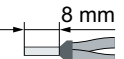
Technical data (Page 419)

NOTICE**Damage to the inverter when using long signal cables**

Using long cables at the inverter's digital inputs and 24 V power supply can lead to overvoltage during switching operations. Overvoltages can damage the inverter.

- If you use cables of more than 30 m at the digital inputs and 24 V power supply, connect an overvoltage protection element between the terminal and the associated reference potential.
We recommend using the Weidmüller overvoltage protection terminal with designation MCZ OVP TAZ DIODE 24VDC.

Table 4-21 Permissible cable and wiring options

Solid or flexible conductors	Finely stranded conductor with non-insulated end sleeve	Finely stranded conductor with partially insulated end sleeve	Two finely stranded conductors with the same cross-section with partially insulated twin end sleeves
 8 mm 0.5 ... 1.5 mm ²	 8 mm 0.5 ... 1.0 mm ²	 8 mm 0.5 mm ²	 8 mm 2 * 0.5 mm ²

Wiring the terminal strip in compliance with EMC

- If you use shielded cables, then you must connect the shield to the mounting plate of the control cabinet or with the shield support of the inverter through a good electrical connection and a large surface area.



Further information about EMC-compliant wiring is available on the Internet:

EMC installation guideline (<http://support.automation.siemens.com/WW/view/en/60612658>)

- Use the shield connection plate of the Control Unit as strain relief.



Overview of Control Units (Page 31)

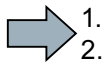
4.5.11 Connecting the temperature contact of the braking resistor

**WARNING**

Danger to life due to fire spreading because of an unsuitable or improperly installed braking resistor

Using an unsuitable or improperly installed braking resistor can cause fires and smoke to develop. Fire and smoke development can cause severe personal injury or material damage.

- Only use braking resistors that are approved for the inverter.
- Install the braking resistor in accordance with regulations.
- Monitor the temperature of the braking resistor.

Procedure

1. Proceed as follows to monitor the braking resistor temperature:

1. Connect the temperature monitoring system of the braking resistor (terminals T1 and T2 on the braking resistor) to a free digital input on the inverter.

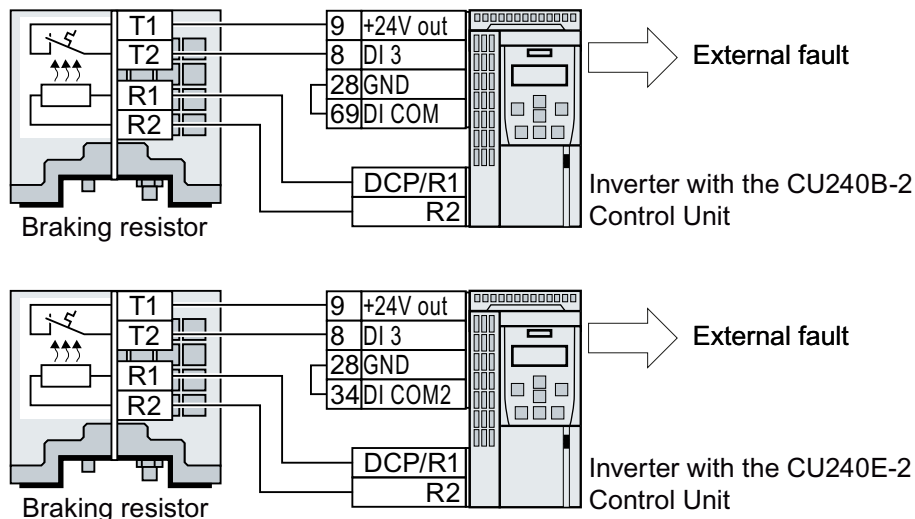


Figure 4-20 Example: Temperature monitoring of the braking resistor via digital input DI 3 on the Control Unit

2. Define the function of the digital input used as an external fault with p2106.
As an example with temperature monitoring via digital input DI 3: p2106 = 722.3.



You have ensured that the temperature is monitored.

4.6 Connecting the inverter to the fieldbus

Fieldbus interfaces of the Control Units

The Control Units are available in different versions for communication with higher-level controls with the subsequently listed fieldbus interfaces:

Fieldbus	Profiles			S7 communication ²⁾	Control Unit
	PROFIdrive	PROFIsafe ¹⁾	PROFInergy ²⁾		
PROFIBUS	✓	✓	---	✓	CU240B-2 DP CU240E-2 DP CU240E-2 DP-F
PROFINET	✓	✓	✓	✓	CU240E-2 PN CU240E-2 PN-F
EtherNet/IP ²⁾	---			---	
USS ²⁾	---			---	CU240B-2 CU240E-2
Modbus RTU ²⁾	---			---	CU240E-2 F

¹⁾ Information on PROFIsafe can be found in the "Safety Integrated" function manual.

²⁾ Information about these fieldbuses, profiles and communication types can be found in the "Fieldbus" function manual.



Overview of the manuals (Page 504)

4.6.1 PROFINET

4.6.1.1 Communication via PROFINET IO and Ethernet

You can either integrate the inverter in a PROFINET network or communicate with the inverter via Ethernet.

The inverter in PROFINET IO operation

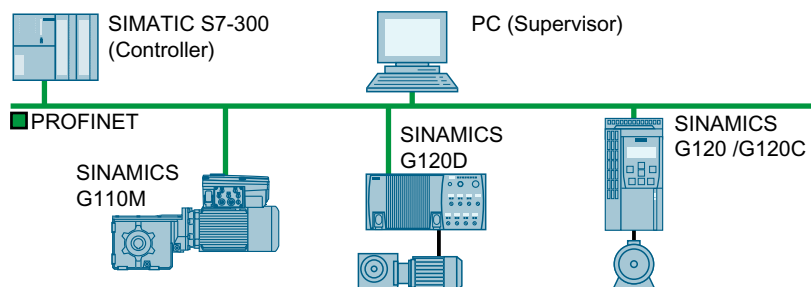


Figure 4-21 The inverter in PROFINET IO operation

The inverter supports the following functions:

- RT
- IRT: The inverter forwards the clock synchronism, but does not support clock synchronism.
- MRP: Media redundancy, impulsed with 200 ms. Requirement: Ring topology
- MRPD: Media redundancy, bumpless. Requirement: IRT and the ring topology created in the control
- Diagnostic alarms in accordance with the error classes specified in the PROFIdrive profile.
- Device replacement without removable data storage medium
- Shared Device for Control Units with fail-safe functions

The inverter as Ethernet node

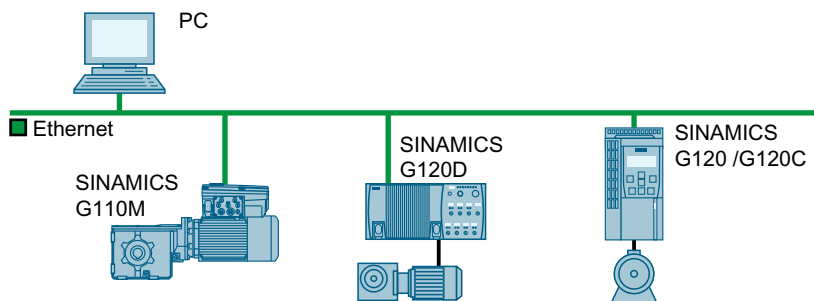


Figure 4-22 The inverter as Ethernet node

Further information on the operation as Ethernet nodes can be found in the Function Manual "Fieldbuses".

 Overview of the manuals (Page 504)

Further information on PROFINET

Further information on PROFINET can be found on the Internet:



- PROFINET – the Ethernet standard for automation (<http://w3.siemens.com/mcms/automation/en/industrial-communications/profinet/Pages/Default.aspx>)
- PROFINET system description (<https://support.industry.siemens.com/cs/ww/en/view/19292127>)

4.6.1.2 Connecting the PROFINET cable to the inverter

Procedure

- ➔ 1. To connect the inverter to a control via PROFINET, proceed as follows:
2.
 1. Integrate the inverter in the bus system (e.g. ring topology) of the control using PROFINET cables and the two PROFINET sockets X150-P1 and X150-P2.
 Fieldbus interface allocation (Page 92)
 The maximum permitted cable length from the previous station and to the next one is 100 m.
 2. Externally supply the inverter with 24 VDC through terminals 31 and 32.
 The external 24 V supply is only required if communications with the control should also run when the line voltage is switched off.
- You have connected the inverter to the control using PROFINET.

Communication with the controller even when the supply voltage on the Power Module is switched off

You must supply the Control Unit with 24 V DC at terminals 31 and 32 if you wish to maintain communication with the control system when the line voltage is switched off.

In the case of brief interruptions of the 24 V power supply, the inverter may signal a fault without communications with the control system being interrupted.

4.6.1.3 What do you have to set for communication via PROFINET?

Configuring PROFINET communication

Proceed as follows to configure the PROFINET communication between the IO Controller and the inverter as IO Device:

1. Configure the IO Controller and the IO Device with an engineering system, e.g. with HW-Config.
 If required, install the GSDML file of the inverter in the engineering system.
 Installing GSDML (Page 124)
2. Load the configuration data into the IO Controller.

Device name

In addition to the MAC address and IP address, PROFINET also uses the device name to identify PROFINET devices. The device name must be unique across the PROFINET network.

To assign the device name, you need an engineering software, e.g. HW-Config or STARTER.

The inverter saves the device name on the inserted memory card.

IP address

In addition to the device name, PROFINET also uses an IP address.

You have the following options to specify the IP address of the inverter:

- You specify the IP address via an engineering software, e.g. via HW-Config or STARTER.
- The IO Controller assigns an IP address to the inverter.

Telegram

Set the same telegram in the inverter as in the IO Controller. Interconnect the telegrams in the control program of the IO Controller with the signals of your choosing.



Drive control via PROFIBUS or PROFINET (Page 196)

Application examples

You can find application examples for PROFINET communication on the Internet:



Controlling the speed of a SINAMICS G120/S120 with S7-300/400 via PROFINET (<https://support.industry.siemens.com/cs/ww/en/view/38844967>)

Controlling the speed of a SINAMICS G110M/G120/G120C/G120D with S7-300/400F via PROFINET, with Safety Integrated (via terminal) and HMI (<https://support.industry.siemens.com/cs/ww/en/view/60441457>)

Controlling the speed of a SINAMICS G110M/G120 (Startdrive) with S7-1500 (TO) via PROFINET, with Safety Integrated (via terminal) and HMI (<https://support.industry.siemens.com/cs/ww/en/view/78788716>)

4.6.1.4 Installing GSDML

Procedure



1. To install the GSDML of the inverter in the engineering system of the controller, proceed as follows:

1. Save the GSDML to your PC .



- From the Internet: GSDML (<https://support.industry.siemens.com/cs/ww/en/view/26641490>)

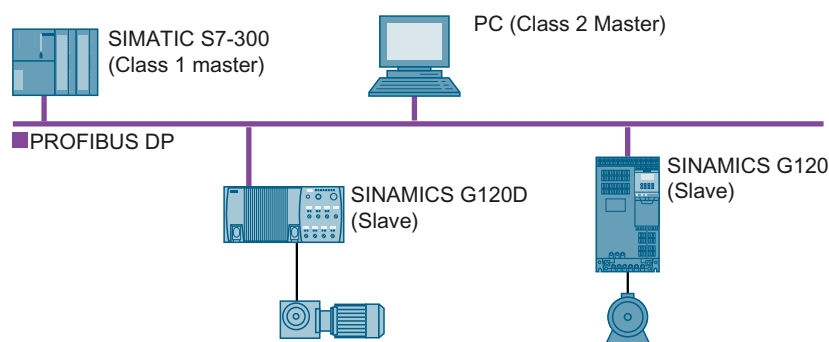
- From your inverter:
Insert a memory card into the inverter.
Set p0804 = 12.
The inverter writes the GSDML as zipped file (*.zip) into directory /SIEMENS/SINAMICS/DATA/CFG on the memory card.

2. Unzip the GSDML file on your computer.
3. Import the GSDML into the engineering system of the controller.



You have now installed the GSDML in the engineering system of the controller.

4.6.2 PROFIBUS



The PROFIBUS DP interface has the following functions:

- Cyclic communication
- Acyclic communication
- Diagnostic alarms



General information on PROFIBUS DP can be found in the Internet:

- PROFIBUS user organization (<http://www.profibus.com/downloads/installation-guide/>)
- Information about PROFIBUS DP (www.siemens.com/profibus)

4.6.2.1 Connecting the PROFIBUS cable to the inverter

Procedure



1. To connect the inverter to a control via PROFIBUS DP, proceed as follows:
 1. Integrate the inverter into the bus system (e.g. line topology) of the control using PROFIBUS cables via socket X126.
 Overview of the interfaces (Page 91)
 The maximum permitted cable length to the previous station and the subsequent one is 100 m at a baud rate of 12 Mbit/s.
 2. Externally supply the inverter with 24 VDC through terminals 31 and 32.
 The external 24 V supply is only required if communications with the control should also run when the line voltage is switched off.



You have now connected the inverter to the control using PROFIBUS DP.

4.6.2.2 What do you have to set for communication via PROFIBUS?

Configuring PROFIBUS communication

Proceed as follows to configure the communication between the PROFIBUS master and the inverter as PROFIBUS slave:

1. Configure the PROFIBUS master and the PROFIBUS slave with an engineering system, e.g. with HW-Config.

If required, install the GSD file of the inverter in the engineering system.

 Installing the GSD (Page 127)

2. Load the configuration data into the PROFIBUS master.

Setting the address

Set the address of the PROFIBUS slave.

 Setting the address (Page 127)

Setting the telegram

Set the telegram in the inverter as in the PROFIBUS master. Interconnect the telegrams in the control program of the PROFIBUS master with the signals of your choosing.

 Drive control via PROFIBUS or PROFINET (Page 196)

Application examples



You can find application examples for PROFIBUS communication on the Internet:

Controlling the speed of a SINAMICS G120/S120 with S7-300/400 via PROFINET (<https://support.industry.siemens.com/cs/ww/en/view/38844967>)

Controlling the speed of a SINAMICS G110M/G120/G120C/G120D with S7-300/400F via PROFINET, with Safety Integrated (via terminal) and HMI (<https://support.industry.siemens.com/cs/ww/en/view/60441457>)

Controlling the speed of a SINAMICS G110M/G120 (Startdrive) with S7-1500 (TO) via PROFINET, with Safety Integrated (via terminal) and HMI (<https://support.industry.siemens.com/cs/ww/en/view/78788716>)

Communication with the controller even when the supply voltage on the Power Module is switched off

You must supply the Control Unit with 24 V DC at terminals 31 and 32 if you wish to maintain communication with the control system when the line voltage is switched off.

In the case of brief interruptions of the 24 V power supply, the inverter may signal a fault without communications with the control system being interrupted.

4.6.2.3 Installing the GSD

Procedure

- ➔ 1. To install the GSD of the inverter in the engineering system of the controller, proceed as follows:
 2.
 1. Save the GSD on your PC via one of the following methods.
 - From the Internet:
GSD (<http://support.automation.siemens.com/WW/view/en/22339653/133100>)
 - From your inverter:
Insert a memory card into the inverter and then set p0804 = 12.
The inverter writes the GSD as zipped file (*.zip) into directory /SIEMENS/SINAMICS/DATA/CFG on the memory card.
 2. Unzip the GSD file on your computer.
 3. Import the GSD in the engineering system of the controller.
- You have now installed the GSD file in the engineering system of the controller.

4.6.2.4 Setting the address

Bit 6 (64)	■
Bit 5 (32)	■
Bit 4 (16)	■
Bit 3 (8)	■
Bit 2 (4)	■
Bit 1 (2)	■
Bit 0 (1)	■
On	Off

Example:

	■
	■
	■
8	■
	■
2	■
	■
= 10	On Off

You have the following options for setting the PROFIBUS address:

- Using the address switch on the Control Unit:
The address switch has priority over the other settings.
- With the parameter p0918 (factory setting: p0918 = 126):
It is only possible to change p0918 if an invalid address is set in the address switch.
- With STARTER or Startdrive:
Setting is only possible if an invalid address is set in the address switch.

Valid addresses: 1 ... 125

➔ Overview of the interfaces (Page 91)

Procedure

- ➔ 1. Proceed as follows to set the PROFIBUS address:
2.
 1. Set the address using one of the subsequently listed options:
 - Via the address switch
 - On an operator panel via p0918
 - With STARTER or Startdrive
After you have changed the address in STARTER, select the button "RAM to ROM"
 2. Switch off the inverter supply voltage.

4.6 Connecting the inverter to the fieldbus

3. Wait until all LEDs on the inverter go dark.
4. Switch on the inverter supply voltage again.
Your settings become active after switching on.

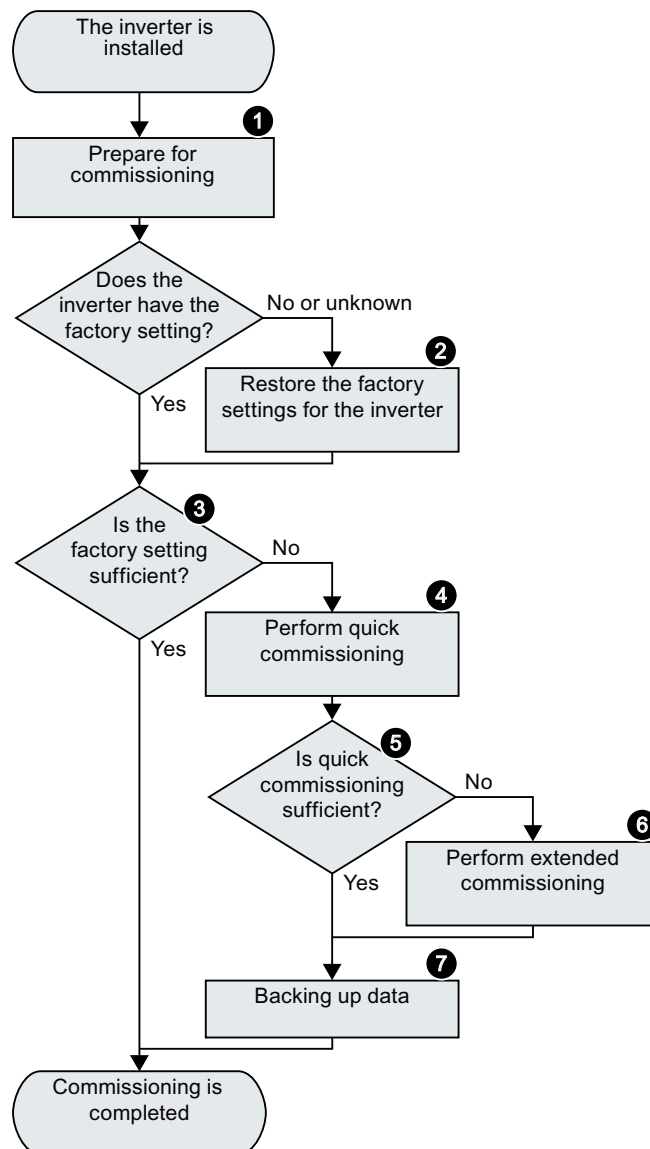


You set the PROFIBUS address.

Commissioning

5.1 Commissioning guidelines

Overview



1. Define the requirements to be met by the drive for your application.
 (Page 131)
2. Restore the factory settings of the inverter if necessary.
 (Page 166)
3. Check if the factory setting of the inverter is sufficient for your application.
 (Page 132)
4. Set the following for quick commissioning of the drive:
 - The closed-loop motor control
 - The inputs and outputs
 - The fieldbus interface (Page 134)
5. Check if additional inverter functions are required for the application.
 (Page 173)
6. If necessary, adapt the drive.
 (Page 173)
7. Save your settings.
 (Page 347)

5.2 Tools to commission the inverter

Operator panel



An operator panel is used to commission, troubleshoot and control the inverter, as well as to back up and transfer the inverter settings.

The **Intelligent Operator Panel (IOP)** is available for snapping onto the inverter, or as handheld with a connecting cable to the inverter. The graphics-capable plain text display of the IOP enables intuitive operation and diagnostics of the inverter.

The IOP is available in two versions:

- With European languages
- With Chinese, English and German

Additional information about the compatibility of the IOP and inverters is available in the Internet:



Compatibility of the IOP and Control Units (<http://support.automation.siemens.com/WW/view/en/67273266>)



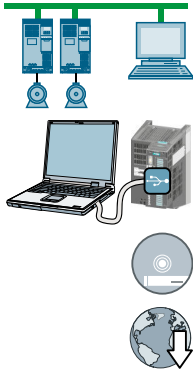
The **Operator Panel BOP-2** for snapping onto the inverter has a two-line display for diagnostics and operating the inverter.

Operating Instructions of the BOP-2 and IOP operator panels:



Overview of the manuals (Page 504)

PC tools



STARTER and **Startdrive** are PC tools that are used to commission, troubleshoot and control the inverter, as well as to back up and transfer the inverter settings. You can connect the PC with the inverter via USB or via the PROFIBUS / PROFINET fieldbus.

Connecting cable (3 m) between PC and inverter: Article number 6SL3255-0AA00-2CA0

STARTER DVD: Article number 6SL3072-0AA00-0AG0

Startdrive DVD: Article number 6SL3072-4CA02-1XG0

Startdrive, system requirements and download (<http://support.automation.siemens.com/WW/view/en/68034568>)

STARTER, system requirements and download (<http://support.automation.siemens.com/WW/view/en/26233208>)

Startdrive tutorial (<http://support.automation.siemens.com/WW/view/en/73598459>)

STARTER videos (<http://www.automation.siemens.com/mcms/mc-drives/en/low-voltage-inverter/sinamics-g120/videos/Pages/videos.aspx>)

5.3 Preparing for commissioning

5.3.1 Collecting motor data

Data for a standard induction motor

Before starting commissioning, you must know the following data:

- **Which motor is connected to the inverter?**
Note down the Article No. of the motor and the motor's nameplate data.
If available, note down the motor code on the motor's nameplate.

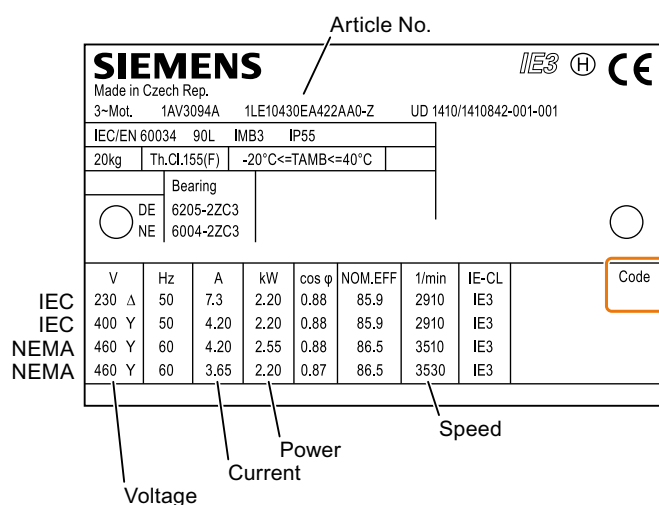


Figure 5-1 Example of the rating plate for a standard induction motor

- **In which region of the world is the motor to be used?**
- Europe IEC: 50 Hz [kW]
- North America NEMA: 60 Hz [hp] or 60 Hz [kW]
- **How is the motor connected?**
Pay attention to the connection of the motor (star connection [Y] or delta connection [Δ]).
Note the appropriate motor data for connecting.

5.3.2 Inverter factory setting

Motor

In the factory, the inverter is set for an induction motor matching the rated power of the Power Module.

Inverter control

You can find the factory settings for the inverter control in the following Chapters:

Inverter interfaces

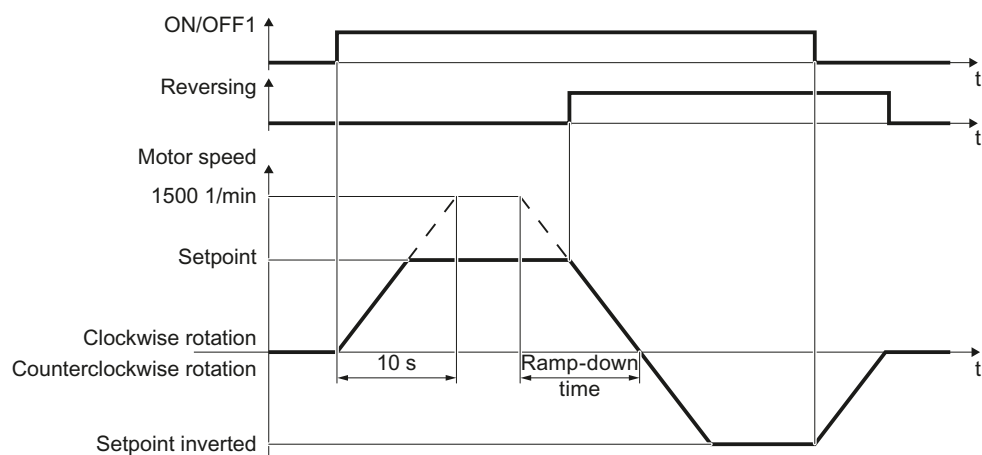
The inputs and outputs and the fieldbus interface of the inverter have specific functions when set to the factory settings.

- ➡ Factory setting of the CU240B-2 interfaces (Page 95)
- ➡ Factory setting of the CU240E-2 interfaces (Page 104)

Switching the motor on and off

The inverter is set in the factory as follows:

- After the ON command, the motor accelerates with a ramp-up time of 10 s (referred to 1500 rpm) to its speed setpoint.
- After the OFF1 command, the motor brakes down to standstill with the ramp-down time.
- The motor direction of rotation reverses with the reversing command.



- Ramp-down time:
- With the PM230 Power Module: 30 s
 - For all other Power Modules: 10 s

Figure 5-2 Switching on and switching off the motor and reversing in the factory setting

The ramp-up and ramp-down times define the maximum motor acceleration when the speed setpoint changes. The ramp-up and ramp-down times are derived from the time between motor standstill and the maximum speed, or between the maximum speed and motor standstill.

Traverse the motor in the jog mode

For an inverter with PROFIBUS or PROFINET interface, operation can be switched over using digital input DI 3. The motor is either switched on and off via the fieldbus – or operated in the jog mode via its digital inputs.

For a control command at the respective digital input, the motor rotates with ± 150 rpm. The same ramp-up and ramp-down times as described above apply.

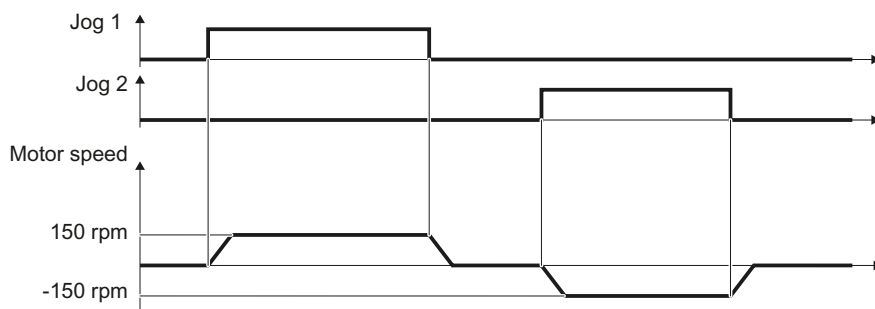


Figure 5-3 Jogging the motor in the factory setting

Minimum and maximum speed

- Minimum speed - factory setting 0 [rpm]
The minimum speed is the lowest speed of the motor independent of the speed setpoint. A minimum speed > 0 is, for example, useful for fans or pumps.
- Maximum speed - factory setting 1500 [rpm]
The inverter limits the motor speed to the maximum speed.

Operate the inverter with the factory setting

In basic applications with a standard induction motor, you can attempt to operate the drive with a rated power of < 18.5 kW without carry out an additional commissioning steps. Check whether the control quality of the drive without commissioning is adequate for the requirements of the application.

We recommend that you configure the drive with the precise motor data.

5.4 Quick commissioning using the BOP-2 operator panel

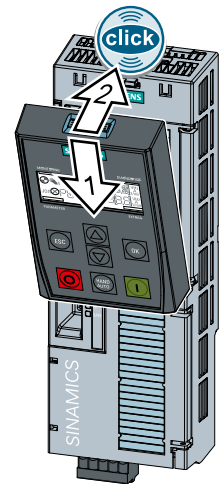
5.4.1 Inserting the BOP-2

Plugging on an operator panel

Procedure



1. To plug an Operator Panel on the Control Unit, proceed as follows:
2.
 1. Locate the lower edge of the Operator Panel into the matching recess of the Control Unit.
 2. Plug the Operator Panel onto the inverter until the latch audibly engages.



You have plugged an operator panel onto the Control Unit.

The operator panel is ready for operation when you connect the inverter to the power supply.

5.4.2 Overview of quick commissioning

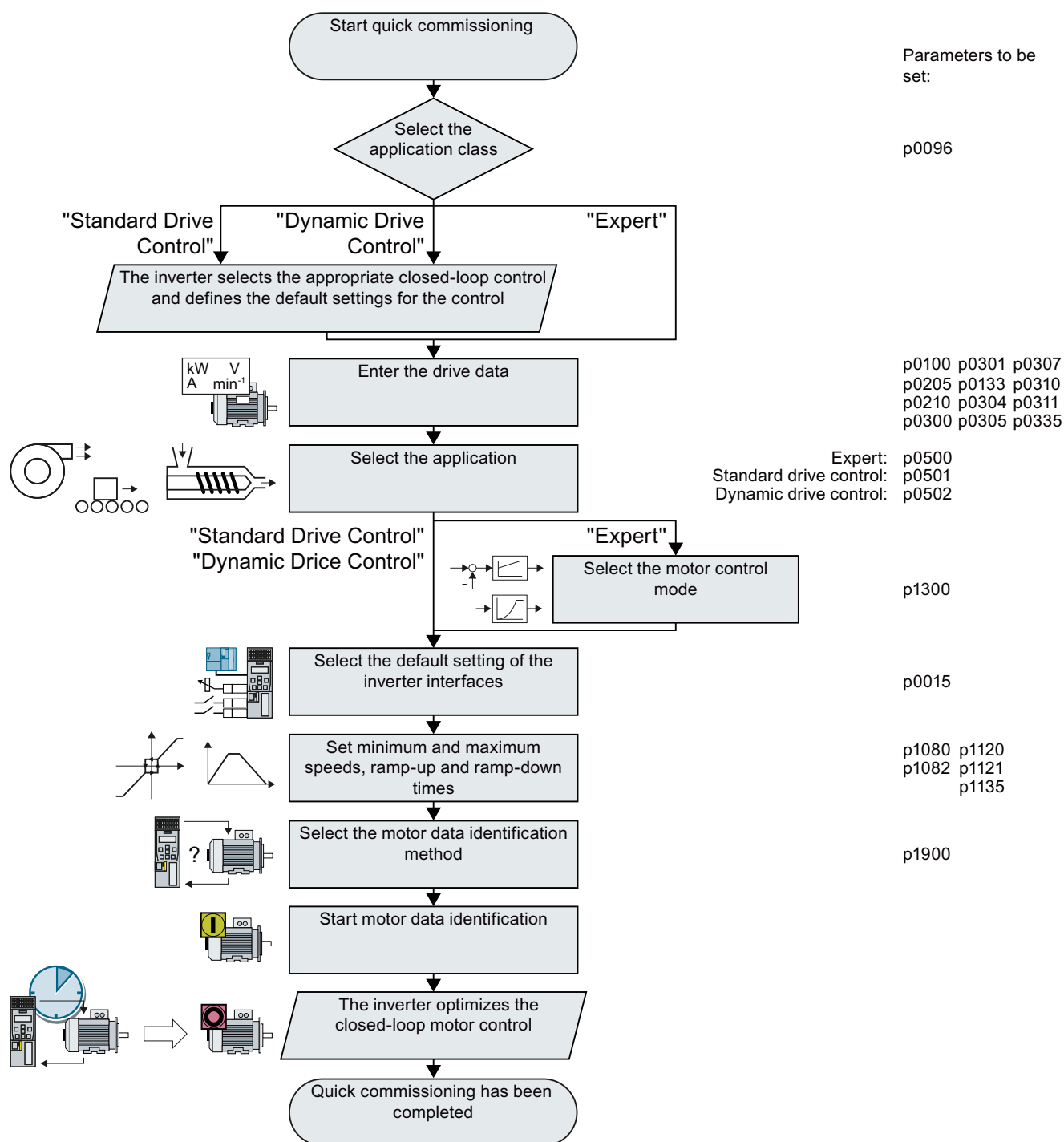
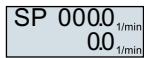


Figure 5-4 Quick commissioning using the BOP-2 operator panel

5.4.3 Start quick commissioning and select the application class

Starting quick commissioning



Preconditions

- The power supply is switched on.
- The operator panel displays setpoints and actual values.

Procedure



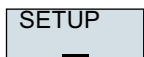
1. Proceed as follows to carry out quick commissioning:
- 2.



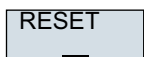
Press the ESC key.



Press one of the arrow keys until the BOP-2 displays the "SETUP" menu.

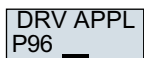


To start quick commissioning, in the "SETUP" menu, press the OK key.



If you wish to restore all of the parameters to the factory setting before the quick commissioning, proceed as follows:

1. Press the OK key.
2. Switchover the display using an arrow key: nO → YES
3. Press the OK key.



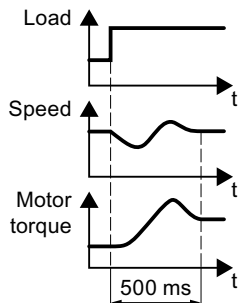
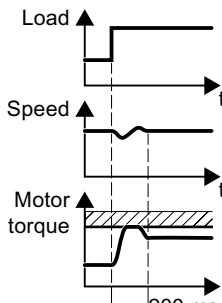
When selecting an application class, the inverter assigns the motor control with the appropriate default settings:

- Standard Drive Control (Page 138)
- Dynamic Drive Control (Page 140)
- Expert (Page 142)

Depending on the Power Module, the inverter may not offer any application class. Also in this case, continue the commissioning as described for "Expert".

Selecting a suitable application class

When selecting an application class, the inverter assigns the appropriate settings to the motor control.

Application class	Standard Drive Control	Dynamic Drive Control
Characteristics	• Typical settling time after a speed change: 100 ms ... 200 ms • Typical settling time after a sudden load change: 500 ms  • Standard Drive Control is suitable for the following requirements: – Motor power ratings < 45 kW – Ramp-up time 0 → Rated speed (dependent on the rated motor power): 1 s (0.1 kW) ... 10 s (45 kW) – Applications with constant load torque without load surges • Standard Drive Control is insensitive to inaccurate motor data settings	• Typical settling time after a speed change: < 100 ms • Typical settling time after a sudden load change: 200 ms  • Dynamic Drive Control controls and limits the motor torque • Torque accuracy that can be achieved: ± 5 % for 15 % ... 100 % of the rated speed • We recommend Dynamic Drive Control for the following applications: – Motor power ratings > 11 kW – For load surges of 10 % ... >100 % of the rated motor torque • Dynamic Drive Control is necessary for a ramp-up time 0 → rated speed (dependent on the rated motor power): < 1 s (0.1 kW) ... < 10 s (132 kW).
Application ex-amples	• Pumps, fans, and compressors with flow characteristic • Wet or dry blasting technology • Mills, mixers, kneaders, crushers, agitators • Horizontal conveyor technology (conveyor belts, roller conveyors, chain conveyors) • Basic spindles	• Pumps and compressors with displacement machines • Rotary furnaces • Extruder • Centrifuge
Motors that can be operated	Induction motors	Induction and synchronous motors
Power Modules that can be operated	PM240-2, PM240P-2	
Max. output frequency	550 Hz	240 Hz

Application class	Standard Drive Control	Dynamic Drive Control
Torque control	Without torque control	Speed control with lower-level torque control
Commissioning	- Contrary to "Dynamic Drive Control" a speed controller does not have to be set - In comparison to the "EXPERT" setting: - Simplified commissioning using preassigned motor data - Reduced number of parameters - Standard Drive Control is preset for Power Modules, frame size A ... frame size C	- Reduced parameter quantity when compared to the "EXPERT" setting: - Dynamic Drive Control is preset for Power Modules, frame size D ... frame size F

5.4.4 Standard Drive Control

EUR/USA
P100

Select the motor standard.

- KW 50HZ: IEC
- HP 60HZ: NEMA
- KW 60HZ: IEC 60 Hz

INV VOLT
P210

Set the inverter supply voltage.

MOT TYPE
P300

Select the motor type. If a 5-digit motor code is stamped on the motor rating plate, select the corresponding motor type with motor code.

Motors without motor code stamped on the rating plate:

- INDUCT: Third-party induction motor
- 1L... IND: 1LE1, 1LG6, 1LA7, 1LA9 induction motors

Motors with motor code stamped on the rating plate:

- 1LE1 IND 100: 1LE1 . 9
- 1PC1 IND: 1PC1
- 1PH8 IND: Induction motor
- 1FP1: Reluctance motor

Depending on the inverter, the motor list in BOP-2 can deviate from the list shown above.

MOT CODE
P301

If you have selected a motor type with motor code, you must now enter the motor code. The inverter assigns the following motor data corresponding to the motor code.

If you do not know the motor code, then you must set the motor code = 0, and enter motor data from p0304 and higher from the rating plate.

87 HZ
P302

87 Hz motor operation The BOP-2 only indicates this step if you selected IEC as the motor standard (EUR/USA, P100 = KW 50HZ).

MOT VOLT
P304

Rated motor voltage

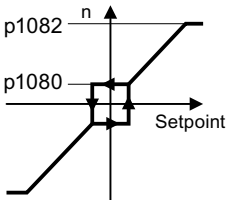
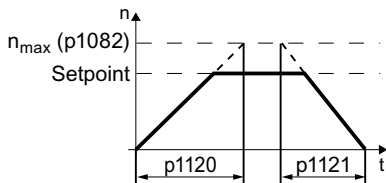
MOT CURR P305__	Rated motor current
MOT POW P307__	Rated motor power
MOT FREQ P310__	Rated motor frequency
MOT RPM P311__	Rated motor speed
MOT COOL P335__	Motor cooling: • SELF: Natural cooling • FORCED: Forced-air cooling • LIQUID: Liquid cooling • NO FAN: Without fan
TEC APPL P501__	Select the basic setting for the motor control: • VEC STD: Constant load; typical applications include conveyor drives • PUMP FAN: Speed-dependent load; typical applications include pumps and fans
MAc PAR P15__	Select the default setting for the interfaces of the inverter that is suitable for your application.  Default settings of the CU240B-2 interfaces (Page 97)  Default settings of the CU240E-2 interfaces (Page 106)
MIN RPM P1080__	
MAX RPM P1082__	
RAMP UP P1120__	
RAMP DWN P1121__	
OFF3 RP P1135__	Ramp-down time after the OFF3 command

Figure 5-5 Minimum and maximum motor speed

Figure 5-6 Ramp-up and ramp-down time of the motor

5.4 Quick commissioning using the BOP-2 operator panel

MOT ID
P1900

Motor data identification Select the method which the inverter uses to measure the data of the connected motor:

- OFF: No motor data identification
- STILL: Default setting: Measure the motor data at standstill. The inverter switches off the motor after the motor data identification has been completed.

FINISH

Complete quick commissioning as follows:

1. Switchover the display using an arrow key: nO → YES
2. Press the OK key.



You have completed quick commissioning.

5.4.5 Dynamic Drive Control

EUR/USA
P100

Select the motor standard.

- KW 50HZ: IEC
- HP 60HZ: NEMA
- KW 60HZ: IEC 60 Hz

INV VOLT
P210

Set the inverter supply voltage.

MOT TYPE
P300

Select the motor type. If a 5-digit motor code is stamped on the motor rating plate, select the corresponding motor type with motor code.

Motors without motor code stamped on the rating plate:

- INDUCT: Third-party induction motor
- 1L... IND: 1LE1, 1LG6, 1LA7, 1LA9 induction motors

Motors with motor code stamped on the rating plate:

- 1LE1 IND 100: 1LE1 . 9
- 1PC1 IND: 1PC1
- 1PH8 IND: Induction motor
- 1FP1: Reluctance motor

Depending on the inverter, the motor list in BOP-2 can deviate from the list shown above.

MOT CODE
P301

If you have selected a motor type with motor code, you must now enter the motor code. The inverter assigns the following motor data corresponding to the motor code.

If you do not know the motor code, then you must set the motor code = 0, and enter motor data from p0304 and higher from the rating plate.

87 HZ

87 Hz motor operation The BOP-2 only indicates this step if you selected IEC as the motor standard (EUR/USA, P100 = KW 50HZ).

MOT VOLT
P304

Rated motor voltage

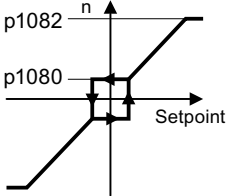
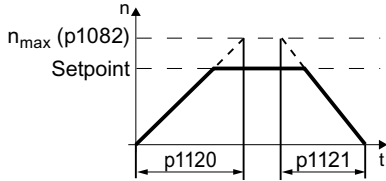
MOT CURR P305__	Rated motor current
MOT POW P307__	Rated motor power
MOT FREQ P310__	Rated motor frequency
MOT RPM P311__	Rated motor speed
MOT COOL P335__	Motor cooling: • SELF: Natural cooling • FORCED: Forced-air cooling • LIQUID: Liquid cooling • NO FAN: Without fan
TEC APPL P502__	Select the basic setting for the motor control: • OP LOOP: Recommended setting for standard applications • CL LOOP: Recommended setting for applications with short ramp-up and ramp-down times. This setting is not suitable for hoisting gear and cranes/lifting gear. • HVY LOAD: Recommended setting for applications with a high break loose torque.
MAc PAr P15__	Select the default setting for the interfaces of the inverter that is suitable for your application.  Default settings of the CU240B-2 interfaces (Page 97)  Default settings of the CU240E-2 interfaces (Page 106)
MIN RPM P1080__	
MAX RPM P1082__	
RAMP UP P1120__	
RAMP DWN P1121__	
OFF3 RP P1135__	Ramp-down time after the OFF3 command

Figure 5-7 Minimum and maximum motor speed

Figure 5-8 Ramp-up and ramp-down time of the motor

5.4 Quick commissioning using the BOP-2 operator panel

MOT ID
P1900

Motor data identification: Select the method which the inverter uses to measure the data of the connected motor:

- OFF: Motor data is not measured.
STIL ROT: Recommended setting: Measure the motor data at standstill and with the motor rotating.
The inverter switches off the motor after the motor data identification has been completed.
- STILL: Default setting: Measure the motor data at standstill.
The inverter switches off the motor after the motor data identification has been completed.
Select this setting if the motor cannot rotate freely – for example, if the traversing range is mechanically limited.
- ROT: Measure the motor data with the motor rotating.
The inverter switches off the motor after the motor data identification has been completed.
- ST RT OP: setting same as STIL ROT.
The motor accelerates to the currently set setpoint after the motor data identification.
- STILL OP: setting same as STILL.
After the motor data identification, the motor accelerates to the currently set setpoint.

FINISH
—

Complete quick commissioning:

- Switch over the display using an arrow key: nO → YES
- Press the OK key.



You have completed quick commissioning.

5.4.6 Expert

EUR/USA
P100

Select the motor standard:

- KW / 50HZ: IEC
- HP / 60HZ: NEMA
- KW / 60HZ: IEC 60 Hz

LOAD TYP
P210

Specify the overload capability of the inverter:

- HIGH OVL: Load cycle with "High Overload"
- LOW OVL: Load cycle with "Low Overload"



Overload capability of the inverter (Page 424)

INV VOLT
P210

Set the supply voltage of the inverter.

MOT TYPE
P300

Select the motor type. If a 5-digit motor code is stamped on the motor rating plate, select the corresponding motor type with motor code.

Motors without motor code stamped on the rating plate:

- INDUCT: Third-party induction motor
- 1L... IND: 1LE1, 1LG6, 1LA7, 1LA9 induction motors

Motors with motor code stamped on the rating plate:

- 1LE1 IND 100: 1LE1 . 9
- 1PC1 IND: 1PC1
- 1PH8 IND: Induction motor
- 1FP1: Reluctance motor

Depending on the inverter, the motor list in BOP-2 can deviate from the list shown above.

MOT CODE
P301__

If you have selected a motor type with motor code, you must now enter the motor code. The inverter assigns the following motor data corresponding to the motor code.

If you do not know the motor code, then you must set the motor code = 0, and enter motor data from p0304 and higher from the rating plate.

87 HZ
__

87 Hz motor operation The BOP-2 only indicates this step if you selected IEC as the motor standard (EUR/USA, P100 = KW 50HZ).

MOT VOLT
P304__

Rated motor voltage

MOT CURR
P305__

Rated motor current

MOT POW
P307__

Rated motor power

MOT FREQ
P310__

Rated motor frequency

MOT RPM
P311__

Rated motor speed

MOT COOL
P335__

Motor cooling:

- SELF: Natural cooling
- FORCED: Forced-air cooling
- LIQUID: Liquid cooling
- NO FAN: Without fan

TEC APPL
P500__

Select the appropriate application:

- VEC STD: In all applications, which do not fit the other setting options.
- PUMP FAN: Applications involving pumps and fans
- SLVC 0HZ: Applications with short ramp-up and ramp-down times. However, this setting is not suitable for hoisting gear and cranes/lifting gear.
- PUMP 0HZ: Setting only for steady-state operation with slow speed changes. We recommend setting VEC STD if load surges in operation cannot be ruled out.

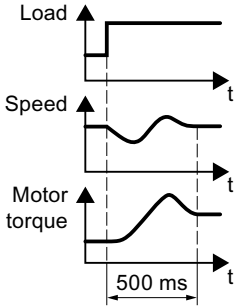
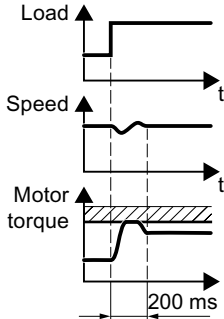
The selection option depends on the Power Module being used. There is no selection option for PM230 Power Modules.

CTRL MOD
P1300

Select the control mode:

- VF LIN: U/f control with linear characteristic
- VF LIN F: Flux current control (FCC)
- VF QUAD: U/f control with square-law characteristic
- SPD N EN: Sensorless vector control

Select a suitable control mode

Control mode	U/f control or flux current control (FCC)	Sensorless vector control
Characteristics	• Typical settling time after a speed change: 100 ms ... 200 ms • Typical settling time after a sudden load change: 500 ms  • The control mode is suitable to address the following requirements: – Motor power ratings < 45 kW – Ramp-up time 0 → rated speed (depending on the motor power rating): 1 s (0.1 kW) ... 10 s (45 kW) – Applications with continuous load torque without sudden load changes • Die Regelungsart is insensitive with respect to imprecise setting of the motor data	• Typical settling time after a speed change: < 100 ms • Typical settling time after a sudden load change: 200 ms  • The control mode controls and limits the motor torque • Torque accuracy that can be achieved: ± 5% for 15% ... 100% of the rated speed • We recommend the control mode for the following applications: – Motor power ratings > 11 kW – For load surges of 10% ... >100% of the rated motor torque • The control mode is necessary for a ramp-up time 0 → Rated speed (dependent on the rated motor power): < 1 s (0.1 kW) ... < 10 s (250 kW).
Application examples	• Pumps, fans, and compressors with flow characteristic • Wet or dry blasting technology • Mills, mixers, kneaders, crushers, agitators • Horizontal conveyor technology (conveyor belts, roller conveyors, chain conveyors) • Basic spindles	• Pumps and compressors with displacement machines • Rotary furnaces • Extruders • Centrifuges
Motors that can be operated	Induction motors	Induction and synchronous motors

Control mode	U/f control or flux current control (FCC)	Sensorless vector control
Power Modules that can be operated	No restrictions	
Max. output frequency	550 Hz	240 Hz
Closed-loop torque control	Without torque control	Torque control with and without higher-level speed control
Commissioning	Contrary to encoderless vector control, the speed controller does not have to be set	

MAc PAr
P15

Select the default setting for the interfaces of the inverter that is suitable for your application.



Default settings of the CU240B-2 interfaces (Page 97)



Default settings of the CU240E-2 interfaces (Page 106)

MIN RPM
P1080

MAX RPM
P1082

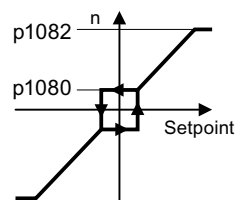


Figure 5-9 Minimum speed of the motor

RAMP UP
P1120

RAMP DWN
P1121

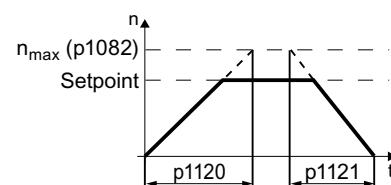


Figure 5-10 Ramp-up and ramp-down time of the motor

OFF3 RP
P1135

Ramp-down time for the OFF3 command

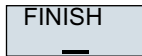
MOT ID
P1900

Motor data identification: Select the method which the inverter uses to measure the data of the connected motor:

- OFF: Motor data is not measured.
- STIL ROT: Recommended setting: Measure the motor data at standstill and with the motor rotating. The inverter switches off the motor after the motor data identification has been completed.
- STILL: Measure the motor data at standstill. The inverter switches off the motor after the motor data identification has been completed.
Select this setting if one of the following cases is applicable:
 - You have selected control mode "SPD N EN", however, the motor cannot freely rotate - for a mechanically limited travel range, for example.
 - You have selected U/f control as control mode, e.g. "VF LIN" or "VF QUAD".

5.4 Quick commissioning using the BOP-2 operator panel

- ROT: Measure the motor data with the motor rotating. The inverter switches off the motor after the motor data identification has been completed.
- ST RT OP: setting same as STIL ROT.
After the motor data identification, the motor accelerates to the currently set setpoint.
- STILL OP: setting same as STILL.
After the motor data identification, the motor accelerates to the currently set setpoint.



Complete quick commissioning:

Switchover the display using an arrow key: nO → YES

Press the OK key.



You have completed quick commissioning.

5.4.7 Identifying the motor data and optimizing the closed-loop control

The inverter has several techniques to automatically identify the motor data and optimize the speed control.

To start the motor data identification routine, you must switch-on the motor via the terminal strip, fieldbus or from the operator panel.

WARNING

Risk of death due to machine motion while motor data identification is active

For the stationary measurement, the motor can make several rotations. The rotating measurement accelerates the motor up to its rated speed. Secure dangerous machine parts before starting motor data identification:

- Before switching on, ensure that nobody is working on the machine or located within its working area.
- Secure the machine's work area against unintended access.
- Lower hanging/suspended loads to the floor.

Preconditions

- You selected a method of motor data identification during quick commissioning, e.g. measuring motor data while the motor is stationary.
When quick commissioning is complete, the inverter issues alarm A07991.
- The motor has cooled down to the ambient temperature.
An excessively high motor temperature falsifies the motor data identification results.



Procedure when using the BOP-2 operator panel



1. To start the motor data identification, proceed as follows:
- 2.



Press the HAND/AUTO key.



The BOP-2 displays the symbol indicating manual operation.



Switch on the motor.



During motor data identification, "MOT-ID" flashes on the BOP-2.



If the inverter again outputs alarm A07991, then it waits for a new ON command to start the rotating measurement.

If the inverter does not output alarm A07991, switch off the motor as described below, and switch over the inverter control from HAND to AUTO.



Switch on the motor to start the rotating measurement.

5.4 Quick commissioning using the BOP-2 operator panel



During motor data identification, "MOT-ID" flashes on the BOP-2.

The motor data identification can take up to 2 minutes depending on the rated motor power.



Depending on the setting, after motor data identification has been completed, the inverter switches off the motor - or it accelerates it to the setpoint.

If required, switch off the motor.



Switch the inverter control from HAND to AUTO.



You have completed the motor data identification.

5.5 Quick commissioning with a PC

The screen forms that are shown in this manual show generally valid examples. The number of setting options available in screen forms depends on the particular inverter type.

5.5.1 Creating a project

Creating a new project

Procedure



1. To create a new project, proceed as follows:
 2.
 1. Start the commissioning software STARTER or Startdrive.
 2. In the menu, select "Project" → "New...".
 3. Specify a name of your choice for the project.
-  You have created a new project.

5.5.2 Transfer inverters connected via USB into the project

Integrating the inverter into the project

Procedure

- ➔
1. Proceed as follows to transfer an inverter connected via USB to your project:
 2.
 1. Switch on the inverter supply voltage.
 2. First insert a USB cable into your PC and then into the inverter.
 3. The PC operating system installs the USB driver when you are connecting the inverter and PC together for the first time.
 4. Press the "Accessible nodes" button.

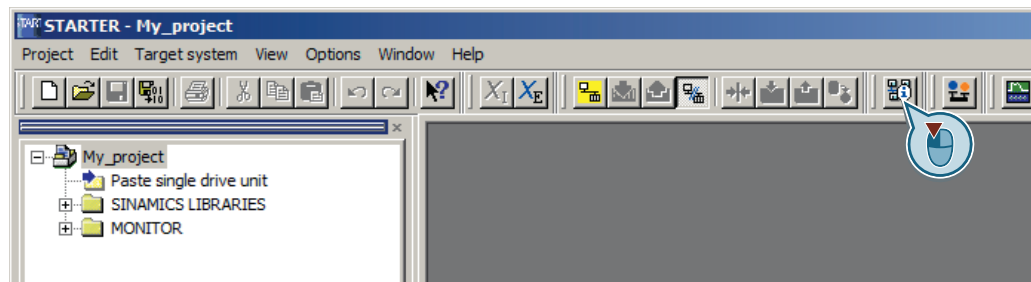


Figure 5-11 "Accessible nodes" in STARTER

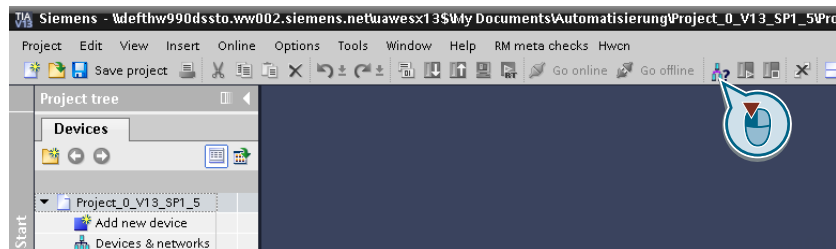


Figure 5-12 "Accessible nodes" in Startdrive

5. When the USB interface is appropriately set, then the "Accessible nodes" screen form shows the inverters that can be accessed.

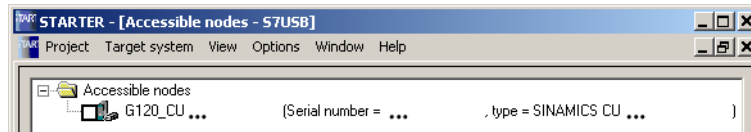


Figure 5-13 Inverters found in STARTER

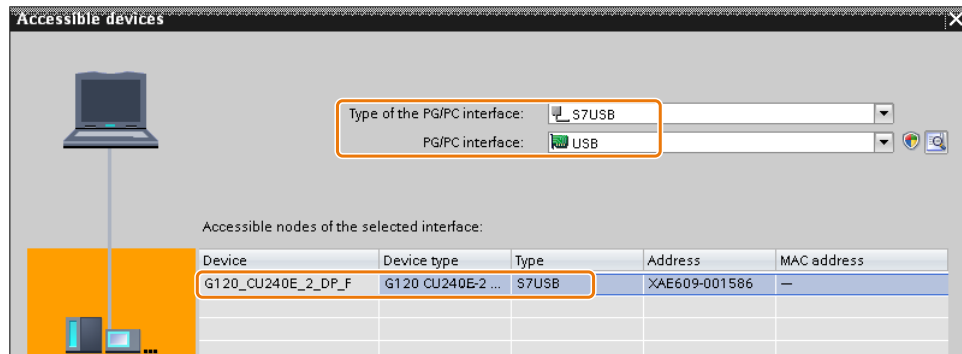


Figure 5-14 Inverters found in Startdrive

If you have not correctly set the USB interface, then the following "No additional nodes found" message is displayed. In this case, follow the description below.

6. Further procedure dependent on the commissioning software used:
- STARTER:
 - Select the inverter ☒.
 - Press the "Accept" button.
 - Startdrive:
 - Transfer the inverter into the project using the menu: "Online - Upload device as new station (hardware and software)".

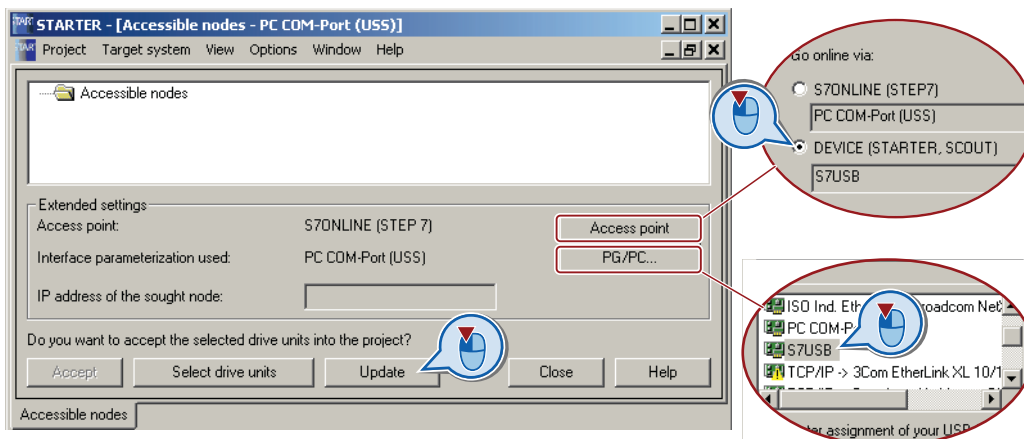
☒ You have transferred an inverter accessible via the USB interface into your project.

Setting the USB interface in STARTER

Procedure



1. Proceed as follows to set the USB interface in STARTER:
2.
 1. Set the "Access point" to "DEVICE (STARTER, Scout)" and the "PG/PC interface" to "S7USB".
 2. Press the "Update" button.



You have set the USB interface.

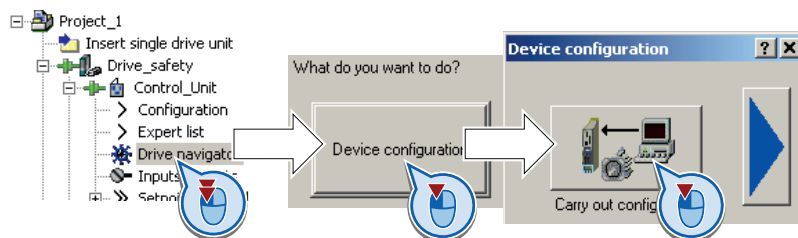
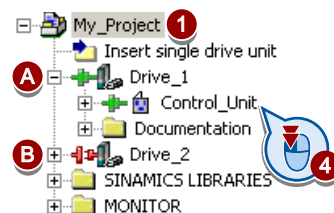
STARTER now shows the inverters connected via USB.

5.5.3 Go online and start quick commissioning

Procedure with STARTER

- ➔ 1. Proceed as follows to start the quick commissioning of the inverter:

2. In the following screen form, select the inverter with which you wish to go online.
3. Download the hardware configuration found online in your project (PG or PC).
Significance of the symbol in front of the inverter:
(A) The inverter is online.
(B) The inverter is offline
4. When you are online, double-click on "Control Unit".
5. Start the commissioning wizard:

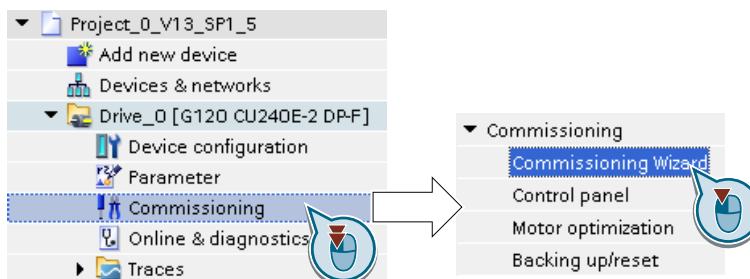


- You have started the quick commissioning of the inverter.

Procedure with Startdrive

- ➔ 1. Proceed as follows to start the quick commissioning of the inverter:

2. In the following screen form, select the inverter with which you wish to go online.
3. Once you are online, select "Commissioning" → "Commissioning Wizard":



- You have started the quick commissioning of the inverter.

5.5.4 Overview of quick commissioning

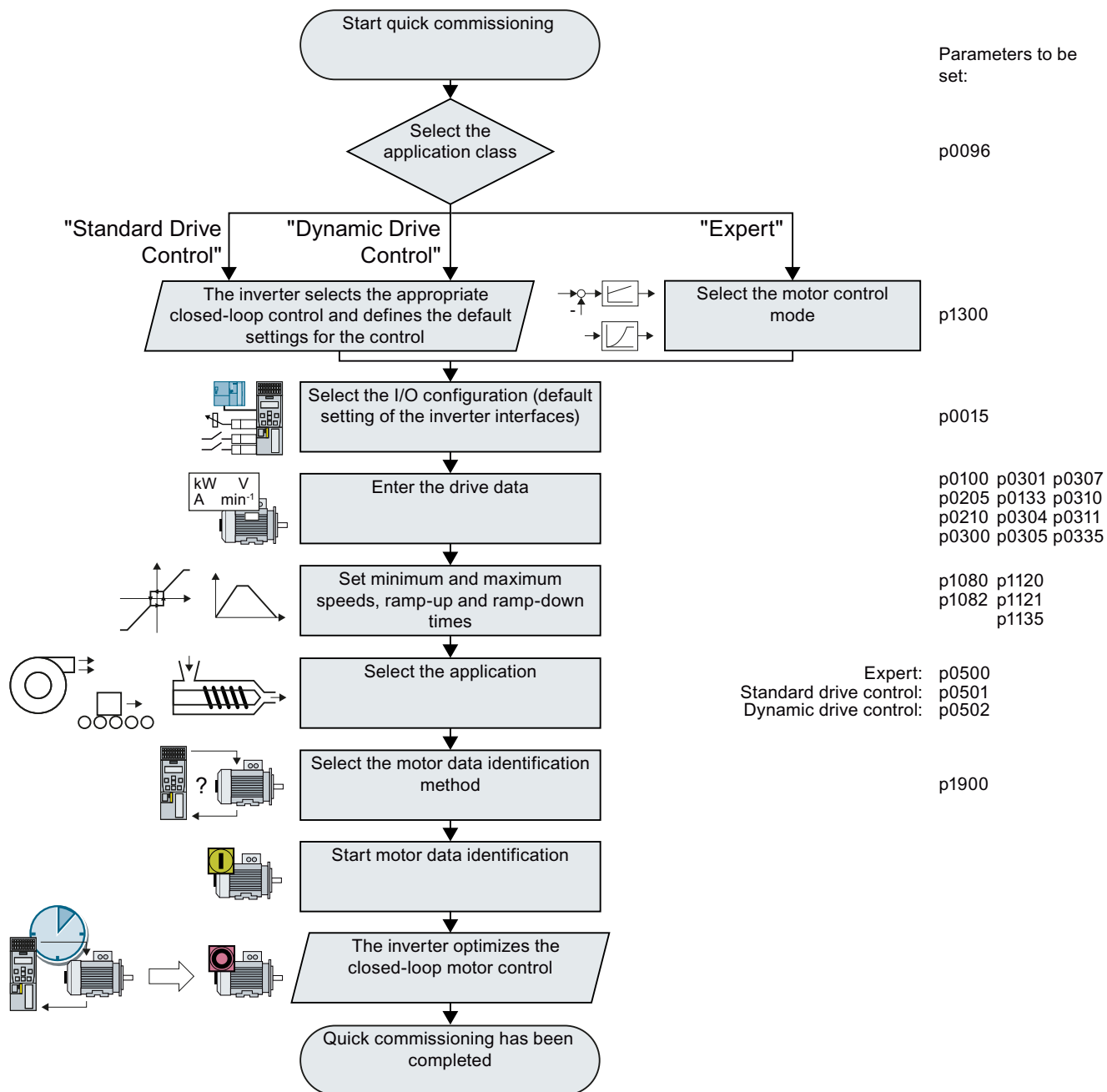


Figure 5-15 Quick commissioning with a PC

5.5.5 Select the application class

Starting quick commissioning

Procedure



1. Proceed as follows to start the quick commissioning:
- 2.

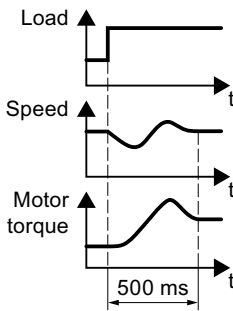
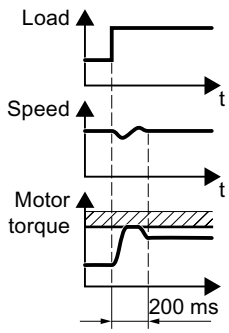
☒ Application class

When selecting an application class, the inverter assigns the motor control with the appropriate default settings:

- [1]  Standard Drive Control (Page 158)
- [2]  Dynamic Drive Control (Page 159)
- [0] Expert - or if no application class is listed:
 Expert (Page 160)

Selecting a suitable application class

When selecting an application class, the inverter assigns the appropriate settings to the motor control.

Application class	Standard Drive Control	Dynamic Drive Control
Characteristics	• Typical settling time after a speed change: 100 ms ... 200 ms • Typical settling time after a sudden load change: 500 ms  • Standard Drive Control is suitable for the following requirements: – Motor power ratings < 45 kW – Ramp-up time 0 → Rated speed (dependent on the rated motor power): 1 s (0.1 kW) ... 10 s (45 kW) – Applications with constant load torque without load surges • Standard Drive Control is insensitive to inaccurate motor data settings	• Typical settling time after a speed change: < 100 ms • Typical settling time after a sudden load change: 200 ms  • Dynamic Drive Control controls and limits the motor torque • Torque accuracy that can be achieved: ± 5 % for 15 % ... 100 % of the rated speed • We recommend Dynamic Drive Control for the following applications: – Motor power ratings > 11 kW – For load surges of 10 % ... >100 % of the rated motor torque • Dynamic Drive Control is necessary for a ramp-up time 0 → rated speed (dependent on the rated motor power): < 1 s (0.1 kW) ... < 10 s (132 kW).
Application ex-amples	• Pumps, fans, and compressors with flow characteristic • Wet or dry blasting technology • Mills, mixers, kneaders, crushers, agitators • Horizontal conveyor technology (conveyor belts, roller conveyors, chain conveyors) • Basic spindles	• Pumps and compressors with displacement machines • Rotary furnaces • Extruder • Centrifuge
Motors that can be operated	Induction motors	Induction and synchronous motors
Power Modules that can be operated	PM240-2, PM240P-2	
Max. output frequency	550 Hz	240 Hz

Application class	Standard Drive Control	Dynamic Drive Control
Torque control	Without torque control	Speed control with lower-level torque control
Commissioning	- Contrary to "Dynamic Drive Control" a speed controller does not have to be set - In comparison to the "EXPERT" setting: - Simplified commissioning using preassigned motor data - Reduced number of parameters - Standard Drive Control is preset for Power Modules, frame size A ... frame size C	- Reduced parameter quantity when compared to the "EXPERT" setting: - Dynamic Drive Control is preset for Power Modules, frame size D ... frame size F

5.5.6 Standard Drive Control

Procedure for application class [1]: Standard Drive Control

- ☒ Defaults of the setpoint Select the I/O configuration to preassign the inverter interfaces.
-  Default settings of the CU240B-2 interfaces (Page 97)
-  Default settings of the CU240E-2 interfaces (Page 106)
- ☒ Drive setting Set the applicable motor standard and the inverter supply voltage.
- ☒ Motor Select your motor.
- ☒ Motor data Enter the motor data according to the rating plate of your motor.
 If you have selected a motor based on its article number, the data has already been entered.
- ☒ Important parameters Set the most important parameters to suit your application.
- ☒ Drive functions Select the application:
- [0] Constant load: Typical applications include conveyor drives
 - [1] Speed-dependent load: Typical applications include pumps and fans
- Motor data identification:
- [0]: No motor data identification
 - [2]: Default setting: Measure the motor data at standstill. The inverter switches off the motor after the motor data identification has been completed.
- ☐ You have completed quick commissioning.

5.5.7 Dynamic Drive Control

Procedure for application class [2]: Dynamic Drive Control

- ☒ Defaults of the setpoint
 - Select the I/O configuration to preassign the inverter interfaces.
 -  Default settings of the CU240B-2 interfaces (Page 97)
 -  Default settings of the CU240E-2 interfaces (Page 106)
 - ☒ Drive setting
 - Set the applicable motor standard and the inverter supply voltage.
 - ☒ Motor
 - Select your motor.
 - ☒ Motor data
 - Enter the motor data according to the rating plate of your motor.
 - If you have selected a motor based on its article number, the data has already been entered.
 - ☒ Important parameters
 - Set the most important parameters to suit your application.
 - ☒ Drive functions
 - Application:
 - [0]: Recommended setting for standard applications.
 - [1]: Recommended setting for applications with ramp-up and ramp-down times < 10 s. This setting is not suitable for hoisting gear and cranes.
 - [5] Recommended setting for applications with a high break loose torque.
 - Motor data identification:
 - [0]: No motor data identification
 - [1]: Recommended setting. Measure the motor data at standstill and with the motor rotating. The inverter switches off the motor after the motor data identification has been completed.
 - [2]: Default setting: Measure the motor data at standstill. The inverter switches off the motor after the motor data identification has been completed.
Select this setting if the motor cannot freely rotate, e.g. for a mechanically limited traversing range.
 - [3]: Measure the motor data while it is rotating. The inverter switches off the motor after the motor data identification has been completed.
 - [11]: The same setting as [1]. The motor accelerates to the currently set setpoint after the motor data identification.
 - [12]: The same setting as [2]. The motor accelerates to the currently set setpoint after the motor data identification.
-  You have completed quick commissioning.

5.5.8 Expert

Procedure without application class or for the application class [0]: Expert

- ☒ Control structure Select the control mode.
- ☒ Defaults of the setpoint Select the I/O configuration to preassign the inverter interfaces.
 -  Default settings of the CU240B-2 interfaces (Page 97)
 -  Default settings of the CU240E-2 interfaces (Page 106)
- ☒ Drive setting Set the applicable motor standard and the inverter supply voltage.

Application:

 - "[0] Load cycle with high overload for applications requiring a high dynamic performance, e.g. conveyor systems.
 - "[1] Load cycle with low overload ..." for applications that do not require a high dynamic performance, e.g. pumps or fans.
 - [6], [7]: Load cycles for applications with encoderless 1FK7 synchronous motors.
- ☒ Motor Select your motor.
- ☒ Motor data Enter the motor data according to the rating plate of your motor.
If you have selected a motor based on its article number, the data has already been entered.
- ☒ Important parameters Set the most important parameters to suit your application.
- ☒ Drive functions Application:
 - [0]: In all applications that do not fall under [1] ... [3]
 - [1]: Applications involving pumps and fans
 - [2]: Applications with short ramp-up and ramp-down times. However, this setting is not suitable for hoisting gear and cranes/lifting gear.
 - [3]: Setting only for steady-state operation with slow speed changes. We recommend setting [1] if load surges in operation cannot be ruled out.

Motor identification:

 - [1]: Recommended setting. Measure the motor data at standstill and with the motor rotating. The inverter switches off the motor after the motor data identification has been completed.
 - [2]: Measure the motor data at standstill. The inverter switches off the motor after the motor data identification has been completed.
Recommended setting for the following cases:
 - You have selected "Speed control" as control mode, however the motor cannot freely rotate, e.g. for mechanically limited traversing sections.
 - You have set "V/f control" as control mode.
 - [3]: Measure the motor data while it is rotating. The inverter switches off the motor after the motor data identification has been completed.

- [11]: The same setting as [1]. The motor accelerates to the currently set setpoint after the motor data identification.
- [12]: The same setting as [2]. The motor accelerates to the currently set setpoint after the motor data identification.

Calculating the motor parameters: Select "Complete calculation".

Set the check mark for "RAM to ROM (save data in the drive)" to save your data in the inverter so that it is not lost when the power fails.

Select "Finish".

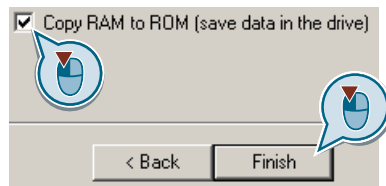


Figure 5-16 Exiting quick commissioning in STARTER

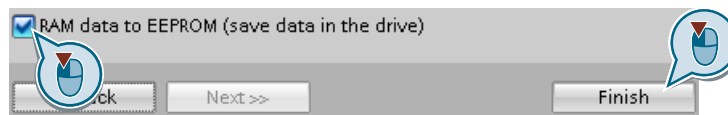
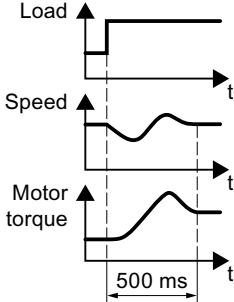
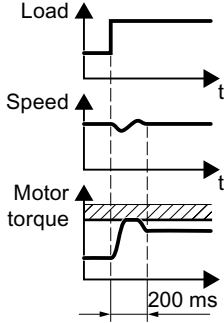


Figure 5-17 Exiting quick commissioning in Startdrive

- You have completed quick commissioning.

Select a suitable control mode

Control mode	U/f control or flux current control (FCC)	Sensorless vector control
Characteristics	- Typical settling time after a speed change: 100 ms ... 200 ms - Typical settling time after a sudden load change: 500 ms  - The control mode is suitable to address the following requirements: - Motor power ratings < 45 kW - Ramp-up time 0 → rated speed (depending on the motor power rating): 1 s (0.1 kW) ... 10 s (45 kW) - Applications with continuous load torque without sudden load changes - Die Regelungsart is insensitive with respect to imprecise setting of the motor data	- Typical settling time after a speed change: < 100 ms - Typical settling time after a sudden load change: 200 ms  - The control mode controls and limits the motor torque - Torque accuracy that can be achieved: $\pm 5\%$ for 15% ... 100% of the rated speed - We recommend the control mode for the following applications: - Motor power ratings > 11 kW - For load surges of 10% ... >100% of the rated motor torque - The control mode is necessary for a ramp-up time 0 → Rated speed (dependent on the rated motor power): < 1 s (0.1 kW) ... < 10 s (250 kW).
Application examples	- Pumps, fans, and compressors with flow characteristic - Wet or dry blasting technology - Mills, mixers, kneaders, crushers, agitators - Horizontal conveyor technology (conveyor belts, roller conveyors, chain conveyors) - Basic spindles	- Pumps and compressors with displacement machines - Rotary furnaces - Extruders - Centrifuges
Motors that can be operated	Induction motors	Induction and synchronous motors
Power Modules that can be operated	No restrictions	
Max. output frequency	550 Hz	240 Hz
Closed-loop torque control	Without torque control	Torque control with and without higher-level speed control
Commissioning	Contrary to encoderless vector control, the speed controller does not have to be set	

5.5.9 Identify motor data

Identify motor data

⚠ WARNING

Danger to life from machine movements while motor data identification is in progress

The stationary measurement can turn the motor a number of revolutions. The rotating measurement accelerates the motor up to the rated speed. Secure dangerous machine parts before starting motor data identification:

- Before switching on, ensure that nobody is working on the machine or located within its working area.
- Secure the machine's working area against unintended access.
- Lower suspended loads to the floor.

Preconditions

- You selected a method of motor data identification during quick commissioning, e.g. measurement of the motor data while the motor is stationary.
When quick commissioning is complete, the inverter issues alarm A07991.
- The motor has cooled down to the ambient temperature.
An excessively high motor temperature distorts the results of motor data identification.
- The PC and inverter are connected to each other online.

Procedure with STARTER

- ➔ 1. To initiate motor data identification and optimize the motor control, proceed as follows:
- 2.

1. Open the control panel.

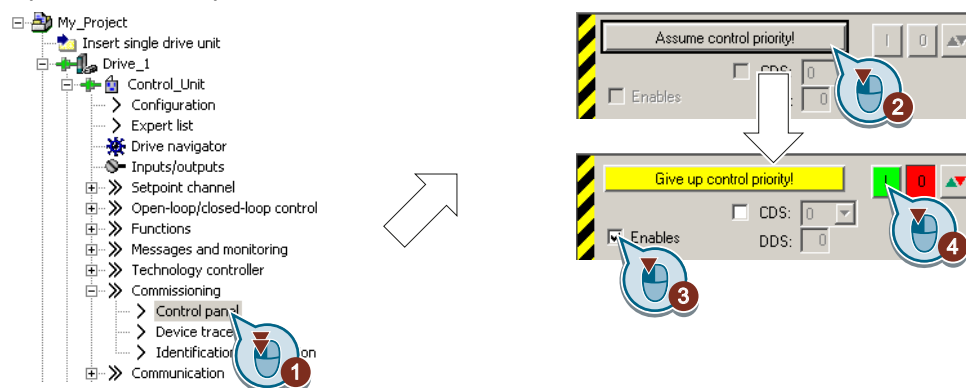


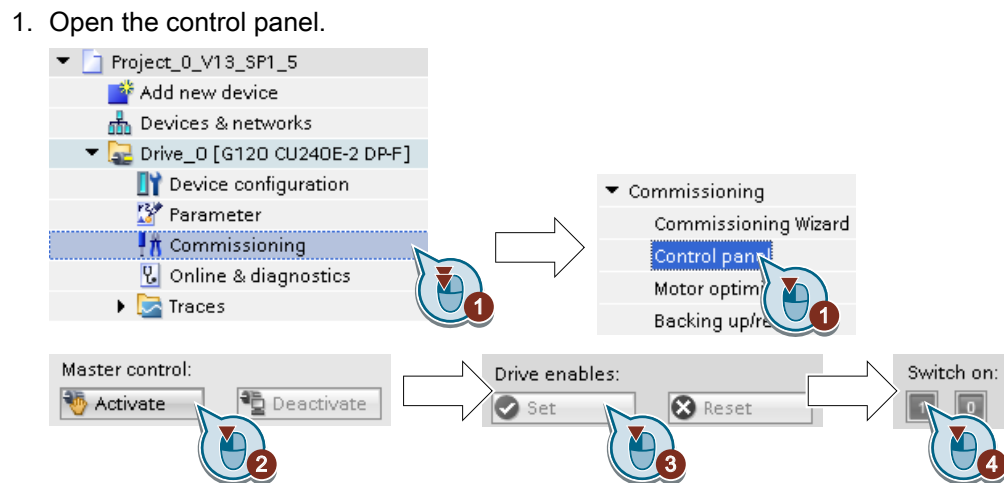
Figure 5-18 Control panel

2. Assume master control for the inverter.
3. Set the "Enable signals"

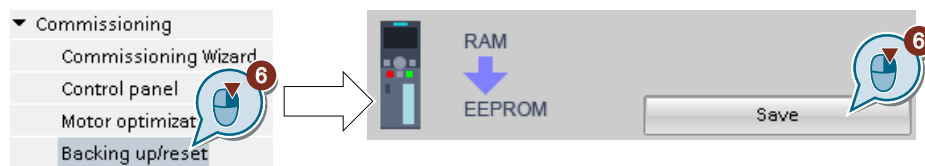
4. Switch on the motor.
The inverter starts the motor data identification. This measurement can take several minutes.
Depending on the setting, after motor data identification has been completed, the inverter switches off the motor - or it accelerates it to the currently set setpoint.
 5. If required, switch off the motor.
 6. Relinquish the master control after the motor data identification.
 7. Press button  (RAM to ROM).
-  You have completed the motor data identification.

Procedure with Startdrive

- ➔ 1. To initiate motor data identification and optimize the motor control, proceed as follows:
- 2.



2. Assume master control for the inverter.
3. Set the "Drive enables"
4. Switch on the motor.
The inverter starts the motor data identification. This measurement can take several minutes.
Depending on the setting, after motor data identification has been completed, the inverter switches off the motor - or it accelerates it to the currently set setpoint.
5. If required, switch off the motor.
6. Relinquish the master control after the motor data identification.
7. Save the settings in the inverter (RAM → EEPROM):



-  You have completed the motor data identification.

Self-optimization of the speed control

If you have selected not only motor data identification but also rotating measurement with self-optimization of the speed control, you must switch on the motor again as described above and wait for the optimization run to finish.

5.6 Restoring the factory setting

When must you reset the inverter to the factory settings?

Reset the inverter to the factory settings in the following cases:

- The line voltage was interrupted during commissioning and you were not able to complete commissioning.
- You can no longer trace the settings that you made during commissioning.
- You do not know whether the inverter was already operational.

Restoring the factory settings when the safety functions are enabled

If you are using the integrated safety functions of the inverter, e.g. "Safe Torque Off", you must reset the safety functions separately from the remaining inverter settings.

The settings of the safety functions are protected by a password.

Settings that are not changed when restoring the factory setting

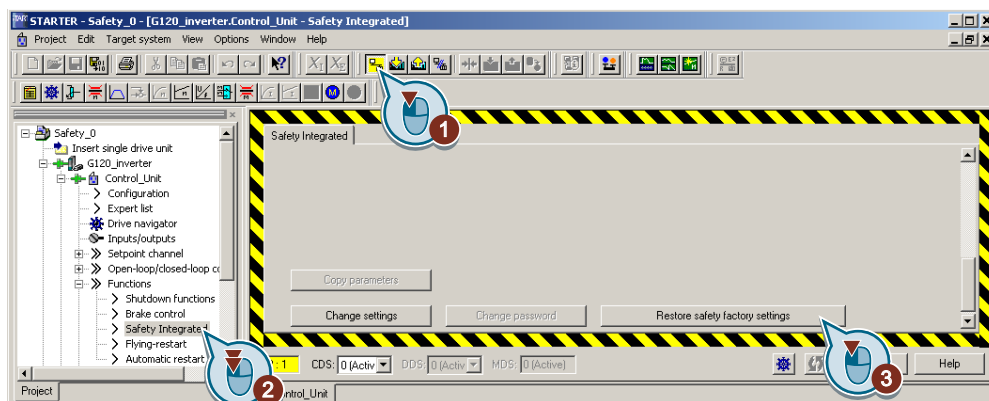
The communication settings and the settings of the motor standard (IEC/NEMA) are kept when restoring the factory setting.

5.6.1 Resetting the safety functions to the factory setting

Procedure with STARTER

- ➔ 1. To reset the safety function settings to the factory setting without changing the standard settings, proceed as follows:

1. Go online.



2. Open the screen form of the safety functions.
 3. Select the button to restore the factory settings.
 4. Enter the password for the safety functions.
 5. Confirm that the parameters have been saved (RAM to ROM).
 6. Go offline.
 7. Switch off the inverter supply voltage.
 8. Wait until all LEDs on the inverter go dark.
 9. Switch on the inverter supply voltage again.

■ You have restored the safety functions in the inverter to the factory settings.

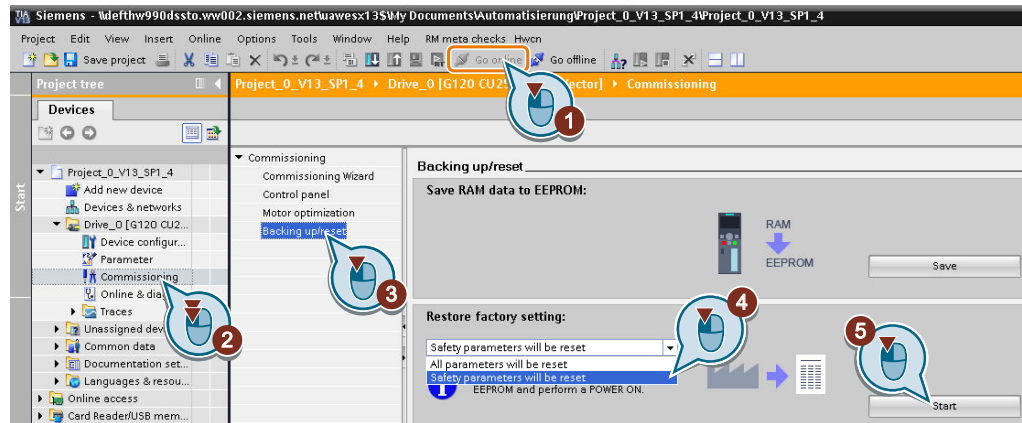
Exception: The password for the safety functions is not reset.

➔ Safety functions password (Page 234)

Procedure with Startdrive

1. To reset the safety function settings to the factory setting without changing the standard settings, proceed as follows:

1. Go online.



2. Select "Commissioning".
3. Select "Backing up/reset".
4. Select "Safety parameters are reset".
5. Press the "Start" button.
6. Enter the password for the safety functions.
7. Confirm that the parameters have been saved (RAM to ROM).
8. Go offline.
9. Switch off the inverter supply voltage.
10. Wait until all LEDs on the inverter go dark.
11. Switch on the inverter supply voltage again.



You have restored the safety functions in the inverter to the factory settings.

Exception: The password for the safety functions is not reset.



Safety functions password (Page 234)

Procedure with an operator panel

Proceed as follows to restore the inverter safety functions to the factory settings:

1. p0010 = 30Set
Activate reset settings.
2. p9761 = ...
Enter the password for the safety functions
3. Start the reset with p0970 = 5.
4. Wait until the inverter sets p0970 = 0.
5. Set p0971 = 1.

6. Wait until the inverter sets $p0971 = 0$.
7. Switch off the inverter supply voltage.
8. Wait until all LEDs on the inverter go dark.
9. Switch on the inverter supply voltage again.



You have restored the safety function settings of your inverter to the factory settings.

5.6.2 Restore the factory settings (without safety functions)

Restore the factory inverter settings

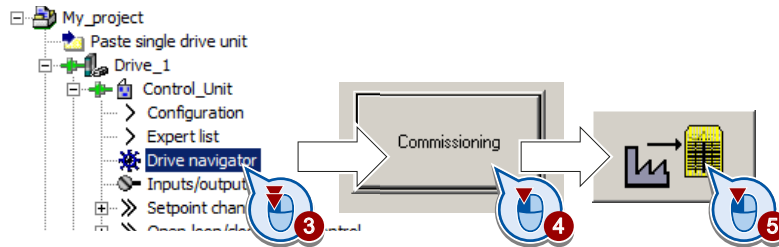


1
2

Procedure with STARTER

Proceed as follows to reset the inverter to factory settings:

1. Select your drive.
2. Go online.
3. Open "Drive Navigator".



4. Select the "Commissioning" button.
5. Press the "Factory setting" button.
6. ☒ In the screen form, select "After loading copy RAM to ROM".
7. Start the reset.
8. Wait until the inverter has been reset to the factory setting.



You have reset the inverter to factory settings.



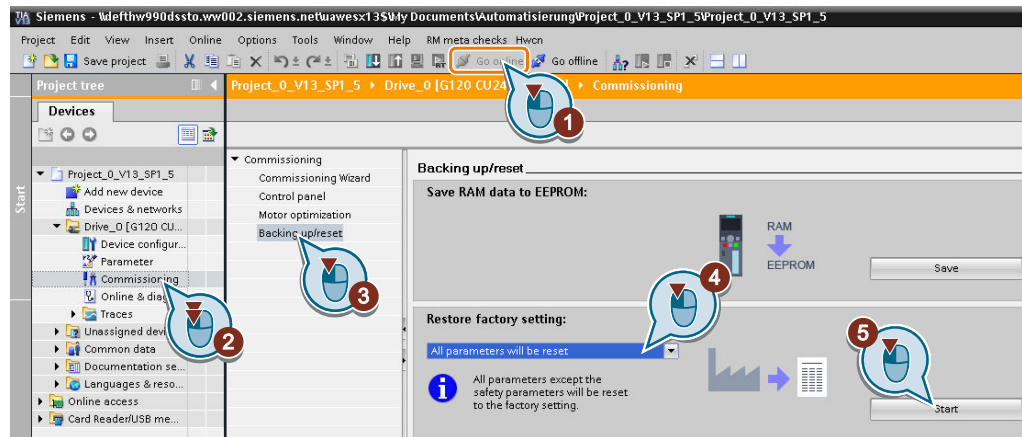
1
2

Procedure with Startdrive

Proceed as follows to reset the inverter to factory settings:

1. Go online.
2. Select "Commissioning".
3. Select "Backing up/reset".
4. Select "All parameters are reset".

5. Press the "Start" button.



6. Wait until the inverter has been reset to the factory setting.

■ You have reset the inverter to factory settings.

Procedure with the BOP-2 operator panel



Proceed as follows to reset the inverter to factory settings:

1. In the "Options" menu, select the "DRVRESET" entry
2. Confirm the reset using the OK key.
3. Wait until the inverter has been reset to the factory setting.

■ You have reset the inverter to factory settings.

Advanced commissioning

6.1 Overview of the inverter functions

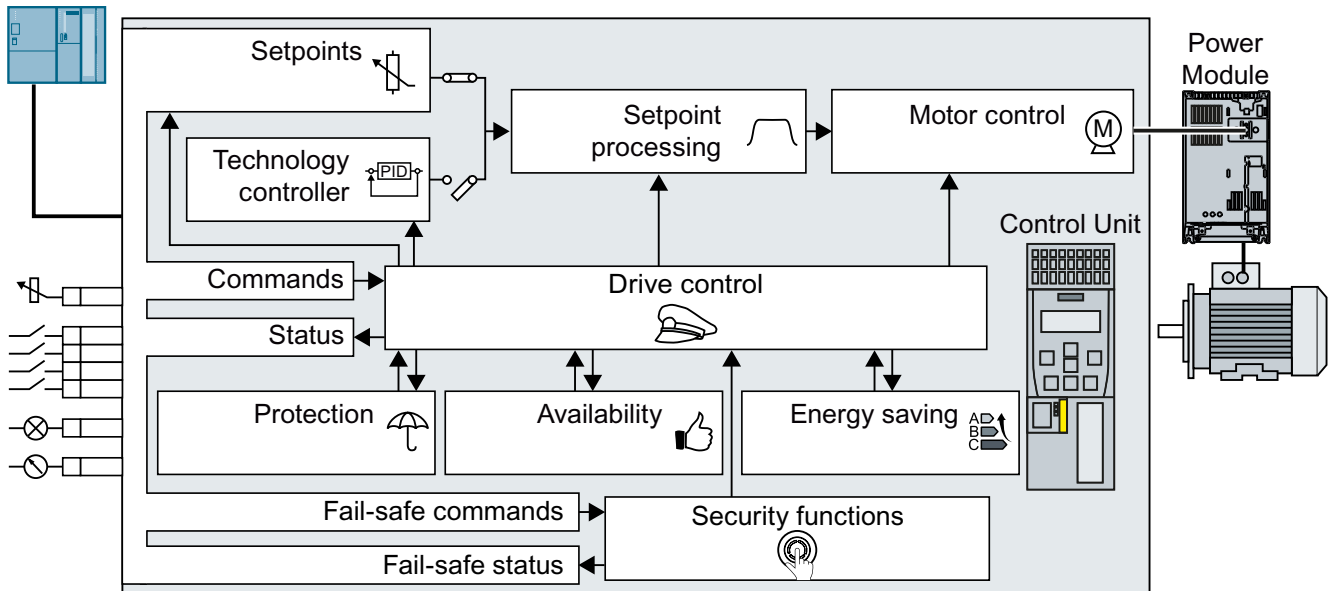


Figure 6-1 Overview of inverter functions

Drive control



The inverter receives its commands from the higher-level control via the terminal strip or the fieldbus interface of the Control Unit. The drive control defines how the inverter responds to the commands.

- Sequence control when switching the motor on and off (Page 176)
- Adapt the default setting of the terminal strip (Page 178)
- Controlling clockwise and counter-clockwise rotation via digital inputs (Page 190)
- Drive control via PROFIBUS or PROFINET (Page 196)
- Drive control via Modbus RTU (Page 212)
- Drive control via USS (Page 215)
- Drive control via Ethernet/IP (Page 218)
- Jogging (Page 219)

The inverter can switch between different settings of the drive control.

- Switching over the drive control (command data set) (Page 220)

The inverter provides a motor holding brake control. The motor holding brake holds the motor in position when it is switched off.



Motor holding brake (Page 222)

The free function blocks permit configurable signal processing within the inverter.



Free function blocks (Page 226)

You can select in which physical units the inverter represents its associated values.



Selecting physical units (Page 227)

Security functions



The safety functions fulfill increased requirements regarding the functional safety of the drive.



Safe Torque Off (STO) safety function (Page 231)

The extended safety functions monitor the drive speed.

The extended safety functions are described in the "Safety Integrated" function manual.



Overview of the manuals (Page 504)

Setpoints and setpoint processing



The setpoint normally defines the motor speed.



Setpoints (Page 251)



The setpoint processing uses a ramp-function generator to prevent speed steps occurring and to limit the speed to a permissible maximum value.



Setpoint calculation (Page 259)

Technology controller



The technology controller controls process variables, e.g. pressure, temperature, level or flow. The motor control receives the setpoint either from the higher-level control or from the technology controller.



PID technology controller (Page 269)

Motor control



The motor closed-loop control ensures that the motor follows the speed setpoint. You can choose between various control modes.



Motor control (Page 275)

The inverter provides several methods to brake the motor electrically. During electrical braking, the motor develops a torque that reduces the speed to standstill.



Electrically braking the motor (Page 302)

Protection of the drive and the driven load



The protection functions prevent damage to the motor, inverter and driven load.



Overcurrent protection (Page 312)

-  Inverter protection using temperature monitoring (Page 313)
-  Motor protection with temperature sensor (Page 316)
-  Motor protection by calculating the temperature (Page 319)
-  Motor and inverter protection by limiting the voltage (Page 322)

The monitoring of the driven load prevents impermissible operating modes, e.g. dry-running of a pump.

-  Monitoring the driven load (Page 324)

Increasing the drive availability



The kinetic buffering converts the kinetic energy of the load into electrical energy to buffer short-term power failures.

-  Kinetic buffering (V_{dc} min control) (Page 339)

The "Flying restart" function permits the fault-free switching on of the motor while it is still turning.

-  Flying restart – switching on while the motor is running (Page 333)

For active automatic restart, after a power failure, the inverter attempts to independently restart the motor and to acknowledge any faults that occur.

-  Automatic restart (Page 335)

Energy saving



If necessary, the main contactor control disconnects the inverter from the power system and so reduces the inverter losses.

-  Line contactor control (Page 341)

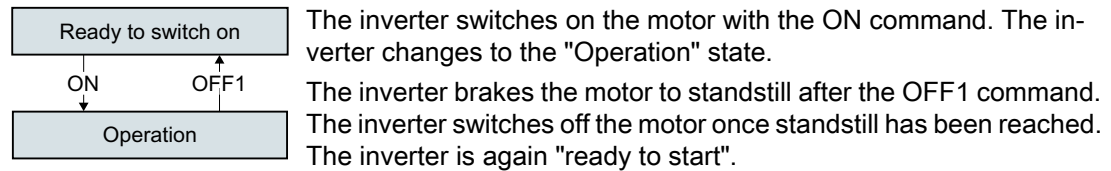
The inverter calculates the amount of energy the controlled inverter operation saves compared with a mechanical flowrate control.

-  Calculating the energy saving for fluid flow machines (Page 343)

6.2 Sequence control when switching the motor on and off



After switching the supply voltage on, the inverter normally goes into the "ready to start" state. In this state, the inverter waits for the command to switch on the motor:



Inverter states and commands for switching the motor on and off

In addition to ON/OFF1, "Ready to switch on" and "Ready", there are no additional inverter states and commands to switch on and switch off the motor:

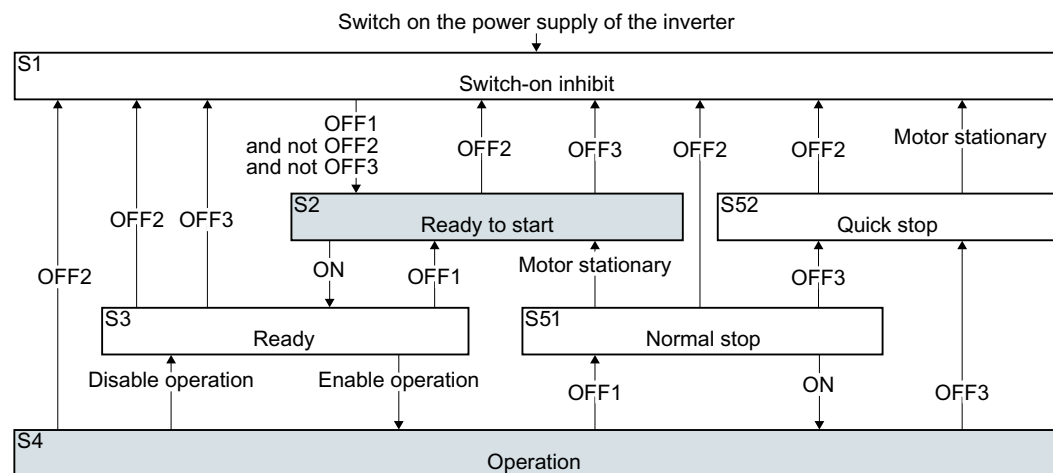


Figure 6-2 Internal sequence control of the inverter when the motor is switched on and off

Table 6-1 Commands for switching the motor on and off

OFF2	The inverter switches off the motor immediately without first braking it.
OFF3	The inverter changes from the "Operation" state to the "Quick stop" state. During "Quick stop", the inverter brakes the motor with the OFF3 ramp-down time. The inverter switches off the motor once standstill has been reached. The command is frequently used for exceptional operating situations where it is necessary to brake the motor very fast, e.g. for collision protection.
Disable operation	The inverter switches the motor OFF.
Enable operation	The inverter switches the motor ON.

Table 6-2 Inverter states

S1	In this state, the inverter does not respond to the ON command. The inverter goes into this state under the following conditions: • ON was active when switching on the inverter. Exception: When the automatic start function is active, ON must be active after switching on the power supply. • OFF2 or OFF3 is selected.
S2	The prerequisites for switching the motor on are satisfied.
S3	The inverter waits for "enable operation".
S4	The motor is switched on.
S51	The inverter brakes the motor with the ramp-down time of the ramp-function generator.
S52	The inverter brakes the motor with the OFF3 ramp-down time.

The abbreviations S1 ... S5b to identify the inverter states are defined in the PROFIdrive profile.

6.3 Adapt the default setting of the terminal strip



In the inverter, the input and output signals are interconnected with specific inverter functions using special parameters. The following parameters are available to interconnect signals:

- Binectors BI and BO are parameters to interconnect binary signals.
- Connectors CI and CO are parameters to interconnect analog signals.

This chapter describes how you adapt the function of individual inverter inputs and outputs using binectors and connectors.

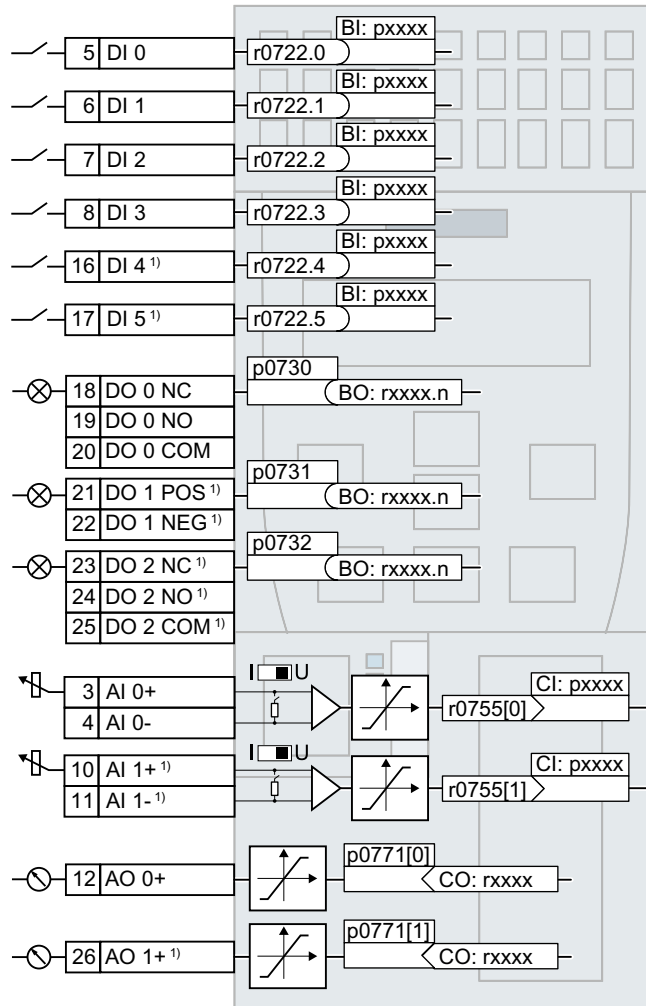
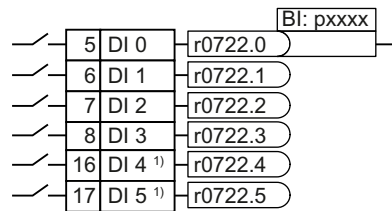


Figure 6-3 Internal interconnection of the inputs and outputs

6.3.1 Digital inputs

Changing the function of a digital input



To change the function of a digital input, you must interconnect the status parameter of the digital input with a binector input of your choice.

Binector inputs are marked with "BI" in the parameter list of the List Manual.

 Interconnecting signals in the inverter (Page 492)

¹⁾Not available with Control Units CU240B-2 and CU240B-2 DP

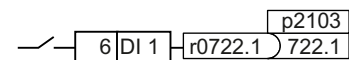
Table 6-3 Binector inputs (BI) of the inverter (selection)

BI	Significance	BI	Significance
p0810	Command data set selection CDS bit 0	p1055	Jog bit 0
p0840	ON/OFF1	p1056	Jog bit 1
p0844	OFF2	p1113	Setpoint inversion
p0848	OFF3	p1201	Flying restart enable signal source
p0852	Enable operation	p2103	1. Acknowledge faults
p1020	Fixed speed setpoint selection bit 0	p2106	External fault 1
p1021	Fixed speed setpoint selection bit 1	p2112	External alarm 1
p1022	Fixed speed setpoint selection bit 2	p2200	Technology controller enable
p1023	Fixed speed setpoint selection bit 3	p3330	Two/three-wire control, control command 1
p1035	Motorized potentiometer setpoint higher	p3331	Two/three-wire control, control command 2
p1036	Motorized potentiometer setpoint lower	p3332	Two/three-wire control, control command 3

A complete list of the binector outputs is provided in the List Manual.

 Overview of the manuals (Page 504)

Changing the function of a digital input - Example



To acknowledge inverter fault messages, using digital input DI 1 you must interconnect DI 1 with the command to acknowledge faults (p2103).

Set p2103 = 722.1.

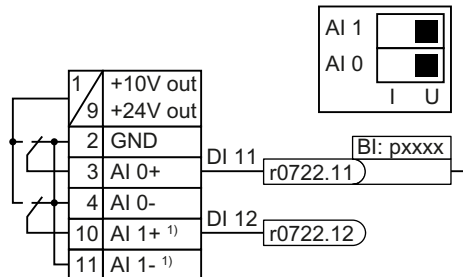
Advanced settings

You can debounce the digital input signal using parameter p0724.

For more information, please see the parameter list and the function block diagrams 2220 f of the List Manual.

 Overview of the manuals (Page 504)

Analog inputs as digital inputs



To use an analog input as additional digital input, you must interconnect the corresponding status parameter r0722.11 or r0722.12 with a binector input of your choice.

You may operate the analog input as digital input with 10 V or with 24 V.

¹ Not available with CU240B-2 and CU240B-2 DP Control Units

NOTICE

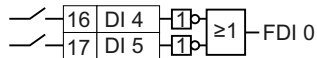
Destruction of the analog input as a result of the excessively high input current

The 10 V or 24 V voltage source will destroy the analog input if the operating mode selector switch is set to "Current input" (I).

- Set the operating mode selector switch of the analog input to voltage (U).

Defining the safety-related input

You require a safety-related input to activate a safety function via the inverter terminal strip.



The inverter combines two digital inputs to create a safety-related input.

You can find additional information on safety-related inputs in the description of the STO safety function.

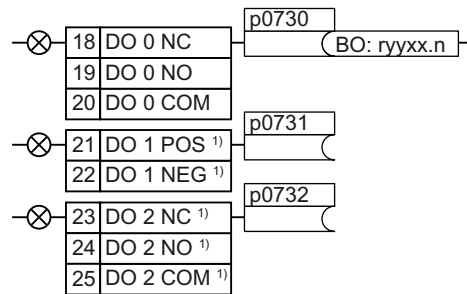
 Safe Torque Off (STO) safety function (Page 231)

Note

Control Units CU240B-2 and CU240B-2 DP do not have a safety-relevant digital input.

6.3.2 Digital outputs

Changing the function of a digital output



To change the function of a digital output, you must interconnect the digital output with a binector output of your choice.

Binector outputs are marked with "BO" in the parameter list of the List Manual.

 Interconnecting signals in the inverter (Page 492)

¹⁾ Not available with Control Units CU240B-2 and CU240B-2 DP

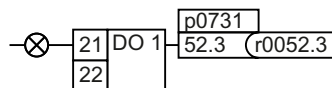
Table 6-4 Frequently used binector outputs (BO) of the inverter

0	Deactivating digital output	r0052.08	0 signal: Deviation, setpoint/actual speed
r0052.00	1 signal: Ready for switching on	r0052.09	1 signal: Control requested
r0052.01	1 signal: Ready for operation	r0052.10	1 signal: Maximum speed (p1082) reached
r0052.02	1 signal: Operation enabled	r0052.11	0 signal: I, M, P limit reached
r0052.03	1 signal: Fault active The inverter inverts signal r0052.03 if it is interconnected to a digital output.	r0052.13	0 signal: Alarm overtemperature motor
		r0052.14	1 signal: Motor CW rotation
r0052.04	0 signal: OFF2 active	r0052.15	0 signal: Alarm inverter overload
r0052.05	0 signal: OFF3 active	r0053.00	1 signal: DC braking active
r0052.06	1 signal: Closing lockout active	r0053.02	1 signal: Speed > minimum speed (p1080)
r0052.07	1 signal: Alarm active	r0053.06	1 signal: Speed ≥ setpoint speed (r1119)

The complete list of binector outputs is provided in the List Manual.

 Overview of the manuals (Page 504)

Changing the function of a digital output - Example



To output inverter fault messages via digital output DO 1, you must interconnect DO1 with these fault messages.

Set p0731 = 52.3

Advanced settings

You can invert the signal of the digital output using parameter p0748.

6.3 Adapt the default setting of the terminal strip

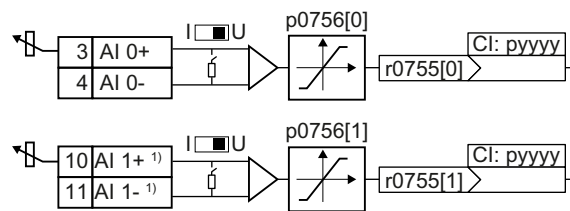
For more information, please see the parameter list and the function block diagrams 2230 f of the List Manual.



Overview of the manuals (Page 504)

6.3.3 Analog inputs

Overview



The parameter p0756[x] and the switch on the inverter specify the analog input type. Define the function of the analog input by interconnecting parameter p0755[x] with a connector input CI of your choice.



Interconnecting signals in the inverter (Page 492)

¹⁾ Not available with Control Units CU240B-2 and CU240B-2 DP

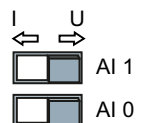
Define the analog input type

The variable speed drive offers a series of default settings, which you can select using parameter p0756:

AI 0	Unipolar voltage input	0 V ... +10 V	p0756[0] =	0
	Unipolar voltage input monitored	+2 V ... +10 V		1
	Unipolar current input	0 mA ... +20 mA		2
	Unipolar current input monitors	+4 mA ... +20 mA		3
	Bipolar voltage input	-10 V ... +10 V		4
	No sensor connected	---		8
AI 1	Unipolar voltage input	0 V ... +10 V	p0756[1] =	0
	Unipolar voltage input monitored	+2 V ... +10 V		1
	Unipolar current input	0 mA ... +20 mA		2
	Unipolar current input monitors	+4 mA ... +20 mA		3
	Bipolar voltage input	-10 V ... +10 V		4
	No sensor connected	---		8

In addition, you must also set the switch associated with the analog input. You can find the switch on the Control Unit behind the lower front door.

- Voltage input: Switch position U (factory setting)
- Current input: Switch position I



Characteristics

If you change the analog input type using p0756, then the inverter automatically selects the appropriate scaling of the analog input. The linear scaling characteristic is defined using two points (p0757, p0758) and (p0759, p0760). Parameters p0757 ... p0760 are assigned to an analog input via their index, e.g. parameters p0757[0] ... p0760[0] belong to analog input 0.

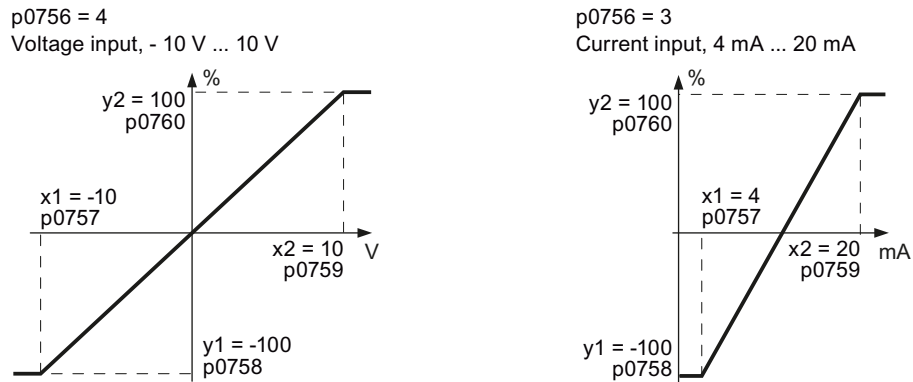


Figure 6-4 Examples for scaling characteristics

Parameter	Description
p0757	x coordinate of the 1st point along the characteristic [p0756 defines the unit]
p0758	y coordinate of the 1st Characteristic point [% of p200x] p200x are the parameters of reference variables, e.g. p2000 is the reference speed
p0759	x coordinate of the 2nd point along the characteristic [p0756 defines the unit]
p0760	y coordinate of the 2nd Characteristic point [% of p200x]
p0761	Wire breakage monitoring response threshold

Adapting the characteristic

You must define your own characteristic if none of the default types match your particular application.

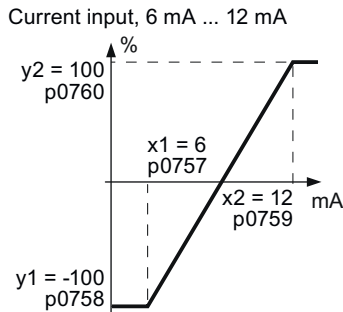
Example

The inverter should convert a 6 mA ... 12 mA signal into the value range -100 % ... 100 % via analog input 0. The wire break monitoring of the inverter should respond when 6 mA is fallen below.

Precondition

You have set analog input 0 as a current input ("I") via the DIP switch on the Control Unit.



Procedure

Set the following parameters to set the analog input as current input with monitoring:

1. set p0756[0] = 3
You have defined analog input 0 as current input with wire break monitoring.
2. Set p0757[0] = 6.0 (x1)
3. Set p0758[0] = -100.0 (y1)
4. Set p0759[0] = 12.0 (x2)
5. Set p0760[0] = 100.0 (y2)
6. Set p0761[0] = 6
An input current < 6 mA results in fault F03505.

Defining the function of an analog input

You define the analog input function by interconnecting a connector input of your choice with parameter p0755. Parameter p0755 is assigned to the particular analog input based on its index, e.g. parameter p0755[0] is assigned to analog input 0.

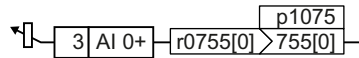
Table 6-5 Frequently used connector inputs (CI) of the inverter

CI	Significance	CI	Significance
p1070	Main setpoint	p2253	Technology controller setpoint 1
p1075	Supplementary setpoint	p2264	Technology controller actual value

A complete list of the connector inputs is provided in the List Manual.



Overview of the manuals (Page 504)

Defining the function of an analog input - example

In order to enter the supplementary setpoint via analog input AI 0, you must interconnect AI 0 with the signal source for the supplementary setpoint.

Set p1075 = 755[0].

Advanced settings**Signal smoothing**

When required, you can smooth the signal, which you read-in via an analog input, using parameter p0753.

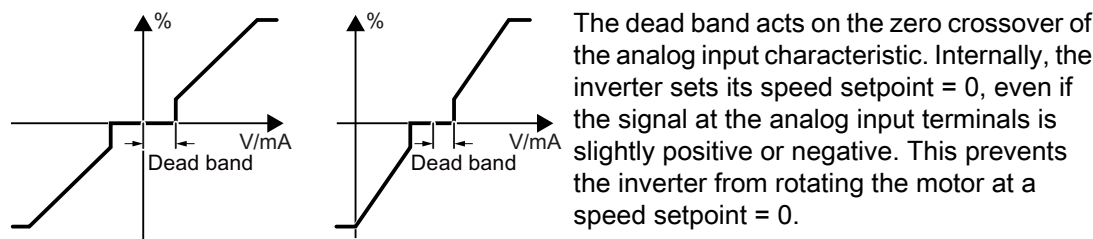
You can find more information in the parameter list and in function diagram 2251 of the List Manual.



Overview of the manuals (Page 504)

Dead band

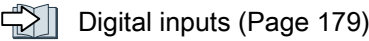
With the control enabled, electromagnetic interference on the signal cable can cause the motor to slowly rotate in one direction in spite of a speed setpoint = 0.



p0764[0]	Analog inputs dead zone, AI 0 (factory setting: 0)
p0764[1]	Analog inputs dead zone, AI 1 (factory setting: 0)

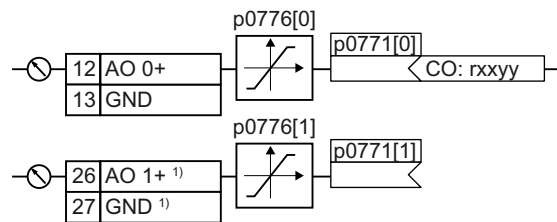
Using an analog input as digital input

An analog input can also be used as digital input.



6.3.4 Analog outputs

Overview



Define the analog output type using parameter p0776.

You define the analog output function by interconnecting parameter p0771 with a connector output CO of your choice.

Connector outputs are marked with "CO" in the parameter list of the List Manual.

 Interconnecting signals in the inverter (Page 492)

¹⁾ Not available with Control Units CU240B-2 and CU240B-2 DP

Defining the analog output type

The inverter offers a series of default settings, which you can select using parameter p0776:

AO 0	Current output (factory setting)	0 mA ... +20 mA	p0776[0] =	0
	Voltage output	0 V ... +10 V		1
	Current output	+4 mA ... +20 mA		2
AO 1	Current output (factory setting)	0 mA ... +20 mA	p0776[1] =	0
	Voltage output	0 V ... +10 V		1
	Current output	+4 mA ... +20 mA		2

Characteristics

If you change the analog output type, then the inverter automatically selects the appropriate scaling of the analog output. The linear scaling characteristic is defined using two points (p0777, p0778) and (p0779, p0780).

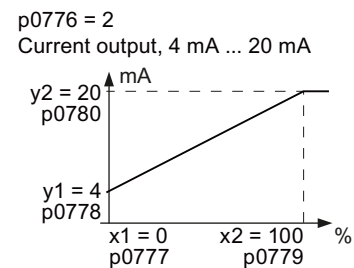
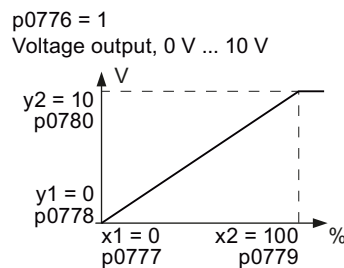


Figure 6-5 Examples for scaling characteristics

6.3 Adapt the default setting of the terminal strip

Parameters p0777 ... p0780 are assigned to an analog output via their index, e.g. parameters p0777[0] ... p0770[0] belong to analog output 0.

Table 6-6 Parameters for the scaling characteristic

Parameter	Description
p0777	x coordinate of the 1st Characteristic point [% of p200x] p200x are the parameters of the reference variables, e.g. p2000 is the reference speed.
p0778	y coordinate of the 1st Characteristic point [V or mA]
p0779	x coordinate of the 2nd Characteristic point [% of p200x]
p0780	y coordinate of the 2nd Characteristic point [V or mA]

Setting the characteristic

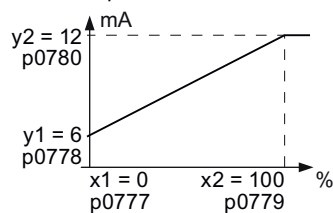
You must define your own characteristic if none of the default types match your particular application.

Example:

Via analog output 0, the inverter should convert a signal in the value range 0 % ... 100 % into an output signal 6 mA ... 12 mA.

Procedure

Current output, 6 mA ... 12 mA



Set the following parameters to set the characteristic to match the example:

1. Set p0776[0] = 2
This defines analog output 0 as a current output.
2. Set p0777[0] = 0.0 (x1)
3. Set p0778[0] = 6.0 (y1)
4. Set p0779[0] = 100.0 (x2)
5. Set p0780[0] = 12.0 (y2)

Defining the function of an analog output

You define the analog output function by interconnecting parameter p0771 with a connector output of your choice. Parameter p0771 is assigned to the particular analog output via its index, e.g. parameter p0771[0] is assigned to analog output 0.

Table 6-7 Connector outputs (CO) of the inverter (selection)

CO	Meaning	CO	Meaning
r0021	Actual speed smoothed	r0026	DC link voltage smoothed
r0024	Output frequency, smoothed	r0027	Absolute actual current smoothed
r0025	Output voltage smoothed		

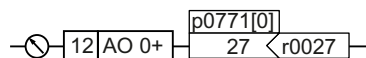
A complete list of the connector outputs is provided in the List Manual.

Additional information is provided in the parameter list and in function diagram 2261 of the List Manual.



Overview of the manuals (Page 504)

Defining the function of an analog output - example



To output the inverter output current via analog output 0, you must interconnect AO 0 with the signal for the output current. Set p0771 = 27.

Advanced settings

You can manipulate the signal that you output via an analog output, as follows:

- Absolute-value generation of the signal (p0775)
- Signal inversion (p0782)

Additional information is provided in the parameter list of the List Manual.

6.4 Controlling clockwise and counter-clockwise rotation via digital inputs



The inverter has a different methods for controlling the motor using two or three commands.

Overview

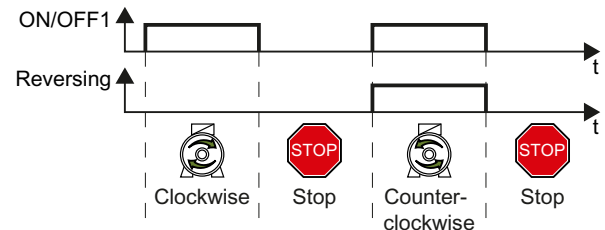
Two wire control, method 1

ON/OFF1:

Switches the motor on or off

Reversing:

Reverses the motor direction of rotation



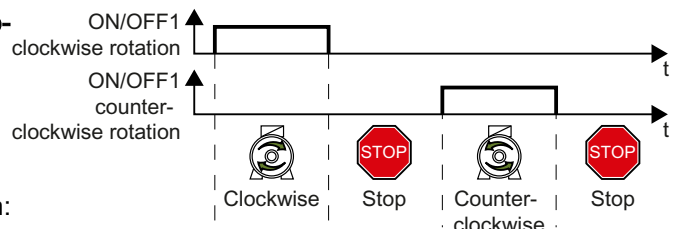
Two wire control, method 2 - and two-wire control, method 3

ON/OFF1 clockwise rotation:

Switches the motor on or off, clockwise rotation

ON/OFF1 counter-clockwise rotation:

Switches the motor on or off, counter-clockwise rotation



Three wire control, method 1

Enable/OFF1:

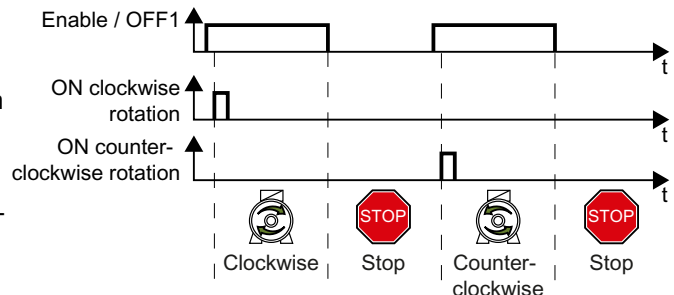
Enables the motor to be switched on or switched off

ON clockwise rotation:

Switches on the motor, clockwise rotation

ON counter-clockwise rotation:

Switches on the motor, counter-clockwise rotation



Three wire control, method 2

Enable/OFF1:

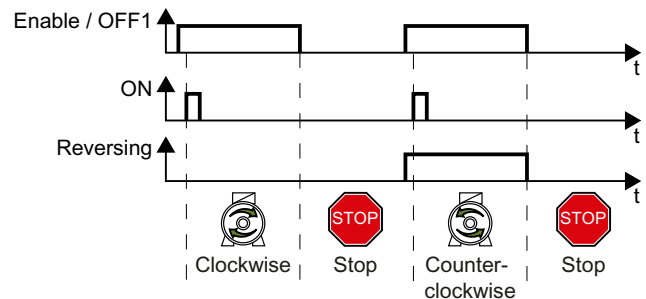
Enables the motor to be switched on or switched off

ON:

Switches on the motor

Reversing:

Reverses the motor direction of rotation



6.4.1 Two-wire control, method 1

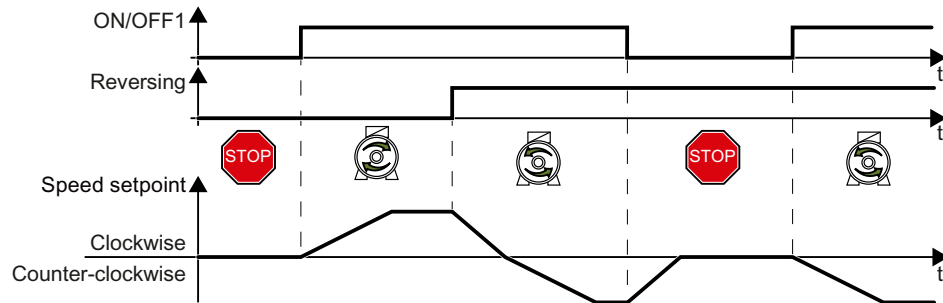


Figure 6-6 Two-wire control, method 1

Command "ON/OFF1" switches the motor on and off. The "Reversing" command inverts the motor direction of rotation.

Table 6-8 Function table

ON/OFF1	Reversing	Function
0	0	OFF1: The motor stops
0	1	
1	0	ON: Clockwise motor rotation
1	1	ON: Counter-clockwise motor rotation

Table 6-9 Select two-wire control, method 1

Parameter	Description
p0015 = 12	Macro drive unit You must carry out quick commissioning in order to set parameter p0015. Assigning digital inputs DI to the commands: DI 0: ON/OFF1 DI 1: Reversing

Table 6-10 Changing the assignment of the digital inputs

Parameter	Description
p0840[0 ... n] = 722.x	BI: ON/OFF1 (ON/OFF1) Example: p0840 = 722.3 $\Rightarrow$ DI 3: ON/OFF1
p1113[0 ... n] = 722.x	BI: Setpoint inversion (reversing)

6.4.2 Two-wire control, method 2

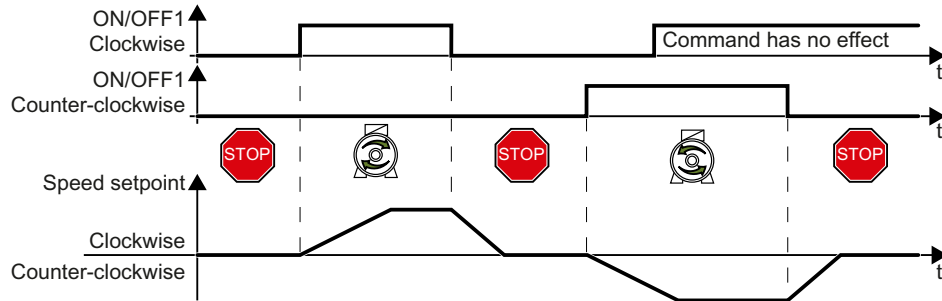


Figure 6-7 Two-wire control, method 2

Commands "ON/OFF1 clockwise rotation" and "ON/OFF1 counter-clockwise rotation" switch on the motor - and simultaneously select a direction of rotation. The inverter only accepts a new command when the motor is at a standstill.

Table 6-11 Function table

ON/OFF1 clockwise rotation	ON/OFF1 counter-clockwise rotation	Function
0	0	OFF1: The motor stops.
1	0	ON: Clockwise motor rotation.
0	1	ON: Counter-clockwise motor rotation.
1	1	ON: The motor direction of rotation is defined by the command that first reaches state "1".

Table 6-12 Select two-wire control, method 2

Parameter	Description
p0015 = 12	Macro drive unit You must carry out quick commissioning in order to set parameter p0015. Assigning digital inputs DI to the commands: DI 0: ON/OFF1 clockwise rotation DI 1: ON/OFF1 counter-clockwise rotation

Table 6-13 Changing the assignment of the digital inputs

Parameter	Description
p3330[0 ... n] = 722.x	BI: 2/3 wire control command 1 (ON/OFF1 clockwise rotation)
p3331[0 ... n] = 722.x	BI: 2/3 wire control command 2 (ON/OFF1 counter-clockwise rotation) Example: p3331 = 722.0 ⇒ DI 0: ON/OFF1 counter-clockwise rotation

6.4.3 Two-wire control, method 3

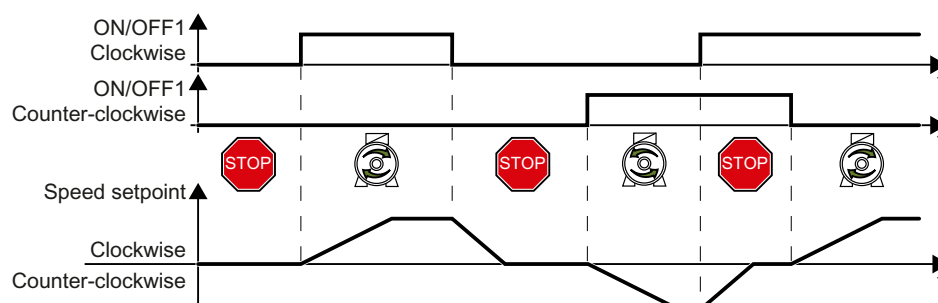


Figure 6-8 Two-wire control, method 3

Commands "ON/OFF1 clockwise rotation" and "ON/OFF1 counter-clockwise rotation" switch on the motor - and simultaneously select a direction of rotation. The inverter accepts a new command at any time, independent of the motor speed.

Table 6-14 Function table

ON/OFF1 clockwise rotation	ON/OFF1 counter-clockwise rotation	Function
0	0	OFF1: The motor stops.
1	0	ON: Clockwise motor rotation.
0	1	ON: Counter-clockwise motor rotation.
1	1	OFF1: The motor stops.

Table 6-15 Select two-wire control, method 3

Parameter	Description
p0015 = 18	Macro drive unit You must carry out quick commissioning in order to set parameter p0015. Assigning digital inputs DI to the commands: DI 0: ON/OFF1 clockwise rotation DI 1: ON/OFF1 counter-clockwise rotation

Table 6-16 Changing the assignment of the digital inputs

Parameter	Description
p3330[0 ... n] = 722.x	BI: 2/3 wire control command 1 (ON/OFF1 clockwise rotation)
p3331[0 ... n] = 722.x	BI: 2/3 wire control command 2 (ON/OFF1 counter-clockwise rotation) Example: p3331 = 722.0 ⇒ DI 0: ON/OFF1 counter-clockwise rotation

6.4.4 Three-wire control, method 1

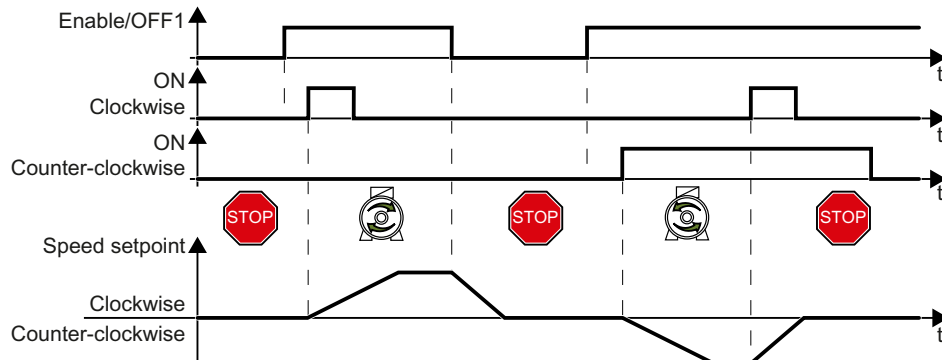


Figure 6-9 Three-wire control, method 1

The "Enable" command is a precondition for switching on the motor. Commands "ON clockwise rotation" and "ON counter-clockwise rotation" switch on the motor - and simultaneously select a direction of rotation. Removing the enable switches the motor off (OFF1).

Table 6-17 Function table

Enable / OFF1	ON clockwise rotation	ON counter-clockwise rotation	Function
0	0 or 1	0 or 1	OFF1: The motor stops.
1	0→1	0	ON: Clockwise motor rotation.
1	0	0→1	ON: Counter-clockwise motor rotation.
1	1	1	OFF1: The motor stops.

Table 6-18 Select three-wire control, method 1

Parameter	Description
p0015 = 19	Macro drive unit You must carry out quick commissioning in order to set parameter p0015. Assigning digital inputs DI to the commands: DI 0: Enable / OFF1 DI 1: ON clockwise rotation DI 2: ON counter-clockwise rotation

Table 6-19 Changing the assignment of the digital inputs

Parameter	Description
p3330[0 ... n] = 722.x	BI: 2/3 wire control command 1 (enable/OFF1)
p3331[0 ... n] = 722.x	BI: 2/3 wire control command 2 (ON clockwise rotation)
p3332[0 ... n] = 722.x	BI: 2/3 wire control command 3 (ON counter-clockwise rotation) Example: p3332 = 722.0 ⇒ DI 0: ON counter-clockwise rotation

6.4.5 Three-wire control, method 2

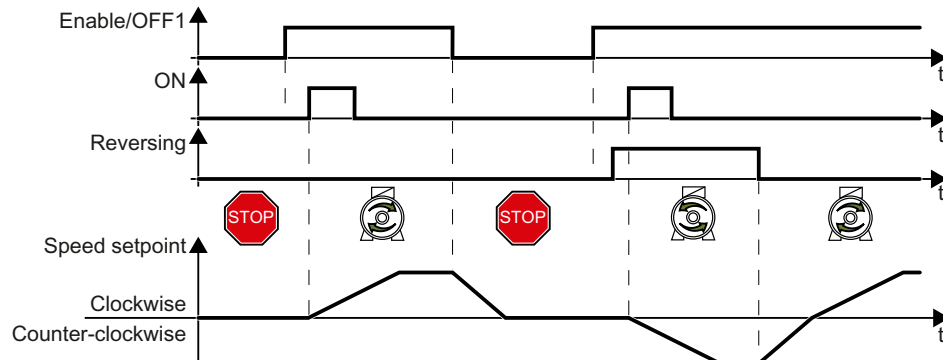


Figure 6-10 Three-wire control, method 2

The "Enable" command is a precondition for switching on the motor. The "ON" command switches the motor on. The "Reversing" command inverts the motor direction of rotation. Removing the enable switches the motor off (OFF1).

Table 6-20 Function table

Enable / OFF1	ON	Reversing	Function
0	0 or 1	0 or 1	OFF1: The motor stops.
1	0→1	0	ON: Clockwise motor rotation.
1	0→1	1	ON: Counter-clockwise motor rotation.

Table 6-21 Select three-wire control, method 2

Parameter	Description
p0015 = 20	Macro drive unit You must carry out quick commissioning in order to set parameter p0015. Assigning digital inputs DI to the commands: DI 0: Enable / OFF1 DI 1: ON DI 2: Reversing

Table 6-22 Changing the assignment of the digital inputs

Parameter	Description
p3330[0 ... n] = 722.x	BI: 2/3 wire control command 1 (enable/OFF1)
p3331[0 ... n] = 722.x	BI: 2/3 wire control command 2 (ON) Example: p3331 = 722.0 ⇒ DI 0: ON command
p3332[0 ... n] = 722.x	BI: 2/3 wire control command 3 (reversing)

6.5 Drive control via PROFIBUS or PROFINET



The send and receive telegrams of the inverter for the cyclic communication are structured as follows:

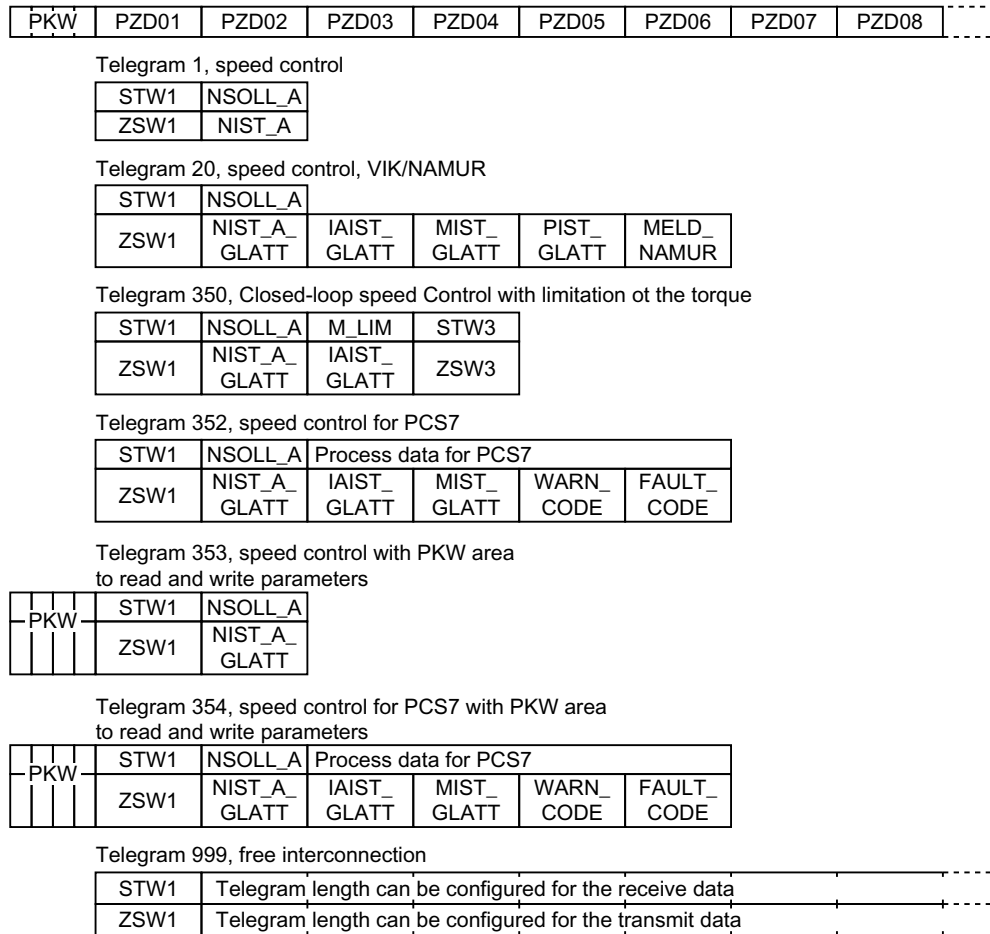


Figure 6-11 Telegrams for cyclic communication

Table 6-23 Explanation of the abbreviations

Abbreviation	Explanation	Abbreviation	Explanation
STW	Control word	MIST_GLATT	Actual smoothed torque
ZSW	Status word	PIST_GLATT	Actual smoothed active power
NSOLL_A	Speed setpoint	M_LIM	Torque limit value
NIST_A	Speed actual value	FAULT_CODE	Fault number
NIST_A_GLATT	Smoothed actual speed value	WARN_CODE	Alarm number
IAIST_GLATT	Smoothed current actual value	MELD_NAMUR	Control word according to the VIK-NAMUR definition

Interconnection of the process data

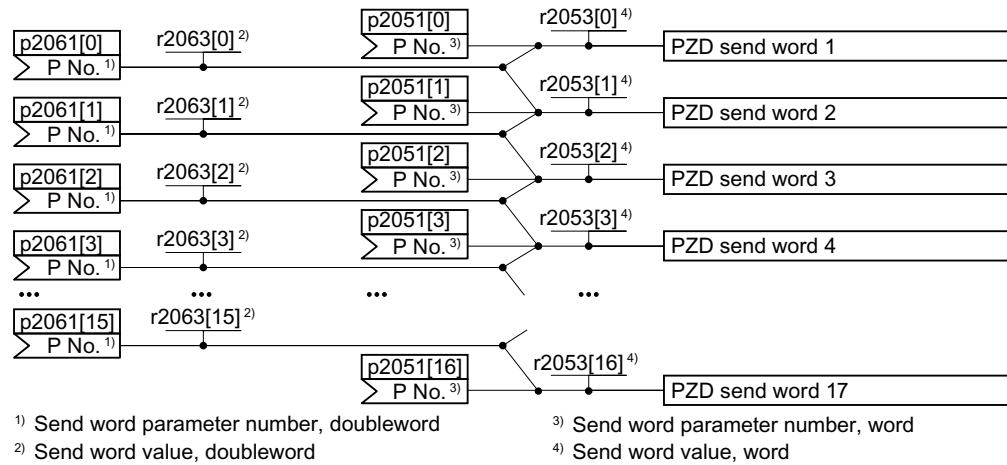


Figure 6-12 Interconnection of the send words

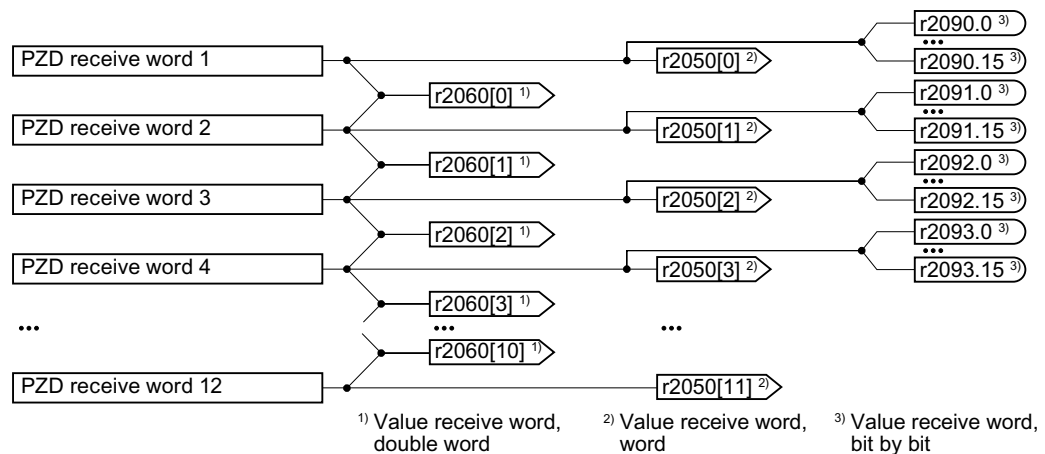


Figure 6-13 Interconnection of the receive words

The telegrams use - with the exception of telegram 999 (free interconnection) - the word-by-word transfer of send and receive data (r2050/p2051).

If you require an individual telegram for your application (e.g. for transferring double words), you can adjust one of the predefined telegrams via parameters p0922 and p2079. For details, please refer to the List Manual, function diagrams 2420 and 2472.

6.5.1 Control and status word 1

Control word 1 (STW1)

Bit	Significance		Explanation	Signal inter-connection in the inverter
	Telegram 20	All other telegrams		
0	0 = OFF1		The motor brakes with the ramp-down time p1121 of the ramp-function generator. The inverter switches off the motor at standstill.	p0840[0] = r2090.0
	0 → 1 = ON		The inverter goes into the "ready" state. If, in addition bit 3 = 1, then the inverter switches on the motor.	
1	0 = OFF2		Switch off the motor immediately, the motor then coasts down to a standstill.	p0844[0] = r2090.1
	1 = No OFF2		The motor can be switched on (ON command).	
2	0 = Quick stop (OFF3)		Quick stop: The motor brakes with the OFF3 ramp-down time p1135 down to standstill.	p0848[0] = r2090.2
	1 = No quick stop (OFF3)		The motor can be switched on (ON command).	
3	0 = Inhibit operation		Immediately switch-off motor (cancel pulses).	p0852[0] = r2090.3
	1 = Enable operation		Switch-on motor (pulses can be enabled).	
4	0 = Disable RFG		The inverter immediately sets its ramp-function generator output to 0.	p1140[0] = r2090.4
	1 = Do not disable RFG		The ramp-function generator can be enabled.	
5	0 = Stop RFG		The output of the ramp-function generator stops at the actual value.	p1141[0] = r2090.5
	1 = Enable RFG		The output of the ramp-function generator follows the setpoint.	
6	0 = Inhibit setpoint		The inverter brakes the motor with the ramp-down time p1121 of the ramp-function generator.	p1142[0] = r2090.6
	1 = Enable setpoint		Motor accelerates with the ramp-up time p1120 to the setpoint.	
7	0 → 1 = Acknowledge faults		Acknowledge fault. If the ON command is still active, the inverter switches to "closing lockout" state.	p2103[0] = r2090.7
8, 9	Reserved			
10	0 = No control via PLC		Inverter ignores the process data from the fieldbus.	p0854[0] = r2090.10
	1 = Control via PLC		Control via fieldbus, inverter accepts the process data from the fieldbus.	
11	1 = Direction reversal		Invert setpoint in the inverter.	p1113[0] = r2090.11
12	Not used			
13	--- ¹⁾	1 = MOP up	Increase the setpoint saved in the motorized potentiometer.	p1035[0] = r2090.13

Bit	Significance		Explanation	Signal inter-connection in the inverter
	Telegram 20	All other telegrams		
14	--- ¹⁾	1 = MOP down	Reduce the setpoint saved in the motorized potentiometer.	p1036[0] = r2090.14
15	CDS bit 0	Reserved	Changes over between settings for different operation interfaces (command data sets).	p0810 = r2090.15

¹⁾ If you change over from another telegram to telegram 20, then the assignment of the previous telegram is kept.

Status word 1 (ZSW1)

Bit	Significance		Comments	Signal inter-connection in the inverter
	Telegram 20	All other telegrams		
0	1 = Ready to start		Power supply switched on; electronics initialized; pulses locked.	p2080[0] = r0899.0
1	1 = Ready		Motor is switched on (ON/OFF1 = 1), no fault is active. With the command "Enable operation" (STW1.3), the inverter switches on the motor.	p2080[1] = r0899.1
2	1 = Operation enabled		Motor follows setpoint. See control word 1, bit 3.	p2080[2] = r0899.2
3	1 = Fault active		The inverter has a fault. Acknowledge fault using STW1.7.	p2080[3] = r2139.3
4	1 = OFF2 inactive		Coast down to standstill is not active.	p2080[4] = r0899.4
5	1 = OFF3 inactive		Quick stop is not active.	p2080[5] = r0899.5
6	1 = Closing lockout active		It is only possible to switch on the motor after an OFF1 followed by ON.	p2080[6] = r0899.6
7	1 = Alarm active		Motor remains switched on; no acknowledgement is necessary.	p2080[7] = r2139.7
8	1 = Speed deviation within the tolerance range		Setpoint / actual value deviation within the tolerance range.	p2080[8] = r2197.7
9	1 = Master control requested		The automation system is requested to accept the inverter control.	p2080[9] = r0899.9
10	1 = Comparison speed reached or exceeded		Speed is greater than or equal to the corresponding maximum speed.	p2080[10] = r2199.1
11	1 = current or torque limit reached	1 = torque limit reached	Comparison value for current or torque has been reached or exceeded.	p2080[11] = r0056.13 / r1407.7
12	--- ¹⁾	1 = Holding brake open	Signal to open and close a motor holding brake.	p2080[12] = r0899.12
13	0 = Alarm, motor overtemperature		--	p2080[13] = r2135.14

Bit	Significance		Comments	Signal inter-connection in the inverter
	Telegram 20	All other telegrams		
14	1 = Motor rotates clockwise		Internal inverter actual value > 0	p2080[14] = r2197.3
	0 = Motor rotates counterclockwise		Internal inverter actual value < 0	
15	1 = CDS display	0 = Alarm, inverter thermal overload		p2080[15] = r0836.0 / r2135.15

- ¹⁾ If you change over from another telegram to telegram 20, then the assignment of the previous telegram is kept.

6.5.2 Control and status word 3

Control word 3 (STW3)

Bit	Significance	Explanation	Signal interconnection in the inverter ¹⁾
	Telegram 350		
0	1 = fixed setpoint bit 0	Selects up to 16 different fixed setpoints.	p1020[0] = r2093.0
1	1 = fixed setpoint bit 1		p1021[0] = r2093.1
2	1 = fixed setpoint bit 2		p1022[0] = r2093.2
3	1 = fixed setpoint bit 3		p1023[0] = r2093.3
4	1 = DDS selection bit 0	Changes over between settings for different motors (drive data sets).	p0820 = r2093.4
5	1 = DDS selection bit 1		p0821 = r2093.5
6	Not used		
7	Not used		
8	1 = technology controller enable	--	p2200[0] = r2093.8
9	1 = enable DC braking	--	p1230[0] = r2093.9
10	Not used		
11	1 = Enable droop	Enable or inhibit speed controller droop.	p1492[0] = r2093.11
12	1 = torque control active 0 = speed control active	Changes over the control mode for vector control.	p1501[0] = r2093.12
13	1 = no external fault 0 = external fault is active (F07860)	--	p2106[0] = r2093.13
14	Not used		
15	1 = CDS bit 1	Changes over between settings for different operation interfaces (command data sets).	p0811[0] = r2093.15

¹⁾ If you switch from telegram 350 to a different one, then the inverter sets all interconnections p1020, ... to "0". Exception: p2106 = 1.

Status word 3 (ZSW3)

Bit	Significance	Description	Signal interconnection in the inverter
0	1 = DC braking active	--	p2051[3] = r0053
1	1 = $ n_{act} > p1226$	Absolute current speed > stationary state detection	
2	1 = $ n_{act} > p1080$	Absolute actual speed > minimum speed	
3	1 = $i_{act} \geq p2170$	Actual current $\geq$ current threshold value	
4	1 = $ n_{act} > p2155$	Absolute actual speed > speed threshold value 2	
5	1 = $ n_{act} \leq p2155$	Absolute actual speed < speed threshold value 2	
6	1 = $ n_{act} \geq r1119$	Speed setpoint reached	
7	1 = DC link voltage $\leq p2172$	Actual DC link voltage $\leq$ threshold value	
8	1 = DC link voltage > p2172	Actual DC link voltage > threshold value	
9	1 = ramp-up or ramp-down completed	Ramp-function generator is not active.	
10	1 = technology controller output at the lower limit	Technology controller output $\leq p2292$	
11	1 = technology controller output at the upper limit	Technology controller output > p2291	
12	Not used		
13	Not used		
14	Not used		
15	Not used		

6.5.3 NAMUR message word

Fault word according to the VIK-NAMUR definition (MELD_NAMUR)

Table 6-24 Fault word according to the VIK-NAMUR definition and interconnection with parameters in the inverter

Bit	Significance	P no.
0	1 = Control Unit signals a fault	p2051[5] = r3113
1	1 = line fault: Phase failure or inadmissible voltage	
2	1 = DC link overvoltage	
3	1 = Power Module fault, e.g. overcurrent or overtemperature	
4	1 = inverter overtemperature	
5	1 = ground fault/phase fault in the motor cable or in the motor	
6	1 = motor overload	
7	1 = communication error to the higher-level control system	
8	1 = fault in a safety-relevant monitoring channel	
10	1 = fault in the internal inverter communication	
11	1 = line fault	
15	1 = other fault	

6.5.4 Data structure of the parameter channel

Structure of the parameter channel

The parameter channel consists of four words. 1. and 2nd word transfer the parameter number and index as well as the type of job (read or write) The 3rd and 4th word contains the parameter contents. The parameter contents can be 16-bit values (such as baud rate) or 32-bit values (e.g. CO parameters).

Bit 11 in the 1st word is reserved and is always assigned 0.

Parameter channel					
PKE (1st word)			IND (2nd word)		PWE (3rd and 4th words)
15 ... 12	11	10 ... 0	15 ... 8	7 ... 0	15 ... 0
AK	S	PNU	Subindex	Page index	PWE 1
	P				PWE 2
	M				

You can find examples of telegrams at the end of this section.

AK: Request and response IDs

Bits 12 ... 15 of the 1st The parameter channel words contain the request and response identifier AK.

Table 6-25 Request identifiers, control → inverter

AK	Description	Response identifier	
		positive	negative
0	No request	0	7 / 8
1	Request parameter value	1 / 2	7 / 8
2	Change parameter value (word)	1	7 / 8
3	Change parameter value (double word)	2	7 / 8
4	Request descriptive element ¹⁾	3	7 / 8
6 ²⁾	Request parameter value (field) ¹⁾	4 / 5	7 / 8
7 ²⁾	Change parameter value (field, word) ¹⁾	4	7 / 8
8 ²⁾	Change parameter value (field, double word) ¹⁾	5	7 / 8
9	Request number of field elements	6	7 / 8

¹⁾ The required element of the parameter is specified in IND (2nd word).

²⁾ The following request IDs are identical: 1 ≡ 6, 2 ≡ 7 3 ≡ 8.

We recommend that you use identifiers 6, 7, and 8.

Table 6-26 Response identifiers, inverter → control

AK	Description
0	No response
1	Transfer parameter value (word)
2	Transfer parameter value (double word)

AK	Description
3	Transfer descriptive element ¹⁾
4	Transfer parameter value (field, word) ²⁾
5	Transfer parameter value (field, double word) ²⁾
6	Transfer number of field elements
7	Inverter cannot process the request. In the most significant word of the parameter channel, the inverter sends an error number to the control, refer to the following table.
8	No master controller status / no authorization to change parameters of the parameter channel interface

¹⁾ The required element of the parameter is specified in IND (2nd word).

²⁾ The required element of the indexed parameter is specified in IND (2nd word).

Table 6-27 Error numbers for response identifier 7

No.	Description
00 hex	Illegal parameter number (access to a parameter that does not exist)
01 hex	Parameter value cannot be changed (change request for a parameter value that cannot be changed)
02 hex	Lower or upper value limit exceeded (change request with a value outside the value limits)
03 hex	Incorrect subindex (access to a subindex that does not exist)
04 hex	No array (access with a subindex to non-indexed parameters)
05 hex	Incorrect data type (change request with a value that does not match the data type of the parameter)
06 hex	Setting not permitted, only resetting (change request with a value not equal to 0 without permission)
07 hex	Descriptive element cannot be changed (change request to a descriptive element error value that cannot be changed)
0B hex	No master control (change request but with no master control, see also p0927.)
0C hex	Keyword missing
11 hex	Request cannot be executed due to the operating state (access is not possible for temporary reasons that are not specified)
14 hex	Inadmissible value (change request with a value that is within the limits but which is illegal for other permanent reasons, i.e. a parameter with defined individual values)
65 hex	Parameter number is currently deactivated (depending on the mode of the inverter)
66 hex	Channel width is insufficient (communication channel is too small for response)
68 hex	Illegal parameter value (parameter can only assume certain values)
6A hex	Request not included / task is not supported (the valid request identifications can be found in table "Request identifications controller → inverter")
6B hex	No change access for a controller that is enabled. (The operating state of the inverter prevents a parameter change)
86 hex	Write access only for commissioning (p0010 = 15) (operating state of the inverter prevents a parameter change)
87 hex	Know-how protection active, access locked
C8 hex	Change request below the currently valid limit (change request to a value that lies within the "absolute" limits, but is however below the currently valid lower limit)

No.	Description
C9 hex	Change request above the currently valid limit (example: a parameter value is too large for the inverter power)
CC hex	Change request not permitted (change is not permitted as the access code is not available)

PNU (parameter number) and page index

The parameter number is located in value PNU in the 1st word of the parameter channel (PKE).

The page index is located in the 2nd word of the parameter channel (IND bit 7 ... 0).

Parameter number	PNU	Page index
0000 ... 1999	0000 ... 1999	0 hex
2000 ... 3999	0000 ... 1999	80 hex
6000 ... 7999	0000 ... 1999	90 hex
8000 ... 9999	0000 ... 1999	20 hex
10000 ... 11999	0000 ... 1999	A0 hex
20000 ... 21999	0000 ... 1999	50 hex
30000 ... 31999	0000 ... 1999	F0 hex
60000 ... 61999	0000 ... 1999	74 hex

Subindex

For indexed parameters, the parameter index is located in subindex (IND Bit 15 ... 8) as hexadecimal value.

PWE: Parameter value or connector

Parameter values or connectors can be located in the PWE.

Table 6-28 Parameter value or connector

	PWE 1	PWE 2	
Parameter value	Bit 15 ... 0	Bit 15 ... 8	Bit 7 ... 0
	0	0	8-bit value
	0	16-bit value	
	32-bit value		
Connector	Bit 15 ... 0	Bit 15 ... 10	Bit 9 ... 0
	Number of the connector	3F hex	The index or bit field number of the connector

6.5.5 Examples for using the parameter channel

Read request: Read out serial number of the Power Module (p7841[2])

To obtain the value of the indexed parameter p7841, you must fill the telegram of the parameter channel with the following data:

- **PKE, bit 12 ... 15 (AK): = 6** (request parameter value (field))
- **PKE, bit 0 ... 10 (PNU): = 1841** (Parameter number without offset)
Parameter number = PNU + offset (page index)
(7841 = 1841 + 6000)
- **IND, bit 8 ... 15 (subindex): = 2** (Index of the parameter)
- **IND, bit 0 ... 7 (page index): = 90 hex** (offset 6000 $\pm$ 90 hex)
- Because you want to read the parameter value, words 3 and 4 in the parameter channel for requesting the parameter value are irrelevant. They should be assigned a value of 0, for example.

Parameter channel																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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0	1	1	0	0	1	1	1	0	0	1	1	0	0	0	1	0	0	0	0	0	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 6-14 Telegram for a read request from p7841[2]

Write request: Change restart mode (p1210)

The restart mode is inhibited in the factory setting (p1210 = 0). In order to activate the automatic restart with "acknowledge all faults and restart for an ON command", p1210 must be set to 26:

- **PKE, bit 12 ... 15 (AK): = 7** (change parameter value (field, word))
- **PKE, bit 0 ... 10 (PNU): = 4BA hex** (1210 = 4BA hex, no offset, as 1210 < 1999)
- **IND, bit 8 ... 15 (subindex): = 0 hex** (parameter is not indexed)
- **IND, bit 0 ... 7 (page index): = 0 hex** (offset 0 corresponds to 0 hex)
- **PWE1, bit 0 ... 15: = 0 hex**
- **PWE2, bit 0 ... 15: = 1A hex** (26 = 1A hex)

Parameter channel																																									
PKE, 1st word								IND, 2nd word								PWE1 - high, 3rd word								PWE2 - low, 4th word																	
15 ... 12		11	10 ... 0						15 ... 8				7 ... 0				15 ... 0								15 ... 0																
AK			Parameter number						Subindex				Page index				Parameter value (bit 16 ... 31)								Parameter value (bit 0 ... 15)																
0	1	1	1	0	1	0	0	1	0	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0

Figure 6-15 Telegram, to activate the automatic restart with p1210 = 26

Write request: Assign digital input 2 with the function ON/OFF1 (p0840[1] = 722.2)

In order to link digital input 2 with ON/OFF1, you must assign parameter p0840[1] (source, ON/OFF1) the value 722.2 (DI 2). To do this, you must fill the telegram of the parameter channel as follows:

- **PKE, bit 12 ... 15 (AK): = 7 hex** (change, parameter value (field, word))
- **PKE, bit 0 ... 10 (PNU): = 348 hex** (840 = 348 hex, no offset, as 840 < 1999)

- IND, bit 8 ... 15 (subindex): = 1 hex (CDS1 = index1)
- IND, bit 0 ... 7 (page index): = 0 hex (offset 0 $\triangleq$ 0 hex)
- PWE1, bit 0 ... 15: = 2D2 hex (722 = 2D2 hex)
- PWE2, bit 10 ... 15: = 3F hex (drive object - for SINAMICS G120, always 63 = 3f hex)
- PWE2, bit 0 ... 9: = 2 hex (index of parameter (DI 2 = 2))

Parameter channel																																																						
PKE, 1st word								IND, 2nd word								PWE1 - high, 3rd word								PWE2 - low, 4th word																														
15 ... 12				11				10 ... 0								15 ... 8				7 ... 0				15 ... 0								15 ... 10				9 ... 0																		
AK								Parameter number								Subindex				Page index				Parameter value								Drive Object				Index																		
0	1	1	1	0	0	1	1	0	1	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	1	0	1	0	1	1	1	1	1	0	0	0	0	0	0	0	1	0

Figure 6-16 Telegram, to assign DI 2 with ON/OFF1

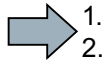
6.5.6 Extend telegrams and change signal interconnection

When you have selected a telegram, the inverter interconnects the corresponding signals with the fieldbus interface. Generally, these interconnections are protected so that they cannot be changed. With the appropriate inverter settings, these interconnections can be changed.

Extend telegram

Every telegram can be extended, by "attaching" additional signals.

Procedure



1. Proceed as follows to extend a telegram:
2.
 1. Using STARTER or an operator panel, set parameter p0922 = 999.
 2. Set parameter p2079 to the appropriate value of the corresponding telegram.
 3. Interconnect additional PZD send words and PZD receive words with signals of your choice via parameters r2050 and p2051.



You have extended the telegram.

Parameter	Description
p0922	PROFIdrive telegram selection
	999: Free telegram (message frame) configuration
p2079	PROFIdrive PZD telegram selection extended
	1: Standard telegram 1, PZD-2/2
	20: Standard telegram 20, PZD-2/6
	350: SIEMENS telegram 350, PZD-4/4
	352: SIEMENS telegram 352, PZD-6/6
	353: SIEMENS telegram 353, PZD-2/2, PKW-4/4
	354: SIEMENS telegram 354, PZD-6/6, PKW-4/4
r2050[0...11]	PROFIdrive PZD receive word Connector output to interconnect the PZD (setpoints) in the word format received from the PROFIdrive controller.
p2051[0...16]	PROFIdrive PZD send word Selection of the PZD (actual values) in the word format to be sent to the PROFIdrive controller.

Selection of the PZD (actual values) in the word format to be sent to the PROFIdrive controller. For further information refer to the function block diagrams 2468 and 2470 of the List Manual.

Freely selecting the signal interconnection of the telegram

The signals in the telegram can be freely interconnected.

Procedure

1. Proceed as follows to change the signal interconnection of a telegram:
2.
 1. Using STARTER or an operator panel, set parameter p0922 = 999.
 2. Using STARTER or an operator panel, set parameter p2079 = 999.
 3. Interconnect additional PZD send words and PZD receive words with signals of your choice via parameters r2050 and p2051.



You have freely interconnected the signals transferred in the telegram.

Parameter	Description
p0922	PROFIdrive telegram selection
	999: Free telegram (message frame) configuration
p2079	PROFIdrive PZD telegram selection extended
	999: Free telegram (message frame) configuration
r2050[0...11]	PROFIdrive PZD receive word Connector output to interconnect the PZD (setpoints) in the word format received from the PROFIdrive controller.
p2051[0...16]	PROFIdrive PZD send word Selection of the PZD (actual values) in the word format to be sent to the PROFIdrive controller.

For further information refer to the function block diagrams 2468 and 2470 of the List Manual.

6.5.7 Slave-to-slave communication

"Direct data exchange" is sometimes called "slave-to-slave communication" or "data exchange broadcast". With direct data exchange, slaves exchange data without any direct involvement of the master.

Further information about the "Direct data exchange" function is provided in the Fieldbus function manual.



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6.5.8 Acyclically reading and writing inverter parameters

The inverter supports the writing and reading of parameters via acyclic communication:

- For PROFIBUS: Up to 240 bytes per write or read request via data set 47
- For PROFINET: Write or read requests via B02E hex and B02F hex

Further information about acyclic communication is provided in the Fieldbus function manual.



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"Reading and writing parameters" application example



Further information is provided in the Internet:

Application examples (<https://support.industry.siemens.com/cs/ww/en/view/29157692>)

6.6 Drive control via Modbus RTU



Modbus RTU is used to transfer cyclic process data and acyclic parameter data between precisely one master and up to 247 slaves. The inverter is always the slave, and sends data when requested to do so by the master. Slave-to-slave communication is not possible.

Settings for Modbus RTU

Parameter	Explanation		
p2020	Fieldbus interface baudrate (Factory setting: 7)	5: 4800 baud 6: 9600 baud 7: 19200 baud 8: 38400 baud 9: 57600 baud	10: 76800 baud 11: 93750 baud 12: 115200 baud 13: 187500 baud
p2021	Fieldbus interface address (Factory setting: 1) Valid addresses: 1 ... 247. The parameter is only active if address 0 is set at the Control Unit address switch. A change only becomes effective after the inverter power supply has been switched off and switched on again.		
p2024	Fieldbus interface times (Factory setting: [0] 1000 ms, [2] 0 ms)	[0] Maximum permissible telegram processing time of the Modbus slave [2] dead time between two telegrams	
r2029	Fieldbus interface error statistics	[0] number of error-free telegrams [1] number of rejected telegrams [2] number of framing errors [3] number of overrun errors	[4] number of parity errors [5] number of starting character errors [6] number of checksum errors [7] number of length errors
p2030 = 2	Fieldbus interface protocol selection: Modbus RTU		
p2031	Fieldbus interface Modbus parity (Factory setting: 2)	0: No parity 1: Odd parity 2: Even parity	
p2040	Fieldbus interface monitoring time (Factory setting: 100 ms) p2040 = 0: The monitoring is deactivated		

Control word 1 (STW1)

Bit	Meaning	Explanation	Signal inter-connection in the inverter
0	0 = OFF1	The motor brakes with the ramp-down time p1121 of the ramp-function generator. The inverter switches off the motor at standstill.	p0840[0] = r2090.0
	0 → 1 = ON	The inverter goes into the "ready" state. If, in addition bit 3 = 1, then the inverter switches on the motor.	

Bit	Meaning	Explanation	Signal inter-connection in the inverter
1	0 = OFF2	Switch off the motor immediately, the motor then coasts down to a standstill.	p0844[0] = r2090.1
	1 = No OFF2	The motor can be switched on (ON command).	
2	0 = Quick stop (OFF3)	Quick stop: The motor brakes with the OFF3 ramp-down time p1135 down to standstill.	p0848[0] = r2090.2
	1 = No quick stop (OFF3)	The motor can be switched on (ON command).	
3	0 = Inhibit operation	Immediately switch-off motor (cancel pulses).	p0852[0] = r2090.3
	1 = Enable operation	Switch-on motor (pulses can be enabled).	
4	0 = Disable RFG	The inverter immediately sets its ramp-function generator output to 0.	p1140[0] = r2090.4
	1 = Do not disable RFG	The ramp-function generator can be enabled.	
5	0 = Stop RFG	The output of the ramp-function generator stops at the actual value.	p1141[0] = r2090.5
	1 = Enable RFG	The output of the ramp-function generator follows the setpoint.	
6	0 = Inhibit setpoint	The inverter brakes the motor with the ramp-down time p1121 of the ramp-function generator.	p1142[0] = r2090.6
	1 = Enable setpoint	Motor accelerates with the ramp-up time p1120 to the setpoint.	
7	0 → 1 = Acknowledge faults	Acknowledge fault. If the ON command is still active, the inverter switches to the "switch-on inhibit" state.	p2103[0] = r2090.7
8, 9	Reserved		
10	0 = No control via PLC	Inverter ignores the process data from the fieldbus.	p0854[0] = r2090.10
	1 = Control via PLC	Control via fieldbus, inverter accepts the process data from the fieldbus.	
11	1 = Direction reversal	Invert setpoint in the inverter.	p1113[0] = r2090.11
12	Reserved		
13	1 = MOP up	Increase the setpoint saved in the motorized potentiometer.	p1035[0] = r2090.13
14	1 = MOP down	Reduce the setpoint saved in the motorized potentiometer.	p1036[0] = r2090.14
15	Reserved		

Status word 1 (ZSW1)

Bit	Meaning	Remarks	Signal inter-connection in the inverter
0	1 = Ready for switching on	Power supply switched on; electronics initialized; pulses locked.	p2080[0] = r0899.0
1	1 = Ready	Motor is switched on (ON/OFF1 = 1), no fault is active. With the command "Enable operation" (STW1.3), the inverter switches on the motor.	p2080[1] = r0899.1

Bit	Meaning	Remarks	Signal inter-connection in the inverter
2	1 = Operation enabled	Motor follows setpoint. See control word 1, bit 3.	p2080[2] = r0899.2
3	1 = Fault active	The inverter has a fault. Acknowledge fault using STW1.7.	p2080[3] = r2139.3
4	1 = OFF2 inactive	Coast down to standstill is not active.	p2080[4] = r0899.4
5	1 = OFF3 inactive	Quick stop is not active.	p2080[5] = r0899.5
6	1 = Switch-on inhibit active	It is only possible to switch on the motor after an OFF1 followed by ON.	p2080[6] = r0899.6
7	1 = Alarm active	Motor remains switched on; no acknowledgment is necessary.	p2080[7] = r2139.7
8	1 = Speed deviation within the tolerance range	Setpoint / actual value deviation within the tolerance range.	p2080[8] = r2197.7
9	1 = Master control requested	The automation system is requested to accept the inverter control.	p2080[9] = r0899.9
10	1 = Comparison speed reached or exceeded	Speed is greater than or equal to the corresponding maximum speed.	p2080[10] = r2199.1
11	1 = Torque limit not reached	Comparison value for current or torque has been fallen below.	p2080[11] = r0056.13 / r1407.7
12	Reserved		p2080[12] = r0899.12
13	0 = Alarm, motor overtemperature	--	p2080[13] = r2135.14
14	1 = Motor rotates clockwise	Internal inverter actual value > 0	p2080[14] = r2197.3
	0 = Motor rotates counter-clockwise	Internal inverter actual value < 0	
15	0 = Alarm, inverter thermal overload		p2080[15] = r2135.15

- ¹⁾ If you change over from another telegram to telegram 20, then the assignment of the previous telegram is kept.

Further information

Further information about Modbus RTU is provided in the "Fieldbus" function manual.



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6.7 Drive control via USS



USS is used to transfer cyclic process data and acyclic parameter data between precisely one master and up to 31 slaves. The inverter is always the slave, and sends data when requested to do so by the master. Slave-to-slave communication is not possible.

Settings for USS

Parameter	Explanation		
p2020	Fieldbus interface baudrate (Factory setting: 8)	4: 2400 baud 5: 4800 baud 6: 9600 baud 7: 19200 baud 8: 38400 baud	9: 57600 baud 10: 76800 baud 11: 93750 baud 12: 115200 baud 13: 187500 baud
p2021	Fieldbus interface address (Factory setting: 0) Valid addresses: 0 ... 30. The parameter is only active if address 0 is set at the Control Unit address switch. A change only becomes effective after the inverter power supply has been switched off and switched on again.		
p2022	Fieldbus interface USS PZD number (Factory setting: 2)		
p2023	Fieldbus interface USS PKW number (Factory setting: 127)	0: PKW 0 words 3: PKW 3 words 4: PKW 4 words 127: PKW variable	
p2024	Fieldbus interface times (Factory setting: [0] 1000 ms, [1] 0 ms, [2] 0 ms)	[0] Maximum permissible telegram processing time of the Modbus slave [1] Character delay time [2] dead time between two telegrams	
r2029	Fieldbus interface error statistics	[0] number of error-free telegrams [1] number of rejected telegrams [2] number of framing errors [3] number of overrun errors	[4] number of parity errors [5] number of starting character errors [6] number of checksum errors [7] number of length errors
p2030 = 1	Fieldbus interface protocol selection: USS		
p2031	Fieldbus interface Modbus parity (Factory setting: 2)	0: No parity 1: Odd parity 2: Even parity	
p2040	Fieldbus interface monitoring time (Factory setting: 100 ms) p2040 = 0: The monitoring is deactivated		

Control word 1 (STW1)

Bit	Meaning	Explanation	Signal inter-connection in the inverter
0	0 = OFF1	The motor brakes with the ramp-down time p1121 of the ramp-function generator. The inverter switches off the motor at standstill.	p0840[0] = r2090.0
	0 → 1 = ON	The inverter goes into the "ready" state. If, in addition bit 3 = 1, then the inverter switches on the motor.	
1	0 = OFF2	Switch off the motor immediately, the motor then coasts down to a standstill.	p0844[0] = r2090.1
	1 = No OFF2	The motor can be switched on (ON command).	
2	0 = Quick stop (OFF3)	Quick stop: The motor brakes with the OFF3 ramp-down time p1135 down to standstill.	p0848[0] = r2090.2
	1 = No quick stop (OFF3)	The motor can be switched on (ON command).	
3	0 = Inhibit operation	Immediately switch-off motor (cancel pulses).	p0852[0] = r2090.3
	1 = Enable operation	Switch-on motor (pulses can be enabled).	
4	0 = Disable RFG	The inverter immediately sets its ramp-function generator output to 0.	p1140[0] = r2090.4
	1 = Do not disable RFG	The ramp-function generator can be enabled.	
5	0 = Stop RFG	The output of the ramp-function generator stops at the actual value.	p1141[0] = r2090.5
	1 = Enable RFG	The output of the ramp-function generator follows the setpoint.	
6	0 = Inhibit setpoint	The inverter brakes the motor with the ramp-down time p1121 of the ramp-function generator.	p1142[0] = r2090.6
	1 = Enable setpoint	Motor accelerates with the ramp-up time p1120 to the setpoint.	
7	0 → 1 = Acknowledge faults	Acknowledge fault. If the ON command is still active, the inverter switches to the "switch-on inhibit" state.	p2103[0] = r2090.7
8, 9	Reserved		
10	0 = No control via PLC	Inverter ignores the process data from the fieldbus.	p0854[0] = r2090.10
	1 = Control via PLC	Control via fieldbus, inverter accepts the process data from the fieldbus.	
11	1 = Direction reversal	Invert setpoint in the inverter.	p1113[0] = r2090.11
12	Reserved		
13	1 = MOP up	Increase the setpoint saved in the motorized potentiometer.	p1035[0] = r2090.13
14	1 = MOP down	Reduce the setpoint saved in the motorized potentiometer.	p1036[0] = r2090.14
15	Reserved		

Status word 1 (ZSW1)

Bit	Meaning	Remarks	Signal inter-connection in the inverter
0	1 = Ready for switching on	Power supply switched on; electronics initialized; pulses locked.	p2080[0] = r0899.0
1	1 = Ready	Motor is switched on (ON/OFF1 = 1), no fault is active. With the command "Enable operation" (STW1.3), the inverter switches on the motor.	p2080[1] = r0899.1
2	1 = Operation enabled	Motor follows setpoint. See control word 1, bit 3.	p2080[2] = r0899.2
3	1 = Fault active	The inverter has a fault. Acknowledge fault using STW1.7.	p2080[3] = r2139.3
4	1 = OFF2 inactive	Coast down to standstill is not active.	p2080[4] = r0899.4
5	1 = OFF3 inactive	Quick stop is not active.	p2080[5] = r0899.5
6	1 = Switch-on inhibit active	It is only possible to switch on the motor after an OFF1 followed by ON.	p2080[6] = r0899.6
7	1 = Alarm active	Motor remains switched on; no acknowledgment is necessary.	p2080[7] = r2139.7
8	1 = Speed deviation within the tolerance range	Setpoint / actual value deviation within the tolerance range.	p2080[8] = r2197.7
9	1 = Master control requested	The automation system is requested to accept the inverter control.	p2080[9] = r0899.9
10	1 = Comparison speed reached or exceeded	Speed is greater than or equal to the corresponding maximum speed.	p2080[10] = r2199.1
11	1 = Torque limit not reached	Comparison value for current or torque has been fallen below.	p2080[11] = r0056.13 / r1407.7
12	Reserved		p2080[12] = r0899.12
13	0 = Alarm, motor overtemperature	--	p2080[13] = r2135.14
14	1 = Motor rotates clockwise	Internal inverter actual value > 0	p2080[14] = r2197.3
	0 = Motor rotates counter-clockwise	Internal inverter actual value < 0	
15	0 = Alarm, inverter thermal overload		p2080[15] = r2135.15

¹⁾ If you change over from another telegram to telegram 20, then the assignment of the previous telegram is kept.

Further information

Further information about USS is provided in the "Fieldbus" function manual.



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6.8 Drive control via Ethernet/IP



EtherNet/IP is an Ethernet-based fieldbus. EtherNet/IP is used to transfer cyclic process data as well as acyclic parameter data.

Settings for Ethernet/IP

Parameter	Explanation		
p2030 = 10	Fieldbus interface protocol selection: Ethernet/IP		
p8920	PN Name of Station		
p8921	PN IP address (Factory setting: 0)		
p8922	PN default gateway (factory setting: 0)		
p8923	PN Subnet Mask (Factory setting: 0)		
p8924	PN DHCP mode (Factory setting: 0)	0: DHCP off 2: DHCP on, identification based on MAC address 3: DHCP on, identification based on Name of Station	
p8925	PN interface configuration (Factory setting: 0)	0: No function 1: Reserved 2: Save the configuration and activate 3: Delete configuration	
p8980	Ethernet/IP profile (Factory setting: 0) A change only becomes effective after the inverter power supply has been switched off and switched on again.		0: SINAMICS 1: ODVA AC/DC
p8982	Ethernet/IP ODVA speed scaling (Factory setting: 128) A change only becomes effective after the inverter power supply has been switched off and switched on again.		
	123: 32	127: 2	131: 0.125
	124: 16	128: 1	132: 0.0625
	125: 8	129: 0.5	133: 0.03125
	126: 4	130: 0.25	

Further information

Further information about USS is provided in the "Fieldbus" function manual.

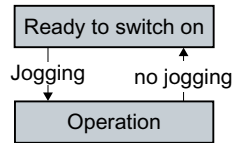


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6.9 Jogging



The "Jog" function is typically used to temporarily move a machine part using local control commands, e.g. a transport conveyor belt, bypassing the higher-level control.



Commands "Jog 1" or "Jog: 2" switch the motor on and off.

The commands are only active when the inverter is in the "Ready for switching on" state.

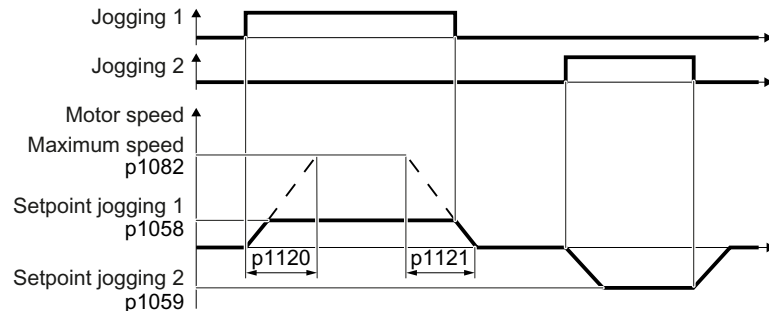


Figure 6-17 Behavior of the motor when "jogging"

After switching on, the motor accelerates to the setpoint, jog 1 or setpoint, jog 2. The two different setpoints can, for example, be assigned to motor clockwise and counter-clockwise rotation.

When jogging, the same ramp-function generator is active as for the ON/OFF1 command.

Jog settings

Parameter	Description	
p1058	Jogging 1 speed setpoint (factory setting 150 rpm)	
p1059	Jogging 2 speed setpoint (factory setting -150 rpm)	
p1082	Maximum speed (factory setting 1500 rpm)	
p1110	Inhibit negative direction	
	=0: Negative direction of rotation is enabled	=1: Negative direction of rotation is inhibited
p1111	Inhibit positive direction	
	=0: Positive direction of rotation is enabled	=1: Positive direction of rotation is inhibited
p1113	Setpoint inversion	
	=0: Setpoint is not inverted	=1: Setpoint is inverted
p1120	Ramp-function generator ramp-up time (factory setting 10 s)	
p1121	Ramp-function generator ramp-down time (factory setting 10 s)	
p1055 = 722.0	Jog bit 0: Select jogging 1 via digital input 0	
p1056 = 722.1	Jog bit 1: Select jogging 2 via digital input 1	

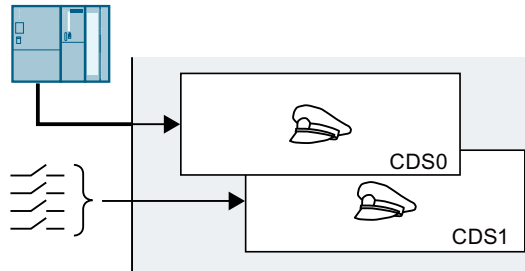
6.10 Switching over the drive control (command data set)



Several applications require the option of switching over the control authority to operate the inverter.

Example: The motor is to be operable either from a central control via the fieldbus or via the local digital inputs of the inverter.

Command data set (CDS)



This means that you can set the inverter control in various ways and toggle between the settings. For instance, as described above, the inverter can either be operated via a fieldbus or via its digital inputs.

The settings in the inverter, which are assigned to a specific master control, are termed the command data set.

You select the command data set using parameter p0810. To do this, you must interconnect parameter p0810 with a control command of your choice, e.g. a digital input.

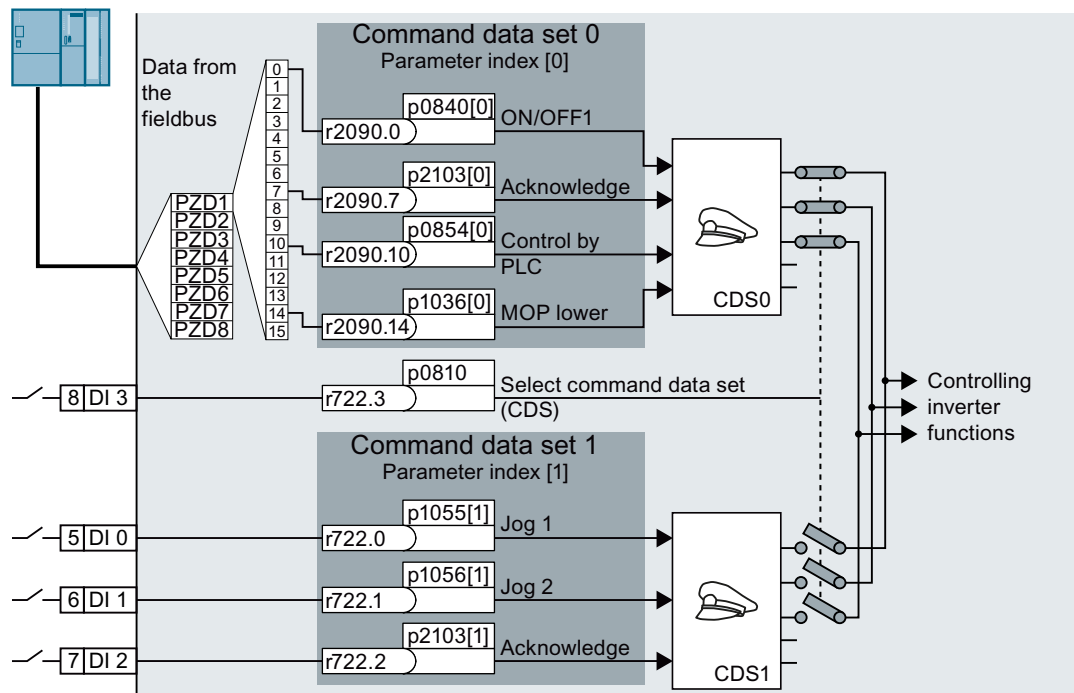


Figure 6-18 Example: Switching over the control via terminal strip to control via PROFIBUS or PROFINET

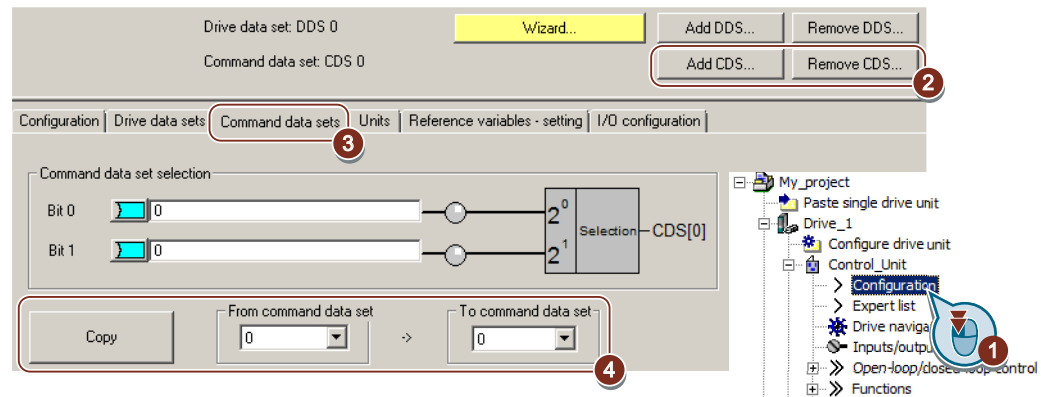
An overview of all the parameters that belong to the command data sets is provided in the List Manual.

Note

It takes approximately 4 ms to toggle between command data sets.

Advanced settings

To change the number of command data sets in STARTER, you must open your STARTER project offline.



- ① If, in the STARTER project tree, you select "Configuration", then you can process the command data sets.
- ② If you require more than two command data sets, then add command data sets using this button or remove them.
- ③, ④ To simplify commissioning several command data sets, under the "Command data sets" tab there is a copy function.

Figure 6-19 Editing command data sets in STARTER

Parameter	Description
p0010 = 15	Drive commissioning: Data sets
p0170	Number of command data sets (factory setting: 2) p0170 = 2, 3, or 4
p0010 = 0	Drive commissioning: Ready
r0050	Displaying the number of the CDS that is currently active
p0809[0]	Number of the command data set to be copied (source)
p0809[1]	Number of the command data set to which the data is to be copied (target)
p0809[2] = 1	Copying is started Once copying has been completed, the inverter sets p0809[2] to 0.
p0810	Command data set selection CDS bit 0
p0811	Command data set selection CDS bit 1
r0050	Displaying the number of the CDS that is currently active

6.11 Motor holding brake



The motor holding brake holds the motor in position when it is switched off. When correctly set, the inverter first switches on the motor and then opens the motor holding brake. When the motor is stationary (zero speed) the inverter closes the motor holding brake, and then switches off the motor.

Function

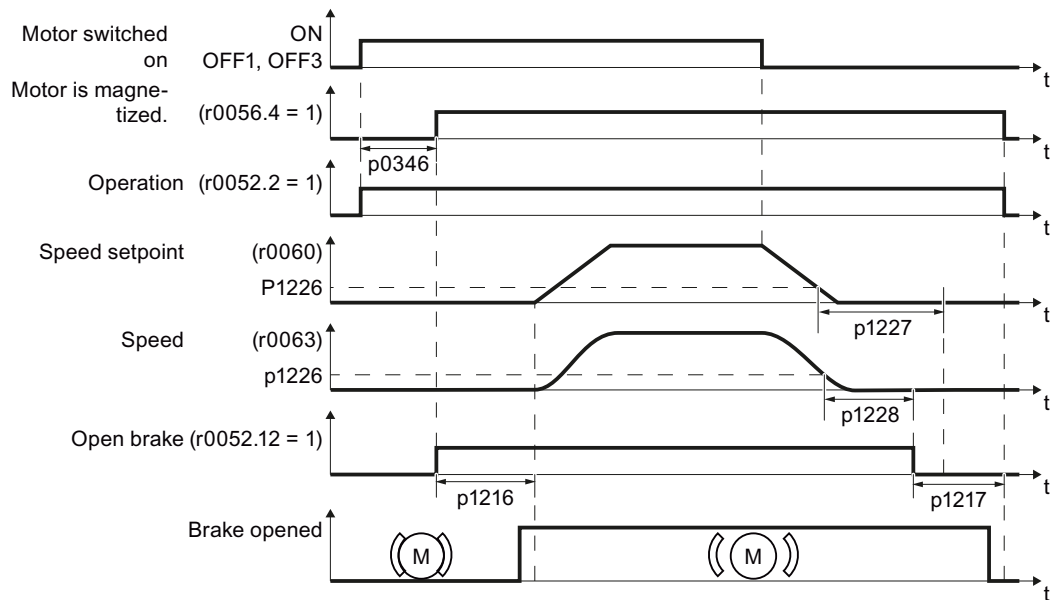


Figure 6-20 Motor holding brake function

Function after an ON command:

1. With the ON command, the inverter switches the motor on.
2. At the end of the "motor excitation build-up time" (p0346), the inverter issues the command to open the brake.
3. The inverter keeps the motor at a standstill until the "motor holding brake opening time" p1216 has ended.
The motor holding brake must be opened within time p1216.
4. The inverter accelerates the motor to the speed setpoint.

Function after an OFF1 or OFF3 command:

1. The inverter brakes the motor down to a standstill and switches it off using the OFF1 or OFF3 command.
2. When braking, the inverter compares the speed setpoint and the actual speed with the "standstill detection speed threshold" p1226:
 - Speed setpoint < p1226: The "standstill detection monitoring time" p1227 starts
 - Current speed < p1226: The "pulse cancellation deceleration time" p1228 starts

3. When the first of the two times (p1227 or p1228) has elapsed, the inverter issues the command to close the brake.
4. After the "motor holding brake closing time" p1217, the inverter switches off the motor. The motor holding brake must close within the time p1217.

Function after an OFF2 command

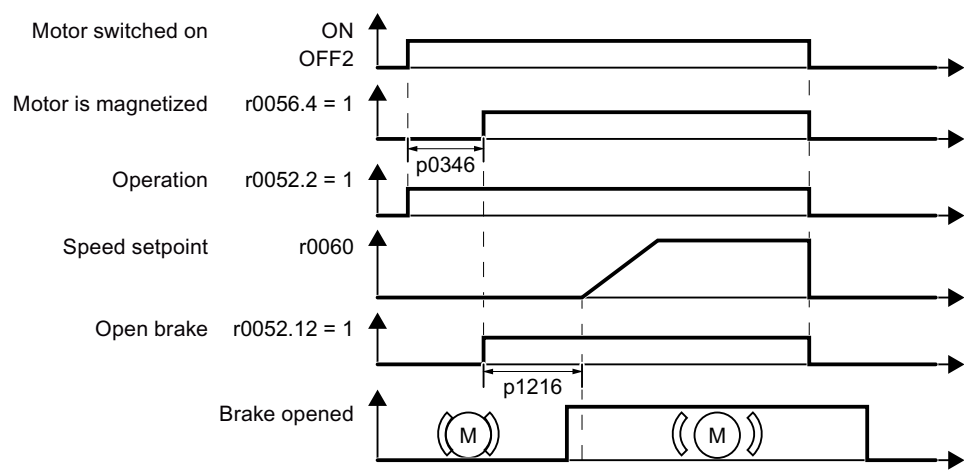


Figure 6-21 Controlling the motor holding brake after OFF2

After an OFF2 command, the inverter issues the signal to immediately close the motor holding brake, irrespective of the motor speed.

Commissioning a motor holding brake



! DANGER

Danger to life due to falling loads

For applications such as lifting equipment, cranes or elevators, there is a danger to life if the "Motor holding brake" function is incorrectly set.

- When commissioning the "Motor holding brake" function, secure any dangerous loads, e.g. by applying the following measures:
 - Lower the load down to the floor
 - Cordon off the hazardous area so that nobody can enter it

Precondition

The motor holding brake is connected to the inverter.

Procedure

To commission the "motor holding brake" function, proceed as follows:



1. Set p1215 = 1.
2. The "Motor holding brake" function" is enabled.
3. Check the magnetizing time p0346; the magnetizing time is pre-assigned during commissioning and must be greater than zero.
4. Find out the mechanical opening and closing times from the technical data of the motor holding brake.
 - Depending on the brake size, brake opening times lie between 25 ms and 500 ms.
 - Depending on the brake size, brake closing times lie between 15 ms and 300 ms.
5. Set the following parameters in the inverter suitably for the mechanical opening and closing times of the motor holding brake:
 - p1216 > mechanical opening time of the motor holding brake
 - p1217 > mechanical closing time of the motor holding brake
6. Switch on the motor.
7. Check the acceleration behavior of the drive immediately after the motor has been switched on:
 - If the motor holding brake opens too late, the inverter will accelerate the motor suddenly against the closed motor holding brake.
Set p1216 larger.
 - If the motor waits too long before accelerating after the motor holding brake has opened, reduce p1216.
8. If the load sags after switching on the motor, then you must increase the motor torque when opening the motor holding brake. Depending on the control mode, you must set different parameters:
 - V/f control (p1300 = 0 to 3):
Increase p1310 in small steps.
Increase p1351 in small steps.
 - Vector control (p1300 ≥ 20):
Increase p1475 in small steps.
9. Switch off the motor.
10. Check the behavior of the drive immediately after the motor has been switched off:
 - If the motor holding brake closes too late, the load briefly sags before the motor holding brake closes.
Set a larger value for p1217.
 - If the motor waits too long before switching off after the motor holding brake has closed, reduce p1217.



You have commissioned the "Motor holding brake" function.

Table 6-29 Setting the control logic of the motor holding brake

Parameter	Description
p1215 = 1	Enable motor holding brake 0 Motor holding brake locked (factory setting) 1 Motor holding brake just like the sequence control 2: Motor holding brake permanently open 3: Motor holding brake just like the sequential control, connected via BICO
p1216	Motor holding brake opening time (factory setting 0.1 s) p1216 > braking signal relay runtimes + brake release time
p1217	Motor holding brake closing time (factory setting 0.1 s) p1217 > braking signal relay runtimes + brake closing time
r0052.12	"Open motor holding brake" command

Table 6-30 Advanced settings

Parameter	Description
p0346	Magnetizing time (factory setting 0 s) During this time the induction motor is magnetized. The inverter calculates this parameter using p0340 = 1 or 3.
p0855	Open motor holding brake (imperative) (factory setting 0)
p0858	Close motor holding brake (imperative) (factory setting 0)
p1226	Stationary state detection speed threshold (factory setting 20 rpm) When braking with OFF1 or OFF3, when the speed falls below this threshold, standstill is detected and the monitoring time p1227 or p1228 is started
p1227	Stationary state detection monitoring time (factory setting 300 s)
p1228	Pulse deletion delay time (factory setting 0.01 s)
p1351	Starting frequency, motor holding brake (factory setting 0%) Setting the frequency set value at the slip compensation output when starting with motor holding brake. When the parameter p1351 is set to > 0, slip compensation is automatically switched on.
p1352	Starting frequency for motor holding brake (factory setting 1351) Setting the signal source for the frequency set value at the slip compensation output when starting with motor holding brake.
p1475	Speed controller torque set value for motor holding brake (factory setting 0) Setting the signal source for the torque set value when starting with motor holding brake.

6.12 Free function blocks

6.12.1 Overview



The free function blocks permit configurable signal processing in the inverter.

The following free function blocks are available:

- AND, OR, XOR, and NOT logic
- RSR (RS flip-flop), DSR (D flip-flop) flip-flops
- Timers MFP (pulse generator), PCL (pulse shortening), PDE (ON delay), PDF (OFF delay), and PST (pulse stretching)
- ADD (adder), SUB (subtractor), MUL (multiplier), DIV (divider), AVA (absolute value generated), NCM (comparator), and PLI (polyline) arithmetic functions
- LIM (limiter), PT1 (smoothing), INT (integrator), DIF (differentiator) controllers
- NSW (analog) BSW (binary) switches
- LVM limit value monitoring

The number of free function blocks in the inverter is limited. You can only use a function block once. The inverter has 3 adders, for instance. If you have already configured three adders, then no other adders are available.

6.12.2 Further information

Application description for the free function blocks



Further information is provided in the Internet:

FAQ (<http://support.automation.siemens.com/WW/view/en/85168215>)

6.13 Selecting physical units

6.13.1 Select the motor standard

Selection options and parameters involved



The inverter represents the motor data corresponding to motor standard IEC or NEMA in different system units: SI units or US units.

Setting the motor standard using p0100 is part of quick commissioning.

Table 6-31 Parameters involved when selecting the motor standard

Parameter	Designation	Motor standard IEC/NEMA, p0100 =		
		0 ¹⁾ IEC motor 50 Hz, SI units	1 NEMA motor 60 Hz, US units	2 NEMA motor 60 Hz, SI units
r0206	Power Module rated power	kW	hp	kW
p0219	Braking resistor braking power	kW	hp	kW
p0307	Rated motor power	kW	hp	kW
p0316	Motor torque constant	Nm/A	lbf ft/A	Nm/A
r0333	Rated motor torque	Nm	lbf ft	Nm
p0341	Motor moment of inertia	kgm ²	lb ft ²	kgm ²
p0344	Motor weight	kg	Lb	kg
r0394	Rated motor power	kW	hp	kW
r1493	Total moment of inertia, scaled	kgm ²	lb ft ²	kgm ²

¹⁾ Factory setting

6.13.2 Selecting the system of units

Some physical units depend on the system of units selected (SI or US), for example the power [kW or hp] or the torque [Nm or lbf ft]. You can select in which system of units the converter represents its physical values.

Options when selecting the system of units

The following options apply when selecting the system of units:

- p0505 = 1: System of units SI (factory setting)
Torque [Nm], power [kW], temperature [°C or K]
- p0505 = 2: Referred system of units/SI
Represented as [%]

- p0505 = 3: US system of units
Torque [lbf ft], power [hp], temperature [°F]
- p0505 = 4: System of units, referred/US
Represented as [%]

Special features

The values for p0505 = 2 and for p0505 = 4 - represented in the converter - are identical. However, the reference to SI or US units is required for internal calculations and to output physical variables.

For variables, which cannot be represented as [%], then the following applies:

p0505 = 1 $\triangleq$ p0505 = 2 and p0505 = 3 $\triangleq$ p0505 = 4.

In the case of variables whose units are identical in the SI system and US system, and which can be displayed as a percentage, the following applies:

p0505 = 1 $\triangleq$ p0505 = 3 and p0505 = 2 $\triangleq$ p0505 = 4.

Reference variables

There is a reference variable in the converter for most parameters with physical units. When the referred representation [%] is set, then the converter scales the physical variables based on the particular reference variable.

When the reference variable changes, then the significance of the scaled value also changes. Example:

- Reference speed = 1500 rpm → fixed speed = 80 % $\triangleq$ 1200 rpm
- Reference speed = 3000 rpm → fixed speed = 80 % $\triangleq$ 2400 rpm

For each parameter you can find the associated reference variable for scaling in the List Manual. Example: r0065 is scaled with reference variable p2000.

If scaling is not specified in the List Manual, then the converter always represents/displays the parameter unscaled (not normalized).

Groups of units

The parameters associated with the selection of a physical unit, belong to different groups of units.

You can find the associated group of units in the List Manual for each parameter. Example: r0333 belongs to unit group 7_4.

An overview of the unit groups and the possible physical units can also be found in the List Manual.



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6.13.3 Selecting the technological unit of the technology controller

Options when selecting the technological unit

p0595 defines in which technological unit the input and output variables of the technology controller are calculated, e.g. [bar], [m³/min] or [kg/h]. More information on this topic is provided in the List Manual.

Reference variable

p0596 defines the reference variable of the technological unit for the technology controller.

Unit group

Parameters involved with p0595 belong to unit group 9_1.

More information on this topic is provided in the List Manual.

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Special features

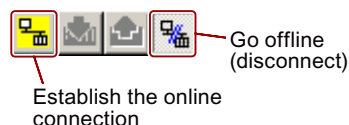
You must optimize the technology controller after changing p0595 or p0596.

6.13.4 Setting the motor standard, system of units and technology unit using STARTER

Precondition

STARTER must be in the offline mode to switch over the units.

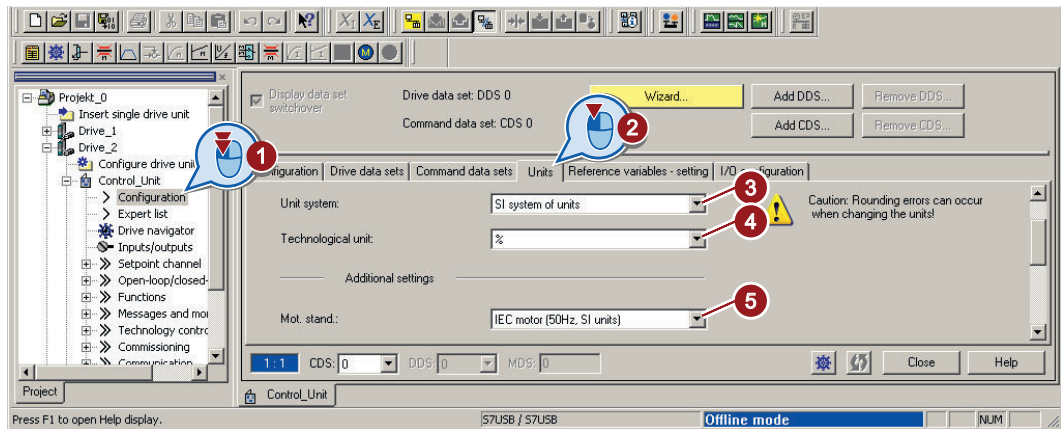
Select the online mode or the offline mode using the adjacent buttons.



STARTER indicates whether you change the settings online in the converter or offline in the PC: **Online mode** / **Offline mode**.

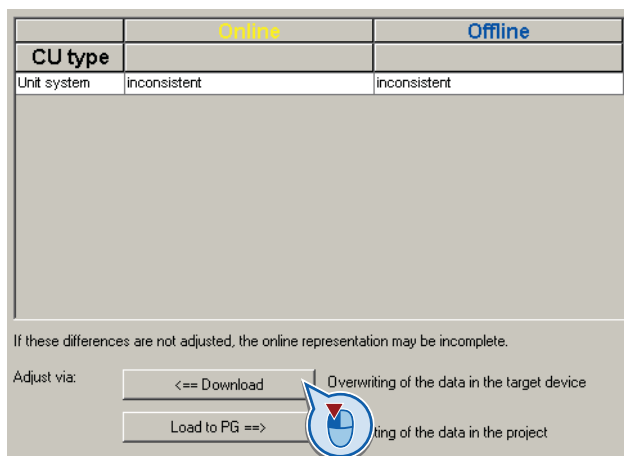
Procedure

- ➔ 1. Proceed as follows to select the motor standard and system of units using STARTER:
2. 1. Select in the "Configuration" project tree.
 2. 2. Select under the "Units" tab.



3. Select the system of units.
4. Select the technological unit of the technology controller.
5. Select the motor standard.
6. Save your settings.
7. Go online.

The inverter signals that offline, other units and process variables are set than in the inverter itself.
8. Accept these settings in the inverter.



- You have selected the motor standard and system of units using STARTER.

6.14 Safe Torque Off (STO) safety function



The operating instructions describe how to commission the STO safety function as basic function for control via a fail-safe digital input.

A description of all the safety functions is provided in the "Safety Integrated" Function Manual:

- The basic functions and the extended functions
- Controlling safety functions via PROFIsafe



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6.14.1 Function description

What is the effect of the STO safety function?

The inverter with active STO function prevents energy supply to the motor. The motor can no longer generate torque on the motor shaft.

Consequently, the STO function prevents the starting of an electrically-driven machine component.

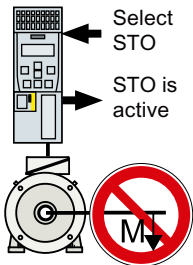


Table 6-32 The STO principle of operation as overview

	Safe Torque Off (STO)	Standard inverter functions linked with STO
1.	The inverter identifies when STO is selected via a fail-safe input or via PROFIsafe.	---
2.	The inverter prevents the energy supply to the motor.	If you use a motor holding brake, the inverter closes the motor holding brake. If you use a line contactor, the inverter opens the line contactor.
3.	The inverter signals "STO is active" via a fail-safe digital output or via PROFIsafe.	---

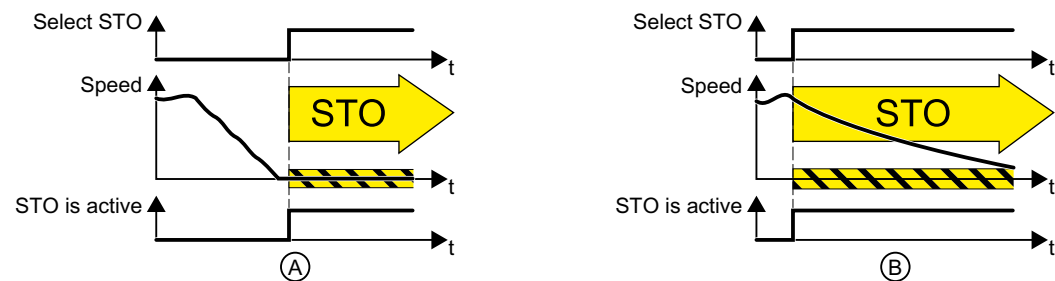


Figure 6-22 Functionality of STO when the motor is at standstill (A) and rotating (B)

If the motor is still rotating (B) when STO is selected, it coasts down to standstill.

The STO safety function is standardized

The STO function is defined in IEC/EN 61800-5-2:

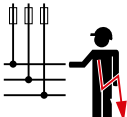
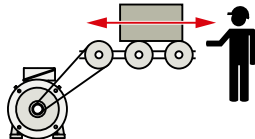
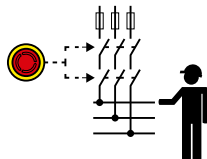
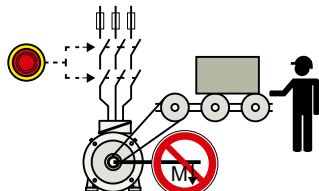
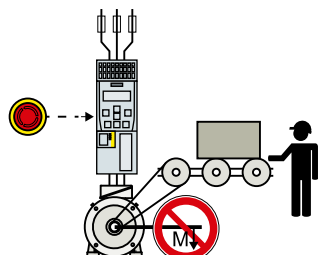
"[...] [The inverter] does not supply any energy to the motor which can generate a torque (or for a linear motor, a force)".

⇒ The STO inverter function conforms to IEC/EN 61800-5-2.

The distinction between Emergency Off and Emergency Stop

"Emergency Off" and "Emergency Stop" are commands that minimize different risks in the machine or plant.

The STO function is suitable for achieving an Emergency Stop but not an Emergency Off.

Risk:	Risk of electric shock: 	Risk of unexpected motion: 
Measure to minimize risk:	Safe switch off Switching off the electric power supply for the installation, either completely or partially.	Safely stop and safely prevent restarting Stopping or preventing the dangerous movement
Command:	Emergency Off	Emergency Stop
Classic solution:	Switch off the power supply: 	Switch off the drive power supply: 
Solution with the STO safety function integrated in the drive:	STO is not suitable for safely switching off an electric voltage.	Select STO:  It is permissible that you switch off the inverter power supply as well. However, switching off the voltage is not required as a risk-reduction measurement.

Application examples for the STO function

The STO function is suitable for applications where the motor is already at a standstill or will come to a standstill in a short, safe period of time through friction. STO does not shorten the run-on of machine components with high inertia.

Examples	Possible solution
When the Emergency Stop button is pressed, a stationary motor should not unintentionally start.	- Interconnect the Emergency Stop pushbutton with a fail-safe inverter digital input. - Select STO via the fail-safe digital input.
A central Emergency Stop button must prevent the unintentional acceleration of several motors that are at a standstill.	- Evaluate the Emergency Stop button in a central control. - Select STO via PROFIsafe.

Prerequisite for STO use

In order to use the STO safety function, the machine manufacturer should have already performed a risk assessment, e.g. in compliance with EN ISO 1050, "Safety of machinery - Principles of risk assessment". The risk assessment must confirm that the inverter is permitted for use in accordance with SIL 2 or PL d.

Restriction by the Power Module

The use of the STO safety function is possible with almost every Power Module. There may be restrictions for a PM230 Power Module:

Power Module	Constraint
PM230 in degree of protection IP55	Article numbers 6SL3223-0DE A : The STO safety function is not possible.
	Article numbers 6SL3223-0DE G : The basic STO function is possible.
PM230 in degree of protection IP20 and Push Through	Article numbers 6SL321 . -1NE L : The STO safety function is not possible.
	Article numbers 6SL321 . -1NE G : The basic STO function is possible.

6.14.2 Commissioning STO

We recommend that you commission the safety functions using the STARTER or Startdrive PC tool.



Tools to commission the inverter (Page 130)

6.14.2.1 Safety functions password

What is the purpose of the password?

The password protects the settings of the safety function from being changed by unauthorized persons.

Does the password need to be set?

The password does not need to be set.

The machine manufacturer decides whether or not a password is required.

The probabilities of failure (PFH) and the certification of the safety functions also apply when no password has been set.

What do I do if I lose the password?

If you no longer know the password but still want to change the settings for safety functions, proceed as follows:

1. Create a new project for the inverter with STARTER or Startdrive.
Leave all the factory setting in the project.
2. Load the project in the inverter.
After loading, the inverter has the factory settings.
3. If a memory card inserted in the inverter, remove it.
4. Recommission the inverter.

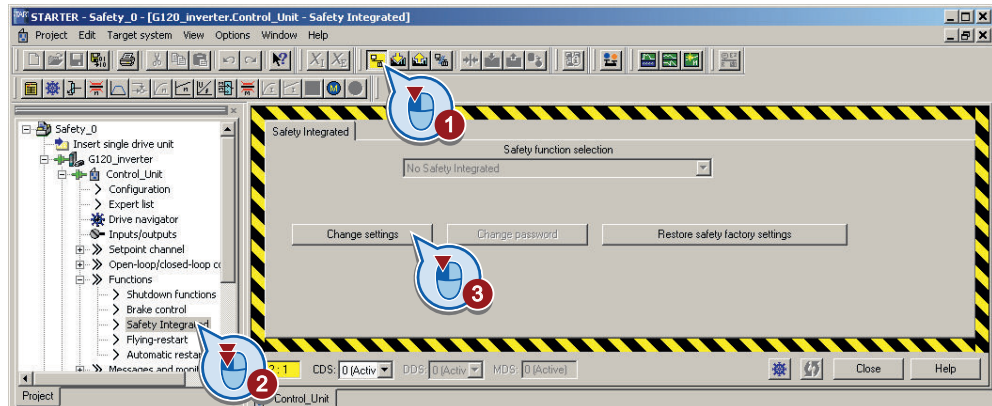
You can obtain more information or learn about alternative procedures from Technical Support.

No.	Description	
p9761	Entering a password (factory setting 0000 hex)	
	0:	No password set
	1 ... FFFF FFFF:	Password is set
p9762	New password	
p9763	Confirm password	

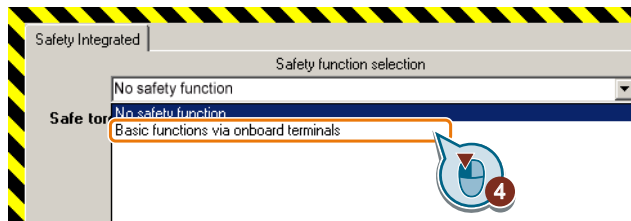
6.14.2.2 Configuring a safety function

Procedure with STARTER

- ➔ 1. To configure the safety functions, proceed as follows:
2. 1. Go online.
 2. 2. Select the "Safety Integrated" function
 3. 3. Select "Change settings".



4. 4. Select "STO via terminal":



- You have completed the following commissioning steps:
- You have started to commission the safety functions.
 - You have selected the basic functions with control via onboard terminals of the inverter.

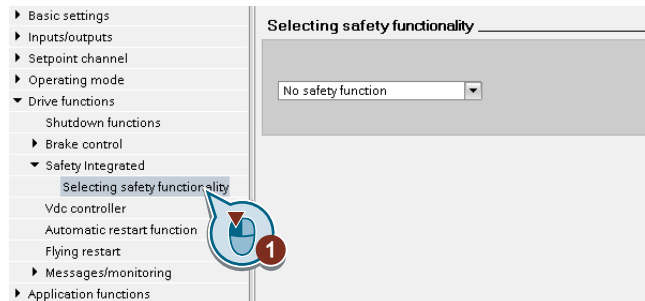
The other selection options are described in the "Safety Integrated" Function Manual.

➔ Overview of the manuals (Page 504)

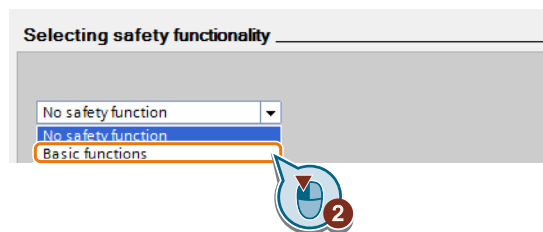
Procedure with Startdrive

1. Proceed as follows to configure the safety functions:

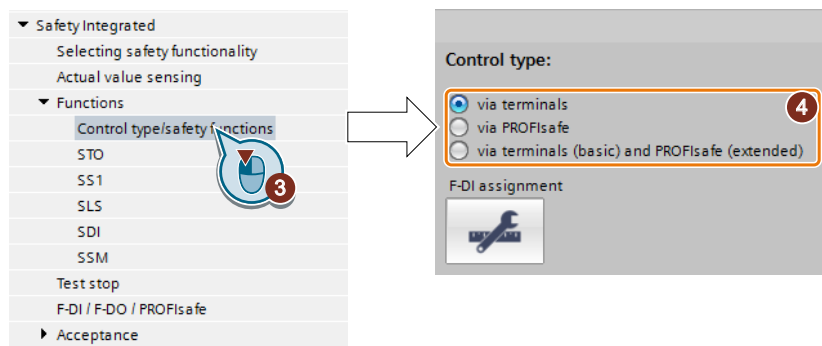
2. 1. Select "Select safety functionality".



2. Enable the safety functions.



3. Controlling the safety functions.



4. Define the interface for controlling the safety functions.



You have configured the safety functions

Parameter	Description	
p0010 = 95	Drive commissioning parameter filter Safety Integrated commissioning	
p9601	Enable functions integrated in the drive (factory setting: 0000 bin)	
	0 hex	None of the safety functions has been released
	1 hex	Basic functions via onboard terminals has been enabled
p9761	Enter a password (factory setting: 0000 hex) Permissible passwords lie in the range 1 ... FFFF FFFF.	
p9762	New password	
p9763	Password confirmation	

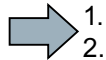
6.14.2.3 Interconnecting the "STO active" signal

If you require the feedback signal "STO active" of the inverter in your higher-level control system, then you must appropriately interconnect the signal.

Precondition

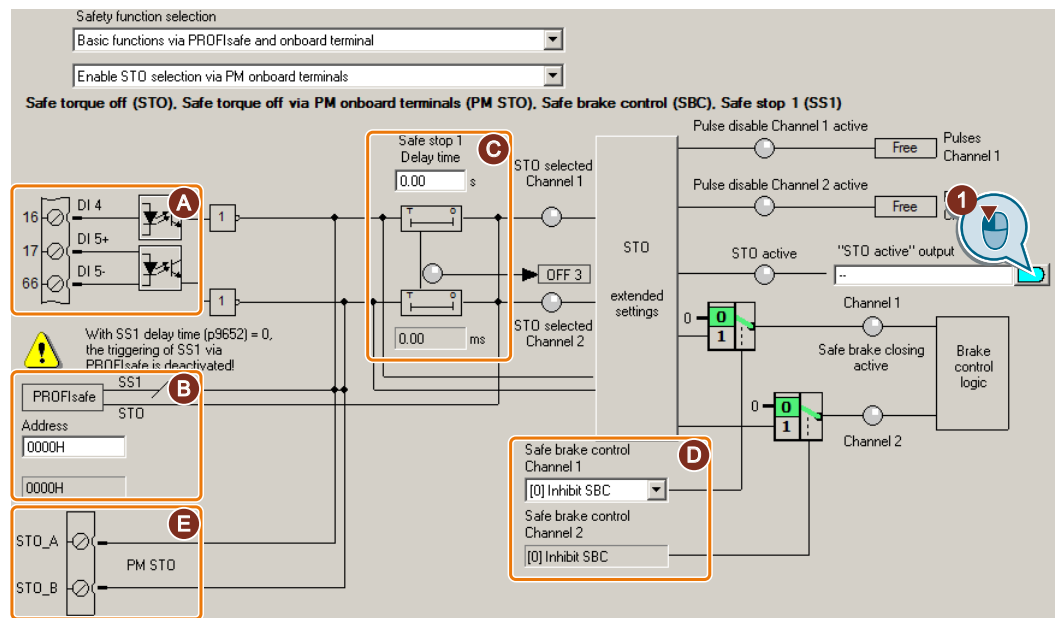
You are online with STARTER or Startdrive.

Procedure with STARTER



1. To interconnect the "STO active" checkback signal, proceed as follows:

1. Select the button for the feedback signal.



The screen form varies depending on the inverter and the interface that has been selected.

- (A) Input terminal or connector for SINAMICS inverters, F0 rail for SIMATIC ET 200pro FC-2.
- (B) PROFIsafe interface
- (C), (D) Delay time for SS1 and enable SBC for an inverter with CU250S-2 Control Unit
- (E) STO via the Power Module terminals for a PM240-2 or PM240P-2, FSD ... FSF Power Module

2. Select the signal that matches your particular application.



You have interconnected the "STO active" checkback signal. After STO has been selected, the inverter signals "STO active" to the higher-level control.

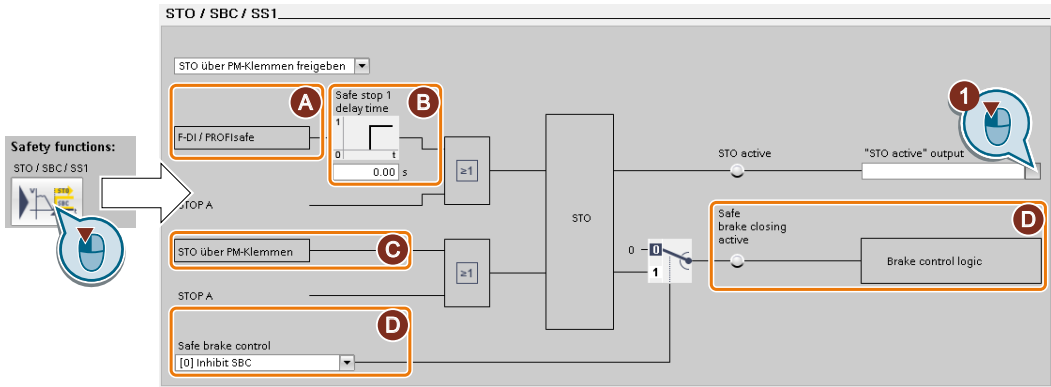
Procedure with Startdrive

- ➔

1.

2.
- To interconnect the "STO active" checkback signal, proceed as follows:

1. Select the button for the feedback signal.



The screen form varies depending on the inverter and the interface that has been selected.

- (A)

Control type
- (B)

Delay time for SS1 and enable of SBC for an inverter with CU250S-2 Control Unit
- (C)

STO via the Power Module terminals for a PM240-2 or PM240P-2, FSD ... FSF Power Module
- (D)

Enable SBC for an inverter with CU250S-2 Control Unit

2. Select the signal that matches your particular application.

■

You have interconnected the "STO active" checkback signal. After STO has been selected, the inverter signals "STO active" to the higher-level control.

Parameter	Description
r9773.01	1 signal: STO is active in the drive

6.14.2.4 Setting the filter for fail-safe digital inputs

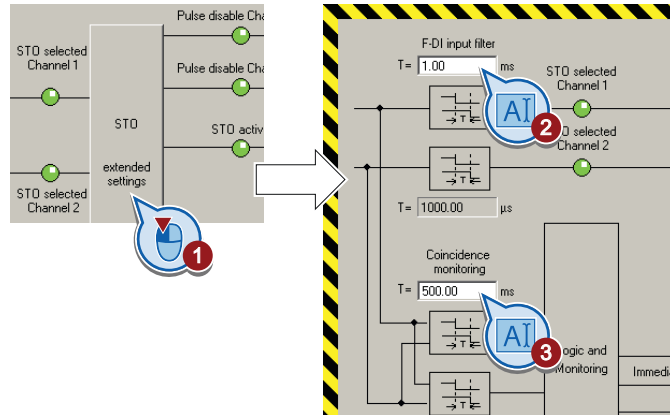
Requirement

You are online with STARTER or Startdrive online.

Procedure with STARTER

- ➔ 1. Proceed as follows to set the input filter and the simultaneity monitoring of the fail-safe digital input:
2.

1. Select the "Extended settings" button.



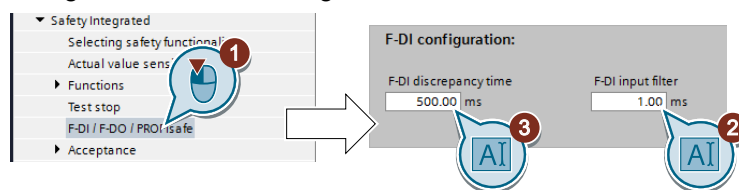
2. Set the debounce time for the F-DI input filter.
3. Set the discrepancy time for the simultaneity monitoring.
4. Close the screen form.

■ You have set the input filter and the simultaneity monitoring of the fail-safe digital input.

Procedure with Startdrive

- ➔ 1. Proceed as follows to set the input filter and the simultaneity monitoring of the fail-safe digital input:
2.

1. Navigate to the filter settings.



2. Set the debounce time for the F-DI input filter.
3. Set the discrepancy time for the simultaneity monitoring.

■ You have set the input filter and the simultaneity monitoring of the fail-safe digital input.

Description of the signal filter

The following filters are available for the fail-safe digital inputs:

- One filter for the simultaneity monitoring
- A filter to suppress short signals, e.g. test pulses.

Set the discrepancy time for the simultaneity monitoring.

The inverter checks that the two input signals of the fail-safe digital input always have the same signal state (high or low).

With electromechanical sensors (e.g. emergency stop buttons or door switches), the two sensor contacts never switch at exactly the same time and are therefore temporarily inconsistent (discrepancy). A permanent discrepancy signifies a fault in the fail-safe digital input circuit, e.g. wire breakage.

When appropriately set, the inverter tolerates brief discrepancies.

The discrepancy time does not extend the inverter response time. The inverter selects its safety function as soon as one of the two F-DI signals changes its state from high to low.

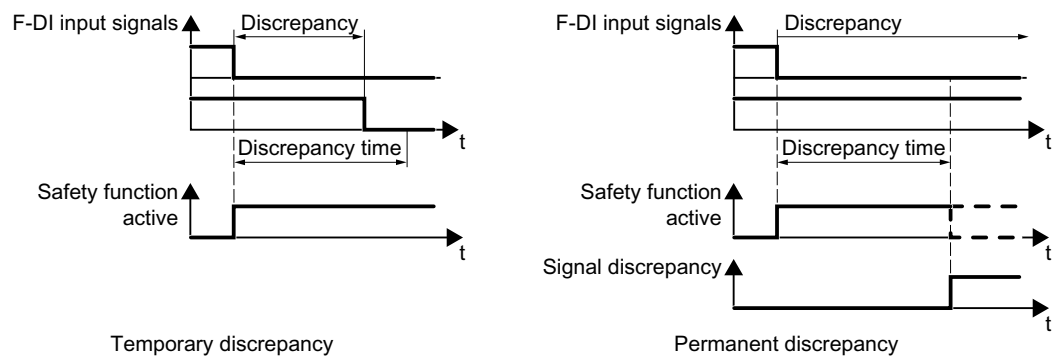


Figure 6-23 Simultaneity monitoring with discrepancy time

Filter to suppress short signals

In the following cases, an immediate inverter response to signal changes of the fail-safe digital inputs is not desirable:

- If a fail-safe digital input of the inverter is interconnected with an electromechanical sensor, signal changes can occur due to contact bounce.
- In order to identify faults due to short-circuit or cross faults, several control modules test their fail-safe digital outputs with "bit pattern tests" (bright/dark test). If a fail-safe digital input of the inverter is interconnected with a fail-safe digital output of an open-loop control module, then the inverter responds with a bit pattern test.
The typical duration of the signal change within a bit pattern test:

- On test: 1 ms
- Off test: 4 ms

If the fail-safe digital input responds to many signal changes within a certain time, then the inverter responds with a fault.

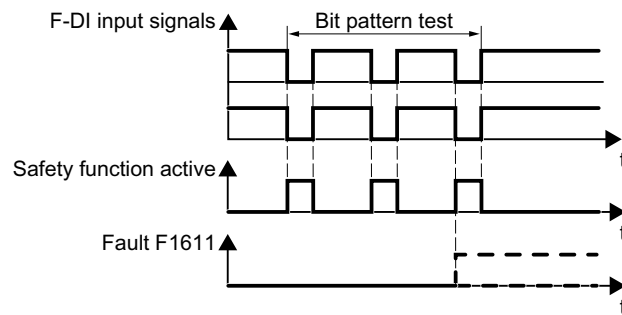


Figure 6-24 Inverter response to a bit pattern test

A filter in the inverter suppresses brief signals as a result of the bit pattern test or contact bounce.

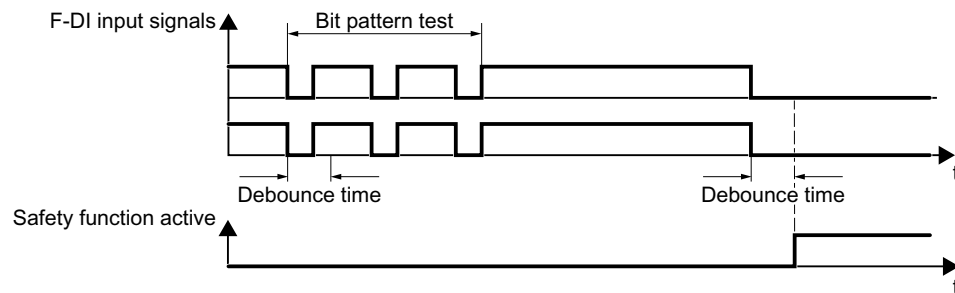


Figure 6-25 Filter to suppress brief signals

The filter extends the response time of the safety function by the debounce time.

Parameter	Description
p9650	F-DI changeover tolerance time (factory setting: 500 ms) Tolerance time to change over the fail-safe digital input for the basic functions.
p9651	STO debounce time (factory setting: 1 ms) Debounce time of the fail-safe digital input for the basic functions.

Debounce times for standard and safety functions

The debounce time p0724 for "standard" digital inputs has no influence on the fail-safe input signals. Conversely, the same applies: The F-DI debounce time does not affect the signals of the "standard" inputs.

If you use an input as a standard input, set the debounce time using parameter p0724 .

If you use an input as a fail-safe input, set the debounce time as described above.

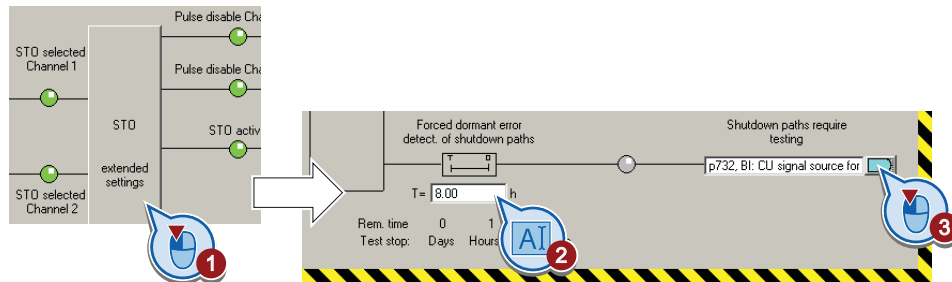
6.14.2.5 Setting the forced checking procedure (test stop)

Requirement

You are online with STARTER or Startdrive online.

Procedure with STARTER

- ➔ 1. To set the forced checking procedure (test stop) of the basic functions, proceed as follows:
2. 1. Select the screen form for setting the forced checking procedure.

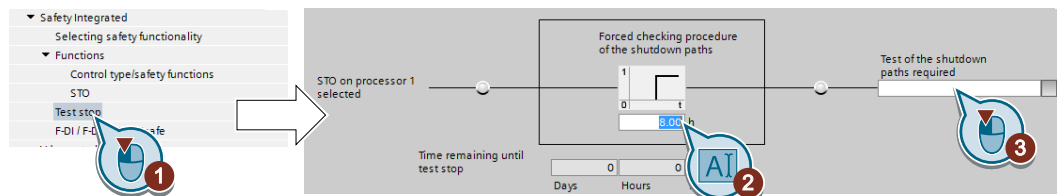


2. Set the monitoring time to a value to match your application.
3. Using this signal, the inverter signals that a forced checking procedure (test stop) is required. Interconnect this signal with an inverter signal of your choice.
4. Close the screen form.

■ You have set the forced checking procedure (test stop) for the basic functions.

Procedure with Startdrive

- ➔ 1. To set the forced checking procedure (test stop) of the basic functions, proceed as follows:
2. 1. Select the screen form for setting the forced checking procedure.



2. Set the monitoring time to a value to match your application.
3. Using this signal, the inverter signals that a forced checking procedure (test stop) is required. Interconnect this signal with an inverter signal of your choice.

■ You have set the forced checking procedure (test stop) for the basic functions.

Description

The forced checking procedure (test stop) of the basic functions is an inverter self test. The inverter checks its circuits to switch off the torque. If you are using the Safe Brake Relay, for a forced checking procedure, the inverter also checks the circuits of this component.

You start the forced checking procedure each time that the STO function is selected.

Using a timer block, the inverter monitors as to whether the forced checking procedure is regularly performed.

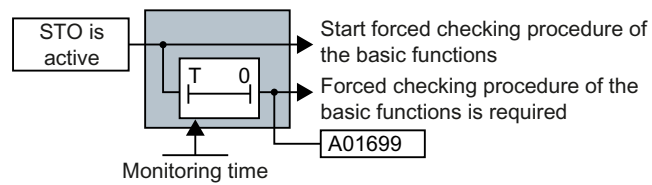


Figure 6-26 Starting and monitoring the forced checking procedure (test stop)

Parameter	Description
p9659	Forced dormant error detection timer (Factory setting: 8 h) Monitoring time for the forced dormant error detection.
r9660	Forced dormant error detection remaining time Displays the remaining time until the forced dormant error detection and testing the safety switch-off signal paths.
r9773.31	1 signal: Forced dormant error detection is required Signals for the higher-level control system.

6.14.2.6 Activating the settings and checking the digital inputs

Activate settings

Requirement

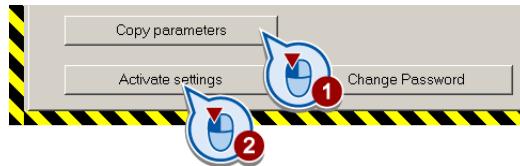
You are online with STARTER or Startdrive online.

Procedure with STARTER



1. To activate the settings for the safety functions, proceed as follows:

2. 1. Press the "Copy parameters" button, to create a redundant image of your inverter settings.

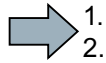


2. Press the "Activate settings" button.
3. If the password is the factory default, you are prompted to change the password.
If you try to set a password that is not permissible, the old password will not be changed.
4. Confirm the prompt for saving your settings (copy RAM to ROM).
5. Switch off the inverter supply voltage.
6. Wait until all LEDs on the inverter go dark (no voltage condition).
7. Switch on the inverter supply voltage again.



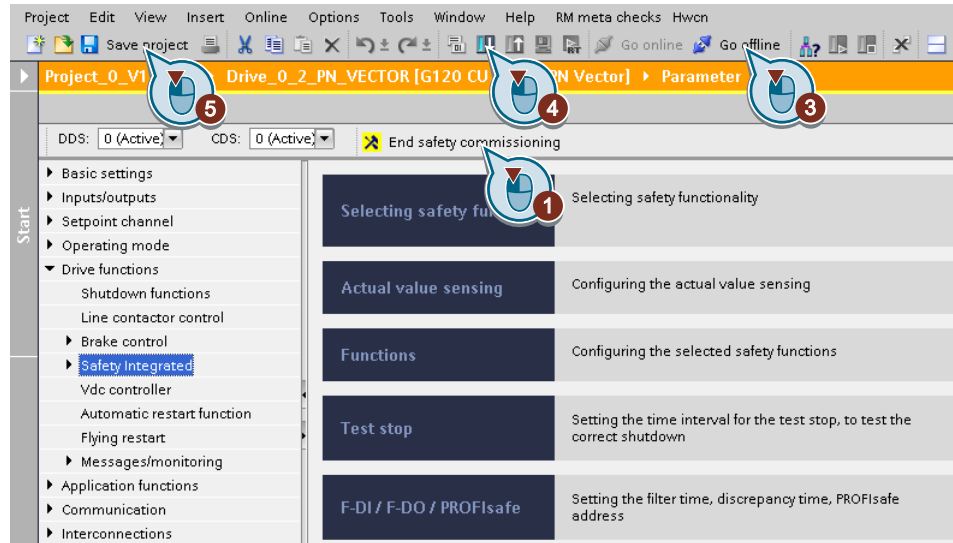
Your settings are now active.

Procedure with Startdrive



1. To activate the settings of the safety functions in the drive, proceed as follows:

1. Click the "End safety commissioning" button.



2. Confirm the prompt for saving your settings (copy RAM to ROM).

3. Disconnect the online connection.

4. Select the "Load from device (software)" button.

5. Save the project.

6. Switch off the inverter supply voltage.

7. Wait until all LEDs on the inverter go dark (no voltage condition).

8. Switch on the inverter supply voltage again.



Your settings are now active.

Parameter	Description
p9700 = D0 hex	SI copy function (factory setting: 0) Start the SI parameter copy function.
p9701 = DC hex	Confirm data change (factory setting: 0) Confirm SI Basic parameter change
p0010 = 0	Drive commissioning parameter filter 0: Ready
p0971 = 1	Save parameter 1: Save the drive object (copy from RAM to ROM) After the inverter has saved the parameters in a non-volatile fashion, then p0971 = 0.

Checking the connection of digital inputs

The simultaneous connection of digital inputs with a safety function and a "standard" function may lead to the drive behaving in unexpected ways.

If you control the safety functions in the inverter via fail-safe digital inputs, then you must check as to whether the fail-safe digital inputs are in some instances interconnected with a "standard" function.

Procedure with STARTER



1. Proceed as follows to check as to whether the fail-safe digital inputs are only used for the safety functions:
2.
 1. In the project navigator, select the inputs/outputs.
 2. Select the screen for the digital inputs.
 3. Remove all interconnections of the digital inputs that you use as fail-safe digital input F-DI:

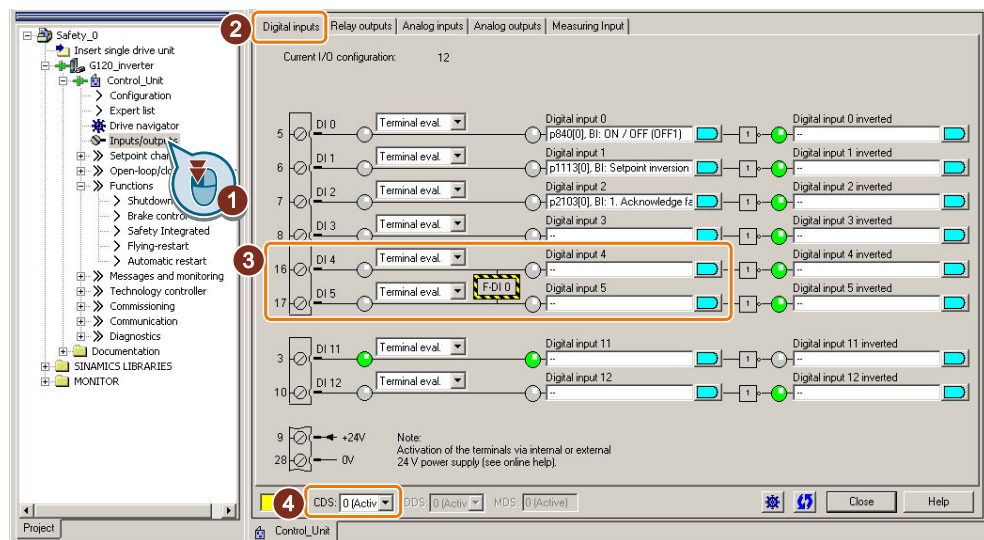


Figure 6-27 Removing the DI 4 and DI 5 digital-input connections

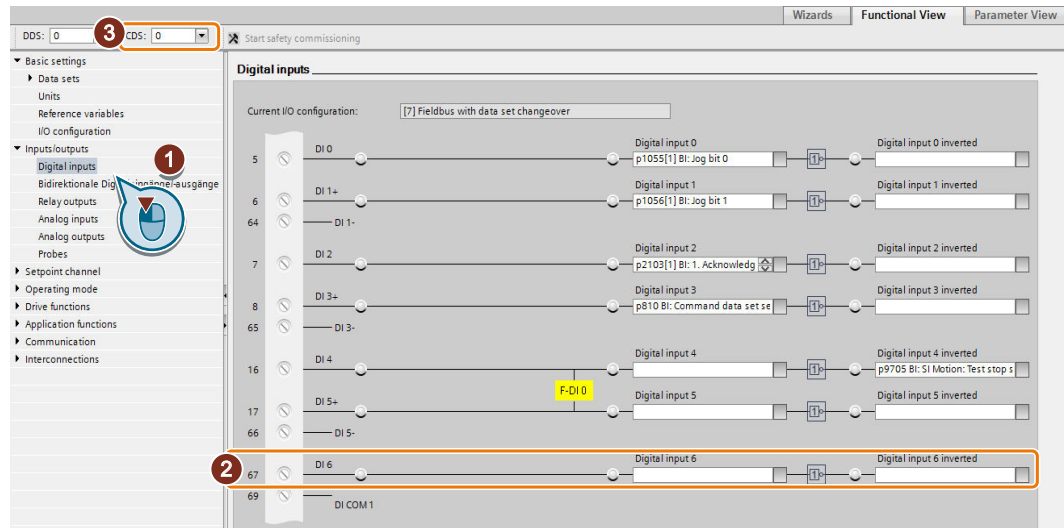
4. You must delete the digital input connections for all CDS if you use the switchover of the command data sets (CDS).
You can find a description of the CDS switchover in the operating instructions.



You have ensured that the fail-safe digital inputs only control the safety functions in the inverter.

Procedure with Startdrive

- ➔ 1. Proceed as follows to check as to whether the fail-safe digital inputs are only used for the safety functions:
2.
 1. Select the screen for the digital inputs.
 2. Remove all interconnections of the digital inputs that you use as fail-safe digital input F-DI:



3. You must delete the digital input connections for all CDS if you use the switchover of the command data sets (CDS).
You can find a description of the CDS switchover in the operating instructions.

■ You have ensured that the fail-safe digital inputs only control the safety functions in the inverter.

6.14.2.7 Acceptance - completion of commissioning

What is an acceptance?

The machine manufacturer is responsible in ensuring that his plant or machine functions perfectly. As a consequence, after commissioning, the machine manufacturer must check those functions or have them checked by specialist personnel, which represent an increased risk of injury or material damage. This acceptance or validation is, for example, also specified in the European machinery directive and essentially comprises two parts:

- Checking the safety-relevant functions and machine parts.
→ **Acceptance test.**
- Generate an "Acceptance report" that describes the test results.
→ **Documentation.**

Supply information for the validation, e.g. the harmonized European standards EN ISO 13849-1 and EN ISO 13849-2.

Acceptance test of the machine or plant

The acceptance test checks whether the safety-relevant functions in the plant or machine function correctly. The documentation of the components used in the safety functions can also provide information about the necessary tests.

Testing the safety-related functions includes, e.g. the following:

- Are all safety equipment such as protective door monitoring devices, light barriers or emergency-off switches connected and ready for operation?
- Does the higher-level control respond as expected to the safety-relevant feedback signals of the inverter?
- Do the inverter settings match the configured safety-relevant function in the machine?

Acceptance test of the inverter

The acceptance test of the inverter is a part of the acceptance test of the entire machine or plant.

The acceptance test of the inverter checks whether the integrated drive safety functions are set up correctly for the planned safety function of the machine.



Acceptance tests for the safety functions (Page 498)

Documentation of the inverter

The following must be documented for the inverter:

- The results of the acceptance test.
- The settings of the integrated drive safety functions.

The STARTER commissioning tool logs the settings of the integrated drive functions, if necessary.



Acceptance - completion of commissioning (Page 248)

The documentation must be signed.

Who may perform the acceptance test of the inverter?

Personnel from the machine manufacturer, who, on account of their technical qualifications and knowledge of the safety functions, are in a position to perform the acceptance test in the correct manner are authorized to perform the acceptance testing of the inverter.

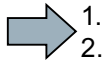
Reduced acceptance test after function expansions

A full acceptance test is necessary only after first commissioning. A reduced acceptance test is sufficient when safety functions are expanded.

Measure	Acceptance test	
	Acceptance test	Documentation
Functional expansion of the machine (additional drive).	Yes. Only check the safety functions of the new drive.	• Supplement machine overview • Supplement inverter data • Add function table • Log the new checksums • Countersignature
Transfer of inverter settings to other identical machines by means of series commissioning.	No. Only check the control of all of the safety functions.	• Add machine description • Check checksums • Check firmware versions

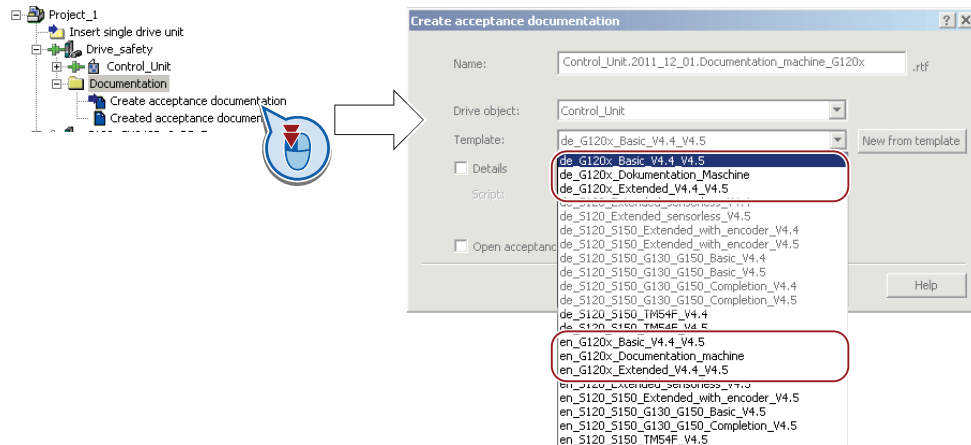
Document for the acceptance

The STARTER provides you with a number of documents to be regarded as a recommendation for the acceptance tests of the safety functions.

Procedure

1. Proceed as follows to create the acceptance documentation for the drive using STARTER:

2. 1. In STARTER, select "Create acceptance documentation":

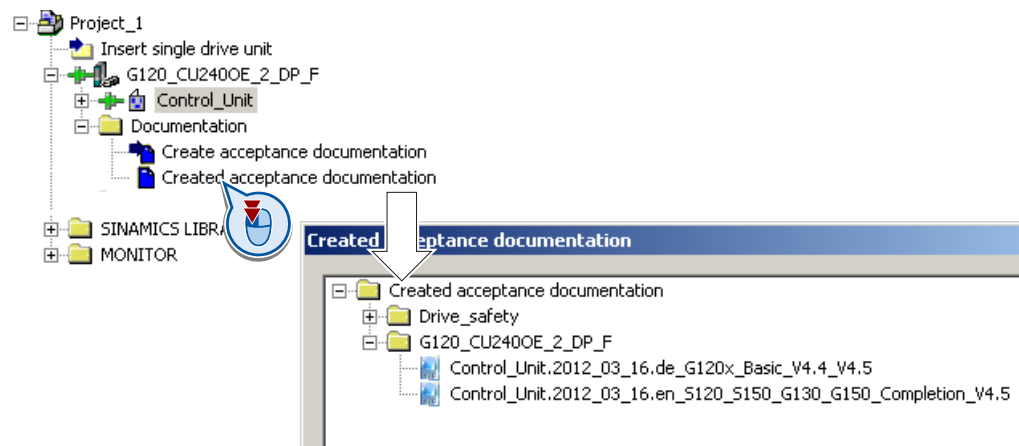


STARTER has templates in German and English.

2. Select the suitable template and create a report for each drive of your machine or system:

- Template for the machine documentation:
de_G120x_Dokumentation_Maschine: German template.
en_G120x_Dokumentation_machine: English template.
- Report of the settings for the basic functions, as of firmware version V4.4:
de_G120x_Basic_V4.4...: German report.
en_G120x_Basic_V4.4...: English report.

3. You load the created reports for archiving and the machine documentation for further processing:



4. Archive the reports and the machine documentation.



You have generated the documents to accept the safety functions.



Acceptance tests for the safety functions (Page 498)

6.15 Setpoints

6.15.1 Overview



The inverter receives its main setpoint from the setpoint source. The main setpoint generally specifies the motor speed.

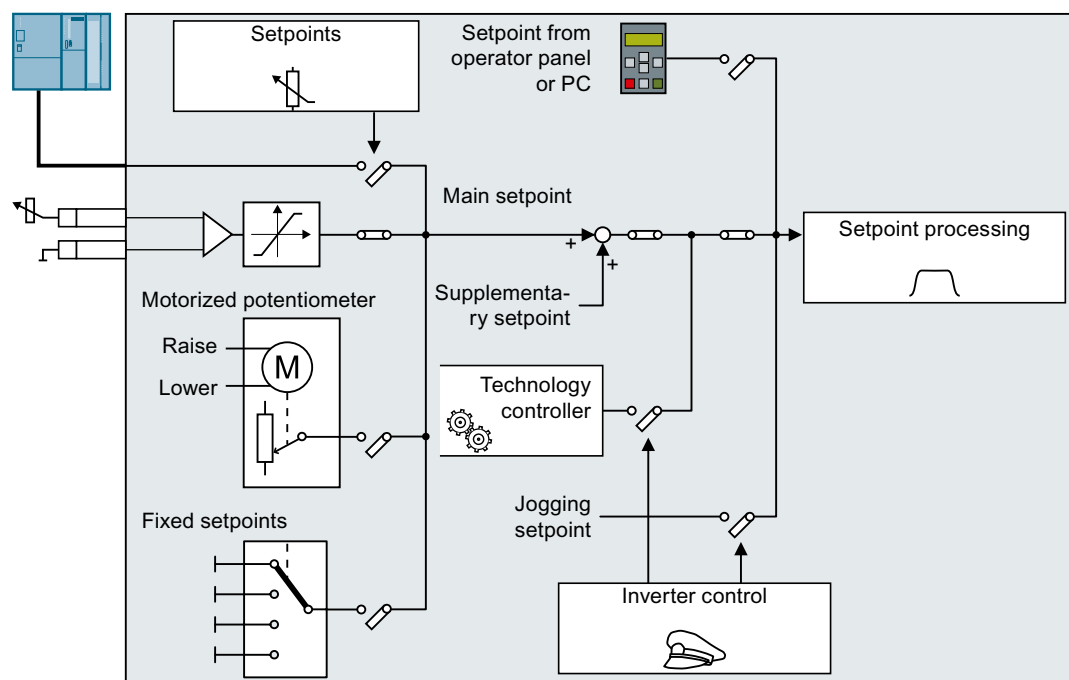


Figure 6-28 Setpoint sources for the inverter

You have the following options when selecting the source of the main setpoint:

- Inverter analog input.
- Inverter fieldbus interface.
- Motorized potentiometer simulated in the inverter.
- Fixed setpoints saved in the inverter.

You have the same selection options when selecting the source of the supplementary setpoint.

Under the following conditions, the inverter switches from the main setpoint to other setpoints:

- When the technology controller is active and appropriately interconnected, its output specifies the motor speed.
- When jogging is active.
- When controlling from an operator panel or the STARTER PC tool.

6.15.2 Analog input as setpoint source

Interconnecting an analog input

If you have selected a pre-assignment without a function of the analog input, then you must interconnect the parameter of the main setpoint with an analog input.



Figure 6-29 Example: Analog input 0 as setpoint source

Table 6-33 Setting with analog input 0 as setpoint source

Parameter	Remark
p1070 = 755[0]	Main setpoint Interconnect the main setpoint with analog input 0
p1075 = 755[0]	Additional setpoint Interconnect the additional setpoint with analog input 0

You must adapt the analog input to the connected signal, e.g. ± 10 V or 4 ... 20 mA.



Analog inputs (Page 183)

6.15.3 Specifying the setpoint via the fieldbus

Interconnecting the fieldbus with the main setpoint

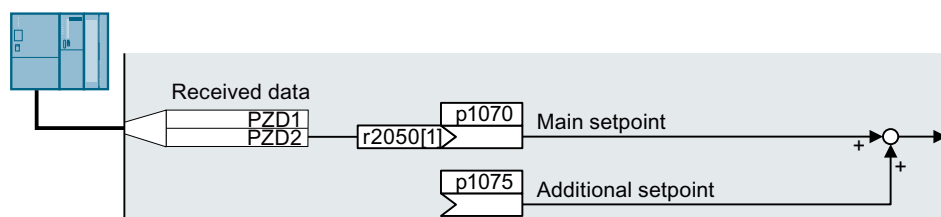


Figure 6-30 Fieldbus as setpoint source

Most standard telegrams receive the speed setpoint as a second process data PZD2.

Table 6-34 Setting the fieldbus as setpoint source

Parameter	Remark
p1070 = 2050[1]	Main setpoint Interconnect the main setpoint with process data PZD2 from the fieldbus.
p1075 = 2050[1]	Additional setpoint Interconnect the additional setpoint with process data PZD2 from the fieldbus.

6.15.4 Motorized potentiometer as setpoint source

The "Motorized potentiometer" function emulates an electromechanical potentiometer. The output value of the motorized potentiometer can be set with the "higher" and "lower" control signals.

Interconnecting the motorized potentiometer (MOP) with the setpoint source

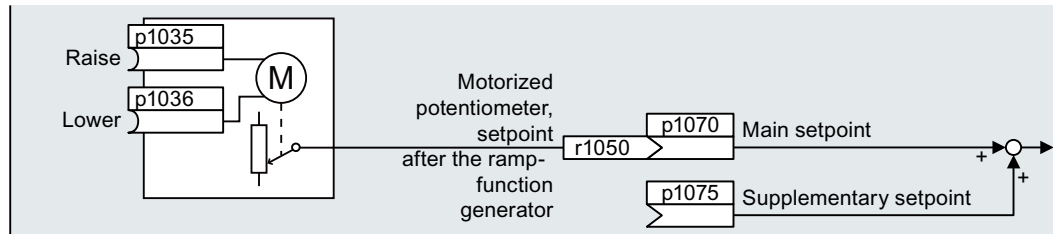


Figure 6-31 Motorized potentiometer as setpoint source

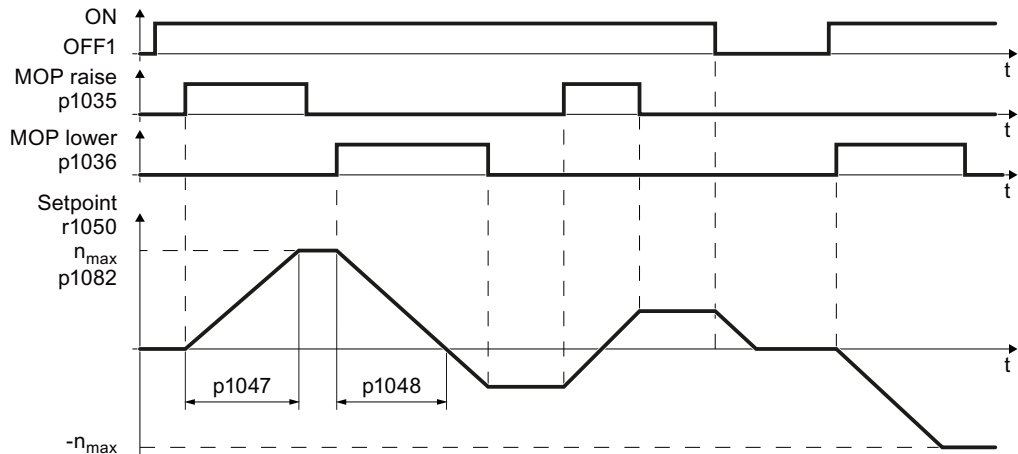


Figure 6-32 Function chart of the motorized potentiometer

Table 6-35 Basic setup of motorized potentiometer

Parameter	Description	
p1035	Motorized potentiometer setpoint higher	Interconnect these commands with signals of your choice.
p1036	Motorized potentiometer setpoint lower	
p1040	MOP start value (factory setting: 0 rpm) Defines the start value [rpm] that is effective when the motor is switched on.	
p1047	MOP ramp-up time (factory setting: 10 s)	
p1048	MOP ramp-down time (factory setting: 10 s)	
r1050	Motorized potentiometer setpoint after the ramp-function generator	
p1070 = 1050	Main setpoint	

Table 6-36 Extended setup of motorized potentiometer

Parameter	Description
p1030	MOP configuration (factory setting: 00110 bin)
	.00 Storage active = 0: After the motor has been switched on, the setpoint = p1040 = 1: After the motor has switched off, the inverter saves the setpoint. After the motor has switched on, the setpoint = the stored value
	.01 Automatic mode, ramp-function generator active (1-signal via BI: p1041) = 0: Ramp-up/ramp-down time = 0 = 1: With ramp-function generator In manual mode (p1041 = 0), the ramp-function generator is always active.
	.02 Initial rounding active 1: With initial rounding. Using the initial rounding function it is possible to enter very small setpoint changes
	.03 Storage in NVRAM active 1: If bit 00 = 1, the setpoint is retained during a power failure
	.04 Ramp-function generator always active 1: The inverter also calculates the ramp-function generator when the motor is switched off
p1037	MOP maximum speed (factory setting: 0 rpm) Automatically pre-assigned when commissioning
p1038	MOP minimum speed (factory setting: 0 rpm) Automatically pre-assigned when commissioning
p1039	Motorized potentiometer, inversion (factory setting: 0) Signal source for inverting the minimum speed / velocity or maximum speed / velocity
p1041	Motorized potentiometer, manual/automatic (factory setting: 0) Signal source for switchover from manual to automatic
p1043	Motorized potentiometer, accept setting value (factory setting: 0) Signal source for accepting the setting value. The motorized potentiometer accepts the setting value p1044 on signal change p1043 = 0 → 1.
p1044	MOP setting value (factory setting: 0) Signal source for the setting value.

For more information about the motorized potentiometer, refer to function diagram 3020 in the List Manual.

6.15.5 Fixed speed setpoint as setpoint source

In many applications after switching on the motor, all that is needed is to run the motor at a constant speed or to switch between different speeds.

Example: After it has been switched on, a conveyor belt only runs with two different velocities.

Interconnecting a fixed speed setpoint with the main setpoint

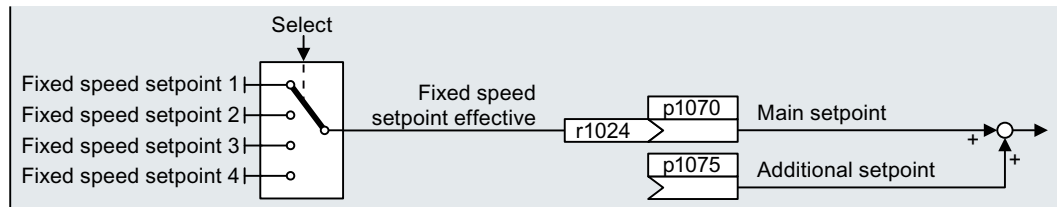


Figure 6-33 Fixed speed setpoint as setpoint source

Table 6-37 Setting the fixed speed setpoint as setpoint source

Parameter	Remark
p1070 = 1024	Main setpoint Interconnect main setpoint with fixed speed setpoint.

Selecting the fixed speed setpoint, either direct or binary

The inverter makes a distinction between two methods when selecting the fixed speed setpoints:

Directly selecting a fixed speed setpoint

You set 4 different fixed speed setpoints. Up to 16 different setpoints are obtained by adding one or several of the four fixed speed setpoints.

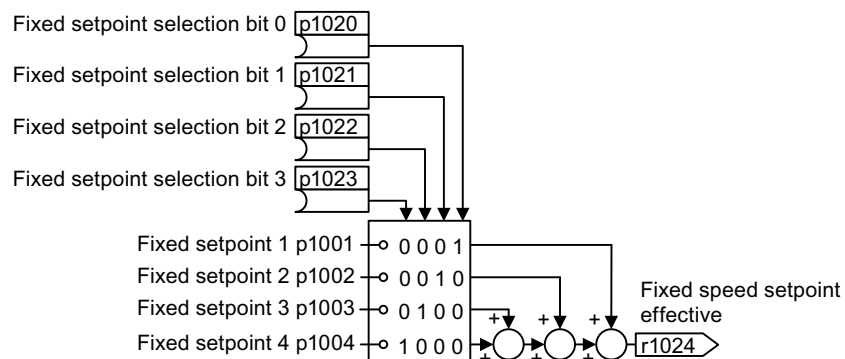


Figure 6-34 Simplified function diagram when directly selecting the fixed speed setpoints

Additional information about direct selection can be found in function diagram 3011 in the List Manual.

Selecting the fixed speed setpoint, binary

You set 16 different fixed speed setpoints. You precisely select one of these 16 fixed speed setpoints by combining four selection bits.

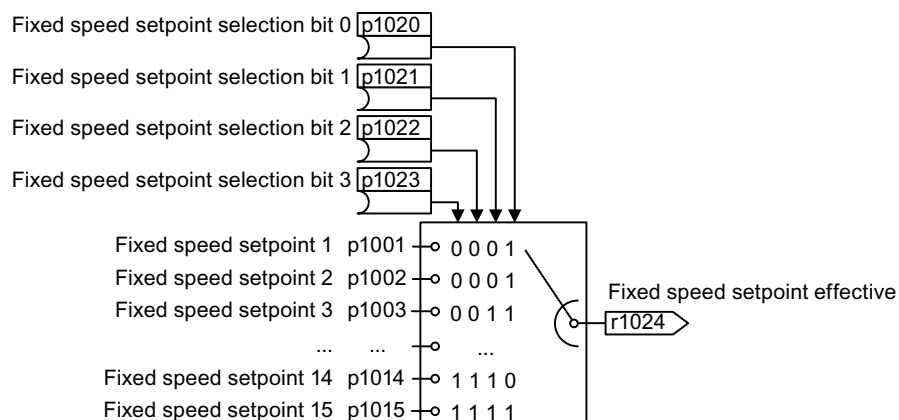


Figure 6-35 Simplified function diagram when selecting the fixed speed setpoints, binary

Additional information about binary selection can be found in function diagram 3010 in the List Manual.

Parameters to set the fixed speed setpoints

Parameter	Description
p1001	Fixed speed setpoint 1 (factory setting: 0 rpm)
p1002	Fixed speed setpoint 2 (factory setting: 0 rpm)
...	...
p1015	Fixed speed setpoint 15 (factory setting: 0 rpm)
p1016	Fixed speed setpoint mode (factory setting: 1)
	1: Direct
	2: Binary
p1020	Fixed speed setpoint selection bit 0 (factory setting: 0)
p1021	Fixed speed setpoint selection bit 1 (factory setting: 0)
p1022	Fixed speed setpoint selection bit 2 (factory setting: 0)
p1023	Fixed speed setpoint selection bit 3 (factory setting: 0)
r1024	Fixed speed setpoint effective
r1025.0	Fixed speed setpoint status
	1 signal Fixed speed setpoint is selected

Example: Directly selecting two fixed speed setpoints

The motor should operate at different speeds as follows:

- The signal on digital input 0 switches the motor on and accelerates it to 300 rpm.
- The signal at digital input 1 accelerates the motor to 2000 rpm.
- The signals at the two digital inputs accelerate the motor to 2300 rpm.

Table 6-38 Settings for the example

Parameter	Description
p1001 = 300.000	Fixed speed setpoint 1 [rpm]
p1002 = 2000.000	Fixed speed setpoint 2 [rpm]
p0840 = 722.0	ON/OFF1: Switches on the motor with digital input 0
p1070 = 1024	Main setpoint: Interconnects the main setpoint with fixed speed setpoint.
p1020 = 722.0	Fixed speed setpoint selection bit 0: Interconnects fixed speed setpoint 1 with digital input 0 (DI 0).
p1021 = 722.1	Fixed speed setpoint selection bit 1: Interconnects fixed speed setpoint 2 with digital input 1 (DI 1).
p1016 = 1	Fixed speed setpoint mode: Directly selects fixed speed setpoints.

Table 6-39 Resulting fixed speed setpoints for the example above

Fixed speed setpoint selected via	Resulting setpoint
DI 0 = 0	Motor stops
DI 0 = 1 and DI 1 = 0	300 rpm
DI 0 = 1 and DI 1 = 1	2300 rpm

6.16 Setpoint calculation

6.16.1 Overview of setpoint processing



The setpoint can be modified as follows using the setpoint processing:

- Invert setpoint to reverse the motor direction of rotation (reversing).
- Inhibit positive or negative direction of rotation, e.g. for conveyor belts, pumps or fans.
- Skip frequency bands to prevent mechanical resonance effects.
The skip frequency band at speed = 0 results in a minimum speed after switching on the motor.
- Limit to a maximum speed to protect the motor and mechanical system.
- Ramp-function generator to accelerate and brake the motor with an optimum torque.

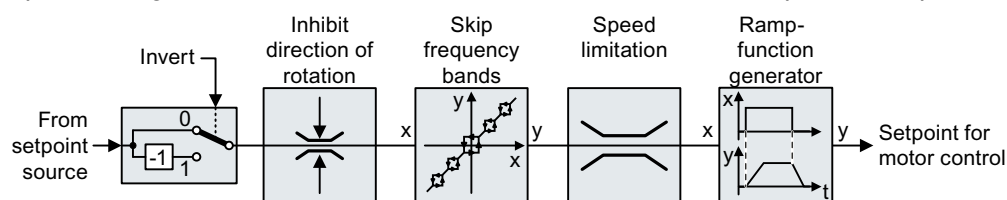
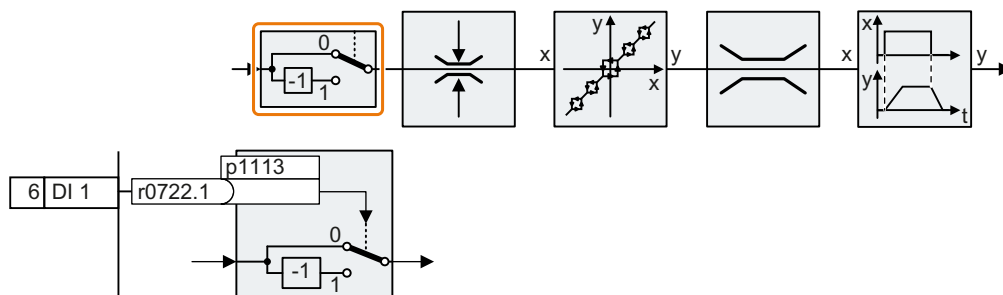


Figure 6-36 Setpoint processing in the converter

6.16.2 Invert setpoint

The inverter provides an option to invert the setpoint sign using a bit. As an example, the setpoint inversion is shown through a digital input.



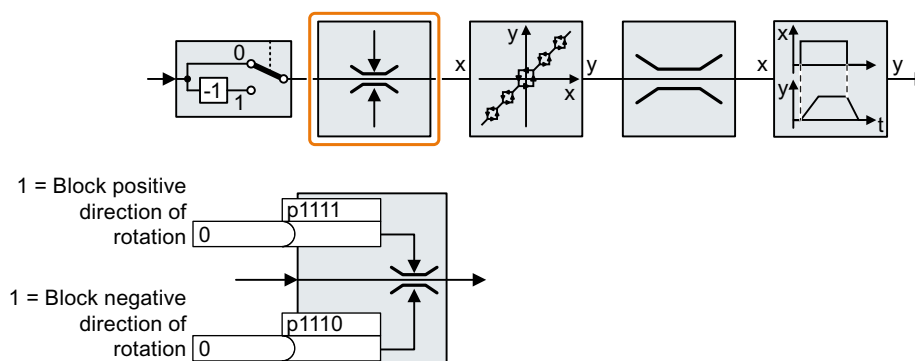
In order to invert the setpoint through the digital input DI 1, connect the parameter p1113 with a binary signal, e.g. the digital input 1.

Table 6-40 Examples of settings to invert the setpoint

Parameter	Remark
p1113 = 722.1	Setpoint inversion Digital input 1 = 0: Setpoint remains unchanged. Digital input 1 = 1: Inverter inverts the setpoint.
p1113 = 2090.11	Invert setpoint via control word 1, bit 11.

6.16.3 Inhibit direction of rotation

In the factory setting of the inverter, both motor directions of rotation are enabled.



Set the corresponding parameter to a value = 1 to permanently block directions of rotation.

Table 6-41 Examples of settings to inhibit the direction of rotation

Parameter	Remark
p1110 = 1	Inhibit negative direction Negative direction is permanently inhibited.
p1110 = 722.3	Inhibit negative direction Digital input 3 = 0: Negative direction of rotation is enabled. Digital input 3 = 1: Negative direction of rotation is inhibited.

6.16.4 Skip frequency bands and minimum speed

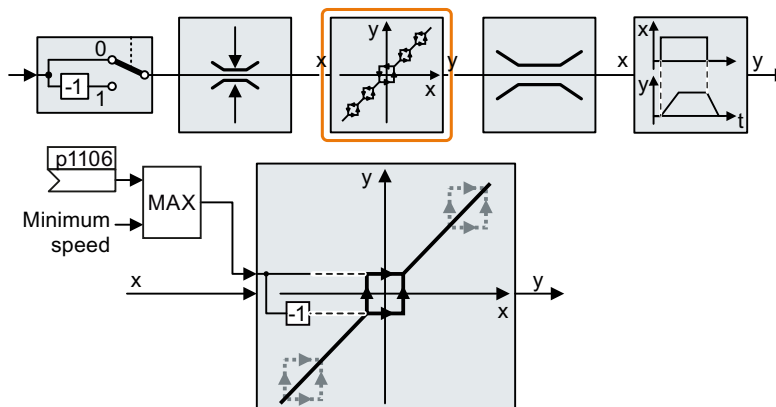
Skip frequency bands

The inverter has four skip frequency bands that prevent continuous motor operation within a specific speed range. Further information is provided in function diagram 3050 of the List Manual.

 Manuals and technical support (Page 504)

Minimum speed

The inverter prevents continuous motor operation at speeds $<$ minimum speed.



Speeds where the absolute value is less than the minimum speed are only possible during motor operation when accelerating or braking.

Table 6-42 Setting the minimum speed

Parameter	Description
p1080	Minimum speed (factory setting: 0 rpm)
p1106	CI: Minimum speed signal source (factory setting: 0) Dynamic specification of the minimum speed

NOTICE

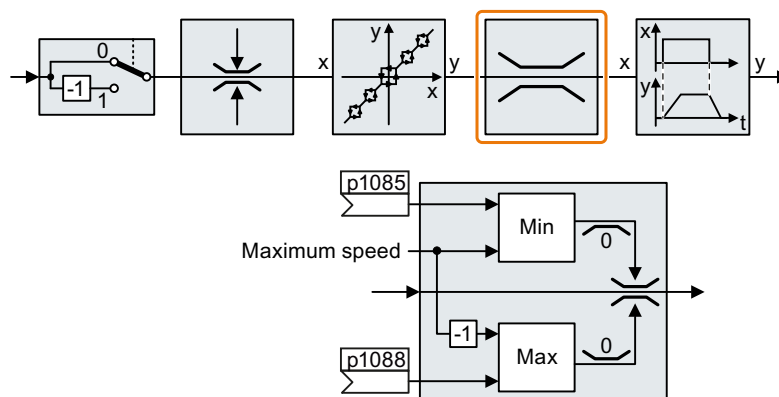
Material damage caused by motor rotating in the incorrect direction

If you are using an analog input as speed setpoint source, then for a setpoint = 0 V, noise voltages can be superimposed on the analog input signal. After the on command, the motor accelerates up to the minimum frequency in the direction of the random polarity of the noise voltage. A motor rotating in the wrong direction can cause significant material damage to the machine or system.

- Inhibit the motor direction of rotation that is not permissible.

6.16.5 Speed limitation

The maximum speed limits the speed setpoint range for both directions of rotation.



The converter generates a message (fault or alarm) when the maximum speed is exceeded.

If you must limit the speed depending on the direction of rotation, then you can define speed limits for each direction.

Table 6-43 Parameters for the speed limitation

Parameter	Description
p1082	Maximum speed (factory setting: 1500 rpm)
p1083	Speed limit, positive direction of rotation (factory setting: 210,000 rpm)
p1085	CI: Speed limit, positive direction of rotation (factory setting: 1083)
p1086	Speed limit, negative direction of rotation (factory setting: -210,000 rpm)
p1088	CI: Speed limit, negative direction of rotation (factory setting: 1086)

6.16.6 Ramp-function generator

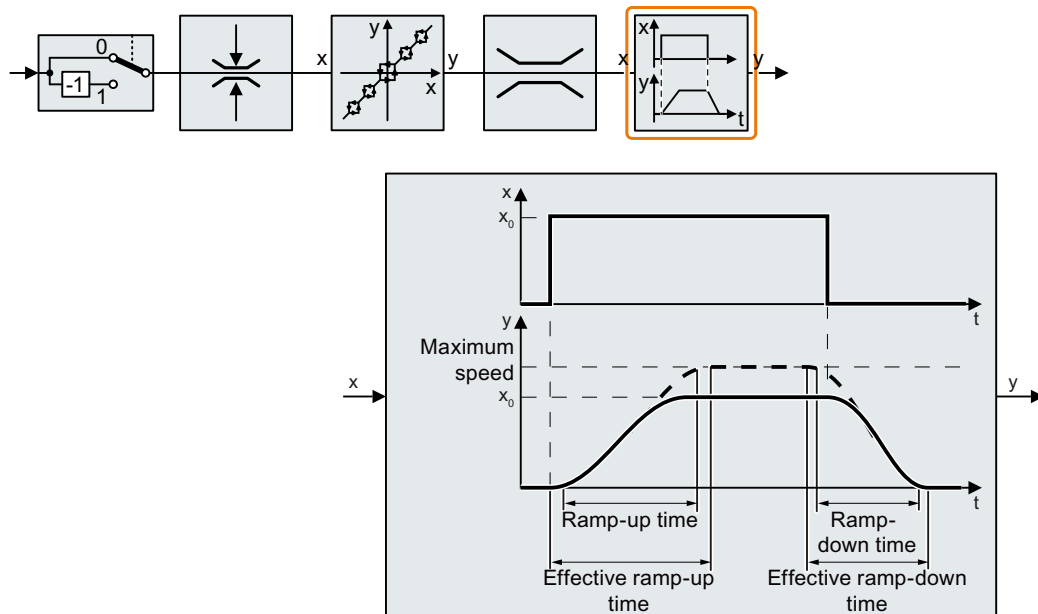
The ramp-function generator in the setpoint channel limits the rate of change of the speed setpoint (acceleration). Reduced acceleration lowers the accelerating torque of the motor. In this case, the motor reduces the load on the mechanical system of the driven machine.

You can select between two different ramp-function generator types:

- **Extended ramp-function generator**
The extended ramp-function generator limits not only the acceleration but also the change in acceleration (jerk) by rounding the setpoint. In this case, the torque does not rise suddenly in the motor.
- **Basic ramp-function generator**
The basic ramp-function generator limits the acceleration, however not the rate the acceleration changes (jerk).

Extended ramp-function generator

The ramp-up and ramp-down times of the extended ramp-function generator can be set independently of each other. The optimum times that you select depend on your particular application in question and can range from just a few 100 ms (e.g. for belt conveyor drives) to several minutes (e.g. for centrifuges).



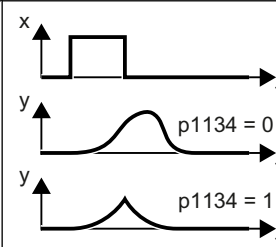
Initial and final rounding permit smooth, jerk-free acceleration and braking.

The ramp-up and ramp-down times of the motor are increased by the rounding times:

- Effective ramp-up time = $p1120 + 0.5 \times (p1130 + p1131)$.
- Effective ramp-down time = $p1121 + 0.5 \times (p1130 + p1131)$.

Table 6-44 Additional parameters to set the extended ramp-function generator

Parameter	Description
p1115	Ramp-function generator selection (factory setting: 1) Select ramp-function generator: 0: Basic ramp-function generator 1: Extended ramp-function generator
p1120	Ramp-function generator, ramp-up time (factory setting: 10 s) Accelerating time in seconds from zero speed up to the maximum speed p1082
p1121	Ramp-function generator, ramp-down time (factory setting: 10 s) Braking time in seconds from the maximum speed down to standstill
p1130	Ramp-function generator initial rounding time (factory setting: 0 s) Initial rounding for the extended ramp-function generator. The value applies for ramp up and ramp down.
p1131	Ramp-function generator final rounding time (factory setting: 0 s) Final rounding for the extended ramp-function generator. The value applies for ramp up and ramp down.
p1134	Ramp-function rounding type (factory setting: 0) 0: Continuous smoothing 1: Discontinuous smoothing
p1135	OFF3 ramp-down time (factory setting 0 s) The quick stop (OFF3) has its own ramp-down time.
p1136	OFF3 initial rounding time (factory setting: 0 s) Initial rounding for OFF3 for the extended ramp-function generator.
p1137	OFF3 final rounding time (factory setting: 0 s) Final rounding for OFF3 for the extended ramp-function generator.



You can find more information in function diagram 3070 and in the parameter list of the List Manual.

Setting the extended ramp-function generator

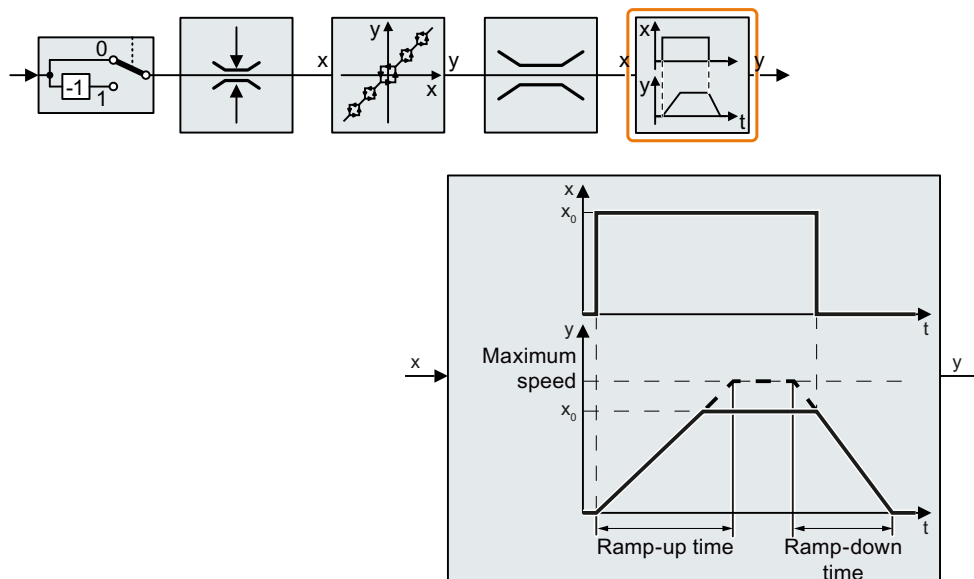
Procedure



1. Proceed as follows to set the extended ramp-function generator:
2.
 1. Enter the highest possible speed setpoint.
 2. Switch on the motor.

3. Evaluate your drive response.
 - If the motor accelerates too slowly, then reduce the ramp-up time.
An excessively short ramp-up time means that the motor will reach its current limiting when accelerating, and will temporarily not be able to follow the speed setpoint. In this case, the drive exceeds the set time.
 - If the motor accelerates too fast, then extend the ramp-up time.
 - Increase the initial rounding if the acceleration is jerky.
 - We recommend that you set the final rounding to the same value as the initial rounding.
 4. Switch off the motor.
 5. Evaluate your drive response.
 - If the motor decelerates too slowly, then reduce the ramp-down time.
The minimum ramp-down time that makes sense depends on your particular application. Depending on the Power Module used, for an excessively short ramp-down time, the converter either reaches the motor current, or the DC link voltage in the converter becomes too high. Depending on the converter setting, the real braking time exceeds the set ramp-down time, or the converter goes into a fault condition when braking.
 - Extend the ramp-down time if the motor is braked too quickly or the converter goes into a fault condition when braking.
 6. Repeat steps 1 ... 5 until the drive behavior meets the requirements of the machine or plant.
-  You have set the extended ramp-function generator.

Basic ramp-function generator



When compared to the extended ramp-function generator, the basic ramp-function generator has no rounding times.

Table 6-45 Parameters for setting the ramp-function generator

Parameter	Description
p1115 = 0	Ramp-function generator selection (factory setting: 1) Select ramp-function generator: 0: Basic ramp-function generator 1: Extended ramp-function generator
p1120	Ramp-function generator, ramp-up time (factory setting: 10 s) Accelerating time in seconds from zero speed up to the maximum speed p1082
p1121	Ramp-function generator, ramp-down time (factory setting: 10 s) Braking time in seconds from the maximum speed down to standstill
p1135	OFF3 ramp-down time (factory setting: 0 s) The quick stop (OFF3) has its own ramp-down time.

Changing the ramp-up and ramp-down times in operation

The ramping up and down time of the ramp-function generator can be changed during operation. The scaling value can come, e.g. from the fieldbus.

Table 6-46 Parameters for setting the scaling

Parameter	Description
p1138	Up ramp scaling (factory setting: 1) Signal source for scaling the acceleration ramp.
p1139	Deceleration ramp scaling (factory setting: 1) Signal source for scaling the deceleration ramp.

Example

In the following example, the higher-level control sets the ramp-up and ramp-down times of the inverter via PROFIBUS.

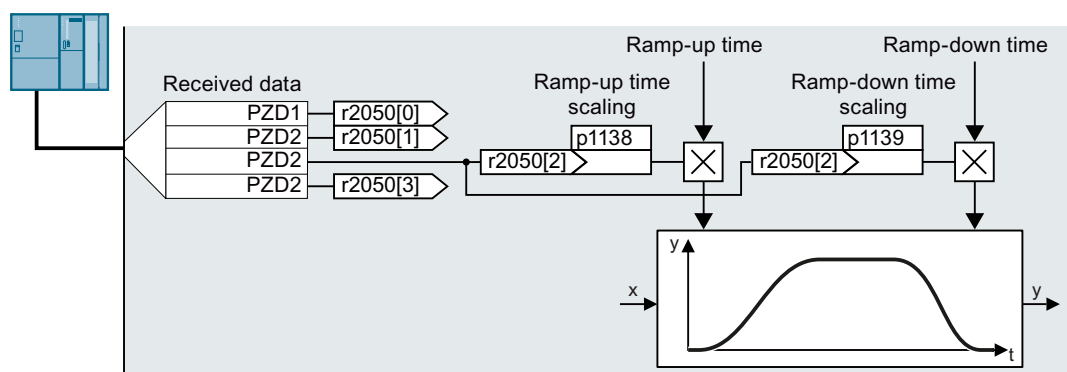


Figure 6-37 Example for changing the ramp-function generator times in operation

Preconditions

- You have commissioned the communication between the inverter and the control system.
- Free telegram 999 has been set in the inverter and in your higher-level control system.
 Extend telegrams and change signal interconnection (Page 209)
- The control sends the scaling value to the inverter in PZD 3.

Procedure



1. To interconnect the scaling of the ramp-up and ramp-down times with PZD receive word 3 from the fieldbus in the inverter, proceed as follows:
 2. 1. Set p1138 = 2050[2].
This means that you have interconnected the scaling factor for the ramp-up time with PZD receive word 3.
 2. Set p1139 = 2050[2].
This means that you have interconnected the scaling factor for the ramp-down time with PZD receive word 3.
-  The inverter receives the value for scaling the ramp-up and ramp-down times via PZD receive word 3.
-  Further information is provided in the Internet:
FAQ (<https://support.industry.siemens.com/cs/ww/en/view/82604741>)

6.17 PID technology controller



The technology controller controls process variables, e.g. pressure, temperature, level or flow.

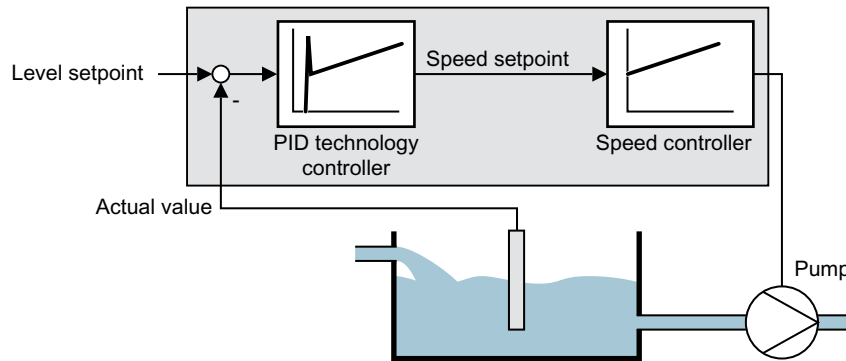
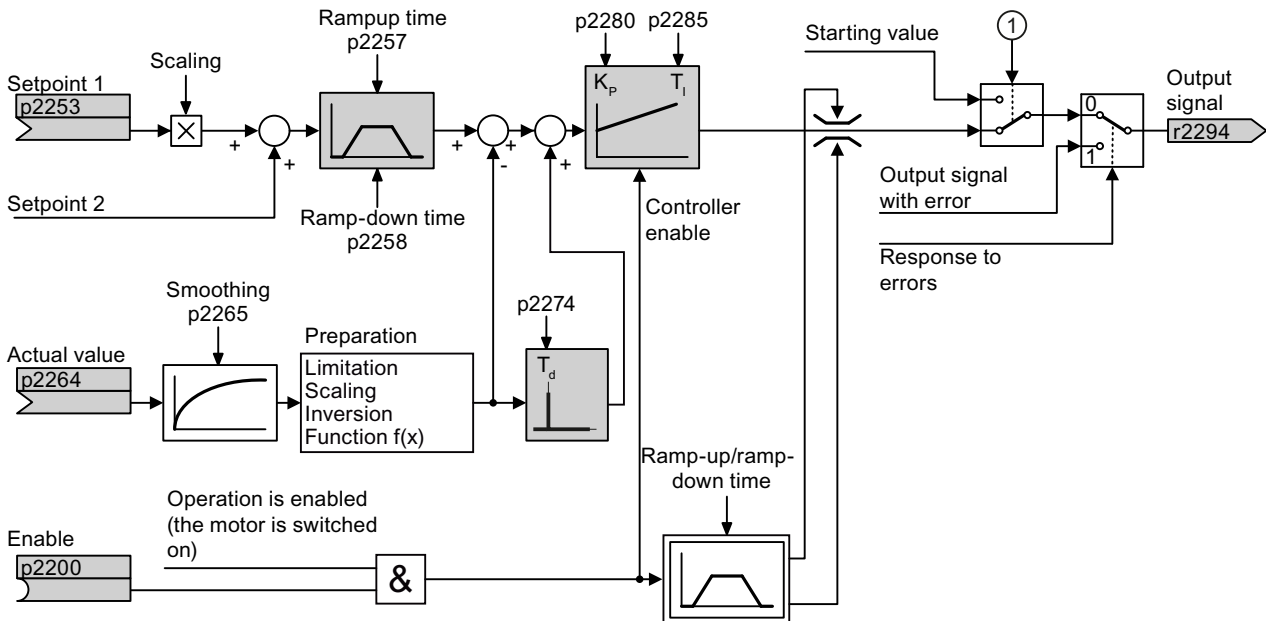


Figure 6-38 Example: Technology controller as a level controller

Simplified representation of the technology controller

The technology controller is implemented as a PID controller (controller with proportional, integral, and derivative action).



- ① The inverter uses the start value when all the following conditions are simultaneously satisfied:
- The technology controller supplies the main setpoint ($p2251 = 0$).
 - The ramp-function generator output of the technology controller has not yet reached the start value.

Figure 6-39 Simplified representation of the technology controller

The settings required as a minimum are marked in gray in the function diagram: Interconnect setpoint and actual values with signals of your choice, set the ramp-function generator and controller parameters K_P , T_i and T_d .

Further information about the following PID controller topics is available in the Internet:

- Setpoint specification: Analog value or fixed setpoint
- Setpoint channel: Scaling, ramp-function generator and filter
- Actual value channel: Filter, limiting and signal processing
- PID controller Principle of operation of the D component, inhibiting the I component and the control sense
- Enable, limiting the controller output and fault response



FAQ (<http://support.automation.siemens.com/WW/view/en/92556266>)

Setting the technology controller

Parameter	Remark
p2200	BI: Technology controller enable (factory setting: 0) 1 signal: Technology controller is enabled.
r2294	CO: Technology controller output signal To interconnect the main speed setpoint with the technology controller output, set p1070 = 2294.
p2253	CI: Technology controller setpoint 1 (factory setting: 0) Setpoint for the technology controller. Example: p2253 = 2224: The inverter interconnects the fixed setpoint p2201 with the setpoint of the technology controller. p2220 = 1: The fixed setpoint p2201 is selected.
p2264	CI: Technology controller actual value (factory setting: 0) Technology controller actual value.
p2257, p2258	Technology controller ramp-up time and ramp-down time (factory setting: 1 s)
p2274	Technology controller differentiation time constant T_d (factory setting: 0.0 s) The differentiation improves the rise time for very slow controlled variables, e.g. a temperature control.
p2280	Technology controller proportional gain K_p (factory setting: 1,0)
p2285	Technology controller integration time (integral time) T_i (factory setting: 30 s)

Advanced settings

Parameter	Remark
Limiting the output of the technology controller In the factory setting, the output of the technology controller is limited to $\pm$ maximum speed. You must change this limit, depending on your particular application. Example: The output of the technology controller supplies the speed setpoint for a pump. The pump should only run in the positive direction.	
p2297	CI: Technology controller maximum limiting signal source (factory setting: 1084)
p2298	CI: Technology controller minimum limiting signal source (factory setting: 2292)
p2291	CO: Technology maximum limiting (factory setting: 100 %)
p2292	CO: Technology minimum limiting (factory setting: 0 %)

Parameter	Remark
Manipulating the actual value of the technology controller	
p2267	Technology controller upper limit actual value (factory setting: 100 %)
p2268	Technology controller lower limit actual value (factory setting: -100 %)
p2269	Technology controller gain actual value (factory setting: 100 %)
p2271	Technology controller actual value inversion (sensor type)
	0: No inversion
	1: Inversion actual value signal If the actual value decreases with increasing motor speed, then p2271 must be set = 1.
p2270	Technology controller actual value function
	0: No function
	1: $\sqrt{\quad}$
	2: x^2
	3: x^3

For further information refer to the function block diagrams 7950 ff of the List Manual.

Autotuning of the PID controller

Autotuning is an inverter function for the automatic optimization of the PID controller.

For active autotuning, the inverter interrupts the connection between the PID controller and the speed controller. Rather than the PID controller output, the autotuning function provides the speed setpoint.

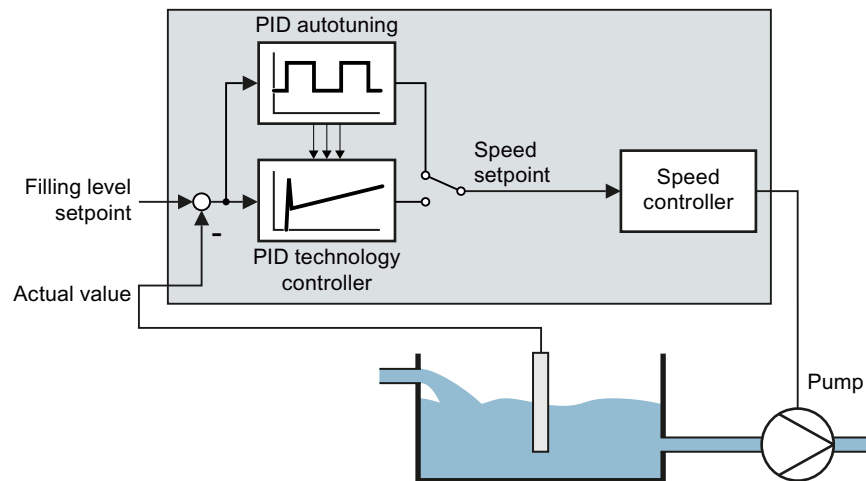


Figure 6-40 Autotuning of the PID controller using the example of a filling-level control

The speed setpoint results from the technology setpoint and a superimposed rectangular signal with amplitude p2355. If actual value = technology setpoint $\pm$ p2355, the autotuning function switches the polarity of the superimposed signal. This causes the inverter to excite the process variable for an oscillation.

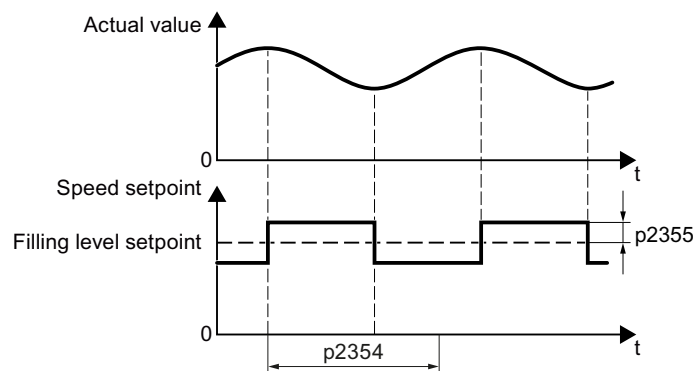


Figure 6-41 Example for speed setpoint and actual process value for autotuning

The inverter calculates the parameters of the PID controller from the determined oscillation frequency.

Autotune the PID controller

Requirements

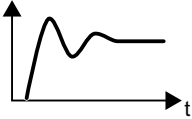
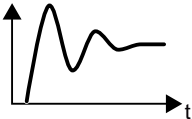
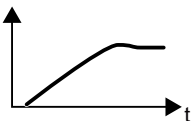
The PID technology controller must be set the same as when used in subsequent operation:

- The actual value is interconnected.
- Scalings, filter and ramp-function generator have been set.
- The PID technology controller is enabled (p2200 = 1 signal)

Procedure



1. Proceed as follows to auto tune the PID controller:
 2.
 1. Select with p2350 the appropriate controller setting.
 2. Switch on the motor.
The inverter signals Alarm A07444.
 3. Wait until alarm A07444 goes away.
The inverter has recalculated parameters p2280, p2274 and p2285.
If the inverter signals fault F07445:
 - If possible, double p2354 and p2355.
 - Repeat the autotuning with the changed parameters.
 4. Back up the calculated values so that they are protected against power failure, e.g. using the BOP-2: OPTIONS → RAM-ROM.
- You have auto tuned the PID controller.

Parameter	Remark
p2350	Enable PID autotuning (factory setting: 0) Automatic controller setting based on the "Ziegler Nichols" method. After completion of the autotuning, the inverter sets p2350 = 0.
	0: No function
	1: Controller setting after completion of the autotuning: The process variable follows the setpoint after a sudden setpoint change (step function) relatively quickly, however with an overshoot. 
	2: Faster controller setting than for p2350 = 1 with larger overshoot of the controlled variable. 
	3: Slower controller setting than for p2350 = 1. Overshoot of the controlled variable is, to a large extent, avoided. 
	4: Controller setting after completion of the autotuning as for p2350 = 1. Optimize only the P and I action of the PID controller. 
p2354	PID autotuning monitoring time (factory setting: 240 s) Monitoring time for the process response. p2354 must be greater than half the period of the process variable oscillation.
p2355	PID autotuning offset (factory setting: 5%) Autotuning offset. p2355 must be sufficiently large so that the amplitude of the process variable oscillation signal can be differentiated from any superimposed noise.

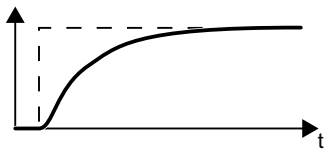
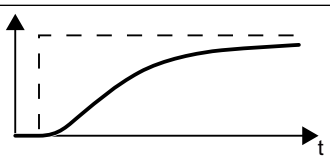
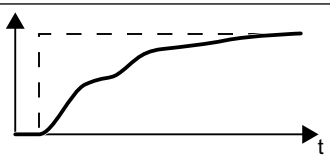
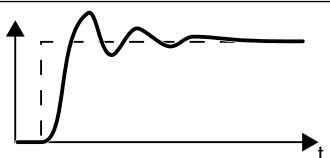
Manually setting the technology controller

Procedure



1. Proceed as follows to manually set the technology controller:

2.
 1. Temporarily set the ramp-up and ramp-down times of the ramp-function generator (p2257 and p2258) to zero.
 2. Enter a setpoint step and monitor the associated actual value, e.g. with the trace function of STARTER.
The slower the response of the process to be controlled, the longer you must monitor the controller response. Under certain circumstances (e.g. for a temperature control), you need to wait several minutes until you can evaluate the controller response.

	Optimum controller response for applications that do not permit any overshoot. The actual value approaches the setpoint without any significant overshoot.
	Optimum controller behavior for fast correction and quick compensation of disturbance components. The actual value approaches the setpoint and slightly overshoots, maximum 10% of the setpoint step.
	The actual value only slowly approaches the setpoint. • Increase the proportional component K_p and reduce the integration time T_i.
	Actual value only slowly approaches the setpoint with slight oscillation. • Increase the proportional component K_p and reduce the rate time T_d (differentiating time).
	The actual value quickly approaches the setpoint, but overshoots too much. • Decrease the proportional component K_p and increase the integration time T_i.

3. Set the ramp-up and ramp-down times of the ramp-function generator back to their original value.



You have now manually set the technology controller.

6.18 Motor control



The inverter has two alternative methods to control (closed loop) the motor speed:

- U/f control
- Vector control with speed controller

6.18.1 Reactor, filter and cable resistance at the inverter output

Correctly setting the components between the inverter and motor

Components between the inverter and the motor influence the closed-loop control quality of the inverter:

- Output reactor or sine-wave filter
In the factory setting, for the motor data identification, the inverter assumes that neither output reactor nor sine wave filter are connected at to the inverter output.
- Motor cable with unusually high cable resistance.
For the motor data identification, the inverter assumes a cable resistance = 20 % of the stator resistance of the cold motor.

You must correctly set the components between the inverter and motor to achieve an optimum closed-loop control quality

Setting the reactor, filter and cable resistance between the inverter and motor

Procedure



1. Proceed as follows to set the reactor, filter and cable resistance between the inverter and motor:

1. Set p0010 = 2.
2. Set the cable resistance in p0352.
3. Set p0230 to the appropriate value.
4. Set p0010 = 0.
5. Carry out the quick commissioning and the motor identification again.



Commissioning (Page 129)



You have set the reactor, filter and cable resistance between the inverter and motor

Parameter

Parameter	Description
p0010	Drive commissioning parameter filter (factory setting: 1) 0: Ready 2: Power unit commissioning
p0230	Drive filter type, motor side (factory setting: 0) 0: No filter 1: Output reactor 2: dv/dt filter 3: Siemens sine-wave filter 4: Sine wave filter, third-party manufacturer
p0350	Motor stator resistance, cold (factory setting: 0 Ω) When selecting a list motor (p0301), p0350 is preset (default setting) and is write protected.
p0352	Cable resistance (Factory setting: 0 Ω) If you set p0352 after the motor data identification, then you must subtract the difference that was changed with p0352, from the stator resistance p0350 or repeat the motor data identification.

Additional information on the parameters is provided in the List Manual.

6.18.2 V/f control

Overview of the U/f control

The U/f control is a closed-loop speed control with the following characteristics:

- The inverter controls the output voltage using the V/f characteristic
- The output frequency is essentially calculated from the speed setpoint and the number of pole pairs of the motor
- The slip compensation corrects the output frequency depending on the load and thus increases the speed accuracy
- Not using a PI controller prevents the speed control from becoming unstable
- In applications in which greater speed accuracy is required, a closed-loop control with load-dependent voltage boost can be selected (flux current control, FCC)

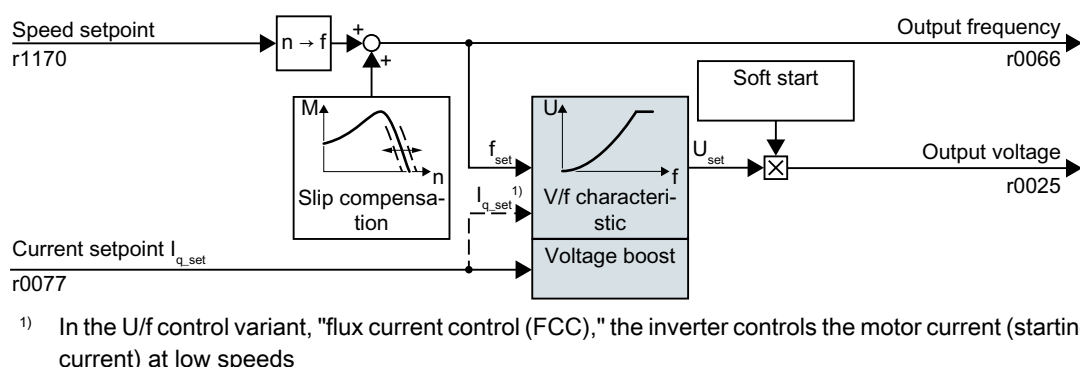


Figure 6-42 Simplified function diagram of the U/f control

One function not shown in the simplified function diagram is the resonance damping for damping mechanical oscillations. You will find the complete function diagrams 6300 et seq. in the List Manual.

For operation of the motor with U/f control, you must set at least the subfunctions shown with a gray background in the figure to adapt them to your application:

- V/f characteristic
- Voltage boost

Default setting after selecting the application class Standard Drive Control

Selecting application class Standard Drive Control in the quick commissioning adapts the structure and the setting options of the U/f control as follows:

- Starting current closed-loop control: At low speeds, a controlled motor current reduces the tendency of the motor to oscillate.
- With increasing speed, transition from closed-loop starting current control into U/f control with voltage boost depending on the load.
- The slip compensation is activated.
- Soft starting is not possible.
- Fewer parameters

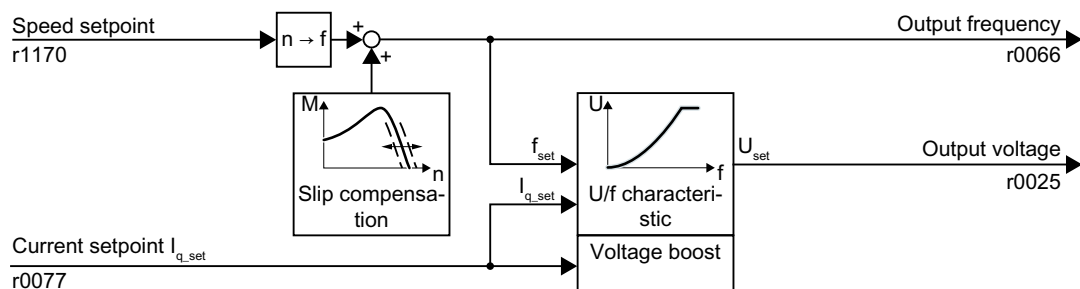
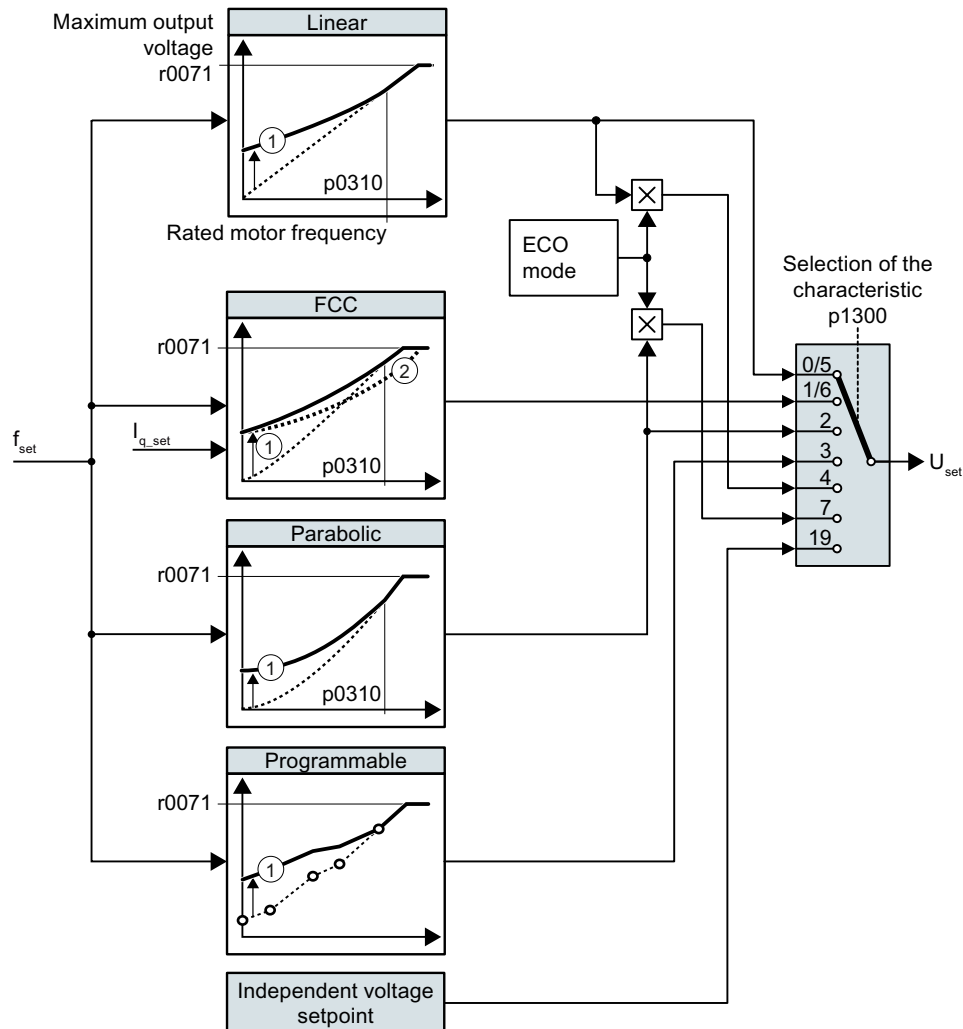


Figure 6-43 Default setting of the U/f control after selecting Standard Drive Control

The complete function diagrams 6850 ff. for application class Standard Drive Control are provided in the List Manual.

6.18.2.1 Characteristics of U/f control

The inverter has different V/f characteristics.



- ① The voltage boost of the characteristic optimizes the speed control at low speeds
- ② With the flux current control (FCC), the inverter compensates for the voltage drop in the stator resistor of the motor

Figure 6-44 Characteristics of V/f control

The inverter increases its output voltage to the maximum possible output voltage. The maximum possible output voltage of the inverter depends on the line voltage.

When the maximum output voltage is reached, the inverter only increases the output frequency. At this point, the motor enters the field weakening range: At constant torque, the slip decreases quadratically as the speed increases.

The value of the output voltage at the rated motor frequency also depends on the following variables:

- Ratio between the inverter size and the motor size
- Line voltage
- Line impedance
- Actual motor torque

The maximum possible output voltage as a function of the input voltage is provided in the technical data.



Technical data (Page 419)

Table 6-47 Linear and parabolic characteristics

Requirement	Application examples	Remark	Characteristic	Parameter
The required torque is independent of the speed	Conveyor belts, roller conveyors, chain conveyors, eccentric worm pumps, compressors, extruders, centrifuges, agitators, mixers	-	Linear	p1300 = 0
		The inverter equalizes the voltage drops across the stator resistance. Recommended for motors less than 7.5 kW. Precondition: You have set the motor data according to the rating plate and have performed the motor identification after quick commissioning.	Linear with Flux Current Control (FCC)	p1300 = 1
The required torque increases with the speed	Centrifugal pumps, radial fans, axial fans	Lower losses in the motor and inverter than for a linear characteristic.	Parabolic	p1300 = 2

Table 6-48 Characteristics for special applications

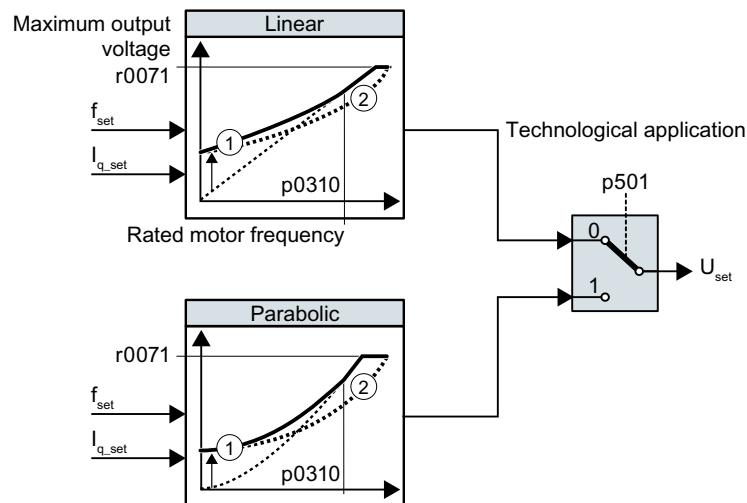
Requirement	Application examples	Remark	Characteristic	Parameter
Applications with a low dynamic response and constant speed	Centrifugal pumps, radial fans, axial fans	If the speed setpoint is reached, and remains unchanged for 5 seconds, then the inverter reduces its output voltage. As a consequence, the ECO mode saves energy with respect to the parabolic characteristic.	ECO mode	p1300 = 4 or p1300 = 7
The inverter must maintain the motor speed constant for the longest possible time.	Drives in the textile sector	When reaching the maximum current limit, the inverter only reduces the output voltage, but not the frequency.	Precise frequency characteristic	p1300 = 5 or p1300 = 6
Freely adjustable U/f characteristic	-	-	Adjustable characteristic	p1300 = 3
U/f characteristic with independent voltage setpoint	-	The interrelationship between the frequency and voltage is not calculated in the inverter, but is specified by the user.	Independent voltage setpoint	p1300 = 19

Additional information on U/f characteristics can be found in the parameter list and in the function diagrams 6300 ff of the List Manual.

Characteristics after selecting the application class Standard Drive Control

Selecting application class Standard Drive Control reduces the number of characteristics and the setting options:

- A linear and a parabolic characteristic are available.
- Selecting a technological application defines the characteristic.
- The following cannot be set - ECO mode, FCC, the programmable characteristic and a specific voltage setpoint.



- ① The closed-loop starting current control optimizes the speed control at low speeds
 ② The inverter compensates the voltage drop across the motor stator resistance

Figure 6-45 Characteristics after selecting Standard Drive Control

Table 6-49 Linear and parabolic characteristics

Requirement	Application examples	Remark	Characteristic	Parameter
The required torque is independent of the speed	Conveyor belts, roller conveyors, chain conveyors, eccentric worm pumps, compressors, extruders, centrifuges, agitators, mixers	-	Linear	p0501 = 0
The required torque increases with the speed	Centrifugal pumps, radial fans, axial fans	Lower losses in the motor and inverter than for a linear characteristic.	Parabolic	p0501 = 1

Additional information on the characteristics can be found in the parameter list and in the function diagrams 6851 ff of the List Manual.

6.18.2.2 Optimizing motor starting

After selection of the U/f characteristic, no further settings are required in most applications.

In the following circumstances, the motor cannot accelerate to its speed setpoint after it has been switched on:

- Load moment of inertia too high
- Load torque too large
- Ramp-up time p1120 too short

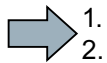
To improve the starting behavior of the motor, a voltage boost can be set for the U/f characteristic at low speeds.

Setting the voltage boost for U/f control

Requirements

- Set the ramp-up time of the ramp-function generator to a value 1 s (< 1 kW) ... 10 s (> 10 kW), depending on the power rating of the motor.
- Increase the starting current in steps of $\leq 5\%$. Excessively high values in p1310 ... p1312 can cause the motor to overheat and switch off (trip) the inverter due to overcurrent. If message A07409 appears, it is not permissible that you further increase the value of any of the parameters.

Procedure



1. Proceed as follows to set the voltage boost:
2.
 1. Switch on the motor with a setpoint of a few revolutions per minute.
 2. Check whether the motor rotates smoothly.
 3. If the motor does not rotate smoothly, or even remains stationary, increase the voltage boost p1310 until the motor runs smoothly.
 4. Accelerate the motor to the maximum speed with maximum load.
 5. Check that the motor follows the setpoint.
 6. If necessary, increase the voltage boost p1311 until the motor accelerates without problem.

In applications with a high break loose torque, you must also increase parameter p1312 in order to achieve a satisfactory motor response.



You have set the voltage boost.

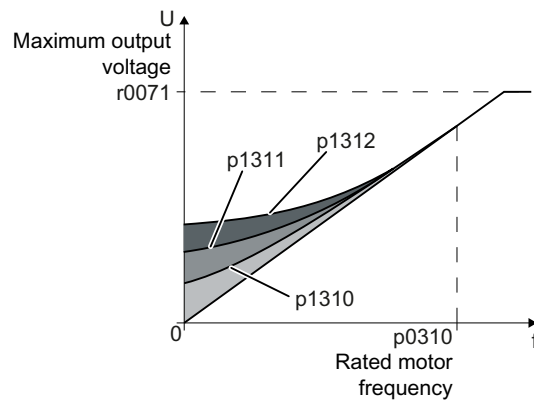


Figure 6-46 The resulting voltage boost using a linear characteristic as example

The inverter boosts the voltage corresponding to the starting currents p1310 ... p1312.

Parameter	Description
p1310	Starting current (voltage boost) permanent (factory setting 50%) Compensates for voltage drops caused by long motor cables and the ohmic losses in the motor.
p1311	Starting current (voltage boost) when accelerating (factory setting 0%) Provides additional torque when the motor accelerates.
p1312	Starting current (voltage boost) when starting (factory setting 0%) Provides additional torque, however, only when the motor accelerates for the first time after it has been switched on ("break loose torque").

You will find more information on this function in the parameter list and in function diagram 6301 in the List Manual.

6.18.2.3 Optimizing the motor startup for application class Standard Drive Control

After selecting application class Standard Drive Control, in most applications no additional settings need to be made.

At standstill, the inverter ensures that at least the rated motor magnetizing current flows. Magnetizing current p0320 approximately corresponds to the no-load current at 50% ... 80% of the rated motor speed.

In the following circumstances, the motor cannot accelerate to its speed setpoint after it has been switched on:

- Load moment of inertia too high
- Load torque too large
- Ramp-up time p1120 too short

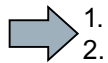
The current can be increased at low speeds to improve the starting behavior of the motor.

Starting current (boost) after selecting the application class Standard Drive Control

Requirements

- Set the ramp-up time of the ramp-function generator to a value 1 s (< 1 kW) ... 10 s (> 10 kW), depending on the power rating of the motor.
- Increase the starting current in steps of $\leq 5\%$. Excessively high values in p1310 ... p1312 can cause the motor to overheat and switch off (trip) the inverter due to overcurrent. If message A07409 appears, it is not permissible that you further increase the value of any of the parameters.

Procedure



1. Proceed as follows to set the voltage boost:
2.
 1. Switch on the motor with a setpoint of a few revolutions per minute.
 2. Check whether the motor rotates smoothly.
 3. If the motor does not rotate smoothly, or even remains stationary, increase the voltage boost p1310 until the motor runs smoothly.
 4. Accelerate the motor to the maximum speed with maximum load.
 5. Check that the motor follows the setpoint.
 6. If necessary, increase the voltage boost p1311 until the motor accelerates without problem.

In applications with a high break loose torque, you must also increase parameter p1312 in order to achieve a satisfactory motor response.



You have set the voltage boost.

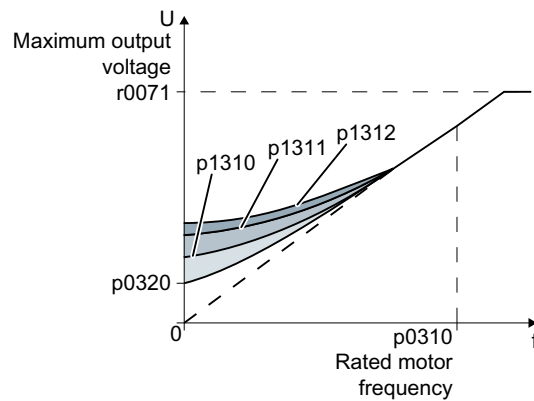


Figure 6-47 The resulting voltage boost using a linear characteristic as example

The inverter boosts the voltage corresponding to the starting currents p1310 ... p1312.

Parameter	Description
p1310	Starting current (voltage boost) permanent (factory setting 50%) Compensates for voltage drops caused by long motor cables and the ohmic losses in the motor. After commissioning, depending on the motor power rating and the technological application p0501, the inverter sets p1310.
p1311	Starting current (voltage boost) when accelerating (factory setting 0%) Provides additional torque when the motor accelerates. After commissioning, depending on the motor power rating and the technological application p0501, the inverter sets p1311.
p1312	Starting current (voltage boost) when starting (factory setting 0%) Provides additional torque, however, only when the motor accelerates for the first time after it has been switched on ("break loose torque").

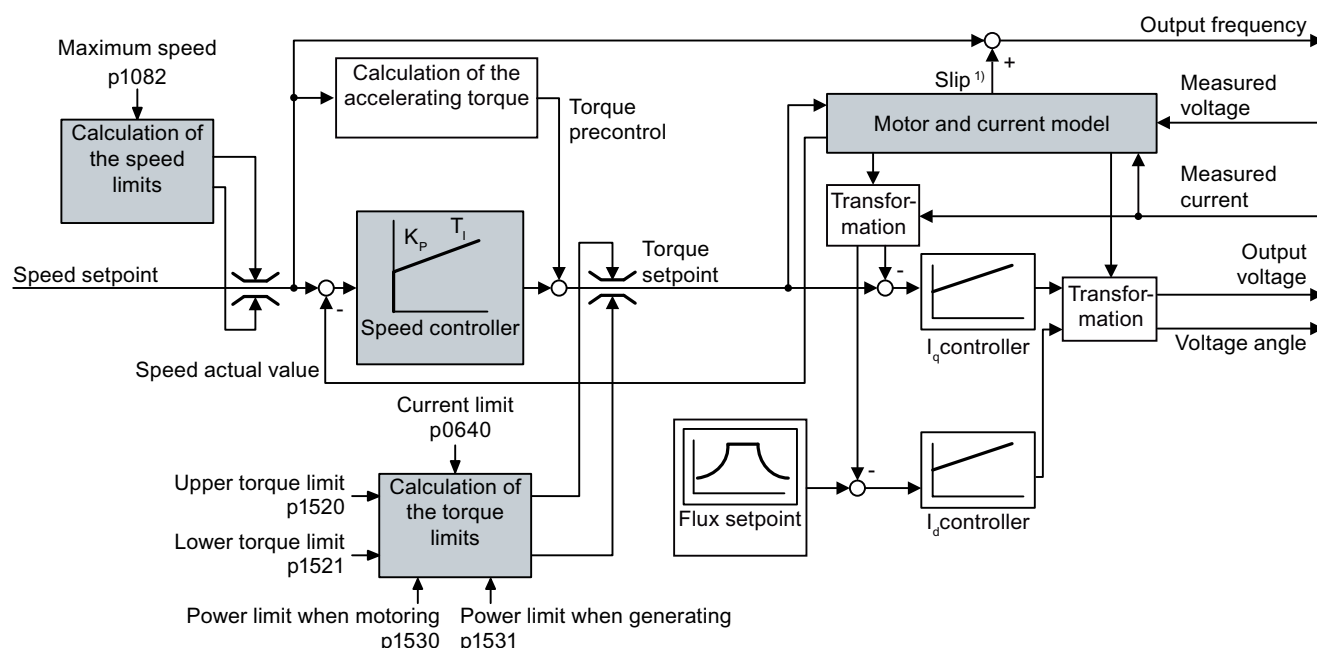
You can find more information about this function in the parameter list and in function diagram 6851 of the List Manual.

6.18.3 Sensorless vector control with speed controller

6.18.3.1 Overview

Overview

The vector control comprises closed-loop current control and a higher-level closed-loop speed control.



¹⁾ For induction motors

Figure 6-48 Simplified function diagram for sensorless vector control with speed controller

All of the function block diagrams 6020 ff. for vector control are provided in the List Manual:

Using the motor model, the inverter calculates the following closed-loop control signals from the measured phase currents and the output voltage:

- Current component I_d
- Current component I_q
- Speed actual value

The setpoint of the current component I_d (flux setpoint) is obtained from the motor data. For speeds above the rated speed, the inverter reduces the flux setpoint along the field weakening characteristic.

When the speed setpoint is increased, the speed controller responds with a higher setpoint for the current component I_q (torque setpoint). The closed-loop control responds to a higher torque setpoint by adding a higher slip frequency to the output frequency. Also in the motor, the higher output frequency results in a higher slip, which is proportional to the accelerating

torque. I_q and I_d controllers keep the motor flux constant using the output voltage, and adjust the matching current component I_q in the motor.

In order to achieve a satisfactory controller response, as a minimum, you must match the subfunctions having a gray background as shown in the diagram above with your particular application.

- **Motor and current model:** In the quick commissioning, correctly set the motor data on the rating plate corresponding to the connection type (Y/ Δ), and carry out the motor data identification routine at standstill.
- **Speed limits and torque limits:** In the quick commissioning, set the maximum speed (p1082) and current limit (p0640) to match your particular application. When exiting quick commissioning, the inverter calculates the torque and power limits corresponding to the current limit. The actual torque limits are obtained from the converted current and power limits and the set torque limits.
- **Speed controller:** Start the rotating measurement of the motor data identification. You must manually optimize the controller if the rotating measurement is not possible.

Default settings after selecting the application class Dynamic Drive Control

Selecting application class Dynamic Drive Control in the quick commissioning adapts the structure of the vector control, and reduces the setting options:

	Vector control after selecting the application class Dynamic Drive Control	Vector control without selecting an application class
Closed-loop torque control without higher-level speed controller	Not possible	Possible
Droop	Not possible	Possible
K_p - and T_i adaptation	Simplified	Advanced
Hold or set the integral component of the speed controller	Not possible	Possible
Acceleration model for pre-control	Default setting	Can be activated
Motor data identification at standstill or with rotating measurement	Shortened, with optional transition into operation	Complete

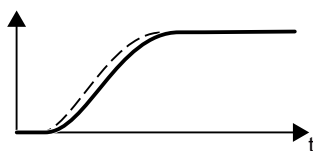
6.18.3.2 Optimizing the closed-loop speed controller

Optimum control response - post optimization not required

Preconditions for assessing the controller response:

- The moment of inertia of the load is constant and does not depend on the speed
- The inverter does not reach the set torque limits during acceleration
- You operate the motor in the range 40 % ... 60 % of its rated speed

If the motor exhibits the following response, the speed control is well set and you do not have to adapt the speed controller manually:

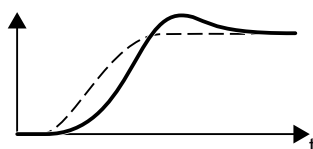


The speed setpoint (broken line) increases with the set ramp-up time and rounding.

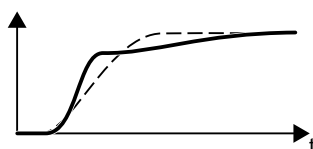
The speed actual value follows the setpoint without any overshoot.

Control optimization required

In some cases, the self optimization result is not satisfactory, or self optimization is not possible as the motor cannot freely rotate.



Initially, the speed actual value follows the speed setpoint with some delay, and then overshoots the speed setpoint.



Initially, the speed actual value increases faster than the speed setpoint. The setpoint passes the actual value before reaching its final value. Finally, the actual value approaches the setpoint without any overshoot.

In these cases we recommend that you manually optimize the speed control.

Optimizing the speed controller

Requirements

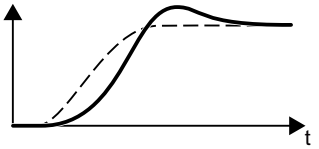
- Torque precontrol is active: p1496 = 100 %.
- The load moment of inertia is constant and independent of the speed.
- The inverter requires 10 % ... 50 % of the rated torque to accelerate.
When necessary, adapt the ramp-up and ramp-down times of the ramp-function generator (p1120 and p1121).
- STARTER and Startdrive have trace functions that allow the speed setpoint and actual value to be recorded.

Procedure



1. To optimize the speed controller, proceed as follows:
1. Switch on the motor.
2. Enter a speed setpoint of approximately 40 % of the rated speed.
3. Wait until the actual speed has stabilized.
4. Increase the setpoint up to a maximum of 60 % of the rated speed.
5. Monitor the associated characteristic of the setpoint and actual speed.

6. Optimize the controller by adapting the ratio of the moments of inertia of the load and motor (p0342):

	Initially, the speed actual value follows the speed setpoint with some delay, and then overshoots the speed setpoint. • Increase p0342
	Initially, the speed actual value increases faster than the speed setpoint. The setpoint passes the actual value before reaching its final value. Finally, the actual value approaches the setpoint without any overshoot. • Reduce p0342

7. Switch off the motor.
8. Set p0340 = 4. The inverter again calculates the speed controller parameters.
9. Switch on the motor.
10. Over the complete speed range check as to whether the speed control operates satisfactorily with the optimized settings.

 You have optimized the speed controller.

When necessary, set the ramp-up and ramp-down times of the ramp-function generator (p1120 and p1121) back to the value before optimization.

Mastering critical applications

The drive control can become unstable for drives with a high load moment of inertia and gearbox backlash or a coupling between the motor and load that can possibly oscillate. In this case, we recommend the following settings:

- Increase the speed actual value smoothing.
- Increase the integral action time T_I : $T_I \geq 4 \times$ speed actual value smoothing.
- If, after these measures, the speed controller does not operate with an adequate dynamic performance, increase the P gain K_P step-by-step.

The most important parameters

Table 6-50 Encoderless speed control

Parameter	Description
p0342	Moment of inertia ratio, total to motor (factory setting: 1.0)
p1496	Acceleration precontrol scaling (factory setting: 0 %) For the rotating measurement of the motor data identification the inverter sets the parameters to 100 %.
p1452	Speed controller speed actual value smoothing time (without encoder) (factory setting: 10 ms)
p1470	Speed controller operation without encoder P gain (factory setting: 0.3)
p1472	Speed controller operation without encoder integral action time (factory setting: 20 ms)

6.18.3.3 Advanced settings

K_p - and T_i adaptation

K_p and T_i adaptation suppress speed control oscillations that may occur. The "rotating measurement" of the motor data identification optimizes the speed controller. If you have performed the rotating measurement, then the K_p - and T_i adaptation has been set.

You can find additional information in the List Manual:

- Vector control with speed controller: Function diagram 6050
- Vector control after presetting the application class Dynamic Drive Control Function diagram 6824

Droop

For mechanically coupled drives, there is the risk that the drives oppose one another: Small deviations in the speed setpoint or actual value of the coupled drives can mean that the drives are operated with significantly different torques.

The droop function ensures even torque distribution between several mechanically coupled drives.

The droop function reduces the speed setpoint as a function of the torque setpoint.

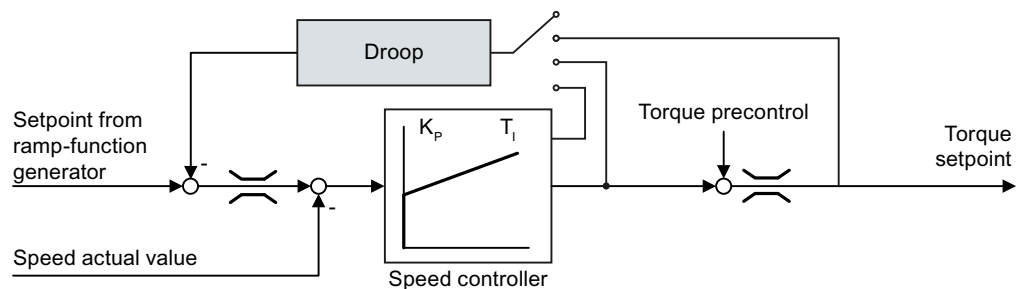


Figure 6-49 Effect of droop in the speed controller

When droop is active, the ramp-function generators of all of the coupled drives must be set to have identical ramp-up and ramp-down times as well as rounding-off.

Par.	Explanation
r1482	Speed controller I torque output
p1488	Droop input source (factory setting: 0) 0: Droop feedback not connected 1: Droop from the torque setpoint 2: Droop from the speed control output 3: Droop from the integral output, speed controller
p1489	Droop feedback scaling (factory setting: 0.05) A value of 0.05 means: At the rated motor torque, the inverter reduces the speed by 5% of the rated motor speed.
r1490	Droop feedback speed reduction
p1492	Droop feedback enable (factory setting: 0)

After selecting application class "Dynamic Drive Control", droop is no longer possible.
You can find additional information in the List Manual, function block diagram 6030.

6.18.3.4 Friction characteristic

Function

In many applications, e.g. applications with geared motors or belt conveyors, the frictional torque of the load is not negligible.

The inverter provides the possibility of precontrolling the torque setpoint, bypassing the speed controller. The precontrol reduces overshooting of the speed after speed changes.

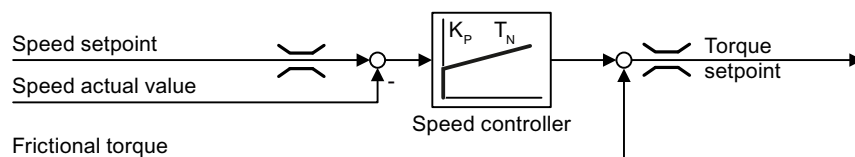


Figure 6-50 Precontrol of the speed controller with frictional torque

The inverter calculates the current frictional torque from a friction characteristic with 10 intermediate points.

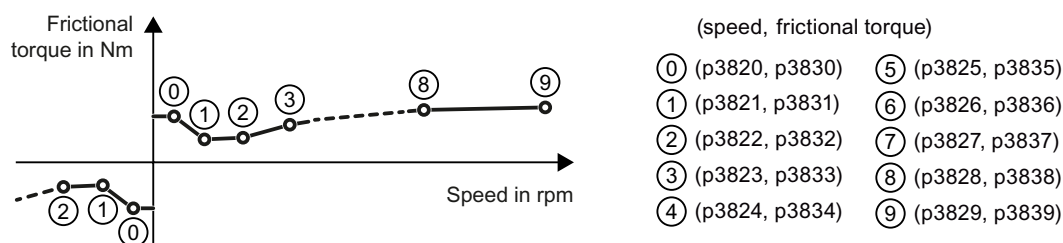


Figure 6-51 Friction characteristic

The intermediate points of the friction characteristic are defined for positive speeds. In the negative direction of rotation, the inverter uses the intermediate points with a negative sign.

Recording a friction characteristic

After quick commissioning, the inverter sets the speeds of the intermediate points to values suitable for the rated speed of the motor. The frictional torque of all intermediate points is still equal to zero. On request, the inverter records the friction characteristic: The inverter accelerates the motor step by step up to the rated speed, measures the frictional torque and writes the frictional torque into the intermediate points of the friction characteristic.

Requirement

The motor is permitted to accelerate up to the rated speed without endangering persons or property.



Procedure

- 1.
- 2.

To record the friction characteristic, proceed as follows:

1. Set P3845 = 1: The inverter accelerates the motor successively in both directions of rotation and averages the measurement results of the positive and negative directions.
2. Switch on the motor (ON/OFF1 = 1).
3. The inverter accelerates the motor.
During measurement, the inverter signals the alarm A07961.
When the inverter has determined all the intermediate points of the friction characteristic without fault code F07963, the inverter stops the motor.



You have recorded the friction characteristic.

Adding friction characteristic for the torque setpoint

If you enable the friction characteristic (p3842 = 1), the inverter adds the output of the friction characteristic r3841 to the torque setpoint.

Parameter

Parameter	Explanation
p3820 ... p2839	Intermediate points of the friction characteristic [rpm; Nm]
r3840	Friction characteristic status word
	.00 1 signal: Friction characteristic OK
	.01 1 signal: Determination of the friction characteristic is active
	.02 1 signal: Determination of the friction characteristic is complete
	.03 1 signal: Determination of the friction characteristic has been aborted
	.08 1 signal: Friction characteristic positive direction
r3841	Friction characteristic, output [Nm]
p3842	Activate friction characteristic 0: Friction characteristic deactivated 1: Friction characteristic activated
p3845	Activate friction characteristic plot (factory setting: 0) 0: Friction characteristic plot deactivated 1: Friction characteristic plot activated, both directions 2: Friction characteristic plot activated, positive direction 3: Friction characteristic plot activated, negative direction
p3846	Friction characteristic plot ramp-up/ramp-down time (factory setting: 10 s) Ramp-up/ramp-down time for automatic plotting of the friction characteristic.
p3847	Friction characteristic plot warm-up period (factory setting: 0 s) At the start of automatic plotting, the inverter accelerates the motor up to the speed = p3829 und keeps the speed constant for this time.

You will find more information in the List Manual.

6.18.3.5 Moment of inertia estimator

Background

From the load moment of inertia and the speed setpoint change, the inverter calculates the accelerating torque required for the motor. Via the speed controller precontrol, the accelerating torque specifies the main percentage of the torque setpoint. The speed controller corrects inaccuracies in the precontrol (feed-forward control).

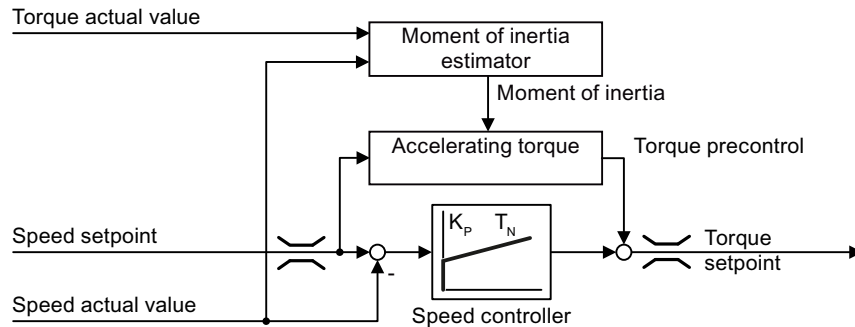


Figure 6-52 Influence of the moment of inertia estimator on the speed control

The more precise the value of the moment of inertia in the inverter, the lower the overshoot after speed changes.

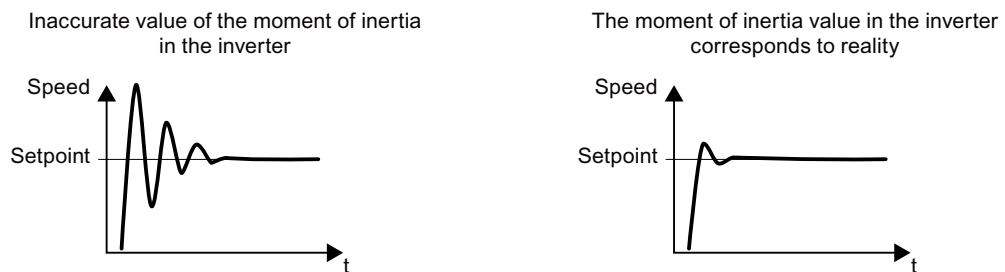


Figure 6-53 Influence of the moment of inertia on the speed

Function

From the actual speed, the actual motor torque and the frictional torque of the load, the inverter calculates the total moment of inertia of the load and motor.

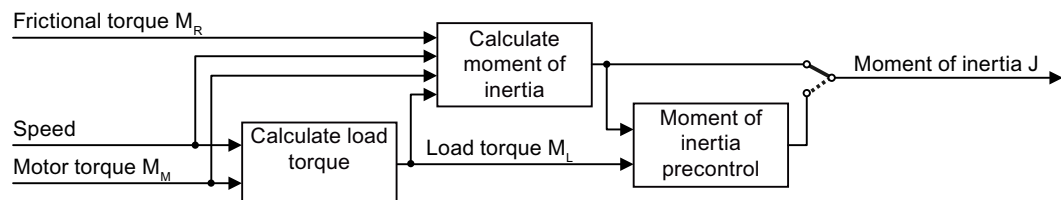


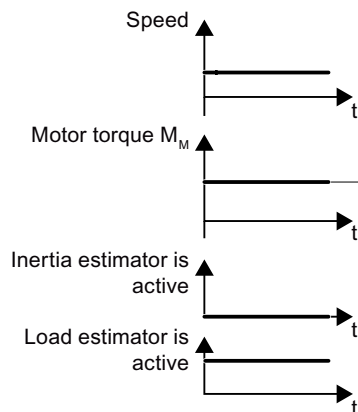
Figure 6-54 Overview of the function of the moment of inertia estimator

When using the moment of inertia estimator, we recommend that you also activate the friction characteristic.



Friction characteristic (Page 291)

Calculating the load torque

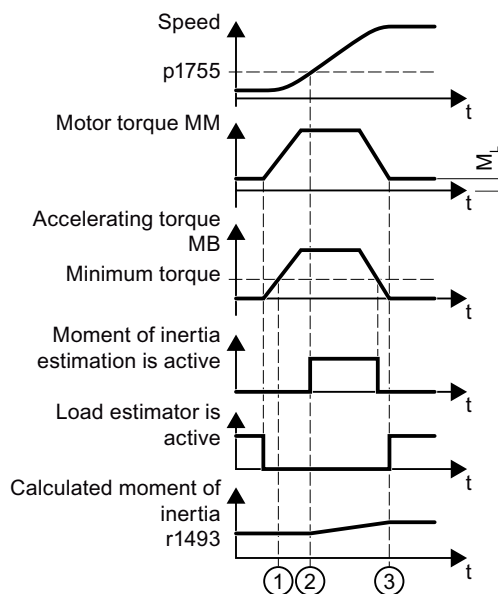


At low speeds, the inverter calculates the load torque M_L from the actual motor torque.

The calculation takes place under the following conditions:

- Speed $\geq p1226$
- Acceleration setpoint $< 8 \text{ 1/s}^2$ (Δ speed change 480 rpm per s)
- Acceleration $\times$ moment of inertia (r1493) $< 0.9 \times p1560$

Calculating the moment of inertia



For higher speed changes, the inverter initially calculates the accelerating torque M_B as difference between the motor torque M_M , load torque M_L and frictional torque M_R :

$$M_B = M_M - M_L - M_R$$

Moment of inertia J of the motor and load is obtained from the accelerating torque M_B and angular acceleration α (α = rate at which the speed changes):

$$J = M_B / \alpha$$

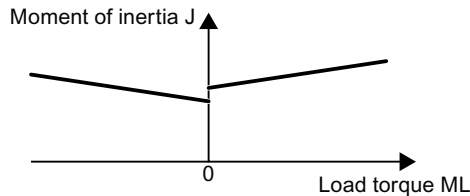
If all of the following conditions are met, the inverter calculates the moment of inertia:

- ① The rated accelerating torque M_B must satisfy the following two conditions:
 - The sign of M_B is the same as the direction of the actual acceleration
 - $M_B > p1560 \times$ rated motor torque (r0333)
- ② speed $> p1755$
- The inverter has calculated the load torque in at least one direction of rotation.
- Acceleration setpoint $> 8 \text{ 1/s}^2$ (Δ speed change 480 rpm per s)
- ③ The inverter calculates the load torque again after acceleration.

Moment of inertia precontrol

In applications where the motor predominantly operates with a constant speed, the inverter can only infrequently calculate the moment of inertia using the function described above. Moment of inertia precontrol is available for situations such as these. The moment of inertia precontrol assumes that there is an approximately linear relationship between the moment of inertia and the load torque.

Example: For a horizontal conveyor, in a first approximation, the moment of inertia depends on the load.



The relationship between load torque and torque is saved in the inverter as linear characteristic.

In a positive direction of rotation:

$$\text{Moment of inertia } J = p5312 \times \text{load torque } M_L + p5313$$

In a negative direction of rotation:

$$\text{Moment of inertia } J = p5314 \times \text{load torque } M_L + p5315$$

You have the following options to determine the characteristic:

- You already know the characteristic from other measurements. In this case, you must set the parameters to known values when commissioning the system.
- The inverter iteratively determines the characteristic by performing measurements while the motor is operational.

Activating the moment of inertia estimator

The moment of inertia estimator is deactivated in the factory setting. $p1400.18 = 0$, $p1400.20 = 0$, $p1400.22 = 0$.

If you performed the rotating measurement for the motor identification during quick commissioning, we recommend leaving the moment of inertia estimator deactivated.

Preconditions

- You have selected encoderless vector control.
- The load torque must be constant whilst the motor accelerates or brakes.
Typical of a constant load torque are conveyor applications and centrifuges, for example. Fan applications, for example, are not permitted.
- The speed setpoint is free from superimposed unwanted signals.
- The motor and load are connected to each other with an interference fit.
Drives with slip between the motor shaft and load are not permitted, e.g. as a result of loose or worn belts.

If the conditions are not met, you must not activate the moment of inertia estimator.

Procedure

1. To activate the moment of inertia estimator, proceed as follows:
2.
 1. Set $p1400.18 = 1$
 2. Check: $p1496 \neq 0$
 3. Activate the acceleration model of the speed controller pre-control: $p1400.20 = 1$.



You have activated the moment of inertia estimator.

The most important settings

Parameter	Explanation	
r0333	Rated motor torque [Nm]	
p0341	Motor moment of inertia (factory setting: 0 kgm ²) The inverter sets the parameter when selecting a listed motor. The parameter is then write-protected.	
p0342	Moment of inertia ratio, total to motor (factory setting: 1) Ratio of moment of inertia load + motor to moment of inertia of motor without load	
p1400	Speed control configuration	
	.18	1 signal: Moment of inertia estimator active
	.20	1 signal: Acceleration model on
	.22	1 signal: Moment of inertia estimator retain value when motor switched off
	0 signal	Moment of inertia estimator reset value to initial value J_0 when motor switched off: $J_0 = p0341 \times p0342 + p1498$ If the load torque can change when the motor is switched off, set $p1400.22 = 0$.
r1407	.24	1 signal: Shortened moment of inertia estimation is active. $p1400.24 = 1$ reduces the duration of the moment of inertia estimation. Disadvantage: If the accelerating torque is not constant while calculating the moment of inertia, the calculation of the moment of inertia using $p1400.24 = 1$ is less precise.
	Status word, speed controller	
	.24	1 signal: Moment of inertia estimator is active
	.25	1 signal: Load estimator is active
	.26	1 signal: Moment of inertia estimator is engaged
	.27	1 signal: Shortened moment of inertia estimation is active.
r1493	Total moment of inertia, scaled $r1493 = p0341 \times p0342 \times p1496$	
p1496	Acceleration precontrol scaling (factory setting: 0%) According to rotating measurement of the motor data identification is $p1496 = 100\%$.	
p1498	Load moment of inertia (factory setting: 0 kgm ²)	

Parameter	Explanation
p1502	Freeze moment of inertia estimator (factory setting: 0) If the load torque changes when accelerating the motor, set this signal to 0.
	0 signal Moment of inertia estimator is active
	1 signal Determined moment of inertia is frozen
p1755	Motor model changeover speed encoderless operation Defines the switchover between open-loop and closed-loop controlled operation of the encoderless vector control. When selecting the closed-loop speed control, the inverter sets $p1755 = 13.3\% \times \text{rated speed}$.

Advanced settings

Parameter	Explanation
p1226	Standstill detection, speed threshold (Factory setting: 20 rpm) The moment of inertia estimator only measures the load torque for speeds $\geq p1226$. p1226 also defines from which speed the inverter switches-off the motor for OFF1 and OFF3.
p1560	Moment of inertia estimator accelerating torque threshold value (factory setting: 10%)
p1561	Moment of inertia estimator change time inertia (factory setting: 500 ms)
p1562	Moment of inertia estimator change time load (factory setting: 10 ms)
	The lower that p1561 or p1562 is, the shorter the moment of inertia estimator measurements. The larger p1561 or p1562 is, the more accurate the results provided by the moment of inertia estimator.
p1563	Moment of inertia estimator load torque positive direction of rotation (factory setting: 0 Nm)
p1564	Moment of inertia estimator load torque negative direction of rotation (factory setting: 0 Nm)
p5310	Moment of inertia precontrol configuration (factory setting: 0000 bin)
	.00 1 signal: Activates calculation of the characteristic (p5312 ... p5315)
	.01 1 signal: Activates moment of inertia precontrol
	p5310.00 = 0, p5310.01 = 0 Deactivating moment of inertia precontrol
	p5310.00 = 1, p5310.01 = 0 Adapting the moment of inertia precontrol
	p5310.00 = 0, p5310.01 = 1 Activating the moment of inertia precontrol. The characteristic of the moment of inertia precontrol remains unchanged.
	p5310.00 = 1, p5310.01 = 1 Activating the moment of inertia precontrol. The inverter adapts the characteristic in parallel.

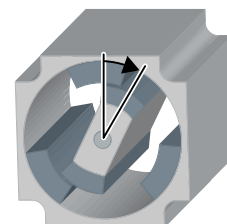
Parameter	Explanation	
r5311	Moment of inertia precontrol status word	
	.00	1 signal: New measuring points for the characteristic of the moment of inertia precontrol are available
	.01	1 signal: New parameters have been calculated
	.02	1 signal: Moment of inertia precontrol active
	.03	1 signal: The characteristic in the positive direction of rotation has been calculated and is ready
	.04	1 signal: The characteristic in the negative direction of rotation has been calculated and is ready
	.05	1 signal: The inverter writes actual results to the parameter
p5312	Moment of inertia precontrol linear positive (factory setting: 0 1/s ²)	In a positive direction of rotation: Moment of inertia = p5312 × load torque + p5313
p5313	Moment of inertia precontrol constant positive (factory setting: 0 kgm ²)	
p5314	Moment of inertia precontrol linear negative (factory setting: 0 1/s ²)	In a negative direction of rotation: Moment of inertia = p5314 × load torque + p5315
p5315	Moment of inertia precontrol constant negative (factory setting: 0 kgm ²)	

6.18.3.6 Pole position identification

The pole position of a synchronous motor

The pole position of a synchronous motor is the deviation between the magnetic axis in the rotor and the magnetic axis in the stator.

The image below shows you the pole position of a synchronous motor in a simplified cross-section.



The inverter must know the pole position of the rotor in the motor in order to be able to control the torque and speed of a synchronous motor.

Pole position identification

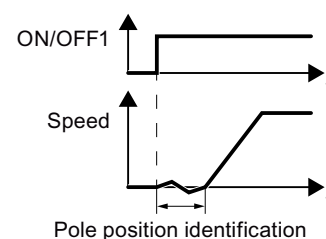
The inverter must measure the pole position for motors not equipped with an encoder, or for encoders, which do not supply the information regarding the pole position.

If you are using a Siemens motor, then the inverter automatically selects the appropriate technique to determine the pole position, and when required starts the pole position identification.

Motor without an encoder

Each time the motor is switched on (ON/OFF1 command), the inverter measures the pole position.

As a result of the measurement, the motor responds to an ON command with a delay of up to 1 second. The motor shaft can rotate slightly during the measurement.



Identifying the pole position using a measurement

6.18.4 Torque control

Torque control is part of the vector control and normally receives its setpoint from the speed controller output. By deactivating the speed controller and directly entering the torque setpoint, the closed-loop speed control becomes closed-loop torque control. The inverter then no longer controls the motor speed, but the torque that the motor generates.

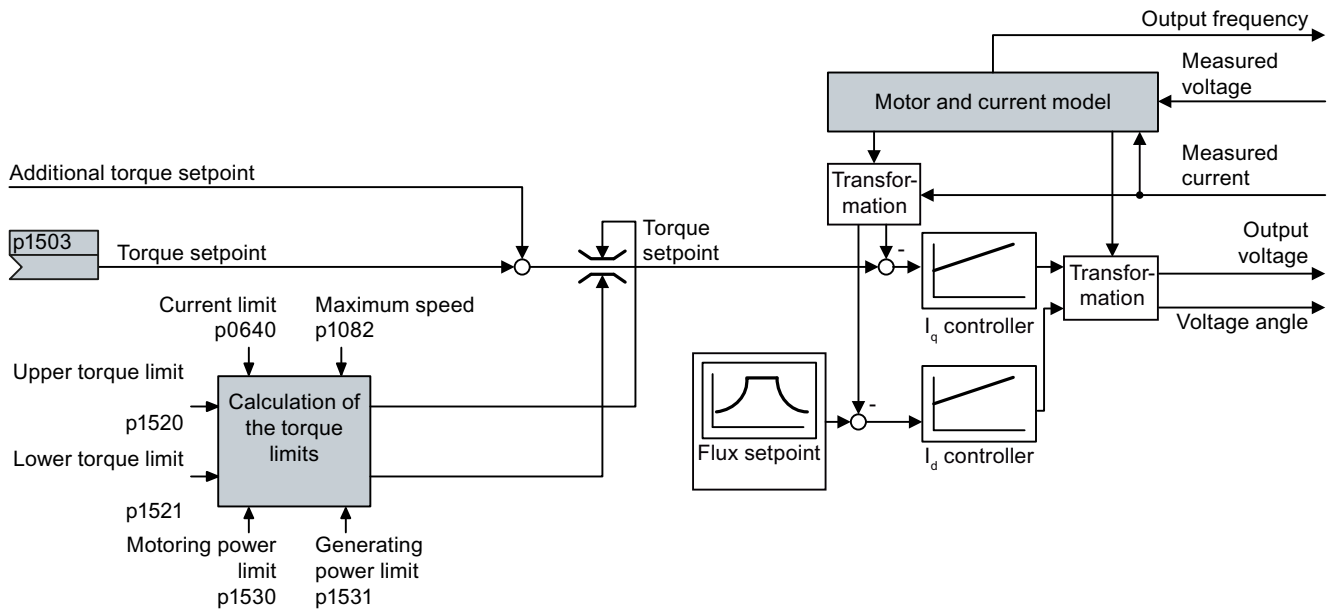


Figure 6-55 Simplified function diagram of the closed-loop torque control

Typical applications for torque control

The torque control is used in applications where the motor speed is specified by the connected driven load. Examples of such applications include:

- Load distribution between master and slave drives:
The master drive is speed controlled, the slave drive is torque controlled.
- Winding machines

The most important settings

Prerequisites for the correct functioning of the torque control:

- You have set the motor data correctly during the quick commissioning
 Commissioning (Page 129)
- You have performed a motor data identification on the cold motor

Parameter	Description
p1300	Control mode: 22: Torque control without speed encoder
p0300 ... p0360	Motor data is transferred from the motor type plate during quick commissioning and calculated with the motor data identification

Parameter	Description
p1511	Additional torque
p1520	Upper torque limit
p1521	Lower torque limit
p1530	Motoring power limit
p1531	Regenerative power limit

Additional information about this function is provided in the parameter list and in function diagrams 6030 onwards in the List Manual.

6.19 Electrically braking the motor

Braking with the motor in generating mode



If the motor brakes the connected load electrically, it will convert the kinetic energy of the motor to electrical energy. The electrical energy E released on braking the load is proportional to the moment of inertia J of the motor and load and to the square of the speed n . The motor attempts to pass the energy on to the inverter.

Main features of the braking functions

DC braking

DC braking prevents the motor from transferring braking energy to the inverter. The inverter impresses a DC current into the motor, therefore braking the motor. The motor converts braking energy E of the load into heat.

- **Advantage:** The motor brakes the load without the inverter having to process regenerative power.
- **Disadvantages:** significant increase in the motor temperature; no defined braking characteristics; no constant braking torque; no braking torque at standstill; braking energy is lost as heat; does not function when the power fails

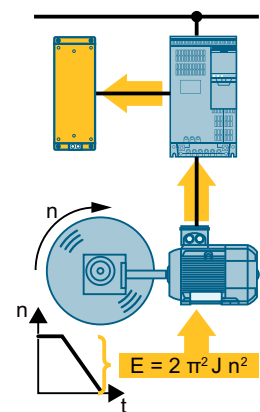
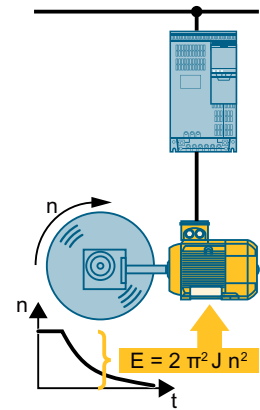
Compound braking

One version of DC braking. The inverter brakes the motor with a defined ramp-down time and superimposes a DC current on the output current.

Dynamic braking

Using a braking resistor, the inverter converts the electrical energy into heat.

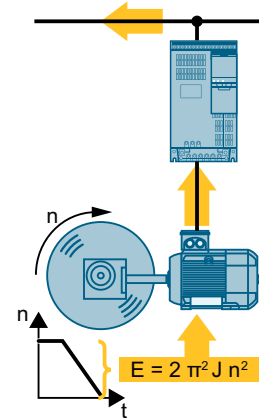
- **Advantages:** defined braking response; motor temperature does not increase any further; constant braking torque
- **Disadvantages:** Braking resistor required; braking energy E is lost in the form of heat



Braking with regenerative feedback into the line supply

The inverter feeds electrical energy back into the line supply (energy recovery).

- *Advantages:* Constant braking torque; the braking energy is not completely converted into heat, but regenerated into the line supply; is suitable for all applications; continuous regenerative operation is possible - e.g. when lowering a suspended load
- *Disadvantage:* Does not function for a power failure



Which Power Module permits which braking method?

Electrical braking methods	Power Modules that can be used
DC braking, compound braking	PM240-2, PM240P-2
Dynamic braking	PM240-2
Braking with energy recovery into the line supply	PM250

6.19.1 DC braking

DC braking is used for applications where the motor must be actively stopped; however, neither an inverter capable of energy recovery nor a braking resistor is available.

Typical applications for DC braking include:

- Centrifuges
- Saws
- Grinding machines
- Conveyor belts

Function

NOTICE

Motor damage caused by overheating during DC braking

The motor will overheat if you use DC braking too frequently or use it for too long. This may damage the motor.

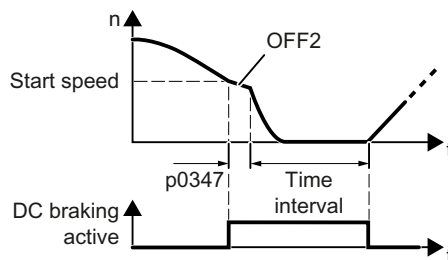
- Monitor the motor temperature.
- Allow the motor to adequately cool down between braking operations.
- If necessary, select another motor braking method.

With DC braking, the inverter outputs an internal OFF2 command for the time that it takes to de-energize the motor p0347 - and then impresses the braking current for the duration of the DC braking.

The DC-braking function is possible only for induction motors.

4 different events initiate DC braking

DC braking when falling below a starting speed



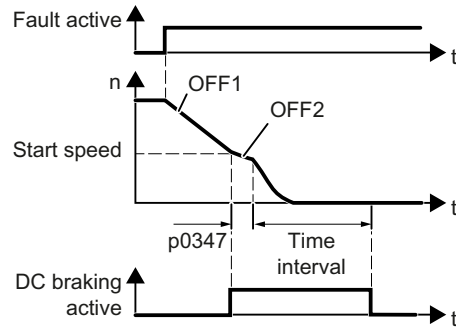
Requirement:

p1230 = 1 and p1231 = 14

Function:

1. The motor speed has exceeded the starting speed.
2. The inverter activates the DC braking as soon as the motor speed falls below the starting speed.

DC braking when a fault occurs



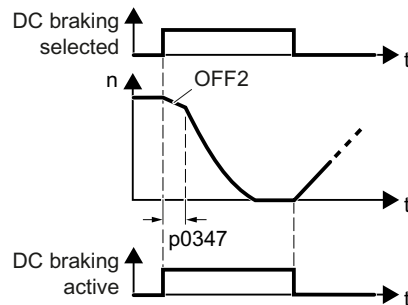
Requirement:

Fault number and fault response are assigned via p2100 and p2101.

Function:

1. A fault occurs, which initiates DC braking as response.
2. The motor brakes along the down ramp to the speed for the start of DC braking.
3. DC braking starts.

DC braking initiated by a control command



Requirement:

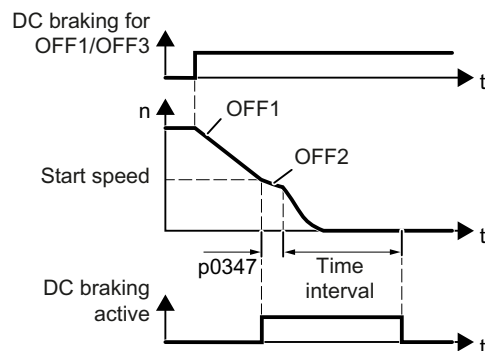
p1231 = 4 and p1230 = control command, e.g. p1230 = 722.3 (control command via DI 3)

Function:

1. The higher-level control issues the command for DC braking, e.g. using DI3: p1230 = 722.3.
2. DC braking starts.

If the higher-level control withdraws the command during DC braking, the inverter interrupts DC braking and the motor accelerates to its setpoint.

DC braking when the motor is switched off



Requirement:

p1231 = 5 or p1230 = 1 and p1231 = 14

Function:

1. The higher-level control switches off the motor (OFF1 or OFF3).
2. The motor brakes along the down ramp to the speed for the start of DC braking.
3. DC braking starts.

Settings for DC braking

Parameter	Description										
p0347	Motor de-excitation time (calculated after quick commissioning) The inverter can trip due to an overcurrent during DC braking if the de-excitation time is too short.										
p1230	DC braking activation (factory setting: 0) Signal source to activate DC braking 0 signal: Deactivated 1 signal: Active										
p1231	Configuring DC braking (factory setting: 0) <table border="1"> <tr> <td>0</td><td>No DC braking</td></tr> <tr> <td>4</td><td>General release for DC braking</td></tr> <tr> <td>5</td><td>DC braking for OFF1/OFF3</td></tr> <tr> <td>14</td><td>DC braking below the starting speed</td></tr> </table>	0	No DC braking	4	General release for DC braking	5	DC braking for OFF1/OFF3	14	DC braking below the starting speed		
0	No DC braking										
4	General release for DC braking										
5	DC braking for OFF1/OFF3										
14	DC braking below the starting speed										
p1232	DC braking braking current (factory setting 0 A)										
p1233	DC braking duration (factory setting 1 s)										
p1234	DC braking start speed (factory setting 210000 rpm)										
r1239	DC braking status word <table border="1"> <tr> <td>.08</td><td>DC braking active</td></tr> <tr> <td>.10</td><td>DC braking ready</td></tr> <tr> <td>.11</td><td>DC braking selected</td></tr> <tr> <td>.12</td><td>DC braking selection internally locked</td></tr> <tr> <td>.13</td><td>DC braking for OFF1/OFF3</td></tr> </table>	.08	DC braking active	.10	DC braking ready	.11	DC braking selected	.12	DC braking selection internally locked	.13	DC braking for OFF1/OFF3
.08	DC braking active										
.10	DC braking ready										
.11	DC braking selected										
.12	DC braking selection internally locked										
.13	DC braking for OFF1/OFF3										

Table 6-51 Configuring DC braking as a response to faults

Parameter	Description
p2100	Set fault number for fault response (factory setting 0) Enter the fault number for which DC braking should be activated, e.g. p2100[3] = 7860 (external fault 1).
p2101 = 6	Fault response setting (factory setting 0) Assigning the fault response: p2101[3] = 6.
The fault is assigned an index of p2100. Assign the same index of p2100 or p2101 to the fault and fault response. The inverter's List Manual lists in the "Faults and alarms" list the possible fault responses for every fault. Entry "DCBRK" means that you may set DC braking as response for this fault.	

6.19.2 Compound braking

Typical applications for compound braking include:

- Centrifuges
- Saws
- Grinding machines
- Horizontal conveyors

For these applications, the motor is normally operated with a constant speed, and is only braked down to standstill after longer periods of time.

Principle of operation

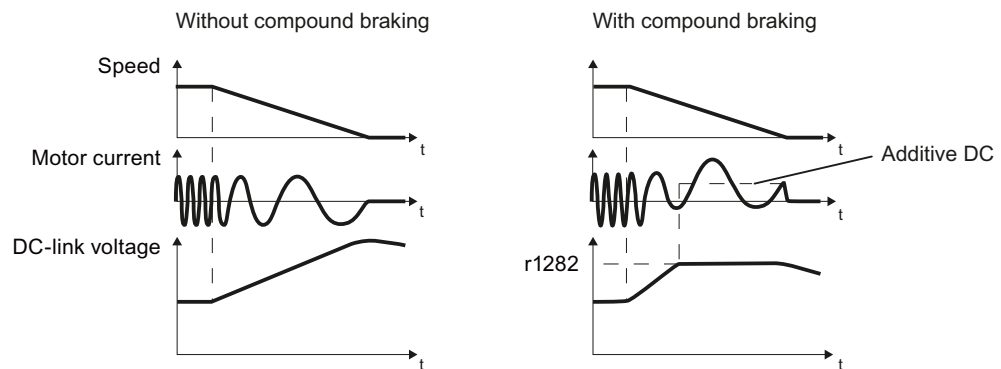


Figure 6-56 Motor brakes with and without active compound braking

Compound braking prevents the DC-link voltage increasing above a critical value. The inverter activates compound braking depending on the DC-link voltage. Above a DC-link voltage threshold (r1282), the inverter adds a DC current to the motor current. The DC current brakes the motor and prevents an excessive increase in the DC-link voltage.

Note

Compound braking is possible only with the U/f control.

Compound braking does not operate in the following cases:

- The "flying restart" function is active
- DC braking is active
- Vector control is selected

Setting and enabling compound braking

Parameter	Description
p3856	Compound braking current (%) With the compound braking current, the magnitude of the DC current is defined, which is additionally generated when stopping the motor for operation with U/f control to increase the braking effect. p3856 = 0 Compound braking locked p3856 = 1 ... 250 Current level of the DC braking current as a % of the rated motor current (p0305) Recommendation: $p3856 < 100\% \times (r0209 - r0331) / p0305 / 2$
r3859.0	Compound-braking status word r3859.0 = 1: Compound braking is active

NOTICE**Motor damage caused by overheating when using compound braking**

The motor will overheat if you use compound braking too frequently or for too long. This may damage the motor.

- Monitor the motor temperature.
- Allow the motor to adequately cool down between braking operations.
- If necessary, select another motor braking method.

6.19.3 Dynamic braking

Typical applications for dynamic braking require continuous braking and acceleration operations or frequent changes of the motor direction of rotation:

- Horizontal conveyors
- Vertical and inclined conveyors
- Hoisting gear

Principle of operation

The DC link voltage increases as soon as the motor supplies regenerative power to the inverter when braking. The regenerative power means that the DC link voltage in the inverter increases. Depending on the DC link voltage, the inverter outputs the regenerative power to the braking resistor through the braking chopper. The braking resistor converts the regenerative power into heat, therefore preventing DC link voltages $> V_{dc_max}$.

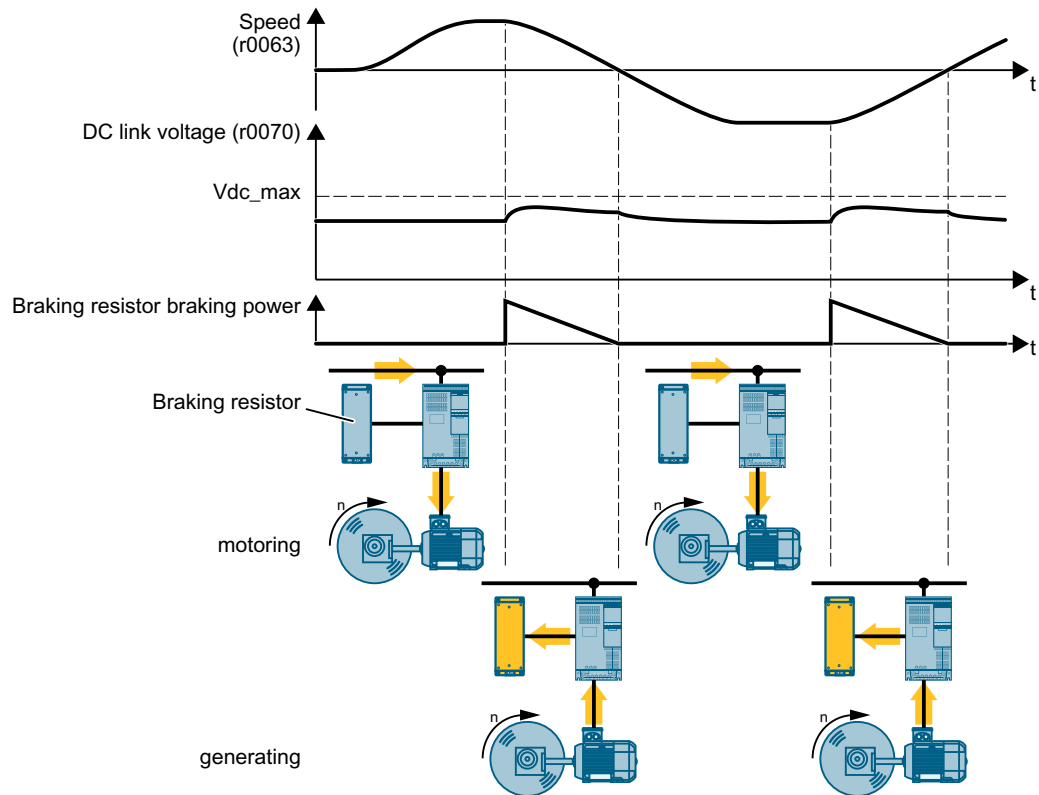
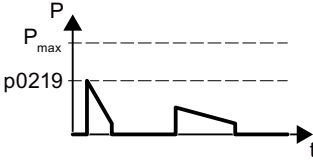


Figure 6-57 Simplified representation of dynamic braking with respect to time

Set dynamic braking

Parameter	Description		
p0219	Braking power of the braking resistor (factory setting: 0 kW) For p0219 > 0, the inverter deactivates the VDC_max controller. For vector control, p0219 specifies the regenerative power limit p1531.  Set with p0219 the maximum braking power that the braking resistor must handle. The inverter extends the ramp-down time of the motor when the braking power is too low. You can find the technical data of the braking resistor in the Power Module hardware installation manual.  Overview of the manuals (Page 504) The SIZER PC tool provides support for calculating the braking power.  Configuring support (Page 506)		
p2106	BI: External fault 1 <table border="1"> <tr> <td>p2106 = 722.x</td><td> Monitor the signal for overtemperature of the braking resistor with digital input x of the inverter.  Connecting the temperature contact of the braking resistor (Page 120) </td></tr> </table>	p2106 = 722.x	Monitor the signal for overtemperature of the braking resistor with digital input x of the inverter.  Connecting the temperature contact of the braking resistor (Page 120)
p2106 = 722.x	Monitor the signal for overtemperature of the braking resistor with digital input x of the inverter.  Connecting the temperature contact of the braking resistor (Page 120)		



An example for configuring a drive with braking resistor is provided in the Internet:

Application example: Engineering and commissioning series lifting equipment/cranes (<https://support.industry.siemens.com/cs/de/en/view/103156155>)

6.19.4 Braking with regenerative feedback to the line

The typical applications for braking with energy recovery (regenerative feedback into the line supply) are as follows:

- Hoist drives
- Centrifuges
- Unwinders

For these applications, the motor must brake for longer periods of time.

The inverter can feed back up to 100% of its rated power into the line supply (referred to "High Overload" base load).



Technical data, PM250 Power Module (Page 467)

Setting the braking with regenerative feedback to the line

Parameter	Description
Limiting the regenerative feedback for V/f control (p1300 < 20)	
p0640	Motor series overload factor (factory setting: 0.00 A, default for quick commissioning) It is only possible to limit the regenerative power with V/f control by limiting the motor current. If the current exceeds this value for longer than 10 s, the inverter shuts down the motor with fault F07806.
Limiting feedback with vector control (p1300 ≥ 20)	
p1531	Power limit generative (factory setting: -0.01 kW) The inverter calculates the parameter based on the quick commissioning or with p0340 = 5.

6.20 Overcurrent protection



The vector control ensures that the motor current remains within the set torque limits.

If you use U/f control, you cannot set any torque limits. The U/f control prevents too high a motor current by influencing the output frequency and the motor voltage (I-max controller).

I_max controller

Requirements

The torque of the motor must decrease at lower speeds, which is the case, for example, with fans.

The load must not drive the motor continuously, e.g. when lowering hoisting gear.

Function

The I-max controller influences the output frequency and the motor voltage.

If the motor current reaches the current limit during acceleration, the I-max controller extends the acceleration operation.

If the load of the motor is so large during stationary operation that the motor current reaches the current limit, the I-max controller reduces the speed and the motor voltage until the motor current is in the permissible range again.

If the motor current reaches the current limit during deceleration, the I-max controller extends the deceleration operation.

Settings

You only have to change the factory settings of the I-max controller if the drive tends to oscillate when it reaches the current limit or if it is shut down due to overcurrent.

Table 6-52 I-max controller parameters

Parameter	Description
p0305	Rated motor current
p0640	Motor current limit
p1340	Proportional gain of the I-max controller for speed reduction
p1341	Integral time of the I-max controller for speed reduction
r0056.13	Status: I-max controller active
r1343	Speed output of the I-max controller Shows the amount to which the I-max controller reduces the speed.

For more information about this function, see function diagram 6300 in the List Manual.

6.21 Inverter protection using temperature monitoring



The inverter temperature is essentially defined by the following effects:

- The ambient temperature
- The ohmic losses increasing with the output current
- Switching losses increasing with the pulse frequency

Monitoring types

The inverter monitors its temperature using the following monitoring types:

- I²t monitoring (alarm A07805, fault F30005)
- Measuring the chip temperature of the Power Module (alarm A05006, fault F30024)
- Measuring the heat sink temperature of the Power Module (alarm A05000, fault F30004)

Inverter response to thermal overload

Parameter	Description
r0036	Power unit overload I²t [%] The I ² t monitoring calculates the inverter utilization based on a current reference value defined in the factory. • Actual current > reference value: r0036 becomes higher. • Actual current < reference value: r0036 becomes lower or remains = 0.
r0037	Power unit temperatures [°C]
p0290	Power unit overload response Factory setting and the ability to be changed depends on the hardware. The dependency is described in the List Manual. A thermal overload is present if the inverter temperature is greater than that specified in p0292. You define how the inverter responds if there is a risk of thermal overload using this parameter. The details are described below.
p0292	Power unit temperature warning threshold (factory setting: Heat sink [0] 5 °C, power semiconductor [1] 15 °C) The value is set as a difference to the shutdown temperature.
p0294	Power unit warning at I²t overload (factory setting: 95 %)

Overload response for p0290 = 0

The inverter responds depending on the control mode that has been set:

- In vector control, the inverter reduces the output current.
- In U/f control, the inverter reduces the speed.

Once the overload condition has been removed, the inverter re-enables the output current or speed.

If the measure cannot prevent an inverter thermal overload, then the inverter switches off the motor with fault F30024.

Overload response for p0290 = 1

The inverter immediately switches off the motor with fault F30024.

Overload response for p0290 = 2

We recommend this setting for drives with square-law torque characteristic, e.g. fans.

The inverter responds in two stages:

1. If you operate the inverter with increased pulse frequency setpoint p1800, then the inverter reduces its pulse frequency starting at p1800.
In spite of the temporarily reduced pulse frequency, the base load output current remains unchanged at the value that is assigned to p1800.

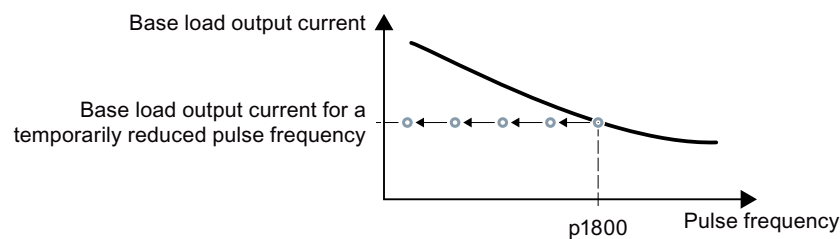


Figure 6-58 Derating characteristic and base load output current for overload

Once the overload condition has been removed, the inverter increases the pulse frequency back to the pulse frequency setpoint p1800.

2. If it is not possible to temporarily reduce the pulse frequency, or the risk of thermal overload cannot be prevented, then stage 2 follows:
 - In vector control, the inverter reduces its output current.
 - In U/f control, the inverter reduces the speed.

Once the overload condition has been removed, the inverter re-enables the output current or speed.

If both measures cannot prevent a power unit thermal overload, then the inverter switches off the motor with fault F30024.

Overload response for p0290 = 3

If you operate the inverter with increased pulse frequency, then the inverter reduces its pulse frequency starting at the pulse frequency setpoint p1800.

In spite of the temporarily reduced pulse frequency, the maximum output current remains unchanged at the value that is assigned to the pulse frequency setpoint. Also see p0290 = 2.

Once the overload condition has been removed, the inverter increases the pulse frequency back to the pulse frequency setpoint p1800.

If it is not possible to temporarily reduce the pulse frequency, or the measure cannot prevent a power unit thermal overload, then the inverter switches off the motor with fault F30024.

Overload response for p0290 = 12

The inverter responds in two stages:

1. If you operate the inverter with increased pulse frequency setpoint p1800, then the inverter reduces its pulse frequency starting at p1800.
There is no current derating as a result of the higher pulse frequency setpoint.
Once the overload condition has been removed, the inverter increases the pulse frequency back to the pulse frequency setpoint p1800.
2. If it is not possible to temporarily reduce the pulse frequency, or the risk of inverter thermal overload cannot be prevented, then stage 2 follows:
 - In vector control, the inverter reduces the output current.
 - In U/f control, the inverter reduces the speed.

Once the overload condition has been removed, the inverter re-enables the output current or speed.

If both measures cannot prevent a power unit thermal overload, then the inverter switches off the motor with fault F30024.

Overload response for p0290 = 13

We recommend this setting for drives with high starting torque, e.g. horizontal conveyors or extruders.

If you operate the inverter with increased pulse frequency, then the inverter reduces its pulse frequency starting at the pulse frequency setpoint p1800.

There is no current derating as a result of the higher pulse frequency setpoint.

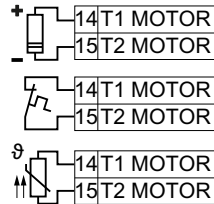
Once the overload condition has been removed, the inverter increases the pulse frequency back to the pulse frequency setpoint p1800.

If it is not possible to temporarily reduce the pulse frequency, or the measure cannot prevent a power unit thermal overload, then the inverter switches off the motor with fault F30024.

6.22 Motor protection with temperature sensor



The inverter can evaluate one of the following sensors to protect the motor against overtemperature:



- KTY84 sensor
- Temperature switch (e.g. bimetallic switch)
- PTC sensor
- Pt1000 sensor

KTY84 sensor

NOTICE

Motor overheating due to incorrectly connected KTY sensor

If a KTY sensor is connected with incorrect polarity, the motor can become damaged due to overheating, as the inverter cannot detect a motor overtemperature condition.

- Connect the KTY sensor with the correct polarity.



Using a KTY sensor, the inverter monitors the motor temperature and the sensor itself for wire-break or short-circuit:

- Temperature monitoring:
The inverter uses a KTY sensor to evaluate the motor temperature in the range from -48°C ... $+248^{\circ}\text{C}$.
Use the p0604 or p0605 parameter to set the temperature for the alarm and fault threshold.
 - Overtemperature alarm (A07910):
- motor temperature $>$ p0604 and p0610 = 0
 - Overtemperature fault (F07011):
The inverter responds with a fault in the following cases:
- motor temperature $>$ p0605
- motor temperature $>$ p0604 and p0610 $\neq$ 0
- Sensor monitoring (A07015 or F07016):
 - Wire-break:
The inverter interprets a resistance $> 2120\ \Omega$ as a wire-break and outputs the alarm A07015. After 100 milliseconds, the inverter changes to the fault state with F07016.
 - Short-circuit:
The inverter interprets a resistance $< 50\ \Omega$ as a short-circuit and outputs the alarm A07015. After 100 milliseconds, the inverter changes to the fault state with F07016.

Temperature switch



The inverter interprets a resistance $\geq 100\ \Omega$ as being an opened temperature switch and responds according to the setting for p0610.

PTC sensor

The inverter interprets a resistance $> 1650 \Omega$ as being an overtemperature and responds according to the setting for p0610.

The inverter interprets a resistance $< 20 \Omega$ as being a short-circuit and responds with alarm A07015. If the alarm is present for longer than 100 milliseconds, the inverter shuts down with fault F07016.

Pt1000 sensor

Using a Pt1000 sensor, the inverter monitors the motor temperature and the sensor itself for wire breakage and/or short-circuit:

- Temperature monitoring:
Using a Pt1000 sensor, the inverter evaluates the motor temperature in the range from -48°C ... $+248^{\circ}\text{C}$.
You set the temperature for the alarm and fault thresholds using parameters p0604 and p0605.
 - Overtemperature alarm (A07910):
- motor temperature $> \text{p0604}$ and $\text{p0610} = 0$
 - Overtemperature fault (F07011):
The inverter responds with a fault in the following cases:
- motor temperature $> \text{p0605}$
- motor temperature $> \text{p0604}$ and $\text{p0610} \neq 0$
- Sensor monitoring (A07015 or F07016):
 - Wire-break:
The inverter interprets a resistance $> 2120 \Omega$ as a wire-break and outputs the alarm A07015. After 100 milliseconds, the inverter changes to the fault state with F07016.
 - Short-circuit:
The inverter interprets a resistance $< 603 \Omega$ as a short-circuit and outputs the alarm A07015. After 100 milliseconds, the inverter changes to the fault state with F07016.

Setting parameters for the temperature monitoring

Parameter	Description
p0335	Motor-cooling method (factory setting: 0) 0: Natural cooling - with fan on the motor shaft 1: Forced ventilation - with a separately driven fan 2: Liquid cooling 128: No fan
p0601	Motor temperature sensor type 0: No sensor (factory setting) 1: PTC 2: KTY84 4: Temperature switch 6: Pt1000
p0604	Mot_temp_mod 2 / sensor alarm threshold (factory setting 130°C) For monitoring the motor temperature using KTY84/Pt1000.

Parameter	Description
p0605	Mot_temp_mod 1/2 / sensor threshold and temperature value (factory setting: 145° C) For monitoring the motor temperature using KTY84/Pt1000.
p0610	Motor overtemperature response (factory setting: 12) Determines the inverter behavior when the motor temperature reaches the alarm threshold p0604. 0: Alarm A07910, no fault. 1: Alarm A07910 and fault F07011. The inverter reduces its current limit. 2: Alarm A07910 and fault F07011. 12: Alarm A07910 and fault F07011. For use of the thermal motor model in parallel to the temperature sensor: After switching off the supply voltage, the inverter saves the most-recently calculated difference to the ambient air temperature. After switching the supply voltage on again, the thermal motor model starts with 90% of the previously saved difference temperature.
p0640	Current limit [A]

Additional information on the motor temperature monitoring can be found in function diagram 8016 of the List Manual.

6.23 Motor protection by calculating the temperature



The inverter calculates the motor temperature based on a thermal motor model with the following properties:

- The thermal motor model detects temperature increases much faster than a temperature sensor.
- After switching off the supply voltage, the inverter saves the most-recently calculated difference to the ambient air temperature (factory setting: p0610 = 12). After switching the supply voltage on again, the thermal motor model starts with 90% of the previously saved difference temperature.

If you are using the thermal motor model together with a temperature sensor, e.g. a Pt1000, then the inverter corrects the model based on the measured temperature.

Thermal motor model 2 for induction motors

The thermal motor model 2 for induction motors is a thermal 3-mass model, consisting of stator core, stator winding and rotor. Thermal motor model 2 calculates the temperatures - both in the rotor as well as in the stator winding.

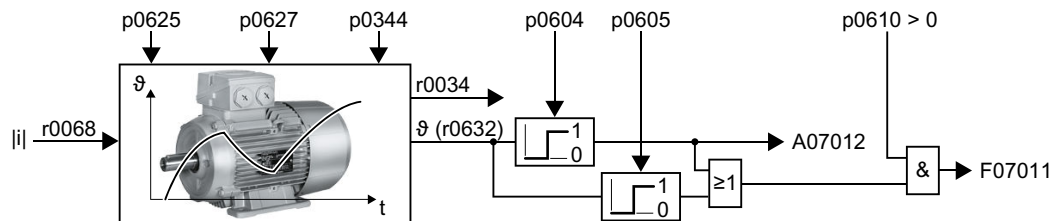


Figure 6-59 Thermal motor model 2 for induction motors

Table 6-53 Thermal motor model 2 for induction motors

Parameter	Description
r0068	CO: Absolute actual current value
p0610	Motor overtemperature response (factory setting: 12)
	0: No response, alarm only, no reduction of I_{max}
	1: Messages, reduction of I_{max}
	2: Messages, no reduction of I_{max}
	12: Messages, no reduction of I_{max} , temperature is saved

Parameter	Description	
p0344	Motor weight (for thermal motor type) (factory setting: 0.0 kg)	After selecting an induction motor (p0300) or a listed induction motor (p0301) during the commissioning, the inverter sets the parameters to values appropriate for the motor. The parameters are write-protected for listed motors (p0301 ≥ 0).
p0604	Mot_temp_mod 2/KTY alarm threshold (factory setting: 130.0° C) Motor temperature > p0604 ⇒ fault F07011.	
p0605	Mot_temp_mod 1/2 threshold (factory setting: 145.0° C) Motor temperature > p0605 ⇒ alarm A07012.	
p0612	Mot_temp_mod activation .01 1 signal: Activate motor temperature model 2 for induction motors .09 1 signal: Activate motor temperature model 2 expansions The inverter sets bit 09 = 1 after commissioning. If you load the parameter settings for firmware version ≤ V4.6 into the inverter, bit 09 = 0 remains.	
p0627	Motor overtemperature, stator winding (factory setting: 80 K)	
p0625	Motor ambient temperature during commissioning (factory setting: 20° C) Specification of the motor ambient temperature in °C at the instant of the motor data identification.	
r0632	Mot_temp_mod stator winding temperature [°C]	

Further information is provided in the function charts 8016 and 8017 of the List Manual.

Thermal motor model 3 for encoderless synchronous motors

The thermal motor model 3 for encoderless synchronous motors 1FK7 or 1FG1 is a thermal 3-mass model, consisting of stator core, stator winding and rotor. Thermal motor model 3 calculates the temperatures - both in the rotor as well as in the stator winding.

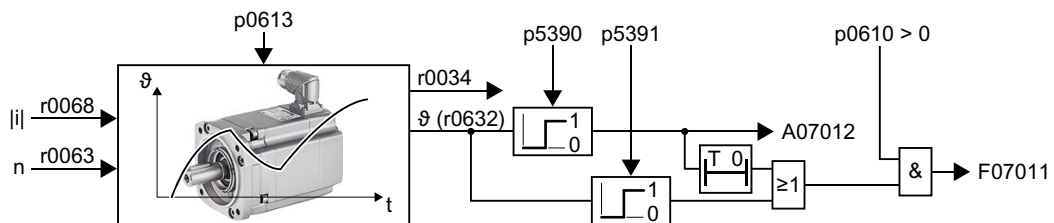


Figure 6-60 Thermal motor model 3 for 1FK7 encoderless synchronous motors

Table 6-54 Thermal motor model 3 for 1FK7 encoderless synchronous motors

Parameter	Description
r0034	CO: Thermal motor load [%]
r0063	CO: Actual speed value [rpm]
r0068	CO: Actual current value [A]
p0610	Motor overtemperature response (factory setting: 12)
0:	No response, alarm only, no reduction of I_{max}
1:	Messages, reduction of I_{max}
2:	Messages, no reduction of I_{max}
12:	Messages, no reduction of I_{max} , temperature is saved

Parameter	Description	
p0612	Mot_temp_mod activation	
	.02	1 signal: Activate motor temperature model 3 for 1FK7 or 1FG1 encoderless synchronous motors
p5390	Mot_temp_mod 1/3 alarm threshold (factory setting: 110.0° C) Motor temperature > p5390 ⇒ alarm A07012.	
p5391	Mot_temp_mod 1/3 fault threshold (factory setting: 120.0° C) Motor temperature > p5391 or motor temperature > p5390, longer than a motor-dependent time calculated by the inverter ⇒ fault F07011.	
p0613	Mot_temp_mod 1/3 ambient air temperature (factory setting: 20° C) Expected motor ambient temperature in °C for motor operation.	
p0625	Motor ambient temperature during commissioning (factory setting: 20° C) Motor ambient temperature in °C at the instant of the motor data identification.	
r0632	Mot_temp_mod stator winding temperature [°C]	

Further information is provided in the function charts 8016 and 8017 of the List Manual.

Thermal motor model 1 for synchronous motors

Further information about thermal motor model 1 for synchronous motors is provided in the function charts 8016 and 8017 of the List Manual.

6.24 Motor and inverter protection by limiting the voltage

What causes an excessively high voltage?



To drive the load, an electric motor converts electrical energy into mechanical energy. If the motor is driven by its load, e.g. due to the load moment of inertia when braking, then the energy flow reverses: The motor temporarily operates as generator, and converts mechanical energy into electrical energy. In inverter operation, the electrical energy flows from the motor into the inverter. If the inverter cannot output the electrical energy supplied by the motor, e.g. to a braking resistor, then the inverter stores the energy in its DC link capacitance. As a consequence, the DC link voltage V_{dc} in the inverter is higher.

An excessively high DC link voltage damages the inverter and also the motor. As a consequence, the inverter monitors its DC link voltage - and when necessary switches off the motor and outputs fault "DC link overvoltage".

Protecting the motor and inverter against overvoltage

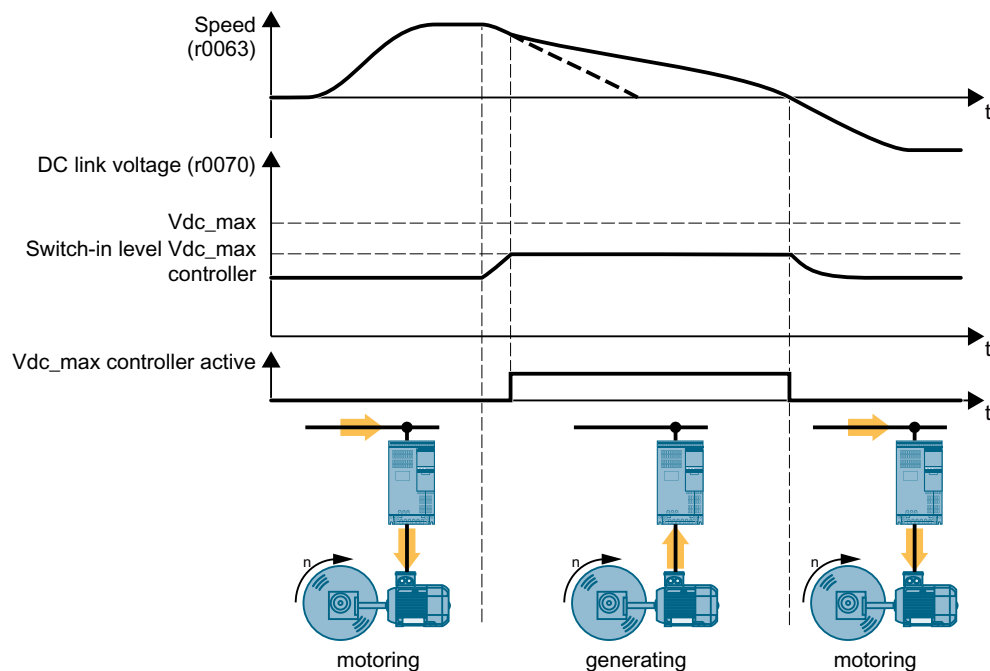


Figure 6-61 Simplified representation of the V_{dc_max} control

The V_{dc_max} control extends the motor ramp-down time when braking. The motor then only feeds so much energy into the inverter to cover the losses in the inverter. The DC link voltage remains in the permissible range.

The V_{dc_max} control is not suitable for applications where the motor is in continuous regenerative operation, e.g. as is the case for cranes and centrifuges.



Electrically braking the motor (Page 302)

The Vdc_max control can be used only with the PM230, PM240-2 and PM240P-2 Power Modules. The Vdc_max control is not required if you use a braking resistor.

PM250 Power Modules feed back regenerative energy into the line supply. Therefore, the Vdc_max control is not required for a PM250 Power Module.

Parameters of the Vdc_max control

The parameters differ depending on the motor control mode.

Parameter for V/f control	Parameter for vector control	Description
p1280 = 1	p1240 = 1	Vdc controller configuration (Factory setting: 1) 1: Vdc controller is enabled
r1282	r1242	Vdc_max control activation level DC link voltage value above which the Vdc_max control is activated
p1283	p1243	Vdc_max control dynamic factor (factory setting: 100 %) Scaling control parameters p1290, p1291 and p1292
p1284	---	Vdc_max controller time threshold Setting the monitoring time of the Vdc_max controller.
p1290	p1250	Vdc_max control proportional gain (factory setting: 1)
p1291	p1251	Vdc_max control integral time (factory setting p1291: 40 ms, p1251: 0 ms)
p1292	p1252	Vdc_max control rate time (factory setting p1292: 10 ms, p1252: 0 ms)
p1294	p1254	Vdc_max control automatic ON level sensing (Factory setting, dependent on the Power Module) Automatically sense switch-on levels of the Vdc_max control. 0: Automatic detection disabled 1: Automatic detection enabled
p0210	p0210	Unit supply voltage If p1254 or p1294 = 0, the inverter uses this parameter to calculate the switch-in thresholds of the Vdc_max control. Set this parameter to the actual value of the input voltage.

For more information about this function, see the List Manual (function diagrams 6320 and 6220).

6.25 Monitoring the driven load



In many applications, the speed and the torque of the motor can be used to determine whether the driven load is in an impermissible operating state. The use of an appropriate monitoring function in the inverter prevents failures and damage to the machine or plant.

Examples:

- For fans or conveyor belts, an excessively low torque can mean a broken drive belt.
- For pumps, insufficient torque can indicate a leakage or dry-running.
- For extruders and mixers, an excessive torque together with low speed can indicate machine blockage.

Functions for monitoring the driven load

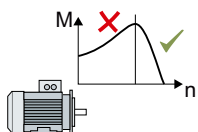
The inverter provides the following means to monitor the driven load via the torque of the motor:

	The stall protection prevents an induction motor from stalling
	The no-load monitoring evaluates the motor current. Insufficient current indicates that the motor and the load are no longer mechanically connected with each other.
	The blocking protection triggers for a motor current that corresponds to the set current limit coupled with motor standstill.
	The torque monitoring assumes that a specific torque is associated with each speed for pumps and fans. Insufficient torque indicates that the motor and the load are no longer mechanically connected. An excessive torque can indicate problems in the mechanical system of the driven load, e.g. a mechanically blocked load.
	Blocking protection, leakage protection and dry-running protection are a monitoring method for pumps or fans. The monitoring combines a torque monitoring with a blocking protection.

Monitoring the driven load with a binary signal:

	The speed monitoring evaluates a periodic binary signal. A signal failure indicates that the motor and the load are no longer mechanically connected with each other.
--	---

6.25.1 Stall protection



If the load of a standard induction motor exceeds the stall torque of the motor, the motor can also stall during operation on the inverter. A stalled motor is stationary and does not develop sufficient torque to accelerate the load.

If the "Motor model fault signal stall detection" r1746 for the time p2178 is present via the "Motor model error threshold stall detection" p1745, the inverter signals "Motor stalled" and fault F07902.

Parameters	Description
r1408	Current controller status word r1408.12 = 1: Motor stalled
p1745	Motor model fault threshold value stall detection (factory setting: 5%) If the motor stalls too early with this setting, then using the trace function you have the option of determining a better value. To do this, set p1745 = 100.
r1746	Motor model fault signal stall detection [%]
p2178	Motor stalled delay time (factory setting: 0.01 s)
r2198	Status word monitoring 2 r2198.07 = 1: Motor stalled

6.25.2 No-load monitoring



In applications with fans, compressors or conveyor belts, an insufficient motor current indicates that the power transmission from the motor to the load is interrupted.

If the motor current for the time p2180 lies below the current level p2179, the inverter signals "output load not available" and alarm A07929.

Parameters	Description
r0068	Actual current value [A] [0] = Unsmoothed [1] = Smoothed with p0045
p2179	Output load detection current limit (Factory setting: 0 A) p2179 = 0: No load detection deactivated
p2180	Output load detection delay time (factory setting: 2000 ms)
r2197	Status word, monitoring functions 1 r2197.11 = 1: Output load not available

6.25.3 Stall protection

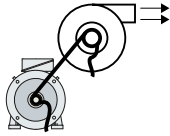


In applications with extruders or mixers, the motor can block for an excessive mechanical load. For a blocked motor, the motor current corresponds to the set current limit without the speed reaching the specified setpoint.

If the speed lies below the speed threshold p2175 for the time p2177 while the motor current reaches the current limit, the inverter signals "Motor blocked" and fault F07900.

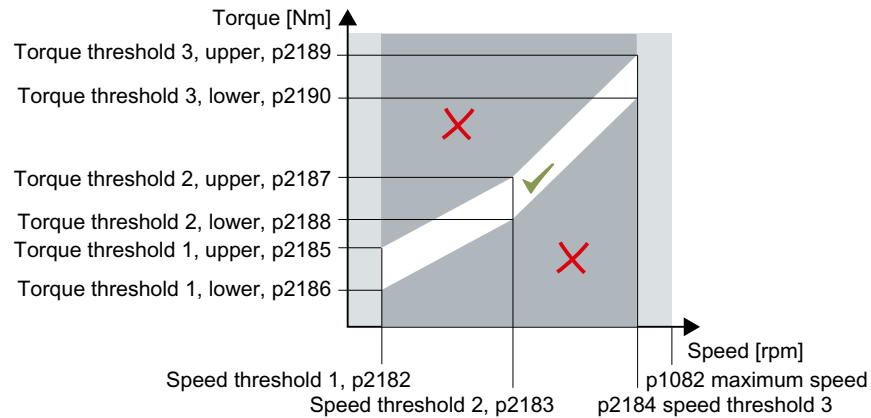
Parameters	Description
r0063	Actual speed value [rpm] [0] = Unsmoothed [1] = Smoothed with p0045
p2175	Motor blocked speed threshold (factory setting: 120 rpm) p2175 = 0: Blocking protection deactivated
p2177	Motor blocked delay time (factory setting: 3 s)
r2198	Status word monitoring 2 r2198.06 = 1: Motor blocked

6.25.4 Torque monitoring



In applications with fans, pumps or compressors with the flow characteristic, the torque follows the speed according to a specific characteristic. An insufficient torque for fans indicates that the power transmission from the motor to the load is interrupted. For pumps, insufficient torque can indicate a leakage or dry-running.

The inverter monitors the torque based on the envelope curve depending on the speed against a lower and upper torque.

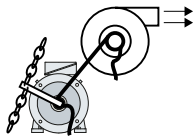


If the torque lies in the impermissible range longer than time p2192, the inverter reacts as specified in p2181.

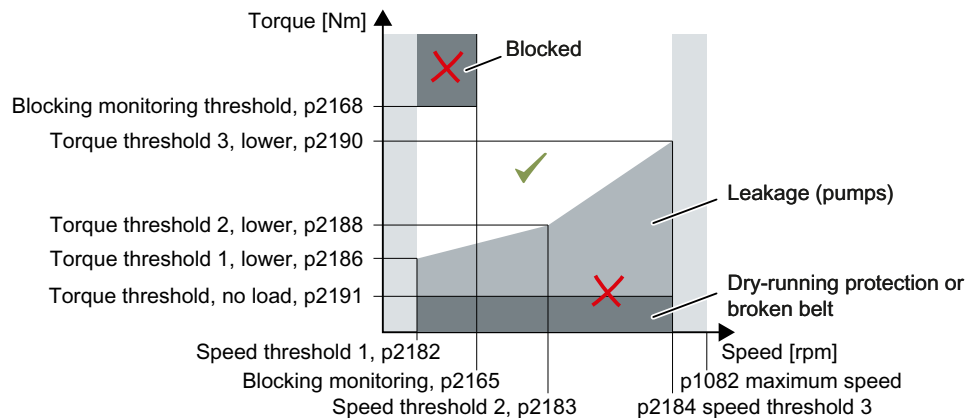
The monitoring is not active below speed threshold 1 and above speed threshold 3.

Parameters	Description	
p2181	Load monitoring response 1: A07920 for torque too low 2: A07921 for torque too high 3: A07922 for torque outside tolerance 4: F07923 for torque too low 5: F07924 for torque too high 6: F07925 for torque outside tolerance	
p2182	Load monitoring speed threshold 1	The inverter interpolates linearly between characteristic points.
p2183	Load monitoring speed threshold 2	
p2184	Load monitoring speed threshold 3	
p2185	Load monitoring torque threshold 1, upper	
p2186	Load monitoring torque threshold 1, lower	
p2187	Load monitoring torque threshold 2, upper	
p2188	Load monitoring torque threshold 2, lower	
p2189	Load monitoring torque threshold 3, upper	
p2190	Load monitoring torque threshold 3, lower	
p2192	Load monitoring delay time Delay time for the message "Leave torque monitoring tolerance band"	
p2193 = 1	Load monitoring configuration (factory setting: 1) 1: Monitoring torque and load drop	

6.25.5 Blocking protection, leakage protection and dry-running protection



In applications with fans, pumps or compressors with the flow characteristic, the torque follows the speed according to a specific characteristic. An insufficient torque for fans indicates that the power transmission from the motor to the load is interrupted. For pumps, insufficient torque can indicate a leakage or dry-running.



If the torque and speed lie in the impermissible range longer than time p2192, the inverter reacts as specified in p2181.

For applications with pumps (p2193 = 4), the inverter detects the following states of the driven load:

- Blocked
- Leakage
- Dry running

For applications with fans or compressors (p2193 = 5), the inverter detects the following states of the driven load:

- Blocked
- Torn belt

The monitoring is not active below speed threshold 1 and above speed threshold 3.

When using the control mode "U/f control" (p1300 < 10), the "Blocking protection" function becomes active when the current limit is reached.

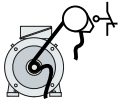


No-load monitoring (Page 325)

Parameters	Description
p2165	Load monitoring blocking monitoring threshold, upper
p2168	Load monitoring blocking monitoring torque threshold

Parameters	Description	
p2181	Load monitoring response	
	7:	A07891: Load monitoring, pump/fan blocked A07892: Load monitoring, pump/fan without load A07893: Load monitoring, pump leakage
	8:	F07894: Load monitoring, pump/fan blocked F07895: Load monitoring, pump/fan without load F07896: Load monitoring, pump leakage
p2182	Load monitoring speed threshold 1 - When using the control mode "U/f control" (p1300 < 10): p2182 > 10% of the rated speed - When using the control mode "encoderless vector control" (p1300 = 20): p2182 > p1755 (switchover speed, motor model)	The inverter interpolates linearly between characteristic points.
p2183	Load monitoring speed threshold 2	
p2184	Load monitoring speed threshold 3	
p2186	Load monitoring torque threshold 1, lower	
p2188	Load monitoring torque threshold 2, lower	
p2190	Load monitoring torque threshold 3, lower	
p2191	Load monitoring torque threshold, no load	
p2192	Load monitoring delay time When using the control mode "encoderless vector control" (p1300 = 20): - For operation with an induction motor: Set p2192 < 1 s or p1750.6 = 1 (the closed-loop control does not change to open-loop controlled mode for blocked motor) - For operation with synchronous and reluctance motors, the inverter sometimes cannot detect a blocked motor.	
p2193 = 4 or 5	Load monitoring configuration (factory setting: 1) 4: Pump monitoring: Differentiation between leakage and dry running 5: Fan monitoring: p2191 has no effect	

6.25.6 Rotation monitoring



The inverter monitors the speed or velocity of a machine component via an electromechanic or electronic encoder, e.g. a proximity switch. Examples of how the function can be used:

- Gearbox monitoring for traction drives and hoisting gear
- Drive belt monitoring for fans and conveyor belts
- Blocking protection for pumps and conveyor belts

The inverter checks whether the encoder consistently supplies a 24 V signal during motor operation. If the encoder signal fails for time p2192, the inverter signals fault F07936.

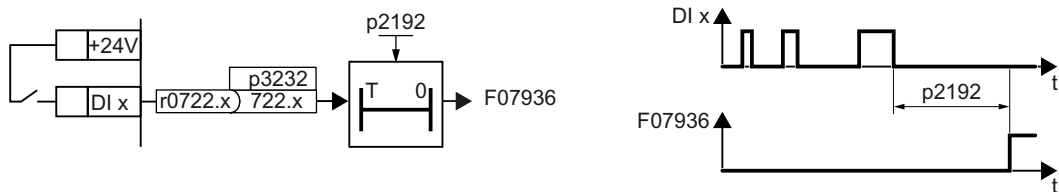
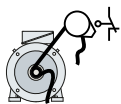


Figure 6-62 Function plan and time response of the speed monitoring

Parameter	Description
p2192	Load monitoring delay time (factory setting 10 s) After the motor is switched on, if the "LOW" signal is present at the associated digital input for longer than this time, the inverter signals a load failure (F07936).
p2193 = 3	Load monitoring configuration (factory setting: 1) 0: Monitoring switched off 1: Monitoring torque and load drop 2: Monitoring speed and load drop 3: Load failure monitoring
p3232	Load monitoring failure detection (factory setting: 1) Connect the load monitoring to a DI x digital input of your choice. p3232 = 722.x

For more information, see the List Manual (the parameter list and function diagram 8013).

6.25.7 Speed deviation monitoring



The inverter calculates and monitors the speed or velocity of a machine component.

Examples of how the function can be used:

- Gearbox monitoring for traction drives and hoisting gear
- Drive belt monitoring for fans and conveyor belts
- Blocking protection for conveyor belts

You require an electronic encoder for the "Speed monitoring" function, e.g. a proximity switch. The inverter analyzes an encoder signal at max. 32 kHz.

The function is only available with the CU240E-2 Control Unit via digital input DI 3.

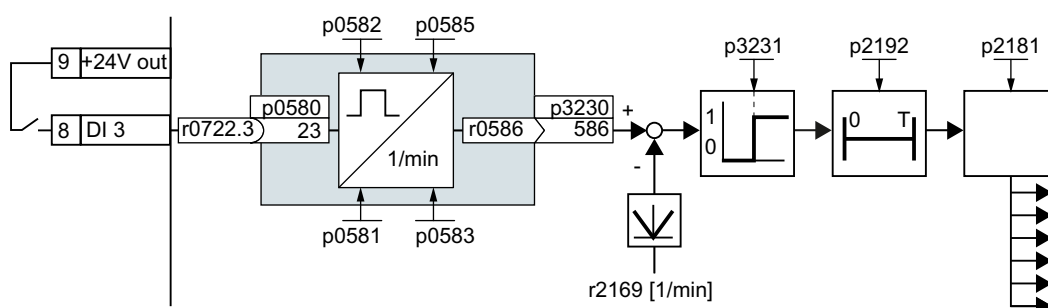


Figure 6-63 Speed deviation monitoring

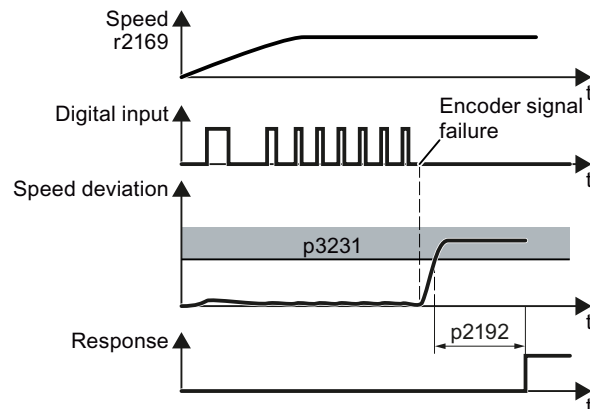


Figure 6-64 Time response of the speed monitoring

The inverter compares speed r0586 with the actual speed value r2169 and signals an excessive deviation between the encoder signal and the motor speed. p2181 specifies the inverter response for an excessive deviation.

Parameter	Description
p0490	Invert probe ¹⁾ (factory setting 0000bin) Using the 3rd bit of the parameter value, you invert the input signals of digital input 3 for the probe.
p0580	Probe ¹⁾ Input terminal (factory setting 0) Connect input of probe with a digital input.

Parameter	Description
p0581	Probe ¹⁾ Edge (factory setting 0) Edge for analyzing the probe signal for measuring the actual speed value 0: 0/1 edge 1: 1/0 edge
p0582	Probe ¹⁾ Pulse per revolution (factory setting 1) Number of pulses per revolution
p0583	Probe ¹⁾ Maximum measurement time (factory setting 10 s) Maximum measurement time for the probe. If there is no new pulse before the maximum measuring time elapses, the inverter sets the actual speed value in r0586 to zero. The time is restarted with the next pulse.
p0585	Probe ¹⁾ Gear ratio (factory setting 1) The inverter multiplies the measured speed by the gear ratio before displaying it in r0586.
r0586	Probe ¹⁾ Actual speed value Result of the speed calculation
p2181	Load monitoring response 1: A07920 for torque too low 2: A07921 for torque too high 3: A07922 for torque outside tolerance 4: F07923 for torque too low 5: F07924 for torque too high 6: F07925 for torque outside tolerance
p2192	Load monitoring delay time (factory setting 10 s) Delay time for evaluating load monitoring.
p2193 = 2	Load monitoring configuration (factory setting: 1) 0: Monitoring switched off 1: Monitoring torque and load drop 2: Monitoring speed and load drop 3: Load failure monitoring
p3230 = 586	Load monitoring actual speed value (factory setting 0) Result of the speed calculation with the evaluation of the speed monitoring.
p3231	Load monitoring speed deviation (factory setting 150 rpm) Permissible speed deviation of load monitoring.

¹⁾ The "Probe" subfunction calculates the speed from the pulse signal of the digital input.

For more information, see the List Manual (the parameter list and function diagram 8013).

6.26 Flying restart – switching on while the motor is running



If you switch on the motor while it is still rotating, without the "Flying restart" function, there is a high probability that a fault will occur as a result of overcurrent (F30001 or F07801). Examples of applications involving an unintentionally rotating motor directly before switching on:

- The motor rotates after a brief line interruption.
- A flow of air turns the fan impeller.
- A load with a high moment of inertia drives the motor.

Principle of operation

The "Flying restart" function comprises the following steps:

1. After the on command, the inverter impresses the search current in the motor and increases the output frequency.
2. When the output frequency reaches the actual motor speed, the inverter waits for the motor excitation build up time.
3. The inverter accelerates the motor to the actual speed setpoint.

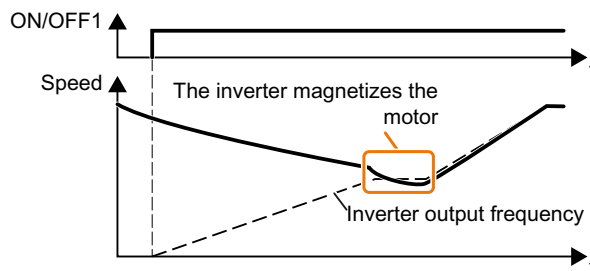


Figure 6-65 Principle of operation of the "flying restart" function

Setting "flying restart" function

Parameter	Description
p1200	Flying restart operating mode (factory setting: 0)
	0 Flying restart is disabled
	1 Flying restart is enabled, search for the motor in both directions, start in the direction of the setpoint
	4 Flying restart is enabled, search for the motor only in the direction of the setpoint

No "Flying restart" function for group drives

It is not permissible that you enable the "Flying restart" function if the inverter is simultaneously driving several motors.

6.26 Flying restart – switching on while the motor is running

Exception: a mechanical coupling ensures that all of the motors always operate with the same speed.

Table 6-55 Advanced settings

Parameter	Description
p0346	Motor excitation build up time Wait time between switching on the motor and enabling the ramp-function generator.
p0347	Motor de-excitation time Within the motor de-excitation time, after an OFF command, the inverter prevents the induction motor from being switched on again. When using a reluctance motor, the inverter ignores the motor de-excitation time.
p1201	Flying restart enable signal source (factory setting: 1) Defines a control command, e.g. a digital input, which enables the flying restart function.
p1202	Flying restart search current (Factory setting depends on the Power Module) Defines the search current with respect to the magnetizing current (r0331), which flows in the motor during the flying restart.
p1203	Flying restart search current factor (Factory setting depends on the Power Module) The value influences the speed with which the output frequency is changed during the flying restart. A higher value results in a longer search time. If the inverter does not find the motor, reduce the search speed (increase p1203).

6.27 Automatic restart



The automatic restart includes two different functions:

- The inverter automatically acknowledges faults.
- After a fault occurs or after a power failure, the inverter automatically switches-on the motor again.

The inverter interprets the following events as power failure:

- The inverter signals fault F30003 (undervoltage in the DC link), after the inverter line voltage has been briefly interrupted.
- All of the inverter power supplies have been interrupted - and all of the energy storage devices in the inverter have discharged to such a level that the inverter electronics fail.

Setting the automatic restart function



WARNING

Danger to life as a result of machine parts unexpectedly starting

When the "automatic restart" function is active ($p1210 > 1$), the motor automatically starts after a power failure. Unexpected movement of machine parts can result in serious injury and material damage.

- Block off hazardous areas within the machine to prevent inadvertent access.

If it is possible that the motor is still rotating for a longer period of time after a power failure or after a fault, then you must also activate the "flying restart" function.



Flying restart – switching on while the motor is running (Page 333)

Using $p1210$, select the automatic restart mode that best suits your application.

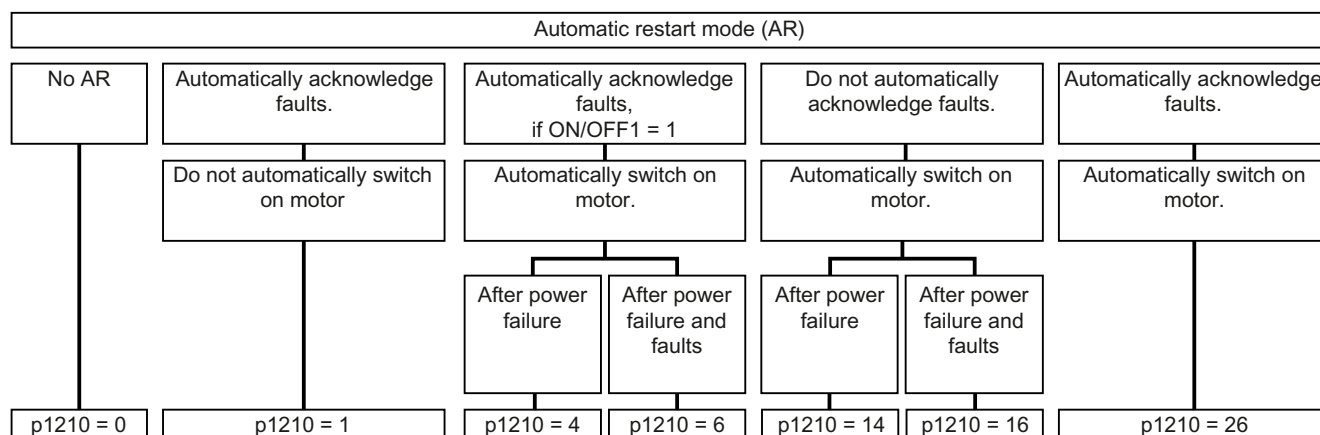
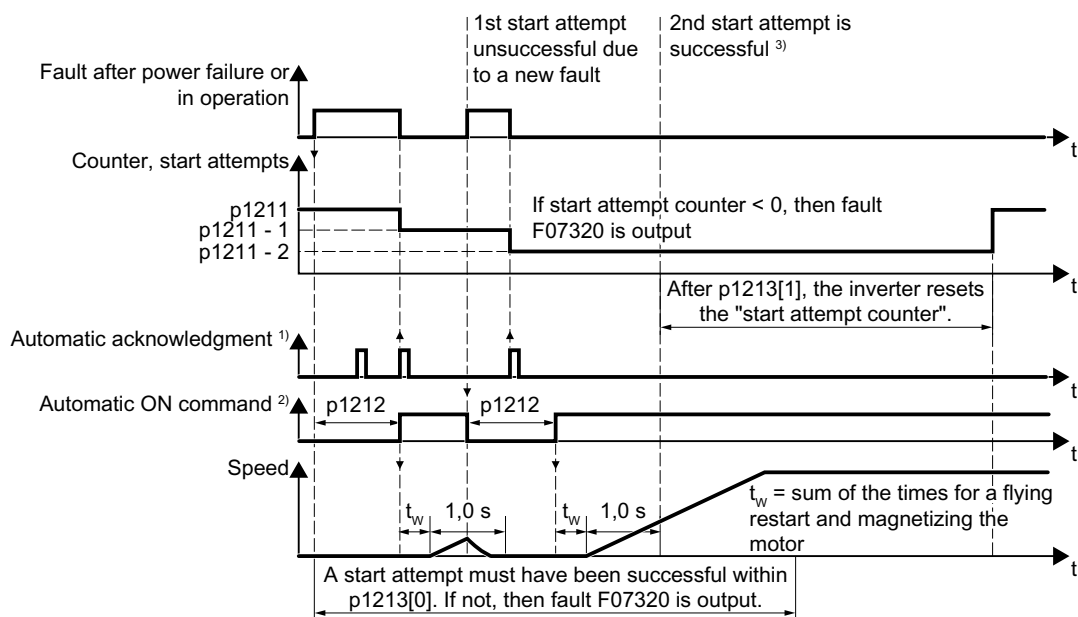


Figure 6-66 Automatic restart modes

The principle of operation of the other parameters is explained in the following diagram and in the table below.



¹⁾ The inverter automatically acknowledges faults under the following conditions:

- p1210 = 1 or 26: Always.
- p1210 = 4 or 6: If the command to switch-on the motor is available at a digital input or via the fieldbus (ON/OFF1 = 1).
- p1210 = 14 or 16: Never.

²⁾ The inverter attempts to automatically switch the motor on under the following conditions:

- p1210 = 1: Never.
- p1210 = 4, 6, 14, 16, or 26: If the command to switch-on the motor is available at a digital input or via the fieldbus (ON/OFF1 = 1).

³⁾ If, after a flying restart and magnetization (r0056.4 = 1) no fault occurs within one second, then the start attempt was successful.

Figure 6-67 Time response of the automatic restart

Parameter for setting the automatic restart

Parameter	Explanation
p1210	Automatic restart mode (factory setting: 0) 0: Disable automatic restart. 1: Acknowledge all faults without restarting. 4: Restart after power failure without further restart attempts. 6: Restart after fault with further restart attempts. 14: Restart after power failure after manual acknowledgement. 16: Restart after fault after manual acknowledgement. 26: Acknowledgement of all faults and restart with ON/OFF1 = 1 command.

Parameter	Explanation
p1211	Automatic restart start attempts (factory setting: 3) This parameter is only effective for the settings p1210 = 4, 6, 14, 16, 26. You define the maximum number of start attempts using p1211. After each successful acknowledgement, the inverter decrements its internal counter of start attempts by 1. p1211 = 0 or 1: The inverter only tries to start once. After an unsuccessful start attempt, the inverter issues fault F07320. p1211 = n, n > 1: The inverter tries to start n-times. The inverter outputs fault F07320 if the nth starting attempt was unsuccessful. The inverter sets the start attempt counter back again to the value of p1211, if one of the following conditions is fulfilled: • After a successful start attempt, the time in p1213[1] has expired. • After fault F07320, switch off the motor (OFF1) and acknowledge the fault. • You change the start value p1211 or the mode p1210.
p1212	Automatic restart wait time start attempt (factory setting: 1.0 s) This parameter is only effective for the settings p1210 = 4, 6, 26. Examples for setting this parameter: 1. After a power failure, a certain time must elapse before the motor can be switched-on, e.g. because other machine components are not immediately ready. In this case, set p1212 longer than the time, after which all of the fault causes have been removed. 2. In operation, the inverter develops a fault condition. The lower you select p1212, then the sooner the inverter attempts to switch-on the motor again.
p1213[0]	Automatic restart monitoring time for restart (factory setting: 60 s) This parameter is only effective for the settings p1210 = 4, 6, 14, 16, 26. With this monitoring function, you limit the time in which the inverter may attempt to automatically switch-on the motor again. The monitoring function starts when a fault is identified and ends with a successful start attempt. If the motor has not successfully started after the monitoring time has expired, fault F07320 is signaled. Set the monitoring time longer than the sum of the following times: + p1212 + Time that the inverter requires to start the motor on the fly. + Motor magnetizing time (p0346) + 1 second You deactivate the monitoring function with p1213 = 0.
p1213[1]	Automatic restart monitoring time to reset the fault counter (factory setting: 0 s) This parameter is only effective for the settings p1210 = 4, 6, 14, 16, 26. Using this monitoring time, you prevent that faults, which continually occur within a certain time period, are automatically acknowledged each time. The monitoring function starts with a successful start attempt and ends after the monitoring time has expired. If, during the monitoring time p1213[1], the inverter made more successful starting attempts than defined in p1211, the inverter interrupts the automatic restart function and signals fault F07320. To switch on the motor again you must acknowledge the fault and switch on the inverter (ON/OFFS1 = 1).

Additional information is provided in the parameter list of the List Manual.

Advanced settings

If you wish to suppress the automatic restart function for certain faults, then you must enter the appropriate fault numbers in p1206[0 ... 9].

Example: p1206[0] = 07331 ⇒ No restart for fault F07331.

Suppressing the automatic restart only functions for the setting p1210 = 6, 16 or 26.



WARNING

Danger to life if an OFF command has no effect.

When the inverter is only controlled via the fieldbus interface, the motor restarts with the setting p1210 = 6, 16, 26 – even if the communication link is currently interrupted. When communication is interrupted, the control system cannot switch off the motor.

- Enter the fault number of the communication error in parameter p1206.
Example:
Fault number F01910 means: Communication via PROFIBUS failed.
Set p1206[n] = 1910 (n = 0 ... 9).

6.28 Kinetic buffering (V_{DC min} control)



Kinetic buffering increases the drive availability. The kinetic buffering utilizes the kinetic energy of the load to buffer line dips and failures. During a line dip, the inverter keeps the motor in the switched-on state for as long as possible. One second is a typical, maximum buffer time.

Preconditions

The following prerequisites must be fulfilled to practically use the "kinetic buffering" function:

- The driven load has a sufficiently high inertia.
- The application allows a motor to be braked during a power failure.

The V_{DC min} control is possible only with the PM240-2 and PM240P-2 Power Modules.

Function

When the line supply dips or is interrupted, the DC link voltage in the inverter decreases. At an adjustable threshold, kinetic buffering intervenes (V_{DC min} control). The V_{DC min} control forces the load to go into slightly regenerative operation. As a consequence, the inverter covers its power loss and the losses in the motor with the kinetic energy of the load. The load speed decreases; however, during kinetic buffering, the DC voltage remains constant. After the line supply returns, the inverter immediately resumes normal operation.

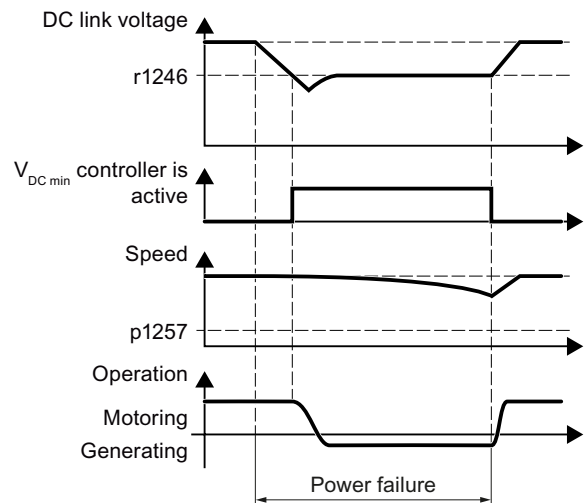


Figure 6-68 Principle mode of operation of kinetic buffering

Parameter	Description
r0056.15	Status word closed-loop control
	0 signal V _{DC min} controller is not active
	1 signal V _{DC min} controller is active (kinetic buffering)
p0210	Device supply voltage (factory setting: 400 V)

Parameter	Description
p1240	V_{DC} controller configuration (factory setting: 1)
	0 Inhibit V_{DC} controller
	1 Enable $V_{DC\ max}$ controller
	2 Enable $V_{DC\ min}$ controller (kinetic buffering)
	3 Enable $V_{DC\ min}$ controller and $V_{DC\ max}$ controller
p1245	$V_{DC\ min}$ controller activation level (kinetic buffering) (factory setting depends on the Power Module, 73 % or 76 %)
r1246	$V_{DC\ min}$ controller activation level [V] $r1246 = p1245 \times \sqrt{2} \times p0210$
p1247	$V_{DC\ min}$ controller dynamic factor (factory setting: 300 %)
p1255	$V_{DC\ min}$ controller time threshold (factory setting: 0 s) Maximum duration of the kinetic buffering. If kinetic buffering lasts longer than that specified in the parameter value, the inverter outputs fault F7406. A value of 0 deactivates the monitoring.
p1257	$V_{DC\ min}$ controller speed threshold (factory setting: 50 rpm) When fallen below, the inverter outputs fault F7405.

6.29 Line contactor control



A line contactor disconnects the inverter from the line supply, and therefore reduces the inverter losses when the motor is not operational.

The inverter can control its own line contactor using a digital output. You must supply the inverter with 24 V so that the line contactor control of the inverter also functions when disconnected from the line supply.

Activate line contactor control

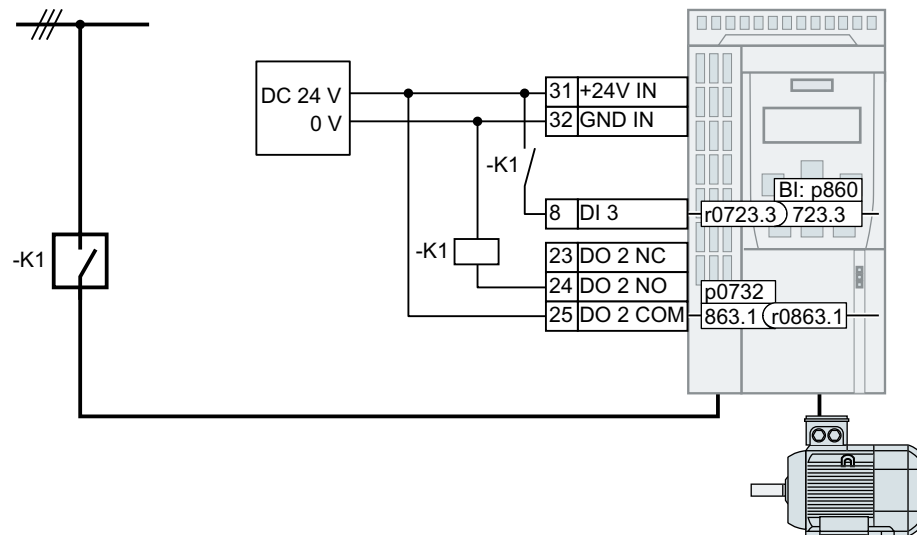


Figure 6-69 Line contactor control via DO 2 with feedback signal via DI 3

In order that the inverter controls line contactor K1 using one of its own digital outputs, you must interconnect the digital output with signal r0863.1 e.g. for DO 2: p0732 = 863.1.

Line contactor control with feedback signal

Interconnect p0860 with the inverted signal of a digital input: p0860 = 723.x.

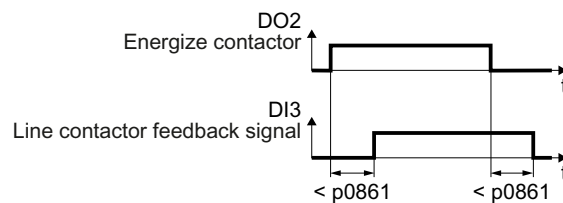


Figure 6-70 Line contactor control via DO 2 with feedback signal via DI 3

If the line contactor feedback signal is not available for longer than the time set in p0861, then the inverter signals fault F07300.

Line contactor control without feedback signal

Interconnect the feedback signal with the signal to control the line contactor: p0860 = 863.1.

Setting the line contactor control

Parameter	Explanation
p0860	Line contactor feedback signal • p0860 = 863.1: no feedback signal (factory setting) • p0860 = 723.x: Feedback signal via DIx
p0861	Line contactor monitoring time (Factory setting: 100 ms) Fault F07300 is output if, for an activated feedback signal, no feedback signal is received via the selected digital input after the time set here has expired.
r0863.1	Drive coupling status/control word Signal to activate the line contactor control
p0867	Line contactor holding time after OFF1 (factory setting: 50 ms) Time for which the line contactor must remain closed after an OFF1.
p0869	Sequence control configuration • p0689 = 0: line contactor immediately opens when the "Safe Torque Off" function becomes active (STO) • p0689 = 1: Line contactor opens when STO is active after the time in p0867 expires

6.30 Calculating the energy saving for fluid flow machines



Fluid flow machines, which mechanically control the flow rate using valves or throttle flaps, operate with a constant speed corresponding to the line frequency.



Figure 6-71 Flow control with pump and throttle connected to a 50 Hz line supply

The lower the flow rate, the poorer the efficiency of the fluid flow machine (pump). The fluid flow machine (pump) has the poorest efficiency when the throttle or valve is completely closed. Further, undesirable effects can occur, for example the formation of vapor bubbles in liquids (cavitation) or the temperature of the medium being pumped can increase.

The inverter controls the flow rate by appropriately varying the speed of the fluid flow machine. As a consequence, the fluid flow machine has the optimum efficiency for each flow rate, and draws less electric power in the partial load range than for closed-loop control concepts based on valves and throttle flaps.

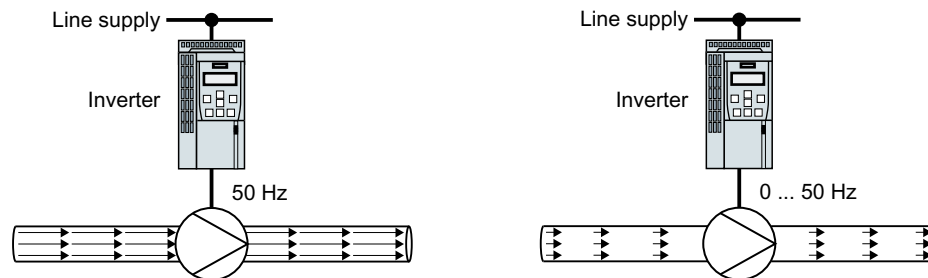
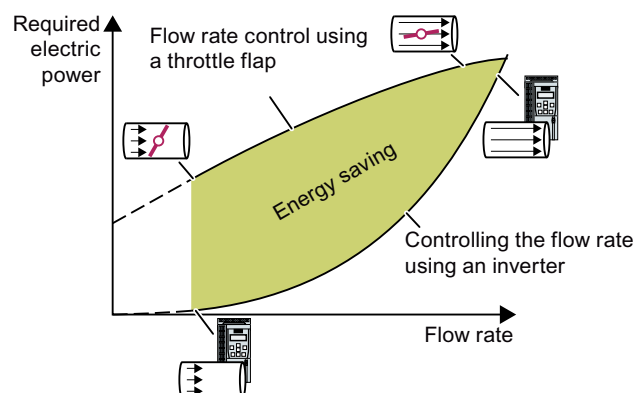


Figure 6-72 Flow control with pump and inverter

Function



The inverter calculates the energy saving from the flow characteristic associated with a mechanical flow control and the measured electric power that is drawn.

The calculation is suitable for centrifugal pumps, fans, radial and axial compressors, for instance.

Parameter	Description
r0039	Energy display [kWh]
	[0] Energy balance Energy usage since the last reset
	[1] Energy drawn since the last reset
	[2] Energy fed back since the last reset
p0040	Reset energy consumption display A signal change 0 → 1 sets r0039[0...2] = 0, r0041 = 0 and r0042 = 0.
r0041	Energy consumption saved (kWh) Energy saved referred to 100 operating hours. For less than 100 operating hours, the inverter interpolates the energy saving to 100 operating hours.
r0042	CO: Process energy display [1 ÷ 1 Wh] For display as process variable. Enable with p0043.
	[0] Energy balance Energy consumption since the last reset.
	[1] Energy drawn since the last reset
	[2] Energy fed back since the last reset.
p0043	BI: Enable energy usage display 1 signal: Process energy display is active in r0042.
p3320 ... p3329	Flow characteristic Power in % Speed in % (Speed, power) ① (p3320, p3321) ② (p3322, p3323) ③ (p3324, p3325) ④ (p3326, p3327) ⑤ (p3328, p3329) Factory setting of the flow characteristic To set the characteristic, you require the following data from the machine manufacturer for each speed interpolation point: • The flow rate of the fluid-flow machine associated with the 5 selected inverter speeds • The power drawn at constant speed associated with the five flow rates corresponds to the line frequency and mechanical throttling of the flow rate.

6.31 Switchover between different settings

There are applications that require different inverter settings.

Example:

You connect different motors to one inverter. Depending on the particular motor, the inverter must operate with the associated motor data and the appropriate ramp-function generator.

Drive data sets (DDS)

You can set several inverter functions differently and then switch over between the different settings.

The associated parameters are indexed (index 0, 1, 2, or 3). Using control commands select one of the four indexes and therefore one of the four saved settings.

The settings in the inverter with the same index are called the drive data set.

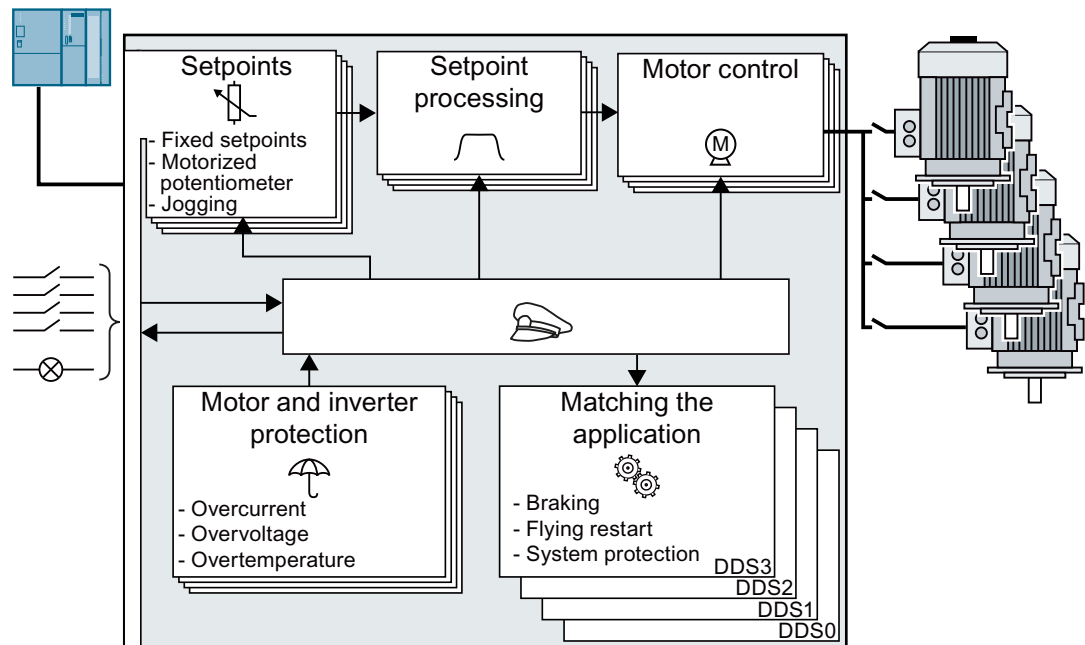


Figure 6-73 Switching over between different settings using drive data sets (DDS)

Using parameter p0180 you can define the number of drive data sets (1 ... 4).

Table 6-56 Selecting the number of **drive data sets**

Parameter	Description
p0010 = 15	Drive commissioning: Data sets
p0180	Drive data sets (DDS) number (factory setting: 1)
p0010 = 0	Drive commissioning: Ready

6.31 Switchover between different settings

Table 6-57 Parameters for switching the drive data sets:

Parameter	Description
p0820[0...n]	Drive data set selection DDS bit 0
p0821[0...n]	Drive data set selection DDS bit 1
	If you use several command data sets CDS, then you must set this parameter for each CDS. The parameters are assigned to a CDS through their index: CDS0: p0820[0], p0821[0] CDS1: p0820[1], p0821[1] ...
p0826	Motor changeover, motor number Each drive data set is assigned a motor number: p0826[0] = motor number for drive data set 0. ... p0826[3] = motor number for drive data set 3. If you operate the same motor with different drive data sets, then you must enter the same motor number in every index of parameter p0826. In this particular case, you can also switch over between the different drive data sets in operation. If you operate different motors on one inverter, then you must number the motors in parameter p0826. In this case, you may only switch over the drive data sets in the "ready for operation" state with the motor switched-off. The switchover time is approx. 50 ms.
r0051	Displaying the number of the DDS that is currently effective

For an overview of all the parameters that belong to the drive data sets and can be switched, see the List Manual.

Table 6-58 Parameters for copying the drive data sets

Parameter	Description
p0819[0]	Source drive data set
p0819[1]	Target drive data set
p0819[2] = 1	Start copy operation

For more information, see the List Manual (the parameter list and function diagram 8565).

Saving the settings and series commissioning

Saving settings outside the inverter

After commissioning, your settings are saved in the inverter so that they are protected against power failure.

We recommend that you additionally back up the settings on a storage medium outside the inverter. Without backup, your settings could be lost if the inverter develops a defect.



Replacing the Control Unit without data backup (Page 404)

The following storage media are available for your settings:

- Memory card
- PG/PC
- Operator panel

Note

Data backup using Operator Panels with USB connection with the PG/PC is not possible

If the inverter is connected with a PG/PC via a USB cable, you cannot backup data to a memory card via an operator panel.

- Disconnect the USB connection between the PG/PC and inverter before you backup data to the memory card via an operator panel.
-

Carrying out series commissioning

Series commissioning is the commissioning of several identical drives.

Precondition

The Control Unit to which the settings are transferred has the same article number and the same or a higher firmware version as the source Control Unit.

Overview

You must proceed as follows to carry out series commissioning:

1. Commission the first inverter.
2. Back up the settings of the first inverter to an external storage medium.
3. Transfer the settings from the first inverter to an additional inverter via the data storage medium.

7.1 Backing up and transferring settings using a memory card

7.1.1 Memory cards

Recommended memory cards



Table 7-1 Memory cards to back up inverter settings

Scope of delivery	Article number
Memory card without firmware	6SL3054-4AG00-2AA0
Memory card with firmware V4.7	6SL3054-7EH00-2BA0
Memory card with firmware V4.7 SP3	6SL3054-7TB00-2BA0
Memory card with firmware V4.7 SP6	6SL3054-7TD00-2BA0

Using memory cards from other manufacturers

The inverter only supports memory cards up to 2 GB. SDHC cards (SD High Capacity) and SDXC cards (SD Extended Capacity) are not permitted.

If you use other SD or MMC memory cards, then you must format the memory card as follows:

- MMC: Format FAT 16
 - Insert the card into your PC's card reader.
 - Command to format the card:
format x: /fs:fat (x: Drive code of the memory card on your PC)
- SD: Format FAT 16 or FAT 32
 - Insert the card into your PC's card reader.
 - Command to format the card:
format x: /fs:fat or format x: /fs:fat32 (x: Drive code of the memory card on your PC.)

Functional restrictions with memory cards from other manufacturers

The following functions are either not possible – or only with some restrictions – when using memory cards from other manufacturers:

- Licensing functions is only possible using the recommended memory cards.
- Know-how protection is only possible with one of the recommended memory cards.
- Under certain circumstances, memory cards from other manufacturers do not support writing or reading data from/to the inverter.

7.1.2 Saving setting on memory card

We recommend that you insert the memory card before switching on the inverter. The inverter always also backs up its settings on an inserted card.

If you wish to back up the inverter settings on a memory card, you have two options:

Automatically backing up

Preconditions

- The inverter power supply has been switched off.
- No USB cable is inserted in the inverter.

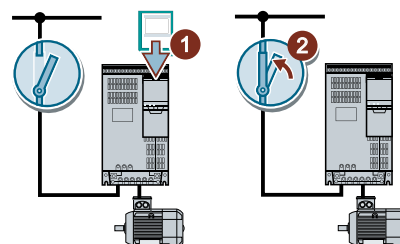
Procedure



1.
2.

Proceed as follows to automatically back up your settings:

1. Insert an empty memory card into the inverter.
2. Switch on the operating voltage of the inverter.



After the power supply has been switched on, the inverter copies its changed settings to the memory card.

Note

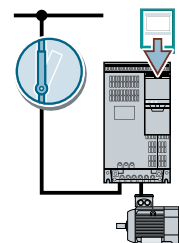
If the memory card is not empty, then the inverter accepts the data from the memory card. This data then overwrites the data in the inverter.

- Only use empty memory cards when automatically backing up your settings for the first time.

Manually backing up

Requirements

- The inverter power supply has been switched on.
- No memory card is inserted in the inverter.



Procedure with STARTER

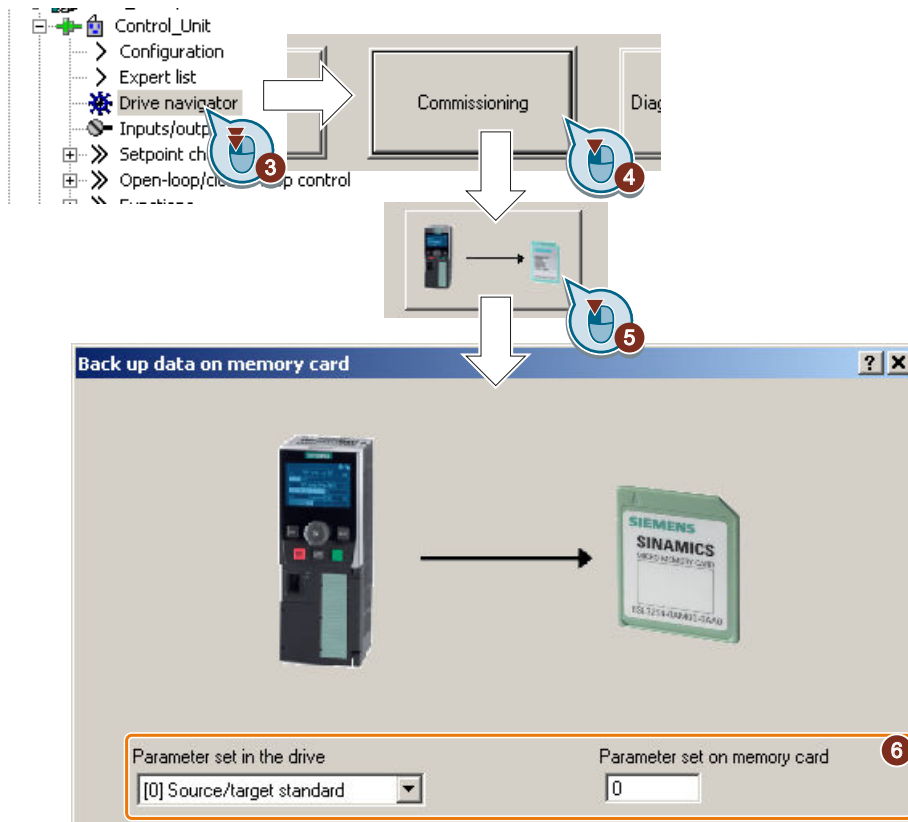


1.
2.

Proceed as follows to back up your settings on a memory card:

1. Go online.
2. Press the “Copy RAM to ROM” button .

3. In your drive, select "Drive Navigator".



4. Select the "Commissioning" button.
5. Select the button to transfer the settings to the memory card.
6. Select the settings as shown in the diagram and start the data backup.
7. Wait until STARTER signals that the data backup has been completed.
8. Close the screen forms.

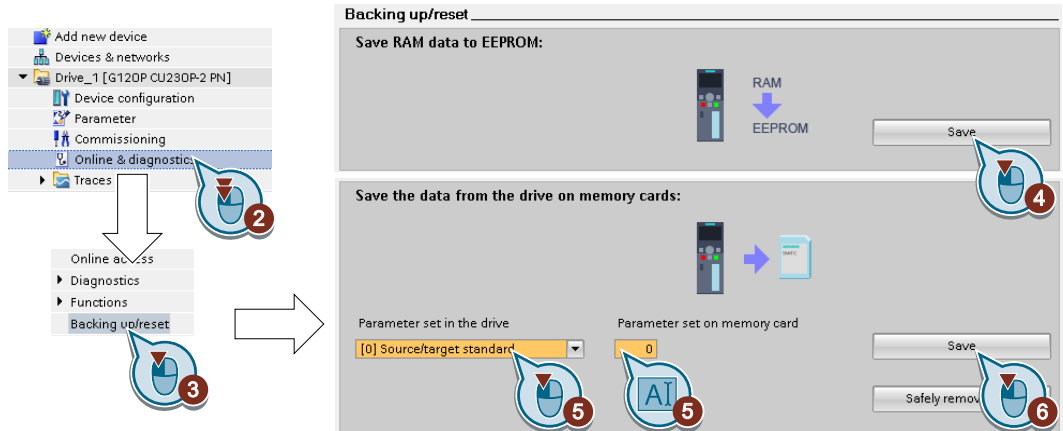
■ You have backed up the settings of the inverter on the memory card.



Procedure with Startdrive

Proceed as follows to back up the inverter settings to a memory card:

1. Go online.
2. Select "Online & diagnostics".



3. Select "Backing up/reset".
4. Back up the settings to the EEPROM of the inverter.
5. Select the settings as shown in the diagram.
6. Start data transfer
7. Wait until Startdrive has signaled that the data backup has been completed.



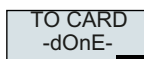
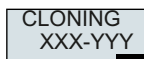
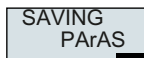
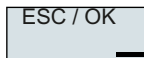
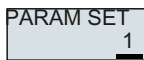
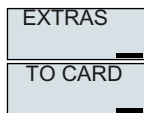
You have backed up the inverter settings to a memory card.

Procedure with BOP-2



Proceed as follows to back up your settings on a memory card:

1. Remove any USB cable that is inserted in the inverter.
2. In the "OPTIONS" menu, select "TO CARD".
3. Set the number of your data backup. You can back up 99 different settings on the memory card.
4. Start data transfer with OK.
5. Wait until the inverter has backed up the settings to the memory card.



You have backed up the settings of the inverter on the memory card.

7.1.3 Transferring the setting from the memory card

Automatically transferring

Precondition

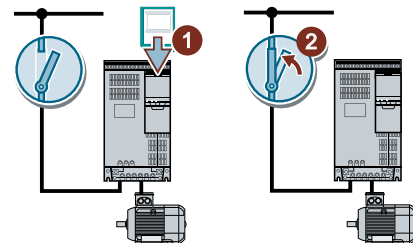
The inverter power supply has been switched off.

Procedure



Proceed as follows to automatically transfer your settings:

1. Insert the memory card into the inverter.
2. Then switch on the inverter power supply.

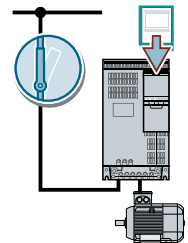


If there is valid parameter data on the memory card, then the inverter accepts the data from the memory card.

Manually transferring

Requirements

- The inverter power supply has been switched on.
- No memory card is inserted in the inverter.



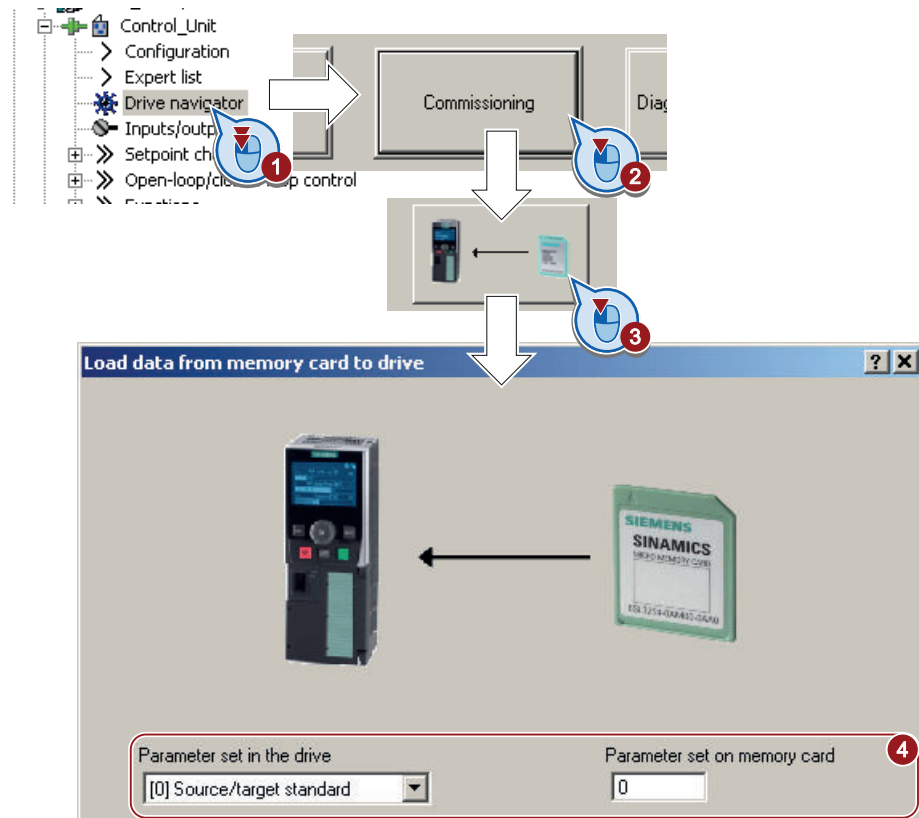
Procedure with STARTER



Proceed as follows to transfer settings from a memory card to the inverter:

1. Go online and in your drive, select the "Drive Navigator".
2. Select the "Commissioning" button.
3. Select the button to transfer the data from the memory card to the inverter.

4. Select the settings as shown in the diagram and start the data backup.



5. Wait until STARTER signals that the data backup has been completed.
6. Close the screen forms.
7. Go offline.
8. Switch off the inverter power supply.
9. Wait until all LEDs on the inverter are dark.
10. Switch on the inverter power supply again.
Your settings become effective after switching on.



You have now transferred your settings from a memory card to the inverter.

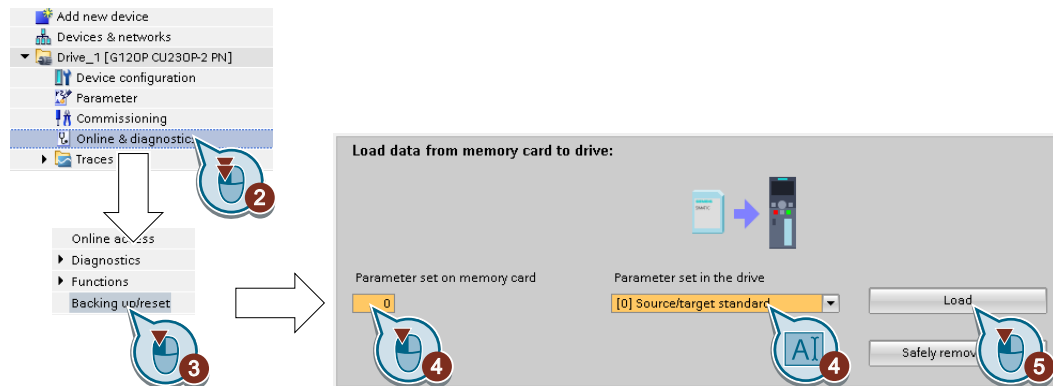
Procedure with Startdrive



Proceed as follows to transfer settings from a memory card to the inverter:

1. Go online.
2. Select "Online & diagnostics".

3. Select "Backing up/reset".



4. Select the settings as shown in the diagram.
 5. Start data transfer
 6. Wait until Startdrive has signaled that the data transfer has been completed.
 7. Go offline.
 8. Switch off the inverter power supply.
 9. Wait until all LEDs on the inverter are dark.
 10. Switch on the inverter power supply again.
- Your settings become effective after switching on.



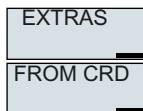
You have now transferred your settings from a memory card to the inverter.

Procedure with the BOP-2

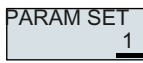


Proceed as follows to transfer settings from a memory card to the inverter:

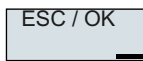
1. Remove any USB cable that is inserted in the inverter.



2. In the "OPTIONS" menu, select "FROM CRD".



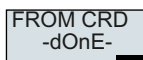
3. Set the number of your data backup. You can back up 99 different settings on the memory card.



4. Start data transfer with OK.



5. Wait until the inverter has transferred the settings from the memory card.



6. Switch off the inverter power supply.
7. Wait until all inverter LEDs are dark.
8. Switch on the inverter power supply again.



You have transferred the settings from the memory card to the inverter.

7.1.4 Safely remove the memory card

NOTICE

Data loss from improper handling of the memory card

If you remove the memory card when the converter is switched on without implementing the "safe removal" function you may destroy the file system on the memory card. The data on the memory card are lost. The memory card will only function again after formatting.

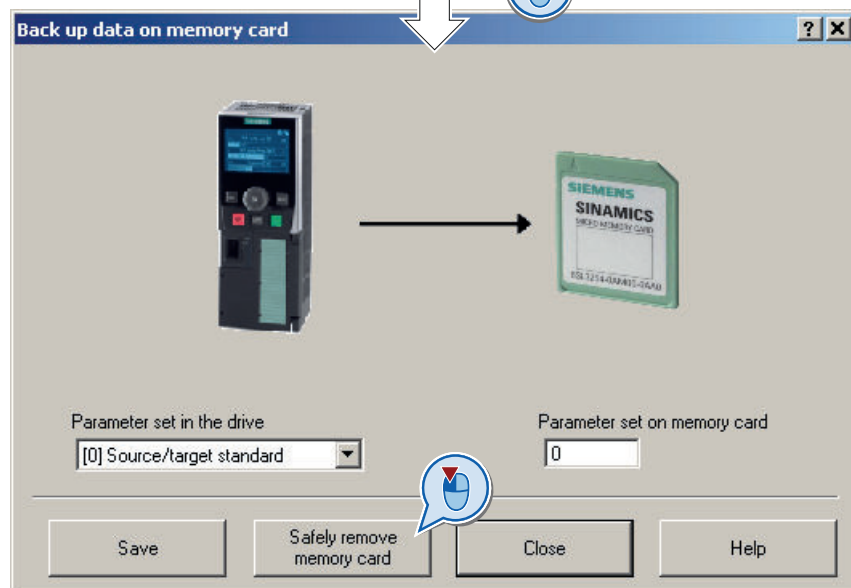
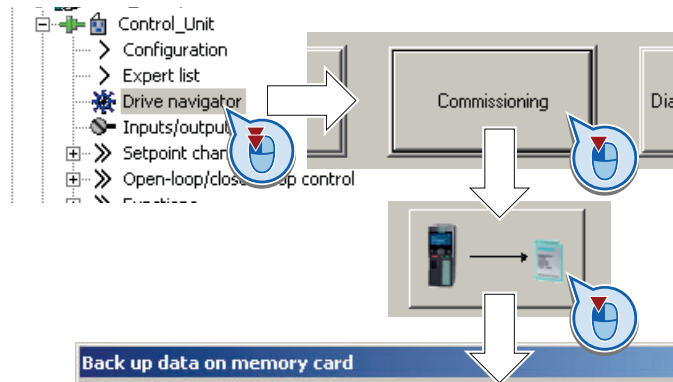
- Only remove the memory card using the "safe removal" function.

Procedure with STARTER



To safely remove the memory card, proceed as follows:

1. Go online.
2. In the Drive Navigator select the following screen form:



3. Click on the button to safely remove the memory card.
STARTER will tell you whether you can remove the memory card from the inverter.



You have now safely removed the memory card from the inverter.

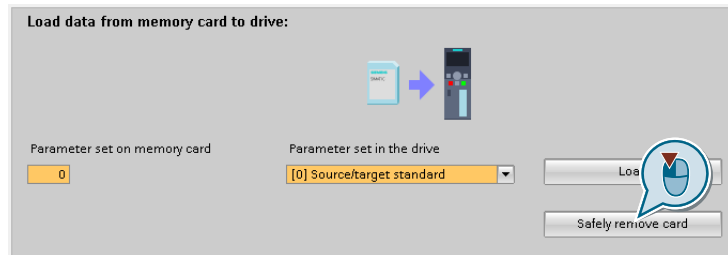
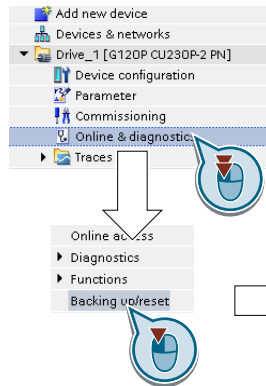
Procedure with Startdrive



1
2

To safely remove the memory card, proceed as follows:

1. In the Drive Navigator select the following screen form:



2. Click on the button to safely remove the memory card.

Startdrive will tell you whether you can remove the memory card from the inverter.



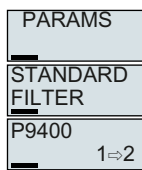
You have now safely removed the memory card from the inverter.

Procedure with the BOP-2



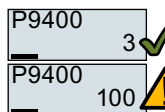
1
2

To safely remove the memory card using BOP-2, proceed as follows:



1. Set p9400 = 2.

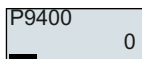
If a memory card is inserted, p9400 = 1.



2. The inverter sets p9400 = 3 or p9400 = 100.

- p9400 = 3: You may remove the memory card from the inverter.

- p9400 = 100: It is not permissible that you remove the memory card. Wait for several seconds and then set p9400 = 2 again.



3. Remove the memory card. After removing the memory card, p9400 = 0.



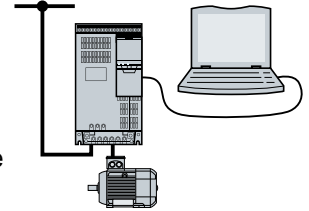
You have now safely removed the memory card using BOP-2.

7.2 Saving the settings to a PC

You can transfer the inverter settings to a PG/PC, or vice versa, the data from a PG/PC to the inverter.

Requirements

- The inverter power supply has been switched on.
- One of the commissioning tools STARTER or Startdrive is installed on the PG/PC.
-  Tools to commission the inverter (Page 130)
- PC and inverter are connected with one another via a USB cable or the fieldbus



Inverter → PC/PG

Procedure with STARTER

- ➔ 1. To back up the settings with STARTER, proceed as follows:
2.
 1. Go online: .
 2. Click the "Download project to PG" button: .
 3. Save the project: .
 4. Wait until STARTER reports that data backup has been completed.
 5. Go offline: .
- You have backed up the settings with STARTER.

Procedure with Startdrive

- ➔ 1. To back up the settings with Startdrive, proceed as follows:
2.
 1. Go online.
 2. Select "Online" > "Upload device to PG/PC."
 3. Back up the project with "Project" > "Save."
 4. Wait until Startdrive reports that data backup has been completed.
 5. Go offline.
- You have backed up the settings with Startdrive.

PC/PG → inverter

The procedure depends on whether you also transfer settings of safety functions or not.

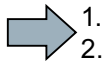
Procedure with STARTER without enabled safety functions

- ➔ 1. To load the settings from the PG to the inverter with STARTER, proceed as follows:
2.
 1. Go online: .
 2. Click the "Download project to target system" button: .

3. Wait until STARTER reports that loading has been completed.
4. To save the data to the non-volatile memory of the inverter, select the "Copy RAM to ROM" button: .
5. Go offline: .

■ You have transferred the settings from the PG to the inverter with STARTER.

Procedure with Startdrive without enabled safety functions



1. To transfer the settings from the PG to the inverter with Startdrive, proceed as follows:

2.
 1. Go online.
 2. Select "Load to device" > "Hardware and software" from the shortcut menu.
 3. Wait until Startdrive reports that loading has been completed.
 4. Go offline.
 5. Confirm the dialog box that then opens by clicking "Yes" to save the data to the non-volatile memory of the inverter (Copy from RAM to ROM).

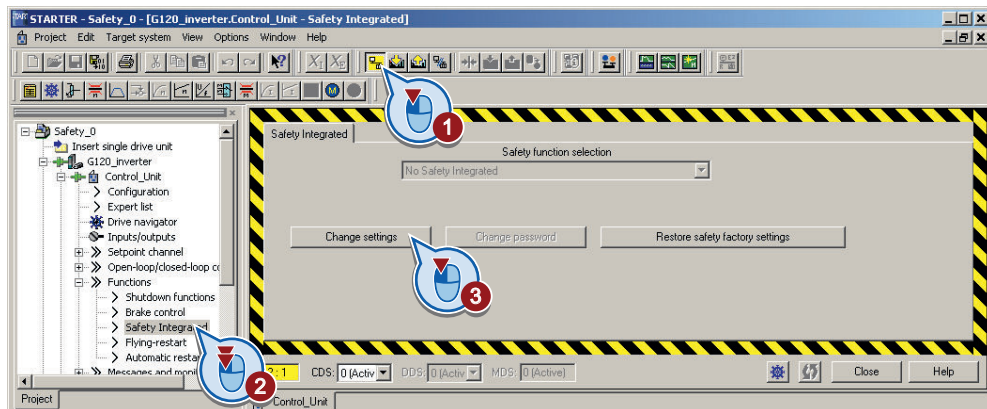
■ You have transferred the settings from the PG to the inverter with Startdrive.

Procedure with STARTER with enabled safety functions



1. To load the settings from the PG to the inverter with STARTER and to activate the safety functions, proceed as follows:

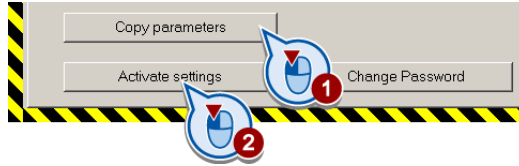
2.
 1. Go online: .
 2. Click the "Download project to target system" button: .
 3. Call the STARTER screen form for the safety functions.



You have transferred the settings from the PG to the inverter.

To activate the safety functions, proceed as follows:

1. Select the "Copy parameter" button.
2. Press the "Activate settings" button.



3. To save the data in the inverter, click the "Copy RAM to ROM" button: .
4. Go offline: .
5. Switch off the inverter power supply.
6. Wait until all LEDs on the inverter go off.
7. Switch on the inverter power supply again. Your settings only become effective after this power-on reset.

 You have transferred the settings from the PG to the inverter with STARTER and activated the safety functions.

Procedure with Startdrive when the safety functions are enabled

- ➔ 1. To transfer the settings from the PG to the inverter with Startdrive and activate the safety functions, proceed as follows:

1. Save the project.
2. Select "Load to device."

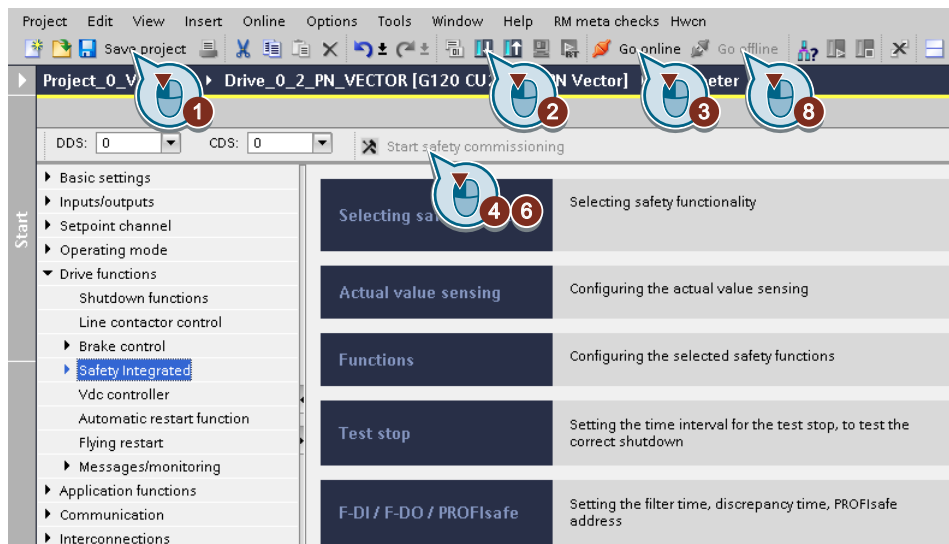


Figure 7-1 Activating settings in Startdrive

3. Connect Startdrive online with the drive.
4. Click the "Start safety commissioning" button.

7.2 Saving the settings to a PC

5. Enter the password for the safety functions.
If the password is the factory default, you are prompted to change the password.
If you try to set a password that is not permissible, the old password will not be changed.
6. Click the "End safety commissioning" button.
7. Confirm the prompt for saving your settings (copy RAM to ROM).
8. Disconnect the online connection.
9. Switch off the inverter supply voltage.
10. Wait until all LEDs on the inverter go dark (no voltage condition).
11. Switch on the inverter supply voltage again.



You have transferred the settings from the PG to the inverter with Startdrive and have activated the safety functions.

7.3 Saving settings to an operator panel

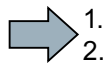
You can transfer the inverter settings to the Operator Panel BOP-2 or vice versa, the data from the BOP-2 to the inverter.

Precondition

The inverter power supply has been switched on.

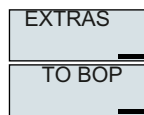
Inverter → BOP-2

Procedure

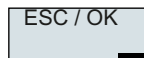


1.
2.

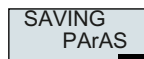
To back up the settings on the BOP-2, proceed as follows:



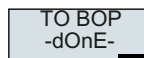
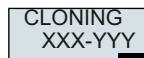
1. In the "OPTIONS" menu, select "TO BOP".



2. Start data transfer with OK.



3. Wait until the inverter has backed up the settings to the BOP-2.



You have backed up the settings on the BOP-2.

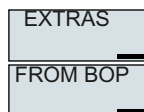
BOP-2 → inverter

Procedure

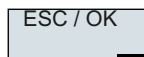


1.
2.

To transfer the settings to the inverter, proceed as follows:



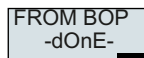
1. In the "OPTIONS" menu, select "FROM BOP".



2. Start data transfer with OK.



3. Wait until the inverter has written the settings to the memory card.



4. Switch off the inverter power supply.

7.3 Saving settings to an operator panel

5. Wait until all inverter LEDs are dark.

6. Switch on the inverter power supply again. Your settings become effective after switching on.



You have transferred the settings to the inverter.

7.4 Other ways to back up settings

In addition to the default setting, the inverter has an internal memory for backing up three other settings.

On the memory card, you can back up 99 other settings in addition to the default setting.



Additional information is available in the Internet: Memory options (<http://support.automation.siemens.com/WW/view/en/43512514>).

7.5 Write protection

The write protection prevents unauthorized changing of the inverter settings. If you are working with a PC tool, such as STARTER, then write protection is only effective online. The offline project is not write-protected.

Write protection is applicable for all user interfaces:

- BOP-2 and IOP operator panels
- STARTER or Startdrive PC tool
- Parameter changes via fieldbus

No password is required for write protection.

Activate and deactivate write protection

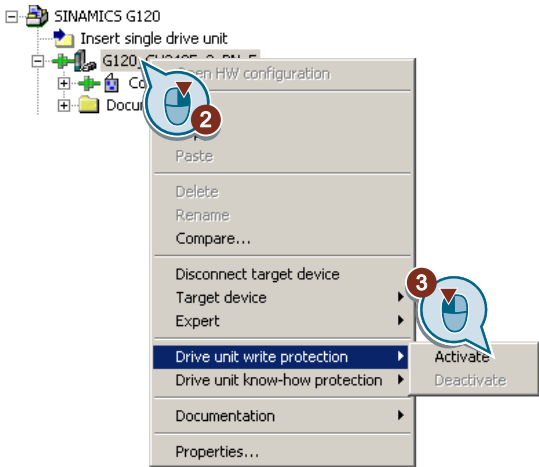
Procedure with STARTER

- 

1.

2.
- Proceed as follows to activate or deactivate the write protection:

 1. Go online.
 2. Open the shortcut menu of the required inverter.
 3. Activate or deactivate write protection.
 4. Press the “Copy RAM to ROM” button to retentively save the settings.



 You have activated or deactivated write protection.

Active write protection can be identified as in the expert list the input fields of adjustable parameters p ... are shaded gray.

Parameters		
r7760	Write protection/know-how protection status	
	.00	1 signal: Write protection active
p7761	Write protection (factory setting: 0)	
	0:	Deactivate write protection
	1:	Activate write protection

Exceptions to write protection

Some functions are excluded from write protection, e.g.:

- Activating/deactivating write protection
- Changing the access level (p0003)
- Saving parameters (p0971)
- Safely removing the memory card (p9400)
- Restoring the factory setting
- Transfer the settings from an external data backup, e.g. upload into the inverter from a memory card.

The parameters that are not write protected are in the List Manual in Section "Parameters for write protection and know-how protection".

Note

Write protection for multimaster fieldbus systems

Via multimaster fieldbus systems, e.g. BACnet, in spite of write protection being activated, parameters can still be changed. So that write protection is also active when accessing via these fieldbuses, you must additionally set p7762 to 1.

In STARTER and Startdrive, this setting is only possible using the expert list.

7.6 Know-how protection

Know-how protection prevents unauthorized reading of the inverter settings.

A password is required to activate or to change know-how protection.

Know-how protection with and without copy protection.

To protect your inverter settings against unauthorized copying, in addition to know-how protection, you can also activate copy protection.

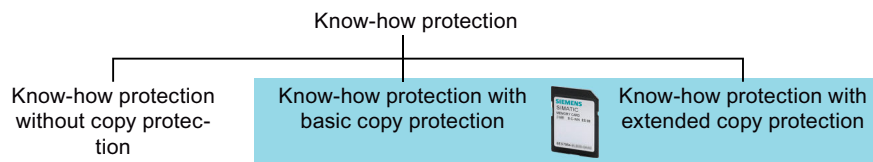


Figure 7-2 Setting options for know-how protection

Know-how protection without copy protection is possible with or without memory card

Know-how protection with copy protection is only possible with a Siemens memory card.



Memory cards (Page 348)

Know-how protection without copy protection

The inverter can be operated with or without memory card. You can transfer inverter settings to other inverters using a memory card, an operator panel STARTER or Startdrive.

Know-how protection with basic copy protection

The inverter can only be operated if the associated memory card with the inverter settings is inserted in the inverter. After replacing an inverter, to be able to operate the new inverter with the settings of the replaced inverter without knowing the password, the memory card must be inserted in the new inverter.

Know-how protection with extended copy protection

The inverter can only be operated if the associated memory card with the inverter settings is inserted in the inverter. It is not possible to insert and use the memory card in another inverter without knowing the password.

Features when know-how protection is active

The active know-how protection provides the following:

- With just a few exceptions, the values of all adjustable parameters p ... are invisible. In STARTER, instead of the parameter values, the text "Know-how protection" is displayed. You can hide know-how protected parameters in the expert list of STARTER using the "Without know-how protection" display filter.
- The values of monitoring parameters r ... remain visible.

- STARTER does not display any screen forms.
- Adjustable parameters cannot be changed using commissioning tools, e.g. an operator panel or Startdrive.

When know-how protection is active, support can only be provided (from Technical Support) after prior agreement from the machine manufacturer (OEM).

Adjustable parameters that can be changed when know-how protection is active

Several adjustable parameters can be read and changed when know-how protection is active. You can find a list of the adjustable parameters that can be read and changed in the List Manual under "KHP_WRITE_NO_LOCK".

In addition, you can define an exception list of adjustable parameters, which end users may change.

Adjustable parameters that can be read when know-how protection is active

Several adjustable parameters can be read but not changed when know-how protection is active. You can find a list of the adjustable parameters that can be read in the List Manual under "KHP_ACTIVE_READ".

Functions that are inhibited when know-how protection is active

Active know-how protection inhibits the following functions:

- Download of inverter settings using STARTER or Startdrive
- Automatic controller optimization
- Stationary or rotating measurement of the motor data identification
- Deleting the alarm history and the fault history
- Generating acceptance documentation for the safety functions

Functions that can be executed when know-how protection is active

The following functions can be executed when know-how protection is active:

- Restoring the factory setting
- Acknowledging faults
- Displaying faults, alarms, fault history and alarm history
- Reading out the diagnostic buffer
- Controlling the inverter via the control panel in STARTER or Startdrive
- Uploading adjustable parameters that can be changed or read when know-how protection is active
- Displaying the generated acceptance documentation for the safety functions

Depending on the know-how protection settings, the trace function in STARTER can also be active when know-how protection is active.

Commissioning know-how protection

Maintain the following sequence:

1. Check as to whether you must extend the exception list.
 List of exceptions (Page 369)
2. Activate the know-how protection.
 Know-how protection (Page 370)

7.6.1 Extending the exception list for know-how protection

In the factory setting, the exception list only includes the password for know-how protection.

Before activating know-how protection, you can additionally enter the adjustable parameters in the exception list, which must still be able to be read and changed by end users – even if know-how protection has been activated.

You do not need to change the exception list, if, with exception of the password, you do not require additional adjustable parameters in the exception list.

Absolute know-how protection

If you remove password p7766 from the exception list, it is no longer possible to enter or change the password for know-how protection.

You must reset the inverter to the factory settings in order to be able to gain access to the inverter adjustable parameters. When restoring the factory settings, you lose what you have configured in the inverter, and you must recommission the inverter.

Extending the exception list

Procedure with STARTER

1. Proceed as follows to extend the exception list for know-how protection:
 2.
 1. Backup the inverter settings using the  button on the PC.
 2. Go offline (.
 3. Using p7763, in the expert list, define the required number of parameters n (n = 1 ... 500) of the exception list.
 4. Save the project.
 5. Go online.
 6. Load the project using the  button in the inverter.
 7. In p7764[0 ... n-1], assign the required parameter numbers to the indices of p7763.
-  You have extended the exception list for know-how protection.

Parameter

Parameter	Description
p7763	KHP OEM exception list number of indices for p7764 (factory setting 1)
p7764	KHP OEM exception list (factory setting [0] 7766, [1 ...499] 0) p7766 is the password for know-how protection

7.6.2 Activating and deactivating know-how protection

Activating know-how protection

Preconditions

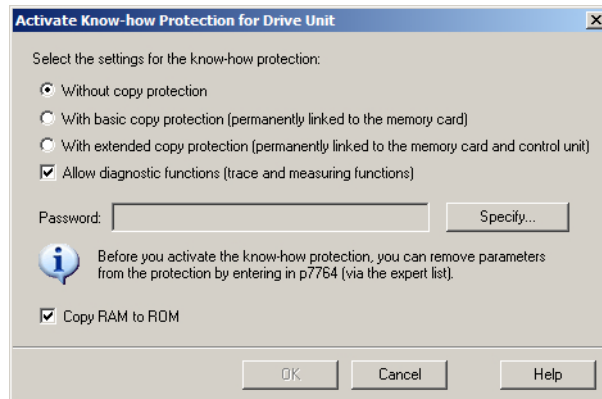
- The inverter has now been commissioned.
- You have generated the exception list for know-how protection.
- To guarantee know-how protection, you must ensure that the project does not remain at the end user as a file.



1. Procedure with STARTER

2. Proceed as follows to activate know-how protection:

1. Go online with STARTER.
If you have generated a project offline on your computer, you must load the project into the inverter and go online.
2. Select the required inverter in the project.
3. In the shortcut menu, select "Know-how protection drive unit/activating".



4. The "Without copy protection" option is active by default. If an appropriate memory card is inserted in the Control Unit, you can select one of two copy protection options:
 - With basic copy protection (permanently linked to the memory card)
 - With extended copy protection (permanently linked to the memory card and Control Unit)

Select the required copy protection option.
5. If, in spite of active know-how protection, you wish to permit diagnostic functions, activate option "Allow diagnostic functions (trace and measuring functions)".
6. Click on "Define"

7. Enter your password. Length of the password: 1 ... 30 characters.
Recommendation for assigning a password:
 - Only use characters from the ASCII set of characters.
If you use arbitrary characters for the password, changing the windows language settings after activating know-how protection can result in problems when subsequently checking a password.
 - For an adequately secure password, the password must have a minimum length of 8 characters, and must include uppercase and lowercase letters as well as a combination of letters, numbers and special characters.
 8. The "Copy RAM to ROM" option is active as standard.
The option must be active in order that the inverter keeps the know-how protection settings after switching off and switching on the power supply.
 9. Click on "OK".
-  You have activated know-how protection.

Preventing data reconstruction from the memory card

As soon as know-how protection has been activated, the inverter only backs up encrypted data to the memory card.

In order to guarantee know-how protection, after activating know-how protection, we recommend that you insert a new, empty memory card. For memory cards that have already been written to, previously backed up data that was not encrypted can be reconstructed.

Changing the password

Procedure with STARTER

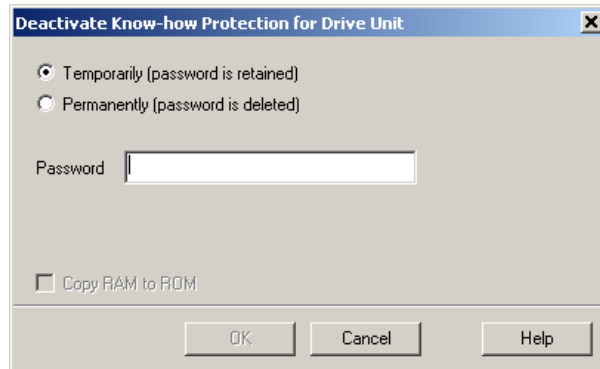
Select the inverter in the project and open the dialog screen form using the shortcut menu "Know-how protection drive unit → Change password ...".

Deactivating know-how protection, deleting a password

Procedure with STARTER

-  1. Proceed as follows to deactivate know-how protection:
2. 1. Go online with STARTER.
 2. Select the required inverter in the project.

3. Using the right-hand mouse key, open the dialog window "Know-how protection drive unit → Deactivate...".



4. Select the required option:
 - Temporary status: Know-how protection is again active after switching off the power supply and switching on again.
 - Final status: Also select "Copy RAM to ROM".
The inverter deletes the password. However, after switching off and switching on the power supply, the password remains deleted.
5. Enter the password for know-how protection.
6. Exit the screen form with OK.



You have deactivated know-how protection.

Parameter

Parameter	Description
r7758[0...19]	KHP Control Unit serial number
p7759[0...19]	KHP Control Unit reference serial number
r7760	Write protection/know-how protection status
	.01 1 signal: Know-how protection active
	.02 1 signal: Know-how protection temporarily unlocked
	.03 1 signal: Know-how protection cannot be deactivated
	.04 1 signal: Extended copy protection active
	.05 1 signal: Basic copy protection active
	.06 1 signal: Trace and measurement functions for diagnostic purposes active
p7765	KHP configuration
p7766[0...29]	KHP password input
p7767[0...29]	KHP password new
p7768[0...29]	KHP password confirmation
p7769[0...20]	KHP memory card reference serial number
r7843[0...20]	Memory card serial number

Alarms, faults and system messages

The inverter has the following diagnostic types:

- LED
The LEDs at the front of the inverter immediately inform you about the most important inverter states.
- System run time
The system run time is the total time that the inverter has been supplied with power since the initial commissioning.
- Alarms and faults
Every alarm and every fault has a unique number.
The inverter signals alarms and faults via the following interfaces:
 - Fieldbus
 - Terminal strip with the appropriate setting
 - Interface to the BOP-2 or IOP operator panel
 - Interface to STARTER or Startdrive
- Identification & maintenance data (I&M)
If requested, the inverter sends data to the higher-level control via PROFIBUS or PROFINET:
 - Inverter-specific data
 - Plant-specific data

8.1 Operating states indicated on LEDs

Table 8-1 Explanation of symbols for the following tables

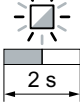
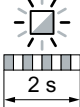
	LED is ON
	LED is OFF
	LED flashes slowly
	LED flashes quickly
	LED flashes with variable frequency

Table 8-2 Basic states

RDY	Explanation
	Temporary state after the supply voltage is switched on.
	The inverter is free of faults
	Commissioning or reset to factory settings
	A fault is active
	Firmware update is active
	Inverter waits until the power supply is switched off and switched on again after a firmware update

Table 8-3 Integrated safety functions

SAFE	Explanation
	One or more safety functions are enabled, but not active.
	One or more safety functions are active and error-free.
	The inverter has detected a safety function fault and initiated a stop response.

Table 8-4 PROFINET fieldbus

LNK	Explanation
	Communication via PROFINET is error-free
	Device naming is active
	No communication via PROFINET

Table 8-5 Fieldbuses via RS 485 interface

BF	Explanation
	Data exchange between the inverter and control system is active
	The fieldbus is active, however, the inverter is not receiving any process data RDY When LED RDY flashes simultaneously: Inverter waits until the power supply is switched off and switched on again after a firmware update
	No fieldbus connection is available RDY When LED RDY flashes simultaneously: Incorrect memory card
	Firmware update failed
	Firmware update is active

Communication via Modbus or USS:

If the fieldbus monitoring is deactivated with p2040 = 0, the BF-LED remains dark, independent of the communication state.

Table 8-6 PROFINET and PROFIBUS fieldbuses

BF	Explanation
	Data exchange between the inverter and control system is active
	Fieldbus interface is not being used
	The fieldbus is improperly configured. RDY When LED RDY flashes simultaneously: Inverter waits until the power supply is switched off and switched on again after a firmware update

8.1 Operating states indicated on LEDs

BF	Explanation	
		No communication with higher-level controller
	RDY 	When LED RDY flashes simultaneously: Incorrect memory card
		Firmware update failed
		Firmware update is active

8.2 System runtime

By evaluating the system runtime of the inverter, you can decide whether you must replace components subject to wear such as fans, motors and gear units.

Principle of operation

The inverter starts the system runtime as soon as the inverter is supplied with power. The system runtime stops when the inverter is switched off.

The system runtime comprises r2114[0] (milliseconds) and r2114[1] (days):

System runtime = r2114[1] × days + r2114[0] × milliseconds

If r2114[0] has reached a value of 86,400,000 ms (24 hours), the inverter sets r2114[0] the value 0 and increases the value of r2114[1] by 1.

Using system runtime, you can track the chronological sequence of faults and alarms over time. When a corresponding message is triggered, the inverter transfers the parameter values r2114 to the corresponding parameters of the alarm or fault buffer.

Parameters	Description
r2114[0]	System runtime (ms)
r2114[1]	System runtime (days)

You cannot reset the system runtime.

8.3 Identification & maintenance data (I&M)

I&M data

The inverter supports the following identification and maintenance (I&M) data.

I&M data	Format	Explanation	Associated parameters	Example for the content
I&M0	u8[64] PROFIBUS u8[54] PROFINET	Inverter-specific data, read only	-	See below
I&M1	Visible String [32]	Plant/system identifier	p8806[0 ... 31]	"ak12-ne.bo2=fu1"
	Visible String [22]	Location code	p8806[32 ... 53]	"sc2+or45"
I&M2	Visible String [16]	Date	p8807[0 ... 15]	"2013-01-21 16:15"
I&M3	Visible String [54]	Any comment	p8808[0 ... 53]	-
I&M4	Octet String[54]	Check signature to track changes for Safety Integrated. This value can be changed by the user. The test signature is reset to the value generated by the machine is p8805 = 0 is used.	p8809[0 ... 53]	Values of r9781[0] and r9782[0]

When requested, the inverter transfers its I&M data to a higher-level control or to a PC/PG with installed STEP 7, STARTER or TIA-Portal.

I&M0

Designation	Format	Example for the content	Valid for PRO-FINET	Valid for PRO-FIBUS
Manufacturer-specific	u8[10]	00 ... 00 hex	---	✓
MANUFACTURER_ID	u16	42d hex (=Siemens)	✓	✓
ORDER_ID	Visible String [20]	„6SL3246-0BA22-1FA0“	✓	✓
SERIAL_NUMBER	Visible String [16]	„T-R32015957“	✓	✓
HARDWARE_REVISION	u16	0001 hex	✓	✓
SOFTWARE_REVISION	char, u8[3]	„V“ 04.70.19	✓	✓
REVISION_COUNTER	u16	0000 hex	✓	✓
PROFILE_ID	u16	3A00 hex	✓	✓
PROFILE_SPECIFIC_TYPE	u16	0000 hex	✓	✓
IM_VERSION	u8[2]	01.02	✓	✓
IM_SUPPORTED	bit[16]	001E hex	✓	✓

8.4 Alarms, alarm buffer, and alarm history

Alarms

Alarms have the following properties:

- Incoming alarms have no direct influence on the inverter.
- Alarms disappear again when the cause is eliminated.
- Alarms do not have to be acknowledged.
- Alarms are displayed as follows:
 - Display via bit 7 in status word 1 (r0052)
 - Display on the operator panel with Axxxxx
 - Display in Startdrive or STARTER

Alarm code or alarm value describe the cause of the alarm.

Alarm buffer

Alarm code		Alarm value		Alarm time received			Alarm time removed	
		I32	float	Days	ms		Days	ms
r2122[0]		r2124[0]	r2134[0]	r2145[0]	r2123[0]	old	r2146[0]	r2125[0]
[1]		[1]	[1]	[1]	[1]	↓ new	[1]	[1]
[2]		[2]	[2]	[2]	[2]		[2]	[2]
[3]		[3]	[3]	[3]	[3]		[3]	[3]
[4]		[4]	[4]	[4]	[4]		[4]	[4]
[5]		[5]	[5]	[5]	[5]		[5]	[5]
[6]		[6]	[6]	[6]	[6]		[6]	[6]
[7]		[7]	[7]	[7]	[7]		[7]	[7]

Figure 8-1 Alarm buffer

The inverter saves incoming alarms in the alarm buffer. An alarm includes an alarm code, an alarm value, and two alarm times:

- Alarm code: r2122
- Alarm value: r2124 in fixed-point format "I32", r2134 in floating-point format "Float"
- Alarm time received = r2145 + r2123
- Alarm time removed = r2146 + r2125

The inverter takes its internal time calculation to save the alarm times.

 System runtime (Page 377)

Up to 8 alarms can be saved in the alarm buffer.

In the alarm buffer, the warnings are sorted according to "Warning time received". If the alarm buffer is completely filled and an additional alarm occurs, then the inverter overwrites the values with Index [7].

Alarm history

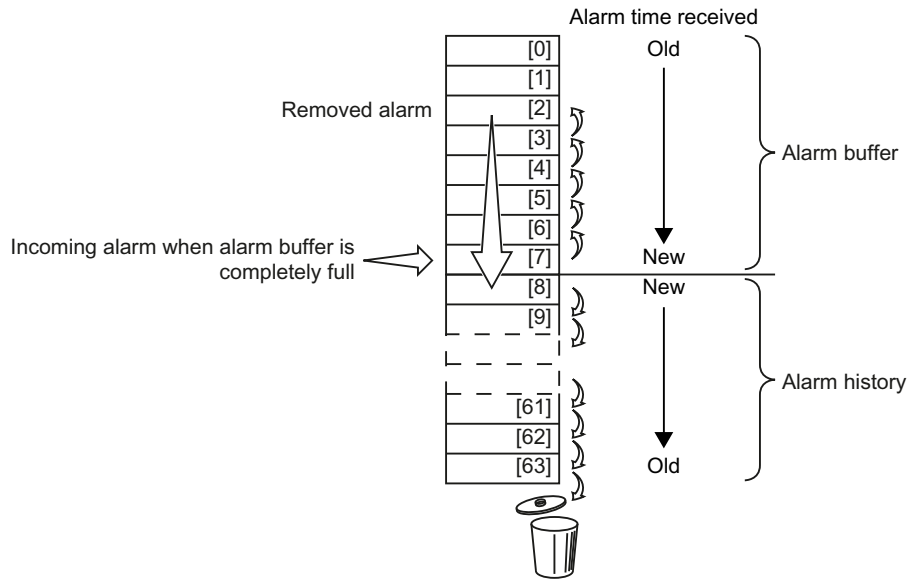


Figure 8-2 Shifting removed alarms into the alarm history

If the alarm buffer is completely filled and an additional alarm occurs, the inverter shifts all removed alarms into the alarm history. The following occurs in detail:

1. To create space after position [8] in the alarm history, the inverter shifts the alarms already stored in the alarm history "down" by one or more positions.
If the alarm history is completely full, the inverter will delete the oldest alarms.
2. The inverter moves the removed alarms from the alarm buffer to the now freed up positions of the alarm history.
Alarms that have not been removed remain in the alarm buffer.
3. The inverter closes gaps in the alarm buffer that occurred when the removed alarms were shifted in the alarm history by shifting the alarms that have not been removed "up".
4. The inverter saves the received alarm as the latest alarm in the alarm buffer.

The alarm history saves up to 56 alarms.

In the alarm history, alarms are sorted according to the "alarm time received". The latest alarm has Index [8].

Parameters of the alarm buffer and the alarm history

Parameter	Description
p2111	Alarm counter Number of alarms that have occurred after the last reset When setting p2111 = 0, all of the alarms that have been removed from the alarm buffer [0...7] are transferred into the alarm history [8...63]
r2122	Alarm code Displays the numbers of the alarms that have occurred

Parameter	Description
r2123	Alarm time received in milliseconds Displays the time in milliseconds when the alarm occurred
r2124	Alarm value Displays additional information about the alarm
r2125	Alarm time removed in milliseconds Displays the time in milliseconds when the alarm was removed
r2145	Alarm time received in days Displays the time in days when the alarm occurred
r2132	Actual alarm code Displays the code of the alarm that last occurred
r2134	Alarm value for float values Displays additional information about the alarm that occurred for float values
r2146	Alarm time removed in days Displays the time in days when the alarm was removed

Extended settings for alarms

Parameter	Description
You can change up to 20 different alarms into a fault or suppress alarms:	
p2118	Setting the message number for the message type Selection of the alarms for which the message type should be changed
p2119	Setting the message type Setting the message type for the selected alarm 1: Fault 2: Alarm 3: No message

You will find details in function diagram 8075 and in the parameter description of the List Manual.

8.5 Faults, alarm buffer and alarm history

Faults

Faults have the following properties:

- In general, a fault leads to the motor being switched off.
- A fault must be acknowledged.
- Faults are displayed as follows:
 - Display in bit 3 of status word 1 (r0052)
 - Display on the operator panel with Fxxxxx
 - Display on the inverter via the LED RDY
 - Display in Startdrive or STARTER

Fault buffer

Fault code		Fault value		Fault time received		Old ↓ New	Fault time removed	
		I32	float	Days	ms		Days	ms
r0945[0]	r0949[0]	r2133[0]		r2130[0]	r0948[0]		r2136[0]	r2109[0]
[1]	[1]	[1]		[1]	[1]		[1]	[1]
[2]	[2]	[2]		[2]	[2]		[2]	[2]
[3]	[3]	[3]		[3]	[3]		[3]	[3]
[4]	[4]	[4]		[4]	[4]		[4]	[4]
[5]	[5]	[5]		[5]	[5]		[5]	[5]
[6]	[6]	[6]		[6]	[6]		[6]	[6]
[7]	[7]	[7]		[7]	[7]		[7]	[7]

Figure 8-3 Fault buffer

The inverter saves incoming faults in the fault buffer. A fault includes a fault code, a fault value, and two fault times:

- Fault code: r0945
The fault code and fault value describe the cause of the fault.
- Fault value: r0949 in fixed-point format "I32", r2133 in floating-point format "Float"
- Fault time received = r2130 + r0948
- Fault time removed = r2136 + r2109

The inverter takes its internal time calculation to save the fault times.

 System runtime (Page 377)

Up to 8 faults can be saved in the fault buffer.

In the fault buffer, the faults are sorted according to "Fault time received". If the fault buffer is completely filled and an additional fault occurs, then the inverter overwrites the values with Index [7].

Acknowledge fault

To acknowledge a fault, you have the following options:

- PROFIdrive control word 1, bit 7 (r2090.7)
- Acknowledging via a digital input
- Acknowledge via the Operator Panel
- Switch off the inverter power supply and switch on again

Faults detected during the inverter-internal monitoring of hardware and firmware can be acknowledged only by switching the supply voltage off and on again. In the list of faults in the List Manual, at the corresponding fault codes you may find the information on limitations when acknowledging.

Fault history

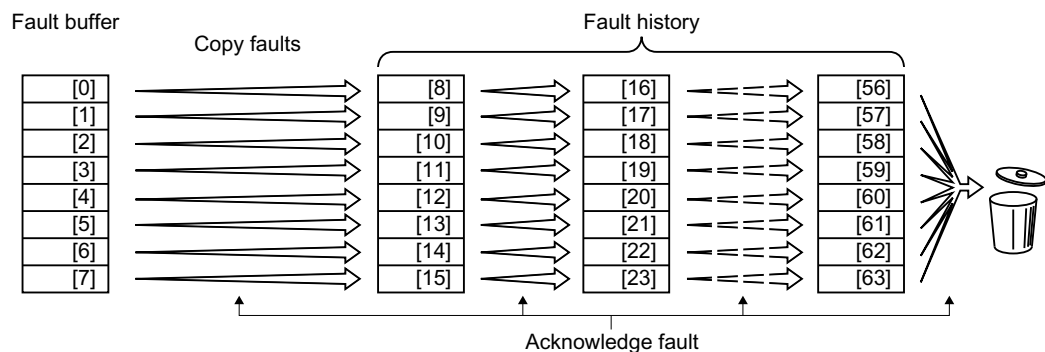


Figure 8-4 Fault history after acknowledging the faults

If at least one of the fault causes in the fault buffer has been removed and you acknowledge the faults, the following takes place:

1. The inverter shifts the values previously saved in the fault history by eight indexes. The inverter deletes the faults that were saved in the indexes [56 ... 63] before the acknowledgement.
2. The inverter copies the contents of the fault buffer to the memory locations [8 ... 15] in the fault history.
3. The inverter deletes the faults that have been removed from the fault buffer. The faults that have not been removed are now saved both in the fault buffer and in the fault history.
4. The inverter writes the time of acknowledgement of the removed faults to "Fault time removed". The "Fault time removed" of the faults that have not been removed retains the value = 0.

The fault history can contain up to 56 faults.

Deleting the fault history

To delete all faults from the fault history, set parameter p0952 to zero.

Parameters of the fault buffer and the fault history

Parameter	Description
r0945	Fault code Displays the numbers of the faults that have occurred
r0948	Fault time received in milliseconds Displays the time in milliseconds when the fault occurred
r0949	Fault value Displays additional information about the fault
p0952	Fault cases, counter A fault case can contain one or several faults. Number of fault cases that have occurred since the last acknowledgement. With p0952 = 0 you delete the fault buffer and the fault history.
r2109	Fault time removed in milliseconds Displays the time in milliseconds when the fault occurred
r2130	Fault time received in days Displays the time in days when the fault occurred
r2131	Actual fault code Displays the code of the oldest fault that is still active
r2133	Fault value for float values Displays additional information about the fault that occurred for float values
r2136	Fault time removed in days Displays the time in days when the fault was removed

Extended settings for faults

Parameter	Description
p2100[0 ... 19]	Setting the fault number for fault response Selecting the faults for which the fault response should be changed. You can modify the motor fault response for up to 20 different fault codes.
p2101[0 ... 19]	Setting, fault response Setting the fault response for the selected fault
p2118[0 ... 19]	Setting the message number for the message type Selection of the message for which the message type should be changed. You can change up to 20 different faults into an alarm, or suppress faults:
p2119[0 ... 19]	Setting the message type Setting the message type for the selected fault 1: Fault 2: Alarm 3: No message

Parameter	Description
p2126[0 ... 19]	Setting the fault number for the acknowledgement mode Selection of the faults for which the acknowledgement type should be changed. You can modify the acknowledgement type for up to 20 different fault codes.
p2127[0 ... 19]	Setting, acknowledgement mode Setting the acknowledgement type for the selected fault 1: Acknowledgement only using POWER ON 2: IMMEDIATE acknowledgement after removing the fault cause

You will find details in function diagram 8075 and in the parameter description of the List Manual.

8.6 List of alarms and faults

Axxxxx Alarm

Fyyyyy: Fault

Table 8-7 The most important alarms and faults

Number	Cause	Remedy
F01000	Software fault in CU	Replacing the Control Unit.
F01001	Floating-point exception	Switch the Control Unit off and on again.
F01015	Software fault in CU	Upgrade firmware or contact technical support.
F01018	Power-up aborted more than once	1. Switch the module off and on again. 2. After this fault has been output, the module is booted with the factory settings. 3. Recommission the inverter.
A01028	Configuration error	Explanation: Parameterization on the memory card has been created with a different type of module (Article number, MLFB). Check the module parameters and recommission if necessary.
F01033	Switching over units: Reference parameter value invalid	Set the value of the reference parameter not equal to 0.0 (p0304, p0305, p0310, p0596, p2000, p2001, p2002, p2003, r2004).
F01034	Switching over units: Calculation of the parameter values after reference value change unsuccessful	Select the value of the reference parameter so that the parameters involved can be calculated in the per unit notation (p0304, p0305, p0310, p0596, p2000, p2001, p2002, p2003, r2004).
F01040	Parameters must be saved	Save parameters (p0971). Switch the Control Unit off and on again.
F01044	Loading memory data card defective	Replace the memory card or the Control Unit.
A01053	System overload measured	The maximum computing power of the Control Unit was exceeded. The following measures reduce the load on the Control Unit: • Use only one data record (CDS and DDS) • Only use the safety features of the basic functions • Deactivate the technology controller • Use the simple ramp-function generator rather than the extended ramp-function generator • Do not use any free function blocks • Reduce the sampling time of the free function blocks
F01054	System limit exceeded	
F01105	CU: Insufficient memory	Reduce number of data sets.
F01122	Frequency at the probe input too high	Reduce the frequency of the pulses at the probe input.
F01205	CU: Time slice overflow	Contact technical support.
F01250	CU hardware fault	Replacing the Control Unit.
F01512	An attempt has been made to establish a conversion factor for scaling which does not exist	Create scaling or check transfer value.
A01590	Motor maintenance interval elapsed	Carry out maintenance and reset the maintenance interval (p0651).
F01600	STOP A triggered	Select the STO safety function and then deselect again.

Number	Cause	Remedy	
F01625	Sign-of-life error in the Safety data	• Check the electrical cabinet design and cable routing for EMC compliance. • Check whether an impermissible voltage is connected at one of the digital outputs. • Check whether a digital output is loaded with an impermissible current. • Check whether additional faults exist and perform diagnostics if applicable. • Select STO safety function and then deselect again. • Switch off the inverter power supply and switch it on again.	
F01650	Acceptance test required	Perform an acceptance test for the safety functions and create the acceptance certificate. To confirm the fault, switch the Control Unit off and on again.	
F01659	Write task for parameter rejected	Cause: The inverter should be reset to the factory setting. However, it is not permissible to reset the safety functions as the safety functions are currently enabled.	
		Remedy with operator panel:	
		p0010 = 30	Parameter reset
		p9761 = ...	If required, enter the password for the safety functions.
		p0970 = 5	Reset start safety parameter. The inverter sets p0970 = 5 once it has reset the parameters.
		Then reset the inverter to the factory setting again.	
F01662	Error, internal communications	• Check the electrical cabinet design and cable routing for EMC compliance. • Check whether an impermissible voltage is connected at one of the digital outputs. • Check whether a digital output is loaded with an impermissible current. If the checks are unsuccessful: • Switch off the inverter power supply and switch it on again • Upgrade the firmware • Contact technical support	
A01666	Static 1 signal at the F-DI for safe acknowledgement	Set the F-DI fail-safe input to a logical 0 signal.	
A01698	Commissioning mode active for safety functions	The inverter revokes the alarm after completion of the safety commissioning.	
A01699	Switch-off signal path test required	After the next deselection of the STO safety function, the inverter revokes the message.	
A01900	PROFIBUS: Configuration telegram faulty	Explanation: A PROFIBUS master is attempting to establish a connection with a faulty configuration telegram. Check the bus configuration on the master and slave side.	

8.6 List of alarms and faults

Number	Cause	Remedy
A01910 F01910	Setpoint timeout	The alarm is generated when p2040 $\neq$ 0 ms and one of the following causes is present: • The bus connection is interrupted • The Modbus master is switched off • Communications error (CRC, parity bit, logical error) • An excessively low value for the fieldbus monitoring time (p2040)
A01920	PROFIBUS: Cyclic connection interrupt	Explanation: The cyclic connection to PROFIBUS master is interrupted. Establish the PROFIBUS connection and activate the PROFIBUS master with cyclic operation.
F03505	Analog input, wire-break	Check the wiring for interruptions. Check the level of the injected signal. The input current measured by the analog input can be read out in r0752.
A03520	Temperature sensor fault	Check that the sensor is connected correctly.
A05000 A05001 A05002 A05004 A05006	Power Module overtemperature	Check the following: - Is the ambient temperature within the defined limit values? - Are the load conditions and duty cycle configured accordingly? - Has the cooling failed?
F06310	Supply voltage (p0210) incorrectly set	Check the set supply voltage and if required change (p0210). Check the line voltage.
F07011	Motor overtemperature	Reduce the motor load. Check the ambient temperature. Check the wiring and connection of the sensor.
A07012	I2t Motor Module overtemperature	Check and if necessary reduce the motor load. Check the motor's ambient temperature. Check the thermal time constant p0611. Check the overtemperature fault threshold p0605.
A07015	Motor temperature sensor alarm	Check that the sensor is connected correctly. Check the parameter assignment (p0601).
F07016	Motor temperature sensor fault	Make sure that the sensor is connected correctly. Check the parameterization (p0601). Deactivate the motor temperature sensor fault evaluation (p0607 = 0).
F07086 F07088	Switching over units: Parameter limit violation	Check the adapted parameter values and if required correct.
F07320	Automatic restart aborted	Increase the number of restart attempts (p1211). The actual number of start attempts is shown in r1214. Increase the wait time in p1212 and/or monitoring time in p1213. Connect an ON command (p0840). Increase the monitoring time of the power unit or switch off (p0857). Reduce the wait time for resetting the fault counter p1213[1] so that fewer faults are registered in the time interval.
A07321	Automatic restart active	Explanation: The automatic restart (AR) is active. During voltage recovery and/or when remedying the causes of pending faults, the drive is automatically switched back on.
F07330	Search current measured too low	Increase the search current (p1202), check the motor connection.

Number	Cause	Remedy
A07400	DC-link voltage maximum controller active	If it is not desirable that the controller intervenes: • Increase the ramp-down times. • Deactivate the Vdc_max control (p1240 = 0 for vector control, p1280 = 0 for U/f control).
A07409	U/f control, current limiting controller active	The alarm automatically disappears after one of the following measures: • Increase the current limit (p0640). • Reduce the load. • Slow down the ramp-ups for the setpoint speed.
F07426	Technology controller actual value limited	• Adapt the limits to the signal level (p2267, p2268). • Check the actual value scaling (p2264).
A07444	PID autotuning is activated	Automatic setting of the PID controller (autotuning) is active (p2350 > 0). The alarm disappears automatically after completion of the autotuning.
F07445	PID autotuning canceled	The inverter has canceled the automatic setting of the PID controller (autotuning) because of a fault. Remedy: Increase p2355 and restart autotuning.
F07801	Motor overcurrent	Check the current limits (p0640). Vector control: Check the current controller (p1715, p1717). U/f control: Check the current limiting controller (p1340 ... p1346). Increase the acceleration ramp (p1120) or reduce the load. Check the motor and motor cables for short-circuit and ground fault. Check the motor regarding the star/delta connection and type plate parameterization. Check the power unit / motor combination. Select the flying restart function (p1200) if switched to rotating motor.
A07805	Drive: Power unit overload I2t	• Reduce the continuous load. • Adapt the load cycle. • Check the assignment of rated currents of the motor and power unit.
F07806	Regenerative power limit exceeded	Increase the deceleration ramp. Reduce the driving load. Use a power unit with higher energy recovery capability. For vector control, the regenerative power limit in p1531 can be reduced until the fault is no longer activated.
F07807	Short-circuit detected	• Check the inverter connection on the motor side for any phase-phase short-circuit. • Rule out that line and motor cables have been interchanged.
A07850 A07851 A07852	External alarm 1 ... 3	The signal for "external alarm 1" has been triggered. Parameters p2112, p2116 and p2117 determine the signal sources for the external alarm 1... 3. Remedy: Remove the causes of these alarms.
F07860 F07861 F07862	External fault 1 ... 3	Remove the external causes for these faults.

8.6 List of alarms and faults

Number	Cause	Remedy
A07891	Load monitoring, pump/fan blocked	- Check the pump/fan for blockage and rectify if necessary. - Check the fan for sluggishness and rectify if necessary. - Adapt the parameterization depending on the load (p2165, p2168).
A07892	Load monitoring, pump/fan without load	- For a pump, check the conveyor medium and provide if necessary. - For a fan, check the belt and replace if necessary - If necessary, increase the torque threshold for detection (p2191).
A07893	Load monitoring, pump leakage	- Rectify the leakage in the pump circuit. - For a false tripping, reduce the torque thresholds of the leakage characteristic (p2186, p2188, p2190).
F07894	Load monitoring, pump/fan blocked	- Check the pump/fan for blockage and rectify if necessary. - Check the fan for sluggishness and rectify if necessary. - Adapt the parameterization depending on the load (p2165, p2168).
F07895	Load monitoring, pump/fan without load	- For a pump, check the conveyor medium and provide if necessary. - For a fan, check the belt and replace if necessary - If necessary, increase the torque threshold for detection (p2191).
F07896	Load monitoring, pump leakage	- Rectify the leakage in the pump circuit. - For a false tripping, reduce the torque thresholds of the leakage characteristic (p2186, p2188, p2190).
F07900	Motor blocked	Check that the motor can run freely. Check the torque limits (r1538 and r1539). Check the parameters of the "Motor blocked" message (p2175, p2177).
F07901	Motor overspeed	Activate the precontrol for the speed limiting controller (p1401 bit 7 = 1). Increase the hysteresis for overspeed signal p2162.
F07902	Motor stalled	Check whether the motor data has been set correctly and perform a motor identification. Check the current limits (p0640, r0067, r0289). If the current limits are too low, the drive cannot be magnetized. Check whether motor cables are disconnected during operation.
A07903	Motor speed deviation	Increase p2163 and/or p2166. Increase the torque, current and power limits.
A07910	Motor overtemperature	Check the motor load. Check the motor's ambient temperature. Check the KTY84 or PT1000 sensor. Check the overtemperatures of the thermal model (p0626 ... p0628).
A07920	Torque/speed too low	The torque deviates from the torque/speed envelope curve. - Check the connection between the motor and the load. - Adapt the parameterization corresponding to the load.
A07921	Torque/speed too high	
A07922	Torque/speed out of tolerance	
F07923	Torque/speed too low	
F07924	Torque/speed too high	- Check the connection between the motor and the load. - Adapt the parameterization corresponding to the load.
A07927	DC braking active	Not required
A07980	Rotary measurement activated	Not required
A07981	No enabling for rotary measurement	Acknowledge pending faults. Establish missing enables (see r00002, r0046).

Number	Cause	Remedy
A07991	Motor identification activated	Switch on the motor and identify the motor data.
F08501	Setpoint timeout	• Check the PROFINET connection. • Set the control into the RUN mode. • If the fault occurs repeatedly, check the monitoring time set p2044.
F08502	Monitoring time, sign-of-life expired	• Check the PROFINET connection.
F08510	Send configuration data not valid	• Check the PROFINET configuration
A08511	Receive configuration data not valid	
A08526	No cyclic connection	• Activate the control with cyclic operation. • Check the parameters "Name of Station" and "IP of Station" (r61000, r61001).
A08565	Consistency error affecting adjustable parameters	Check the following: • IP address, subnet mask or default gateway is not correct. • IP address or station name used twice in the network. • Station name contains invalid characters.
F08700	Communications error	A CAN communications error has occurred. Check the following: • Bus cable • Baud rate (p8622) • Bit timing (p8623) • Master Start the CAN controller manually with p8608 = 1 after the cause of the fault has been resolved!
F13100	Know-how protection: Copy protection error	The know-how protection and the copy protection for the memory card are active. An error occurred when checking the memory card. • Insert a suitable memory card and switch the inverter power supply temporarily off and then on again (POWER ON). • Deactivate the copy protection (p7765).
F13101	Know-how protection: Copy protection cannot be activated	Insert a valid memory card.
F30001	Overcurrent	Check the following: • Motor data, if required, carry out commissioning • Motor connection method (Y / Δ) • U/f operation: Assignment of rated currents of motor and Power Module • Line quality • Make sure that the line commutating reactor is connected properly • Power cable connections • Power cables for short-circuit or ground fault • Power cable length • Line phases If this doesn't help: • U/f operation: Increase the acceleration ramp • Reduce the load • Replace the power unit

8.6 List of alarms and faults

Number	Cause	Remedy
F30002	DC-link voltage overvoltage	Increase the ramp-down time (p1121). Set the rounding times (p1130, p1136). Activate the DC-link voltage controller (p1240, p1280). Check the line voltage (p0210). Check the line phases.
F30003	DC-link voltage undervoltage	Check the line voltage (p0210).
F30004	Inverter overtemperature	Check whether the inverter fan is running. Check whether the ambient temperature is in the permissible range. Check whether the motor is overloaded. Reduce the pulse frequency.
F30005	I2t inverter overload	Check the rated currents of the motor and Power Module. Reduce the current limit p0640. When operating with U/f characteristic: Reduce p1341.
F30011	Line phase failure	Check the inverter's input fuses. Check the motor feeder cables.
F30015	Motor cable phase failure	Check the motor cables. Increase the ramp-up or ramp-down time (p1120).
F30021	Ground fault	• Check the power cable connections. • Check the motor. • Check the current transformer. • Check the cables and contacts of the brake connection (a wire might be broken).
F30022	Power Module: Monitoring V_{CE}	Check or replace Power Module.
F30027	Time monitoring for DC-link pre-charging	Check the line voltage at the input terminals. Check the line voltage setting (p0210).
F30035	Overtemperature, intake air	• Check whether the fan is running. • Check the fan filter elements. • Check whether the ambient temperature is in the permissible range.
F30036	Overtemperature, inside area	
F30037	Rectifier overtemperature	See F30035 and, in addition: • Check the motor load. • Check the line phases
A30049	Internal fan defective	Check the internal fan and if required replace.
F30052	Incorrect Power Module data	Replace the Power Module or upgrade the Control Unit firmware.
F30053	Fault in the FPGA data	Replace the Power Module.
F30059	Internal fan defective	Check the internal fan and if required replace.
F30074	Communications fault between Control Unit and Power Module	There is a communications fault between the Control Unit and the Power Module. Possible causes: • The Control Unit may have been removed or inserted incorrectly. • The external 24 V Control Unit power supply has dipped to $\leq 95\%$ of the rated voltage for ≤ 3 ms
A30502	DC-link overvoltage	• Check the unit supply voltage (p0210). • Check the dimensioning of the line reactor.
F30600	STOP A triggered	Select STO safety function and then deselect again.

Number	Cause	Remedy
F30662	CU hardware fault	Switch the Control Unit off and on again, upgrade the firmware or contact technical support.
F30664	CU power up aborted	Switch the Control Unit off and on again, upgrade the firmware or contact technical support.
F30850	Software fault in the Power Module	Replace Power Module or contact technical support.
A30920	Temperature sensor fault	Check that the sensor is connected correctly.
A50001	PROFINET configuration error	A PROFINET control is attempting to establish a connection with an incorrect configuration telegram. Check whether "Shared Device" is activated (p8929 = 2).
A50010	PROFINET name of station invalid	Correct the name of station (p8920) and activate (p8925 = 2).
A50020	PROFINET: Second control missing	"Shared Device" is activated (p8929 = 2). However, only the connection to a PROFINET control is available.

For further information, please refer to the List Manual.



Overview of the manuals (Page 504)

Corrective maintenance

9.1 Spare parts compatibility

Continuous development within the scope of product maintenance

Inverter components are being continuously developed within the scope of product maintenance. Product maintenance includes, for example, measures to increase the ruggedness or hardware changes which become necessary as components are discontinued.

These further developments are "spare parts-compatible" and do not change the article number.

In the scope of such spare parts-compatible ongoing development, plug connector or connection positions are sometimes slightly modified. This does not cause any problems when the components are properly used. Please take this fact into consideration in special installation situations (e.g. allow sufficient reserve regarding the cable length).

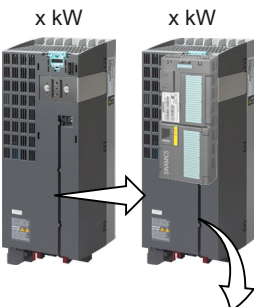
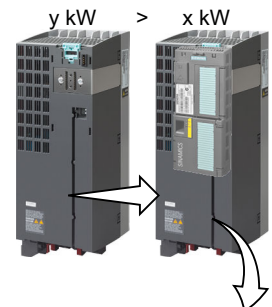
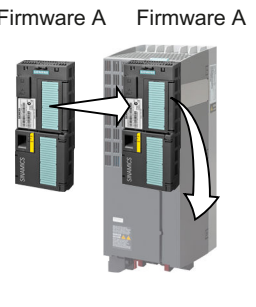
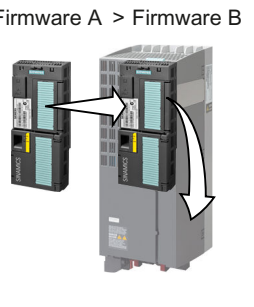
9.2 Replacing inverter components

9.2.1 Overview of replacing converter components

Permissible replacement of components

In the event of a long-term function fault, you must replace the Power Module or Control Unit. The inverter's Power Module and Control Unit can be replaced independently of each other.

In the following cases you will need to replace the inverter:

Replacing the Power Module		Replacing the Control Unit	
Replacement: - Same type - Same power rating	Replacement: - Same type - Same frame size - Higher power rating	Replacement: - Same type - Same firmware version	Replacement: - Same type - higher firmware version (e.g. replace FW V4.2 by FW V4.3)
			
	Power Module and motor must be adapted to one another (ratio of motor and Power Module rated power > 1/4)	After replacing the Control Unit, you must restore the inverter's settings.	

 **WARNING**

Risk of injury due to uncontrolled drive motion

Replacing inverters of different types can result in uncontrolled motion of the drive.

- In all cases that are not permitted according to the table above, recommission the drive after replacing an inverter.

Special issue relating to communication via PROFINET: Device replacement without removable data storage medium

The inverter supports the PROFINET functionality, replacing the device without data storage medium. After replacing the Control Unit, the inverter is automatically assigned its device name from the IO controller.

Independent of this, after replacing the inverter, you must transfer the settings of the old inverter to the new inverter.

Details of the device replacement without removable storage medium can be found in the Internet:



PROFINET system description (<http://support.automation.siemens.com/WW/view/en/19292127>).

9.2.2 Replacing a Control Unit with enabled safety function

Replacing a Control Unit with data backup on a memory card

If you use a memory card with firmware, after the replacement, you obtain a precise copy (firmware and settings) of the replaced Control Unit.

Precondition

You have a memory card with the actual settings of the Control unit to be replaced.

Procedure



1.
2.

To replace the Control Unit, proceed as follows:

1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
2. Remove the signal cables from the Control Unit.
3. Remove the defective Control Unit.
4. Remove the memory card from the old Control Unit and insert it in the new Control Unit.
5. Mount the new Control Unit on the Power Module. The new Control Unit must have the same article number and the same or higher firmware version as the Control Unit that was replaced.
6. Reconnect the signal cables of the Control Unit.
7. Switch on the line voltage again.
8. The inverter loads the settings from the memory card.
9. Check what the inverter reports after loading.
 - Alarm A01028:
The loaded settings are not compatible with the inverter.
Clear the alarm with p0971 = 1 and recommission the drive.
 - Fault F01641:
Acknowledge the message.
Perform a **reduced** acceptance test.
 Reduced acceptance after component replacement and firmware change
(Page 416)

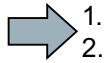


You have replaced the Control Unit and transferred the safety function settings from the memory card to the new Control Unit.

Replacing a Control Unit with data backup in STARTER

Precondition

You have backed up the actual settings of the Control Unit to be replaced to a PC using STARTER.

Procedure

1. To replace the Control Unit, proceed as follows:
2.
 1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
 2. Remove the signal cables of the Control Unit.
 3. Remove the defective Control Unit.
 4. Mount the new Control Unit on the Power Module.
 5. Reconnect the signal cables of the Control Unit.
 6. Switch on the line voltage again.
 7. Open the project in the PC the matches the drive.
 8. Go online and transfer the settings from the PC to the inverter by pressing the  button. The inverter signals faults after the download. Ignore these faults, as they will be automatically acknowledged by the following steps.
 9. Select the screen form for the safety functions.
 10. Select the "Change settings" button.
 11. Select the "Activate settings" button.
 12. Save your settings (copy RAM to ROM).
 13. Switch off the inverter power supply.
 14. Wait until all LEDs on the inverter go dark.
 15. Switch on the inverter power supply again.
 16. Perform a **reduced** acceptance test.



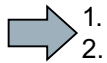
Reduced acceptance after component replacement and firmware change
(Page 416)



You have replaced the Control Unit and transferred the safety function settings from the PC to the new Control Unit.

Replacing a Control Unit with data backup in Startdrive**Precondition**

You have backed up the actual settings of the Control Unit to be replaced to a PC using Startdrive.

Procedure

1. To replace the Control Unit, proceed as follows:
2.
 1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
 2. Remove the signal cables of the Control Unit.
 3. Remove the defective Control Unit.
 4. Mount the new Control Unit on the Power Module.

5. Reconnect the signal cables of the Control Unit.
6. Switch on the line voltage again.
7. Open the project in the PC the matches the drive.
8. Select "Load to device".
9. Connect Startdrive online with the drive.
The inverter signals faults after the download. Ignore these faults, as they will be automatically acknowledged by the following steps.
10. Press the "Start safety commissioning" button.
11. Enter the password for the safety functions.
12. Confirm the prompt for saving your settings (copy RAM to ROM).
13. Disconnect the online connection.
14. Switch off the inverter power supply.
15. Wait until all LEDs on the inverter go dark.
16. Switch on the inverter power supply again.
17. Perform a **reduced** acceptance test.
 Reduced acceptance after component replacement and firmware change
(Page 416)

-  You have replaced the Control Unit and transferred the safety function settings from the PC to the new Control Unit.

Replacing the Control Unit with data backup in the operator Panel

Precondition

You have backed up the actual settings of the Control Unit to be replaced to an Operator Panel.

Procedure

-  1. To replace the Control Unit, proceed as follows:
1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
 2. Remove the signal cables of the Control Unit.
 3. Remove the defective Control Unit.
 4. Mount the new Control Unit on the Power Module.
 5. Reconnect the signal cables of the Control Unit.
 6. Switch on the line voltage again.
 7. Plug the Operator Panel into the Control Unit or connect the Operator Panel handheld device with the inverter.
 8. Transfer the settings from the Operator Panel to the inverter.
 9. Wait until the transfer is complete.

10. After loading, check whether the inverter outputs alarm A01028.

- Alarm A01028:
The loaded settings are not compatible with the inverter.
Clear the alarm with p0971 = 1 and recommission the drive.
- No alarm A01028: Proceed with the next step.

11. Switch off the inverter power supply.

12. Wait until all LEDs on the inverter go dark.

13. Switch on the inverter power supply again.

The inverter reports the faults F01641, F01650, F01680 and F30680. Ignore these faults, as they will be automatically acknowledged by the following steps.

14. Set p0010 to 95.

15. Set p9761 to the safety password.

16. Set p9701 to AC hex.

17. Set p0010 to 0.

18. Back up the settings so they are not lost when the power fails:

- BOP-2 in the menu "EXTRAS" - "RAM-ROM".
- IOP in the menu "SAVE RAM TO ROM".

19. Switch off the inverter power supply.

20. Wait until all LEDs on the inverter go dark.

21. Switch on the inverter power supply again.

22. Perform a **reduced** acceptance test.



Reduced acceptance after component replacement and firmware change
(Page 416)



You have replaced the Control Unit and transferred the safety function settings from the Operator Panel to the new Control Unit.

9.2.3 Replacing the Control Unit without the safety functions enabled

Replacing a Control Unit with data backup on a memory card

If you use a memory card with firmware, after the replacement, you obtain a precise copy (firmware and settings) of the replaced Control Unit.

Precondition

You have a memory card with the actual settings of the Control unit to be replaced.

Procedure



1. To replace the Control Unit, proceed as follows:
2.
 1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
 2. Remove the signal cables from the Control Unit.
 3. Remove the defective Control Unit.
 4. Mount the new Control Unit on the Power Module. The new Control Unit must have the same article number and the same or higher firmware version as the Control Unit that was replaced.
 5. Remove the memory card from the old Control Unit and insert it in the new Control Unit.
 6. Reconnect the signal cables of the Control Unit.
 7. Connect up the line voltage again.
 8. The inverter loads the settings from the memory card.
 9. After loading, check whether the inverter outputs alarm A01028.
 - Alarm A01028:
The loaded settings are not compatible with the inverter.
Clear the alarm with p0971 = 1, and recommission the drive.
 - No alarm A01028:
The inverter accepts the settings that have been loaded.



You have successfully replaced the Control Unit.

Replacing a Control Unit with data backup in the PC

Precondition

You have backed up the actual settings of the Control Unit to be replaced to a PC using STARTER.

Procedure



1. To replace the Control Unit, proceed as follows:
2.
 1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
 2. Remove the signal cables of the Control Unit.

3. Remove the defective Control Unit.
4. Mount the new Control Unit on the Power Module.
5. Reconnect the signal cables of the Control Unit.
6. Connect up the line voltage again.
7. Open the right project for the drive in STARTER.
8. Go online and transfer the settings from the PC to the inverter by pressing the  button.
The inverter signals faults after the download.
Ignore these faults, as they will be automatically acknowledged by the following steps.
9. Save your settings (copy RAM to ROM).



You have successfully replaced the Control Unit.

Replacing the Control Unit with data backup in the operator Panel

Precondition

You have backed up the actual settings of the Control Unit to be replaced to an operator panel.

Procedure



1. To replace the Control Unit, proceed as follows:
2.
 1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
 2. Remove the signal cables of the Control Unit.
 3. Remove the defective Control Unit.
 4. Mount the new Control Unit on the Power Module.
 5. Reconnect the signal cables of the Control Unit.
 6. Connect up the line voltage again.
 7. Plug the Operator Panel into the Control Unit or connect the Operator Panel handheld device with the inverter.
 8. Transfer the settings from the Operator Panel to the inverter.
 9. Wait until the transfer is complete.
 10. After loading, check whether the inverter outputs alarm A01028.
 - Alarm A01028:
The loaded settings are not compatible with the inverter.
Clear the alarm with p0971 = 1 and recommission the drive.
 - No alarm A01028: Proceed with the next step.
 11. Back up the settings so they are not lost when the power fails:
 - BOP-2 in the menu "EXTRAS" - "RAM-ROM".
 - IOP in the menu "SAVE RAM TO ROM".



You have replaced the Control Unit and transferred the safety function settings from the operator panel to the new Control Unit.

9.2.4 Replacing the Control Unit without data backup

If you do not backup the settings, then you must recommission the drive after replacing the Control Unit.

Procedure



1. To replace the Control Unit without backed-up settings, proceed as follows:
2.
 1. Disconnect the line voltage to the Power Module and (if installed) the external 24 V supply or the voltage for the digital outputs of the Control Unit.
 2. Remove the signal cables of the Control Unit.
 3. Remove the defective Control Unit.
 4. Mount the new Control Unit on the Power Module.
 5. Reconnect the signal cables of the Control Unit.
 6. Switch on the line voltage again.
 7. Recommission the drive.



The Control Unit replacement has been completed after the drive has been successfully commissioned.

9.2.5 Replacing a Control Unit with active know-how protection

Replacing devices with know-how protection without copy protection

In the case of know-how protection without copy protection, the inverter settings can be transferred to another inverter using a memory card.



Saving setting on memory card (Page 349)



Transferring the setting from the memory card (Page 352)

Replacing devices with know-how protection with copy protection

The know-how protection with copy protection hides the inverter settings and also prevents the duplication of the inverter settings.

If the inverter settings can neither be copied nor forwarded, a recommissioning is required after inverter replacement.

To avoid the recommissioning, you must use a Siemens memory card, and the machine manufacturer must have an identical prototype machine that it uses as sample.

There are two options for replacing the device:

Option 1: The machine manufacturer only knows the serial number of the new inverter

1. The end customer provides the machine manufacturer with the following information:
 - For which machine must the inverter be replaced?
 - What is the serial number (r7758) of the new inverter?
2. The machine manufacturer performs the following steps online on the prototype machine:
 - Deactivating know-how protection
 Activating and deactivating know-how protection (Page 370)
 - Enter the serial number of the new inverter in p7759.
 - Enter the serial number of the inserted memory card as reference serial number in p7769.
 - Activate know-how protection with copy protection. "Copy RAM to ROM" must be activated.
 Activating and deactivating know-how protection (Page 370)
 - Write the configuration with p0971 = 1 to the memory card.
 - Send the memory card to the end customer.
3. The end customer inserts the memory card and switches on the power supply for the inverter.

When powering up, the inverter checks the serial numbers of the card and when there is a match, the inverter goes into the "ready to start" state.

If the numbers do not match, then the inverter signals fault F13100 (no valid memory card).

Option 2: The machine manufacturer knows the serial number of the new inverter and the serial number of the memory card

1. The end customer provides the machine manufacturer with the following information:
 - For which machine must the inverter be replaced?
 - What is the serial number (r7758) of the new inverter?
 - What is the serial number of the memory card?
2. The machine manufacturer performs the following steps online on the prototype machine:
 - Deactivating know-how protection
 Activating and deactivating know-how protection (Page 370)
 - Enter the serial number of the new inverter in p7759.
 - Enter the serial number of the customer's memory card as reference serial number in p7769.
 - Activate know-how protection with copy protection. "Copy RAM to ROM" must be activated.
 Activating and deactivating know-how protection (Page 370)
 - Write the configuration with p0971 = 1 to the memory card.
 - Copy the encrypted project from the card to the associated PC.
 - Send the encrypted project to the end customer, e.g. via e-mail.
3. The end customer copies the project to the Siemens memory card that belongs to the machine, inserts it in the inverter and switches on the power supply for the inverter.

When powering up, the inverter checks the serial numbers of the card and when there is a match, the inverter goes into the "ready to start" state.

If the numbers do not match, then the inverter signals fault F13100 (no valid memory card).

9.2.6 Replacing a Power Module with enabled safety function



DANGER

Danger from touching energized Power Module connections

After switching off the line voltage, it will take up to 5 minutes until the capacitors in the Power Module are sufficiently discharged for the residual voltage to be safe. Death or serious injury will result if energized parts are touched.

- Check the safe isolation of the Power Module connections before carrying out installation work.

NOTICE

Material damage from swapping the motor's connection lines

The direction in which the motor rotates switches if you swap the two phases of the motor line. A motor running backwards may damage the machine or installation.

- Connect the three phases of the motor lines in the right order.
- After replacing the Power Module, check the direction of motor rotation.

Procedure



1. To replace the Power Module, proceed as follows:
 2.
 1. Switch off the line voltage of the Power Module.
You can leave any external 24 V supply to the Control Unit switched on.
 2. Remove the connecting cables of the Power module.
 3. Remove the Control Unit from the Power Module.
 4. Replace the Power Module.
 5. Mount the Control Unit onto the new Power Module.
 6. Connect up the new Power Module using the connecting cables.
 7. Switch on the line supply and, if necessary, the 24 V supply for the Control Unit.
 8. The inverter reports F01641.
 9. Perform a reduced acceptance test.
 Reduced acceptance after component replacement and firmware change (Page 416)
-  You have successfully replaced the Power Module.

9.2.7 Replacing a Power Module without the safety function being enabled

Procedure



1. Proceed as follows to exchange a Power Module:
2.
 1. Switch off the supply voltage to the Power Module.
You do not have to switch off an external 24 V power supply for the Control Unit if one is being used.
 2. Remove the connecting cables of the Power Module.
 3. Remove the Control Unit from the Power Module.
 4. Replace the old Power Module with the new Power Module.
 5. Mount the Control Unit onto the new Power Module.
 6. Connect up the new Power Module using the connecting cables.

NOTICE

Material damage caused by the improper direction of motor rotation

The direction in which the motor rotates switches if you exchange the two phases of the motor line. For work machines designed for only one direction of rotation, an improper direction of rotation can damage the work machine. Work machines with only one permissible direction of rotation include certain compressors, saws and pumps.

- Connect the three phases of the motor lines in the right order.
- After replacing the Power Module, check the direction of motor rotation.

7. Switch on the line supply and, if being used, the 24 V supply of the Control Unit.



You have successfully replaced the Power Module.

9.3 Firmware upgrade and downgrade

Preparing a memory card for a firmware upgrade or downgrade

Procedure

1. Proceed as follows to prepare a memory card for the firmware upgrade or downgrade:
2.
 1. Download the required firmware to your PC from the Internet.
Download (<https://support.industry.siemens.com/cs/ww/en/view/67364620>)
 2. Extract the files to a directory of your choice on your PC.
 3. Transfer the unzipped files into the root directory of the memory card.

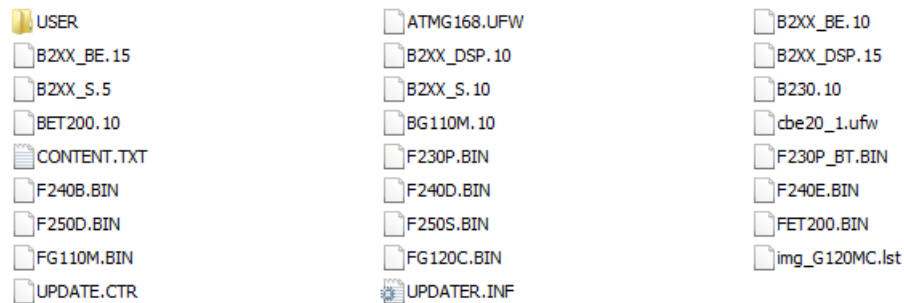


Figure 9-1 Example of memory card contents after the file transfer

Depending on the firmware, the filenames and the number of files may differ from the display above.

The "USER" directory does not exist on unused memory cards. After the memory card is plugged in for the first time, the inverter creates a new "USER" directory.

- You have prepared the memory card for the firmware upgrade or downgrade.

Memory cards that can be ordered:  Memory cards (Page 348)

Overview of firmware upgrades and downgrades

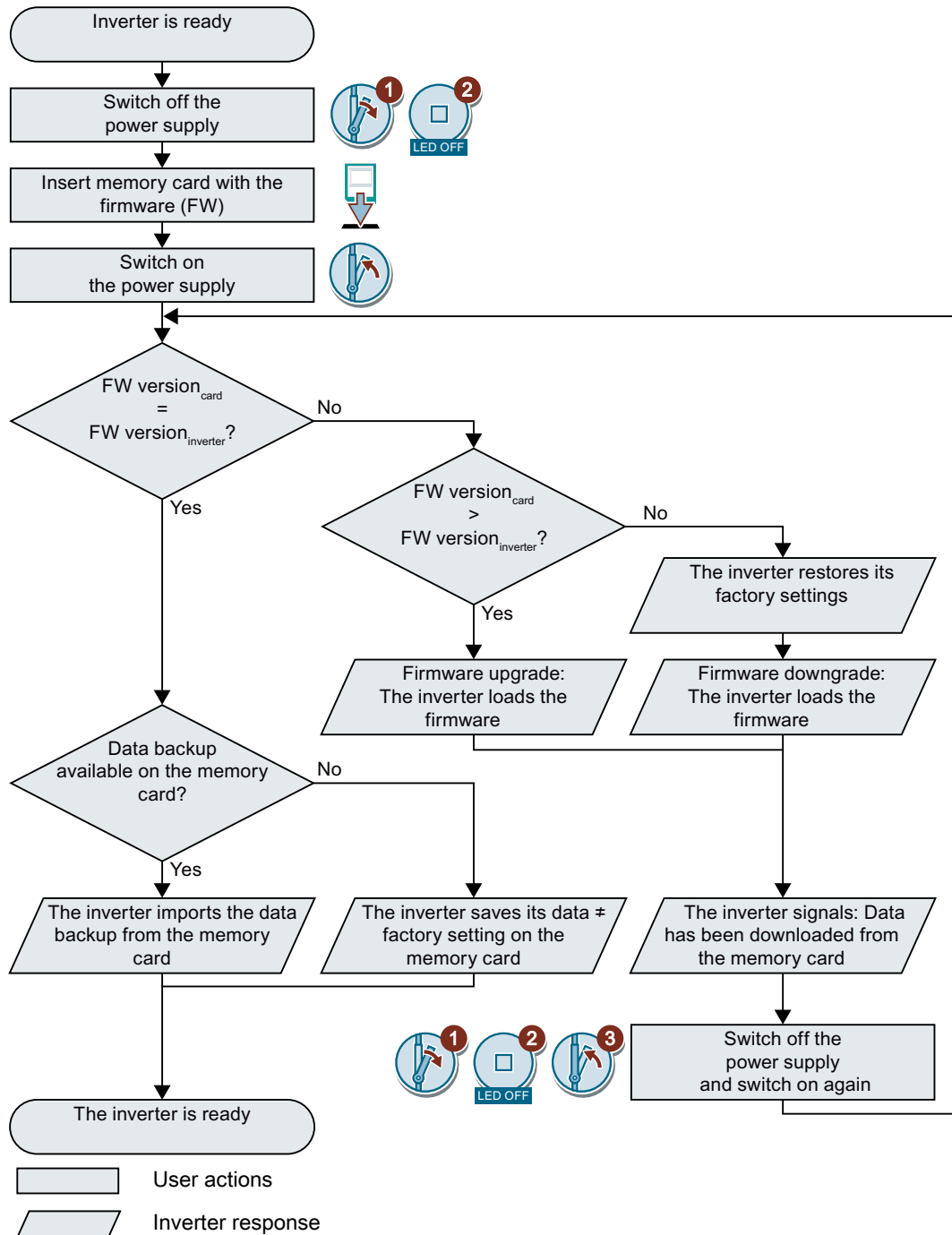


Figure 9-2 Overview of the firmware upgrade and firmware downgrade

9.3.1 Upgrading the firmware

When upgrading the firmware, you replace the inverter firmware by a later version. Only update the firmware to a later version if you require the expanded functional scope of the newer version.

Precondition

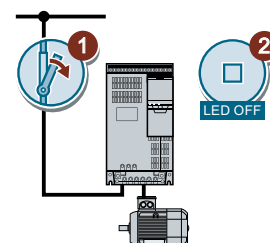
- The firmware version of your inverter is at least V4.5.
- Inverter and memory card have different firmware versions.

Procedure

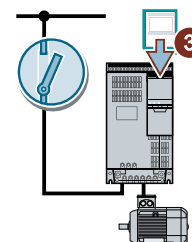


1. Proceed as follows to upgrade the inverter firmware to a later version:

1. Switch off the inverter power supply.
2. Wait until all LEDs on the inverter are dark.



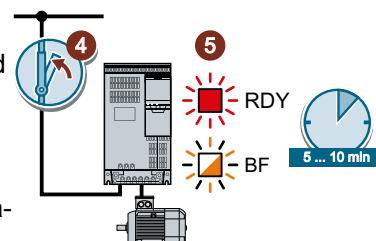
3. Insert the card with the matching firmware into the inverter slot until it latches into place.



4. Switch on the inverter power supply again.
5. The inverter transfers the firmware from the memory card into its memory.

The transfer takes approximately 5 ... 10 minutes.

While data is being transferred, the LED RDY on the inverter stays red. The LED BF flashes orange with a variable frequency.

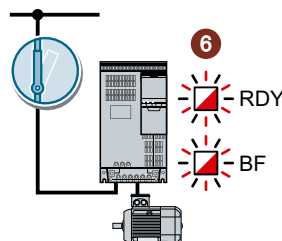


6. At the end of the transfer, the LED RDY and BF slowly flash red (0.5 Hz).

Power supply failure during transfer

The inverter firmware will be incomplete if the power supply fails during the transfer.

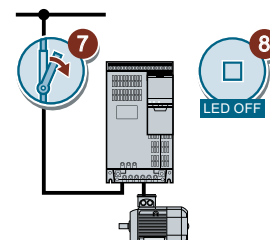
- Start again with step 1 of the instructions.



7. Switch off the inverter power supply.
8. Wait until all LEDs on the inverter are dark.

Decide whether you will withdraw the memory card from the inverter:

- You remove the memory card:
⇒ The inverter keeps its settings.



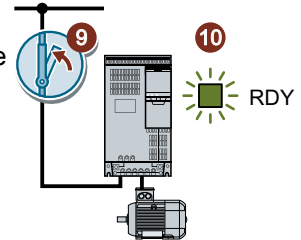
- You leave the memory card in the inverter:
 - ⇒ If the memory card still does not have a data backup of the inverter settings, in step 9 the inverter writes its settings to the memory card.
 - ⇒ If the memory card already includes a data backup, the inverter imports the settings from the memory card in step 9.

9. Switch on the inverter power supply again.

10. If the firmware upgrade was successful, after several seconds the inverter LED RDY turns green.

If the memory card is still inserted, depending on the previous content of the memory card, one of the two following cases has occurred:

- The memory card contains a data backup:
 - ⇒ The inverter has taken the settings from the memory card.
- There was no data backup on the memory card:
 - ⇒ The inverter has written its settings to the memory card.



■ You have upgraded the inverter firmware.

Memory cards with license

If the memory card includes a license, e.g. for the basic positioner, then the memory card must remain inserted after the firmware has been updated.

9.3.2 Firmware downgrade

When downgrading the firmware, you replace the inverter firmware by an older version. Only downgrade the firmware to an older version if, after replacing an inverter, you require the same firmware in all of your inverters.

Precondition

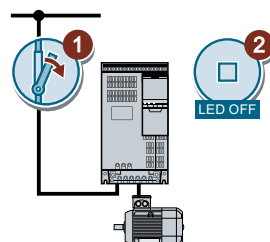
- The firmware version of your inverter is at least V4.6.
- Inverter and memory card have different firmware versions.
- You have backed up your settings on the memory card, in an Operator Panel or in a PC.

Procedure

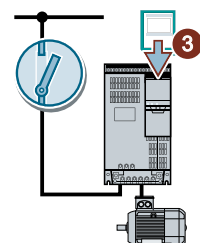


1. Proceed as follows to downgrade the inverter firmware to an older version:

1. Switch off the inverter power supply.
2. Wait until all LEDs on the inverter are dark.



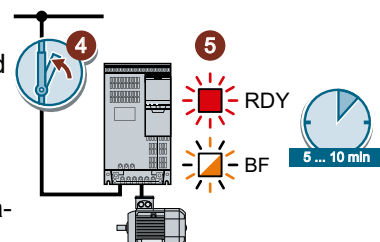
3. Insert the card with the matching firmware into the inverter slot until it latches into place.



4. Switch on the inverter power supply again.
5. The inverter transfers the firmware from the memory card into its memory.

The transfer takes approximately 5 ... 10 minutes.

While data is being transferred, the LED RDY on the inverter stays red. The LED BF flashes orange with a variable frequency.

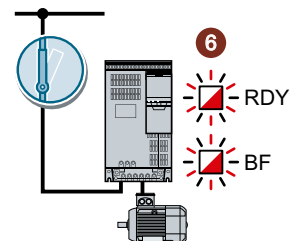


6. At the end of the transfer, the LED RDY and BF slowly flash red (0.5 Hz).

Power supply failure during transfer

The inverter firmware will be incomplete if the power supply fails during the transfer.

- Start again with Step 1 of these instructions.



7. Switch off the inverter power supply.

8. Wait until all LEDs on the inverter are dark.

Decide whether you want to withdraw the memory card from the inverter:

- The memory card contains a data backup:
⇒ The inverter has taken the settings from the memory card.
- There was no data backup on the memory card:
⇒ The inverter has the factory setting.

9. Switch on the inverter power supply again.

10. If the firmware downgrade was successful, after several seconds the inverter LED RDY turns green.

If the memory card is still inserted, depending on the previous content of the memory card, one of the two following cases has occurred:

- The memory card contains a data backup:
⇒ The inverter has taken the settings from the memory card.
- There was no data backup on the memory card:
⇒ The inverter has the factory setting.

11. If the memory card did not contain a data backup of the inverter settings, then you must transfer your settings to the inverter from another data backup.

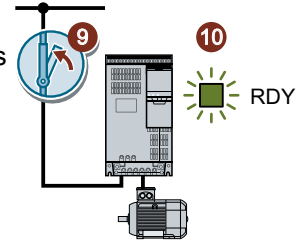
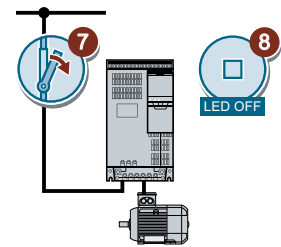
 Saving the settings and series commissioning (Page 347)



You have replaced the inverter's firmware by an older version.

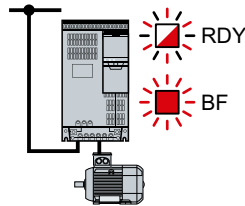
Memory cards with license

If the memory card includes a license, e.g. for the basic positioner, then the memory card must remain inserted after the firmware has been updated.



9.3.3 Correcting an unsuccessful firmware upgrade or downgrade

How does the inverter signal an unsuccessful upgrade or downgrade?



The inverter signals an unsuccessful firmware upgrade or downgrade by a quickly flashing LED RDY and the lit LED BF.

Correcting an unsuccessful upgrade or downgrade

You can check the following to correct an unsuccessful firmware upgrade or downgrade:

- Does the firmware version of your inverter fulfill the preconditions?
 - For an upgrade, as a minimum V4.5.
 - For a downgrade, as a minimum V4.6.
- Have you correctly inserted the card?
- Does the card contain the correct firmware?
- Repeat the appropriate procedure.

9.4 Reduced acceptance after component replacement and firmware change

After a component has been replaced or the firmware updated, a reduced acceptance test of the safety functions must be performed.

Measure	Acceptance test	
	Acceptance test	Documentation
Replacing the Control Unit.	No. Only check the direction of rotation of the motor.	• Supplement inverter data • Log the new checksums • Countersignature
Replacing the Power Module.		Supplement the hardware version in the inverter data
Replace the motor with an identical pole pair number		No change.
Replace the gearbox with an identical ratio		
Replacing safety-related peripherals (e.g. Emergency Stop switch).	No. Only check the control of the safety functions that are influenced by the components that have been replaced.	No change.
Inverter firmware update.	No.	• Supplement firmware version in the inverter data • Log the new checksums • Countersignature.

9.5 If the converter no longer responds

If the inverter no longer responds

For example, when loading an incorrect file from the memory card, the inverter can go into a state where it can no longer respond to commands from the operator panel or from a higher-level control system. In this case, you must reset the inverter to its factory setting and recommission it. This inverter state is manifested in two different ways:

Case 1

- The motor is switched off.
- You cannot communicate with the inverter, either via the operator panel or other interfaces.
- The LEDs flicker and after 3 minutes the inverter has still not powered up.

Procedure

- 
1. Proceed as follows to restore the inverter factory settings:
 2.
 1. Remove the memory card if one is inserted in the inverter.
 2. Switch off the inverter power supply.
 3. Wait until all LEDs on the inverter go dark. Then switch on the inverter power supply again.
 4. Repeat steps 2 and 3 as often as required until the inverter outputs fault F01018:
 5. Set p0971 = 1.
 6. Switch off the inverter power supply.
 7. Wait until all LEDs on the inverter go dark. Then switch on the inverter power supply again.
The inverter now powers up with the factory settings.
 8. Recommission the inverter.
- 
- You have restored the inverter factory settings.

Case 2

- The motor is switched off.
- You cannot communicate with the inverter, either via the operator panel or other interfaces.
- The LEDs flash and are dark - this process is continually repeated.

Procedure

- 
1. Proceed as follows to restore the inverter factory settings:
 2.
 1. Remove the memory card if one is inserted in the inverter.
 2. Switch off the inverter power supply.
 3. Wait until all LEDs on the inverter go dark. Then switch on the inverter power supply again.
 4. Wait until the LEDs flash orange.
 5. Repeat steps 2 and 3 as often as required until the inverter outputs fault F01018.
 6. Now set p0971 = 1.

9.5 If the converter no longer responds

7. Switch off the inverter power supply.
8. Wait until all LEDs on the inverter go dark. Then switch on the inverter power supply again. The inverter now powers up with the factory settings.
9. Recommission the inverter.



You have restored the inverter factory settings.

The motor cannot be switched-on

If the motor cannot be switched-on, then check the following:

- Is a fault present?
If there is, then remove the fault cause and acknowledge the fault.
- Has the inverter been completely commissioned $p0010 = 0$?
If not, the inverter is e.g. still in a commissioning state.
- Is the inverter reporting the "ready to start" status ($r0052.0 = 1$)?
- Is the inverter missing some enable signals ($r0046$)?
- How does the inverter receive its setpoint and commands?
Digital inputs, analog inputs or fieldbus?

Technical data

10.1 Technical data, CU240B-2 Control Unit

Feature	Data	
Fieldbus interface	CU240B-2	With RS485 interface for the following protocols: • USS • Modbus RTU
	CU240B-2 DP	With PROFIBUS interface
Operating voltage	You have two options for the Control Unit power supply: • Supply from the Power Module • External supply via terminals 31 and 32 with 20.4 ... 28.8 VDC. Use a power supply with protective extra low voltage (PELV according to EN 61800-5-1), Class 2. The 0 V of the power supply must be connected to the PE of the plant/system through a low-ohmic connection. The power supply is electrically isolated from the control terminals.	
Current consumption	max 0.5 A	
Power loss	5.0 W	Plus power loss of the output voltages.
Output voltages	+24 V out (terminal 9), 18 V ... 28.8 V, max. 100 mA	
	+10 V out (terminals 1 and 35), 9.5 V ... 10.5 V, max. 10 mA	
Setpoint resolution	0.01 Hz	
Digital inputs	4 (DI 0 ... DI 3)	• Low < 5 V, high > 11 V • Electrically isolated • 30 V maximum input voltage • 5.5 mA current consumption • SIMATIC-compatible • PNP/NPN switchable • 10 ms response time for debounce time p0724 = 0.
Analog input	1 (AI 0)	• Differential input • 12-bit resolution • 13 ms ± 1 ms response time • Switchable: – 0 V ... 10 V or -10 V ... +10 V (typical power consumption: 0.1 mA, voltage < 35 V) – 0 mA ... 20 mA (120 Ω input resistance, voltage < 10 V, current < 80 mA) • If AI 0 has been configured as additional digital input: Voltage < 35 V, low < 1.6 V, high > 4.0 V, 13 ms ± 1 ms response time for debounce time p0724 = 0.

Technical data

10.1 Technical data, CU240B-2 Control Unit

Feature	Data	
Digital output	1 (DO 0)	- Relay output, 30 V DC / max. 0.5 A for ohmic loads - Update time 2 ms
Analog output	1 (AO 0)	0 V ... 10 V or 0 mA ... 20 mA - Reference potential: "GND" 16-bit resolution 4 ms update time
Temperature sensor	PTC	- Short-circuit monitoring 22 Ω - Switching threshold 1650 Ω
	KTY84	- Short-circuit monitoring < 50 Ω - Wire-break > 2120 Ω
	Pt1000	- Short-circuit monitoring < 603 Ω - Wire-break > 2120 Ω
	Temperature switch with isolated contact.	
USB interface	Mini-B	
Dimensions (WxHxD)	73 mm × 199 mm × 39 mm	The depth specification is valid when mounting on the Power Module.
Weight	0.49 kg	
Memory cards	Slot for SD or MMC memory cards  Memory cards (Page 348)	
Operating temperature	-10° C ... 55° C	Without inserted Operator Panel
	0° C ... 50° C	With inserted BOP-2 or IOP Operator Panel
	Observe any possible restrictions regarding the operating temperature as a result of the Power Module.	
Storage temperature	- 40° C ... 70° C	
Relative humidity	< 95%	Condensation is not permissible.

10.2 Technical data, CU240E-2 Control Unit

Feature	Data	
Fieldbus interface	CU240E-2, CU240E-2 F	With RS485 interface for the following protocols: • USS • Modbus RTU
	CU240E-2 DP, CU240E-2 DP-F	With PROFIBUS interface
	CU240E-2 PN, CU240E-2 PN-F	With PROFINET interface
Operating voltage	You have two options for the Control Unit power supply: • Supply from the Power Module • External supply via terminals 31 and 32 with 20.4 ... 28.8 VDC. Use a power supply with protective extra low voltage (PELV according to EN 61800-5-1), Class 2. The 0 V of the power supply must be connected to the PE of the plant/system through a low-ohmic connection. The power supply is electrically isolated from the control terminals.	
Current consumption	max 0.5 A	
Power loss	5.0 W	Plus power loss of the output voltages.
Output voltages	+24 V out (terminal 9), 18 V ... 28.8 V, max. 100 mA	
	+10 V out (terminals 1 and 35), 9.5 V ... 10.5 V, max. 10 mA	
Setpoint resolution	0.01 Hz	
Digital inputs	6 (DI 0 ... DI 5)	• Low < 5 V, high > 11 V • Electrically isolated • 30 V maximum input voltage • 5.5 mA current consumption • SIMATIC-compatible • PNP/NPN switchable • 10 ms response time for debounce time p0724 = 0.
Pulse input	1 (DI 3)	Maximum frequency 32 kHz
Analog inputs	2 (AI 0, AI 1)	• Differential input • 12-bit resolution • 13 ms ± 1 ms response time • AI 0 and AI 1 can be switched over: – 0 V ... 10 V or -10 V ... +10 V (typical power consumption: 0.1 mA, voltage < 35 V) – 0 mA ... 20 mA (120 Ω input resistance, voltage < 10 V, current < 80 mA) • If AI 0 and AI 1 are configured as supplementary digital inputs: Voltage < 35 V, low < 1.6 V, high > 4.0 V, 13 ms ± 1 ms response time for debounce time p0724 = 0.



Article numbers:
Overview of Control Units (Page 31)

Feature	Data	
Digital outputs	3 (DO 0 ... DO 2)	- DO 0: Relay output, 30 VDC / max. 0.5 A with resistive load - DO 1: Transistor output, 30 VDC / max. 0.5 A with resistive load, protection against incorrect voltage polarity. - DO 2: Relay output, 30 VDC / max. 0.5 A with resistive load. 2 ms update time For applications which require UL certification, the voltage at DO 0 must not exceed 30 VDC referred to ground potential and must be supplied via a grounded class 2 power supply.
Analog outputs	2 (AO 0, AO 1)	0 V ... 10 V or 0 mA ... 20 mA - Reference potential: "GND" 16-bit resolution 4 ms update time
Temperature sensor	PTC	- Short-circuit monitoring 22 Ω - Switching threshold 1650 Ω
	KTY84	- Short-circuit monitoring < 50 Ω - Wire-break > 2120 Ω
	Pt1000	- Short-circuit monitoring < 603 Ω - Wire-break > 2120 Ω
	Temperature switch with isolated contact.	
Fail-safe digital input	1 (DI 4 and DI 5)	- If you have enabled the basic functions of the safety functions, DI 4 and DI 5 form the fail-safe digital input. - Maximum input voltage 30 V, 5.5 mA - Response time: - Typical: 5 ms + debounce time p9651 - Typical, if debounce time = 0: 6 ms - Worst-case scenario: 15 ms + debounce time - Worst case, if debounce time = 0: 16 ms Extended safety function data can be found in the "Safety Integrated" function manual.  Overview of the manuals (Page 504)
PFH	5 × 10E-8	Probability of failure of the safety functions (Probability of Failure per Hour)
USB interface	Mini-B	
Dimensions (WxHxD)	73 mm × 199 mm × 39 mm	The depth specification is valid when mounting on the Power Module.
Weight	0.49 kg	
Memory cards	Slot for SD or MMC memory cards,  Memory cards (Page 348)	

Feature	Data	
Operating temperature	-10° C ... 55° C	CU240E-2, CU240E-2 F, CU240E-2 DP, CU240E-2 DP-F Without inserted Operator Panel
	-10° C ... 53° C	CU240E-2 PN, CU240E-2 PN-F Without inserted Operator Panel
	0° C ... 50° C	With inserted BOP-2 or IOP Operator Panel
	Observe any possible restrictions regarding the operating temperature as a result of the Power Module.	
Storage temperature	- 40° C ... 70° C	
Relative humidity	< 95%	Condensation is not permissible.

10.3 Overload capability of the inverter

Overload capability is the property of the inverter to temporarily supply a current that is higher than the rated current to accelerate a load. Two typical load cycles are defined to clearly demonstrate the overload capability: "Low Overload" and "High Overload"

Definitions

Base load

Constant load between the accelerating phases of the drive

Low Overload

- **LO base load input current**
Permissible input current for a "Low Overload" load cycle
- **LO base load output current**
Permissible output current for a "Low Overload" load cycle
- **LO base load power**
Rated power based on the LO base load output current

High Overload

- **HO base load input current**
Permissible input current for a "High Overload" load cycle
- **HO base load output current**
Permissible output current for a "High Overload" load cycle
- **HO base load power**
Rated power based on the HO base load output current

If not specified otherwise, the power and current data in the technical data always refer to a load cycle according to Low Overload.

We recommend the "SIZER" engineering software to select the inverter.



You will find additional information about SIZER on the Internet: Download SIZER (<http://support.automation.siemens.com/WW/view/en/10804987/130000>).

Load cycles and typical applications:

"Low Overload" load cycle

The "Low Overload" load cycle assumes a uniform base load with low requirements placed on brief accelerating phases. Typical applications when designing according to "Low Overload" include:

- Pumps, fans and compressors
- Wet or dry blasting technology
- Mills, mixers, kneaders, crushers, agitators
- Basic spindles
- Rotary kilns
- Extruders

"High Overload" load cycle

The "High Overload" load cycle permits, for reduced base load, dynamic accelerating phases. Typical applications when designing according to "High Overload" include:

- Horizontal and vertical conveyor technology (conveyor belts, roller conveyors, chain conveyors)
- Centrifuges
- Escalators/moving stairways
- Lifters/Lowerers
- Elevators
- Gantry cranes
- Cable railways
- Storage and retrieval machines

10.4 Technical data, PM240-2 Power Module

Protective organs for the Power Module

The fuses listed in the following tables are examples of suitable fuses.



Additional components for branch protection are available in the Internet:

Branch protection and short-circuit strength according to UL and IEC (<https://support.industry.siemens.com/cs/ww/en/view/109479152>)

10.4.1 High overload - low overload PM240-2

Typical inverter load cycles

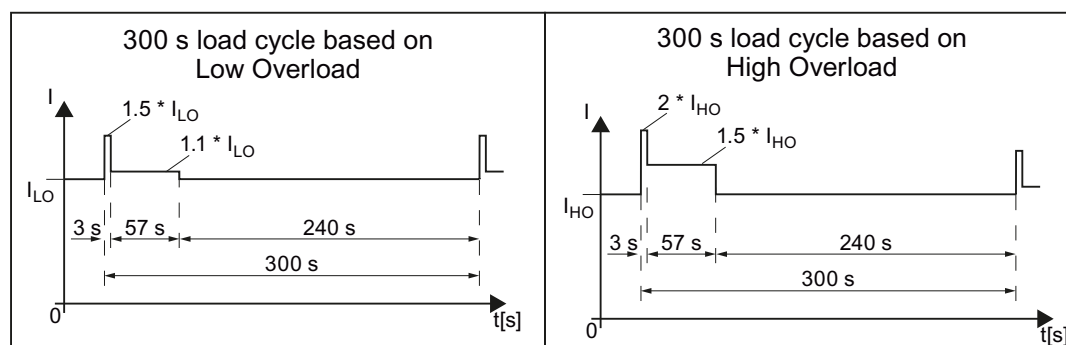


Figure 10-1 "Low Overload" and "High Overload" load cycles

Property	Version
Ambient conditions for transport in the transport packaging	
Climatic ambient conditions	- 40° C ... + 70° C, according to Class 2K4 to EN 60721-3-2 maximum humidity 95% at 40° C
Mechanical ambient conditions	- FSA ... FSC: Shock and vibration permissible according to 1M2 to EN 60721-3-2 - FSD ... FSF: Shock and vibration permissible according to 2M3 to EN 60721-3-2
Protection against chemical substances	Protected according to Class 2C2 to EN 60721-3-2
Biological ambient conditions	Suitable according to Class 2B1 to EN 60721-3-2
Ambient conditions for long-term storage in the product packaging	
Climatic ambient conditions	- 25° C ... + 55° C, according to Class 1K3 to EN 60721-3-1
Protection against chemical substances	Protected according to Class 1C2 to EN 60721-3-1
Biological ambient conditions	Suitable according to class 1B1 to EN 60721-3-1
Ambient conditions in operation	

Property	Version
Installation altitude	Up to 1000 m above sea level without limitations  Restrictions for special ambient conditions (Page 474)
Climatic ambient conditions ¹⁾	- FSA ... FSC ambient operating temperature²⁾ - For operation according to Low Overload: -10° C ... +40° C - For operation according to High Overload: -10° C ... +50° C  Restrictions for special ambient conditions (Page 474) - FSD ... FSF ambient operating temperature²⁾ - For operation according to Low Overload: -20° C ... +40° C - For operation according to High Overload: -20° C ... +50° C  Restrictions for special ambient conditions (Page 474) - Relative humidity: 5 ... 95%, condensation not permitted - Oil mist, salt mist, ice formation, condensation, dripping water, spraying water, splashing water and water jets are not permitted
Mechanical ambient conditions	Shock and vibration permissible, according to Class 3M1 to EN 60721-3-3 Vibration test during operation according to IEC 60068-2-6 Test Fc (sinusoidal) 0 ... 57 Hz: 0.075 mm deflection amplitude 57 ... 150 Hz: 1 g acceleration amplitude 10 frequency cycles per axis Shock test according to IEC 60068-2-27 Test Ea (half-sine) 5 g peak acceleration 30 ms duration 3 shocks in all three axes in both directions
Protection against chemical substances	- FSA ... FSC: Protected according to 3C2 to EN 60721-3-3 - FSD ... FSF: 3C3 protection according to EN 60721-3-3
Biological ambient conditions	Suitable according to 3B1 to EN 60721-3-3
Pollution	Suitable for environments with degree of pollution 2 according to EN 61800-5-1
Cooling	Forced air cooling AF, according to EN 60146
Cooling air	Clean and dry air

¹⁾ Increased ruggedness regarding temperature range and relative humidity; therefore better than 3K3 according to EN 60721-3-3

²⁾ Observe the permissible ambient operating temperatures for the Control Unit and the Operator Panel (IOP or BOP-2).

10.4.2 General technical data, 200 V inverters

Property	Version
Line voltage	FSA ... FSC 200 V ... 240 V 1-ph. AC $\pm 10\%$ 0.55 kW ... 4 kW - LO 0.37 kW ... 3 kW - HO
	200 V ... 240 V 3-ph. AC $\pm 10\%$ 0.55 kW ... 7.5 kW - LO 0.37 kW ... 5.5 kW - HO
	FSD ... FSF 200 V ... 240 V 3-ph. AC $\pm 10\%$ (in operation $-20\% < 1 \text{ min}$)
Line supply configurations	Grounded TN/TT line supplies or non-grounded IT line supplies  Connecting the line supply, motor, and inverter components (Page 69)
Line impedance	FSA ... FSC $2\% \leq U_k < 4\%$. For $U_k < 2\%$, we recommend a line reactor, or a Power Module with the next higher power rating.
	FSD ... FSF No restrictions
Power factor λ	FSA ... FSC 0.7 without line reactor for $U_k \geq 2\%$ 0.85 with line reactor for $U_k < 2\%$
	FSD ... FSF 0.95
Output voltage	0 V ... 0.95 3-ph. AC $\times$ input voltage
Input frequency	50 Hz ... 60 Hz, $\pm 3 \text{ Hz}$
Output frequency	0 ... 550 Hz, depending on the control mode
Inrush current	$< \text{LO base load input current}$
Overvoltage category according to EN 61800-5-1	III for line supplies
Pulse frequency	4 kHz (factory setting), Adjustable as follows in 2 kHz steps: • 2 kHz ... 16 kHz for devices with an LO base load output from 0.55 kW ... 30 kW. • 2 kHz ... 8 kHz for devices with an LO base load output 37 kW and higher If you increase the pulse frequency, the inverter reduces the maximum output current.
Short-circuit current rating (SCCR) and branch protection	$\leq 100 \text{ kA rms}$  Branch protection and short-circuit strength according to UL and IEC (https://support.industry.siemens.com/cs/ww/en/view/109479152)
Braking methods	DC braking, compound braking, dynamic braking with integrated braking chopper
Degree of protection according to EN 60529	IP20
	IP55 PT devices outside the control cabinet
Protection class according to EN 61800-5-1	The inverters are devices with protection class I
Touch protection according to EN 50274	DGUV regulation 3 when used for the intended purpose
Cooling in compliance with EN 60146	Forced air cooling AF

10.4.3 Specific technical data, 200 V inverters

Table 10-1 PM240-2, IP20, Frame Size A, 1-ph./3-ph. AC 200 V ... 240 V

Article no. without filter Article no. with filter	6SL3210-1PB13-0UL0 6SL3210-1PB13-0AL0	6SL3210-1PB13-8UL0 6SL3210-1PB13-8AL0
LO base load output	0.55 kW	0.75 kW
LO base load input current, 1-ph. AC	7.5 A	9.6 A
LO base load input current, 3-ph. AC	4.2 A	5.5 A
LO base load output current	3.2 A	4.2 A
HO base load output	0.37 kW	0.55 kW
HO base load input current, 1-ph. AC	6.6 A	8.4 A
HO base load input current, 3-ph. AC	3.0 A	4.2 A
HO base load output current	2.3 A	3.2 A
Fuse according to IEC Fuse according to UL, class J	3NA3805 (16 A) 15 A	3NA3805 (16 A) 15 A
Power loss	0.04 kW	0.04 kW
Required cooling air flow	5 l/s	5 l/s
Weight without filter	1.4 kg	1.4 kg
Weight with filter	1.6 kg	1.6 kg

Table 10-2 PM240-2, PT, Frame Size A, 1-ph./3-ph. AC 200 V ... 240 V

Article no. without filter Article no. with filter	6SL3211-1PB13-8UL0 6SL3211-1PB13-8AL0
LO base load output	0.75 kW
LO base load input current, 1-ph. AC	9.6 A
LO base load input current, 3-ph. AC	5.5 A
LO base load output current	4.2 A
HO base load output	0.55 kW
HO base load input current, 1-ph. AC	8.4 A
HO base load input current, 3-ph. AC	4.2 A
HO base load output current	3.2 A
Fuse according to IEC Fuse according to UL, class J	3NA3 805 (16 A) 15 A
Power loss	0.04 kW
Required cooling air flow	5 l/s
Weight without filter	1.8 kg
Weight with filter	2.0 kg

Table 10-3 PM240-2, IP20, Frame Size B, 1-ph./3-ph. AC 200 V ... 240 V

Article no. without filter Article no. with filter	6SL3210-1PB15-5UL0 6SL3210-1PB15-5AL0	6SL3210-1PB17-4UL0 6SL3210-1PB17-4AL0	6SL3210-1PB21-0UL0 6SL3210-1PB21-0AL0
LO base load output	1.1 kW	1.5 kW	2.2 kW
LO base load input current, 1-ph. AC	13.5 A	18.1 A	24.0 A
LO base load input current, 3-ph. AC	7.8 A	9.7 A	13.6 A
LO base load output current	6 A	7.4 A	10.4 A
HO base load output	0.75 kW	1.1 kW	1.5 kW
HO base load input current, 1-ph. AC	11.8 A	15.8 A	20.9 A
HO base load input current, 3-ph. AC	5.5 A	7.8 A	9.7 A
HO base load output current	4.2 A	6 A	7.4 A
Fuse according to IEC Fuse according to UL, class J	3NA3812 (32 A) 35 A	3NA3812 (32 A) 35 A	3NA3812 (32 A) 35 A
Power loss	0.05 kW	0.07 kW	0.12 kW
Required cooling air flow	9.2 l/s	9.2 l/s	9.2 l/s
Weight without filter	2.8 kg	2.8 kg	2.8 kg
Weight with filter	3.1 kg	3.1 kg	3.1 kg

Table 10-4 PM240-2, PT, Frame Size B, 1-ph./3-ph. AC 200 V ... 240 V

Article no. without filter Article no. with filter	6SL3211-1PB21-0UL0 6SL3211-1PB21-0AL0
LO base load output	2.2 kW
LO base load input current, 1-ph. AC	24.0 A
LO base load input current, 3-ph. AC	13.6 A
LO base load output current	10.4 A
HO base load output	1.5 kW
HO base load input current, 1-ph. AC	20.9 A
HO base load input current, 3-ph. AC	9.7 A
HO base load output current	7.4 A
Fuse according to IEC Fuse according to UL, class J	3NA3812 (32 A) 35 A
Power loss	0.12 kW ¹⁾
Required cooling air flow	9.2 l/s
Weight without filter	3.4 kg
Weight with filter	3.7 kg

1) approx. 0.08 through the heatsink

10.4 Technical data, PM240-2 Power Module

Table 10-5 PM240-2, IP 20, Frame Size C, 1-ph./3-ph. AC 200 V ... 240 V

Article no. without filter Article no. with filter	6SL3210-1PB21-4UL0 6SL3210-1PB21-4AL0	6SL3210-1PB21-8UL0 6SL3210-1PB21-8AL0
LO base load output	3 kW	4 kW
LO base load input current, 1-ph. AC	35.9 A	43.0 A
LO base load input current, 3-ph. AC	17.7 A	22.8 A
LO base load output current	13.6 A	17.5 A
HO base load output	2.2 kW	3 kW
HO base load input current, 1-ph. AC	31.3 A	37.5 A
HO base load input current, 3-ph. AC	13.6 A	17.7 A
HO base load output current	10.4 A	13.6 A
Fuse according to IEC Fuse according to UL, class J	3NA3820 (50 A) 50 A	3NA3820 (50 A) 50 A
Power loss	0.14 kW	0.18 kW
Required cooling air flow	18.5 l/s	18.5 l/s
Weight without filter	5.0 kg	5.0 kg
Weight with filter	5.2 kg	5.2 kg

Table 10-6 PM240-2, PT, Frame Size C, 1-ph./3-ph. AC 200 V ... 240 V

Article no. without filter Article no. with filter	6SL3211-1PB21-8UL0 6SL3211-1PB21-8AL0
LO base load output	4 kW
LO base load input current, 1-ph. AC	43.0 A
LO base load input current, 3-ph. AC	22.8 A
LO base load output current	17.5 A
HO base load output	3 kW
HO base load input current, 1-ph. AC	37.5 A
HO base load input current, 3-ph. AC	17.7 A
HO base load output current	13.6 A
Fuse according to IEC Fuse according to UL, class J	3NA3820 (50 A) 50 A
Power loss	0.18 kW ¹⁾
Required cooling air flow	18.5 l/s
Weight without filter	5.9 kg
Weight with filter	6.2 kg

1) approx. 0.09 through the heatsink

Table 10-7 PM240-2, IP 20, Frame Size C, 3-ph. AC 200 V ... 240 V

Article no. without filter Article no. with filter	6SL3210-1PC22-2UL0 6SL3210-1PC22-2AL0	6SL3210-1PC22-8UL0 6SL3210-1PC22-8AL0
LO base load output	5.5 kW	7.5 kW
LO base load input current	28.6 A	36.4 A
LO base load output current	22.0 A	28.0 A
HO base load output	4 kW	5.5 kW
HO base load input current	22.8 A	28.6 A
HO base load output current	17.5 A	22.0 A
Fuse according to IEC	3NA3820 (50 A)	3NA3820 (50 A)
Fuse according to UL, class J	50 A	50 A
Power loss	0.2 kW	0.26 kW
Required cooling air flow	18.5 l/s	18.5 l/s
Weight without filter	5.0 kg	5.0 kg
Weight with filter	5.2 kg	5.2 kg

Table 10-8 PM240-2, IP20, Frame Size D, 3-ph. AC 200 V ... 240 V

Article no. without filter	6SL3210-1PC24-2UL0	6SL3210-1PC25-4UL0	6SL3210-1PC26-8UL0
LO base load output	11 kW	15 kW	18.5 kW
LO base load input current	40 A	51 A	64 A
LO base load output current	42 A	54 A	68 A
HO base load output	7.5 kW	11 kW	15 kW
HO base load input current	36 A	43 A	56 A
HO base load output current	35 A	42 A	54 A
Fuse according to IEC/UL	3NA3822 (63 A)	3NA3824 (80 A)	3NA3830 (100 A)
Fuse according to IEC/UL, Class J	60 A	70 A	90 A
Power loss	0.45 kW	0.61 kW	0.82 kW
Required cooling air flow	55 l/s	55 l/s	55 l/s
Weight	17 kg	17 kg	17 kg

10.4 Technical data, PM240-2 Power Module

Table 10-9 PM240-2, IP20, Frame Size E, 3-ph. AC 200 V ... 240 V

Article no. without filter	6SL3210-1PC28-0UL0	6SL3210-1PC31-1UL0
LO base load output	22 kW	30 kW
LO base load input current	76 A	98 A
LO base load output current	80 A	104 A
HO base load output	18.5 kW	22 kW
HO base load input current	71 A	83 A
HO base load output current	68 A	80 A
Fuse according to IEC/UL	3NA3830 (100 A)	3NA3836 (160 A)
Fuse according to IEC/UL, Class J	100 A	150 A
Power loss	0.92 kW	1.28 kW
Required cooling air flow	83 l/s	83 l/s
Weight	26 kg	26 kg

Table 10-10 PM240-2, IP20, Frame Size F, 3-ph. AC 200 V ... 240 V

Article no. without filter	6SL3210-1PC31-3UL0	6SL3210-1PC31-6UL0	6SL3210-1PC31-8UL0
LO base load output	37 kW	45 kW	55 kW
LO base load input current	126 A	149 A	172 A
LO base load output current	130 A	154 A	178 A
HO base load output	30 kW	37 kW	45 kW
HO base load input current	110 A	138 A	164 A
HO base load output current	104 A	130 A	154 A
Fuse according to IEC/UL	3NA3140 (200 A)	3NA3140 (200 A)	3NA3142 (224 A)
Fuse according to IEC/UL, Class J	175 A	200 A	225 A
Power loss	1.38 kW	1.72 kW	2.09 kW
Required cooling air flow	153 l/s	153 l/s	153 l/s
Weight	57 kg	57 kg	57 kg

10.4.4 Current derating depending on the pulse frequency, 200 V inverters

Article number	LO power [kW]	LO base load output current [A]							
Pulse frequency [kHz]		2	4 *)	6	8	10	12	14	16
6SL3210-1PB13-0 . L0	0.55	3.2	3.2	2.7	2.2	1.9	1.6	1.4	1.3
6SL321 . -1PB13-8 . L0	0.75	4.2	4.2	3.6	2.9	2.5	2.1	1.9	1.7
6SL3211-1PB15-5 . L0	1.1	6	6	5.1	4.2	3.6	3	2.7	2.4
6SL3210-1PB17-4 . L0	1.5	7.4	7.4	6.3	5.2	4.4	3.7	3.3	3
6SL321 . -1PB21-0 . L0	2.2	10.4	10.4	8.8	7.3	6.2	5.2	4.7	4.2
6SL3210-1PB21-4 . L0	3	13.6	13.6	11.6	9.5	8.2	6.8	6.1	5.4
6SL321 . -1PB21-8 . L0	4	17.5	17.5	14.9	12.3	10.5	8.8	7.9	7
6SL3210-1PC22-2 . L0	5.5	22	22	18.7	15.4	13.2	11	9.9	8.8
6SL3210-1PC22-8 . L0	7.5	28	28	23.8	19.6	16.8	14	12.6	11.2
6SL3210-1PC24-2 . L0	11	42	42	35.7	29.4	25.2	21	18.9	16.8
6SL3210-1PC25-4 . L0	15	54	54	45.9	37.8	32.4	27	24.3	21.6
6SL3210-1PC26-8 . L0	18.5	68	68	57.8	47.6	40.8	34	30.6	27.2
6SL3210-1PC28-0 . L0	22	80	80	68	56	48	40	36	32
6SL3210-1PC31-1 . L0	30	104	104	88.4	72.8	62.4	52	46.8	41.6
6SL3210-1PC31-3 . L0	37	130	130	110.5	91	---	---	---	---
6SL3210-1PC31-6 . L0	45	154	154	130.9	107.8	---	---	---	---
6SL3210-1PC31-8 . L0	55	178	178	151.3	124.6	---	---	---	---

*) Factory setting

The permissible motor cable length depends on the particular cable type and the pulse frequency that has been selected

10.4.5 General technical data, 400 V inverters

Property	Version
Line voltage	FSA ... FSC 380 V ... 480 V 3-ph. AC $\pm 10\%$
	FSD ... FSF 3-ph. AC 380 V ... 480 V $\pm 10\%$ (in operation $-20\% < 1 \text{ min}$)
Line supply configurations	Grounded TN/TT line supplies or non-grounded IT line supplies  Connecting the line supply, motor, and inverter components (Page 69)
Line impedance	FSA ... FSC $1\% \leq U_k < 4\%$, for values smaller than 1%, we recommend a line reactor, or a Power Module with the next higher power rating.
	FSD ... FSF No restrictions
Power factor λ	FSA ... FSC 0.7 without line reactor for $U_k \geq 1\%$ 0.85 with line reactor for $U_k < 1\%$
	FSD ... FSF > 0.9
Output voltage	3-ph. AC 0 V ... 0.95 x input voltage (max.)
Input frequency	50 Hz ... 60 Hz, $\pm 3 \text{ Hz}$
Output frequency	0 ... 550 Hz, depending on the control mode
Inrush current	$< \text{LO base load input current}$
Overvoltage category according to EN 61800-5-1	III for line supplies
Pulse frequency	Factory setting • 4 kHz for devices with an LO base load power $< 75 \text{ kW}$ • 2 kHz for devices with an LO base load power $\geq 75 \text{ kW}$ Can be adjusted in 2 kHz steps as follows: • 2 kHz ... 16 kHz for devices with an LO base load power $< 55 \text{ kW}$ • 2 kHz ... 8 kHz for devices with an LO base load output of 55 kW ... 90 kW • 2 kHz ... 4 kHz for devices with an LO base load output $\geq 110 \text{ kW}$ If you increase the pulse frequency, the inverter reduces the maximum output current.
Short-circuit current rating (SCCR) and branch protection	$\leq 100 \text{ kA rms}$  Branch protection and short-circuit strength according to UL and IEC (https://support.industry.siemens.com/cs/ww/en/view/109479152)
Braking methods	DC braking, compound braking, dynamic braking with integrated braking chopper
Degree of protection according to EN 60529	IP20
	IP55 PT devices outside the control cabinet
Protection class according to EN 61800-5-1	The inverters are devices with protection class I
Touch protection according to EN 50274	DGUV regulation 3 when used for the intended purpose
Cooling in compliance with EN 60146	Forced air cooling AF

10.4.6 Specific technical data, 400 V inverters

Table 10-11 PM240-2, IP20, Frame Size A, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1PE11-8UL1	6SL3210-1PE12-3UL1	6SL3210-1PE13-2UL1
Article no. with filter	6SL3210-1PE11-8AL1	6SL3210-1PE12-3AL1	6SL3210-1PE13-2AL1
LO base load output	0.55 kW	0.75 kW	1.1 kW
LO base load input current	2.3 A	2.9 A	4.1 A
LO base load output current	1.7 A	2.2 A	3.1 A
HO base load output	0.37 kW	0.55 kW	0.75 kW
HO base load input current	2.0 A	2.6 A	3.3 A
HO base load output current	1.3 A	1.7 A	2.2 A
Fuse according to IEC	3NA3805 (16 A)	3NA3805 (16 A)	3NA3805 (16 A)
Fuse according to UL, class J	10 A	10 A	15 A
Power loss	0.04 kW	0.04 kW	0.04 kW
Required cooling air flow	5 l/s	5 l/s	5 l/s
Weight without filter	1.3 kg	1.3 kg	1.3 kg
Weight with filter	1.5 kg	1.5 kg	1.5 kg

Table 10-12 PM240-2, IP20, Frame Size A, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1PE14-3UL1	6SL3210-1PE16-1UL1	6SL3210-1PE18-0UL1
Article no. with filter	6SL3210-1PE14-3AL1	6SL3210-1PE16-1AL1	6SL3210-1PE18-0AL1
LO base load output	1.5 kW	2.2 kW	3.0 kW
LO base load input current	5.5 A	7.7 A	10.1 A
LO base load output current	4.1 A	5.9 A	7.7 A
HO base load output	1.1 kW	1.5 kW	2.2 kW
HO base load input current	4.7 A	6.1 A	8.8 A
HO base load output current	3.1 A	4.1 A	5.9 A
Fuse according to IEC	3NA3805 (16 A)	3NA3805 (16 A)	3NA3805 (16 A)
Fuse according to UL, class J	20 A	30 A	30 A
Power loss	0.07 kW	0.1 kW	0.12 kW
Required cooling air flow	5 l/s	5 l/s	5 l/s
Weight without filter	1.4 kg	1.4 kg	1.4 kg
Weight with filter	1.6 kg	1.6 kg	1.6 kg

10.4 Technical data, PM240-2 Power Module

Table 10-13 PM240-2, PT, Frame Size A, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3211-1PE18-0UL1
Article no. with filter	6SL3211-1PE18-0AL1
LO base load output	3.0 kW
LO base load input current	10.1 A
LO base load output current	7.7 A
HO base load output	2.2 kW
HO base load input current	8.8 A
HO base load output current	5.9 A
Fuse according to IEC	3NA3805 (16 A)
Fuse according to UL, class J	30 A
Power loss without filter	0.12 kW ¹⁾
Required cooling air flow	7 l/s
Weight without filter	1.8 kg
Weight with filter	2.0 kg

1) approx. 0.1 kW through the heatsink

Table 10-14 PM240-2, IP20, Frame Size B, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1PE21-1UL0	6SL3210-1PE21-4UL0	6SL3210-1PE21-8UL0
Article no. with filter	6SL3210-1PE21-1AL0	6SL3210-1PE21-4AL0	6SL3210-1PE21-8AL0
LO base load output	4.0 kW	5.5 kW	7.5 kW
LO base load input current	13.3 A	17.2 A	22.2 A
LO base load output current	10.2 A	13.2 A	18.0 A
HO base load output	3.0 kW	4.0 kW	5.5 kW
HO base load input current	11.6 A	15.3 A	19.8 A
HO base load output current	7.7 A	10.2 A	13.2 A
Fuse according to IEC	3NA3812 (32 A)	3NA3812 (32 A)	3NA3812 (32 A)
Fuse according to UL, class J	35 A	35 A	35 A
Power loss	0.11 kW	0.15 kW	0.2 kW
Required cooling air flow	9.2 l/s	9.2 l/s	9.2 l/s
Weight without filter	2.9 kg	2.9 kg	3.0 kg
Weight with filter	3.1 kg	3.1 kg	3.2 kg

Table 10-15 PM240-2, PT, Frame Size B, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3211-1PE21-8UL0
Article no. with filter	6SL3211-1PE21-8AL0
LO base load output	7.5 kW
LO base load input current	22.2 A
LO base load output current	18.0 A
HO base load output	5.5 kW
HO base load input current	19.8 A
HO base load output current	13.7 A
Fuse according to IEC	3NA3812 (32 A)
Fuse according to UL, class J	35 A
Power loss	0.2 kW ¹⁾
Required cooling air flow	9.2 l/s
Weight without filter	3.6 kg
Weight with filter	3.9 kg

1) approx. 0.16 kW through the heatsink

Table 10-16 PM240-2, IP20, Frame Size C, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1PE22-7UL0	6SL3210-1PE23-3UL0
Article no. with filter	6SL3210-1PE22-7AL0	6SL3210-1PE23-3AL0
LO base load output	11.0 kW	15.0 kW
LO base load input current	32.6 A	39.9 A
LO base load output current	26.0 A	32.0 A
HO base load output	7.5 kW	11.0 kW
HO base load input current	27.0 A	36.0 A
HO base load output current	18.0 A	26.0 A
Fuse according to IEC	3NA3820 (50 A)	3NA3820 (50 A)
Fuse according to UL, class J	50 A	50 A
Power loss	0.3 kW	0.37 kW
Required cooling air flow	18.5 l/s	18.5 l/s
Weight without filter	4.7 kg	4.8 kg
Weight with filter	5.3 kg	5.4 kg

10.4 Technical data, PM240-2 Power Module

Table 10-17 PM240-2, PT, Frame Size C, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3211-1PE23-3UL0
Article no. with filter	6SL3211-1PE23-3AL0
LO base load output	15.0 kW
LO base load input current	39.9 A
LO base load output current	32.0 A
HO base load output	11.0 kW
HO base load input current	36.0 A
HO base load output current	26.0 A
Fuse according to IEC	3NA3820 (50 A)
Fuse according to UL, class J	50 A
Power loss	0.37 kW ¹⁾
Required cooling air flow	18.5 l/s
Weight without filter	5.8 kg
Weight with filter	6.3 kg

1) approx. 0.3 kW through the heatsink

Table 10-18 PM240-2, IP20, Frame Size D, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1PE23-8UL0	6SL3210-1PE24-5UL0	6SL3210-1PE26-0UL0
Article no. with filter	6SL3210-1PE23-8AL0	6SL3210-1PE24-5AL0	6SL3210-1PE26-0AL0
LO base load output	18.5 kW	22 kW	30 kW
LO base load input current	36 A	42 A	57 A
LO base load output current	38 A	45 A	60 A
HO base load output	15 kW	18.5 kW	22 kW
HO base load input current	33 A	38 A	47 A
HO base load output current	32 A	38 A	45 A
Fuse according to IEC/UL	3NA3822 (63 A)	3NA3824 (80 A)	3NA3830 (100 A)
Fuse according to IEC/UL, Class J	60 A	70 A	90 A
Power loss without filter	0.57 kW	0.70 kW	0.82 kW
Power loss with filter	0.58 kW	0.71 kW	0.83 kW
Required cooling air flow	55 l/s	55 l/s	55 l/s
Weight without filter	16 kg	16 kg	17 kg
Weight with filter	17.5 kg	17.5 kg	18.5 kg

Table 10-19 PM240-2, IP20, Frame Size D, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1PE27-5UL0
Article no. with filter	6SL3210-1PE27-5AL0
LO base load output	37 kW
LO base load input current	70 A
LO base load output current	75 A
HO base load output	30 kW
HO base load input current	62 A
HO base load output current	60 A
Fuse according to IEC/UL	3NA3830 (100 A)
Fuse according to IEC/UL, Class J	100 A
Power loss without filter	1.09 kW
Power loss with filter	1.10 kW
Required cooling air flow	55 l/s
Weight without filter	17 kg
Weight with filter	18.5 kg

Table 10-20 PM240-2, IP20, Frame Size E, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1PE28-8UL0	6SL3210-1PE31-1UL0
Article no. with filter	6SL3210-1PE28-8AL0	6SL3210-1PE31-1AL0
LO base load output	45 kW	55 kW
LO base load input current	86 A	104 A
LO base load output current	90 A	110 A
HO base load output	37 kW	45 kW
HO base load input current	78 A	94 A
HO base load output current	75 A	90 A
Fuse according to IEC/UL	3NA3832 (125 A)	3NA3836 (160 A)
Fuse according to IEC/UL, Class J	125 A	150 A
Power loss without filter	1.29 kW	1.65 kW
Power loss with filter	1.30 kW	1.67 kW
Required cooling air flow	83 l/s	83 l/s
Weight without filter	26 kg	26 kg
Weight with filter	28 kg	28 kg

10.4 Technical data, PM240-2 Power Module

Table 10-21 PM240-2, IP20, Frame Size F, 3-ph. AC 380 V ... 480 V

Article no. without filter Article no. with filter	6SL3210-1PE31-5UL0 6SL3210-1PE31-5AL0	6SL3210-1PE31-8UL0 6SL3210-1PE31-8AL0	6SL3210-1PE32-1UL0 6SL3210-1PE32-1AL0
LO base load output	75 kW	90 kW	110 kW
LO base load input current	140 A	172 A	198 A
LO base load output current	145 A	178 A	205 A
HO base load output	55 kW	75 kW	90 kW
HO base load input current	117 A	154 A	189 A
HO base load output current	110 A	145 A	178 A
Fuse according to IEC/UL	3NA3140 (200 A)	3NA3142 (224 A)	3NA3250 (300 A)
Fuse according to IEC/UL, Class J	200 A	250 A	300 A
Power loss without filter	1.91 kW	2.46 kW	2.28 kW
Power loss with filter	1.93 kW	2.48 kW	2.30 kW
Required cooling air flow	153 l/s	153 l/s	153 l/s
Weight without filter	57 kg	57 kg	61 kg
Weight with filter	63 kg	63 kg	65 kg

Table 10-22 PM240-2, IP20, Frame Size F, 3-ph. AC 380 V ... 480 V

Article no. without filter Article no. with filter	6SL3210-1PE32-5UL0 6SL3210-1PE32-5AL0
LO base load output	132 kW
LO base load input current	242 A
LO base load output current	250 A
HO base load output	110 kW
HO base load input current	218 A
HO base load output current	205 A
Fuse according to IEC/UL	3NA3252 (315 A)
Fuse according to IEC/UL, Class J	350 A
Power loss without filter	2.98 kW
Power loss with filter	3.02 kW
Required cooling air flow	153 l/s
Weight without filter	61 kg
Weight with filter	65 kg

10.4.7 Current derating depending on the pulse frequency, 400 V inverters

Article number	LO power [kW]	LO base load output current [A]							
Pulse frequency [kHz]		2	4 *)	6	8	10	12	14	16
6SL3210-1PE11-8 . L1	0.55	1.7	1.7	1.4	1.2	1	0.9	0.8	0.7
6SL3210-1PE12-3 . L1	0.75	2.2	2.2	1.9	1.5	1.3	1.1	1	0.9
6SL3211-1PE13-2 . L1	1.1	3.1	3.1	2.6	2.2	1.9	1.6	1.4	1.2
6SL3210-1PE14-3 . L1	1.5	4.1	4.1	3.5	2.9	2.5	2.1	1.8	1.6
6SL3210-1PE16-1 . L1	2.2	5.9	5.9	5	4.1	3.5	3	2.7	2.4
6SL321 . -1PE18-0 . L1	3	7.7	7.7	6.5	5.4	4.6	3.9	3.5	3.1
6SL3210-1PE21-1 . L0	4	10.2	10.2	8.7	7.1	6.1	5.1	4.6	4.1
6SL3210-1PE21-4 . L0	5.5	13.2	13.2	11.2	9.2	7.9	6.6	5.9	5.3
6SL321 . -1PE21-8 . L0	7.5	18	18	15.3	12.6	10.8	9	8.1	7.2
6SL3210-1PE22-7 . L0	11	26	26	22.1	18.2	15.6	13	11.7	10.4
6SL321 . -1PE23-3 . L0	15	32	32	27.2	22.4	19.2	16	14.4	12.8
6SL3210-1PE23-8 . L0	18.5	38	38	32.3	26.6	22.8	19	17.1	15.2
6SL3210-1PE24-5 . L0	22	45	45	38.3	31.5	27	22.5	20.3	18
6SL3210-1PE26-0 . L0	30	60	60	51	42	36	30	27	24
6SL3210-1PE27-5 . L0	37	75	75	63.8	52.5	45	37.5	33.8	30
6SL3210-1PE28-8 . L0	45	90	90	76.5	63	54	45	40.5	36
6SL3210-1PE31-1 . L0	55	110	110	93.5	77	---	---	---	---
Pulse frequency [kHz]		2 *)	4	6	8	10	12	14	16
6SL3210-1PE31-5 . L0	75	145	145	123.3	101.5	---	---	---	---
6SL3210-1PE31-8 . L0	90	178	178	151.3	124.6	---	---	---	---
6SL3210-1PE32-1 . L0	110	205	143.5	---	---	---	---	---	---
6SL3210-1PE32-5 . L0	132	250	175	---	---	---	---	---	---

*) Factory setting

The permissible motor cable length depends on the particular cable type and the pulse frequency that has been selected

10.4.8 General technical data, 690 V inverters

Property	Version
Line voltage	- for systems according to IEC: 3-ph. AC 500 V ... 690 V $\pm$ 10% (in operation -20% < 1 min) - for systems according to UL: 3-ph. AC 500 V ... 600 V $\pm$ 10% (in operation -20% < 1 min) filtered devices only with Slash Rating (600Y/347V AC)
Line supply configurations	Grounded TN/TT line supplies or non-grounded IT line supplies  Connecting the line supply, motor, and inverter components (Page 69)
Line impedance	No restrictions
Power factor λ	> 0.9
Output voltage	3-ph. AC 0 V ... 0.95 x input voltage (max.)
Input frequency	50 Hz ... 60 Hz, $\pm$ 3 Hz
Output frequency	0 ... 550 Hz, depending on the control mode
Inrush current	< LO base load input current
Overvoltage category according to EN 61800-5-1	III for line supplies
Pulse frequency	2 kHz (factory setting), can be adjusted to 4 kHz If you increase the pulse frequency, the inverter reduces the maximum output current.
Short-circuit current rating (SCCR) and branch protection	$\leq$ 100 kA rms  Branch protection and short-circuit strength according to UL and IEC (https://support.industry.siemens.com/cs/ww/en/view/109479152)
Braking methods	DC braking, compound braking, dynamic braking with integrated braking chopper
Degree of protection according to EN 60529	IP20; must be installed in a control cabinet
Protection class according to EN 61800-5-1	The inverters are devices with protection class I
Touch protection according to EN 50274	DGUV regulation 3 when used for the intended purpose
Cooling in compliance with EN 60146	Forced air cooling AF

10.4.9 Specific technical data, 690 V inverters

Table 10-23 PM240-2, IP20, Frame Size D, 3-ph. AC 500 V ... 690 V

Article no. without filter Article no. with filter	6SL3210-1PH21-4UL0 6SL3210-1PH21-4AL0	6SL3210-1PH22-0UL0 6SL3210-1PH22 -0AL0	6SL3210-1PH22-3UL0 6SL3210-1PH22 -3AL0
LO base load output	11 kW	15 kW	18.5 kW
LO base load input current	14 A	18 A	22 A
LO base load output current	14 A	19 A	23 A
HO base load output	7.5 kW	11 kW	15 kW
HO base load input current	11 A	14 A	20 A
HO base load output current	11 A	14 A	19 A
Fuse according to IEC/UL Fuse according to IEC/UL, Class J	3NA3807-6 (20 A) 20 A	3NA3810-6 (25 A) 25 A	3NA3812-6 (32 A) 30 A
Power loss without filter	0.35 kW	0.44 kW	0.52 kW
Power loss with filter	0.35 kW	0.45 kW	0.52 kW
Required cooling air flow	55 l/s	55 l/s	55 l/s
Weight without filter	17 kg	17 kg	17 kg
Weight with filter	18.5 kg	18.5 kg	18.5 kg

Table 10-24 PM240-2, IP20, Frame Size D, 3-ph. AC 500 V ... 690 V

Article no. without filter Article no. with filter	6SL3210-1PH22-7UL0 6SL3210-1PH22-7AL0	6SL3210-1PH23-5UL0 6SL3210-1PH23 -5AL0	6SL3210-1PH24-2UL0 6SL3210-1PH24-2AL0
LO base load output	22 kW	30 kW	37 kW
LO base load input current	25 A	33 A	40 A
LO base load output current	27 A	35 A	42 A
HO base load output	18.5 kW	22 kW	30 kW
HO base load input current	24 A	28 A	36 A
HO base load output current	23 A	27 A	35 A
Siemens fuse according to IEC/UL Fuse according to IEC/UL, Class J	3NA3817-6KJ (40 A) 35 A	3NA3820-6KJ (50 A) 45 A	3NA3822-6 (63 A) 60 A
Power loss without filter	0.60 kW	0.77 kW	0.93 kW
Power loss with filter	0.60 kW	0.78 kW	0.94 kW
Required cooling air flow	55 l/s	55 l/s	55 l/s
Weight without filter	17 kg	17 kg	17 kg
Weight with filter	18.5 kg	18.5 kg	18.5 kg

10.4 Technical data, PM240-2 Power Module

Table 10-25 PM240-2, IP20, Frame Size E, 3-ph. AC 500 V ... 690 V

Article no. without filter Article no. with filter	6SL3210-1PH25-2UL0 6SL3210-1PH25-2AL0	6SL3210-1PH26-2UL0 6SL3210-1PH26-2AL0
LO base load output	45 kW	55 kW
LO base load input current	50 A	59 A
LO base load output current	52 A	62 A
HO base load output	37 kW	45 kW
HO base load input current	44 A	54 A
HO base load output current	42 A	52 A
Siemens fuse according to IEC/UL Fuse according to IEC/UL, Class J	3NA3824-6 (80A) 80 A	3NA3824-6 (80A) 80 A
Power loss without filter	1.07 kW	1.30 kW
Power loss with filter	1.08 kW	1.31 kW
Required cooling air flow	83 l/s	83 l/s
Weight without filter	26 kg	26 kg
Weight with filter	28 kg	28 kg

Table 10-26 PM240-2, IP20, Frame Size F, 3-ph. AC 500 V ... 690 V

Article no. without filter Article no. with filter	6SL3210-1PH28-0UL0 6SL3210-1PH28-0AL0	6SL3210-1PH31-0UL0 6SL3210-1PH31-0AL0	6SL3210-1PH31-2UL0 6SL3210-1PH31-2AL0
LO base load output	75 kW	90 kW	110 kW
LO base load input current	78 A	97 A	111 A
LO base load output current	80 A	100 A	115 A
HO base load output	55 kW	75 kW	90 kW
HO base load input current	66 A	85 A	106 A
HO base load output current	62 A	80 A	100 A
Siemens fuse according to IEC/UL Fuse according to IEC/UL, Class J	3NA3830-6 (100 A) 100 A	3NA3132-6 (125 A) 125 A	3NA3136-6 (160 A) 150 A
Power loss without filter	1.37 kW	1.74 kW	1.95 kW
Power loss with filter	1.38 kW	1.76 kW	1.97 kW
Required cooling air flow	153 l/s	153 l/s	153 l/s
Weight without filter	60 kg	60 kg	60 kg
Weight with filter	64 kg	64 kg	64 kg

Table 10-27 PM240-2, IP20, Frame Size F, 3-ph. AC 500 V ... 690 V

Article no. without filter	6SL3210-1PH31-4UL0
Article no. with filter	6SL3210-1PH31-4AL0
LO base load output	132 kW
LO base load input current	137 A
LO base load output current	142 A
HO base load output	110 kW
HO base load input current	122 A
HO base load output current	115 A
Siemens fuse according to IEC/UL	3NA3140-6 (200 A)
Fuse according to IEC/UL, Class J	200 A
Power loss without filter	2.48 kW
Power loss with filter	2.51 kW
Required cooling air flow	153 l/s
Weight without filter	60 kg
Weight with filter	64 kg

10.4.10 Current derating depending on the pulse frequency, 690 V inverters

Article number	LO power [kW]	LO base load output current [A]	
Pulse frequency [kHz]		2 *)	4
6SL3210-1PH21-4 . L0	11	14	8.4
6SL3210-1PH22-0 . L0	15	19	11.4
6SL3210-1PH22-3 . L0	18.5	23	13.8
6SL3210-1PH22-7 . L0	22	27	16.2
6SL3210-1PH23-5 . L0	30	35	21
6SL3210-1PH24-2 . L0	37	42	25.2
6SL3210-1PH25-2 . L0	45	52	31.2
6SL3210-1PH26-2 . L0	55	62	37.2
6SL3210-1PH28-0 . L0	75	80	48
6SL3210-1PH31-0 . L0	90	100	60
6SL3210-1PH31-2 . L0	110	115	69
6SL3210-1PH31-4 . L0	132	142	85.2

*) Factory setting

The permissible motor cable length depends on the particular cable type and the pulse frequency that has been selected

10.5 Technical Data, PM240P-2 Power Module

Protective organs for the Power Module

The fuses listed in the following tables are examples of suitable fuses.



Additional components for branch protection are available in the Internet:

Branch protection and short-circuit strength according to UL and IEC (<https://support.industry.siemens.com/cs/ww/en/view/109479152>)

Typical inverter load cycles

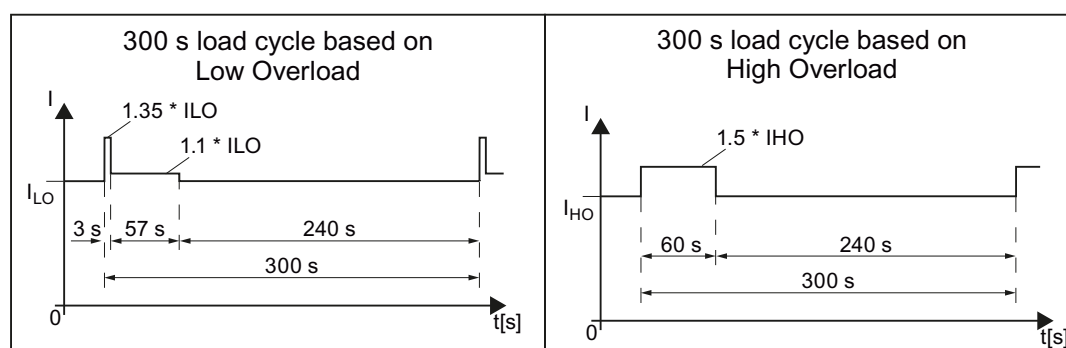


Figure 10-2 "Low Overload" and "High Overload" load cycles

10.5.1 Ambient conditions

Property	Version
Ambient conditions for transport in the transport packaging	
Climatic ambient conditions	- 40° C ... + 70° C, according to Class 2K4 to EN 60721-3-2 maximum humidity 95% at 40° C
Mechanical ambient conditions	Shock and vibration permissible according to 2M3 to EN 60721-3-2
Protection against chemical substances	Protected according to Class 2C2 to EN 60721-3-2
Biological ambient conditions	Suitable according to Class 2B1 to EN 60721-3-2
Ambient conditions for long-term storage in the product packaging	
Climatic ambient conditions	- 25° C ... + 55° C, according to Class 1K3 to EN 60721-3-1
Protection against chemical substances	Protected according to Class 1C2 to EN 60721-3-1
Biological ambient conditions	suitable according to class 1B1 to EN 60721-3-1
Ambient conditions in operation	

Property	Version
Installation altitude	Up to 1000 m above sea level without derating, > 1000 m  Restrictions for special ambient conditions (Page 474)
Climatic ambient conditions ¹⁾	- Frame sizes FSD ... FSF temperature range ²⁾ - in operation acc. to LO: -20° C ... +40° C - in operation acc. to HO: -20° C ... +50° C - for higher temperatures  Restrictions for special ambient conditions (Page 474) - Relative humidity: 5 ... 95%, condensation not permitted - Oil mist, salt mist, ice formation, condensation, dripping water, spraying water, splashing water and water jets are not permitted
Mechanical ambient conditions	- Vibration levels permissible according to Class 3M1 to EN 60721-3-3 - Shock permissible according to Class 3M1 to EN 60721-3-3
Protection against chemical substances	Protected according to 3C2 to EN 60721-3-3
Biological ambient conditions	Suitable according to 3B1 to EN 60721-3-3
Pollution	Suitable for environments with degree of pollution 2 according to EN 61800-5-1
Cooling	Forced air cooling AF, according to EN 60146
Cooling air	Clean and dry air

¹⁾ Increased ruggedness regarding temperature range and relative humidity; therefore better than 3K3 according to EN 60721-3-3

²⁾ Observe the permissible ambient temperatures for the Control Unit and possibly the Operator Panel (IOP or BOP-2).

10.5.2 General technical data, 400 V inverters

Property	Version
Line voltage	3 AC 380 V ... 480 V $\pm 10\%$ (in operation $-20\% < 1$ min)
Line system configurations	Grounded TN/TT line systems or non-grounded IT line systems
Line impedance	$U_k < 4\%$, line reactor is not required
Power factor λ	> 0.9
Output voltage	3 AC 0 V ... 0.95 x input voltage (max.)
Input frequency	50 Hz ... 60 Hz, ± 3 Hz
Output frequency	0 ... 550 Hz, depending on the control mode
Inrush current	$< LO$ base load input current
Overvoltage category according to EN 61800-5-1	III for line supplies
Pulse frequency	Factory setting • 4 kHz for devices with an LO base load power < 75 kW • 2 kHz for devices with an LO base load power ≥ 75 kW Can be adjusted in 2 kHz steps as follows: • 2 kHz ... 16 kHz for devices with an LO base load power < 55 kW • 2 kHz ... 8 kHz for devices with an LO base load power ≥ 55 kW • 2 kHz ... 4 kHz for devices with an LO base load power ≥ 110 kW If you increase the pulse frequency, the inverter reduces the maximum output current.
Short-circuit current rating (SCCR) and branch protection	≤ 100 kA rms  Branch protection and short-circuit strength according to UL and IEC (https://support.industry.siemens.com/cs/ww/en/view/109479152)
Braking methods	DC braking, compound braking
Degree of protection according to EN 60529	IP20 Must be installed in a control cabinet
Protection class according to EN 61800-5-1	The inverters are devices with protection class I
Touch protection according to EN 50274	DGUV regulation 3 when used for the intended purpose
Cooling in compliance with EN 60146	Forced air cooling AF

10.5.3 Specific technical data, 400 V inverters

Table 10-28 PM240P-2, IP20, Frame Size D, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1RE24-5UL0	6SL3210-1RE26-0UL0	6SL3210-1RE27-5UL0
Article no. with filter	6SL3210-1RE24-5AL0	6SL3210-1RE26-0AL0	6SL3210-1RE27-5AL0
LO base load power	22 kW	30 kW	37 kW
LO base load input current	42 A	57 A	70 A
LO base load output current	45 A	60 A	75 A
HO base load power	18.5 kW	22 kW	30 kW
HO base load input current	38 A	47 A	62 A
HO base load output current	38 A	45 A	60 A
Siemens fuse according to IEC/UL	3NE1820-0 / 80 A	3NE1021-0 / 100 A	3NE1021-0 / 100 A
Fuse according to IEC/UL, Class J	70 A	90 A	100 A
Power loss without filter	0.68 kW	0.76 kW	1.01 kW
Power loss with filter	0.68 kW	0.77 kW	1.02 kW
Cooling air flow required	55 l/s	55 l/s	55 l/s
Weight without filter	16 kg	17 kg	17 kg
Weight with filter	17.5 kg	18.5 kg	18.5 kg

Table 10-29 PM240P-2, IP20, Frame Size E, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1RE28-8UL0	6SL3210-1RE31-1UL0
Article no. with filter	6SL3210-1RE28-8AL0	6SL3210-1RE31-1AL0
LO base load power	45 kW	55 kW
LO base load input current	86 A	104 A
LO base load output current	90 A	110 A
HO base load power	37 kW	45 kW
HO base load input current	78 A	94 A
HO base load output current	75 A	90 A
Siemens fuse according to IEC/UL	3NE1022-0 / 125 A	3NE1224-0 / 160 A
Fuse according to IEC/UL, Class J	125 A	150 A
Power loss without filter	1.19 kW	1.54 kW
Power loss with filter	1.2 kW	1.55 kW
Required cooling air flow	83 l/s	83 l/s
Weight without filter	26 kg	26 kg
Weight with filter	28 kg	28 kg

Table 10-30 PM240P-2, IP20, Frame Size F, 3-ph. AC 380 V ... 480 V

Article no. without filter	6SL3210-1RE31-5UL0	6SL3210-1RE31-8UL0	6SL3210-1RE32-1UL0
Article no. with filter	6SL3210-1RE31-5AL0	6SL3210-1RE31-8AL0	6SL3210-1RE32-1AL0
LO base load power	75 kW	90 kW	110 kW
LO base load input current	140 A	172 A	198 A

10.5 Technical Data, PM240P-2 Power Module

Article no. without filter Article no. with filter	6SL3210-1RE31-5UL0 6SL3210-1RE31-5AL0	6SL3210-1RE31-8UL0 6SL3210-1RE31-8AL0	6SL3210-1RE32-1UL0 6SL3210-1RE32-1AL0
LO base load output current	145 A	178 A	205 A
HO base load power	55 kW	75 kW	90 kW
HO base load input current	117 A	154 A	189 A
HO base load output current	110 A	145 A	178 A
Siemens fuse according to IEC/UL	3NE1225-0 / 200 A	3NE1227-0 / 250 A	3NE1230-0 / 315 A
Fuse according to IEC/UL, Class J	200 A	250 A	300 A
Power loss without filter	1.95 kW	2.54 kW	2.36 kW
Power loss with filter	1.97 kW	2.56 kW	2.38 kW
Required cooling air flow	153 l/s	153 l/s	153 l/s
Weight without filter	57 kg	57 kg	61 kg
Weight with filter	63 kg	63 kg	65 kg

Table 10-31 PM240P-2, IP20, Frame Size F, 3-ph. AC 380 V ... 480 V

Article no. without filter Article no. with filter	6SL3210-1RE32-5UL0 6SL3210-1RE32-5AL0
LO base load power	132 kW
LO base load input current	242 A
LO base load output current	250 A
HO base load power	110 kW
HO base load input current	218 A
HO base load output current	205 A
Siemens fuse according to IEC/UL	3NE1331-0 / 350 A
Fuse according to IEC/UL, Class J	350 A
Power loss without filter	3.09 kW
Power loss with filter	3.12 kW
Required cooling air flow	153 l/s
Weight without filter	61 kg
Weight with filter	65 kg

10.5.4 Current derating depending on the pulse frequency, 400 V inverters

Article number	LO power [kW]	LO base load output current [A]							
Pulse frequency [kHz]		2	4 *)	6	8	10	12	14	16
6SL3210-1RE24-5 . L0	22	45	45	38.3	31.5	27	22.5	20.3	18
6SL3210-1RE26-0 . L0	30	60	60	51	42	36	30	27	24
6SL3210-1RE27-5 . L0	37	75	75	63.8	52.5	45	37.5	33.8	30
6SL3210-1RE28-8 . L0	45	90	90	76.5	63	54	45	40.5	36
6SL3210-1RE31-1 . L0	55	110	110	93.5	77	---	---	---	---
Pulse frequency [kHz]		2 *)	4	6	8	10	12	14	16
6SL3210-1RE31-5 . L0	75	145	145	123.3	101.5	---	---	---	---
6SL3210-1RE31-8 . L0	90	178	178	151.3	124.6	---	---	---	---
6SL3210-1RE32-1 . L0	110	205	143.5	---	---	---	---	---	---
6SL3210-1RE32-5 . L0	132	250	175	---	---	---	---	---	---

*) Factory setting

The permissible motor cable length depends on the particular cable type and the pulse frequency that has been selected

10.5.5 General technical data, 690 V inverters

Property	Version
Line voltage	3 AC 500 V ... 690 V $\pm$ 10% (in operation -20% < 1 min) with Class J fuses, maximum 600 V
Line system configurations	Grounded TN/TT line systems or non-grounded IT line systems
Line impedance	$U_k < 4\%$, line reactor is not required
Power factor λ	> 0.9
Output voltage	3 AC 0 V ... 0.95 $\times$ input voltage (max.)
Input frequency	50 Hz ... 60 Hz, $\pm$ 3 Hz
Output frequency	0 ... 550 Hz, depending on the control mode
Inrush current	$< LO$ base load input current
Overvoltage category according to EN 61800-5-1	III for line supplies
Pulse frequency	2 kHz (factory setting), can be adjusted to 4 kHz If you increase the pulse frequency, the inverter reduces the maximum output current.
Short-circuit current rating (SCCR) and branch protection	≤ 100 kA rms  Branch protection and short-circuit strength according to UL and IEC (https://support.industry.siemens.com/cs/ww/en/view/109479152)
Braking methods	DC braking, compound braking
Degree of protection according to EN 60529	IP20; must be installed in a control cabinet
Protection class according to EN 61800-5-1	The inverters are devices with protection class I
Touch protection according to EN 50274	DGUV regulation 3 when used for the intended purpose
Cooling in compliance with EN 60146	Forced air cooling AF

10.5.6 Specific technical data, 690 V inverters

Note

Power data

The power ratings. Specified in the subsequent tables refer to an input voltage of 690 V. The power is correspondingly reduced for lower input voltages.



Restrictions for special ambient conditions (Page 474).

Table 10-32 PM240P-2, IP20, frame size D, 3 AC 500 V ... 690 V

Article no. without filter Article no. with filter	6SL3210-1RH21-4UL0 6SL3210-1RH21-4AL0	6SL3210-1RH22-0UL0 6SL3210-1RH22-0AL0	6SL3210-1RH22-3UL0 6SL3210-1RH22-3AL0
LO base load power	11 kW	15 kW	18.5 kW
LO base load input current	14 A	18 A	22 A
LO base load output current	14 A	19 A	23 A
HO base load power	7.5 kW	11 kW	15 kW
HO base load input current	11 A	14 A	20 A
HO base load output current	11 A	14 A	19 A
Siemens fuse according to IEC/UL Fuse according to IEC/UL, Class J	3NE1815-0 / 25 A 20 A	3NE1815-0 / 25 A 25 A	3NE1803-0 / 35 A 30 A
Power loss	0.32 kW	0.41 kW	0.48 kW
Cooling air flow required	55 l/s	55 l/s	55 l/s
Weight without filter	17 kg	17 kg	17 kg
Weight with filter	18.5 kg	18.5 kg	18.5 kg

Table 10-33 PM240P-2, IP20, frame size D, 3 AC 500 V ... 690 V

Article no. without filter Article no. with filter	6SL3210-1RH22-7UL0 6SL3210-1RH22-7AL0	6SL3210-1RH23-5UL0 6SL3210-1RH23-5AL0	6SL3210-1RH24-2UL0 6SL3210-1RH24-2AL0
LO base load power	22 kW	30 kW	37 kW
LO base load input current	25 A	33 A	40 A
LO base load output current	27 A	35 A	42 A
HO base load power	18.5 kW	22 kW	30 kW
HO base load input current	24 A	28 A	36 A
HO base load output current	23 A	27 A	35 A
Siemens fuse according to IEC/UL Fuse according to IEC/UL, Class J	3NE1803-0 / 35 A 35 A	3NE1817-0 / 50 A 50 A	3NE1818-0 / 63 A 60 A
Power loss without filter	0.56 kW	0.72 kW	0.88 kW
Power loss with filter	0.56 kW	0.73 kW	0.88 kW
Cooling air flow required	55 l/s	55 l/s	55 l/s

10.5 Technical Data, PM240P-2 Power Module

Article no. without filter	6SL3210-1RH22-7UL0	6SL3210-1RH23-5UL0	6SL3210-1RH24-2UL0
Article no. with filter	6SL3210-1RH22-7AL0	6SL3210-1RH23-5AL0	6SL3210-1RH24-2AL0
Weight without filter	17 kg	17 kg	17 kg
Weight with filter	18.5 kg	18.5 kg	18.5 kg

Table 10-34 PM240P-2, IP20, frame sizes E, 3 AC 500 V ... 690 V

Article no. without filter	6SL3210-1RH25-2UL0	6SL3210-1RH26-2UL0
Article no. with filter	6SL3210-1RH25-2AL0	6SL3210-1RH26-2AL0
LO base load power	45 kW	55 kW
LO base load input current	50 A	59 A
LO base load output current	52 A	62 A
HO base load power	37 kW	45 kW
HO base load input current	44 A	54 A
HO base load output current	42 A	52 A
Siemens fuse according to IEC/UL	3NA1820-0 / 80 A	3NE1820-0 / 80 A
Fuse according to IEC/UL, Class J	80 A	80 A
Power loss without filter	1.00 kW	1.21 kW
Power loss with filter	1.00 kW	1.22 kW
Required cooling air flow	83 l/s	83 l/s
Weight without filter	26 kg	26 kg
Weight with filter	28 kg	28 kg

Table 10-35 PM240-2, IP20, frame size F, 3 AC 500 V ... 690 V

Article no. without filter	6SL3210-1RH28-0UL0	6SL3210-1RH31-0UL0	6SL3210-1RH31-2UL0
Article no. with filter	6SL3210-1RH28-0AL0	6SL3210-1RH31-0AL0	6SL3210-1RH31-2AL0
LO base load power	75 kW	90 kW	110 kW
LO base load input current	78 A	97 A	111 A
LO base load output current	80 A	100 A	115 A
HO base load power	55 kW	75 kW	90 kW
HO base load input current	66 A	85 A	106 A
HO base load output current	62 A	80 A	100 A
Siemens fuse according to IEC/UL	3NE1021-0 / 100 A	3NE1022-0 / 125 A	3NE1224-0 / 160 A
Fuse according to IEC/UL, Class J	110 A	150 A	150 A
Power loss without filter	1.34 kW	1.71 kW	2 kW
Power loss with filter	1.35 kW	1.72 kW	2.02 kW
Required cooling air flow	153 l/s	153 l/s	153 l/s
Weight without filter	60 kg	60 kg	60 kg
Weight with filter	64 kg	64 kg	64 kg

10.5 Technical Data, PM240P-2 Power Module

Table 10-36 PM240-2, IP20, frame size F, 3 AC 500 V ... 690 V

Article no. without filter	6SL3210-1RH31-4UL0
Article no. with filter	6SL3210-1RH31-4AL0
LO base load power	132 kW
LO base load input current	137 A
LO base load output current	142 A
HO base load power	110 kW
HO base load input current	122 A
HO base load output current	115 A
Siemens fuse according to IEC/UL	3NE1225-0 / 200 A
Fuse according to IEC/UL, Class J	200 A
Power loss without filter	2.56 kW
Power loss with filter	2.59 kW
Required cooling air flow	153 l/s
Weight without filter	60 kg
Weight with filter	64 kg

10.5.7 Current derating depending on the pulse frequency, 690 V inverters

Article number	LO power [kW]	LO base load output current [A]	
Pulse frequency [kHz]		2	4
6SL3210-1RH21-4 . L0		14	8.4
6SL3210-1RH22-0 . L0		19	11.4
6SL3210-1RH22-3 . L0		23	13.8
6SL3210-1RH22-7 . L0		27	16.2
6SL3210-1RH23-5 . L0		35	21
6SL3210-1RH24-2 . L0		42	25.2
6SL3210-1RH25-2 . L0		52	31.2
6SL3210-1RH26-2 . L0		62	37.2
6SL3210-1RH28-0 . L0		80	48
6SL3210-1RH31-0 . L0		100	60
6SL3210-1RH31-2 . L0		115	69
6SL3210-1RH31-4 . L0		142	85.2

The permissible motor cable length depends on the cable type and the selected pulse frequency.

10.6 Technical data, PM230 Power Module

Typical inverter load cycles

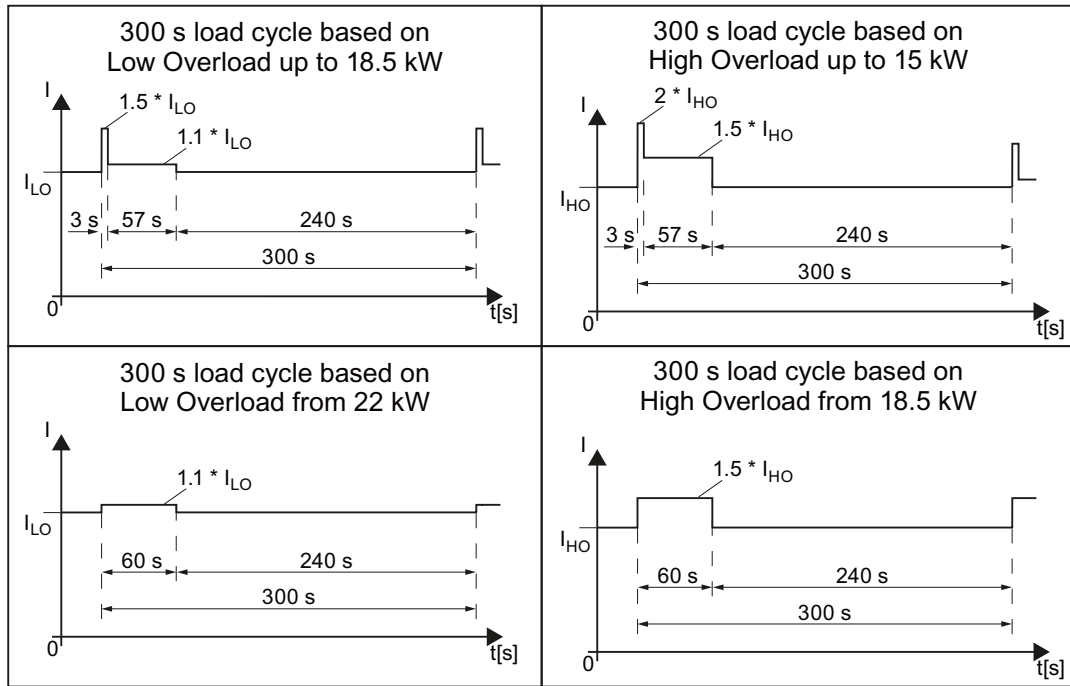


Figure 10-3 Duty cycles, "High Overload" and "Low Overload"

10.6.1 Ambient conditions

Property	Version
Ambient conditions for transport in the transport packaging	
Climatic ambient conditions	- 40 °C ... + 70 °C, according to Class 2K4 to EN 60721-3-2 maximum humidity 95 % at 40 °C
Mechanical ambient conditions	FSA ... FSC: Shock and vibration permissible according to 1M2 to EN 60721-3-2 FSD ... FSF: Shock and vibration permissible according to 2M3 to EN 60721-3-2
Protection against chemical substances	Protected according to Class 2C2 to EN 60721-3-2
Biological environmental conditions	Suitable according to Class 2B1 to EN 60721-3-2
Ambient conditions for long-term storage in the product packaging	
Climatic ambient conditions	- 25 °C ... + 55 °C, according to Class 1K3 to EN 60721-3-1
Protection against chemical substances	Protected according to Class 1C2 to EN 60721-3-1
Biological environmental conditions	Suitable according to Class 1B1 according to EN 60721-3-1

Property	Version
Ambient conditions in operation	
Installation altitude	Up to 1000 m above sea level without derating, > 1000 m  Restrictions for special ambient conditions (Page 474)
Climatic ambient conditions ¹⁾	- Temperature range without derating²⁾ – LO base load power: 0 °C...40 °C – HO base load power: 0 °C...40 °C - For higher temperatures.  Restrictions for special ambient conditions (Page 474) - Relative humidity: 5 ... 95%, condensation not permitted - Oil mist, salt mist, ice formation, condensation, dripping water, spraying water, splashing water and water jets are not permitted
Mechanical ambient conditions	FSA ... FSF: Vibration levels permissible according to Class 3M1 to EN 60721-3-3 FSA ... FSC: Shock, permissible according to Class 3M2 to EN 60721-3-3 FSD ... FSF: Shock permissible according to Class 3M1 to EN 60721-3-3
Protection against chemical substances	protected according to 3C2 acc. to EN 60721-3-3
Biological environmental conditions	suitable according to 3B1 acc. to EN 60721-3-3
Pollution	Suitable for environments with degree of pollution 2 according to EN 61800-5-1
Cooling	Forced air cooling AF, according to EN 60146
Cooling air	clean and dry air

¹⁾ Increased ruggedness regarding temperature range and relative humidity; therefore better than Class 3K3 to EN 60721-3-3

²⁾ Observe the permissible ambient temperatures for the Control Unit and possibly the operator panel (IOP or BOP-2).

10.6.2 General technical data, PM230

Property	Version	
Line voltage	3-phase 380 ... 480 VAC $\pm$ 10%	
Output voltage	3-phase 0 VAC ... input voltage x 0.95 (max.)	
Input frequency	50 Hz ... 60 Hz, $\pm$ 3 Hz	
Output frequency	0 Hz ... 550 Hz, depending on the control mode	
Power factor λ	0.9	
Line impedance	Uk $\leq$ 1%, line reactor not permissible	
Inrush current	< LO base load input current	
Pulse frequency (factory setting)	4 kHz The pulse frequency can be increased in 2 kHz steps up to 16 kHz (up to 8 kHz for 55 kW and 75 kW). An increase in the pulse frequency results in a lower output current.	
Electromagnetic compatibility	Devices with filters in compliance with EN 61800-3: 2004 are suitable for Category C2 environments.	
Braking methods	DC braking	
Degree of protection	IP20 built-in units PT devices	IP20 when mounted in a control cabinet IP54 on the control cabinet wall
Operating temperature at	LO base load power without derating	0° C ... +40° C
	HO base load power without derating	0° C ... +50° C
	LO/HO base load power with derating:	Up to 60° C
	 Restrictions for special ambient conditions (Page 474)	
Storage temperature	-40° C ... +70° C	
Relative humidity	< 95% - condensation not permissible	
Pollution	Protected according to pollution degree 2 according to EN 61800-5-1: 2007	
Environmental requirements	Protected against damaging chemical substances according to environmental class 3C2 according to EN 60721-3-3: 1995	
Shock and vibration	- Long-term storage in the transport packaging according to Class 1M2 according to EN 60721-3-1: 1997 - Transport in the transport packaging according to Class 2M3 according to EN 60721-3-2: 1997 - Vibration during operation according to Class 3M2 according to EN 60721-3-3: 1995	
Installation altitude	without derating: up to 1000 m above sea level with derating: up to 4000 m above sea level	 Restrictions for special ambient conditions (Page 474)
Permissible short-circuit current	Frame size D ... F: 65 kA ¹⁾	
Overvoltage category	Supply circuits: Non-supply circuits:	Overvoltage category III Overvoltage category II
Standards	UL ^{1),2)} , CE, C-tick The drive only satisfies the UL requirements when UL-certified fuses are used.	

¹⁾ If fuse-protected with a listed Class J or 3NE1 fuse, rated voltage 600 VAC with the rated current of the specific inverter.

²⁾ UL available soon for frame sizes D ... F

10.6.3 Detailed technical data, PM230

Table 10-37 PM230, IP20, frame size A, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3210-1NE11-3UG1	6SL3210-1NE11-7UG1	6SL3210-1NE12-2UG1
Article no. - with filter	6SL3210-1NE11-3AG1	6SL3210-1NE11-7AG1	6SL3210-1NE12-2AG1
LO base load power	0.37 kW	0.55 kW	0.75 kW
LO base load input current	1.3 A	1.8 A	2.3 A
LO base load output current	1.3 A	1.7 A	2.2 A
HO base load power	0.25 kW	0.37 kW	0.55 kW
HO base load input current	0.9 A	1.3 A	1.8 A
HO base load output current	0.9 A	1.3 A	1.7 A
Fuse according to IEC / UL	3NE1813-0	3NE1813-0	3NE1813-0
Fuse according to UL, Class J	2 A	4 A	4 A
Circuit breaker 3RV2711-1KD10	12.5 A	12.5 A	12.5 A
Power loss	0.04 kW	0.04 kW	0.05 kW
Required cooling air flow	1.5 l/s	1.5 l/s	4.5 l/s
Weight without filter	1.4 kg	1.4 kg	1.4 kg
Weight with filter	1.6 kg	1.6 kg	1.6 kg

Table 10-38 PM230, IP20, frame size A, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3210-1NE13-1UG1	6SL3210-1NE14-1UG1	6SL3210-1NE15-8UG1
Article no. - with filter	6SL3210-1NE13-1AG1	6SL3210-1NE14-1AG1	6SL3210-1NE15-8AG1
LO base load power	1.1 kW	1.5 kW	2.2 kW
LO base load input current	3.2 A	4.2 A	6.1 A
LO base load output current	3.1 A	4.1 A	5.9 A
HO base load power	0.75 kW	1.1 kW	1.5 kW
HO base load input current	2.3 A	3.2 A	4.2 A
HO base load output current	2.2 A	3.1 A	4.1 A
Fuse according to IEC / UL	3NE1813-0	3NE1813-0	3NE1813-0
Fuse according to UL, Class J	6 A	6 A	10 A
Power loss	0.06 kW	0.07 kW	0.08 kW
Circuit breaker N3RV2711-1KD10	12.5 A	12.5 A	12.5 A
Required cooling air flow	4.5 l/s	4.5 l/s	4.5 l/s
Weight without filter	1.4 kg	1.4 kg	1.4 kg
Weight with filter	1.6 kg	1.6 kg	1.6 kg

Table 10-39 PM230, IP20, frame size A, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3210-1NE17-7UG1
Article no. - with filter	6SL3210-1NE17-7AG1
LO base load power	3 kW
LO base load input current	8.0 A

10.6 Technical data, PM230 Power Module

Article no. - without filter	6SL3210-1NE17-7UG1
Article no. - with filter	6SL3210-1NE17-7AG1
LO base load output current	7.7 A
HO base load power	2.2 kW
HO base load input current	6.1 A
HO base load output current	5.9 A
Fuse according to IEC / UL	3NE1813-0
Fuse according to UL, Class J	10 A
Circuit breaker N3RV2711-1KD10	12.5 A
Power loss	0.11 kW
Required cooling air flow	4.5 l/s
Weight without filter	1.4 kg
Weight with filter	1.6 kg

Table 10-40 PM230, PT, frame size A, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3211-1NE17-7UG1
Article no. - with filter	6SL3211-1NE17-7AG1
LO base load power	3 kW
LO base load input current	8.0 A
LO base load output current	7.7 A
HO base load power	2.2 kW
HO base load input current	6.1 A
HO base load output current	5.9 A
Fuse according to IEC / UL	3NE1813-0
Fuse according to UL, Class J	10 A
Power loss	0.11 kW
Required cooling air flow	4.5 l/s
Weight without filter	1.7 kg
Weight with filter	1.9 kg

Table 10-41 PM230, IP20, frame size B, 3-phase 380 ... 480 VAC

Article no. - without filter	6SL3210-1NE21-0UG1	6SL3210-1NE21-3UG1	6SL3210-1NE21-8UG1
Article no. - with filter	6SL3210-1NE21-0AG1	6SL3210-1NE21-3AG1	6SL3210-1NE21-8AG1
LO base load power	4 kW	5.5 kW	7.5 kW
LO base load input current	10.5 A	13.6 A	18.6 A
LO base load output current	10.2 A	13.2 A	18 A
HO base load power	3 kW	4 kW	5.5 kW
HO base load input current	8.0 A	10.5 A	13.6 A
HO base load output current	7.7 A	10.2 A	13.2 A
Fuse according to IEC / UL	3NE1813-0	3NE1814-0	3NE1815-0
Fuse according to UL, Class J	15 A	20 A	25 A

Article no. - without filter	6SL3210-1NE21-0UG1	6SL3210-1NE21-3UG1	6SL3210-1NE21-8UG1
Article no. - with filter	6SL3210-1NE21-0AG1	6SL3210-1NE21-3AG1	6SL3210-1NE21-8AG1
Power loss	0.12 kW	0.15 kW	0.22 kW
Required cooling air flow	9.2 l/s	9.2 l/s	9.2 l/s
Weight without filter	2.8 kg	2.8 kg	2.8 kg
Weight with filter	3 kg	3 kg	3 kg

Table 10-42 PM230, PT, frame size B, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3211-1NE21-8UG1
Article no. - with filter	6SL3211-1NE21-8AG1
LO base load power	7.5 kW
LO base load input current	18.6 A
LO base load output current	18 A
HO base load power	5.5 kW
HO base load input current	13.6 A
HO base load output current	13.2 A
Fuse according to IEC / UL	3NE1815-0
Fuse according to UL, Class J	25 A
Power loss	0.22 kW
Required cooling air flow	9.2 l/s
Weight without filter	3.4 kg
Weight with filter	3.6 kg

Table 10-43 PM230, IP20, frame size C, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3210-1NE22-6UG1	6SL3210-1NE23-2UG1	6SL3210-1NE23-8UG1
Article no. - with filter	6SL3210-1NE22-6AG1	6SL3210-1NE23-2AG1	6SL3210-1NE23-8AG1
LO base load power	11 kW	15 kW	18.5 kW
LO base load input current	26.9 A	33.1 A	39.2 A
LO base load output current	26 A	32 A	38 A
HO base load power	7.5 kW	11 kW	15 kW
HO base load input current	18.6 A	26.9 A	33.1 A
HO base load output current	18 A	26 A	32 A
Fuse according to IEC / UL	3NE1803-0	3NE1817-0	3NE1817-0
Fuse according to UL, Class J	35 A	45 A	50 A
Power loss	0.3 kW	0.35 kW	0.45 kW
Required cooling air flow	18.5 l/s	18.5 l/s	18.5 l/s
Weight without filter	4.5 kg	4.5 kg	4.5 kg
Weight with filter	5.1 kg	5.1 kg	5.1 kg

10.6 Technical data, PM230 Power Module

Table 10-44 PM230, PT, frame size C, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3211-1NE23-8UG1
Article no. - with filter	6SL3211-1NE23-8AG1
LO base load power	18.5 kW
LO base load input current	39.2 A
LO base load output current	38 A
HO base load power	15 kW
HO base load input current	33.1 A
HO base load output current	32 A
Fuse according to IEC / UL	3NE1817-0
Fuse according to UL, Class J	50 A
Power loss	0.45 kW
Required cooling air flow	18.5 l/s
Weight without filter	5.4 kg
Weight with filter	6 kg

Table 10-45 PM230, IP20, frame size D, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3210-1NE24-5UL0	6SL3210-1NE26-0UL0
Article no. - with filter	6SL3210-1NE24-5AL0	6SL3210-1NE26-0AL0
LO base load power	22 kW	30 kW
LO base load input current	42 A	56 A
LO base load output current	45 A	60 A
HO base load power	18.5 kW	22 kW
HO base load input current	36 A	42 A
HO base load output current	38 A	45 A
Fuse according to IEC / UL	3NE1818-0	3NE1820-0
Power loss	0.52 kW	0.68 kW
Required cooling air flow	80 l/s	80 l/s
Weight without filter	11 kg	11 kg
Weight with filter	14 kg	14 kg

Table 10-46 PM230, IP20, frame size E, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3210-1NE27-5UL0	6SL3210-1NE28-8UL0
Article no. - with filter	6SL3210-1NE27-5AL0	6SL3210-1NE28-8AL0
LO base load power	37 kW	45 kW
LO base load input current	70 A	84 A
LO base load output current	75 A	90 A
HO base load power	30 kW	37 kW
HO base load input current	56 A	70 A
HO base load output current	60 A	75 A
Fuse according to IEC / UL	3NE1021-0	3NE1022-0

Article no. - without filter	6SL3210-1NE27-5UL0	6SL3210-1NE28-8UL0
Article no. - with filter	6SL3210-1NE27-5AL0	6SL3210-1NE28-8AL0
Power loss	0.99 kW	1.2 kW
Required cooling air flow	80 l/s	80 l/s
Weight without filter	15 kg	15 kg
Weight with filter	22 kg	22 kg

Table 10-47 PM230, IP20, frame size F, 3 AC 380 V ... 480 V

Article no. - without filter	6SL3210-1NE31-1UL0	6SL3210-1NE31-5UL0
Article no. - with filter	6SL3210-1NE31-1AL0	6SL3210-1NE31-5AL0
LO base load power	55 kW	75 kW
LO base load input current	102 A	135 A
LO base load output current	110 A	145 A
HO base load power	45 kW	55 kW
HO base load input current	84 A	102 A
HO base load output current	90 A	110 A
Fuse according to IEC / UL	3NE1224-0	3NE1225-0
Power loss	1.4 kW	1.9 kW
Required cooling air flow	150 l/s	150 l/s
Weight without filter	33 kg	33 kg
Weight with filter	48 kg	48 kg

10.6.4 Current reduction depending on pulse frequency

Current derating depending on the pulse frequency

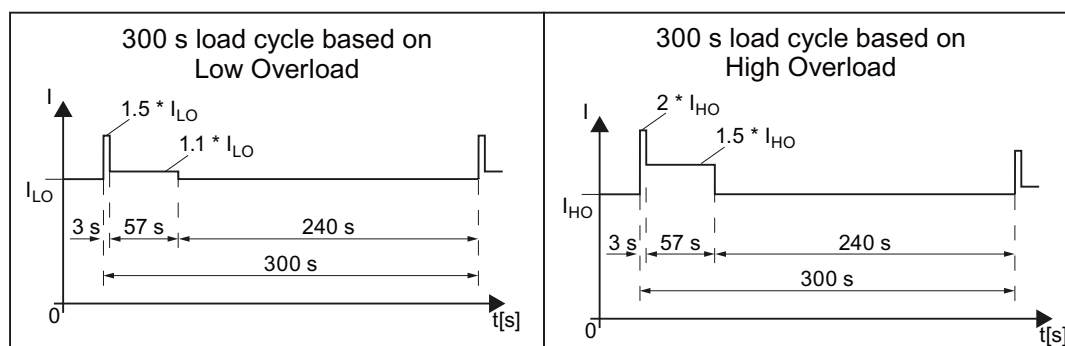
LO base load	Output base-load current at a pulse frequency of							
	2 kHz	4 kHz	6 kHz	8 kHz	10 kHz	12 kHz	14 kHz	16 kHz
kW	A	A	A	A	A	A	A	A
0.37	--	1.30	1.11	0.91	0.78	0.65	0.59	0.52
0.55	--	1.70	1.45	1.19	1.02	0.85	0.77	0.68
0.75	--	2.20	1.87	1.54	1.32	1.10	0.99	0.88
1.1	--	3.10	2.64	2.17	1.86	1.55	1.40	1.24
1.5	--	4.10	3.49	2.87	2.46	2.05	1.85	1.64
2.2	--	5.90	5.02	4.13	3.54	2.95	2.66	2.36
3.0	--	7.70	6.55	5.39	4.62	3.85	3.47	3.08
4.0	--	10.20	8.67	7.14	6.12	5.10	4.59	4.08
5.5	--	13.20	11.22	9.24	7.92	6.60	5.94	5.28
7.5	--	18.00	15.30	12.60	10.80	9.00	8.10	7.20
11.0	--	26.00	22.10	18.20	15.60	13.00	11.70	10.40
15.0	--	32.00	27.20	22.40	19.20	16.00	14.40	12.80
18.5	--	38.00	32.30	26.60	22.80	19.00	17.10	15.20
22	--	45.00	38.25	31.50	27.00	22.50	20.25	18.00
30	--	60.00	51.00	42.00	36.00	30.00	27.00	24.00
37	--	75.00	63.75	52.50	45.00	37.50	33.75	30.00
45	--	90.00	76.50	63.00	54.00	45.00	40.50	36.00
55	--	110.0	93.50	77.00	66.00	55.00	49.50	44.00
75	--	145.0	123.3	101.5	--	--	--	--

The permissible cable length to the motor also depends on the cable type and the selected pulse frequency.

10.7 Technical data, PM250 Power Module

10.7.1 High Overload - Low Overload

Typical inverter load cycles



10.7.2 Ambient conditions

Ambient conditions during operation

Property	Version
Ambient conditions for transport in the transport packaging	
Climatic ambient conditions	- 40° C ... + 70° C, according to Class 2K4 to EN 60721-3-2 maximum humidity 95% at 40° C
Mechanical ambient conditions	FSC: Shock and vibration permissible according to 1M2 to EN 60721-3-2 FSD ... FSF: Shock and vibration permissible according to 2M3 to EN 60721-3-2
Protection against chemical substances	Protected according to Class 2C2 to EN 60721-3-2
Biological ambient conditions	Suitable according to Class 2B1 to EN 60721-3-2
Ambient conditions for long-term storage in the product packaging	
Climatic ambient conditions	- 25° C ... + 55° C, according to Class 1K3 to EN 60721-3-1
Protection against chemical substances	Protected according to Class 1C2 to EN 60721-3-1
Biological ambient conditions	Suitable according to class 1B1 to EN 60721-3-1
Ambient conditions in operation	
Installation altitude	Up to 1000 m above sea level without limitations
	 Restrictions for special ambient conditions (Page 474)

Property	Version
Climatic ambient conditions ¹⁾	- Ambient operating temperature ²⁾ - For operation according to Low Overload: 0° C ... +40° C - For operation according to High Overload: 0° C ... +50° C  Restrictions for special ambient conditions (Page 474) - Relative humidity: 5 ... 95%, condensation not permitted - Oil mist, salt mist, ice formation, condensation, dripping water, spraying water, splashing water and water jets are not permitted
Mechanical ambient conditions	- FSC ... FSF: Vibration levels permissible according to Class 3M1 to EN 60721-3-3 - FSC: Shock, permissible according to Class 3M2 to EN 60721-3-3 - FSD ... FSF: Shock permissible according to Class 3M1 to EN 60721-3-3
Protection against chemical substances	Protected according to 3C2 to EN 60721-3-3
Biological ambient conditions	Suitable according to 3C2 to EN 60721-3-3
Pollution	Suitable for environments with degree of pollution 2 according to EN 61800-5-1, condensation not permitted
Cooling	Forced air cooling AF, according to EN 60146
Cooling air	Clean and dry air

¹⁾ Increased ruggedness regarding temperature range and relative humidity; therefore better than 3K3 according to EN 60721-3-3

²⁾ Observe the permissible ambient temperatures for the Control Unit and possibly the Operator Panel (IOP or BOP-2).

Approvals

- UL, cUL, CE, c-tick, SEMI F47.

Use UL-certified fuses in order to comply with UL requirements.

10.7.3 General technical data, PM250

Property	Version
Line voltage	3-phase 380 ... 480 VAC $\pm$ 10%
Output voltage	3-phase 0 VAC ... input voltage x 0.87 (max.)
Input frequency	50 Hz ... 60 Hz, $\pm$ 3 Hz
Output frequency	0 ... 550 Hz, depending on the control mode
Power factor λ	0.9
Inrush current	< LO base load input current
Pulse frequency (factory setting)	4 kHz The pulse frequency can be adjusted up to 16 kHz in 2 kHz steps. The higher the pulse frequency, the lower the available output current.  Current reduction depending upon pulse frequency (Page 472)
Electromagnetic compatibility	The devices comply with EN 61800-3: 2004 suitable for Category C2 and C3 environments.
Braking methods	DC braking, energy recovery (up to 100% of the output power)
Degree of protection	IP20 built-in devices (they must be installed in a switch cabinet)

10.7.4 Specific technical data, PM250

Note

The values for Low Overload (LO) are identical with those of the rated values.

Table 10-48 PM250, IP20, Frame Size C, 3-ph. AC 380 V ... 480 V

Article no.	6SL3225-0BE25-5AA1	6SL3225-0BE27-5AA1	6SL3225-0BE31-1AA1
LO base load output	7.5 kW	11 kW	15 kW
LO base load input current	18 A	25 A	32 A
LO base load output current	18 A	25 A	32 A
HO base load output	5.5 kW	7.5 kW	11 kW
HO base load input current	13.2 A	19 A	26 A
HO base load output current	13.2 A	19 A	26 A
Fuse	20 A, Class J	32 A, Class J	35 A, Class J
Power loss	0.24 kW	0.30 kW	0.31 kW
Required cooling air flow	38 l/s	38 l/s	38 l/s
Weight	7.5 kg	7.5 kg	7.5 kg

Table 10-49 PM250, IP20, Frame Size D, 3-ph. AC 380 V ... 480 V

Article no.	6SL3225-0BE31-5AA0	6SL3225-0BE31-8AA0	6SL3225-0BE32-2AA0
LO base load output	18.5 kW	22 kW	30 kW
LO base load input current	36 A	42 A	56 A
LO base load output current	38 A	45 A	60 A
HO base load output	15 kW	18.5 kW	22 kW
HO base load input current	30 A	36 A	42 A
HO base load output current	32 A	38 A	45 A
Fuse according to IEC	3NA3820	3NA3822	3NA3824
Fuse according to UL	50 A, Class J 3NE1817-0	63 A, Class J 3NE1818-0	80 A, Class J 3NE1820-0
Power loss	0.44 kW	0.55 kW	0.72 kW
Required cooling air flow	22 l/s	22 l/s	39 l/s
Weight	15 kg	15 kg	16 kg

Table 10-50 PM250, IP20, Frame Size E, 3-ph. AC 380 V ... 480 V

Article no.	6SL3225-0BE33-0AA0	6SL3225-0BE33-7AA0
LO base load output	37 kW	45 kW
LO base load input current	70 A	84 A
LO base load output current	75 A	90 A

Article no.	6SL3225-0BE33-0AA0	6SL3225-0BE33-7AA0
HO base load output	30 kW	37 kW
HO base load input current	56 A	70 A
HO base load output current	60 A	75 A
Fuse according to IEC	3NA3830	3NA3832
Fuse according to UL	100 A, Class J 3NE1821-0	125 A, Class J 3NE1822-0
Power loss	1.04 kW	1.2 kW
Required cooling air flow	22 l/s	39 l/s
Weight	21 kg	21 kg

Table 10-51 PM250, IP20, Frame size F, 3-ph. AC 380 V ... 480 V

Article no.	6SL3225-0BE34-5AA0	6SL3225-0BE35-5AA0	6SL3225-0BE37-5AA0
LO base load output	55 kW	75 kW	90 kW
LO base load input current	102 A	135 A	166 A
LO base load output current	110 A	145 A	178 A
HO base load output	45 kW	55 kW	75 kW
HO base load input current	84 A	102 A	135 A
HO base load output current	90 A	110 A	145 A
Fuse according to IEC	3NA3836	3NA3140	3NA3144
Fuse according to UL	160 A, Class J 3NE1824-0	200 A, Class J 3NE1825-0	250 A, Class J 3NE1827-0
Power loss	1.5 kW	2.0 kW	2.4 kW
Required cooling air flow	94 l/s	94 l/s	117 l/s
Weight	51 kg	51 kg	51 kg

10.7.5 Current reduction depending upon pulse frequency

Relationship between pulse frequency and current reduction

Table 10-52 Current reduction depending on pulse frequency

Rated Power (LO)	Base load current (LO)	Base load current (LO) at pulse frequency of					
		4 kHz	6 kHz	8 kHz	10 kHz	12 kHz	14 kHz
kW	A	A	A	A	A	A	A
0,55	1,7						
0,75	2,2						
1,1	3,1						
1,5	4,1						
2,2	5,9						
3	7,7						
4	10.2						
5,5	13.2						
7.5	18.0	12.5	11.9	10.6	9.20	7.90	6.60
11	25.0	18.1	17.1	15.2	13.3	11.4	9.50
15	32.0	24.7	23.4	20.8	18.2	15.6	12.8
18.5	38.0	32.3	26.6	22.8	19.0	17.1	15.2
22	45.0	38.3	31.5	27.0	22.5	20.3	18.0
30	60.0	51.0	42.0	36.0	30.0	27.0	24.0
37	75.0	63.8	52.5	45.0	37.5	33.8	30.0
45	90.0	76.5	63.0	54.0	45.0	40.5	36.0
55	110	93.5	77.0	--	--	--	--
75	145	123	102	--	--	--	--
90	178	151	125	--	--	--	

10.8 Data regarding the power loss in partial load operation



You can find data regarding power loss in partial load operation in the Internet:

Partial load operation (<http://support.automation.siemens.com/WW/view/en/94059311>)

10.9 Restrictions for special ambient conditions

Permissible line supplies dependent on the installation altitude

- For installation altitudes ≤ 2000 m above sea level, it is permissible to connect the inverter to any of the line supplies that are specified for it.
- For installation altitudes 2000 m ... 4000 m above sea level, the following applies:
 - Connection to a TN line system with grounded neutral point is permissible.
 - TN systems with grounded line conductor are not permitted.
 - The TN line system with grounded neutral point can also be supplied using an isolation transformer.
 - The phase-to-phase voltage does not have to be reduced.

Note

Using Power Modules connected to TN line supplies with voltages ≥ 600 V for installation altitudes 2000 m ... 4000 m

For voltages ≥ 600 V, the TN line supply must have a grounded neutral point established using an isolating transformer.

10.9.1 PM240-2 and PM240P-2 Power Modules: Current reduction as a function of the installation altitude and ambient temperature

Current reduction as a function of the installation altitude and ambient temperature

At installation altitudes above 1000 m and temperatures higher than 40° C (low overload) or 50° C (high overload), the inverter output current is reduced. The following tables contain the details.

Table 10-53 Maximum permitted output current for loading according to LO

Installation altitude [m] up to	Ambient temperature [°C]								
	20	25	30	35	40	45	50	55	60
	Output current in [%] when loading to LO								
1000	100					93	85	76	66
1500	100				95	88	81	72	63
2000	100			97	90	83	77	68	59
2500	100		98	91	85	79	72	64	56
3000	100	98	92	86	80	74	68	60	53
3500	98	92	86	81	75	69	64	57	50
4000	91	86	81	75	70	65	60	53	46

Table 10-54 Maximum permitted output current when loading according to HO

Installation altitude [m] up to	Ambient temperature [°C]								
	20	25	30	35	40	45	50	55	60
	Output current in [%] when loading to HO								
1000	100							87	74
1500	100						95	83	70
2000	100				99	95	90	78	67
2500	100			98	94	89	85	74	63
3000	100		96	92	88	84	80	70	59
3500	98	94	90	86	83	79	75	65	56
4000	91	88	84	81	77	74	70	61	52

Also observe the maximum permissible ambient operating temperatures for the Control Unit and possibly the Operator Panel.

10.9.2 PM230 and PM250 Power Modules: Current derating depending on the ambient air temperature

Current derating depending on the ambient air temperature

The Control Unit and Operator Panel can restrict the maximum permissible operating ambient temperature of the Power Module.

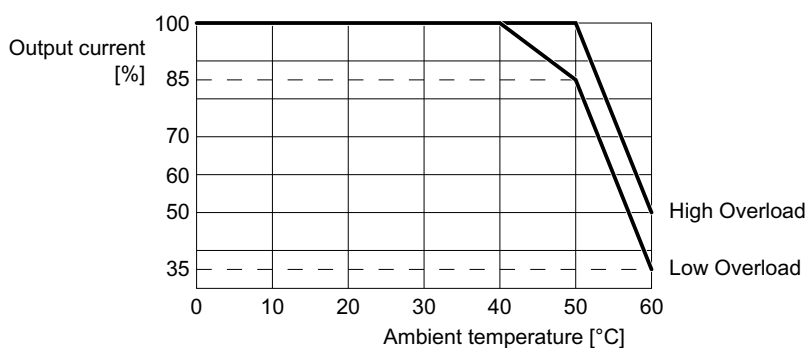


Figure 10-4 Characteristic for the PM230 Power Module

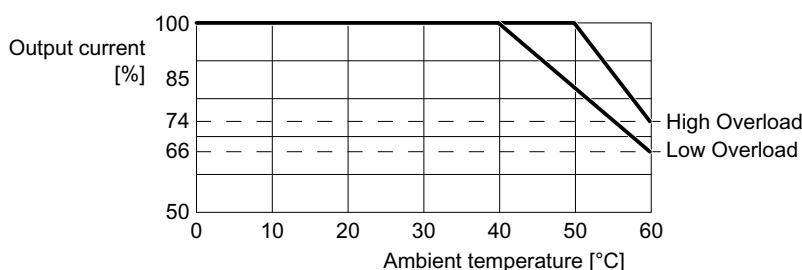


Figure 10-5 Characteristic for the PM250 Power Module

Current derating depending on the installation altitude

The permissible inverter output current is reduced above an installation altitude of 1000 m.

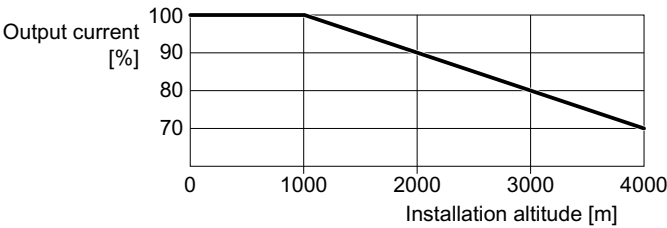


Figure 10-6 Characteristic for the PM250 Power Module

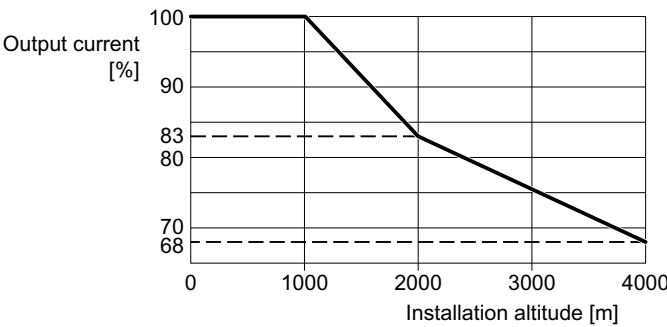


Figure 10-7 Characteristic for the PM230 Power Module

Appendix

A.1 New and extended functions

A.1.1 Firmware version 4.7 SP6

Table A-1 New functions and function changes in firmware 4.7 SP6

	Function	SINAMICS								
			G120					G120D		
		G110M	G120C	CU230P-2	CU240B-2	CU240E-2	CU250S-2	CU240D-2	CU250D-2	ET 200pro FC-2
1	Support for the Power Module PM240-2, FSF frame sizes	-	-	✓	✓	✓	✓	-	-	-
	Support of PM240P-2 Power Modules frame sizes FSD ... FSF	-	-	✓	✓	✓	-	-	-	-
	Support of safety function Safe Torque Off (STO)via the terminals of the PM240-2 Power Module, frame size FSF and PM240P-2 Power Module FSD ... FSF	-	-	-	-	✓	✓	-	-	-
	You can find additional information in the "Safety Integrated" function manual.  Overview of the manuals (Page 504)									
2	Support for Power Module PM330 JX frame size	-	-	✓	-	-	-	-	-	-
3	Support for 1PC1 induction motors	✓	✓	✓	✓	✓	✓	✓	✓	✓
4	The control of synchronous reluctance takes into account the inductance of the output reactor.	-	-	✓	-	-	-	-	-	-
5	Support of motor temperature sensor Pt1000	✓	✓	✓	✓	✓	✓	✓	✓	✓
6	New p4621 parameter for disabling PTC short-circuit monitoring	-	-	-	-	-	-	✓	✓	✓
7	Revision of the thermal motor model for protecting the motor against damage due to overheating in the stator or rotor	✓	✓	✓	✓	✓	✓	✓	✓	✓
8	Changing the quick commissioning in the "Standard Drive Control" application class: The motor data identification is no longer permanently set to p1900 = 12; instead, users select the appropriate motor data identification. Factory setting: p1900 = 2.	-	✓	✓	✓	✓	✓	-	-	-
9	The free function blocks are also available in the SINAMICS G120C.	✓	✓	✓	✓	✓	✓	✓	-	-



Changes in the current manual (Page 5)

A.1.2 Firmware version 4.7 SP3

Table A-2 New functions and function changes in firmware 4.7 SP3

	Function	SINAMICS							
		G120						G120D	ET 200pro FC-2
		G110M	G120C	CU230P-2	CU240B-2	CU240E-2	CU250S-2	CU240D-2	CU250D-2
1	PM240-2 Power Modules, frame sizes FSD and FSE are supported	-	-	✓	✓	✓	✓	-	-
	The Safety Integrated basic function Safe Torque Off (STO) is supported via the terminals of the PM240-2 Power Module, frame sizes FSD and FSE	-	-	-	-	✓	✓	-	-
2	Revised PM230 Power Module with new article numbers supported: - IP55 degree of protection: 6SL3223-0DE G . - IP20 degree of protection and Push Through: 6SL321 . -1NE G . You can find additional information in the "Safety Integrated" function manual.  Overview of the manuals (Page 504)	-	-	✓	✓	✓	-	-	-
	The Safety Integrated basic function Safe Torque Off (STO) is supported with the revised PM230 Power Module	-	-	-	-	✓	-	-	-
3	PM330 Power Module, frame size HX is supported	-	-	✓	-	-	-	-	-
4	1FP1 reluctance motors are supported	-	-	✓	-	-	-	-	-
5	Encoderless 1FK7 synchronous motors are supported ¹⁾ Operation with 1FK7 synchronous motors without encoder has already been released with firmware V4.7 for SINAMICS G120D with CU240D-2 Control Units.	-	✓	-	✓	✓	✓	✓ ¹⁾	-
6	Encoderless 1FG1 geared synchronous motors are supported	-	✓	-	✓	✓	✓	✓	-
7	Selection list for 1PH8 induction motors in the STARTER and Startdrive commissioning wizard	-	✓	✓	✓	✓	✓	-	-
8	Updated selection list for 1LE1 induction motors in the STARTER and Startdrive commissioning wizard	✓	✓	✓	✓	✓	✓	✓	✓
9	Motor support expanded with 1LE1, 1LG6, 1LA7 and 1LA9 induction motors	✓	-	-	-	-	-	-	-
10	Speed and position control obtain their respective actual value from an SSI encoder with incremental tracks. The output signals of the encoder are available as encoder 2 for position control and timer 1 for speed control.	-	-	-	-	-	✓	-	✓
11	Power Module with temperature-controlled fan	✓	-	-	-	-	-	-	-
12	SINAMICS "Standard Drive Control" and "Dynamic Drive Control" application classes to simplify commissioning and increase the degree of ruggedness of the closed-loop motor control. The SINAMICS application classes are available with the following inverters: - SINAMICS G120C - SINAMICS G120 with PM240, PM240-2 and PM330 Power Modules	-	✓	✓	✓	✓	✓	-	-
13	Moment of inertia estimator with moment of inertia precontrol to optimize the speed controller in operation	✓	✓	-	✓	✓	✓	✓	✓

	Function	SINAMICS								
				G120			G120D			
14	Friction torque characteristic with automatic plotting to optimize the speed controller	✓	✓	-	✓	✓	✓	✓	✓	✓
15	Automatic optimization of the technology controller	-	-	✓	✓	✓	-	-	-	-
16	The sign of the system deviation for the additional, free technology controller can be switched over. A new parameter defines the sign of the system deviation matching the particular application, e.g. for cooling or heating applications.	-	-	✓	-	-	-	-	-	-
17	The technology controller output can be enabled and disabled during operation	-	✓	✓	✓	✓	✓	-	-	-
18	Ramp-function generator remains active with enabled technology controller	-	-	✓	-	-	-	-	-	-
19	Line contactor control using a digital output of the inverter to save energy when the motor is switched off	✓	✓	✓	✓	✓	✓	✓	✓	-
20	Fast flying restart for PM330 Power Modules: The "Flying restart" function does not have to wait for the motor demagnetization time, and identifies the motor speed without requiring a search operation.	-	-	✓	-	-	-	-	-	-
21	Load torque monitoring extended to include the following functions: • Protection against blocking, leakage and dry running operation in pump applications • Protection against blocking and broken belts in fan applications	✓	-	✓	✓	✓	-	-	-	-
22	Automatic switchover of the real time clock from daylight saving time (summer time) to standard time (winter time).	-	-	✓	-	-	-	-	-	-
23	New or revised default settings of the interfaces: p0015 macros 110, 112 and 120	-	-	✓	-	-	-	-	-	-
24	Expansion of the temperature sensors to include DIN-Ni1000 for analog inputs AI 2 and AI 3	-	-	✓	-	-	-	-	-	-
25	Communication via AS-Interface. Default setting of the communication via AS-i: p0015 macros 30, 31, 32 and 34	✓	-	-	-	-	-	-	-	-
26	Communication expansion via Modbus: Adjustable parity bit, access to parameters and analog inputs	✓	✓	✓	✓	✓	✓	-	-	-
27	Extending communication via BACnet: Access to parameters and analog inputs	-	-	✓	-	-	-	-	-	-
28	The bus error LED for communication via USS and Modbus can be switched off	✓	✓	✓	✓	✓	✓	-	-	-
29	Default of the minimum speed to 20% of the rated motor speed	-	-	✓	-	-	-	-	-	-
30	For commissioning with an operator panel, the inverter automatically backs up the measured data retentively in the ROM after identification of the motor data.	✓	✓	✓	✓	✓	✓	✓	✓	✓
31	The result of the energy savings calculation for flow machines is available as a connector	✓	✓	✓	✓	✓	✓	✓	✓	✓
32	New "ppm" unit (parts per million) for unit switching	✓	✓	✓	✓	✓	✓	✓	✓	✓
33	Displaying speeds during commissioning via operator panel in units of Hz instead of rpm. Conversion from Hz to rpm via p8552	-	-	✓	-	-	-	-	-	-
34	Voltage-dependent current limit for 600V devices of Power Module PM330 and PM240-2	-	-	✓	✓	✓	✓	-	-	-

A.1.3 Firmware version 4.7

Table A-3 New functions and function changes in Firmware 4.7

	Function	SINAMICS							
				G120			G120D		
		G110M	G120C	CU230P-2	CU240B-2	CU240E-2	CU250S-2	CU240D-2	CU250D-2
1	Supporting the identification & maintenance datasets (I&M1 ... 4)	✓	✓	✓	✓	✓	✓	✓	✓
2	Fall in pulse rate with increased drive power required by the motor The inverter temporarily lowers the pulse frequency if required when the motor is started up, and simultaneously increases the current limit.	✓	✓	✓	✓	✓	✓	✓	✓
3	S7 communication - Direct data exchange between the inverter and human-machine interface (HMI). - Increase in communication performance with the engineering tools and support of the S7 routing	✓	✓	✓	✓	✓	✓	✓	✓
4	The basic functions of Safety Integrated are unrestrictedly available in all control types with 1FK7 encoderless permanent-field synchronous motors	-	-	-	-	-	-	✓	-
5	Direct selection of the 1FK7 encoderless permanent-field synchronous motors using the Article No. with allocated code number It is not necessary to input individual motor data	-	-	-	-	-	-	✓	-
6	Pulse input as source of setpoint value The inverter calculates its speed setpoint from a sequence of pulses at the digital input.	-	-	-	-	-	✓	-	-
7	Dynamic IP address assignment (DHCP) and temporary device names for PROFINET	✓	✓	✓	-	✓	✓	✓	✓
8	PROFInergy Slave profile 2 and 3	✓	✓	✓	-	✓	✓	✓	✓
9	Uniform behavior for component replacement After a component is replaced, an inverter with activated Safety Integrated will report what type of component has been replaced using a unique code.	✓	✓	-	-	✓	✓	✓	✓
10	Improved direct-component control in PM230 Optimized efficiency for pump and fan applications	-	-	✓	-	-	-	-	-
11	Rounding down of BACnet and macros	-	-	✓	-	-	-	-	-

A.1.4 Firmware version 4.6 SP6

Table A-4 New functions and function changes in firmware 4.6 SP6

	Function	SINAMICS						
			G120				G120D	
		G120C	CU230P-2	CU240B-2	CU240E-2	CU250S-2	CU240D-2	CU250D-2
1	Support for the new Power Modules • PM330 IP20 GX	-	✓	-	-	-	-	-

A.1.5 Firmware version 4.6

Table A-5 New functions and function changes in Firmware 4.6

	Function	SINAMICS					
		G120				G120D	
		G120C	CU230P-2	CU240B-2	CU240E-2	CU250S-2	CU240D-2 CU250D-2
1	Support for the new Power Modules - PM240-2 IP20 FSB ... FSC - PM240-2 in through-hole technology FSB ... FSC	-	✓	✓	✓	✓	-
2	Support for the new Power Modules PM230 in through-hole technology FSD ... FSF	-	✓	✓	✓	-	-
3	Motor data preassignment for the 1LA/1LE motors via code number During quick commissioning with the operator panel, set the motor data using a code number	✓	✓	✓	✓	✓	✓
4	Extension to communication via CANopen CAN velocity, ProfilTorque, SDO channel for each axis, system test with CodeSys, suppression of ErrorPassiv alarm	✓	✓	-	-	✓	-
5	Extension to communication via BACnet Multistate value objects for alarms, commandable AO objects, objects for configuring the PID controller	-	✓	-	-	-	-
6	Communication via EtherNet/IP	✓	✓	-	✓	✓	✓
7	Skip frequency band for analog input A symmetrical skip frequency band can be set for each analog input around the 0 V range.	✓	✓	✓	✓	✓	-
8	Changing the control of the motor holding brake	✓	-	✓	✓	✓	-
9	Safety function SBC (Safe Brake Control) Secure control of a motor holding brake when using the "Safe Brake Module" option	-	-	-	-	✓	-
10	Safety function SS1 (Safe Stop 1) without speed monitoring	-	-	-	-	✓	-
11	Straightforward selection of standard motors Selection of 1LA... and 1LE... motors with an operator panel using a list containing code numbers	✓	✓	✓	✓	✓	✓
12	Firmware update via memory card	✓	✓	✓	✓	✓	✓
13	Safety info channel BICO source r9734.0...14 for the status bits of the extended safety functions	-	-	-	✓	✓	✓
14	Diagnostic alarms for PROFIBUS	✓	✓	✓	✓	✓	✓

A.1.6 Firmware version 4.5

Table A-6 New functions and function changes in Firmware 4.5

	Function	SINAMICS					
		G120				G120D	
		G120C	CU230P-2	CU240B-2	CU240E-2	CU240D-2	CU250D-2
1	Support for the new Power Modules: • PM230 IP20 FSA ... FSF • PM230 in a push-through FSA ... FSC	-	✓	✓	✓	-	-
2	Support for the new Power Modules: • PM240-2 IP20 FSA • PM240-2 in push-through FSA	-	✓	✓	✓	-	-
3	New Control Units with PROFINET support	✓	✓	-	✓	✓	✓
4	Support of the PROFlenergy profile	✓	✓	-	✓	✓	✓
5	Shared device support via PROFINET	✓	✓	-	✓	✓	✓
6	Write protection	✓	✓	✓	✓	✓	✓
7	Know-how protection	✓	✓	✓	✓	✓	✓
8	Adding a second command data set (CDS0 → CDS0 ... CDS1) (All other inverters have four command data sets)	✓	-	-	-	-	-
9	Position control and basic positioner	-	-	-	-	-	✓
10	Support of an HTL encoder	-	-	-	-	✓	✓
11	Support of an SSI encoder	-	-	-	-	-	✓
12	Fail-safe digital output	-	-	-	-	✓	✓

A.2 Handling the BOP 2 operator panel

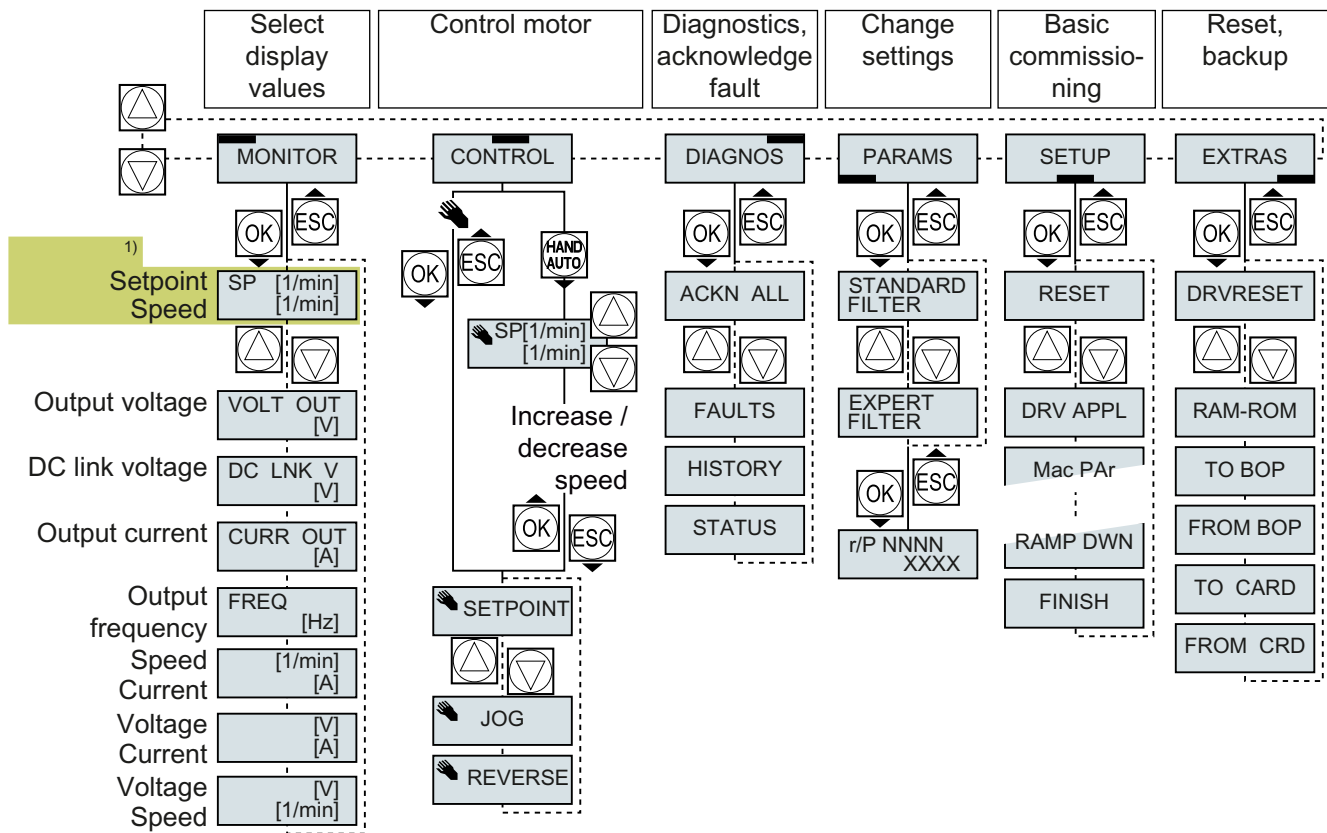


Figure A-1 Menu of the BOP-2

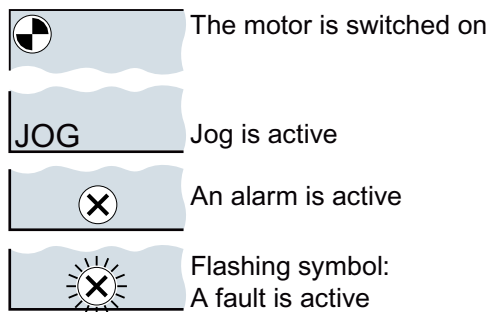


Figure A-2 Other keys and symbols of the BOP-2

Procedure for switching the motor on and off via the operator panel:

1. Press MANUAL AUTO
2. Master control of the inverter is released via the BOP-2
3. Switch on motor
4. Switch off the motor

A.2.1 Changing settings using BOP-2

Changing settings using BOP-2

You can modify the settings of your inverter by changing the values of the its parameters. The inverter only permits changes to "write" parameters. Write parameters begin with a "P", e.g. P45.

The value of a read-only parameter cannot be changed. Read-only parameters begin with an "r", for example: r2.

Procedure



1. To change write parameters using the BOP-2, proceed as follows:

1. Select the menu to display and change parameters.
Press the OK key.

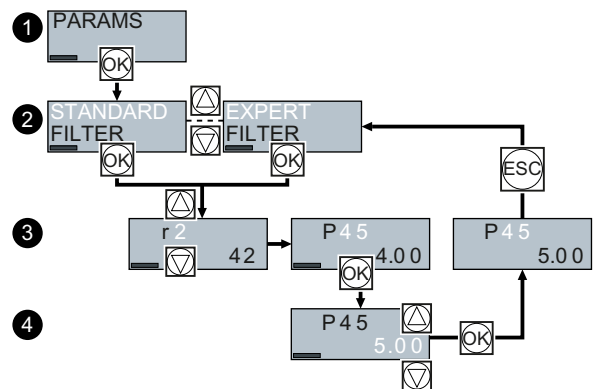
2. Select the parameter filter using the arrow keys.
Press the OK key.

– STANDARD: The inverter only displays the most important parameters.

– EXPERT: The inverter displays all of the parameters.

3. Select the required number of a write parameter using the arrow keys.
Press the OK key.

4. Select the value of the write parameter using the arrow keys.
Accept the value with the OK key.



You have now changed a write parameter using the BOP-2.

The inverter saves all the changes made using the BOP-2 so that they are protected against power failure.

A.2.2 Changing indexed parameters

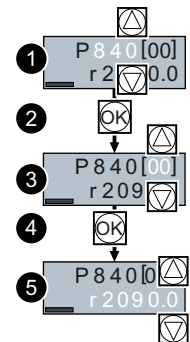
Changing indexed parameters

For indexed parameters, several parameter values are assigned to a parameter number. Each of the parameter values has its own index.

Procedure

- ➔ 1. To change an indexed parameter, proceed as follows:

1. Select the parameter number.
2. Press the OK key.
3. Set the parameter index.
4. Press the OK key.
5. Set the parameter value for the selected index.



■ You have now changed an indexed parameter.

A.2.3 Directly entering the parameter number and value

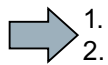
Directly select the parameter number

The BOP-2 offers the possibility of setting the parameter number digit by digit.

Precondition

The parameter number is flashing in the BOP-2 display.

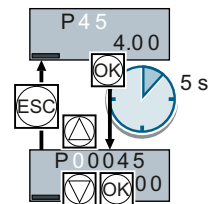
Procedure



- 1.
- 2.

To select the parameter number directly, proceed as follows:

1. Press the OK button for longer than five seconds.
2. Change the parameter number digit-by-digit.
If you press the OK button then the BOP-2 jumps to the next digit.
3. If you have entered all of the digits of the parameter number, press the OK button.



You have now entered the parameter number directly.

Entering the parameter value directly

The BOP-2 offers the option of setting the parameter value digit by digit.

Precondition

The parameter value flashes in the BOP-2 display.

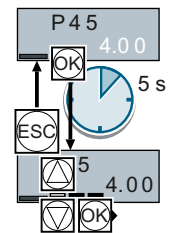
Procedure



- 1.
- 2.

To select the parameter value directly, proceed as follows:

1. Press the OK button for longer than five seconds.
2. Change the parameter value digit-by-digit.
If you press the OK button then the BOP-2 jumps to the next digit.
3. If you have entered all of the digits of the parameter value, press the OK button.

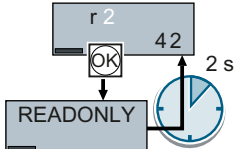
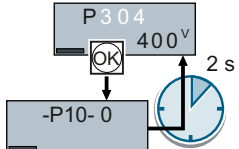
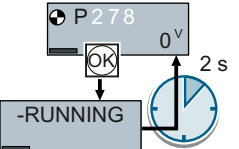


You have now entered the parameter value directly.

A.2.4 A parameter cannot be changed

When cannot you change a parameter?

The inverter indicates why it currently does not permit a parameter to be changed:

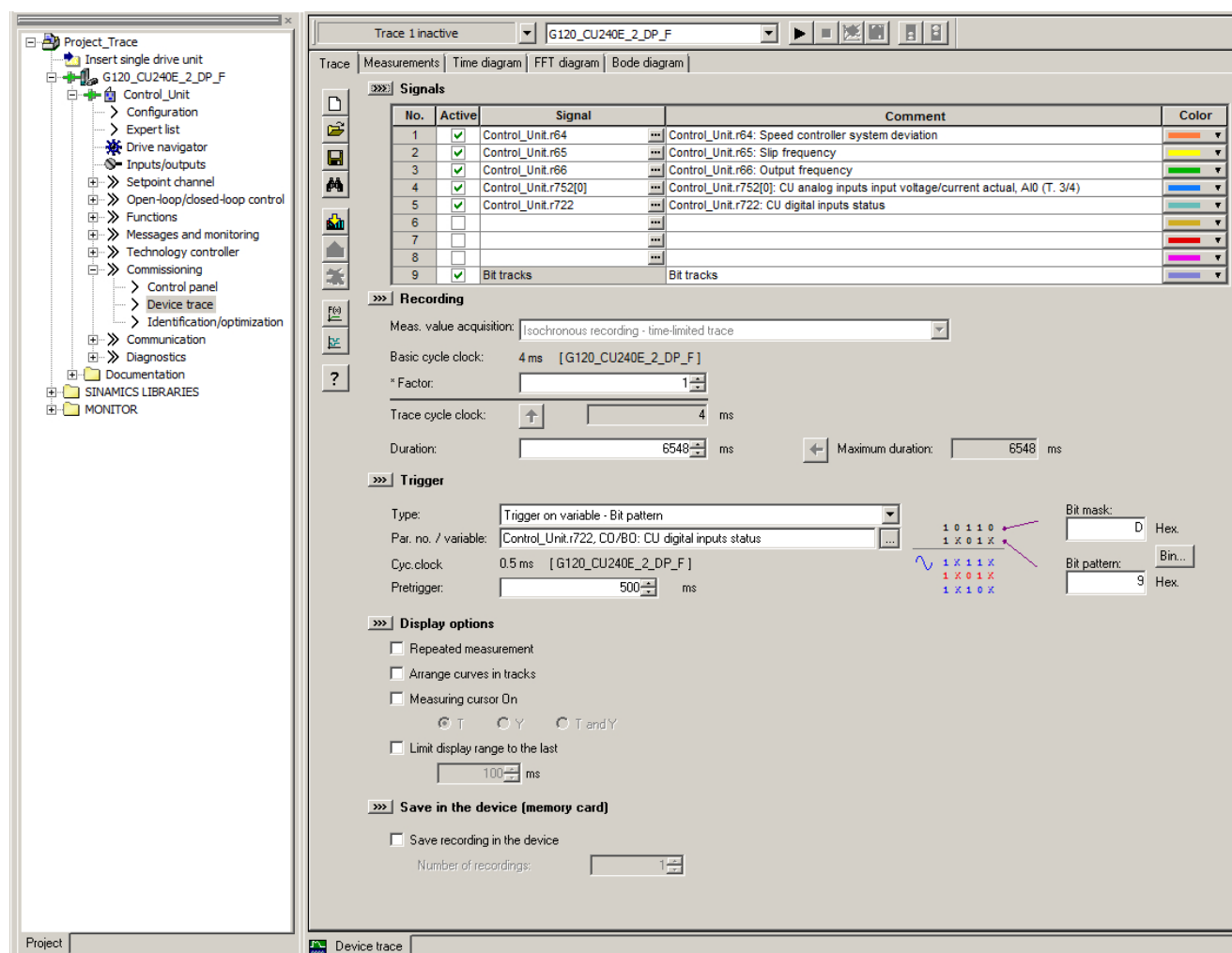
Read parameters cannot be adjusted	The parameter can only be adjusted during quick commissioning.	A parameter can only be adjusted when the motor is switched off
		

The operating state in which you can change a parameter is provided in the List Manual for each parameter.

A.3 The device trace in STARTER

Description

The device trace graphically displays inverter signals with respect to time.



Signals

In two settings that are independent of one another, using  you can interconnect eight signals each.

Recording

You can start a measurement as frequently as you require. As long as you do not exit START, the results remain under the "Measurements" tab with data and time. When terminating STARTER or under the "Measurements" tab, you can save the measurement results in the *.trc format.

If you require more than two settings for your measurements, you can either save the individual settings in the project or export them in *.clg format, and load or import them, if necessary.

You can record individual bits of a parameter (e.g. r0722. 1) by allocating the relevant bit using "bit track" ().

Using the mathematical function () you can define a curve, for example, the difference between the speed setpoint and the speed actual value.

The device trace shows "individual bits" or "mathematical functions" as signal No. 9.

Recording cycle and duration

The device trace records data in a CU-dependent basic cycle clock. The maximum recording duration depends on the number of recorded signals and the trace clock cycle.

Proceed as follows to extend the recording duration:

1. Multiply the trace clock cycle by an integral number.
2. Accept the displayed maximum duration using .

Alternatively, you can also specify the measurement period and then calculate the trace clock cycle of STARTER using .

Trigger (condition to start the device trace)

The device trace starts as soon as you press the  (start trace) button.

Using the button , you can define another trigger to start the device trace.

The pretrigger defines the time in which the signals are traced before the trigger condition. As a consequence, the trigger condition traces itself.

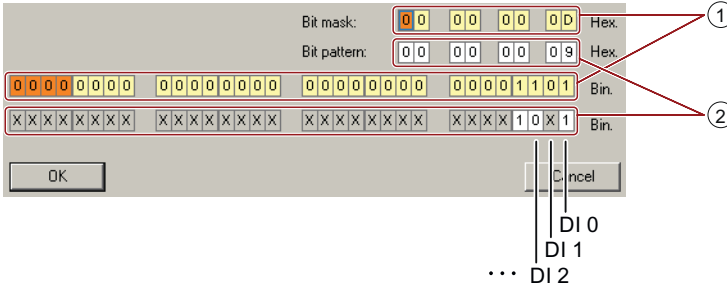
Example of a bit pattern as trigger:

You must define the pattern and value of a bit parameter for the trigger. To do so, proceed as follows:

Using , select "Trigger to variable - bit pattern"

Using , select the bit parameter

Using , open the screen form in which you set the bits and their values for the start condition



- ① Select the bits for the trace trigger, upper row hex format, lower row binary format
- ② Define the bits for the trace trigger, upper row hex format, lower row binary format

Figure A-3 Trigger as bit pattern of r0722 (status of the digital inputs)

In the example, the trace starts if digital inputs DI 0 and DI 3 are high, and DI 2 is low. The state of the other digital inputs is not relevant for the trigger condition.

Further, you can either set an alarm or fault as start condition.

Display options

In this area, you can set how the measurement results are displayed.

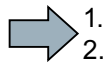
- Repeat measurement
This places the measurements that you wish to perform at different times above one other.
- Arrange curves in tracks
This defines whether the device trace will show all measured values on a common zero line or on separate zero lines.
- Measuring cursor On
This allows you to analyze the measuring intervals in more detail.

Save to device (memory card)

In this area, you define whether the device trace will save successive measurements in directory /USER/SINAMICS/DATA/TRACE on a inserted memory card.

Displaying the measurements backed up on the memory card

Procedure



1. To display the measurements backed up on the memory card, proceed as follows:

1. Insert the memory card into a card reader.
2. Go to the "Measurements" tab in the device trace.
3. Open the read out ACX files with the "Open measurements" button.



STARTER shows the measurements backed up on the memory card.

A.4 Interconnecting signals in the inverter

A.4.1 Fundamentals

The following functions are implemented in the inverter:

- Open-loop and closed-loop control functions
- Communication functions
- Diagnosis and operating functions

Every function comprises one or several blocks that are interconnected with one another.

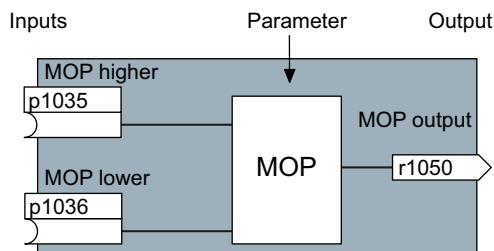


Figure A-4 Example of a block: Motorized potentiometer (MOP)

Most of the blocks can be adapted to specific applications using parameters.

You cannot change the signal interconnection within the block. However, the interconnection between blocks can be changed by interconnecting the inputs of a block with the appropriate outputs of another block.

The signal interconnection of the blocks is realized, contrary to electric circuitry, not using cables, but in the software.

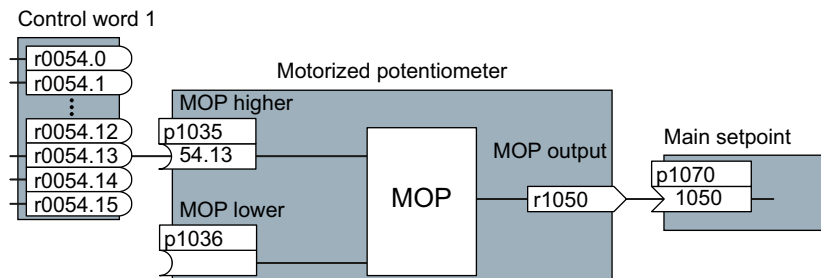


Figure A-5 Example: Signal interconnection of two blocks for digital input 0

Binectors and connectors

Connectors and binectors are used to exchange signals between the individual blocks:

- Connectors are used to interconnect "analog" signals (e.g. MOP output speed)
- Binectors are used to interconnect digital signals (e.g. "Enable MOP up" command)

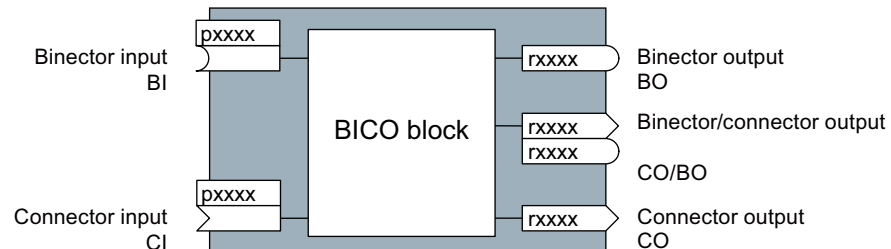


Figure A-6 Symbols for binector and connector inputs and outputs

Binector/connector outputs (CO/BO) are parameters that combine more than one binector output in a single word (e.g. r0052 CO/BO: status word 1). Each bit in the word represents a digital (binary) signal. This summary reduces the number of parameters and simplifies parameter assignment.

Binector or connector outputs (CO, BO or CO/BO) can be used more than once.

Interconnecting signals

When must you interconnect signals in the inverter?

If you change the signal interconnection in the inverter, you can adapt the inverter to a wide range of requirements. This does not necessarily have to involve highly complex functions.

Example 1: Assign a different function to a digital input.

Example 2: Switch the speed setpoint from the fixed speed to the analog input.

Principle when connecting BICO blocks using BICO technology

When interconnecting the signal, the following principle applies: **Where does the signal come from?**

An interconnection between two BICO blocks consists of a connector or a binector and a BICO parameter. The input of a block must be assigned the output of a different block: In the BICO parameters, enter the parameter numbers of the connector/binector that should supply its output signal to the BICO parameter.

How much care is required when you change the signal interconnection?

Note which changes you make. A subsequent analysis of the set signal interconnections is possible only by evaluating the parameter list.

We recommend that you use the STARTER and Startdrive commissioning tools for setting the signal interconnections.

Where can you find additional information?

- This manual suffices for assigning a different meaning to the digital inputs.
- The parameter list in the List Manual is sufficient for more complex signal interconnections.
- The function diagrams in the List Manual provide a complete overview of the factory setting for the signal interconnections and the setting options.

A.4.2 Example**Moving a basic control logic into the inverter**

A conveyor system is to be configured in such a way that it can only start when two signals are present simultaneously. These could be the following signals, for example:

- The oil pump is running (the required pressure level is not reached, however, until after 5 seconds)
- The protective door is closed

To implement this task, you must insert free function blocks between digital input 0 and the command to switch on the motor (ON/OFF1).

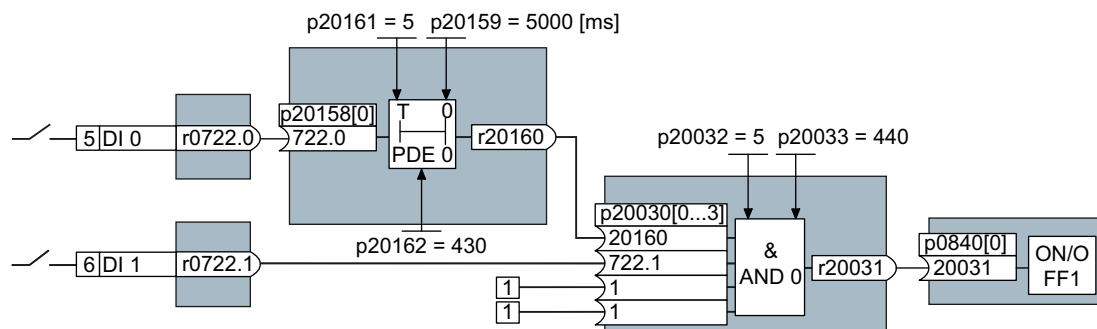


Figure A-7 Example: Signal interconnection for control logic

The signal of digital input 0 (DI 0) is fed through a time block (PDE 0) and is interconnected with the input of a logic block (& AND 0). The signal of digital input 1 (DI 1) is interconnected to the second input of the logic block. The logic block output issues the ON/OFF1 command to switch-on the motor.

Setting the control logic

Parameter	Description
p20161 = 5	The time block is enabled by assigning to runtime group 5 (time slice of 128 ms)
p20162 = 430	Run sequence of the time block within runtime group 5 (processing before the AND logic block)
p20032 = 5	The AND logic block is enabled by assigning to runtime group 5 (time slice of 128 ms)

Parameter	Description
p20033 = 440	Run sequence of the AND logic block within runtime group 5 (processing after the time block)
p20159 = 5000.00	Setting the delay time [ms] of the time module: 5 seconds
p20158 = 722.0	Connect the status of DI 0 to the input of the time block r0722.0 = Parameter that displays the status of digital input 0.
p20030[0] = 20160	Interconnect the timer block to the 1st input of the AND
p20030[1] = 722.1	Interconnect the status of DI 1 with the 2nd AND input r0722.1 = Parameter that displays the status of digital input 1.
p0840 = 20031	Interconnect the AND output to ON/OFF1

Explanation of the example using the ON/OFF1 command

Parameter p0840[0] is the input of the "ON/OFF1" block of the inverter. Parameter r20031 is the output of the AND block. To interconnect ON/OFF1 with the output of the AND block, set p0840 = 20031.

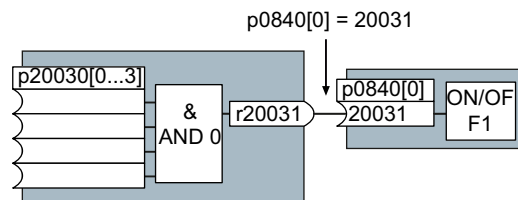


Figure A-8 Interconnecting blocks by setting p0840[0] = 20031

A.5 Connecting a fail-safe digital input

The following examples show the interconnection of a fail-safe digital input corresponding to PL d according to EN 13849-1 and SIL2 according to IEC61508. You can find additional examples and information in the "Safety Integrated" function manual.

Special requirements placed on EMC-compliant installation

Use shielded signal cables. Connect the shield at both conductor ends.

In order to connect two or more inverter terminals, use the shortest possible jumpers directly at the terminals themselves.

PM-switching and PP-switching fail-safe digital outputs

The inverter allows a fail-safe PM-switching digital output as well as a PP-switching fail-safe digital output to be connected.

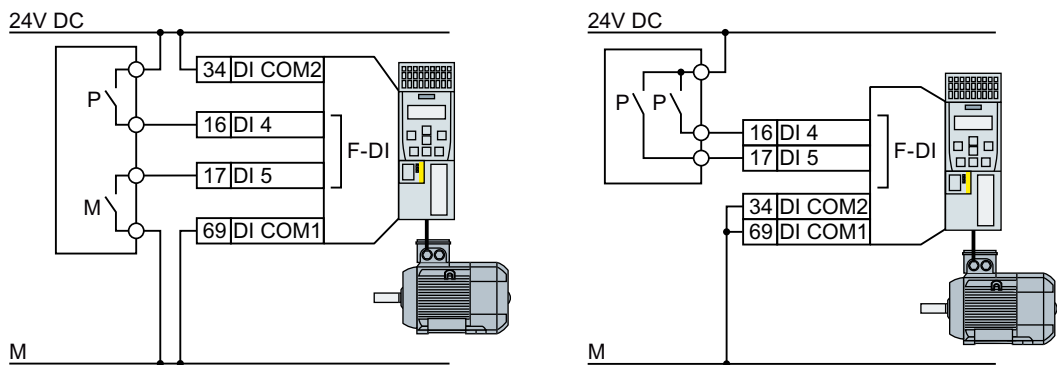


Figure A-9 Connecting a PM-switching and PP-switching fail-safe digital output

Connection examples

The following examples comply with PL d according to EN 13849-1 and SIL2 according to IEC 61508 for the case that all components are installed in one control cabinet.

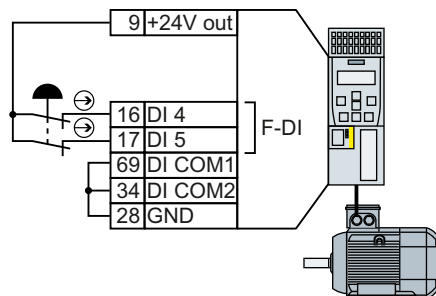


Figure A-10 Connecting a sensor, e.g. Emergency Stop mushroom pushbutton or limit switch

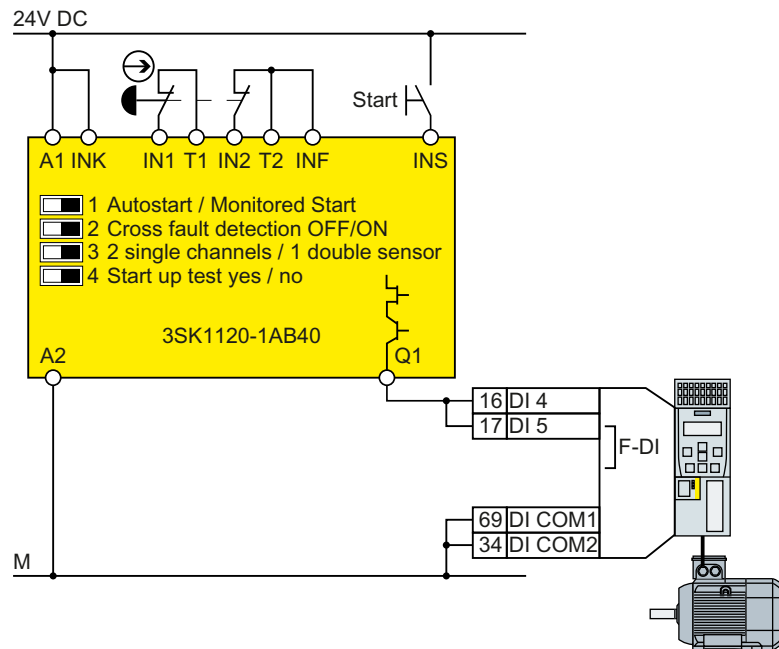


Figure A-11 Connecting a safety relay, e.g. SIRIUS 3SK11

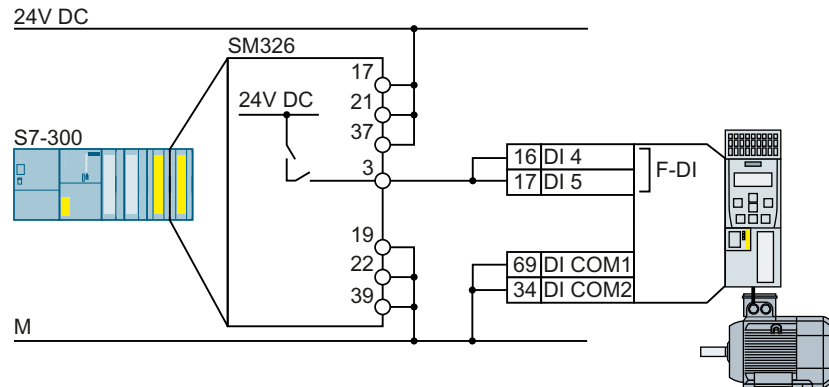


Figure A-12 Connecting an F digital output module, e.g. SIMATIC F digital output module

The Safety Integrated function manual provides additional connection options and connections in separate control cabinets.

 Overview of the manuals (Page 504)

A.6 Acceptance tests for the safety functions

A.6.1 Recommended acceptance test

The following descriptions for the acceptance test are recommendations that illustrate the principle of acceptance. You may deviate from these recommendations if you check the following once you have completed commissioning:

- Correct assignment of the interfaces of each converter with the safety function:
 - Fail-safe inputs
 - PROFIsafe address
- Correct setting of the STO safety function.

Note

Perform the acceptance test with the maximum possible velocity and acceleration in order to test the expected maximum braking distances and braking times.

Note

Non-critical alarms

The following alarms are issued following each system ramp-up and are not critical for acceptance:

- A01697
 - A01796
-

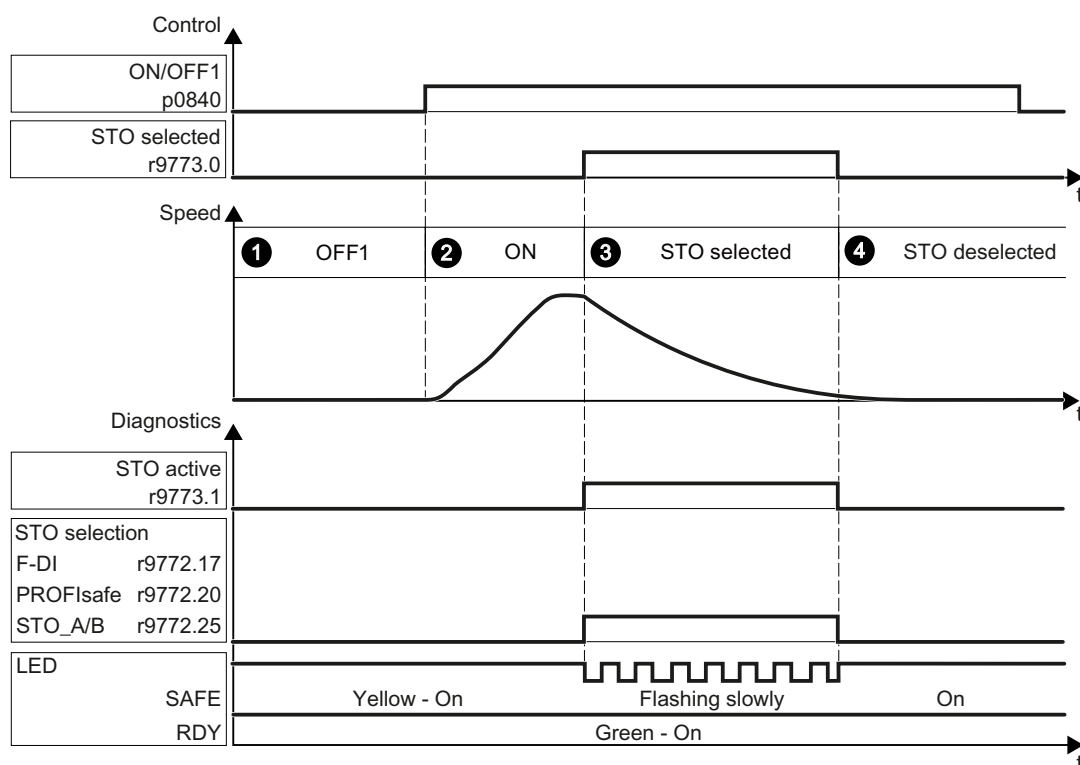


Figure A-13 Acceptance test for STO (basic functions)

Procedure

- ➔ 1. To perform an acceptance test of the STO function as part of the basic functions, proceed as follows:
2.

			Status
1.	The inverter is ready		
	The inverter signals neither faults nor alarms of the safety functions (r0945[0...7], r2122[0...7]).		
	STO is not active (r9773.1 = 0).		
2.	Switch on motor		
	2.1.	Enter a speed setpoint ≠ 0.	
	2.2.	Switch on the motor (ON command).	
	2.3.	Check that the correct motor is running.	

				Status
3.	Select STO			
3.1.	Select STO while the motor is running. <i>Test each configured activation, e.g. via digital inputs and PROFIsafe.</i>			
3.2.	Check the following:			
	When controlled via PROFIsafe	When controlled via a fail-safe F-DI digital input	When controlled via STO_A and STO_B terminals on a PM240-2 or PM240P-2 Power Module	
	The inverter signals the following: "STO selection via PROFIsafe" (r9772.20 = 1)	The inverter signals the following: "STO Selection via terminal" (r9772.17 = 1)	The inverter signals the following: "STO Selection via terminal on Power Module" (r9772.25 = 1)	
	If a mechanical brake is not available, the motor coasts down. A mechanical brake brakes the motor and holds it to ensure that it remains at standstill.			
	The inverter signals neither faults nor alarms of the safety functions (r0945[0...7], r2122[0...7]).			
	The inverter signals the following: "STO is selected" (r9773.0 = 1). "STO is active" (r9773.1 = 1).			
4.	Deselect STO			
4.1.	Deselect STO.			
4.2.	Check the following:			
	STO is not active (r9773.1 = 0).			
	The inverter signals neither faults nor alarms of the safety functions (r0945[0...7], r2122[0...7]).			



You have performed the acceptance test of the STO function.

A.6.2 Machine documentation

Machine or plant description

Designation	
Type	
Serial number	
Manufacturer	
End customer	
Overview diagram of the machine and/or system:	

Inverter data

The inverter data include the hardware version of the safety-relevant inverter.

Labeling the drive	Article number and hardware version of the inverter

Function table

The active safety functions depending on the operating mode and safety equipment are shown in the function table.

Operating mode	Safety equipment	Drive	Selected safety function	Checked

Table A-7 Example of a function table

Operating mode	Safety equipment	Drive	Selected safety function	Checked
<i>Automatic</i>	<i>Protective door closed</i>	<i>Conveyor belt</i>	---	---
	<i>Protective door open</i>	<i>Conveyor belt</i>	<i>STO</i>	
	<i>Emergency Stop button pressed</i>	<i>Conveyor belt</i>	<i>STO</i>	

Acceptance test reports

File name of the acceptance reports	

Data backup

Data	Storage medium			Holding area
	Archiving type	Designation	Date	
Acceptance test reports				
PLC program				
Circuit diagrams				

Countersignatures**Commissioning engineer**

The commissioning engineer confirms that the tests and checks listed above have been correctly executed.

Date	Name	Company/dept.	Signature
...	...	...	...

Machine manufacturer

The machine OEM confirms the correctness of the settings documented above.

Date	Name	Company/dept.	Signature
...	...	...	...

A.6.3 Documenting the settings for the basic functions, firmware V4.4 ... V4.7 SP6

Drive = <pDO-NAME_v>

Table A-8 Firmware version

Name	Number	Value
Control Unit firmware version	r18	<r18_v>
SI version, safety functions integrated in the drive (processor 1)	r9770	<r9770_v>

Table A-9 Monitoring cycle

Name	Number	Value
SI monitoring clock cycle (processor 1)	r9780	<r9780_v>

Table A-10 Checksums

Name	Number	Value
SI module identifier, Control Unit	r9670	<r9670_v>
SI module identifier, Power Module	r9672	<r9672_v>
SI reference checksum SI parameters (processor 1)	p9799	<p9799_v>
SI reference checksum SI parameters (processor 2)	p9899	<p9899_v>

Table A-11 Settings of the safety functions

Name	Number	Value
SI enable, functions integrated in the drive	p9601	<p9601_v>
<i>Only for the CU250S-2 Control Unit</i> SI enable safe brake control	p9602	<p9602_v>
SI PROFIsafe address	p9610	<p9610_v>
F-DI switch over discrepancy time	p9650	<p9650_v>
SI STO debounce time	p9651	<p9651_v>
<i>Only for the CU250S-2 Control Unit</i> SI Safe Stop 1 delay time	p9652	<p9652_v>
SI forced dormant error detection timer	p9659	<p9659_v>
SI forced checking procedure STO via PM terminals time	p9661	<p9661_v>

Table A-12 Safety logbook

Name	Number	Value
SI change control checksum	r9781[0]	<r9781[0]_v>
SI change control checksum	r9781[1]	<r9781[1]_v>
SI change control time stamp	r9782[0]	<r9782[0]_v>
SI change control time stamp	r9782[1]	<r9782[1]_v>

A.7 Manuals and technical support

A.7.1 Overview of the manuals



Manuals with additional information that can be downloaded

- CU240B/E-2 Compact Operating Instructions (<https://support.industry.siemens.com/cs/ww/en/view/109477361>)
 Commissioning the inverter.
- CU240B/E-2 operating instructions (<https://support.industry.siemens.com/cs/ww/en/view/109478828>)
 Installing, commissioning and maintaining the inverter. Advanced commissioning (this manual)
- EMC installation guideline (<http://support.automation.siemens.com/WW/view/en/60612658>)
 EMC-compliant switch cabinet design, potential equalization and cable routing
- "Safety Integrated" function manual (<https://support.industry.siemens.com/cs/ww/en/view/109477367>)
 Configuring PROFI-safe. Installing, commissioning and operating fail-safe functions of the inverter.
- "Fieldbus" function manual (<https://support.industry.siemens.com/cs/ww/en/view/109477369>)
 Configuring fieldbuses.
- CU240B/E-2 List Manual (<https://support.industry.siemens.com/cs/ww/en/view/109477251>)
 Parameter list, alarms and faults. Graphic function diagrams
- BOP-2 operating instructions (<https://support.industry.siemens.com/cs/ww/en/view/42185248>)
 Using the Operator Panel.
- IOP operating instructions (<https://support.industry.siemens.com/cs/ww/en/view/109478559>)
 Using the Operator Panel. Mounting the door mounting kit for IOP.
- Application manual IOP (<https://support.industry.siemens.com/cs/ww/en/view/109483443>)
 The commissioning wizards in the IOP

- Power Module Installation Manual (<https://support.industry.siemens.com/cs/ww/en/ps/13224/man>)
Installing Power Modules, reactors and filters. Technical data, maintenance
 
- Accessories manual (<https://support.industry.siemens.com/cs/ww/en/ps/13225/man>)
Installation descriptions for inverter components, e.g. line reactors and line filters. The printed installation descriptions are supplied together with the components.


Finding the most recent edition of a manual

If there are multiple editions of a manual, select the latest edition:

> Manual Fieldbus systems: PROFINET, PROFIBUS, EtherNet/IP, CANopen, USS, Bacnet, Modbus, P1

04/2014, FW V4.7.3 Function manual, A5E34229197B AA

For products: 6SL3544-0MB02-1PA0, 6SL3244-0BB13-1FA0, ... > All products

 View of this manual

04/2015, FW V4.7.3

04/2015, FW V4.7.3

04/2014, FW V4.7.3

08/11/2014
ID: 99685159
★★★★☆ (3)

Configuring a manual

Further information about the configurability of manuals is available in the Internet:



MyDocumentationManager (<https://www.industry.siemens.com/topics/global/en/planning-efficiency/documentation/Pages/default.aspx>).

Select "Display and configure" and add the manual to your "mySupport-documentation":

Function manual
Function Manual
Article number of the documentation: A5E34229197B AA
Description / topic: 04/2014, FW V4.7,

 Show and configure
 Download (5644 KB)



mySupport Cockpit

> Add to mySupport favorites
> Add to mySupport documentation
> Fav

Not all manuals can be configured.

The configured manual can be exported in RTF, PDF or XML format.

A.7.2 Configuring support

Catalog

Ordering data and technical information for SINAMICS G inverters.



Catalogs for download or online catalog (Industry Mall):



Everything about SINAMICS G120 (www.siemens.com/sinamics-g120)

SIZER

The configuration tool for SINAMICS, MICROMASTER and DYNAVERT T drives, motor starters, as well as SINUMERIK, SIMOTION controllers and SIMATIC technology



SIZER on DVD:

Article number: 6SL3070-0AA00-0AG0



Download SIZER (<http://support.automation.siemens.com/WW/view/en/10804987/130000>)

EMC (electromagnetic compatibility) technical overview

Standards and guidelines, EMC-compliant control cabinet design



EMC overview (<https://support.industry.siemens.com/cs/ww/en/view/103704610>)

EMC Guidelines configuration manual

EMC-compliant control cabinet design, potential equalization and cable routing



EMC installation guideline (<http://support.automation.siemens.com/WW/view/en/60612658>)

Safety Integrated for novices technical overview

Application examples for SINAMICS G drives with Safety Integrated



Safety Integrated for novices (<https://support.industry.siemens.com/cs/ww/en/view/80561520>)

A.7.3 Product Support



You can find additional information on the product and more in the Internet under (<http://www.siemens.com/automation/service&support>)

This address provides the following:

- Actual product information (product memorandums), FAQs (frequently asked questions), downloads.
- The Newsletter contains the latest information on the products you use.
- The Knowledge Manager (Intelligent Search) helps you find the documents you need.
- Users and specialists from around the world share their experience and knowledge in the Forum.
- You can find your local representative for Automation & Drives via our contact database under "Contact & Partner".
- Information about local service, repair, spare parts and much more can be found under "Services".

A.8 Mistakes and improvements

If you come across any mistakes when reading this manual or if you have any suggestions for how it can be improved, then please send your suggestions to the following address or by E-mail:

Siemens AG
Digital Factory
Motion Control
Postfach 3180
91050 Erlangen, Germany

E-mail (<mailto:docu.motioncontrol@siemens.com>)

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